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Math

Contents

I	Set Theory	7
1	Binary Relations	8
1.1	Definition and Basic Properties	8
1.1.1	Examples of Binary Relations	9
1.1.2	Operations on Binary Relations	9
1.1.3	The Complement of a Binary Relation	10
1.1.4	Converse of a Binary Relation	11
1.2	Composition of Binary Relations	13
1.2.1	Basic Properties of Composition	13
1.2.2	Additional Properties of Composition	15
1.2.3	Restriction and Corestriction, Direct and Inverse Images	17
2	Heterogenous Binary Relations	25
2.1	Definition	25
2.2	Classification	25
2.3	Induced Functions on Power Sets	33
3	Homogeneous Relations	35
3.1	Definition	35
3.2	Reflexivity, Symmetry, and Transitivity	35
3.3	Special Types of Homogeneous Relations	37
3.4	Properties of Homogeneous Relations	37
3.4.1	Preorders and Induced Equivalence Relations	38
3.4.2	Relationship Between Strict and Non-Strict Partial Orders	38
II	Linear Algebra	40
4	Modules	41
4.1	Rings	41
4.2	Modules	42
5	Foundations of Vector Spaces	43
5.1	Vector Spaces	44
5.2	Minkowski Structure on $\mathcal{P}(V)$	50
5.3	Vector Subspaces	54
5.4	Span	59
5.5	Linear Independence	63
5.6	Internal Direct Sums	66
5.7	Bases and Dimension	70

5.8	Ordered Bases and Coordinates	79
5.9	Flags and Chains of Subspaces	82
5.9.1	Ordered Bases and Set-Theoretic Requirements	82
5.9.2	Flags	82
5.9.3	Flags from Ordered Bases	83
5.9.4	From a Complete Flag Back to a Basis: Jumps	85
5.10	Chapter Compendium	87
6	Linear Maps	88
6.1	Linear Transformations	89
6.2	Universal Property of Vector Spaces	92
6.3	Fundamental Subspaces	95
6.4	Hom Spaces as Vector Spaces	97
6.5	Endomorphism Algebras	99
6.6	Dimension of Hom Spaces	100
6.7	Injectivity, Surjectivity, and Bijectivity	101
6.8	The Rank-Nullity Theorem	110
6.9	Chapter Compendium	112
7	Matrices	114
7.1	Matrices	115
7.1.1	Why Matrices?	115
7.1.2	Linear Maps Between Coordinate Spaces and Their Grids	115
7.1.3	Transport of Structure	118
7.1.4	Practical Examples	120
7.2	Matrices as Linear Maps	121
7.3	Coordinates and Actions: A Pattern Beyond Linear Algebra	123
7.3.1	The Set Picture: Enumerations and Permutations	123
7.3.2	The Same Pattern in Vector Spaces	125
7.4	The Matrix of a Linear Map	127
7.4.1	Columns and the Coordinate Formula	127
7.4.2	Composition and Change of Bases	128
7.4.3	Invariance of Rank and Nullity	131
7.4.4	Variance, and a Forward Reference	131
7.4.5	Examples	132
7.5	Invertibility	133
7.5.1	Invertible Matrices	133
7.5.2	The Inverse of the Matrix of a Map	133
7.5.3	Worked Example	134
8	Subspace Representations	135
8.1	Equivalence and Similarity of Linear Maps	136
8.1.1	Worked Example	140
8.2	Reduction: Gaussian Elimination and Normal Forms	141

9 Quotient Spaces and Isomorphism Theorems	142
9.1 Quotient Space Construction	143
9.2 Examples of Quotient Spaces	146
9.3 Dimension of Quotient Spaces	148
9.4 Universal Property of Quotient Spaces	149
9.5 Isomorphism Theorems	151
9.5.1 First Isomorphism Theorem	151
9.5.2 Second Isomorphism Theorem	152
9.5.3 Third Isomorphism Theorem	154
9.5.4 Fourth Isomorphism Theorem (Lattice Correspondence)	156
9.6 Chapter Compendium	160
10 Products, Hom Spaces, and Dual Spaces	161
10.1 Direct Products and Coproducts	162
10.1.1 The Categorical Product	162
10.1.2 The Categorical Coproduct	163
10.1.3 Finite Case: Biproduct Structure	166
10.1.4 Dimensions of Products and Coproducts	168
10.2 Internal vs External Direct Sums	171
10.3 Hom-Product Isomorphisms	172
10.4 Free Vector Space Characterizations	176
10.5 Dimension of Hom Spaces	176
10.6 Dual Spaces	178
10.7 Dual Basis	179
10.8 Annihilators	181
10.9 Hyperplanes	183
10.10 Dual of a Linear Map	185
10.11 Bidual and Canonical Isomorphism	186
10.12 Chapter Compendium	187
11 Endomorphisms and Spectral Theory	189
12 Multilinear Algebra and Determinants	190
13 Normed and Inner Product Spaces	191
III Algebra III	192
14 3. Prime and Irreducible elements	193
IV Measure Theory	196
15 Set-Theoretic Foundations	197
15.1 π -Systems	197
15.2 Algebras of Sets	199
15.3 Monotone Classes	201
15.4 σ -Algebras	203
15.4.1 The Borel Hierarchy	205

15.5 Set Decomposition Lemmas	207
15.6 λ -Systems (Dynkin Systems)	208
15.7 Monotone Class Lemma	211
15.8 Dynkin's π - λ Theorem	213
16 Contents	216
16.1 Contentable Spaces	216
16.2 Definition of Content	216
16.3 Jordan Content on $\mathbb{R}^n$	217
16.4 Inner and Outer Content	219
16.4.1 Jordan Inner and Outer Content	219
17 The Riemann Integral	223
17.1 Riemann/Jordan Simple Functions	223
17.2 Upper and Lower Riemann Sums	223
17.3 Riemann Integral	224
17.4 Riemann Double Integral	224
17.5 Limitations of Riemann Integration	225
18 Measures	226
18.1 Measurable Spaces	226
18.2 Definition of Measure	226
18.3 Outer Measures	228
18.4 Complete Measures and Completion	228
18.5 Carathéodory's Theorem	231
18.6 Lebesgue Measure	233
18.7 Regularity	233
18.8 Completion	234
18.9 Uniqueness of Measures	234
19 Measurable Functions	236
19.1 Definition and Basic Properties	236
19.1.1 Pullback and Pushforward σ -Algebras	237
19.1.2 Pushforward Measure	237
19.2 Simple Functions	239
19.3 σ -Simple Functions	240
19.4 σ -Simple Function Approximation	240
19.5 Simple Function Approximation	242
20 The Lebesgue Integral	244
20.1 Integration of Positive Functions	244
20.2 Monotone Convergence Theorem	248
20.3 Fatou's Lemma	249
20.4 Integrable Functions	249
20.5 Dominated Convergence Theorem	250
20.6 Change of Variables	251
20.7 Riemann vs Lebesgue	252

21 Product Measures	254
21.1 Product σ -Algebras	254
21.2 Sections of Sets and Functions	257
21.3 Product Measures	259
21.4 Tonelli's Theorem	262
21.5 Fubini's Theorem	265
21.6 Consequences of Fubini-Tonelli	268
21.6.1 Young's Inequality	269
21.6.2 Layer Cake Formula	270
22 The Radon-Nikodym Theorem	273
22.1 Signed Measures	273
22.2 Absolute Continuity and Singularity	275
22.3 The Radon-Nikodym Theorem	275
22.4 Properties of the Radon-Nikodym Derivative	277
23 L^p Spaces	279
23.1 Definitions	279
23.2 Young's Inequality	280
23.3 Hölder's Inequality	280
23.4 Minkowski's Inequality	281
23.5 Completeness	283
24 Convergence in Measure	285
24.1 Definition	285
24.2 Relations Between Convergence Modes	285
24.3 Riesz's Theorem	285
24.4 Vitali Convergence	285
25 Applications	287
25.1 Differentiation Under the Integral	287
25.2 Gaussian Integral	287
26 Exercises	288
26.1 σ -Algebras and Borel Sets	288
26.2 Measures	288
26.3 Carathéodory and Lebesgue Measure	289
26.4 Riemann vs Lebesgue	289
26.5 Measurable Functions	290
26.6 Convergence Theorems	290
26.7 L^p Spaces	291
26.8 Product Measures	291
26.9 Convergence in Measure	292
26.10 Parameter Integrals	292
26.11 Modes of Convergence	292
26.12 Simple Functions and Integration	293

Part I

Set Theory

CHAPTER 1

Binary Relations

1.1 Definition and Basic Properties

The following are equivalent for any set R :

- $(\exists A) (\exists B) (R \subseteq A \times B)$.
- $\forall w \in R: (\exists x) (\exists y) (w = (x, y))$.

Proof:

($\Rightarrow$) Assume $(\exists A) (\exists B) (R \subseteq A \times B)$.

Let $w \in R$ be arbitrary. Since $R \subseteq A \times B$, we have $w \in A \times B$. By definition of Cartesian product, $w = (a, b)$ for some $a \in A$ and $b \in B$.

Taking $x = a$ and $y = b$, we have shown that $\forall w \in R: (\exists x) (\exists y) (w = (x, y))$.

($\Leftarrow$) Assume $\forall w \in R: (\exists x) (\exists y) (w = (x, y))$.

We need to find sets A and B such that $R \subseteq A \times B$.

By the **Axiom of Replacement**, since for each $w \in R$ there exists a unique first component x such that $w = (x, y)$ for some y , we can form the set:

$$A := \{x \mid (\exists y) ((x, y) \in R)\}$$

Similarly, we can form the set:

$$B := \{y \mid (\exists x) ((x, y) \in R)\}$$

More precisely, by the Axiom of Replacement applied to the function $f(x) = \text{"first component of } x\text{"}$ on the set R , we obtain A . Similarly for B using the function $g(x) = \text{"second component of } x\text{"}$.

Now let $w \in R$. By assumption, $(\exists x) (\exists y) (w = (x, y))$. By construction of A and B using replacement, we have $x \in A$ and $y \in B$, so $w = (x, y) \in A \times B$.

Therefore $R \subseteq A \times B$, which proves $(\exists A) (\exists B) (R \subseteq A \times B)$. QED

Remark:

Notice that the sets A and B are not unique, but they must obey $\mathcal{D}(R) \subseteq A$ and $\mathcal{R}(R) \subseteq B$.

Definition 1.1.1 (Binary Relations and Graphs of Binary Relations):

2 A **graph of a binary relation** is a set R that satisfies one of the following equivalent conditions above. A **binary relation** is a tuple (A, B, R) such that $R \subseteq A \times B$.

We will denote graphs of binary relations with capital letters, such as R, S etc. and binary relations with lowercase letters, such as r, s etc.

Definition 1.1.2 (Domain and Range for a Graph of a Binary Relation):

Let R be a graph of a binary relation.
The **domain** of R , denoted $\mathcal{D}(R)$, is the set:

$$\mathcal{D}(R) := \{x \mid (\exists y) ((x, y) \in R)\}$$

The **range** of R , denoted $\mathcal{R}(R)$, is the set:

$$\mathcal{R}(R) := \{y \mid (\exists x) ((x, y) \in R)\}$$

1.1.1 Examples of Binary Relations**Example 1.1.3 (Identity Relation):**

Let A be any set. The **identity graph** on A is defined as:

$$\text{Id}_A := \{(a, a) \mid a \in A\}$$

This is the identity relation $\text{id}_A := (A, A, \text{Id}_A)$ where $\text{Id}_A \subseteq A \times A$.

Example 1.1.4 (Empty Relation):

Let A and B be any sets. The **empty relation** from A to B is:

$$\emptyset \subseteq A \times B$$

This gives us the binary relation $(A, B, \emptyset)$.

Example 1.1.5 (Universal Relation):

Let A and B be any sets. The **universal relation** from A to B is:

$$\nabla_{A,B} := A \times B$$

This gives us the binary relation $(A, B, \nabla_{A,B})$ where every element of A is related to every element of B .

Remark:

These three examples represent the extreme cases:

- The identity relation relates each element only to itself
- The empty relation relates no elements
- The universal relation relates every possible pair of elements

1.1.2 Operations on Binary Relations

Since binary relations are defined as tuples (A, B, R) where $R \subseteq A \times B$, we can define set-theoretic operations on them when they share the same domain and codomain.

Definition 1.1.6 (Arbitrary Union of Binary Relations):

Let $\{r_i\}_{i \in I} = \{(A, B, R_i)\}_{i \in I}$ be a family of binary relations with the same domain A and codomain B . Their **union**, denoted $\bigcup_{i \in I} r_i$, is defined as:

$$\bigcup_{i \in I} r_i := \left(A, B, \bigcup_{i \in I} R_i \right)$$

Definition 1.1.7 (Arbitrary Intersection of Binary Relations):

Let $\{r_i\}_{i \in I} = \{(A, B, R_i)\}_{i \in I}$ be a family of binary relations with the same domain A and codomain B . Their **intersection**, denoted $\bigcap_{i \in I} r_i$, is defined as:

$$\bigcap_{i \in I} r_i := \left(A, B, \bigcap_{i \in I} R_i \right)$$

Definition 1.1.8 (Subset Relation for Binary Relations):

Let $r = (A, B, R)$ and $s = (A, B, S)$ be binary relations with the same domain A and codomain B . We say $r \subseteq s$ if and only if $R \subseteq S$.

1.1.3 The Complement of a Binary Relation**Definition 1.1.9 (Complement of a binary relation):**

Let $r = (A, B, R)$ be a binary relation. The **complement** of r , denoted r^c , is the binary relation (A, B, R^c) where:

$$R^c := \nabla_{A,B} \setminus R = \{(x, y) \in A \times B \mid (x, y) \notin R\}$$

Remark:

Note that the complement operation is only well-defined for binary relations (tuples with specified domain and codomain), not for arbitrary graphs of binary relations, since we need the universal set $A \times B$ to form the complement.

Proposition 1.1.10 (Properties of Complement for Binary Relations):

Let $\{r_i\}_{i \in I} = \{(A, B, R_i)\}_{i \in I}$ be a family of binary relations with the same domain and codomain. Then:

1. **Involution:** $(r^c)^c = r$ for any $r = (A, B, R)$
2. **De Morgan's Laws:** $(\bigcup_{i \in I} r_i)^c = \bigcap_{i \in I} r_i^c$ and $(\bigcap_{i \in I} r_i)^c = \bigcup_{i \in I} r_i^c$
3. **Antimonotonicity:** If $r = (A, B, R)$ and $s = (A, B, S)$ with $R \subseteq S$ then $s^c \subseteq r^c$
4. **Universal and Empty:** The universal relation $(A, B, \nabla_{A,B})$ has complement $(A, B, \emptyset)$, and vice versa

Proof:

(1) **Involution:**

$$\begin{aligned} (r^c)^c &= (A, B, ((A \times B) \setminus R)^c) \\ &= (A, B, (A \times B) \setminus ((A \times B) \setminus R)) \\ &= (A, B, R) = r \end{aligned}$$

(2) **De Morgan's Laws:** For the first law:

$$\begin{aligned} \left(\bigcup_{i \in I} r_i \right)^c &= \left(A, B, \left(\bigcup_{i \in I} R_i \right)^c \right) \\ &= \left(A, B, (A \times B) \setminus \left(\bigcup_{i \in I} R_i \right) \right) \\ &= \left(A, B, \bigcap_{i \in I} ((A \times B) \setminus R_i) \right) \\ &= \left(A, B, \bigcap_{i \in I} R_i^c \right) = \bigcap_{i \in I} r_i^c \end{aligned}$$

For the second law:

$$\begin{aligned}
 \left(\bigcap_{i \in I} r_i \right)^c &= \left(A, B, \left(\bigcap_{i \in I} R_i \right)^c \right) \\
 &= \left(A, B, (A \times B) \setminus \left(\bigcap_{i \in I} R_i \right) \right) \\
 &= \left(A, B, \bigcup_{i \in I} ((A \times B) \setminus R_i) \right) \\
 &= \left(A, B, \bigcup_{i \in I} R_i^c \right) = \bigcup_{i \in I} r_i^c
 \end{aligned}$$

(3) Antimonotonicity: Assume $r = (A, B, R)$ and $s = (A, B, S)$ with $R \subseteq S$. Let $(x, y) \in S^c$. Then $(x, y) \in A \times B$ and $(x, y) \notin S$. Since $R \subseteq S$, if (x, y) were in R then it would be in S , which is not the case. Therefore $(x, y) \notin R$, so $(x, y) \in R^c$.

(4) Universal and Empty: Direct from the definition of complement.

QED

1.1.4 Converse of a Binary Relation

Definition 1.1.11 (Converse of the graph of a binary relation):

Let R be a graph of a binary relation. The **converse** (or **transpose**) of R , denoted R^T or R^{-1} , is defined as:

$$R^T := \{(y, x) \mid (x, y) \in R\}$$

Definition 1.1.12 (Converse of a binary relation):

Let $r = (A, B, R)$ be a binary relation. The **converse** of r , denoted r^T , is the binary relation (B, A, R^T) .

Proposition 1.1.13 (Properties of Converse):

Let $\{R_i\}_{i \in I}$ be a family of graphs of binary relations. Then:

1. **Involution:** $(R^T)^T = R$ for any graph R
2. **Domain and Range:** $\mathcal{D}(R^T) = \mathcal{R}(R)$ and $\mathcal{R}(R^T) = \mathcal{D}(R)$ for any graph R
3. **Arbitrary Union:** $(\bigcup_{i \in I} R_i)^T = \bigcup_{i \in I} R_i^T$
4. **Arbitrary Intersection:** $(\bigcap_{i \in I} R_i)^T = \bigcap_{i \in I} R_i^T$
5. **Monotonicity:** If $R \subseteq S$ then $R^T \subseteq S^T$
6. **Identity:** $\text{Id}_A^T = \text{Id}_A$ for any set A
7. **Empty and Universal:** $\emptyset^T = \emptyset$ and $(\nabla_{A,B})^T = \nabla_{B,A}$

Proof:

(1) Involution: For any (x, y) :

$$\begin{aligned}
 (x, y) \in (R^T)^T &\Leftrightarrow (y, x) \in R^T \\
 &\Leftrightarrow (x, y) \in R
 \end{aligned}$$

Therefore $(R^T)^T = R$.

(2) Domain and Range:

$$\begin{aligned}
\mathcal{D}(R^T) &= \{x \mid (\exists y) ((x, y) \in R^T)\} \\
&= \{x \mid (\exists y) ((y, x) \in R)\} \\
&= \{x \mid x \in \mathcal{R}(R)\} = \mathcal{R}(R)
\end{aligned}$$

Similarly, $\mathcal{R}(R^T) = \mathcal{D}(R)$.

(3) Arbitrary Union: For any (x, y) :

$$\begin{aligned}
(x, y) \in \left(\bigcup_{i \in I} R_i \right)^T &\Leftrightarrow (y, x) \in \bigcup_{i \in I} R_i \\
&\Leftrightarrow \exists i \in I : (y, x) \in R_i \\
&\Leftrightarrow \exists i \in I : (x, y) \in R_i^T \\
&\Leftrightarrow (x, y) \in \bigcup_{i \in I} R_i^T
\end{aligned}$$

(4) Arbitrary Intersection: For any (x, y) :

$$\begin{aligned}
(x, y) \in \left(\bigcap_{i \in I} R_i \right)^T &\Leftrightarrow (y, x) \in \bigcap_{i \in I} R_i \\
&\Leftrightarrow \forall i \in I : (y, x) \in R_i \\
&\Leftrightarrow \forall i \in I : (x, y) \in R_i^T \\
&\Leftrightarrow (x, y) \in \bigcap_{i \in I} R_i^T
\end{aligned}$$

(5) Monotonicity: Assume $R \subseteq S$. Let $(x, y) \in R^T$. Then $(y, x) \in R$. Since $R \subseteq S$, we have $(y, x) \in S$, which implies $(x, y) \in S^T$. Therefore $R^T \subseteq S^T$.

(6) Identity: For any (x, y) :

$$\begin{aligned}
(x, y) \in (\text{Id}_A)^T &\Leftrightarrow (y, x) \in \text{Id}_A \\
&\Leftrightarrow y = x \text{ and } y \in A \\
&\Leftrightarrow x = y \text{ and } x \in A \\
&\Leftrightarrow (x, y) \in \text{Id}_A
\end{aligned}$$

(7) Empty and Universal: For the empty relation: $(x, y) \in \emptyset^T \Leftrightarrow (y, x) \in \emptyset \Leftrightarrow \text{false} \Leftrightarrow (x, y) \in \emptyset$.

For the universal relation: $(x, y) \in (A \times B)^T \Leftrightarrow (y, x) \in A \times B \Leftrightarrow y \in A \text{ and } x \in B \Leftrightarrow (x, y) \in B \times A$. QED

Proposition 1.1.14 (Properties of Converse for Binary Relations):

Let $\{r_i\}_{i \in I} = \{(A, B, R_i)\}_{i \in I}$ be a family of binary relations with the same domain and codomain. Then:

1. **Involution:** $(r^T)^T = r$ for any relation r
2. **Domain and Codomain:** $\mathcal{D}(r^T) = \mathcal{R}(r)$ and $\mathcal{C}(r^T) = \mathcal{D}(r)$ for any relation r
3. **Arbitrary Union:** $(\bigcup_{i \in I} r_i)^T = \bigcup_{i \in I} r_i^T$
4. **Arbitrary Intersection:** $(\bigcap_{i \in I} r_i)^T = \bigcap_{i \in I} r_i^T$

5. **Monotonicity:** If $r \subseteq s$ then $r^T \subseteq s^T$
6. **Identity:** $\text{id}_A^T = \text{id}_A$
7. **Empty and Universal:** $\emptyset_{A,B}^T = \emptyset_{B,A}$ and $(\nabla_{A,B})^T = \nabla_{B,A}$

Proof:

These properties follow directly from the corresponding properties for graphs of binary relations, since the converse operation is defined componentwise on the underlying graph. QED

Example 1.1.15:

Let $A = \{1, 2\}$ and $B = \{a, b\}$. Consider the graph of a binary relation:

$$R = \{(1, a), (1, b), (2, a)\} \subseteq A \times B$$

Then its converse is:

$$R^T = \{(a, 1), (b, 1), (a, 2)\} \subseteq B \times A$$

Notice that $\mathcal{D}(R) = \{1, 2\} = \mathcal{R}(R^T)$ and $\mathcal{R}(R) = \{a, b\} = \mathcal{D}(R^T)$.

1.2 Composition of Binary Relations

Definition 1.2.1 (Composition of graphs of binary relations):

Let R and S be graphs of binary relations. The **composition** of R and S , denoted $S \circ R$, is defined as:

$$S \circ R := \{(a, c) \mid (\exists b) ((a, b) \in R \text{ and } (b, c) \in S)\}$$

Definition 1.2.2 (Composition of binary relations):

Let $r = (A, B, R)$ and $s = (B, C, S)$ be binary relations. The **composition** of r and s , denoted $s \circ r$, is the binary relation $(A, C, S \circ R)$.

Remark:

Note the order: $s \circ r$ means we first apply relation r , then relation s . This follows the standard function composition convention.

The composition of graphs of binary relations is defined for all graphs, but the composition of binary relations is only defined when the codomain of the first relation matches the domain of the second relation.

Example 1.2.3:

Let $A = \{1, 2\}$, $B = \{a, b, c\}$, and $C = \{x, y\}$. Consider:

$$R = \{(1, a), (1, b), (2, c)\}$$

$$S = \{(a, x), (b, x), (c, y)\}$$

Then:

$$S \circ R = \{(1, x), (2, y)\}$$

1.2.1 Basic Properties of Composition

Proposition 1.2.4 (Basic Properties of Composition for graphs):

Let R , S , and T be graphs of binary relations. Then:

1. $\mathcal{D}(S \circ R) \subseteq \mathcal{D}(R)$

2. $\mathcal{R}(S \circ R) \subseteq \mathcal{R}(S)$
3. $S \circ R = \{(a, c) \mid \exists b \in \mathcal{R}(R) \cap \mathcal{D}(S) : (a, b) \in R \text{ and } (b, c) \in S\}$
4. **Associativity:** $(T \circ S) \circ R = T \circ (S \circ R)$
5. **Identity:** $\text{Id}_B \circ R = R = R \circ \text{Id}_A$ when $R \subseteq A \times B$
6. **Converse of Composition:** $(S \circ R)^T = R^T \circ S^T$

Proof:

(1) Let $a \in \mathcal{D}(S \circ R)$. Then $\exists c (a, c) \in S \circ R$, so $\exists b (a, b) \in R$ and $(b, c) \in S$. Thus $a \in \mathcal{D}(R)$.

(2) Let $c \in \mathcal{R}(S \circ R)$. Then $\exists a (a, c) \in S \circ R$, so $\exists b (a, b) \in R$ and $(b, c) \in S$. Thus $c \in \mathcal{R}(S)$.

(3) By definition:

$$S \circ R = \{(a, c) \mid (\exists b) ((a, b) \in R \text{ and } (b, c) \in S)\}$$

If $(a, b) \in R$ then $b \in \mathcal{R}(R)$, and if $(b, c) \in S$ then $b \in \mathcal{D}(S)$. Thus:

$$(\exists b) ((a, b) \in R \text{ and } (b, c) \in S) \Leftrightarrow \exists b \in \mathcal{R}(R) \cap \mathcal{D}(S) : (a, b) \in R \text{ and } (b, c) \in S$$

Therefore the equality holds.

(4) **Associativity:** For any (a, d) :

$$\begin{aligned} (a, d) \in (T \circ S) \circ R &\Leftrightarrow (\exists b) ((a, b) \in R \text{ and } (b, d) \in T \circ S) \\ &\Leftrightarrow (\exists b) ((a, b) \in R \text{ and } (\exists c) ((b, c) \in S \text{ and } (c, d) \in T)) \\ &\Leftrightarrow (\exists c) ((\exists b) ((a, b) \in R \text{ and } (b, c) \in S) \text{ and } (c, d) \in T) \\ &\Leftrightarrow (\exists c) ((a, c) \in S \circ R \text{ and } (c, d) \in T) \\ &\Leftrightarrow (a, d) \in T \circ (S \circ R) \end{aligned}$$

(5) **Identity:** First part: Assume $R \subseteq A \times B$. For any (a, b) :

$$\begin{aligned} (a, b) \in \text{Id}_A \circ R &\Leftrightarrow (\exists x) ((a, x) \in \text{Id}_A \text{ and } (x, b) \in R) \\ &\Leftrightarrow \exists x \in A : (a = x \text{ and } (x, b) \in R) \\ &\Leftrightarrow (a, b) \in R \end{aligned}$$

Similarly for $R \circ \text{Id}_B = R$.

(6) **Converse of Composition:** For any (a, c) :

$$\begin{aligned} (a, c) \in (S \circ R)^T &\Leftrightarrow (c, a) \in S \circ R \\ &\Leftrightarrow (\exists b) ((c, b) \in R \text{ and } (b, a) \in S) \\ &\Leftrightarrow (\exists b) ((b, c) \in R^T \text{ and } (a, b) \in S^T) \\ &\Leftrightarrow (a, c) \in R^T \circ S^T \end{aligned}$$

QED

Corollary 1.2.5 (Basic Properties of Composition for binary relations):

Let r , s , and t be binary relations where the compositions are defined. Then:

1. **Associativity:** $(t \circ s) \circ r = t \circ (s \circ r)$
2. **Identity:** If $\text{id}_B = (B, B, \text{Id}_B)$ is the identity relation on B , then:

- $\text{id}_B \circ r = r$ for any relation $r = (A, B, R)$
- $s \circ \text{id}_B = s$ for any relation $s = (B, C, S)$

3. **Converse of Composition:** $(s \circ r)^T = r^T \circ s^T$

Proof:

These properties follow immediately from the corresponding properties for graphs of binary relations, since the composition of binary relations $(A, C, S \circ R)$ is defined in terms of the composition of their underlying graphs $S \circ R$. QED

1.2.2 Additional Properties of Composition

Proposition 1.2.6 (Algebraic Properties of Composition for graphs):

Let R, S, T be graphs of binary relations, and let $\{R_i\}_{i \in I}$ and $\{S_i\}_{i \in I}$ be families of graphs of binary relations where the compositions are defined. Then:

1. **Distributivity over Union:**

- Left distributivity: $S \circ (\bigcup_{i \in I} R_i) = \bigcup_{i \in I} S \circ R_i$
- Right distributivity: $(\bigcup_{i \in I} S_i) \circ R = \bigcup_{i \in I} S_i \circ R$

2. **Subdistributivity over Intersection:**

- Left subdistributivity: $S \circ (\bigcap_{i \in I} R_i) \subseteq \bigcap_{i \in I} S \circ R_i$
- Right subdistributivity: $(\bigcap_{i \in I} S_i) \circ R \subseteq \bigcap_{i \in I} S_i \circ R$

3. **Monotonicity:**

- If $R_1 \subseteq R_2$ then $S \circ R_1 \subseteq S \circ R_2$ (Left monotonicity)
- If $S_1 \subseteq S_2$ then $S_1 \circ R \subseteq S_2 \circ R$ (Right monotonicity)

4. **Antimonotonicity with Complement:**

- $S^c \circ R \subseteq (S \circ R)^c$
- $S \circ R^c \subseteq (S \circ R)^c$

Proof:

(1) **Distributivity over Union:**

For left distributivity:

$$\begin{aligned}
 (a, c) \in S \circ \left(\bigcup_{i \in I} R_i \right) &\Leftrightarrow (\exists b) ((a, b) \in \bigcup_{i \in I} R_i \text{ and } (b, c) \in S) \\
 &\Leftrightarrow (\exists b) ((\exists i \in I : (a, b) \in R_i) \text{ and } (b, c) \in S) \\
 &\Leftrightarrow \exists i \in I : (\exists b) ((a, b) \in R_i \text{ and } (b, c) \in S) \\
 &\Leftrightarrow \exists i \in I : (a, c) \in S \circ R_i \\
 &\Leftrightarrow (a, c) \in \bigcup_{i \in I} S \circ R_i
 \end{aligned}$$

Right distributivity follows similarly.

(2) **Subdistributivity over Intersection:**

For left subdistributivity:

$$\begin{aligned}
 (a, c) \in S \circ \left(\bigcap_{i \in I} R_i \right) &\Leftrightarrow (\exists b) ((a, b) \in \bigcap_{i \in I} R_i \text{ and } (b, c) \in S) \\
 &\Leftrightarrow (\exists b) ((\forall i \in I : (a, b) \in R_i) \text{ and } (b, c) \in S) \\
 &\Rightarrow \forall i \in I : (\exists b) ((a, b) \in R_i \text{ and } (b, c) \in S) \\
 &\Leftrightarrow \forall i \in I : (a, c) \in S \circ R_i \\
 &\Leftrightarrow (a, c) \in \bigcap_{i \in I} S \circ R_i
 \end{aligned}$$

Right subdistributivity follows similarly. Note that equality doesn't hold in general.

(3) Monotonicity:

If $R_1 \subseteq R_2$:

$$\begin{aligned}
 (a, c) \in S \circ R_1 &\Leftrightarrow (\exists b) ((a, b) \in R_1 \text{ and } (b, c) \in S) \\
 &\Rightarrow (\exists b) ((a, b) \in R_2 \text{ and } (b, c) \in S) \quad (\text{since } R_1 \subseteq R_2) \\
 &\Leftrightarrow (a, c) \in S \circ R_2
 \end{aligned}$$

If $S_1 \subseteq S_2$:

$$\begin{aligned}
 (a, c) \in S_1 \circ R &\Leftrightarrow (\exists b) ((a, b) \in R \text{ and } (b, c) \in S_1) \\
 &\Rightarrow (\exists b) ((a, b) \in R \text{ and } (b, c) \in S_2) \quad (\text{since } S_1 \subseteq S_2) \\
 &\Leftrightarrow (a, c) \in S_2 \circ R
 \end{aligned}$$

(4) Antimonotonicity with Complement:

For the first part:

$$\begin{aligned}
 (a, c) \in S^c \circ R &\Leftrightarrow (\exists b) ((a, b) \in R \text{ and } (b, c) \in S^c) \\
 &\Leftrightarrow (\exists b) ((a, b) \in R \text{ and } (b, c) \notin S) \\
 &\Rightarrow \neg(\forall b) ((a, b) \in R \Rightarrow (b, c) \in S) \\
 &\Leftrightarrow (a, c) \notin S \circ R \\
 &\Leftrightarrow (a, c) \in (S \circ R)^c
 \end{aligned}$$

For the second part:

$$\begin{aligned}
 (a, c) \in S \circ R^c &\Leftrightarrow (\exists b) ((a, b) \in R^c \text{ and } (b, c) \in S) \\
 &\Leftrightarrow (\exists b) ((a, b) \notin R \text{ and } (b, c) \in S) \\
 &\Rightarrow \neg(\forall b) ((b, c) \in S \Rightarrow (a, b) \in R) \\
 &\Leftrightarrow (a, c) \notin S \circ R \\
 &\Leftrightarrow (a, c) \in (S \circ R)^c
 \end{aligned}$$

QED

Proposition 1.2.7 (Algebraic Properties of Composition for Binary Relations):

Let $r_i = (A, B, R_i)$, $s_i = (B, C, S_i)$, and $t = (C, D, T)$ be binary relations (with matching domains and codomains for composition). Let $\{r_i\}_{i \in I}$ be a family of binary relations from A to B , and $\{s_i\}_{i \in I}$ be a family of binary relations from B to C . Then:

1. Distributivity over Union:

- Left distributivity: $s \circ (\bigcup_{i \in I} r_i) = \bigcup_{i \in I} s \circ r_i$
- Right distributivity: $(\bigcup_{i \in I} s_i) \circ r = \bigcup_{i \in I} s_i \circ r$

2. Subdistributivity over Intersection:

- Left subdistributivity: $s \circ (\bigcap_{i \in I} r_i) \subseteq \bigcap_{i \in I} s \circ r_i$
- Right subdistributivity: $(\bigcap_{i \in I} s_i) \circ r \subseteq \bigcap_{i \in I} s_i \circ r$

3. Monotonicity:

- If $r_1 \subseteq r_2$ then $s \circ r_1 \subseteq s \circ r_2$ (Left monotonicity)
- If $s_1 \subseteq s_2$ then $s_1 \circ r \subseteq s_2 \circ r$ (Right monotonicity)

4. Antimonotonicity with Complement:

- $s^c \circ r \subseteq (s \circ r)^c$
- $s \circ r^c \subseteq (s \circ r)^c$

where the complement is taken relative to the same domain and codomain.

Proof:

All properties follow from the corresponding properties for the underlying graphs, since composition, union, intersection, and complement of binary relations are defined in terms of their graphs. QED

Remark:

The antimonotonicity properties show that composition is "complement-reversing" in each argument. Note that these are inclusions rather than equalities - the reverse inclusions do not hold in general.

1.2.3 Restriction and Corestriction, Direct and Inverse Images**Definition 1.2.8 (Restriction):**

Let R be a graph of a binary relation and A be a set. The **restriction** of R to A , denoted $R|_A$, is defined as:

$$R|_A := \{(x, y) \in R \mid x \in A\} = R \cap (A \times \mathcal{R}(R))$$

If $r = (A, B, R)$ is a binary relation, then the restriction of R to $A' \subseteq A$, denoted $r|_{A'}$, is just $(A', B, R|_{A'})$.

Definition 1.2.9 (Corestriction):

Let R be a graph of a binary relation and B be a set. The **corestriction** of R to B , denoted $R|^B$, is defined as:

$$R|^B := \{(x, y) \in R \mid y \in B\} = R \cap (\mathcal{D}(R) \times B)$$

If $r = (A, B, R)$ is a binary relation, then the corestriction of R to $B' \subseteq B$, denoted $r|^{B'}$, is just $(A, B', R|^{B'})$.

Remark:

The notation works to both restrict and corestrict at the same time.

$$R|_A^B := \{(x, y) \in R \mid x \in A \text{ and } y \in B\} = R \cap (A \times B)$$

If $r = (A, B, R)$ is a binary relation and $A' \subseteq A$ and $B' \subseteq B$, denoted $r|_{A'}^{B'}$, is just $(A', B', R|_{A'}^{B'})$.

Definition 1.2.10 (Direct Image):

Let R be a graph of a binary relation and A be a set. The **direct image** of A under R , denoted $R[A]$, is defined as:

$$R[A] := \{y \in \mathcal{R}(R) \mid \exists x \in A : (x, y) \in R\} = \mathcal{R}(R|_A)$$

If $r = (A, B, R)$ is a binary relation, then the direct image of A under r , denoted $r[A]$, is just $r[A] := R[A]$.

Definition 1.2.11 (Inverse Image):

Let R be a graph of a binary relation and B be a set. The **inverse image** of B under R , denoted $R^{\leftarrow}[B]$, is defined as:

$$R^{\leftarrow}[B] := \{x \in \mathcal{D}(R) \mid \exists y \in B : (x, y) \in R\} = \mathcal{D}(R|_B^B)$$

If $r = (A, B, R)$ is a binary relation, then the inverse image of B under r , denoted $r^{\leftarrow}[B]$, is just $r^{\leftarrow}[B] := R^{\leftarrow}[B]$.

Remark (Composition via Restriction):

For $r = (A, B, R)$, $s = (C, D, S)$ with $B \subseteq C$, we can define:

$$s \circ r := (A, D, (S|_{\mathcal{R}(R)}^D) \circ (R|_A^{\mathcal{D}(S)}))$$

This equals $(A, D, S \circ R)$ since:

$$S \circ R = (S|_{\mathcal{R}(R)}^D) \circ (R|_A^{\mathcal{D}(S)})$$

Proposition 1.2.12 (Unification via Composition):

Let R be a graph of a binary relation and let A, B be sets. Then:

1. $R|_A = R \circ \text{Id}_A$
2. $R|_B^B = \text{Id}_B \circ R$
3. $R[A] = \mathcal{R}(R \circ \text{Id}_A)$
4. $R^{\leftarrow}[B] = \mathcal{D}(\text{Id}_B \circ R)$

Proof:

(1) **Restriction:**

$$\begin{aligned} R \circ \text{Id}_A &= \{(x, z) \mid (\exists y)((x, y) \in \text{Id}_A \text{ and } (y, z) \in R)\} \\ &= \{(x, z) \mid \exists y \in A : x = y \text{ and } (y, z) \in R\} \\ &= \{(x, z) \mid x \in A \text{ and } (x, z) \in R\} \\ &= R|_A \end{aligned}$$

(2) **Corestriction:**

$$\begin{aligned} \text{Id}_B \circ R &= \{(x, z) \mid (\exists y)((x, y) \in R \text{ and } (y, z) \in \text{Id}_B)\} \\ &= \{(x, z) \mid \exists y \in B : (x, y) \in R \text{ and } y = z\} \\ &= \{(x, y) \mid (x, y) \in R \text{ and } y \in B\} \\ &= R|_B^B \end{aligned}$$

(3) **Direct Image:** From property (1):

$$\begin{aligned} R[A] &= \{y \mid \exists x \in A : (x, y) \in R\} \\ &= \mathcal{R}(R|_A) \\ &= \mathcal{R}(R \circ \text{Id}_A) \end{aligned}$$

(4) **Inverse Image:** From property (2):

$$\begin{aligned} R^{\leftarrow}[B] &= \{x \mid \exists y \in B : (x, y) \in R\} \\ &= \mathcal{D}(R|_B) \\ &= \mathcal{D}(\text{Id}_B \circ R) \end{aligned}$$

QED

Corollary 1.2.13 (Duality via Converse):

The operations of restriction and corestriction are dual via converse, as are direct and inverse images:

1. $R^T|_B = (R|_B)^T$
2. $R^T|_A = (R|_A)^T$
3. $R|_B = (R^T|_B)^T$
4. $R|_A = (R^T|_A)^T$
5. $R^T[A] = R^{\leftarrow}[A]$
6. $R^{T\leftarrow}[B] = R[B]$
7. $R[A] = R^{T\leftarrow}[A]$
8. $R^{\leftarrow}[B] = R^T[B]$

Proof:

These follow directly from the definitions and properties of converse operations:

(1): By definition of restriction and converse:

$$\begin{aligned} R^T|_B &= \{(x, y) \in R^T \mid x \in B\} \\ &= \{(x, y) \mid (y, x) \in R \text{ and } x \in B\} \\ &= \{(y, x) \mid (y, x) \in R \text{ and } x \in B\}^T \\ &= (R|_B)^T \end{aligned}$$

(2): By definition of corestriction and converse:

$$\begin{aligned} R^T|_A &= \{(x, y) \in R^T \mid y \in A\} \\ &= \{(x, y) \mid (y, x) \in R \text{ and } y \in A\} \\ &= \{(y, x) \mid (y, x) \in R \text{ and } y \in A\}^T \\ &= (R|_A)^T \end{aligned}$$

(3) and (4): These follow from (1) and (2) by applying converse to both sides and using the involution property $(R^T)^T = R$.

(5): By definition:

$$\begin{aligned} R^T[A] &= \{y \mid \exists x \in A : (x, y) \in R^T\} \\ &= \{y \mid \exists x \in A : (y, x) \in R\} \\ &= R^{\leftarrow}[A] \end{aligned}$$

(6): By definition:

$$\begin{aligned} R^{\top \leftarrow} [B] &= \{x \mid \exists y \in B : (x, y) \in R^{\top}\} \\ &= \{x \mid \exists y \in B : (y, x) \in R\} \\ &= R[B] \end{aligned}$$

(7) and (8): These follow from (5) and (6) by applying converse to both sides and using the involution property $(R^{\top})^{\top} = R$. QED

Remark:

This proposition shows that restriction, corestriction, direct images, and inverse images are all special cases of composition with identity relations. This unifies these seemingly different operations under the single framework of relation composition, demonstrating the fundamental role of composition in binary relation theory.

The corollary further shows the elegant duality between these operations via the converse operation, illustrating the deep symmetries inherent in the theory of binary relations. Specifically, corestriction is the restriction of the converse and dually, while direct image is the inverse image of the converse and dually.

Corollary 1.2.14 (Properties of Domain and Range):

Let $\{R_i\}_{i \in I}$ be a family of graphs of binary relations. Then:

1. **Distributivity over union:**

- $\mathcal{D}(\bigcup_{i \in I} R_i) = \bigcup_{i \in I} \mathcal{D}(R_i)$
- $\mathcal{R}(\bigcup_{i \in I} R_i) = \bigcup_{i \in I} \mathcal{R}(R_i)$

2. **Subdistributivity over intersection:**

- $\mathcal{D}(\bigcap_{i \in I} R_i) \subseteq \bigcap_{i \in I} \mathcal{D}(R_i)$
- $\mathcal{R}(\bigcap_{i \in I} R_i) \subseteq \bigcap_{i \in I} \mathcal{R}(R_i)$

Proof:

(1) **Distributivity over union:**

$$\begin{aligned} x \in \mathcal{D}(\bigcup_{i \in I} R_i) &\iff \exists y : ((x, y) \in \bigcup_{i \in I} R_i) \\ &\iff \exists y : (\exists i \in I : (x, y) \in R_i) \\ &\iff \exists i \in I : \exists y : ((x, y) \in R_i) \\ &\iff \exists i \in I : x \in \mathcal{D}(R_i) \\ &\iff x \in \bigcup_{i \in I} \mathcal{D}(R_i) \end{aligned}$$

The proof for range follows from $\mathcal{R}(\bigcup_{i \in I} R_i) = \mathcal{D}(\bigcup_{i \in I} R_i^{\top}) = \mathcal{D}(\bigcup_{i \in I} R_i^{\top}) = \bigcup_{i \in I} \mathcal{D}(R_i^{\top}) = \bigcup_{i \in I} \mathcal{R}(R_i)$.

(2) **Subdistributivity over intersection:**

$$\begin{aligned} x \in \mathcal{D}(\bigcap_{i \in I} R_i) &\iff \exists y : ((x, y) \in \bigcap_{i \in I} R_i) \\ &\iff \exists y : (\forall i \in I : (x, y) \in R_i) \\ &\implies \forall i \in I : \exists y : ((x, y) \in R_i) \\ &\iff \forall i \in I : x \in \mathcal{D}(R_i) \\ &\iff x \in \bigcap_{i \in I} \mathcal{D}(R_i) \end{aligned}$$

The proof for range follows from $\mathcal{R}(\bigcap_{i \in I} R_i) = \mathcal{D}(\bigcap_{i \in I} R_i^T) = \mathcal{D}(\bigcap_{i \in I} R_i^T) \subseteq \bigcap_{i \in I} \mathcal{D}(R_i^T) = \bigcap_{i \in I} \mathcal{R}(R_i)$.
QED

Corollary 1.2.15 (Properties of Direct and Inverse Images):

Let R and S be graphs of binary relations, and let A, B be sets. Then:

1. **Monotonicity in the relation:**

- If $R \subseteq S$ then $R[A] \subseteq S[A]$
- If $R \subseteq S$ then $R^{\leftarrow}[B] \subseteq S^{\leftarrow}[B]$

2. **Monotonicity in the set:**

- If $A_1 \subseteq A_2$ then $R[A_1] \subseteq R[A_2]$
- If $B_1 \subseteq B_2$ then $R^{\leftarrow}[B_1] \subseteq R^{\leftarrow}[B_2]$

3. **Distributivity over union of sets:**

- $R[(\bigcup_{i \in I} A_i)] = \bigcup_{i \in I} R[A_i]$
- $R^{\leftarrow}[(\bigcup_{i \in I} B_i)] = \bigcup_{i \in I} R^{\leftarrow}[B_i]$

4. **Subdistributivity over intersection of sets:**

- $R[(\bigcap_{i \in I} A_i)] \subseteq \bigcap_{i \in I} R[A_i]$
- $R^{\leftarrow}[(\bigcap_{i \in I} B_i)] \subseteq \bigcap_{i \in I} R^{\leftarrow}[B_i]$

5. **Distributivity over union of relations:**

- $(\bigcup_{i \in I} R_i)[A] = \bigcup_{i \in I} R_i[A]$
- $(\bigcup_{i \in I} R_i)^{\leftarrow}[B] = \bigcup_{i \in I} R_i^{\leftarrow}[B]$

6. **Subdistributivity over intersection of relations:**

- $(\bigcap_{i \in I} R_i)[A] \subseteq \bigcap_{i \in I} R_i[A]$
- $(\bigcap_{i \in I} R_i)^{\leftarrow}[B] \subseteq \bigcap_{i \in I} R_i^{\leftarrow}[B]$

7. **Antimonotonicity with complement:**

- $R^c[A] \subseteq R[A]^c$
- $R^{c\leftarrow}[B] \subseteq R^{\leftarrow}[B]^c$

Proof:

All properties follow from the representation of images as compositions with identity relations and the algebraic properties of composition.

(1) **Monotonicity in the relation:** From the representation $R[A] = \mathcal{R}(R \circ \text{Id}_A)$ and the right monotonicity of composition, if $R \subseteq S$ then $R \circ \text{Id}_A \subseteq S \circ \text{Id}_A$, so their ranges satisfy $R[A] \subseteq S[A]$. Similarly for inverse images using $R^{\leftarrow}[B] = \mathcal{D}(\text{Id}_B \circ R)$ and left monotonicity.

(2) **Monotonicity in the set:** If $A_1 \subseteq A_2$, then $\text{Id}_{A_1} \subseteq \text{Id}_{A_2}$. By left monotonicity of composition, $R \circ \text{Id}_{A_1} \subseteq R \circ \text{Id}_{A_2}$, so their ranges satisfy $R[A_1] \subseteq R[A_2]$. Similarly for inverse images using right monotonicity.

(3) Distributivity over union of sets:

$$\begin{aligned}
R \left[\left(\bigcup_{i \in I} A_i \right) \right] &= \mathcal{R}(R \circ \text{Id}_{\bigcup_i A_i}) \\
&= \mathcal{R}(R \circ \left(\bigcup_i \text{Id}_{A_i} \right)) \\
&= \mathcal{R}(\bigcup_i R \circ \text{Id}_{A_i}) \\
&= \bigcup_i \mathcal{R}(R \circ \text{Id}_{A_i}) \\
&= \bigcup_i R[A_i]
\end{aligned}$$

Similarly for inverse images using left distributivity of composition.

(4) Subdistributivity over intersection of sets:

$$\begin{aligned}
R \left[\left(\bigcap_{i \in I} A_i \right) \right] &= \mathcal{R}(R \circ \text{Id}_{\bigcap_i A_i}) \\
&= \mathcal{R}(R \circ \left(\bigcap_i \text{Id}_{A_i} \right)) \\
&\subseteq \mathcal{R}(\bigcap_i R \circ \text{Id}_{A_i}) \\
&\subseteq \bigcap_i \mathcal{R}(R \circ \text{Id}_{A_i}) \\
&= \bigcap_i R[A_i]
\end{aligned}$$

Similarly for inverse images using left subdistributivity of composition.

(5) Distributivity over union of relations:

$$\begin{aligned}
\left(\bigcup_{i \in I} R_i \right) [A] &= \mathcal{R}(\left(\bigcup_i R_i \right) \circ \text{Id}_A) \\
&= \mathcal{R}(\bigcup_i R_i \circ \text{Id}_A) \\
&= \bigcup_i \mathcal{R}(R_i \circ \text{Id}_A) \\
&= \bigcup_i R_i[A]
\end{aligned}$$

Similarly for inverse images using right distributivity.

(6) Subdistributivity over intersection of relations:

$$\begin{aligned}
\left(\bigcap_{i \in I} R_i\right) [A] &= \mathcal{R}\left(\left(\bigcap_i R_i\right) \circ \text{Id}_A\right) \\
&\subseteq \mathcal{R}\left(\bigcap_i R_i \circ \text{Id}_A\right) \\
&\subseteq \bigcap_i \mathcal{R}(R_i \circ \text{Id}_A) \\
&= \bigcap_i R_i [A]
\end{aligned}$$

Similarly for inverse images using right subdistributivity.

(7) Antimonotonicity with complement: From the antimonotonicity of composition:

$$\begin{aligned}
R^c [A] &= \mathcal{R}(R^c \circ \text{Id}_A) \\
&\subseteq \mathcal{R}((R \circ \text{Id}_A)^c) \\
&\subseteq \mathcal{R}(R \circ \text{Id}_A)^c \\
&= R [A]^c
\end{aligned}$$

Similarly for inverse images using the other antimonotonicity property of composition. Note that the last inclusion holds because for any set X , $\mathcal{R}(X^c) \subseteq \mathcal{R}(X)^c$ when the complement is taken relative to a universal set. QED

Proposition 1.2.16 (Properties of Images for Binary Relations):

Let $r = (A, B, R)$ and $s = (A, B, S)$ be binary relations with the same domain and codomain. Then:

1. Monotonicity in the relation:

- If $R \subseteq S$ then $r [A] \subseteq s [A]$
- If $R \subseteq S$ then $r^{\leftarrow} [B] \subseteq s^{\leftarrow} [B]$

2. Monotonicity in the set:

- If $A_1 \subseteq A_2$ then $r [A_1] \subseteq r [A_2]$
- If $B_1 \subseteq B_2$ then $r^{\leftarrow} [B_1] \subseteq r^{\leftarrow} [B_2]$

3. Distributivity over union of sets:

- $r [(\bigcup_{i \in I} A_i)] = \bigcup_{i \in I} r [A_i]$
- $r^{\leftarrow} [(\bigcup_{i \in I} B_i)] = \bigcup_{i \in I} r^{\leftarrow} [B_i]$

4. Subdistributivity over intersection of sets:

- $r [(\bigcap_{i \in I} A_i)] \subseteq \bigcap_{i \in I} r [A_i]$
- $r^{\leftarrow} [(\bigcap_{i \in I} B_i)] \subseteq \bigcap_{i \in I} r^{\leftarrow} [B_i]$

5. Distributivity over union of relations:

- $(\bigcup_{i \in I} r_i) [A] = \bigcup_{i \in I} r_i [A]$
- $(\bigcup_{i \in I} r_i)^{\leftarrow} [B] = \bigcup_{i \in I} r_i^{\leftarrow} [B]$

6. Subdistributivity over intersection of relations:

- $(\bigcap_{i \in I} r_i) [A] \subseteq \bigcap_{i \in I} r_i [A]$
- $(\bigcap_{i \in I} r_i)^{\leftarrow} [B] \subseteq \bigcap_{i \in I} r_i^{\leftarrow} [B]$

7. Antimonotonicity with complement:

- $r^c [A] \subseteq B \setminus r [A]$
- $r^{c\leftarrow} [B] \subseteq A \setminus r^{\leftarrow} [B]$

Proof:

The first six properties follow directly from the corresponding properties for graphs of binary relations. For the antimonotonicity properties:

Let $r = (A, B, R)$. Then $r^c = (A, B, (A \times B) \setminus R)$.

For direct image:

$$\begin{aligned}
 r^c [A] &= \{y \in B \mid \exists x \in A : (x, y) \in (A \times B) \setminus R\} \\
 &= \{y \in B \mid \exists x \in A : (x, y) \notin R\} \\
 &\subseteq \{y \in B \mid y \notin r [A]\} \\
 &= B \setminus r [A]
 \end{aligned}$$

Similarly for inverse image:

$$\begin{aligned}
 r^{c\leftarrow} [B] &= \{x \in A \mid \exists y \in B : (x, y) \in (A \times B) \setminus R\} \\
 &= \{x \in A \mid \exists y \in B : (x, y) \notin R\} \\
 &\subseteq \{x \in A \mid x \notin r^{\leftarrow} [B]\} \\
 &= A \setminus r^{\leftarrow} [B]
 \end{aligned}$$

QED

CHAPTER 2

Heterogenous Binary Relations

2.1 Definition

Let $r = (A, B, R)$ be a binary relation. If $A = B$, we call r a homogenous relation. If we do not care whether $A = B$ we say r is heterogenous.

2.2 Classification

We classify binary relations based on how they relate elements from their domain to their codomain. The following four fundamental properties capture different aspects of this relationship.

Definition 2.2.1 (Univalent Relation):

Let $r = (A, B, R)$ be a binary relation. We say r is **univalent** (or **right-unique**) if every element in the domain is related to at most one element in the codomain:

$$\forall a \in A : \forall b_1, b_2 \in B : (a, b_1) \in R \text{ and } (a, b_2) \in R \implies b_1 = b_2$$

Equivalently, if $(a, b_1) \in R$ and $(a, b_2) \in R$, then $b_1 = b_2$.

If $r = (A, B, R)$ we also say the graph of binary relations R is univalent.

Definition 2.2.2 (Total Relation):

Let $r = (A, B, R)$ be a binary relation. We say r is **total** (or sometimes **left-total**) if every element in the domain A is related to at least one element in the codomain B :

$$\forall a \in A : \exists b \in B : (a, b) \in R$$

Equivalently, $\mathcal{D}(R) = A$.

Definition 2.2.3 (Injective Relation):

Let $r = (A, B, R)$ be a binary relation. We say r is **injective** (or **left-unique**) if every element in the codomain is related to at most one element in the domain:

$$\forall b \in B : \forall a_1, a_2 \in A : (a_1, b) \in R \text{ and } (a_2, b) \in R \implies a_1 = a_2$$

Equivalently, if $(a_1, b) \in R$ and $(a_2, b) \in R$, then $a_1 = a_2$.

Definition 2.2.4 (Surjective Relation):

Let $r = (A, B, R)$ be a binary relation. We say r is **surjective** (or **right-total**) if every element in the codomain B is related to at least one element in the domain A :

$$\forall b \in B : \exists a \in A : (a, b) \in R$$

Equivalently, $\mathcal{R}(R) = B$.

Definition 2.2.5 (Functions as Relation):

Let $r = (A, B, R)$ be a binary relation. We say r is a **function** if it is both total and univalent. That is:

1. Every element in A is related to at least one element in B (total)
2. Every element in A is related to at most one element in B (univalent)

Equivalently, r is a function $\iff \forall a \in A : \exists! b \in B (a, b) \in R$.

Remark:

If $r = (A, B, R)$ we also say the graph of binary relations R is univalent. If $r = (A, B, R)$ we also say the graph of binary relations R is injective. We do not say a graph of a binary relation is total or surjective since more information on the domain/codomain is needed.

Remark:

Terminology:

These properties can be combined in various ways:

- A relation that is total but not univalent relates each element of A to one or more elements of B
- A relation that is univalent but not total relates some (but not necessarily all) elements of A to exactly one element of B (this is sometimes called a **partial function**)
- A relation that is both a function and injective is called an **injective function**
- A relation that is both a function and surjective is called a **surjective function**
- A relation that is a function, injective, and surjective is called a **bijective function** or **bijection**
- A relation that is surjective but not a function can relate elements of A to multiple elements of B , as long as every element of B is covered

Example 2.2.6 (Basic Examples):

Let A be any nonempty set.

1. The **identity relation** $\text{id}_A = (A, A, \text{Id}_A)$ is total, univalent, injective, and surjective (hence a bijection).
2. The **empty relation** $(A, B, \emptyset)$ is univalent and injective, but not total (unless $A = \emptyset$) and not surjective (unless $B = \emptyset$).
3. The **universal relation** $(A, B, A \times B)$ is total and surjective, but not univalent (unless $|A| \leq 1$) and not injective (unless $|B| \leq 1$).

Example 2.2.7:

Let $A = \{1, 2, 3\}$ and $B = \{a, b, c\}$.

1. $R_1 = \{(1, a), (2, b), (3, c)\}$ is total, univalent, injective, surjective, hence a bijective function.
2. $R_2 = \{(1, a), (1, b), (2, a), (3, c)\}$ is total and surjective, but not univalent (hence not a function) and not injective.
3. $R_3 = \{(1, a), (2, b)\}$ is univalent and injective but not total and not surjective (hence not a function).
4. $R_4 = \{(1, a), (2, a), (3, b)\}$ is total and univalent (hence a function) but not injective and not surjective.
5. $R_5 = \{(1, a), (2, a), (2, b)\}$ is neither total nor univalent nor injective nor surjective nor a function.

6. $R_6 = \{(1, a), (2, b), (3, c), (3, a)\}$ is total and surjective but not univalent (hence not a function) and not injective.
7. $R_7 = \{(1, a), (1, b), (2, b), (3, c)\}$ is total, univalent, and surjective (hence a surjective function) but not injective.

Proposition 2.2.8 (Characterization via Converse):

Let $r = (A, B, R)$ be a binary relation. Then:

1. r is injective if and only if r^T is univalent
2. r is univalent if and only if r^T is injective
3. r is surjective if and only if r^T is total
4. r is total if and only if r^T is surjective

Proof:

(1) r is injective means:

$$\forall b \in B : \forall a_1, a_2 \in A : (a_1, b) \in R \text{ and } (a_2, b) \in R \implies a_1 = a_2$$

This is equivalent to:

$$\forall b \in B : \forall a_1, a_2 \in A : (b, a_1) \in R^T \text{ and } (b, a_2) \in R^T \implies a_1 = a_2$$

which means r^T is univalent.

(2) This follows by applying (1) to r^T and using the involution property $(r^T)^T = r$.

(3) r is surjective means:

$$\forall b \in B : \exists a \in A ((a, b) \in R)$$

This is equivalent to:

$$\forall b \in B : \exists a \in A ((b, a) \in R^T)$$

which means r^T is total.

(4) This follows by applying (3) to r^T and using the involution property $(r^T)^T = r$. QED

Remark:

The characterization via converse establishes a fundamental duality between properties of relations:

- Any universal property that holds for univalent relations automatically transfers to injective relations via the converse, and vice versa. If we prove that all univalent relations satisfy some property P , then all injective relations satisfy the "converse version" of P , since r injective means r^T univalent.
- Similarly, any universal property of total relations automatically transfers to surjective relations via the converse, and vice versa.

This duality principle allows us to prove half as many theorems while obtaining twice as many results. For instance, if we prove a composition property for univalent relations, we immediately obtain the corresponding result for injective relations by applying the theorem to the converse.

Proposition 2.2.9 (Equivalent definitions):

Let $r = (A, B, R)$ be a binary relation. Then:

1. r is univalent $\iff R \circ R^T \subseteq \text{Id}_B$

2. r is total $\iff R^T \circ R \supseteq \text{Id}_A$
3. r is injective $\iff R^T \circ R \subseteq \text{Id}_A$
4. r is surjective $\iff R \circ R^T \supseteq \text{Id}_B$

Proof:

(1) Univalent:

$(\implies)$ Assume r is univalent. Let $(b_1, b_2) \in R \circ R^T$. Then $\exists a \in A : ((b_1, a) \in R^T \text{ and } (a, b_2) \in R)$, which means $\exists a \in A : ((a, b_1) \in R \text{ and } (a, b_2) \in R)$. Since r is univalent, we have $b_1 = b_2$. Therefore $(b_1, b_2) = (b_1, b_1) \in \text{Id}_B$, so $R \circ R^T \subseteq \text{Id}_B$.

$(\impliedby)$ Assume $R \circ R^T \subseteq \text{Id}_B$. Suppose $(a, b_1) \in R$ and $(a, b_2) \in R$. Then $(b_1, a) \in R^T$ and $(a, b_2) \in R$, so $(b_1, b_2) \in R \circ R^T$. Since $R \circ R^T \subseteq \text{Id}_B$, we have $(b_1, b_2) \in \text{Id}_B$, which means $b_1 = b_2$. Therefore r is univalent.

(2) Total:

$(\implies)$ Assume r is total. Let $a \in A$ be arbitrary. Since r is total, $\exists b \in B : ((a, b) \in R)$. Then $(a, b) \in R$ and $(b, a) \in R^T$, so $(a, a) \in R^T \circ R$. Since a was arbitrary, $(a, a) \in R^T \circ R$ for all $a \in A$. Therefore $\text{Id}_A \subseteq R^T \circ R$.

$(\impliedby)$ Assume $R^T \circ R \supseteq \text{Id}_A$. Let $a \in A$ be arbitrary. Then $(a, a) \in \text{Id}_A \subseteq R^T \circ R$. By definition of composition, $\exists b \in B : ((a, b) \in R \text{ and } (b, a) \in R^T)$. Since $(b, a) \in R^T$ is equivalent to $(a, b) \in R$, the first condition already gives us $(a, b) \in R$ for some $b \in B$. Therefore r is total.

(3) Injective:

$(\implies)$ Assume r is injective. Let $(a_1, a_2) \in R^T \circ R$. Then $\exists b \in B : ((a_1, b) \in R \text{ and } (b, a_2) \in R^T)$, which means $\exists b \in B : ((a_1, b) \in R \text{ and } (a_2, b) \in R)$. Since r is injective, we have $a_1 = a_2$. Therefore $(a_1, a_2) = (a_1, a_1) \in \text{Id}_A$, so $R^T \circ R \subseteq \text{Id}_A$.

$(\impliedby)$ Assume $R^T \circ R \subseteq \text{Id}_A$. Suppose $(a_1, b) \in R$ and $(a_2, b) \in R$. Then $(a_1, b) \in R$ and $(b, a_2) \in R^T$, so $(a_1, a_2) \in R^T \circ R$. Since $R^T \circ R \subseteq \text{Id}_A$, we have $(a_1, a_2) \in \text{Id}_A$, which means $a_1 = a_2$. Therefore r is injective.

(4) Surjective:

$(\implies)$ Assume r is surjective. Let $b \in B$ be arbitrary. Since r is surjective, $\exists a \in A : ((a, b) \in R)$. Then $(a, b) \in R$ and $(b, a) \in R^T$, so $(b, b) \in R \circ R^T$. Since b was arbitrary, $(b, b) \in R \circ R^T$ for all $b \in B$. Therefore $\text{Id}_B \subseteq R \circ R^T$.

$(\impliedby)$ Assume $R \circ R^T \supseteq \text{Id}_B$. Let $b \in B$ be arbitrary. Then $(b, b) \in \text{Id}_B \subseteq R \circ R^T$. By definition of composition, $\exists a \in A : ((b, a) \in R^T \text{ and } (a, b) \in R)$. Since $(b, a) \in R^T$ is equivalent to $(a, b) \in R$, the second condition already gives us $(a, b) \in R$ for some $a \in A$. Therefore r is surjective. QED

Remark:

Note that the proofs of (3) and (4) follow the same structure as (1) and (2) respectively, which is expected given the duality established by the characterization via converse: r is injective if and only if r^T is univalent, and r is surjective if and only if r^T is total.

Proposition 2.2.10 (Alternate Equivalent Conditions for Univalence, Totality, Injectivity, and Surjectivity):

Let $r = (A, B, R)$ be a binary relation. Let $\{s_i\}_{i \in I}$ be a family of relations and $\{B_i\}_{i \in I}$ be a family of sets, where $|I| \geq 2$ when specified. Then:

1. **Total:** r is total $\iff r^{\leftarrow}[B] = A$.
2. **Surjective:** r is surjective $\iff r[A] = B$.
3. **Univalent:** The following are equivalent:
 - r is univalent.
 - $(\bigcap_{i \in I} s_i) \circ r = \bigcap_{i \in I} s_i \circ r$ for any family of relations $\{s_i\}_{i \in I}$ with $|I| \geq 2$.
 - $r^{\leftarrow}[\bigcap_{i \in I} B_i] = \bigcap_{i \in I} r^{\leftarrow}[B_i]$ for any family of sets $\{B_i\}_{i \in I}$ with $|I| \geq 2$.

- $r^{\leftarrow} [B_1 \cap B_2] = r^{\leftarrow} [B_1] \cap r^{\leftarrow} [B_2]$ for any sets B_1, B_2 .

4. **Injective:** The following are equivalent:

- r is injective.
- $r \circ (\bigcap_{i \in I} s_i) = \bigcap_{i \in I} r \circ s_i$ for any family of relations $\{s_i\}_{i \in I}$ with $|I| \geq 2$.
- $r [\bigcap_{i \in I} A_i] = \bigcap_{i \in I} r [A_i]$ for any family of sets $\{A_i\}_{i \in I}$ with $|I| \geq 2$.
- $r [A_1 \cap A_2] = r [A_1] \cap r [A_2]$ for any sets A_1, A_2 .

Proof:

(1) **Total:**

($\Rightarrow$) Assume r is total. By definition, this means $\mathcal{D}(R) = A$. We know that $r^{\leftarrow} [B] = \mathcal{D}(\text{Id}_B \circ R)$. Since $R \subseteq A \times B$, the range of R is a subset of B , so $\text{Id}_B \circ R = R$. Thus $r^{\leftarrow} [B] = \mathcal{D}(R) = A$.

($\Leftarrow$) Assume $r^{\leftarrow} [B] = A$. By definition of inverse image, $r^{\leftarrow} [B] = \{x \in A \mid \exists y \in B : (x, y) \in R\} = \mathcal{D}(R)$. Since $\mathcal{D}(R) = A$, r is total.

(2) **Surjective:**

This follows from (1) by duality. r is surjective $\iff r^T$ is total $\iff r^{T\leftarrow} [A] = B \iff r [A] = B$.

(3) **Univalent:**

We prove the equivalences in a cycle: univalent $\implies$ composition $\implies$ domain-intersection $\implies$ arbitrary intersection $\implies$ binary intersection $\implies$ univalent.

(*Univalent $\implies$ Composition*): Assume r is univalent. We always have $(\bigcap_{i \in I} s_i) \circ r \subseteq \bigcap_{i \in I} s_i \circ r$. For the reverse inclusion, let $(x, z) \in \bigcap_{i \in I} s_i \circ r$. Then $\forall i \in I : (x, z) \in s_i \circ r$, so $\forall i \in I : \exists y : ((x, y) \in R \text{ and } (y, z) \in s_i)$.

Since r is univalent, for a given x , the y such that $(x, y) \in R$ is unique (if it exists). Let this unique element be y . Then we must have $(y, z) \in s_i$ for all $i \in I$, so $(y, z) \in \bigcap_{i \in I} s_i$. Thus $(x, z) \in (\bigcap_{i \in I} s_i) \circ r$.

(*Composition $\implies$ Domain-Intersection*): Assume $(\bigcap_{i \in I} s_i) \circ r = \bigcap_{i \in I} s_i \circ r$ for any family of relations. We show that $\mathcal{D}(\bigcap_{i \in I} s_i \circ r) = \bigcap_{i \in I} \mathcal{D}(s_i \circ r)$.

We always have $\mathcal{D}(\bigcap_{i \in I} s_i \circ r) \subseteq \bigcap_{i \in I} \mathcal{D}(s_i \circ r)$. For the reverse inclusion, let $x \in \bigcap_{i \in I} \mathcal{D}(s_i \circ r)$. Then $\forall i \in I : x \in \mathcal{D}(s_i \circ r)$, so $\forall i \in I : \exists z_i : ((x, z_i) \in s_i \circ r)$. By the composition assumption, $(\bigcap_{i \in I} s_i) \circ r = \bigcap_{i \in I} s_i \circ r$. Taking domains: $\mathcal{D}((\bigcap_{i \in I} s_i) \circ r) = \mathcal{D}(\bigcap_{i \in I} s_i \circ r)$. Since $x \in \bigcap_{i \in I} \mathcal{D}(s_i \circ r)$ and the $\subseteq$ direction holds, we need to show $x \in \mathcal{D}(\bigcap_{i \in I} s_i \circ r)$. From $\forall i \in I : \exists z_i : ((x, z_i) \in s_i \circ r)$ and univalence of r (which follows from the composition property), all z_i must be equal. Let z be this common value. Then $(x, z) \in s_i \circ r$ for all i , so $(x, z) \in \bigcap_{i \in I} s_i \circ r$, thus $x \in \mathcal{D}(\bigcap_{i \in I} s_i \circ r)$.

(*Domain-Intersection $\implies$ Arbitrary Intersection*): Assume $\mathcal{D}(\bigcap_{i \in I} s_i \circ r) = \bigcap_{i \in I} \mathcal{D}(s_i \circ r)$ for any family of relations. Taking $s_i = \text{Id}_{B_i}$:

$$\begin{aligned} r^{\leftarrow} \left[\bigcap_{i \in I} B_i \right] &= \mathcal{D}(\text{Id}_{\bigcap_{i \in I} B_i} \circ R) \\ &= \mathcal{D}((\bigcap_{i \in I} \text{Id}_{B_i}) \circ r) \\ &= \mathcal{D}(\bigcap_{i \in I} \text{Id}_{B_i} \circ r) \\ &= \bigcap_{i \in I} \mathcal{D}(\text{Id}_{B_i} \circ r) \\ &= \bigcap_{i \in I} r^{\leftarrow} [B_i] \end{aligned}$$

(*Arbitrary Intersection $\implies$ Binary Intersection*): Trivial, take $I = \{1, 2\}$.

(*Binary Intersection $\implies$ Univalent*): Assume $r^{\leftarrow} [B_1 \cap B_2] = r^{\leftarrow} [B_1] \cap r^{\leftarrow} [B_2]$ for all B_1, B_2 . Suppose r is not univalent. Then $\exists x, y_1, y_2 : ((x, y_1) \in R \text{ and } (x, y_2) \in R \text{ and } y_1 \neq y_2)$.

Let $B_1 = \{y_1\}$ and $B_2 = \{y_2\}$. Then $B_1 \cap B_2 = \emptyset$, so $r^{\leftarrow}[B_1 \cap B_2] = \emptyset$. But $x \in r^{\leftarrow}[B_1]$ and $x \in r^{\leftarrow}[B_2]$, so $x \in r^{\leftarrow}[B_1] \cap r^{\leftarrow}[B_2]$. Contradiction.

(4) Injective:

This follows from (3) by duality. r is injective $\iff r^T$ is univalent. Applying (3) to r^T :

- r^T is univalent $\iff (\bigcap_{i \in I} s_i) \circ r^T = \bigcap_{i \in I} s_i \circ r^T$. Taking converses: $r \circ (\bigcap_{i \in I} s_i^T) = \bigcap_{i \in I} r \circ s_i^T$, which is equivalent to the composition statement for injective relations.
- r^T is univalent $\iff r^{T\leftarrow}[\bigcap_{i \in I} B_i] = \bigcap_{i \in I} r^{T\leftarrow}[B_i]$. Since $r^{T\leftarrow}[B] = r[B]$, this gives $r[\bigcap_{i \in I} A_i] = \bigcap_{i \in I} r[A_i]$.
- Similarly for the binary intersection case.

QED

Proposition 2.2.11 (Composition Properties):

Let $r = (A, B, R)$ and $s = (B, C, S)$ be binary relations. Then:

1. If r and s are both univalent, then $s \circ r$ is univalent
2. If r and s are both total, then $s \circ r$ is total
3. If r and s are both injective, then $s \circ r$ is injective
4. If r and s are both surjective, then $s \circ r$ is surjective
5. If r and s are both functions, then $s \circ r$ is a function

Proof:

(1) Univalent: Assume r and s are both univalent. We need to show $s \circ r$ is univalent, i.e., $(S \circ R) \circ (S \circ R)^T \subseteq \text{Id}_C$.

By the converse of composition property, $(S \circ R)^T = R^T \circ S^T$. Therefore:

$$\begin{aligned} (S \circ R) \circ (S \circ R)^T &= (S \circ R) \circ (R^T \circ S^T) \\ &= S \circ (R \circ R^T) S^T \quad (\text{by associativity}) \\ &\subseteq S \circ (\text{Id}_B) S^T \quad (\text{since } r \text{ is univalent}) \\ &= S \circ S^T \\ &\subseteq \text{Id}_C \quad (\text{since } s \text{ is univalent}) \end{aligned}$$

Therefore $s \circ r$ is univalent.

(2) Total: Assume r and s are both total. We need to show $s \circ r$ is total, i.e., $(S \circ R)^T \circ S \circ R \supseteq \text{Id}_A$.

By the converse of composition property:

$$\begin{aligned} (S \circ R)^T \circ S \circ R &= (R^T \circ S^T) \circ (S \circ R) \\ &= R^T \circ (S^T \circ S \circ R) \quad (\text{by associativity}) \\ &\supseteq R^T \circ (\text{Id}_B) R \quad (\text{since } s \text{ is total}) \\ &= R^T \circ R \\ &\supseteq \text{Id}_A \quad (\text{since } r \text{ is total}) \end{aligned}$$

Therefore $s \circ r$ is total.

(3) Injective: Assume r and s are both injective. We need to show $s \circ r$ is injective, i.e., $(S \circ R)^T \circ S \circ R \subseteq \text{Id}_A$.

By the converse of composition property:

$$\begin{aligned}
 (S \circ R)^T \circ S \circ R &= (R^T \circ S^T) \circ (S \circ R) \\
 &= R^T \circ (S^T \circ S \circ R) \quad (\text{by associativity}) \\
 &\subseteq R^T \circ (\text{Id}_B)R \quad (\text{since } s \text{ is injective}) \\
 &= R^T \circ R \\
 &\subseteq \text{Id}_A \quad (\text{since } r \text{ is injective})
 \end{aligned}$$

Therefore $s \circ r$ is injective.

(4) Surjective: Assume r and s are both surjective. We need to show $s \circ r$ is surjective, i.e., $(S \circ R) \circ (S \circ R)^T \supseteq \text{Id}_C$.

By the converse of composition property:

$$\begin{aligned}
 (S \circ R) \circ (S \circ R)^T &= (S \circ R) \circ (R^T \circ S^T) \\
 &= S \circ (R \circ R^T)S^T \quad (\text{by associativity}) \\
 &\supseteq S \circ (\text{Id}_B)S^T \quad (\text{since } r \text{ is surjective}) \\
 &= S \circ S^T \\
 &\supseteq \text{Id}_C \quad (\text{since } s \text{ is surjective})
 \end{aligned}$$

Therefore $s \circ r$ is surjective.

(5) Functions: This follows immediately from (1) and (2), since a function is a relation that is both univalent and total. QED

Remark:

These composition properties show that the classes of total, univalent, injective, surjective, and functions relations are all closed under composition. In particular:

- The composition of functions is a function
- The composition of injective functions is injective
- The composition of surjective functions is surjective
- The composition of bijections is a bijection

Proposition 2.2.12 (Necessary Conditions for Composition Properties):

Let $r = (A, B, R)$ and $s = (B, C, S)$ be binary relations. Then:

1. **Totality:** If $s \circ r$ is total, then r is total.
2. **Surjectivity:** If $s \circ r$ is surjective, then s is surjective.
3. **Univalence:** If $s \circ r$ is univalent, then $s|_{\mathcal{R}(R)}$ is univalent.
4. **Injectivity:** If $s \circ r$ is injective, then $r|_{\mathcal{D}(S)}$ is injective.

Proof:

(1) Totality:

Assume $s \circ r$ is total. Let $a \in A$ be arbitrary. Since $s \circ r$ is total, $\exists c \in C : (a, c) \in S \circ R$. By definition of composition, $\exists b \in B : (a, b) \in R$ and $(b, c) \in S$. Thus $\exists b \in B : (a, b) \in R$, so r is total.

(2) Surjectivity:

This follows from (1) by duality. $s \circ r$ is surjective $\iff (s \circ r)^T$ is total $\iff r^T \circ s^T$ is total $\implies s^T$ is total $\iff s$ is surjective.

(3) Univalence:

Assume $s \circ r$ is univalent. We need to show $s|_{\mathcal{R}(R)}$ is univalent. Let $b_1, b_2 \in \mathcal{R}(R)$ and $c \in C$ such that $(b_1, c) \in S$ and $(b_2, c) \in S$. Since $b_1, b_2 \in \mathcal{R}(R)$, there exist $a_1, a_2 \in A$ such that $(a_1, b_1) \in R$ and $(a_2, b_2) \in R$.

Then $(a_1, c) \in S \circ R$ (via b_1) and $(a_2, c) \in S \circ R$ (via b_2). Since $s \circ r$ is univalent, we have $a_1 = a_2$. Let $a = a_1 = a_2$. Then $(a, b_1) \in R$ and $(a, b_2) \in R$.

Now, if r were univalent, we would have $b_1 = b_2$ immediately. However, we cannot assume r is univalent. Instead, we observe that for any $b \in \mathcal{R}(R)$ and $c \in C$, if $(b, c) \in S$, then for any a with $(a, b) \in R$, we have $(a, c) \in S \circ R$. Since $s \circ r$ is univalent, this c is uniquely determined by a . But we need to show uniqueness in b for fixed c .

Actually, let's reconsider. Suppose $(b_1, c), (b_2, c) \in S$ with $b_1, b_2 \in \mathcal{R}(R)$. Pick any a_1 with $(a_1, b_1) \in R$ and any a_2 with $(a_2, b_2) \in R$. If $a_1 = a_2 = a$, then $(a, c) \in S \circ R$ via both b_1 and b_2 . Since composition is well-defined, this is fine. But if $a_1 \neq a_2$, then both (a_1, c) and (a_2, c) are in $S \circ R$, which contradicts univalence of $s \circ r$ only if they're related to the same element in A , which they're not.

The correct statement is: if $(b, c_1), (b, c_2) \in S$ with $b \in \mathcal{R}(R)$, then $c_1 = c_2$. Let a be such that $(a, b) \in R$. Then $(a, c_1), (a, c_2) \in S \circ R$. Since $s \circ r$ is univalent, $c_1 = c_2$. Therefore $s|_{\mathcal{R}(R)}$ is univalent.

(4) Injectivity:

This follows from (3) by duality. $s \circ r$ is injective $\iff (s \circ r)^T$ is univalent $\iff r^T \circ s^T$ is univalent $\implies r^T|_{\mathcal{R}(S^T)}$ is univalent. Since $\mathcal{R}(S^T) = \mathcal{D}(S)$ and $r^T|_{\mathcal{D}(S)} = (r|_{\mathcal{D}(S)})^T$, we have $(r|_{\mathcal{D}(S)})^T$ is univalent, which means $r|_{\mathcal{D}(S)}$ is injective. QED

Proposition 2.2.13 (Characterization of Isomorphisms in Rel):

Let $r = (A, B, R)$ be a binary relation. Then r is an isomorphism in the category **Rel** (i.e., there exists $s = (B, A, S)$ such that $r \circ s = \text{id}_B$ and $s \circ r = \text{id}_A$) if and only if r is total, univalent, injective, and surjective.

Proof:

($\implies$) Assume r is an isomorphism with inverse $s = (B, A, S)$. Then $r \circ s = \text{id}_B$ and $s \circ r = \text{id}_A$.

Since id_A is total and injective, and id_B is surjective and univalent, we apply the Necessary Conditions proposition to both compositions:

From $s \circ r = \text{id}_A$:

- r is total (item 1).
- s is surjective (item 2), so $\mathcal{R}(S) = A$.
- $r|_{\mathcal{D}(S)}$ is injective (item 4).

From $r \circ s = \text{id}_B$:

- r is surjective (item 2).
- s is total (item 1), so $\mathcal{D}(S) = B$.
- $r|_{\mathcal{R}(S)}$ is univalent (item 3).

Since $\mathcal{D}(S) = B$ and $\mathcal{R}(S) = A$, we have $r|_B = r$ is injective and $r|_A = r$ is univalent. Thus r has all four properties.

($\impliedby$) Assume r is total, univalent, injective, and surjective. We show that $s = r^T$ is the inverse of r .

By the Equivalent definitions proposition:

$$\begin{aligned} R \circ R^T &\supseteq \text{Id}_B && \text{(since } r \text{ is surjective)} \\ R \circ R^T &\subseteq \text{Id}_B && \text{(since } r \text{ is univalent)} \end{aligned}$$

Therefore $r \circ r^T = \text{id}_B$.

Similarly:

$$\begin{aligned} R^T \circ R &\supseteq \text{Id}_A \quad (\text{since } r \text{ is total}) \\ R^T \circ R &\subseteq \text{Id}_A \quad (\text{since } r \text{ is injective}) \end{aligned}$$

Therefore $r^T \circ r = \text{id}_A$.

Thus r is an isomorphism with inverse r^T .

QED

Remark:

This proposition shows that in the category **Rel**, isomorphisms are precisely the bijective functions. The forward direction demonstrates the power of the Necessary Conditions proposition: by analyzing the composition with the inverse, we can deduce all four properties. The backward direction shows that the converse relation provides the inverse when all four properties hold.

2.3 Induced Functions on Power Sets

Every binary relation $r = (A, B, R)$ induces two functions between the power sets of its domain and codomain.

Proposition 2.3.1 (Induced Functions from Relations):

Let $r = (A, B, R)$ be a binary relation. Then r induces:

1. A function $r_* : \mathcal{P}(A) \rightarrow \mathcal{P}(B)$ defined by $r_*(N) = r[N]$ for all $N \subseteq A$.
2. A function $r^* : \mathcal{P}(B) \rightarrow \mathcal{P}(A)$ defined by $r^*(M) = r^{\leftarrow}[M]$ for all $M \subseteq B$.

Proof:

Both are well-defined functions since for any subset of the domain/codomain, the direct/inverse image is uniquely determined and is a subset of the codomain/domain respectively. QED

Remark:

Not all functions from $\mathcal{P}(A)$ to $\mathcal{P}(B)$ arise from binary relations. We now characterize precisely which functions can be induced by relations.

Example 2.3.2 (Functions Not Induced by Relations):

Let $A = \{1, 2\}$ and $B = \{a, b\}$. Consider the function $f : \mathcal{P}(A) \rightarrow \mathcal{P}(B)$ defined by:

$$\begin{aligned} f(\emptyset) &= \{a\} \\ f(\{1\}) &= \{b\} \\ f(\{2\}) &= \{a\} \\ f(\{1, 2\}) &= \{a, b\} \end{aligned}$$

This function cannot be induced by any relation $r = (A, B, R)$ because $f(\emptyset) = \{a\} \neq \emptyset$, but for any relation, $r[\emptyset] = \emptyset$.

Proposition 2.3.3 (Characterization of Functions Induced by Relations):

A function $f : \mathcal{P}(A) \rightarrow \mathcal{P}(B)$ is induced by some binary relation $r = (A, B, R)$ (i.e., $f = r_*$) if and only if f distributes over arbitrary unions:

$$f\left(\bigcup_{i \in I} N_i\right) = \bigcup_{i \in I} f(N_i)$$

for any family $\{N_i\}_{i \in I}$ of subsets of A .

Proof:

($\Rightarrow$) Assume $f = r_*$ for some relation $r = (A, B, R)$.

By the properties of direct images established earlier, $r[\bigcup_{i \in I} N_i] = \bigcup_{i \in I} r[N_i]$.

($\Leftarrow$) Assume f distributes over arbitrary unions. Define $R = \{(a, b) \in A \times B \mid b \in f(\{a\})\}$. We claim that $f = r_*$ where $r = (A, B, R)$.

For any $N \subseteq A$, we can write $N = \bigcup_{a \in N} \{a\}$. Then:

$$\begin{aligned} r[N] &= \{b \in B \mid \exists a \in N : (a, b) \in R\} \\ &= \{b \in B \mid \exists a \in N : b \in f(\{a\})\} \\ &= \bigcup_{a \in N} f(\{a\}) \\ &= f\left(\bigcup_{a \in N} \{a\}\right) \quad (\text{by distributivity over unions}) \\ &= f(N) \end{aligned}$$

Thus $f = r_*$.

QED

Remark:

Note that the property $f(\emptyset) = \emptyset$ is automatically satisfied since taking $I = \emptyset$ in the arbitrary union property gives $f\left(\bigcup_{i \in \emptyset} N_i\right) = f(\emptyset) = \bigcup_{i \in \emptyset} f(N_i) = \emptyset$.

CHAPTER 3

Homogeneous Relations

3.1 Definition

A binary relation $r = (A, A, R)$ where the domain and codomain are the same set is called a **homogeneous relation** or **endorelation** on A .

Homogeneous relations have special properties that do not apply to general heterogeneous relations. In this section, we explore these properties.

3.2 Reflexivity, Symmetry, and Transitivity

Definition 3.2.1 (Reflexive Relation):

A homogeneous relation $r = (A, A, R)$ is **reflexive** if every element is related to itself:

$$\forall a \in A: (a, a) \in R$$

Equivalently, $\text{Id}_A \subseteq R$.

Definition 3.2.2 (Irreflexive Relation):

A homogeneous relation $r = (A, A, R)$ is **irreflexive** if no element is related to itself:

$$\forall a \in A: (a, a) \notin R$$

Equivalently, $R \cap \text{Id}_A = \emptyset$.

Definition 3.2.3 (Symmetric Relation):

A homogeneous relation $r = (A, A, R)$ is **symmetric** if whenever a is related to b , then b is related to a :

$$\forall a, b \in A: (a, b) \in R \implies (b, a) \in R$$

Equivalently, $R \subseteq R^T$ in which case $R = R^T$.

Definition 3.2.4 (Asymmetric Relation):

A homogeneous relation $r = (A, A, R)$ is **asymmetric** if whenever a is related to b , then b is not related to a :

$$\forall a, b \in A: (a, b) \in R \implies (b, a) \notin R$$

Equivalently, $R \cap R^T = \emptyset$.

Definition 3.2.5 (Antisymmetric Relation):

A homogeneous relation $r = (A, A, R)$ is **antisymmetric** if whenever a is related to b and b is related to a , then $a = b$:

$$\forall a, b \in A: (a, b) \in R \text{ and } (b, a) \in R \implies a = b$$

Equivalently, $R \cap R^T \subseteq \text{Id}_A$.

Definition 3.2.6 (Transitive Relation):

A homogeneous relation $r = (A, A, R)$ is **transitive** if whenever a is related to b and b is related to c , then a is related to c :

$$\forall a, b, c \in A : (a, b) \in R \text{ and } (b, c) \in R \implies (a, c) \in R$$

Equivalently, $R \circ R \subseteq R$.

Definition 3.2.7 (Connected Relation):

A homogeneous relation $r = (A, A, R)$ is **connected** (or **complete**) if for any two distinct elements, at least one is related to the other:

$$\forall a, b \in A : a \neq b \implies (a, b) \in R \text{ or } (b, a) \in R$$

Equivalently, $R \cup R^T \cup \text{Id}_A = A \times A$.

Definition 3.2.8 (Strongly Connected Relation):

A homogeneous relation $r = (A, A, R)$ is **strongly connected** (or **total**) if for any two elements (not necessarily distinct), at least one is related to the other:

$$\forall a, b \in A : (a, b) \in R \text{ or } (b, a) \in R$$

Equivalently, $R \cup R^T = A \times A$.

Definition 3.2.9 (Trichotomous Relation):

A homogeneous relation $r = (A, A, R)$ is **trichotomous** if for any two elements, exactly one of the following holds: a is related to b , b is related to a , or $a = b$:

$$\forall a, b \in A : (a, b) \in R \oplus (b, a) \in R \oplus a = b$$

where $\oplus$ denotes exclusive or. Equivalently, R , R^T , and Id_A are pairwise disjoint and $R \cup R^T \cup \text{Id}_A = A \times A$.

Definition 3.2.10 (Minimal Element Property (Non-Strict)):

A homogeneous relation $r = (A, A, R)$ has the **minimal element property** if every nonempty subset of A has a minimal element with respect to R :

$$\forall S \subseteq A : S \neq \emptyset \implies \exists m \in S : \forall s \in S : (s, m) \in R \implies s = m$$

Definition 3.2.11 (Minimal Element Property (Strict)):

A homogeneous relation $r = (A, A, R)$ has the **strict minimal element property** if every nonempty subset of A has a strict minimal element with respect to R :

$$\forall S \subseteq A : S \neq \emptyset \implies \exists m \in S : \forall s \in S : (s, m) \notin R$$

Definition 3.2.12 (Minimum Element Property (Non-Strict)):

A homogeneous relation $r = (A, A, R)$ has the **minimum element property** if every nonempty subset of A has a minimum element with respect to R :

$$\forall S \subseteq A : S \neq \emptyset \implies \exists m \in S : \forall s \in S : (m, s) \in R$$

Definition 3.2.13 (Minimum Element Property (Strict)):

A homogeneous relation $r = (A, A, R)$ has the **strict minimum element property** if every nonempty subset of A has a strict minimum element with respect to R :

$$\forall S \subseteq A : S \neq \emptyset \implies \exists m \in S : \forall s \in S : s \neq m \implies (m, s) \in R$$

3.3 Special Types of Homogeneous Relations

Definition 3.3.1 (Preorder Relation):

A homogeneous relation $r = (A, A, R)$ is a **preorder relation** if it is reflexive and transitive.

Definition 3.3.2 (Equivalence Relation):

A homogeneous relation $r = (A, A, R)$ is an **equivalence relation** if it is reflexive, symmetric, and transitive.

Definition 3.3.3 ((Non-Strict) Partial Order):

A homogeneous relation $r = (A, A, R)$ is a **partial order** if it is reflexive, antisymmetric, and transitive.

Definition 3.3.4 (Strict Partial Order):

A homogeneous relation $r = (A, A, R)$ is a **strict partial order** if it is irreflexive, asymmetric, and transitive.

Definition 3.3.5 (Total Order):

A homogeneous relation $r = (A, A, R)$ is a **total order** (or **linear order**) if it is a partial order that is also connected. Equivalently, it is reflexive, antisymmetric, transitive, and connected.

Remark:

These definitions form a hierarchy of increasingly structured relations:

- Equivalence relations partition sets into equivalence classes
- Partial orders provide a notion of comparison without requiring all elements to be comparable
- Total orders extend partial orders by requiring all elements to be comparable
- Well-orders are total orders with the additional property that every subset has a least element

3.4 Properties of Homogeneous Relations

Proposition 3.4.1 (Asymmetry Implies Irreflexivity):

Let $r = (A, A, R)$ be an asymmetric relation. Then r is irreflexive.

Proof:

Suppose for contradiction that there exists $a \in A$ such that $(a, a) \in R$. By asymmetry, since $(a, a) \in R$, we must have $(a, a) \notin R$, which is a contradiction. Therefore, r is irreflexive. QED

Proposition 3.4.2 (Irreflexivity and Asymmetry for Transitive Relations):

Let $r = (A, A, R)$ be a transitive relation. Then r is irreflexive if and only if r is asymmetric.

Proof:

($\Leftarrow$) This direction follows from the previous proposition and does not require transitivity.

($\Rightarrow$) Assume r is transitive and irreflexive. We prove that r is asymmetric.

Suppose $(a, b) \in R$. We need to show $(b, a) \notin R$.

Suppose for contradiction that $(b, a) \in R$. Since r is transitive and we have both $(a, b) \in R$ and $(b, a) \in R$, we obtain $(a, a) \in R$. But this contradicts irreflexivity. Therefore, $(b, a) \notin R$, and thus r is asymmetric. QED

Remark:

These propositions reveal an important relationship:

- Asymmetry always implies irreflexivity (no additional assumptions needed)
- For transitive relations, irreflexivity implies asymmetry (transitivity is essential here)

- Therefore, for transitive relations, irreflexivity and asymmetry are equivalent properties

This establishes that the definition of a strict partial order (irreflexive, asymmetric, transitive) has some redundancy—we could equivalently define it as irreflexive and transitive, or as asymmetric and transitive.

Remark (Weakening Partial Orders): 1. A (non-strict) partial order can be weakened to a preorder by removing the antisymmetry requirement. That is, if r is reflexive, antisymmetric, and transitive, then removing antisymmetry yields a preorder (reflexive and transitive).

2. A strict partial order cannot be weakened in an analogous way. Removing asymmetry from an irreflexive, asymmetric, transitive relation yields an irreflexive, transitive relation, which is not a standard structure like a preorder.

This illustrates a fundamental difference between strict and non-strict partial orders: (non-strict) partial orders fit into a natural hierarchy

$$\text{Partial Order} \xrightarrow{\text{drop antisymmetry}} \text{Preorder}$$

while strict partial orders do not admit such a natural weakening to a well-known structure.

3.4.1 Preorders and Induced Equivalence Relations

A preorder is a weaker structure than a partial order, lacking antisymmetry. However, every preorder naturally induces an equivalence relation that captures when two elements are "equivalent" with respect to the ordering.

Proposition 3.4.3 (Equivalence Relation Induced by a Preorder):

Let $r_{\leq} = (A, A, R_{\leq})$ be a preorder. Define the relation $R_{\sim} = \{(a, b) \in A \times A \mid (a, b) \in R_{\leq} \text{ and } (b, a) \in R_{\leq}\}$, and let $r_{\sim} = (A, A, R_{\sim})$. Then $r_{\sim}$ is an equivalence relation.

In other words, if we define $a \sim b$ to mean $a \leq b$ and $b \leq a$, then $\sim$ is an equivalence relation.

Proof:

We need to verify that $r_{\sim}$ is reflexive, symmetric, and transitive.

Reflexivity: For any $a \in A$, since $r_{\leq}$ is reflexive, we have $(a, a) \in R_{\leq}$. Therefore, $(a, a) \in R_{\leq}$ and $(a, a) \in R_{\leq}$, which means $(a, a) \in R_{\sim}$.

Symmetry: Suppose $(a, b) \in R_{\sim}$. Then $(a, b) \in R_{\leq}$ and $(b, a) \in R_{\leq}$. By symmetry of the definition, we also have $(b, a) \in R_{\leq}$ and $(a, b) \in R_{\leq}$, which means $(b, a) \in R_{\sim}$.

Transitivity: Suppose $(a, b) \in R_{\sim}$ and $(b, c) \in R_{\sim}$. Then:

- $(a, b) \in R_{\leq}$ and $(b, a) \in R_{\leq}$
- $(b, c) \in R_{\leq}$ and $(c, b) \in R_{\leq}$

Since $r_{\leq}$ is transitive, from $(a, b) \in R_{\leq}$ and $(b, c) \in R_{\leq}$, we obtain $(a, c) \in R_{\leq}$.

Similarly, from $(c, b) \in R_{\leq}$ and $(b, a) \in R_{\leq}$, we obtain $(c, a) \in R_{\leq}$.

Therefore, $(a, c) \in R_{\leq}$ and $(c, a) \in R_{\leq}$, which means $(a, c) \in R_{\sim}$.

QED

3.4.2 Relationship Between Strict and Non-Strict Partial Orders

There is a natural correspondence between strict and non-strict partial orders on a set. We explore this bidirectional relationship.

Proposition 3.4.4 (From Non-Strict to Strict Partial Order):

Let $r_{\leq} = (A, A, R_{\leq})$ be a (non-strict) partial order. Define $R_{<} = R_{\leq} \setminus \text{Id}_A = \{(a, b) \in R_{\leq} \mid a \neq b\}$ and let $r_{<} = (A, A, R_{<})$. Then $r_{<}$ is a strict partial order.

In other words, if $a \leq b$ denotes $(a, b) \in R_{\leq}$, and we define $a < b$ to mean $a \leq b$ and $a \neq b$, then $<$ is a strict partial order.

Proof:

By definition, $R_<$ excludes all diagonal elements from $R_\leq$, so it is irreflexive. We need to verify that $r_<$ is also transitive (which by our previous proposition will guarantee it is also asymmetric, completing the proof).

Transitivity: Suppose $(a, b) \in R_<$ and $(b, c) \in R_<$. Then $(a, b) \in R_\leq$, $(b, c) \in R_\leq$, $a \neq b$, and $b \neq c$.

Since $R_\leq$ is transitive, $(a, c) \in R_\leq$. We need to show $a \neq c$ to conclude $(a, c) \in R_<$.

Suppose for contradiction that $a = c$. Then $(a, b) \in R_\leq$ and $(b, a) \in R_\leq$ (since $(b, c) = (b, a)$). By antisymmetry of $R_\leq$, we have $a = b$. But this contradicts $a \neq b$. Therefore, $a \neq c$, and thus $(a, c) \in R_<$.

Since $r_<$ is irreflexive and transitive, by our earlier proposition it is also asymmetric. Therefore, $r_<$ is a strict partial order. QED

Proposition 3.4.5 (From Strict to Non-Strict Partial Order):

Let $r_< = (A, A, R_<)$ be a strict partial order. Define $R_\leq = R_< \cup \text{Id}_A$ and let $r_\leq = (A, A, R_\leq)$. Then $r_\leq$ is a (non-strict) partial order.

In other words, if $a < b$ denotes $(a, b) \in R_<$, and we define $a \leq b$ to mean $a < b$ or $a = b$, then $r_\leq$ is a (non-strict) partial order.

Proof:

We need to verify that $r_\leq$ is reflexive, antisymmetric, and transitive.

Reflexivity: For any $a \in A$, we have $(a, a) \in \text{Id}_A \subseteq R_\leq$.

Antisymmetry: Suppose $(a, b) \in R_\leq$ and $(b, a) \in R_\leq$. We need to show $a = b$.

If either $(a, b) \in \text{Id}_A$ or $(b, a) \in \text{Id}_A$, then $a = b$. Otherwise, both $(a, b) \in R$ and $(b, a) \in R$, which contradicts the asymmetry of R . Therefore, $a = b$.

Transitivity: Suppose $(a, b) \in R_\leq$ and $(b, c) \in R_\leq$. We analyze by cases:

- If $(a, b) \in \text{Id}_A$, then $a = b$, so $(a, c) = (b, c) \in R_\leq$.
- If $(b, c) \in \text{Id}_A$, then $b = c$, so $(a, c) = (a, b) \in R_\leq$.
- If $(a, b) \in R$ and $(b, c) \in R$, then by transitivity of R , we have $(a, c) \in R \subseteq R_\leq$.

In all cases, $(a, c) \in R_\leq$. QED

Remark:

These two propositions establish a bijective correspondence between strict and (non-strict) partial orders on a set. Given either one, we can uniquely recover the other. Moreover, these transformations are inverses:

- Starting with a (non-strict) partial order $\leq$, constructing $<$, and then reconstructing the partial order gives back $\leq$.
- Starting with a strict partial order $<$, constructing $\leq$, and then reconstructing the strict order gives back $<$.

Part II

Linear Algebra

CHAPTER 4

Modules

4.1 Rings

Definition 4.1.1 (Ring):

A **ring** is a set R together with two binary operations, addition $(+)$ and multiplication $(\cdot)$, usually denoted as a tuple $(R, +, \cdot)$ such that:

Addition Structure:

1. $(R, +)$ is an abelian group:
 - **Associativity:** $(a + b) + c = a + (b + c)$ for all $a, b, c \in R$
 - **Commutativity:** $a + b = b + a$ for all $a, b \in R$
 - **Identity:** $\exists 0 \in R \forall a \in R a + 0 = 0 + a = a$
 - **Inverses:** $\forall a \in R \exists (-a) \in R a + (-a) = (-a) + a = 0$

Multiplication Structure:

2. **Associativity:** $(ab)c = a(bc)$ for all $a, b, c \in R$

Distributive Laws:

3. **Left distributivity:** $a(b + c) = ab + ac$ for all $a, b, c \in R$
4. **Right distributivity:** $(a + b)c = ac + bc$ for all $a, b, c \in R$

Definition 4.1.2 (Ring with Unity):

A ring R is called a **ring with unity** (or **unital ring**) if there exists an element $1 \in R$ such that $1a = a1 = a$ for all $a \in R$. The element 1 is called the **multiplicative identity** or **unity**.

Example 4.1.3:

Common examples of rings include:

1. $\mathbb{Z}, \mathbb{Q}, \mathbb{R}, \mathbb{C}$ with ordinary addition and multiplication
2. $\mathbb{Z}_n$ (integers modulo n) for any positive integer n
3. Matrix rings $M_n(R)$ for any ring R
4. Polynomial rings $R[x]$ for any ring R

4.2 Modules

Definition 4.2.1 (Left Module):

Let R be a ring (with unity). A **left R -module** (or **left module over R**), denoted as ${}_R M$, is a set M together with an operation $+$: $M \times M \rightarrow M$ a scalar multiplication operation $\cdot$: $R \times M \rightarrow M$ such that for all $r, s \in R$ and all $m, n \in M$:

Addition Structure:

1. $(M, +)$ is an abelian group:

- **Associativity:** $(a + b) + c = a + (b + c)$ for all $a, b, c \in M$
- **Commutativity:** $a + b = b + a$ for all $a, b \in M$
- **Identity:** $\exists 0 \in M : \forall a \in M : a + 0 = 0 + a = a$
- **Inverses:** $\forall a \in M : \exists (-a) \in M : a + (-a) = (-a) + a = 0$

Scalar Multiplication Structure:

1. **Associativity:** $r(sm) = (rs)m$
2. **Unity:** $1 \cdot m = m$ (if R is unital and 1 is the unity of R)
3. **Left distributivity:** $r(m + n) = rm + rn$
4. **Right distributivity:** $(r + s)m = rm + sm$

Definition 4.2.2 (Right Module):

Let R be a ring with unity. A **right R -module** (or **right module over R**), denoted as M_R , is a set M together with an operation $+$: $M \times M \rightarrow M$ a scalar multiplication operation $\cdot$: $R \times M \rightarrow M$ such that for all $r, s \in R$ and all $m, n \in M$:

Addition Structure:

1. $(M, +)$ is an abelian group:

- **Associativity:** $(a + b) + c = a + (b + c)$ for all $a, b, c \in M$
- **Commutativity:** $a + b = b + a$ for all $a, b \in M$
- **Identity:** $\exists 0 \in M : \forall a \in M : a + 0 = 0 + a = a$
- **Inverses:** $\forall a \in M : \exists (-a) \in M : a + (-a) = (-a) + a = 0$

1. **Associativity:** $(mr)s = m(rs)$
2. **Unity:** $m \cdot 1 = m$ (if R is unital and 1 is the unity of R)
3. **Left distributivity:** $(m + n)r = mr + nr$

Remark:

When R is a commutative ring, there is no distinction between left and right modules, and we simply speak of **R -modules**.

Example 4.2.3:

Important examples of modules include:

1. Every vector space over a field F is an F -module
2. Every abelian group is a $\mathbb{Z}$ -module
3. For any ring R , the ring R itself is both a left and right R -module
4. If R is a ring and I is an ideal of R , then R/I is an R -module

CHAPTER 5

Foundations of Vector Spaces

5.1 Vector Spaces

Intuition (Chapter Overview):

The algebraic structure of a vector space is minimal by design: a set of objects (vectors) that can be added together and scaled by elements of a field, subject to the familiar rules of arithmetic. Yet this sparse axiom set supports a remarkably rich theory. The key concepts — subspace, span, linear independence, basis, dimension — arise naturally from asking progressively sharper questions about how vectors combine:

- Which subsets are *closed* under taking linear combinations? (Subspaces.)
- What is the smallest such subset containing a given collection? (Span.)
- When does every vector in a span have a *unique* expression? (Linear independence.)
- What is the most economical generating set? (Basis.)
- Is the size of a basis an invariant? (Dimension.)

Each concept refines the previous one, and together they provide the vocabulary for the entire subject. This chapter develops them in sequence.

Remark (Categorical Perspective):

Linear algebra studies the category $K\text{-Vect}$, where the objects are K -vector spaces and the morphisms are K -linear transformations. This categorical perspective unifies the treatment and motivates many constructions we will encounter.

Definition 5.1.1 (Vector Space):

Let K be a field. A K -vector space (or **vector space over K**) is a set V equipped with two operations:

- **Vector addition:** $+: V \times V \rightarrow V$
- **Scalar multiplication:** $\cdot: K \times V \rightarrow V$

satisfying the following axioms for all $u, v, w \in V$ and $a, b \in K$:

Abelian group axioms (for addition):

1. **Associativity:** $(u + v) + w = u + (v + w)$
2. **Commutativity:** $u + v = v + u$
3. **Identity:** $\exists 0 \in V : \forall v \in V : v + 0 = v$
4. **Inverses:** $\forall v \in V : \exists (-v) \in V : v + (-v) = 0$

Ring action axioms (for scalar multiplication):

5. **Compatibility:** $a(bv) = (ab)v$
6. **Identity:** $1_K \cdot v = v$
7. **Distributivity over vector addition:** $a(u + v) = au + av$
8. **Distributivity over scalar addition:** $(a + b)v = av + bv$

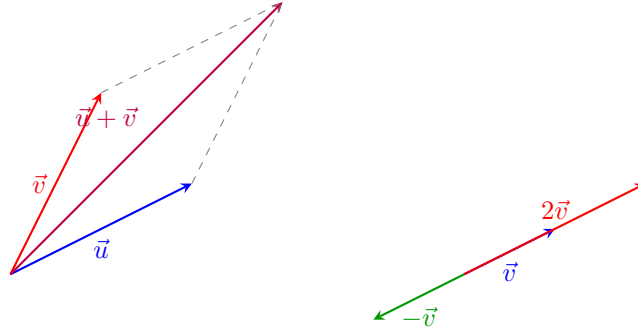
Elements of V are called **vectors**, and elements of K are called **scalars**.

Example 5.1.2 (Geometric Vectors in $\mathbb{R}^2$ and $\mathbb{R}^3$):

The prototypical example of a vector space is the set of geometric vectors in 2D or 3D space. We identify points with their **position vectors**—arrows from the origin to the point.

Addition follows the **tip-to-toe rule**: to add $\vec{u}$ and $\vec{v}$, place the tail of $\vec{v}$ at the tip of $\vec{u}$; the sum $\vec{u} + \vec{v}$ is the arrow from the origin to the tip of $\vec{v}$. Equivalently, this is the **parallelogram rule**.

Scalar multiplication scales the vector: $a\vec{v}$ stretches (if $|a| > 1$) or shrinks (if $|a| < 1$) the vector, and reverses direction if $a < 0$.

**Example 5.1.3 (Coordinate Spaces K^n):**

For any field K and positive integer n , the set

$$K^n = \{(x_0, \dots, x_{n-1}) \mid x_i \in K\}$$

forms a K -vector space with pointwise operations:

$$\begin{aligned} (x_0, \dots, x_{n-1}) + (y_0, \dots, y_{n-1}) &= (x_0 + y_0, \dots, x_{n-1} + y_{n-1}) \\ a \cdot (x_0, \dots, x_{n-1}) &= (ax_0, \dots, ax_{n-1}) \end{aligned}$$

More formally, $K^n = K^{\{0,1,\dots,n-1\}}$ —the set of functions from the ordinal $n = \{0, 1, \dots, n-1\}$ to K . We identify the n -tuple $(x_0, \dots, x_{n-1})$ with the function $i \mapsto x_i$.

Remark (Geometric Interpretation of K^n):

For $n = 1, 2, 3$ over $K = \mathbb{R}$, we identify points in $\mathbb{R}^n$ with **position vectors**: arrows from the origin 0 to the point. This identification is bijective:

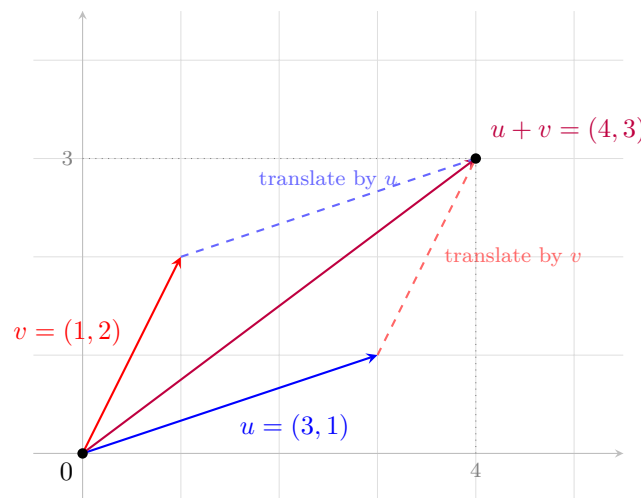
$$(x_0, \dots, x_{n-1}) \longleftrightarrow \text{arrow from } 0 \text{ to } (x_0, \dots, x_{n-1})$$

Under this identification, the vector space operations on $\mathbb{R}^n$ correspond exactly to the geometric operations on arrows described in Example 5.1.2.

Why addition matches the parallelogram rule. Let $e_0 = (1, 0)$ and $e_1 = (0, 1)$ be the standard basis vectors. Any $u = (u_0, u_1)$ decomposes as $u = u_0e_0 + u_1e_1$, and similarly $v = v_0e_0 + v_1e_1$. Then by the vector space axioms:

$$u + v = (u_0e_0 + u_1e_1) + (v_0e_0 + v_1e_1) = (u_0 + v_0)e_0 + (u_1 + v_1)e_1$$

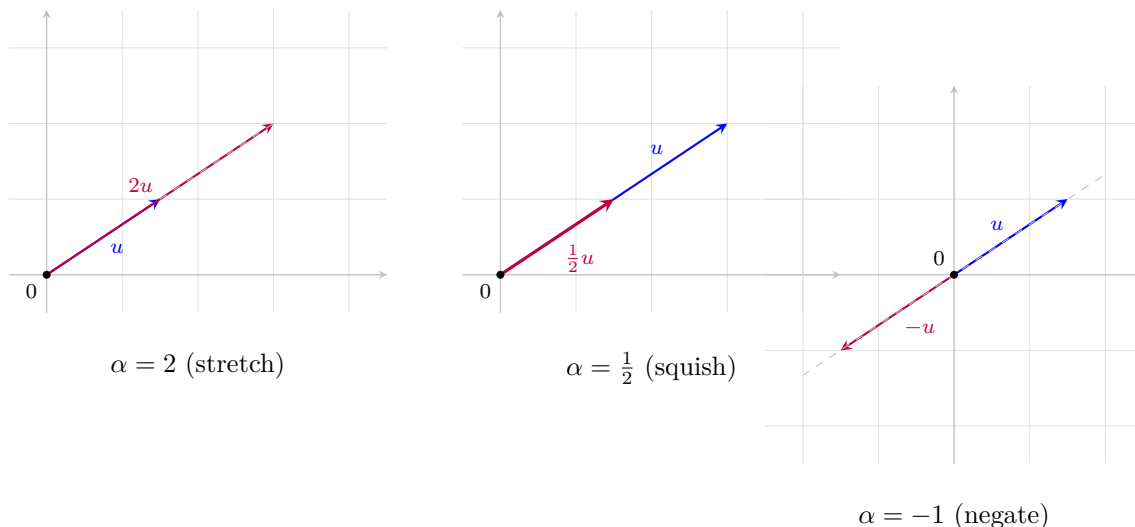
which corresponds to the point $(u_0 + v_0, u_1 + v_1)$. Geometrically, this is the fourth vertex of the parallelogram formed by 0, u , and v : translating the arrow v so that its tail is at the tip of u places its tip at $(u_0 + v_0, u_1 + v_1)$.



Why scalar multiplication matches scaling. For $\alpha \in \mathbb{R}$ and $u = u_0e_0 + u_1e_1$, the scalar multiplication axioms give:

$$\alpha \cdot u = \alpha(u_0e_0 + u_1e_1) = (\alpha u_0)e_0 + (\alpha u_1)e_1$$

which corresponds to the point $(\alpha u_0, \alpha u_1)$. Geometrically, this is the point at fraction α along the ray from 0 through u : it stretches the arrow by $|\alpha|$ and reverses direction if $\alpha < 0$. The parametric line $t \mapsto t \cdot u = (tu_0, tu_1)$ confirms this.



This geometric intuition guides much of linear algebra, even in higher dimensions and over abstract fields where visualization is not possible.

Example 5.1.4 (Function Spaces K^I):

For any set I , the set of all functions $f : I \rightarrow K$ forms a K -vector space K^I with pointwise operations:

$$\begin{aligned}(f + g)(i) &= f(i) + g(i) \\ (af)(i) &= a \cdot f(i)\end{aligned}$$

For each $i \in I$, define the **indicator function** $e_i : I \rightarrow K$ by

$$e_i(j) = \delta_{ij} = \begin{cases} 1_K & \text{if } j = i \\ 0_K & \text{if } j \neq i \end{cases}$$

These indicator functions will play a central role once we introduce bases.

Example 5.1.5 (Functions with Finite Support $K^{(I)}$):

The set of functions with **finite support**

$$K^{(I)} = \{f \in K^I \mid \{i \in I \mid f(i) \neq 0\} \text{ is finite}\}$$

is a subspace of K^I . Every $f \in K^{(I)}$ can be written as a finite sum

$$f = \sum_{i \in \text{supp}(f)} f(i) \cdot e_i$$

where $\text{supp}(f) = \{i \in I \mid f(i) \neq 0\}$ is the support of f .

Note that $K^{(I)} = K^I$ if and only if I is finite.

Remark (Column Vector Notation for K^n):

For $n \in \mathbb{N}$, the elements of $K^n = K^{\{0,1,\dots,n-1\}}$ are functions from $\{0,1,\dots,n-1\}$ to K , i.e., finite-length sequences. We denote these as **column vectors**:

$$f \longleftrightarrow \begin{pmatrix} f(0) \\ f(1) \\ \vdots \\ f(n-1) \end{pmatrix}$$

For instance, the indicator functions in K^n are denoted

$$e_0 = \begin{pmatrix} 1 \\ 0 \\ \vdots \\ 0 \end{pmatrix}, \quad e_1 = \begin{pmatrix} 0 \\ 1 \\ \vdots \\ 0 \end{pmatrix}, \quad \dots, \quad e_{n-1} = \begin{pmatrix} 0 \\ 0 \\ \vdots \\ 1 \end{pmatrix}$$

and a general element $(a_0, a_1, \dots, a_{n-1}) \in K^n$ is written as $\begin{pmatrix} a_0 \\ a_1 \\ \vdots \\ a_{n-1} \end{pmatrix}$.

Remark (Two Structures on K^I and $K^{(I)}$):

The spaces K^I and $K^{(I)}$ carry two fundamentally different structures simultaneously:

- **As a K -vector space:** equipped with pointwise addition and scalar multiplication. The operations that respect this structure are K -linear maps.
- **As a set of functions $I \rightarrow K$** (a “data structure”): a collection of I -indexed families of scalars. Operations that respect this structure include evaluation at a point ($f \mapsto f(i)$), support ($f \mapsto \text{supp}(f)$), entrywise absolute value (when K is ordered), sorting entries, lexicographic ordering, and restriction to subsets of I .

The two structures coexist on the same underlying set, but they are logically independent. Most operations from the second list are *not* K -linear (e.g., $f \mapsto |f(i)|$ fails linearity over $\mathbb{C}$; sorting entries is not even well-defined as a vector space operation). Conversely, a K -linear map $T : K^{(I)} \rightarrow K^{(J)}$ need not respect any “data” property of tuples — it scrambles entries according to its matrix.

When we write $(1, 0, 0) \in \mathbb{R}^3$, we are silently using both structures: as a *vector* it can be added and scaled; as a *triple of real numbers* it can be compared componentwise, plugged into a formula, or printed on a page. Linear algebra concerns itself with the first structure; computation (by hand or machine) requires the second. The coordinate isomorphism (Theorem 5.8.4) and the Representation Theorem (Theorem 7.1.2) are the bridges that let us pass between them.

Example 5.1.6 (Polynomial Spaces):

For a field K , the polynomial ring $K[X]$ is a K -vector space. We define important subspaces based on degree constraints:

- $K[X]_{\leq n} = \{p \in K[X] \mid \deg p \leq n\}$ (polynomials of degree at most n)
- $K[X]_{< n} = \{p \in K[X] \mid \deg p < n\}$ (polynomials of degree less than n)

Example 5.1.7 (Matrix Spaces):

For positive integers m, n , the set $\text{Mat}_{m,n}(K)$ of $m \times n$ matrices with entries in K forms a K -vector space with entrywise operations:

$$\begin{aligned}(A + B)_{ij} &= A_{ij} + B_{ij} \\ (aA)_{ij} &= a \cdot A_{ij}\end{aligned}$$

Example 5.1.8 (Field Extensions):

When L is a field extension of K (i.e., $K \subseteq L$ as fields), then L is naturally a K -vector space with the restriction of multiplication in L as scalar multiplication.

- $\mathbb{C}$ is a 2-dimensional $\mathbb{R}$ -vector space (basis: $\{1, i\}$)
- $\mathbb{R}$ is an infinite-dimensional $\mathbb{Q}$ -vector space

Proposition 5.1.9 (Basic Properties):

Let V be a K -vector space. The following properties hold for all $v \in V$ and $a \in K$:

1. $0_K \cdot v = 0_V$
2. $a \cdot 0_V = 0_V$
3. $(-1_K) \cdot v = -v$
4. $av = 0_V \implies a = 0_K \text{ or } v = 0_V$

Proof:

We prove each property:

- (1) We have $0_K \cdot v = (0_K + 0_K) \cdot v = 0_K \cdot v + 0_K \cdot v$. Adding $-(0_K \cdot v)$ to both sides yields $0_V = 0_K \cdot v$.
- (2) We have $a \cdot 0_V = a \cdot (0_V + 0_V) = a \cdot 0_V + a \cdot 0_V$. Adding $-(a \cdot 0_V)$ to both sides yields $0_V = a \cdot 0_V$.
- (3) We have $v + (-1_K) \cdot v = 1_K \cdot v + (-1_K) \cdot v = (1_K + (-1_K)) \cdot v = 0_K \cdot v = 0_V$. By uniqueness of additive inverses, $(-1_K) \cdot v = -v$.
- (4) Suppose $av = 0_V$ and $a \neq 0_K$. Since K is a field, a^{-1} exists. Then $v = 1_K \cdot v = (a^{-1}a) \cdot v = a^{-1}(av) = a^{-1} \cdot 0_V = 0_V$. QED

Proposition 5.1.10 (Cancellation Laws):

Let V be a K -vector space. Then:

1. If $av = bv$ and $v \neq 0_V$, then $a = b$.
2. If $av = aw$ and $a \neq 0_K$, then $v = w$.

Proof:

(1) From $av = bv$ we get $av - bv = 0_V$, hence $(a - b)v = 0_V$. Since $v \neq 0_V$, by Proposition 5.1.9(4), we have $a - b = 0_K$, so $a = b$.

(2) From $av = aw$ we get $a(v - w) = 0_V$. Since $a \neq 0_K$, by Proposition 5.1.9(4), we have $v - w = 0_V$, so $v = w$. QED

Remark (Notation Abuse):

We use 0 to denote both the zero scalar 0_K and the zero vector 0_V . Context will always make the intended meaning clear.

5.2 Minkowski Structure on $\mathcal{P}(V)$

Remark (Why Set-Level Operations?):

Before studying subspaces, it is useful to have a calculus of *subsets* of a vector space. The Minkowski sum $A + B = \{a + b \mid a \in A, b \in B\}$ and the scalar-set product $LA = \{\lambda a \mid \lambda \in L, a \in A\}$ lift the vector space operations from individual vectors to subsets. These operations will reappear throughout the subject:

- The **sum of subspaces** $U + W$ is a Minkowski sum.
- **Cosets** $v + W$ are translates of a subspace.
- The **subspace criterion** can be stated as $KU + KU = U$.

We establish the algebraic rules (associativity, distributivity, monotonicity) here so they are available when needed later. The key subtlety is that distributivity is only *partial* for sets, unlike for individual vectors — a point illustrated by explicit counterexamples below.

Definition 5.2.1 (Minkowski Sum):

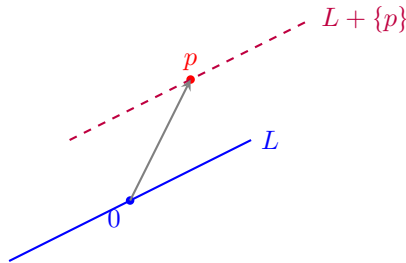
Let V be a K -vector space and $A, B \subseteq V$. The **Minkowski sum** of A and B is

$$A + B = \{a + b \mid a \in A, b \in B\}$$

Example 5.2.2 (Minkowski Sum Examples):

Consider the following examples:

1. In $\mathbb{R}$: $\{1, 2\} + \{10, 20\} = \{11, 21, 12, 22\}$
2. In $\mathbb{R}$: $[0, 1] + [0, 1] = [0, 2]$
3. In $\mathbb{R}^2$: If L is a line through the origin and p is a point, then $L + \{p\}$ is a line parallel to L passing through p .



For singletons, we use the notation $u + U := \{u\} + U$ (translation of U by u).

Definition 5.2.3 (Minkowski Scalar-Set Product):

Let $L \subseteq K$ and $A \subseteq V$. The **Minkowski scalar-set product** is

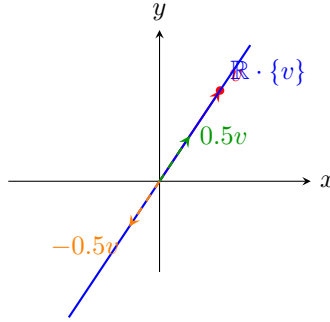
$$L \cdot A = \{\lambda a \mid \lambda \in L, a \in A\}$$

Example 5.2.4 (Minkowski Product Examples):

Consider the following examples:

1. In $\mathbb{R}$: $\{1, 2\} \cdot \{3\} = \{3, 6\}$
2. In $\mathbb{R}$: $[0, 1] \cdot [0, 1] = [0, 1]$

3. In $\mathbb{R}^n$: For $v \neq 0$, $\mathbb{R} \cdot \{v\} = \{\lambda v \mid \lambda \in \mathbb{R}\}$ is the line through the origin in direction v .



For singletons, we write $\lambda \cdot A := \{\lambda\} \cdot A$ (scalar multiplication of the set).

Proposition 5.2.5 (Monoid Structure of $(\mathcal{P}(V), +)$):

The Minkowski sum satisfies:

1. **Associativity:** $(A + B) + C = A + (B + C)$
2. **Commutativity:** $A + B = B + A$
3. **Identity:** $\{0\} + A = A$
4. **Absorbing element:** $\emptyset + A = \emptyset$

Proof:

Properties (1)–(3) follow directly from the corresponding axioms of vector addition. For (4), note that $\emptyset + A = \{a + b \mid a \in \emptyset, b \in A\} = \emptyset$ since there are no elements in $\emptyset$. QED

Proposition 5.2.6 (Monoid Action of $(\mathcal{P}(K), \cdot)$ on $\mathcal{P}(V)$):

The Minkowski scalar-set product satisfies:

1. **Identity:** $\{1_K\} \cdot A = A$
2. **Compatibility:** $(L \cdot M) \cdot A = L \cdot (M \cdot A)$
3. **Absorbing:** $\emptyset \cdot A = \emptyset$ and $L \cdot \emptyset = \emptyset$
4. **Zero properties:** $L \cdot \{0\} = \{0\}$ if $L \neq \emptyset$; $\{0\} \cdot A = \{0\}$ if $A \neq \emptyset$

Proof:

- (1) We have $\{1_K\} \cdot A = \{1_K \cdot a \mid a \in A\} = \{a \mid a \in A\} = A$.
- (2) Both sides equal $\{\lambda\mu a \mid \lambda \in L, \mu \in M, a \in A\}$.
- (3) Follows from the empty set having no elements.
- (4) If $L \neq \emptyset$, then $L \cdot \{0\} = \{\lambda \cdot 0 \mid \lambda \in L\} = \{0\}$. Similarly for $\{0\} \cdot A$. QED

Proposition 5.2.7 (Monotonicity):

Minkowski operations are monotone:

1. If $A \subseteq B$, then $A + C \subseteq B + C$
2. If $A \subseteq B$, then $LA \subseteq LB$
3. If $L \subseteq M$, then $LA \subseteq MA$

Proof:

Each property follows immediately: elements of the LHS are also elements of the RHS since $A \subseteq B$ or $L \subseteq M$. QED

Proposition 5.2.8 (Partial Distributivity):

For all $L, M \subseteq K$ and $A, B \subseteq V$:

1. $L(A + B) \subseteq LA + LB$
2. $(L + M)A \subseteq LA + MA$

Proof:

(1) Let $x \in L(A + B)$. Then $x = \lambda(a + b)$ for some $\lambda \in L$, $a \in A$, $b \in B$. By distributivity in V : $x = \lambda a + \lambda b \in LA + LB$.

(2) Let $x \in (L + M)A$. Then $x = (\lambda + \mu)a$ for some $\lambda \in L$, $\mu \in M$, $a \in A$. By distributivity in V : $x = \lambda a + \mu a \in LA + MA$. QED

Example 5.2.9 (Failure of Reverse Inclusions):

The reverse inclusions of Proposition 5.2.8 do not hold in general:

1. $LA + LB \not\subseteq L(A + B)$: Take $L = \{1, 2\}$, $A = B = \{1\}$ in $\mathbb{R}$. Then $L(A + B) = \{1, 2\} \cdot \{2\} = \{2, 4\}$, but $LA + LB = \{1, 2\} + \{1, 2\} = \{2, 3, 4\}$. We have $3 \in LA + LB$ but $3 \notin L(A + B)$.
2. $LA + MA \not\subseteq (L + M)A$: Take $L = M = \{1\}$, $A = \{0, 1\}$ in $\mathbb{R}$. Then $(L + M)A = \{2\} \cdot \{0, 1\} = \{0, 2\}$, but $LA + MA = \{0, 1\} + \{0, 1\} = \{0, 1, 2\}$. We have $1 \in LA + MA$ but $1 \notin (L + M)A$.

Proposition 5.2.10 (Full Distributivity Conditions):

The reverse inclusions hold under these conditions:

1. If $L = \{\lambda\}$ is a singleton, then $\lambda A + \lambda B = \lambda(A + B)$
2. If $A = \{a\}$ is a singleton, then $La + Ma = (L + M)a$
3. If $KB = B$ and $0 \in A$, then $LA + LB = L(A + B)$
4. If $KA = A$, $A + A = A$, and $L + M \neq \{0\}$, then $LA + MA = (L + M)A$

Proof:

(1) Let $x \in \lambda A + \lambda B$, so $x = \lambda a + \lambda b$ for some $a \in A$, $b \in B$. Then $x = \lambda(a + b) \in \lambda(A + B)$.

(2) Let $x \in La + Ma$, so $x = \lambda a + \mu a$ for some $\lambda \in L$, $\mu \in M$. Then $x = (\lambda + \mu)a \in (L + M)a$.

(3) Assume $KB = B$ and $0 \in A$. Let $x = \lambda a + \mu b \in LA + LB$.

Case $\lambda \neq 0$: Since $KB = B$, we have $(\mu/\lambda)b \in B$. Thus $x = \lambda(a + (\mu/\lambda)b) \in L(A + B)$.

Case $\lambda = 0$: Then $x = \mu b = \mu(0 + b) \in L(A + B)$ since $0 \in A$.

(4) Assume $KA = A$, $A + A = A$, $L + M \neq \{0\}$. Let $x = \lambda a + \mu a' \in LA + MA$.

Case $\lambda + \mu \neq 0$: Since $KA = A$ and $A + A = A$, $\frac{\lambda}{\lambda + \mu}a + \frac{\mu}{\lambda + \mu}a' \in A$. Thus $x = (\lambda + \mu) \cdot \left(\frac{\lambda}{\lambda + \mu}a + \frac{\mu}{\lambda + \mu}a' \right) \in (L + M)A$.

Case $\lambda + \mu = 0$: Then $\mu = -\lambda$ and $x = \lambda(a - a')$. Since $L + M \neq \{0\}$, there exist $\lambda' \in L$ and $\mu' \in M$ such that $\lambda' + \mu' \neq 0$. Now $a - a' \in A + A = A$, so $KA = A$ gives $\frac{\lambda}{\lambda' + \mu'}(a - a') \in A$. Thus $x = (\lambda' + \mu') \cdot \frac{\lambda}{\lambda' + \mu'}(a - a') \in (L + M)A$. QED

Proposition 5.2.11 (Distributivity over Unions):

For arbitrary families:

1. $A + \bigcup_{i \in I} B_i = \bigcup_{i \in I} (A + B_i)$

$$2. L \cdot \bigcup_{i \in I} A_i = \bigcup_{i \in I} LA_i$$

$$3. (\bigcup_{i \in I} L_i) \cdot A = \bigcup_{i \in I} L_i A$$

Proof:

(1) We have $x \in A + \bigcup_{i \in I} B_i$ iff $x = a + b$ with $a \in A$ and $b \in B_j$ for some j , iff $x \in A + B_j$ for some j , iff $x \in \bigcup_{i \in I} (A + B_i)$.

(2) and (3) are analogous. QED

Proposition 5.2.12 (Sub-Distributivity over Intersections):

For arbitrary families:

$$1. A + \bigcap_{i \in I} B_i \subseteq \bigcap_{i \in I} (A + B_i)$$

$$2. L \cdot \bigcap_{i \in I} A_i \subseteq \bigcap_{i \in I} LA_i$$

$$3. (\bigcap_{i \in I} L_i) \cdot A \subseteq \bigcap_{i \in I} L_i A$$

Proof:

(1) If $b \in \bigcap_{i \in I} B_i$, then $b \in B_i$ for all i , so $a + b \in A + B_i$ for all i , hence $a + b \in \bigcap_{i \in I} (A + B_i)$. QED

Example 5.2.13 (Counterexamples for Reverse Intersection Inclusions):

The reverse inclusions fail:

1. $(A + B) \cap (A + C) \not\subseteq A + (B \cap C)$: Take $A = \{0, 1\}$, $B = \{0\}$, $C = \{1\}$ in $\mathbb{R}$. Then $B \cap C = \emptyset$, so $A + (B \cap C) = \emptyset$. But $(A + B) \cap (A + C) = \{0, 1\} \cap \{1, 2\} = \{1\} \neq \emptyset$.

2. $LA \cap LB \not\subseteq L(A \cap B)$: Take $L = \{1, 2\}$, $A = \{1\}$, $B = \{2\}$ in $\mathbb{R}$. Then $A \cap B = \emptyset$, so $L(A \cap B) = \emptyset$. But $LA \cap LB = \{1, 2\} \cap \{2, 4\} = \{2\} \neq \emptyset$.

Proposition 5.2.14 (Singleton Equality for Intersections):

For singletons:

$$1. \{a\} + \bigcap_{i \in I} B_i = \bigcap_{i \in I} (\{a\} + B_i)$$

$$2. \{\lambda\} \cdot \bigcap_{i \in I} A_i = \bigcap_{i \in I} \{\lambda\} A_i$$

Proof:

(1) If $x \in \bigcap_{i \in I} (\{a\} + B_i)$, then for each i , $x = a + b_i$ for some $b_i \in B_i$. But $b_i = x - a$ is the same for all i , so $x - a \in \bigcap_{i \in I} B_i$, hence $x \in \{a\} + \bigcap_{i \in I} B_i$.

(2) is analogous (for $\lambda \neq 0$; if $\lambda = 0$, both sides equal $\{0\}$). QED

5.3 Vector Subspaces

Remark (Why Subspaces?):

A vector space may be large or complicated, but its internal structure is revealed by its subspaces — subsets that are themselves vector spaces under the inherited operations. Lines and planes through the origin in $\mathbb{R}^3$ are the prototypical examples: they are the “flat” subsets that pass through zero. Every fundamental construction in linear algebra — kernel, image, span, eigenspace — produces a subspace. Understanding the lattice of all subspaces of V (how they intersect, sum, and nest) is therefore central to understanding V itself.

The **subspace criterion** below reduces the verification to a single condition: closure under linear combinations. This is the test used in practice.

Definition 5.3.1 (Vector Subspace):

Let V be a K -vector space. A subset $U \subseteq V$ is a **K -vector subspace** of V (written $U \leq_K V$) if U is itself a K -vector space under the inherited operations from V .

Remark (The Subspace Lattice $\text{Sub}(V)$):

We write $\text{Sub}(V)$ for the **set of all K -vector subspaces of V** , ordered by inclusion. It is a *complete lattice*: any family of subspaces has a meet (intersection) and a join (sum):

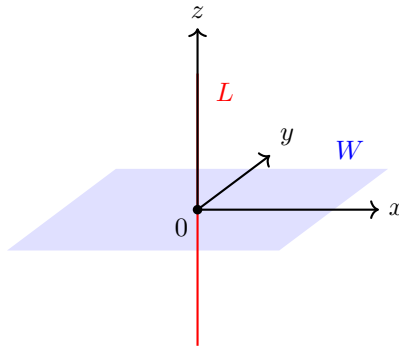
$$\bigwedge_{i \in I} U_i = \bigcap_{i \in I} U_i, \quad \bigvee_{i \in I} U_i = \sum_{i \in I} U_i.$$

The bottom element is $\{0\}$ and the top element is V . This lattice structure underlies the theory of flags (Section 5.9).

Example 5.3.2 (Subspaces in Low Dimensions):

The following are subspaces:

1. Lines through the origin in $\mathbb{R}^2$
2. Lines and planes through the origin in $\mathbb{R}^3$
3. $K[X]_{\leq n} \leq_K K[X]$ (polynomials of degree at most n)



Example 5.3.3 (Non-Subspaces):

The following are *not* subspaces:

1. A line NOT through the origin in $\mathbb{R}^2$ (fails: $0 \notin U$)
2. The first quadrant $\{(x, y) \mid x \geq 0, y \geq 0\}$ in $\mathbb{R}^2$ (fails: not closed under scalar multiplication by -1)
3. The unit circle $\{(x, y) \mid x^2 + y^2 = 1\}$ in $\mathbb{R}^2$ (fails: $0 \notin U$ and not closed under addition)

Proposition 5.3.4 (Subspace Criterion):

A subset $U \subseteq V$ is a subspace if and only if:

1. $U \neq \emptyset$
2. $\forall a, b \in K : \forall u, v \in U : au + bv \in U$

Proof:

($\Rightarrow$) If U is a subspace, then U is a vector space, so $0 \in U$. Closure under linear combinations follows from the vector space axioms.

($\Leftarrow$) Assume (1) and (2). We verify that U inherits all vector space axioms from V :

- **Closure under addition:** Take $a = b = 1$ in (2).
- **Closure under scalar multiplication:** Take $b = 0$ in (2).
- **Contains zero:** Taking $a = b = 0$ and any $u \in U$ gives $0 \in U$.
- **Additive inverses:** Taking $a = -1$, $b = 0$ gives $-u \in U$.
- All other axioms are inherited from V .

QED

Proposition 5.3.5 (Minkowski Characterization of Subspaces):

For a subset $U \subseteq V$, the following are equivalent:

1. U is a subspace
2. $U \neq \emptyset$, $U + U = U$, and $KU = U$
3. $U \neq \emptyset$ and $KU + KU = U$

where $KU = \{au \mid a \in K, u \in U\}$ denotes scalar multiplication of the set.

Proof:

(1) $\Rightarrow$ (2): If U is a subspace, then:

- $U \neq \emptyset$ since $0 \in U$.
- $U + U \subseteq U$ by closure under addition. Also $U \subseteq U + U$ since $u = u + 0$ for any $u \in U$.
- $KU \subseteq U$ by closure under scalar multiplication. Also $U \subseteq KU$ since $u = 1 \cdot u$ for any $u \in U$.

Thus $U \neq \emptyset$, $U + U = U$, and $KU = U$.

(2) $\Rightarrow$ (3): Assume $U \neq \emptyset$, $U + U = U$, and $KU = U$. Then $U \neq \emptyset$ trivially, and $KU + KU = U + U = U$.

(3) $\Rightarrow$ (1): Assume $U \neq \emptyset$ and $KU + KU = U$. We verify the subspace criterion:

- $0 \in U$: Since $U \neq \emptyset$, pick $u \in U$ and any $\lambda \in K$. Then $0 = \lambda u + (-\lambda)u \in KU + KU = U$.
- Closure under linear combinations: For any $a, b \in K$ and $u, v \in U$: $au + bv \in KU + KU = U$.

QED

Example 5.3.6 (Trivial Subspaces):

Every vector space V has two **trivial subspaces**:

- The zero subspace $\{0\}$

- The whole space V

A subspace U with $\{0\} \subsetneq U \subsetneq V$ is called **proper** or **nontrivial**.

Definition 5.3.7 (Linear Combination):

Let V be a K -vector space and $S \subseteq V$. A **linear combination** of vectors in S is a **finite** sum

$$\sum_{i \in n} a_i v_i$$

where $n \in \mathbb{N}$, $a_i \in K$, and $v_i \in S$.

The empty sum (when $n = 0$) is defined to be 0.

Example 5.3.8 (Linear Combinations):

Consider the following examples:

1. $3v_1 + 2v_2$ is a linear combination of $\{v_1, v_2\}$
2. The empty sum 0 is a linear combination of any set (including $\emptyset$)
3. In $\mathbb{R}^3$: $(1, 2, 3) = 1 \cdot e_0 + 2 \cdot e_1 + 3 \cdot e_2$ where e_i are standard vectors

Remark (Finiteness is Essential):

Infinite series like $\sum_{i \in \mathbb{N}} a_i v_i$ are **not** linear combinations in the algebraic sense. Such series require topological structure (limits) to define.

As a counterexample: in $K^{\mathbb{N}}$, let $e_i = (0, \dots, 0, 1, 0, \dots)$ with 1 in position i . Then the constant sequence $(1, 1, 1, \dots)$ is **not** a (finite) linear combination of the e_i 's, even though "intuitively" it equals $\sum_{i \in \mathbb{N}} e_i$.

Proposition 5.3.9 (Subspace iff Closed Under Linear Combinations):

$U \leq_K V$ if and only if U is nonempty and closed under all finite linear combinations:

$$\forall F \subseteq U, F \text{ finite} : \forall (a_v)_{v \in F} \in K^F : \sum_{v \in F} a_v v \in U$$

Proof:

This is a reformulation of the subspace criterion. The condition for two vectors $au + bv \in U$ implies (by induction) that any finite linear combination is in U . Conversely, the linear combination condition with $|F| \leq 2$ gives the criterion. QED

Proposition 5.3.10 (Linear Combinations of Indicator Functions):

Let I be any set and let $(e_i)_{i \in I}$ be the indicator functions in K^I . Then the set of all linear combinations of the e_i 's is precisely $K^{(I)}$. In particular, if I is infinite, then there exist elements of K^I that are not linear combinations of the e_i 's.

Proof:

We show both inclusions.

($\subseteq$) Let g be a linear combination of the e_i 's. Then $g = \sum_{j \in n} a_j e_{i_j}$ for some $n \in \mathbb{N}$, scalars $a_j \in K$, and distinct indices $i_0, \dots, i_{n-1} \in I$. For any $k \in I$:

$$g(k) = \sum_{j \in n} a_j e_{i_j}(k) = \sum_{j \in n} a_j \delta_{i_j k} = \begin{cases} a_j & \text{if } k = i_j \text{ for some } j \in \{0, \dots, n-1\} \\ 0 & \text{otherwise} \end{cases}$$

Hence $\text{supp}(g) \subseteq \{i_0, \dots, i_{n-1}\}$, which is finite. Therefore $g \in K^{(I)}$.

($\supseteq$) Let $f \in K^{(I)}$. Then $\text{supp}(f) = \{i_0, \dots, i_{n-1}\}$ is finite. Define $g = \sum_{k \in n} f(i_k) \cdot e_{i_k}$. For any $j \in I$:

$$g(j) = \sum_{k \in n} f(i_k) \cdot e_{i_k}(j) = \sum_{k \in n} f(i_k) \cdot \delta_{i_k j} = \begin{cases} f(i_k) & \text{if } j = i_k \text{ for some } k \in \{0, \dots, n-1\} \\ 0 & \text{otherwise} \end{cases}$$

Since $j \notin \text{supp}(f)$ implies $f(j) = 0 = g(j)$, and $j = i_k \in \text{supp}(f)$ implies $g(j) = f(i_k) = f(j)$, we conclude $f = g$. Hence f is a linear combination of the e_i 's.

For the last claim: if I is infinite, then the constant-1 function $f(i) = 1$ for all $i \in I$ has $\text{supp}(f) = I$, which is infinite. By the first inclusion, f is not a linear combination of the e_i 's. QED

Proposition 5.3.11 (Intersection of Subspaces):

Let $(U_i)_{i \in I}$ be a family of subspaces of V . Then $\bigcap_{i \in I} U_i$ is a subspace of V .

Proof:

Let $U = \bigcap_{i \in I} U_i$.

- $U \neq \emptyset$: Since $0 \in U_i$ for all i , we have $0 \in U$.
- **Closure:** If $u, v \in U$ and $a, b \in K$, then $u, v \in U_i$ for all i . Since each U_i is a subspace, $au + bv \in U_i$ for all i , hence $au + bv \in U$.

QED

Proposition 5.3.12 (Union of Subspaces):

Let $U, W \leq_K V$. Then $U \cup W$ is a subspace if and only if $U \subseteq W$ or $W \subseteq U$.

Proof:

($\Leftarrow$) If $U \subseteq W$, then $U \cup W = W$, which is a subspace. Similarly if $W \subseteq U$.

($\Rightarrow$) Suppose $U \cup W$ is a subspace but $U \not\subseteq W$ and $W \not\subseteq U$. Then there exist $u \in U \setminus W$ and $w \in W \setminus U$. Since $U \cup W$ is a subspace, $u + w \in U \cup W$.

Case 1: $u + w \in U$. Then $w = (u + w) - u \in U$, contradicting $w \notin U$.

Case 2: $u + w \in W$. Then $u = (u + w) - w \in W$, contradicting $u \notin W$.

In either case we get a contradiction. QED

Proposition 5.3.13 (Minkowski Sum of Subspaces):

If $U, W \leq_K V$, then $U + W \leq_K V$.

Proof:

We verify the subspace criterion:

- $U + W \neq \emptyset$: Since $0 \in U$ and $0 \in W$, we have $0 = 0 + 0 \in U + W$.
- **Closure:** Let $x = u_1 + w_1$ and $y = u_2 + w_2$ be in $U + W$ (with $u_i \in U$, $w_i \in W$). For $a, b \in K$:

$$ax + by = a(u_1 + w_1) + b(u_2 + w_2) = (au_1 + bu_2) + (aw_1 + bw_2) \in U + W$$

since $au_1 + bu_2 \in U$ and $aw_1 + bw_2 \in W$.

QED

Definition 5.3.14 (Sum of Subspaces):

For a family $(U_i)_{i \in I}$ of subspaces of V , the **sum** is

$$\sum_{i \in I} U_i = \left\{ \sum_{i \in F} u_i \mid F \subseteq I \text{ finite, } \forall i \in F : u_i \in U_i \right\}$$

This is the set of all finite sums with one summand from each subspace in some finite subset of the family.

Proposition 5.3.15 (Sum of Subspaces is a Subspace):

For any family $(U_i)_{i \in I}$ of subspaces, $\sum_{i \in I} U_i$ is a subspace of V .

Proof:

Let $S = \sum_{i \in I} U_i$.

- $S \neq \emptyset$: The empty sum (taking $F = \emptyset$) gives $0 \in S$.
- **Closure:** Let $x = \sum_{i \in F} u_i$ and $y = \sum_{j \in G} v_j$ be in S . For $a, b \in K$:

$$ax + by = \sum_{i \in F} au_i + \sum_{j \in G} bv_j = \sum_{k \in F \cup G} w_k$$

where $w_k = au_k + bv_k$ if $k \in F \cap G$, $w_k = au_k$ if $k \in F \setminus G$, and $w_k = bv_k$ if $k \in G \setminus F$. Each $w_k \in U_k$, so $ax + by \in S$.

QED

Proposition 5.3.16 (Finite Sums Equal Minkowski Sums):

For a finite family $(U_0, \dots, U_{n-1})$ of subspaces:

$$\sum_{i \in n} U_i = U_0 + U_1 + \dots + U_{n-1}$$

where the right side is the iterated Minkowski sum.

Proof:

By induction on n . For $n = 2$: both sides equal $\{u_1 + u_2 \mid u_1 \in U_1, u_2 \in U_2\}$. The inductive step follows from associativity of the Minkowski sum.

QED

5.4 Span

Remark (Motivation):

Given a collection of vectors S , we want the smallest subspace that contains all of them — the subspace “generated by” S . This is the **span** of S , and it admits two equivalent descriptions that serve different purposes:

- **Top-down** (intersection): $\text{span}_K(S)$ is the intersection of all subspaces containing S . This definition is clean and immediately shows that the span is a subspace, but it is hard to compute with.
- **Bottom-up** (linear combinations): $\text{span}_K(S)$ is the set of all finite linear combinations of elements in S . This description is constructive and is how we actually work with spans.

The equivalence of these two viewpoints (Proposition 5.4.4) is one of the first nontrivial facts of the theory.

Geometrically, the span of one nonzero vector is a line through the origin; the span of two non-collinear vectors is a plane through the origin; and so on. In general, $\text{span}_K(S)$ is the smallest “flat” subset of V that passes through the origin and contains every vector in S .

Definition 5.4.1 (Span):

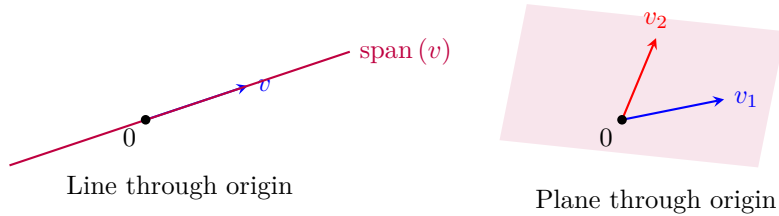
Let V be a K -vector space and $S \subseteq V$. The **span** (or **linear span**) of S is the smallest subspace of V containing S :

$$\text{span}_K(S) = \bigcap \{U \leq_K V \mid S \subseteq U\}$$

Example 5.4.2 (Spans in Coordinate Spaces):

Consider the following examples:

1. $\text{span}_{\mathbb{R}}(\{(1, 0)\}) = \text{the } x\text{-axis in } \mathbb{R}^2$
2. $\text{span}_{\mathbb{R}}(\{(1, 0), (0, 1)\}) = \mathbb{R}^2$
3. $\text{span}_K(\{1, X, X^2\}) = K[X]_{\leq 2}$ (polynomials of degree at most 2)

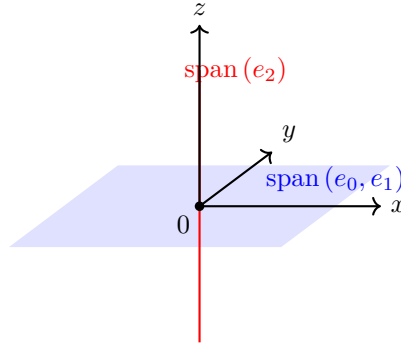


Example 5.4.3 (Spans in $\mathbb{R}^3$):

In $\mathbb{R}^3$, consider the standard basis vectors $e_0 = (1, 0, 0)$, $e_1 = (0, 1, 0)$, $e_2 = (0, 0, 1)$:

1. **Line:** $\text{span}_{\mathbb{R}}(\{e_2\}) = \{(0, 0, z) \mid z \in \mathbb{R}\}$ (the z -axis)
2. **Plane:** $\text{span}_{\mathbb{R}}(\{e_0, e_1\}) = \{(x, y, 0) \mid x, y \in \mathbb{R}\}$ (the xy -plane)
3. **Whole space:** $\text{span}_{\mathbb{R}}(\{e_0, e_1, e_2\}) = \mathbb{R}^3$

Non-example: $\{(1, 0, 0), (2, 0, 0)\}$ spans only a line, not a plane, because $(2, 0, 0) = 2 \cdot (1, 0, 0)$ is redundant.

**Proposition 5.4.4 (Constructive Characterization of Span):**

For any $S \subseteq V$:

$$\text{span}_K(S) = \left\{ \sum_{i \in n} a_i v_i \mid n \in \mathbb{N}, a_i \in K, v_i \in S \right\}$$

That is, $\text{span}_K(S)$ equals the set of all finite K -linear combinations of elements in S .

Proof:

Let L denote the set of all finite linear combinations of elements in S .

($\text{span}_K(S) \subseteq L$): We show L is a subspace containing S , hence contains $\text{span}_K(S)$.

- $L \neq \emptyset$: The empty sum gives $0 \in L$.
- **Closure**: If $x = \sum_{i \in n} a_i v_i$ and $y = \sum_{j \in m} b_j w_j$ are in L , then for $\alpha, \beta \in K$:

$$\alpha x + \beta y = \sum_{i \in n} \alpha a_i v_i + \sum_{j \in m} \beta b_j w_j \in L$$

- $S \subseteq L$: Each $v \in S$ equals $1 \cdot v \in L$.

By definition of span, $\text{span}_K(S) \subseteq L$.

($L \subseteq \text{span}_K(S)$): Let U be any subspace with $S \subseteq U$. Every linear combination of elements in S lies in U (by Proposition 5.3.9), so $L \subseteq U$. Thus $L \subseteq \bigcap \{U : S \subseteq U\} = \text{span}_K(S)$. QED

Proposition 5.4.5 (Properties of Span):

The span operation satisfies:

1. $\text{span}_K(S)$ is a subspace of V
2. $S \subseteq \text{span}_K(S)$
3. If $S \subseteq U$ where $U \leq_K V$, then $\text{span}_K(S) \subseteq U$
4. **Monotonicity**: $S \subseteq T \implies \text{span}_K(S) \subseteq \text{span}_K(T)$
5. **Idempotence**: $\text{span}_K(\text{span}_K(S)) = \text{span}_K(S)$
6. **Subspace criterion**: U is a subspace iff $\text{span}_K(U) = U$
7. **Union**: $\text{span}_K(S \cup T) = \text{span}_K(\text{span}_K(S) \cup \text{span}_K(T))$

Proof:

- (1) Follows from Proposition 5.3.11.
- (2) Each $v \in S$ is contained in every subspace containing S .
- (3) This is the definition: $\text{span}_K(S)$ is the intersection of all such U .
- (4) If $S \subseteq T$, then every subspace containing T also contains S , so $\text{span}_K(S) \subseteq \text{span}_K(T)$.
- (5) $\text{span}_K(S)$ is a subspace containing itself, so $\text{span}_K(\text{span}_K(S)) \subseteq \text{span}_K(S)$. By (2), $\text{span}_K(S) \subseteq \text{span}_K(\text{span}_K(S))$.
- (6) $(\Rightarrow)$ If U is a subspace, then U is the smallest subspace containing U , so $\text{span}_K(U) = U$. $(\Leftarrow)$ If $\text{span}_K(U) = U$, then U is a subspace by (1).
- (7) By monotonicity, $\text{span}_K(S) \cup \text{span}_K(T) \subseteq \text{span}_K(S \cup T)$, so $\text{span}_K(\text{span}_K(S) \cup \text{span}_K(T)) \subseteq \text{span}_K(S \cup T)$. Conversely, $S \cup T \subseteq \text{span}_K(S) \cup \text{span}_K(T)$, so $\text{span}_K(S \cup T) \subseteq \text{span}_K(\text{span}_K(S) \cup \text{span}_K(T))$. QED

Corollary 5.4.6 (Span of Empty Set and Zero):

The following hold:

1. $\text{span}_K(\emptyset) = \{0\}$
2. $\text{span}_K(\{0\}) = \{0\}$

Proof:

- (1) The only linear combination of elements from $\emptyset$ is the empty sum, which equals 0. Thus $\text{span}_K(\emptyset) = \{0\}$.
- (2) Every linear combination of 0 has the form $\sum_{i \in n} a_i \cdot 0 = 0$. Thus $\text{span}_K(\{0\}) = \{0\}$. QED

Proposition 5.4.7 (Sum as Span):

For subspaces $U, W \leq_K V$:

$$U + W = \text{span}_K(U \cup W)$$

Proof:

$(U + W \subseteq \text{span}_K(U \cup W))$: Every element of $U + W$ has the form $u + w$ with $u \in U$, $w \in W$. This is a linear combination of elements in $U \cup W$, hence in $\text{span}_K(U \cup W)$.

$(\text{span}_K(U \cup W) \subseteq U + W)$: Since $U, W \subseteq U + W$ (take $w = 0$ or $u = 0$), we have $U \cup W \subseteq U + W$. Since $U + W$ is a subspace (Proposition 5.3.13), $\text{span}_K(U \cup W) \subseteq U + W$. QED

Proposition 5.4.8 (Arbitrary Sums as Span):

For a family $(U_i)_{i \in I}$ of subspaces:

$$\sum_{i \in I} U_i = \text{span}_K\left(\bigcup_{i \in I} U_i\right)$$

Proof:

Let $S = \sum_{i \in I} U_i$ denote the subspace sum.

$(S \subseteq \text{span}_K(\bigcup_{i \in I} U_i))$: Every element of S has the form $\sum_{j \in F} u_j$ where $F \subseteq I$ is finite and $u_j \in U_j$ for each $j \in F$. Since each $u_j \in U_j \subseteq \bigcup_{i \in I} U_i$, this is a finite linear combination (with coefficients 1) of elements in $\bigcup_{i \in I} U_i$. Hence $\sum_{j \in F} u_j \in \text{span}_K(\bigcup_{i \in I} U_i)$.

$(\text{span}_K(\bigcup_{i \in I} U_i) \subseteq S)$: First, $\bigcup_{i \in I} U_i \subseteq S$: for any $u \in U_j$, we have $u = \sum_{i \in \{j\}} u$ where $\{j\}$ is a finite subset of I and $u \in U_j$, so $u \in S$.

Since S is a subspace (Proposition 5.3.15) containing $\bigcup_{i \in I} U_i$, by the minimality of span, $\text{span}_K(\bigcup_{i \in I} U_i) \subseteq S$. QED

Proposition 5.4.9 (Sum of Spans):

For a family $(S_i)_{i \in I}$ of subsets of V :

$$\sum_{i \in I} \text{span}_K(S_i) = \text{span}_K\left(\bigcup_{i \in I} S_i\right)$$

Proof:

By Proposition 5.4.8:

$$\sum_{i \in I} \text{span}_K(S_i) = \text{span}_K\left(\bigcup_{i \in I} \text{span}_K(S_i)\right)$$

Since $S_i \subseteq \text{span}_K(S_i)$, we have $\bigcup_i S_i \subseteq \bigcup_i \text{span}_K(S_i)$. Each $\text{span}_K(S_i) \subseteq \text{span}_K\left(\bigcup_j S_j\right)$ by monotonicity, so $\bigcup_i \text{span}_K(S_i) \subseteq \text{span}_K\left(\bigcup_j S_j\right)$.

By idempotence:

$$\text{span}_K\left(\bigcup_i \text{span}_K(S_i)\right) = \text{span}_K\left(\bigcup_i S_i\right)$$

QED

Definition 5.4.10 (Generating System):

A subset $G \subseteq V$ is a **generating system** (or **spanning set**) for V if $\text{span}_K(G) = V$.

Example 5.4.11 (Generating Systems):

Consider the following examples:

1. $\{(1, 0), (0, 1)\}$ generates $\mathbb{R}^2$ (standard basis)
2. $\{(1, 0), (0, 1), (1, 1)\}$ also generates $\mathbb{R}^2$ (redundant generator)
3. $\{1, X, X^2, \dots\}$ generates $K[X]$
4. Every vector space has a generating system: $V = \text{span}_K(V)$

Proposition 5.4.12 (V Generates V):

Every vector space V is a generating system for itself: $\text{span}_K(V) = V$.

Proof:

Since V is a subspace containing V , we have $\text{span}_K(V) \subseteq V$. By Proposition 5.4.5(2), $V \subseteq \text{span}_K(V)$. Hence $\text{span}_K(V) = V$. QED

Proposition 5.4.13 (Superset of Generating is Generating):

If G generates V and $G \subseteq H \subseteq V$, then H generates V .

Proof:

By monotonicity: $V = \text{span}_K(G) \subseteq \text{span}_K(H) \subseteq V$. QED

Proposition 5.4.14 ($\text{span}_K(\{e_i\}_{i \in I}) = K^{(I)}$):

For any index set I , the span of the indicator functions equals the space of functions with finite support:

$$\text{span}_K(\{e_i\}_{i \in I}) = K^{(I)}$$

Proof:

This is an immediate reformulation of Proposition 5.3.10 in the language of span: the set of all linear combinations of the e_i 's is precisely $K^{(I)}$, and by the constructive characterization of span (Proposition 5.4.4), this set equals $\text{span}_K(\{e_i\}_{i \in I})$. QED

Corollary 5.4.15:

If I is infinite, then $\text{span}_K(\{e_i\}_{i \in I}) = K^{(I)} \subsetneq K^I$. In particular, the indicator functions $(e_i)_{i \in I}$ do **not** generate K^I .

Proof:

By Proposition 5.4.14, $\text{span}_K(\{e_i\}_{i \in I}) = K^{(I)}$. Since I is infinite, the constant-1 function $f(i) = 1$ for all $i \in I$ lies in $K^I \setminus K^{(I)}$ (as shown in Proposition 5.3.10), so $K^{(I)} \subsetneq K^I$. QED

5.5 Linear Independence

Remark (Intuition):

Not all vectors contribute equally to a span. Some vectors are “redundant”—they can be expressed in terms of others. Linear independence captures the notion that each vector in a set genuinely adds new directions.

Example 5.5.1 (Linear Independence and Dependence):

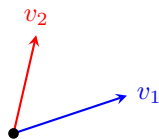
Consider the following examples in $\mathbb{R}^2$ and $\mathbb{R}^3$:

Linearly independent sets:

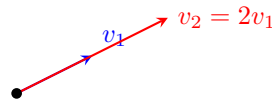
- $\{(1, 0), (0, 1)\}$ in $\mathbb{R}^2$
- $\{1, X, X^2\}$ in $K[X]$
- $\{e_0, e_1, e_2\}$ (standard basis) in $\mathbb{R}^3$

Linearly dependent sets:

- $\{(1, 0), (2, 0)\}$ in $\mathbb{R}^2$ (one is a scalar multiple of the other)
- $\{(1, 0, 0), (0, 1, 0), (1, 1, 0)\}$ in $\mathbb{R}^3$ (the third is the sum of the first two)



Linearly independent



Linearly dependent

Example 5.5.2 (Linear Independence in $\mathbb{R}^3$):

Consider the following cases in $\mathbb{R}^3$:

Independent (3 vectors spanning $\mathbb{R}^3$):

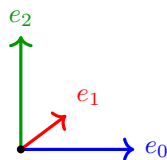
- $\{e_0, e_1, e_2\} = \{(1, 0, 0), (0, 1, 0), (0, 0, 1)\}$ — the standard basis
- $\{(1, 1, 0), (1, 0, 1), (0, 1, 1)\}$ — three non-coplanar vectors

Dependent (more than $\dim = 3$ vectors):

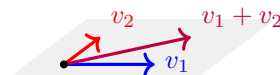
- $\{e_0, e_1, e_2, (1, 1, 1)\}$ — any 4 vectors in $\mathbb{R}^3$ are dependent

Dependent (coplanar vectors):

- $\{(1, 0, 0), (0, 1, 0), (1, 1, 0)\}$ — three vectors in the xy -plane satisfy $(1, 1, 0) = (1, 0, 0) + (0, 1, 0)$



Independent



Dependent (coplanar)

Intuition (Linear Dependence on a Set):

Let $S \subseteq V$ and $v \in S$. Intuitively, we say v is **linearly dependent on** $S \setminus \{v\}$ if $v \in \text{span}_K(S \setminus \{v\})$ —that is, v can be expressed as a linear combination of the other vectors in S . Linear dependence is formally defined below as “not linearly independent.”

Example 5.5.3 (Dependence Examples):

In $\mathbb{R}^3$ with standard basis $\{e_0, e_1, e_2\}$:

- $(1, 2, 3)$ is linearly dependent on $\{e_0, e_1, e_2\}$ because $(1, 2, 3) = 1 \cdot e_0 + 2 \cdot e_1 + 3 \cdot e_2 \in \text{span}_{\mathbb{R}}(\{e_0, e_1, e_2\})$.
- e_0 is linearly *independent* of $\{e_1, e_2\}$ because $e_0 = (1, 0, 0) \notin \text{span}_{\mathbb{R}}(\{e_1, e_2\}) = \{(0, y, z) \mid y, z \in \mathbb{R}\}$.

Definition 5.5.4 (Unique Representation):

Let $L \subseteq V$. A vector $v \in V$ has **unique representation** in terms of L if there exists a **unique** finite subset $F \subseteq L$ and **unique nonzero** scalars $(a_\ell)_{\ell \in F}$ such that

$$v = \sum_{\ell \in F} a_\ell \ell$$

The requirement that scalars be nonzero ensures uniqueness: otherwise we could add $0 \cdot w$ for any $w \in L \setminus F$.

Theorem 5.5.5 (Characterizations of Linear Independence):

Let $L \subseteq V$. The following are equivalent:

- (i) $\forall v \in L: \text{span}_K(\{v\}) \cap \text{span}_K(L \setminus \{v\}) = \{0\}$
- (i') $\forall v \in L: \text{span}_K(L \setminus \{v\}) \subsetneq \text{span}_K(L)$
- (ii) Every $v \in \text{span}_K(L)$ has unique representation in terms of L
- (iii) Some $v \in \text{span}_K(L)$ has unique representation
- (iv) 0 has unique representation (i.e., if $0 = \sum_{i \in n} a_i \ell_i$ with $\ell_i \in L$ distinct and $a_i \neq 0$, then $n = 0$)

Proof:

(i') $\Leftrightarrow$ (i): We have $\text{span}_K(\{v\}) \cap \text{span}_K(L \setminus \{v\}) = \{0\}$ iff $v \notin \text{span}_K(L \setminus \{v\})$ (since $\text{span}_K(\{v\}) = \{\lambda v \mid \lambda \in K\}$). This is equivalent to $\text{span}_K(L \setminus \{v\}) \neq \text{span}_K(L)$ (since $\text{span}_K(L) = \text{span}_K((L \setminus \{v\}) \cup \{v\})$).

(ii) $\Rightarrow$ (iii): Trivial.

(iii) $\Rightarrow$ (iv): Suppose $v \in \text{span}_K(L)$ has unique representation $v = \sum_{\ell \in F} a_\ell \ell$. If $0 = \sum_{\ell \in G} b_\ell \ell$ with $b_\ell \neq 0$, then

$$v = v + 0 = \sum_{\ell \in F} a_\ell \ell + \sum_{\ell \in G} b_\ell \ell$$

For this to be the unique representation of v , we need $G = \emptyset$.

(iv) $\Rightarrow$ (ii): Suppose $v = \sum_{\ell \in F} a_\ell \ell = \sum_{\ell \in G} b_\ell \ell$ are two representations (with nonzero coefficients). Then

$$0 = \sum_{\ell \in F} a_\ell \ell - \sum_{\ell \in G} b_\ell \ell$$

By (iv), all coefficients must be zero after combining like terms. This means $F = G$ and $a_\ell = b_\ell$ for all $\ell \in F$.

(i) $\Rightarrow$ (iv) (requires field axioms): Suppose $0 = \sum_{i \in n} a_i \ell_i$ with $a_i \neq 0$ and $n \geq 1$. Then $\ell_0 = -\frac{1}{a_0} \sum_{i \in n, i \neq 0} a_i \ell_i \in \text{span}_K(L \setminus \{\ell_0\})$, contradicting (i').

(iv) $\Rightarrow$ (i) (does not require field axioms): Suppose $v \in \text{span}_K(\{v\}) \cap \text{span}_K(L \setminus \{v\})$ with $v \neq 0$. Then $v = \lambda v$ for some $\lambda \in K$ and $v = \sum_{\ell \neq v} b_\ell \ell$. Thus $0 = (\lambda - 1)v - \sum_{\ell \neq v} b_\ell \ell$. If $v \notin L \setminus \{v\}$ (the summands), this is a nontrivial relation, contradicting (iv). QED

Definition 5.5.6 (Linearly Independent Set):

A subset $L \subseteq V$ is **linearly independent** if it satisfies any (hence all) of the equivalent conditions in Theorem 5.5.5.

A set that is not linearly independent is called **linearly dependent**.

Proposition 5.5.7 (Zero Vector Implies Dependence):

If $0 \in L$, then L is linearly dependent.

Proof:

The relation $1 \cdot 0 = 0$ is a nontrivial linear combination (one term with nonzero coefficient), violating condition (iv). QED

Proposition 5.5.8 (Empty Set is Linearly Independent):

The empty set $\emptyset$ is linearly independent.

Proof:

Condition (iv) is vacuously satisfied: there are no nontrivial linear combinations of elements from the empty set. QED

Proposition 5.5.9 (Subset of Linearly Independent is Linearly Independent):

If L is linearly independent and $M \subseteq L$, then M is linearly independent.

Proof:

Any nontrivial linear combination of elements in M would be a nontrivial linear combination of elements in L , contradicting the independence of L . QED

Proposition 5.5.10 (Finite Character of Linear Independence):

A set $L \subseteq V$ is linearly independent if and only if every finite subset of L is linearly independent.

Proof:

($\Rightarrow$): Proposition 5.5.9. ($\Leftarrow$): Every nontrivial linear combination $\sum_{i \in n} a_i \ell_i = 0$ involves only finitely many elements $\{\ell_0, \dots, \ell_{n-1}\} \subseteq L$. If this finite subset is linearly independent, all $a_i = 0$, contradicting nontriviality. QED

Proposition 5.5.11 (Extending a Linearly Independent Set):

Let L be linearly independent and $v \in V$. Then $L \cup \{v\}$ is linearly independent if and only if $v \notin \text{span}_K(L)$.

Proof:

($\Rightarrow$) Suppose $L \cup \{v\}$ is linearly independent. By condition (i), $\text{span}_K(\{v\}) \cap \text{span}_K(L) = \{0\}$. Since $v \in \text{span}_K(\{v\})$, if $v \in \text{span}_K(L)$, then $v \in \{0\}$, so $v = 0$. But $\{0\} \cup L$ contains 0, and $0 = 0 \cdot 0$ (if $0 \in L$) or $0 = 1 \cdot 0$ gives a nontrivial relation. Hence $v \notin \text{span}_K(L)$.

($\Leftarrow$) Suppose $v \notin \text{span}_K(L)$. If $0 = av + \sum_{\ell \in F} a_\ell \ell$ with $a \neq 0$ or some $a_\ell \neq 0$:

- If $a \neq 0$: $v = -\frac{1}{a} \sum_{\ell \in F} a_\ell \ell \in \text{span}_K(L)$, contradiction.
- If $a = 0$: $0 = \sum_{\ell \in F} a_\ell \ell$ is nontrivial, contradicting independence of L .

QED

5.6 Internal Direct Sums

Remark (Motivation):

A sum $U_1 + U_2 + \cdots + U_k$ of subspaces always produces a subspace, but the decomposition of a vector $v = u_1 + u_2 + \cdots + u_k$ may not be unique: different choices of $u_i \in U_i$ can give the same v . When the decomposition *is* unique for every vector in the sum, we have a **direct sum** — the strongest and most useful way subspaces can combine.

Direct sums are the subspace-level analogue of linear independence: linear independence says each *vector* in a set contributes a genuinely new direction; a direct sum says each *subspace* in a family contributes a genuinely new piece of the space. The two notions are formally connected by Proposition 5.6.7: a set of vectors is linearly independent if and only if the one-dimensional subspaces they span form a direct sum family.

In practice, direct sum decompositions appear whenever we decompose a space into independent components — coordinate axes, eigenspaces, invariant subspaces — and the uniqueness of the decomposition is what makes computations in the components unambiguous.

Remark (Duality with Linear Independence):

The characterizations of internal direct sums are **similar** to those of linear independence:

- **Linear independence:** unique representation of vectors as linear combinations of **vectors** in L
- **Direct sum:** unique representation of vectors as sums of **subspace elements** from $(U_i)_{i \in I}$

Definition 5.6.1 (Unique Representation in a Family of Subspaces):

Given a family $(U_i)_{i \in I}$ of subspaces of V , a vector v has **unique representation** in the family if there exists a **unique** finite subset $F \subseteq I$ and **unique nonzero** vectors $(u_i)_{i \in F}$ with $u_i \in U_i$ such that

$$v = \sum_{i \in F} u_i$$

The requirement that components be nonzero ensures uniqueness: otherwise we could add $0 \in U_j$ for any j .

Theorem 5.6.2 (Characterizations of Internal Direct Sums):

Let $(U_i)_{i \in I}$ be a family of subspaces of V . The following are equivalent:

- $\forall j \in I: U_j \cap \sum_{i \in I, i \neq j} U_i = \{0\}$
- Every $v \in \sum_{i \in I} U_i$ has unique representation in the family
- Some $v \in \sum_{i \in I} U_i$ with $v \neq 0$ has unique representation
- 0 has unique representation (i.e., if $0 = \sum_{i \in F} u_i$ with $u_i \in U_i$ nonzero, then $F = \emptyset$)

Proof:

(ii) $\Rightarrow$ (iii): Trivial (take any nonzero v ; note $v = 0$ always has unique representation with $F = \emptyset$).

(iii) $\Rightarrow$ (iv): Let $v \neq 0$ have unique representation $v = \sum_{i \in F} u_i$ with $u_i \neq 0$. If $0 = \sum_{j \in G} w_j$ with $w_j \neq 0$, then

$$v = v + 0 = \sum_{i \in F} u_i + \sum_{j \in G} w_j$$

For uniqueness, we need $G = \emptyset$.

(iv) $\Rightarrow$ (ii): Suppose $v = \sum_{i \in F} u_i = \sum_{j \in G} w_j$ with $u_i, w_j \neq 0$. Then $0 = \sum_{i \in F} u_i - \sum_{j \in G} w_j$. For $k \in F \cap G$: the coefficient is $u_k - w_k \in U_k$. For $k \in F \setminus G$: the coefficient is $u_k \in U_k$. For $k \in G \setminus F$: the coefficient is $-w_k \in U_k$. By (iv), all these must be zero, so $F = G$ and $u_i = w_i$.

(i) $\Rightarrow$ (iv) (requires field axioms for extraction): Suppose $0 = \sum_{i \in F} u_i$ with $u_j \neq 0$ for some $j \in F$. Then $u_j = -\sum_{i \in F, i \neq j} u_i \in \sum_{i \in I, i \neq j} U_i$. Also $u_j \in U_j$, so $u_j \in U_j \cap \sum_{i \in I, i \neq j} U_i = \{0\}$ by (i). This contradicts $u_j \neq 0$.

(iv) $\Rightarrow$ (i): Suppose $u \in U_j \cap \sum_{i \in I, i \neq j} U_i$ with $u \neq 0$. Then $u = \sum_{i \neq j} u_i$ for some $u_i \in U_i$. Rearranging: $0 = (-u) + \sum_{i \neq j} u_i$ where $-u \in U_j$. The component in U_j is $-u \neq 0$, contradicting (iv). QED

Definition 5.6.3 (Internal Direct Sum Family):

A family $(U_i)_{i \in I}$ of subspaces is an **internal direct sum family** if it satisfies the equivalent conditions of Theorem 5.6.2. We write $\bigoplus_{i \in I} U_i$ for the sum when the family is a direct sum family.

Proposition 5.6.4 (Two-Subspace Criterion):

For two subspaces $U, W \leq_K V$:

$$U + W \text{ is a direct sum} \iff U \cap W = \{0\}$$

When this holds, we write $U \oplus W$ for the direct sum.

Proof:

Taking $I = \{0, 1\}$ with $U_0 = U$, $U_1 = W$, condition (i) becomes:

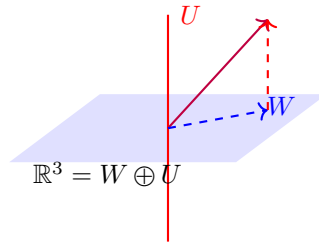
- $U \cap W = \{0\}$
- $W \cap U = \{0\}$

These are equivalent, giving the criterion. QED

Example 5.6.5 (Direct Sum Decompositions):

Consider the following direct sum decompositions:

1. $\mathbb{R}^3 = (\text{xy-plane}) \oplus (\text{z-axis})$
2. $K^n = \bigoplus_{i \in n} K \cdot e_i$ (direct sum of coordinate axes)
3. $K[X] = K[X]_{\text{even}} \oplus K[X]_{\text{odd}}$ (even/odd degree polynomials)

**Example 5.6.6 (Non-Direct Sum):**

Two distinct lines through the origin in $\mathbb{R}^3$ do NOT decompose $\mathbb{R}^3$: their sum is only a plane, not all of $\mathbb{R}^3$.

However, two distinct lines through the origin in $\mathbb{R}^2$ DO form a direct sum decomposition of $\mathbb{R}^2$ (assuming they are not the same line).

Proposition 5.6.7 (Linear Independence and Direct Sum):

Let $F \subseteq V$ and let $(A_i)_{i \in I}$ be a partition of F into linearly independent subsets. Then the following are equivalent:

1. F is linearly independent
2. $(\text{span}_K(A_i))_{i \in I}$ is a direct sum family

Proof:

(1) $\Rightarrow$ (2): Suppose F is linearly independent. We show $(\text{span}_K(A_i))_{i \in I}$ satisfies the direct sum property (iv): if $0 = \sum_{i \in G} u_i$ with $u_i \in \text{span}_K(A_i)$ nonzero, then $G = \emptyset$.

Each $u_i \in \text{span}_K(A_i)$ can be written as $u_i = \sum_{v \in B_i} a_{i,v}v$ for some finite $B_i \subseteq A_i$ with $a_{i,v} \neq 0$. Thus:

$$0 = \sum_{i \in G} \sum_{v \in B_i} a_{i,v}v$$

Since (A_i) partitions F , all these v are distinct elements of F . By linear independence of F (Theorem 5.5.5(iv)), all $a_{i,v} = 0$. But we assumed $a_{i,v} \neq 0$, so each $B_i = \emptyset$, hence each $u_i = 0$. This contradicts $u_i \neq 0$, so $G = \emptyset$.

(2) $\Rightarrow$ (1): Suppose $(\text{span}_K(A_i))_{i \in I}$ is a direct sum family. To prove F is linearly independent, consider any relation $0 = \sum_{v \in G} a_v v$ with $G \subseteq F$ finite and scalars $a_v \in K$; we show that every $a_v = 0$.

Group the terms by the block of the partition to which each vector belongs: for each $i \in I$ put $G_i = G \cap A_i$ and

$$u_i = \sum_{v \in G_i} a_v v \in \text{span}_K(A_i).$$

Since (A_i) partitions F , the sets G_i are pairwise disjoint with union G , so

$$\sum_{i \in I} u_i = \sum_{v \in G} a_v v = 0$$

(a finite sum, as only finitely many G_i are nonempty). By the defining property of a direct sum family (characterization (iv)), each $u_i = 0$.

Now fix $i \in I$. Then $\sum_{v \in G_i} a_v v = u_i = 0$ is a finite linear combination of the block A_i , which is linearly independent *by hypothesis*; hence $a_v = 0$ for every $v \in G_i$. As this holds for all i and the G_i cover G , we conclude $a_v = 0$ for every $v \in G$. Therefore F is linearly independent. QED

Corollary 5.6.8 (Singleton Partition Case):

Let $F \subseteq V$ with $0 \notin F$. Then F is linearly independent if and only if $(\text{span}_K(\{v\}))_{v \in F}$ is a direct sum family with sum $\text{span}_K(F)$.

Proof:

Since $0 \notin F$, each singleton block $\{v\}$ (for $v \in F$) is linearly independent; hence $(\{v\})_{v \in F}$ is a partition of F into linearly independent subsets, and Proposition 5.6.7 applies to give the stated equivalence. The hypothesis $0 \notin F$ cannot be dropped: a block $\text{span}_K(\{0\}) = \{0\}$ satisfies the direct-sum condition vacuously, so for $F \ni 0$ the family would be a direct sum family even though F is linearly dependent. Finally, the sum of the family is

$$\sum_{v \in F} \text{span}_K(\{v\}) = \text{span}_K\left(\bigcup_{v \in F} \{v\}\right) = \text{span}_K(F)$$

(sum of spans equals the span of the union), independently of directness. QED

Proposition 5.6.9 (Direct Sum Characterization via Linear Independence):

A family $(U_i)_{i \in I}$ is a direct sum family if and only if for any linearly independent sets $F_i \subseteq U_i$, the union $\bigcup_{i \in I} F_i$ is linearly independent.

Proof:

($\Rightarrow$) Suppose (U_i) is a direct sum family. Let $F_i \subseteq U_i$ be linearly independent for each i . Suppose $0 = \sum_k a_k v_k$ where $v_k \in \bigcup_{i \in I} F_i$ with $a_k \neq 0$. Group by index: $0 = \sum_i (\sum_{v \in F_i} a_v v)$. Each inner sum $u_i = \sum_{v \in F_i} a_v v \in \text{span}_K(F_i) \subseteq U_i$. By the direct sum property, each $u_i = 0$. By linear independence of F_i , all $a_v = 0$, so the only representation of 0 is the empty sum. Hence $\bigcup_{i \in I} F_i$ is linearly independent.

($\Leftarrow$) We prove (U_i) satisfies the direct sum property (iv): if $0 = \sum_{i \in G} u_i$ with $u_i \in U_i$ nonzero, then $G = \emptyset$.

Suppose $0 = \sum_{i \in G} u_i$ with $u_i \neq 0$ for all $i \in G$. For each $i \in G$, the singleton $\{u_i\} \subseteq U_i$ is linearly independent (since $u_i \neq 0$). By assumption, $\bigcup_{i \in G} \{u_i\}$ is linearly independent. But $0 = \sum_{i \in G} 1 \cdot u_i$ is a nontrivial representation of zero, contradicting linear independence. Hence $G = \emptyset$. QED

Definition 5.6.10 (Direct Sum Decomposition of V):

We say V **decomposes** as a direct sum $V = \bigoplus_{i \in I} U_i$ if:

1. $(U_i)_{i \in I}$ is a direct sum family
2. $V = \sum_{i \in I} U_i$

Equivalently: every $v \in V$ has unique representation in the family.

5.7 Bases and Dimension

Remark (Why Bases?):

A basis is the central concept of linear algebra. It is a set of vectors that is simultaneously linearly independent (no redundancy) and generating (no gaps). Equivalently, it is a set in which every vector in the space has a **unique** expression as a finite linear combination.

The existence of a basis (guaranteed by Zorn's lemma) has far-reaching consequences:

- It reduces the study of an arbitrary vector space to the study of the coordinate space $K^{(I)}$, since every space with a basis of cardinality $|I|$ is isomorphic to $K^{(I)}$.
- It reduces the study of linear maps to the study of their values on basis vectors (the universal property, Theorem 6.2.1).
- It produces a numerical invariant — the **dimension** — that classifies vector spaces up to isomorphism.

The fact that any two bases of the same space have the same cardinality (Theorem 5.7.14) is what makes dimension well-defined. This is ultimately a consequence of the Steinitz exchange lemma (Theorem 5.7.12).

Definition 5.7.1 (Basis):

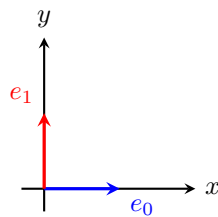
Let V be a K -vector space. A subset $B \subseteq V$ is a **basis** for V if B is both linearly independent and a generating system for V .

A basis of infinite cardinality is sometimes called a **Hamel basis** to distinguish it from topological bases used in functional analysis.

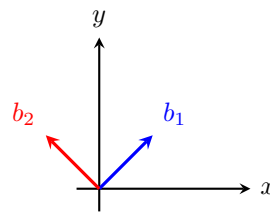
Example 5.7.2 (Standard Bases):

Consider the following examples of bases:

1. $\{(1, 0), (0, 1)\}$ is the standard basis for $\mathbb{R}^2$
2. $\{(1, 1), (1, -1)\}$ is a non-standard basis for $\mathbb{R}^2$
3. $\{1, X, X^2, \dots\}$ is a basis for $K[X]$
4. $\{e_0, \dots, e_{n-1}\}$ is the standard basis for K^n



Standard basis



Rotated basis

Example 5.7.3 (Bases in $\mathbb{R}^3$):

In $\mathbb{R}^3$, consider the following:

Bases (linearly independent + spanning):

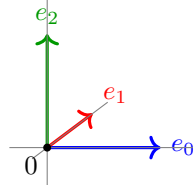
- $\{e_0, e_1, e_2\} = \{(1, 0, 0), (0, 1, 0), (0, 0, 1)\}$ — standard basis
- $\{(1, 1, 0), (1, 0, 1), (0, 1, 1)\}$ — non-standard basis (verify: determinant $\neq 0$)
- $\{(1, 0, 0), (1, 1, 0), (1, 1, 1)\}$ — upper triangular basis

Not a basis (spanning but dependent):

- $\{e_0, e_1, e_2, (1, 1, 1)\}$ — 4 vectors in $\mathbb{R}^3$, so dependent

Not a basis (independent but not spanning):

- $\{e_0, e_1\}$ — only spans the xy -plane, not all of $\mathbb{R}^3$



Standard basis of $\mathbb{R}^3$

Proposition 5.7.4 (Basis as Direct Sum Decomposition):

B is a basis for V if and only if $V = \bigoplus_{v \in B} \text{span}_K(\{v\})$.

Proof:

By Proposition 5.6.7, B is linearly independent iff $(\text{span}_K(\{v\}))_{v \in B}$ is a direct sum family.

Moreover, B generates V iff $V = \text{span}_K(B) = \sum_{v \in B} \text{span}_K(\{v\})$.

Combining: B is a basis iff the family is a direct sum family with sum V .

QED

Remark (Observations from Previous Sections):

From our earlier work:

- The empty set $\emptyset$ is always linearly independent (Proposition 5.5.8)
- The whole space V is always a generating system ($\text{span}_K(V) = V$)

Proposition 5.7.5 (Maximal/Minimal Characterizations):

Let V be a K -vector space. Then:

1. A linearly independent set is maximal (under inclusion) iff it is a basis
2. A generating set is minimal (under inclusion) iff it is a basis

Proof:

(1):

($\Rightarrow$) Let L be maximal linearly independent. Suppose L does not generate V . Then there exists $v \in V \setminus \text{span}_K(L)$. By Proposition 5.5.11, $L \cup \{v\}$ is linearly independent, contradicting maximality of L .

($\Leftarrow$) If L is a basis, it is linearly independent. If $L \subsetneq M$ with M linearly independent, pick $v \in M \setminus L$. Since L generates V , $v \in \text{span}_K(L)$, so M is not linearly independent by Proposition 5.5.11. Contradiction.

(2):

($\Rightarrow$) Let G be minimal generating. Suppose G is linearly dependent. Then some $g \in G$ satisfies $g \in \text{span}_K(G \setminus \{g\})$. By Proposition 5.4.5(7), $\text{span}_K(G) = \text{span}_K(G \setminus \{g\})$, so $G \setminus \{g\}$ generates V , contradicting minimality.

($\Leftarrow$) If G is a basis, it generates V . If $H \subsetneq G$ also generates V , pick $g \in G \setminus H$. Then $g \in \text{span}_K(H) \subseteq \text{span}_K(G \setminus \{g\})$, contradicting linear independence. QED

Theorem 5.7.6 (Basis Extension Theorem):

Let L be a linearly independent subset and G a generating set with $L \subseteq G$. Then there exists a basis B with $L \subseteq B \subseteq G$.

Proof:

Consider the poset

$$\mathcal{F} = \{M \mid L \subseteq M \subseteq G, M \text{ linearly independent}\}$$

ordered by inclusion. This is nonempty since $L \in \mathcal{F}$.

Chains have upper bounds: Let $(M_\alpha)_{\alpha \in A}$ be a chain in $\mathcal{F}$. Set $M = \bigcup_{\alpha \in A} M_\alpha$. Then:

- $L \subseteq M \subseteq G$ since each M_α satisfies this.
- M is linearly independent: any finite subset $F \subseteq M$ is contained in some M_α (since chains are totally ordered), hence F is linearly independent.

Thus $M \in \mathcal{F}$ is an upper bound.

By Zorn's Lemma, $\mathcal{F}$ has a maximal element B .

B is a basis: B is linearly independent by construction. We show B generates V . Suppose $g \in G \setminus \text{span}_K(B)$. Then $B \cup \{g\}$ is linearly independent (Proposition 5.5.11) and satisfies $L \subseteq B \cup \{g\} \subseteq G$, contradicting maximality. Thus $G \subseteq \text{span}_K(B)$, so $V = \text{span}_K(G) \subseteq \text{span}_K(B)$. QED

Corollary 5.7.7 (Fundamental Corollaries of Basis Extension):

The following hold for any K -vector space V :

1. **Existence:** Every vector space has a basis (take $L = \emptyset$, $G = V$)
2. **Extension:** Any linearly independent set can be extended to a basis (take $G = V$)
3. **Extraction:** Any generating set contains a basis (take $L = \emptyset$)

Definition 5.7.8 (Complementary Subspaces):

Let $U \leq_K V$. A subspace $W \leq_K V$ is a **complement** of U in V if $V = U \oplus W$, i.e., if:

1. $U + W = V$ (the subspaces span V)
2. $U \cap W = \{0\}$ (the subspaces meet trivially)

We also say U and W are **complementary subspaces** of V .

Proposition 5.7.9 (Existence of Complements):

Every subspace $U \leq_K V$ has a complement in V .

Proof:

Let B_U be a basis for U . By the Basis Extension Theorem (Theorem 5.7.6), B_U extends to a basis $B = B_U \cup C$ for V , where $B_U \cap C = \emptyset$ (take $C = B \setminus B_U$).

Let $W = \text{span}_K(C)$. We show $V = U \oplus W$:

- $U + W = V$: Since $B = B_U \cup C$ spans V , and B_U spans U while C spans W , every $v \in V$ can be written as a sum of elements from U and W .
- $U \cap W = \{0\}$: Let $v \in U \cap W$. Then $v = \sum_{b \in F_U} a_b b$ for some finite $F_U \subseteq B_U$, and $v = \sum_{c \in F_C} d_c c$ for some finite $F_C \subseteq C$. Thus $\sum_{b \in F_U} a_b b - \sum_{c \in F_C} d_c c = 0$. Since $B = B_U \cup C$ is linearly independent and $B_U \cap C = \emptyset$, all coefficients are zero. Hence $v = 0$.

QED

Corollary 5.7.10 (Non-Uniqueness of Complements):

Complements are generally **not unique**. Different extensions of a basis for U yield different complements.

Example 5.7.11 (Multiple Complements):

In $\mathbb{R}^2$, let $U = \text{span}_{\mathbb{R}}((1, 0))$ (the x -axis).

- $W_1 = \text{span}_{\mathbb{R}}((0, 1))$ (the y -axis) is a complement of U
- $W_2 = \text{span}_{\mathbb{R}}((1, 1))$ (the line $y = x$) is also a complement of U
- In fact, $\text{span}_{\mathbb{R}}((a, b))$ is a complement of U for any $b \neq 0$

All these give $\mathbb{R}^2 = U \oplus W_i$, illustrating non-uniqueness.

Theorem 5.7.12 (Steinitz Exchange Lemma):

Let L be linearly independent and S a spanning set for V . Then there exists an injection $f : L \rightarrow S$ such that $(S \setminus f[L]) \cup L$ spans V .

Proof:

Consider the poset

$$\mathcal{P} = \{(L_0, S_0) \mid L_0 \subseteq L, S_0 \subseteq S, \exists g : L_0 \rightarrow S_0 : g \text{ a bijection}, (S \setminus S_0) \cup L_0 \text{ spans } V\}$$

ordered by $(L_0, S_0) \leq (L_1, S_1)$ iff $L_0 \subseteq L_1$ and $S_0 \subseteq S_1$. This is nonempty since $(\emptyset, \emptyset) \in \mathcal{P}$ (with S spanning).

Chains have upper bounds: Let (L_α, S_α) be a chain. Set $\bar{L} = \bigcup_\alpha L_\alpha$ and $\bar{S} = \bigcup_\alpha S_\alpha$. The bijections $g_\alpha : L_\alpha \rightarrow S_\alpha$ combine to a bijection $\bar{L} \rightarrow \bar{S}$. For spanning: any $v \in V$ is a finite linear combination from $(S \setminus S_\alpha) \cup L_\alpha$ for some α , hence from $(S \setminus \bar{S}) \cup \bar{L}$.

By Zorn's Lemma, there exists a maximal element $(\bar{L}, \bar{S})$ with bijection $\bar{g} : \bar{L} \rightarrow \bar{S}$.

Claim: $\bar{L} = L$. Suppose $u \in L \setminus \bar{L}$. Since $(S \setminus \bar{S}) \cup \bar{L}$ spans V :

$$u = \sum_{\ell \in \bar{L}} a_\ell \ell + \sum_{s \in F} b_s s$$

for some finite $F \subseteq S \setminus \bar{S}$. If all $b_s = 0$, then $u \in \text{span}_K(\bar{L}) \subseteq \text{span}_K(L \setminus \{u\})$, contradicting linear independence of L . So some $b_{s_0} \neq 0$. Then:

$$s_0 = \frac{1}{b_{s_0}} \left(u - \sum_{\ell} a_\ell \ell - \sum_{s \neq s_0} b_s s \right)$$

Thus $(S \setminus (\bar{S} \cup \{s_0\})) \cup (\bar{L} \cup \{u\})$ spans V (we can express s_0 in terms of the others). Setting $L' = \bar{L} \cup \{u\}$ and $S' = \bar{S} \cup \{s_0\}$ with $g'(u) = s_0$ extends our pair, contradicting maximality.

Thus $\bar{L} = L$, and $f = \bar{g} : L \rightarrow \bar{S} \subseteq S$ is the required injection. QED

Corollary 5.7.13 (Cardinality Bound):

If L is linearly independent and S spans V , then $|L| \leq |S|$.

Proof:

The injection $f : L \rightarrow S$ from Theorem 5.7.12 gives $|L| \leq |S|$. QED

Remark (Finite-Dimensional Case and Choice):

If V has a **finite** generating set, both the Basis Extension Theorem and Steinitz Exchange Lemma hold **without** the Axiom of Choice. Zorn's Lemma for finite posets reduces to exhaustive search and is provable in ZF.

Theorem 5.7.14 (Dimension Invariance):

Any two bases of a vector space V have the same cardinality.

Proof:

Let B_1 and B_2 be bases for V . Since B_1 is linearly independent and B_2 spans, $|B_1| \leq |B_2|$. Since B_2 is linearly independent and B_1 spans, $|B_2| \leq |B_1|$. By Cantor-Bernstein, $|B_1| = |B_2|$. QED

Definition 5.7.15 (Dimension):

The **dimension** of a K -vector space V , denoted $\dim_K(V)$, is the common cardinality of all bases of V .

For infinite-dimensional spaces, dimension is an actual cardinal number (e.g., $\aleph_0$, $\mathfrak{c}$), **not** simply “ ∞ ”.

Corollary 5.7.16 (Dimension Zero and Trivial Spaces):

$\dim_K(V) = 0$ if and only if $V = \{0\}$.

Proof:

$V = \{0\}$ has basis $\emptyset$ with $|\emptyset| = 0$. Conversely, if $\dim_K(V) = 0$, then every basis has 0 elements, so $\emptyset$ is a basis and $V = \text{span}_K(\emptyset) = \{0\}$. QED

Corollary 5.7.17 (Subspace with Equal Dimension):

If $U \leq_K V$ with V finite-dimensional and $\dim_K(U) = \dim_K(V)$, then $U = V$.

Proof:

Let B be a basis for U . Then $|B| = \dim_K(U) = \dim_K(V)$. Since B is linearly independent in V and $|B| = \dim_K(V)$, B is a maximal linearly independent set in V (Proposition 5.7.5), hence a basis for V . Thus $V = \text{span}_K(B) = U$. QED

Proposition 5.7.18 (Dimensions of Finite-Dimensional Spaces):

The following hold:

1. $\dim_K(K^n) = n$, since $\{e_0, \dots, e_{n-1}\}$ is a basis.
2. $\dim_K(K[X]_{\leq n}) = n + 1$, since $\{1, X, \dots, X^n\}$ is a basis.
3. $\dim_K(\text{Mat}_{m,n}(K)) = mn$, since the elementary matrices E_{ij} form a basis.

Proof:

In each case, we exhibit a basis and count its elements.

(1): The standard vectors $e_i = (0, \dots, 0, 1, 0, \dots, 0)$ with 1 in position i span K^n (any $(a_0, \dots, a_{n-1}) = \sum_{i \in n} a_i e_i$) and are linearly independent (evaluate at position i to see coefficient a_i is determined).

(2): The monomials $1, X, \dots, X^n$ span $K[X]_{\leq n}$ by definition. They are linearly independent: if $\sum_{i=0}^n a_i X^i = 0$ as polynomials, then all $a_i = 0$.

(3): The elementary matrices E_{ij} (with 1 in position (i, j) and 0 elsewhere) span $\text{Mat}_{m,n}(K)$ (any matrix $A = \sum_{i,j} A_{ij} E_{ij}$) and are linearly independent (the (i, j) -entry determines the coefficient of E_{ij}). There are mn such matrices. QED

Proposition 5.7.19 (Dimension of Polynomial Space):

$\dim_K(K[X]) = \aleph_0$.

Proof:

The set $\{1, X, X^2, \dots\}$ is a basis:

- **Spanning:** Every polynomial is a finite linear combination of monomials.
- **Linear independence:** If $\sum_{i=0}^n a_i X^i = 0$, then all $a_i = 0$ (polynomials are equal iff all coefficients are equal).

This basis is countably infinite, so $\dim_K(K[X]) = |\mathbb{N}| = \aleph_0$. QED

Proposition 5.7.20 (Cardinality Comparisons):

For any K -vector space V :

- If L is linearly independent, then $|L| \leq \dim_K(V)$
- If G generates V , then $\dim_K(V) \leq |G|$

Proof:

Immediate from Corollary 5.7.13 and the existence of a basis. QED

Theorem 5.7.21 (Grassmann Formula):

For subspaces $U, W \leq_K V$:

$$\dim_K(U + W) + \dim_K(U \cap W) = \dim_K(U) + \dim_K(W)$$

Proof:

Let B_0 be a basis for $U \cap W$. Extend B_0 to a basis B_U for U and to a basis B_W for W (using Theorem 5.7.6). Write $B_U = B_0 \cup C_U$ and $B_W = B_0 \cup C_W$ (disjoint unions) where C_U, C_W are the extension parts.

Claim: $B_0 \cup C_U \cup C_W$ is a basis for $U + W$ (where the unions are disjoint).

Spanning: Any $u + w \in U + W$ with $u \in U, w \in W$ can be written as a linear combination of $B_U \cup B_W = B_0 \cup C_U \cup C_W$.

Linear independence: Suppose

$$\sum_{b \in B_0} a_b b + \sum_{c \in C_U} a_c c + \sum_{d \in C_W} a_d d = 0$$

The sum $v = \sum_{b \in B_0} a_b b + \sum_{c \in C_U} a_c c \in U$ equals $-\sum_{d \in C_W} a_d d \in W$. Thus $v \in U \cap W = \text{span}_K(B_0)$. But $v = \sum_{b \in B_0} a_b b + \sum_{c \in C_U} a_c c$ with $C_U \cap B_0 = \emptyset$. By linear independence of $B_U = B_0 \cup C_U$, all $a_c = 0$. Thus $v = \sum_{b \in B_0} a_b b \in \text{span}_K(B_0)$. Then $-\sum_{d \in C_W} a_d d = v \in \text{span}_K(B_0)$. By linear independence of $B_W = B_0 \cup C_W$, all $a_d = 0$. Finally, $\sum_{b \in B_0} a_b b = 0$ implies all $a_b = 0$ by linear independence of B_0 .

Now count:

$$\begin{aligned} \dim_K(U + W) &= |B_0| + |C_U| + |C_W| \\ \dim_K(U) &= |B_0| + |C_U| \\ \dim_K(W) &= |B_0| + |C_W| \\ \dim_K(U \cap W) &= |B_0| \end{aligned}$$

Adding: $\dim_K(U + W) + \dim_K(U \cap W) = 2|B_0| + |C_U| + |C_W| = \dim_K(U) + \dim_K(W)$. QED

Corollary 5.7.22 (Direct Sum Dimension):

If $U \cap W = \{0\}$, then $\dim_K(U \oplus W) = \dim_K(U) + \dim_K(W)$.

Proof:

From Grassmann with $\dim_K(U \cap W) = \dim_K(\{0\}) = 0$. QED

Proposition 5.7.23 (All Complements Have the Same Dimension):

Let $U \leq_K V$. If W_1 and W_2 are both complements of U in V (i.e. $V = U \oplus W_1 = U \oplus W_2$), then $\dim_K(W_1) = \dim_K(W_2)$.

Proof:

The additive identity $\dim_K(U) + \dim_K(W_1) = \dim_K(V) = \dim_K(U) + \dim_K(W_2)$ (Corollary 5.7.22) does *not* permit cancelling $\dim_K(U)$: for infinite cardinals, $\kappa + \mu = \kappa + \nu$ does not imply $\mu = \nu$ (for instance $\aleph_0 + 1 = \aleph_0 + 2$). Instead we construct an explicit bijection between W_1 and W_2 that carries a basis to a basis.

The projection onto W_1 along U . Since $V = U \oplus W_1$, Theorem 5.6.2 gives every $v \in V$ a *unique* representation $v = u + w$ with $u \in U$ and $w \in W_1$. Define a function $\varphi: V \rightarrow W_1$ by $\varphi(v) = w$, the W_1 -component of v .

φ preserves the vector space operations. Let $v = u + w$ and $v' = u' + w'$ be the decompositions of $v, v' \in V$ (so $u, u' \in U$ and $w, w' \in W_1$), and let $\lambda \in K$. Then

$$v + v' = (u + u') + (w + w'), \quad \lambda v = (\lambda u) + (\lambda w),$$

where $u + u', \lambda u \in U$ and $w + w', \lambda w \in W_1$. By uniqueness of the decomposition these *are* the representations of $v + v'$ and λv , so

$$\varphi(v + v') = w + w' = \varphi(v) + \varphi(v'), \quad \varphi(\lambda v) = \lambda w = \lambda \varphi(v).$$

Two immediate values. If $w \in W_1$ then $w = 0 + w$ is its decomposition, so $\varphi(w) = w$. If $u \in U$ then $u = u + 0$, so $\varphi(u) = 0$. Conversely, $\varphi(v) = 0$ means the W_1 -component of v vanishes, i.e. $v = u \in U$. Hence

$$\varphi(v) = 0 \iff v \in U. \quad (*)$$

The restriction of φ to W_2 is a bijection onto W_1 . Let $g := \varphi|_{W_2}: W_2 \rightarrow W_1$.

Injective. Suppose $w \in W_2$ with $g(w) = 0$. By (*), $w \in U$, so $w \in U \cap W_2 = \{0\}$ (as $V = U \oplus W_2$), giving $w = 0$.

Surjective. Let $w_1 \in W_1$. Since $V = U \oplus W_2$, write $w_1 = u + w_2$ with $u \in U$ and $w_2 \in W_2$. Applying φ and using $\varphi(u) = 0$ together with $\varphi(w_1) = w_1$,

$$w_1 = \varphi(w_1) = \varphi(u) + \varphi(w_2) = 0 + g(w_2) = g(w_2),$$

so w_1 lies in the image of g .

Transporting a basis. Let B be a basis of W_2 , and set $g[B] = \{g(b) \mid b \in B\}$. We show $g[B]$ is a basis of W_1 with $|g[B]| = |B|$.

Since g is injective, the vectors $g(b)$ for $b \in B$ are pairwise distinct, so $b \mapsto g(b)$ is a bijection of B onto $g[B]$ and $|g[B]| = |B|$.

Spanning. Let $w_1 \in W_1$. By surjectivity $w_1 = g(w_2)$ for some $w_2 \in W_2$, and $w_2 = \sum_{i \in F} \lambda_i b_i$ for a finite F and distinct $b_i \in B$. As g preserves the operations,

$$w_1 = g\left(\sum_{i \in F} \lambda_i b_i\right) = \sum_{i \in F} \lambda_i g(b_i) \in \text{span}_K(g[B]).$$

Linear independence. Suppose $\sum_{i \in F} \mu_i g(b_i) = 0$ for a finite F and distinct $b_i \in B$. Then $g(\sum_{i \in F} \mu_i b_i) = 0$, so $\sum_{i \in F} \mu_i b_i = 0$ by injectivity of g , whence all $\mu_i = 0$ by linear independence of B .

Therefore $g[B]$ is a basis of W_1 , and

$$\dim_K(W_1) = |g[B]| = |B| = \dim_K(W_2). \quad \text{QED}$$

Definition 5.7.24 (Codimension):

Let $U \leq_K V$. The **codimension** of U in V is

$$\text{codim}_K(U; V) := \dim_K(W)$$

where W is any complement of U in V (well-defined by Proposition 5.7.23). Equivalently, $\text{codim}_K(U; V) = \dim_K(V) - \dim_K(U)$ when V is finite-dimensional.

Proposition 5.7.25 (Basis Union Lemma):

If $(U_i)_{i \in I}$ is a direct sum family and B_i is a basis for U_i , then $\bigcup_{i \in I} B_i$ is a basis for $\bigoplus_{i \in I} U_i$.

Proof:

Spanning: Each $u_i \in U_i$ is in $\text{span}_K(B_i)$, so any finite sum $\sum_{i \in F} u_i$ (with $F \subseteq I$ finite) is in $\text{span}_K(\bigcup_{i \in I} B_i)$.

Linear independence: By Proposition 5.6.9, the union of linearly independent sets from a direct sum family is linearly independent. QED

Corollary 5.7.26 (Arbitrary Direct Sum Dimension):

For a direct sum decomposition:

$$\dim_K \left(\bigoplus_{i \in I} U_i \right) = \sum_{i \in I} \dim_K(U_i)$$

(where the sum is cardinal addition for infinite families).

Theorem 5.7.27 (Inclusion-Exclusion for Dimension):

For a finite family $(U_0, \dots, U_{n-1})$ of finite dimensional subspaces:

$$\dim_K \left(\sum_{i \in n} U_i \right) = \sum_{\emptyset \neq J \subseteq \{0, \dots, n-1\}} (-1)^{|J|+1} \dim_K \left(\bigcap_{j \in J} U_j \right)$$

Proof:

By induction on n . The base case $n = 2$ is the Grassmann formula.

For the inductive step, apply Grassmann to U_{n-1} and $\sum_{i \in n-1} U_i$, then use the identity $U_{n-1} \cap \sum_{i \in n-1} U_i = \sum_{i \in n-1} (U_{n-1} \cap U_i)$ (which holds for subspaces of an ambient space). QED

Theorem 5.7.28 (Cardinal Properties of Vector Spaces):

Let $V \neq \{0\}$ be a non-trivial K -vector space with basis B . Then:

1. **Lower bound:** $\max(|K|, |B|) \leq |V|$
2. **Bijection with disjoint union:** There is a bijection

$$V \sim \bigcup_{n \in \mathbb{N}} (K \setminus \{0\})^n \times \binom{B}{n}$$

where the union is disjoint. Thus:

$$|V| = \sum_{n \in \mathbb{N}} (|K| - 1)^n \cdot \binom{|B|}{n}$$

3. **Finite case:** If $|B| = m$ is finite, then $|V| = |K|^m$.
4. **Infinite case:** If $\max(|K|, |B|) = \kappa \geq \aleph_0$, then $|V| = \kappa$.

Proof:

(1): The map $k \mapsto k \cdot b_0$ embeds K into V for any fixed $b_0 \in B$. The map $b \mapsto 1 \cdot b$ embeds B into V .

(2): We construct an explicit bijection. Let $v \in V$. Since B is a basis, there exists a **unique** finite subset $F_v \subseteq B$ (the **support** of v) and **unique nonzero** scalars $(a_b)_{b \in F_v}$ such that

$$v = \sum_{b \in F_v} a_b \cdot b$$

Define the map $\phi : V \rightarrow \bigcup_{n \in \mathbb{N}} (K \setminus \{0\})^n \times \binom{B}{n}$ by:

$$\phi(v) = ((a_b)_{b \in F_v}, F_v) \in (K \setminus \{0\})^{|F_v|} \times \binom{B}{|F_v|}$$

where $|F_v| = n$ is the size of the support.

Injectivity: If $\phi(v) = \phi(w)$, then $F_v = F_w$ and $(a_b)_{b \in F_v} = (c_b)_{b \in F_w}$, so $v = \sum_{b \in F_v} a_b b = \sum_{b \in F_w} c_b b = w$.

Surjectivity: Given $(a_0, \dots, a_{n-1}) \in (K \setminus \{0\})^n$ and $\{b_0, \dots, b_{n-1}\} \in \binom{B}{n}$, define $v = \sum_{i \in n} a_i b_i$. Then $\phi(v) = ((a_0, \dots, a_{n-1}), \{b_0, \dots, b_{n-1}\})$.

The union is disjoint because vectors have a unique support size. Taking cardinalities:

$$|V| = \sum_{n \in \mathbb{N}} |(K \setminus \{0\})^n| \cdot \left| \binom{B}{n} \right| = \sum_{n \in \mathbb{N}} (|K| - 1)^n \cdot \binom{|B|}{n}$$

(3): When B is finite with $|B| = m$, we have $\binom{m}{n} = 0$ for $n > m$. By the binomial theorem:

$$|V| = \sum_{n=0}^m (|K| - 1)^n \cdot \binom{m}{n} = ((|K| - 1) + 1)^m = |K|^m$$

(4): Let $\kappa = \max(|K|, |B|) \geq \aleph_0$. From (2):

$$|V| = \sum_{n \in \mathbb{N}} (|K| - 1)^n \cdot \binom{|B|}{n} \leq \sum_{n \in \mathbb{N}} \kappa^n \cdot \kappa^n = \sum_{n \in \mathbb{N}} \kappa^{2n} = \aleph_0 \cdot \kappa = \kappa$$

Combined with (1): $\kappa \leq |V| \leq \kappa$, so $|V| = \kappa$.

QED

Proposition 5.7.29 (Infinite-Dimensional Examples):

The following are infinite-dimensional vector spaces with dimension $\mathfrak{c}$ (continuum):

1. $\dim_{\mathbb{Q}}(\mathbb{R}) = \mathfrak{c}$ (the reals as a $\mathbb{Q}$ -vector space)
2. $\dim_{\mathbb{R}}(C([0, 1], \mathbb{R})) = \mathfrak{c}$ (continuous real functions on $[0, 1]$)

Proof:

These results follow from the Cardinal Properties (Theorem 5.7.28).

(1): Let B be a basis for $\mathbb{R}$ as a $\mathbb{Q}$ -vector space. Since $|\mathbb{Q}| = \aleph_0 < \mathfrak{c} = |\mathbb{R}|$, we have $|B| > \aleph_0$. By item (4) of Cardinal Properties with $\kappa = \max(\aleph_0, |B|) = |B|$:

$$|\mathbb{R}| = |B|$$

Since $|\mathbb{R}| = \mathfrak{c}$, we have $\dim_{\mathbb{Q}}(\mathbb{R}) = |B| = \mathfrak{c}$.

(2): We show $\mathfrak{c} \leq \dim_{\mathbb{R}}(C([0, 1], \mathbb{R})) \leq \mathfrak{c}$.

Upper bound: The evaluation map $f \mapsto (f(q))_{q \in [0, 1] \cap \mathbb{Q}}$ embeds $C([0, 1], \mathbb{R})$ into $\mathbb{R}^{\mathbb{Q} \cap [0, 1]}$ (since continuous functions are determined by their values on a dense set). Thus:

$$|C([0, 1], \mathbb{R})| \leq |\mathbb{R}^{\mathbb{Q}}| = \mathfrak{c}^{\aleph_0} = \mathfrak{c}$$

By Cardinal Properties (item 4), $\dim_{\mathbb{R}}(C([0, 1], \mathbb{R})) \leq \mathfrak{c}$.

Lower bound: The functions $\{f_a : x \mapsto e^{ax} \mid a \in \mathbb{R}\}$ are linearly independent (a nontrivial relation $\sum_{i \in n} c_i e^{a_i x} = 0$ with distinct a_i would contradict the Wronskian being nonzero). Thus $\dim_{\mathbb{R}}(C([0, 1], \mathbb{R})) \geq \mathfrak{c}$.

QED

5.8 Ordered Bases and Coordinates

Remark (Motivation):

A basis B determines unique coefficients for each vector, but without an ordering on B , we cannot write these coefficients as a *column vector*. Writing is inherently order-preserving, so to represent vectors in matrix form, we need **ordered bases**.

Definition 5.8.1 (Ordered Basis):

An **ordered basis** of V is a basis B equipped with a **total order** $<$ on it. A **well-ordered basis** is the special case in which $<$ is a well-order. For finite B the two notions coincide. When B is finite, we write $\mathcal{B} = (b_0, \dots, b_{n-1})$ as an n -tuple. When B is infinite, we write $\mathcal{B} = (b_i)_{i \in I}$ for some totally ordered index set I .

Remark (Total Order vs. Well-Order — What Each Is For):

The order on a basis is *auxiliary structure*, and the different layers of the theory need different amounts of it. Total order is the right default.

- **A total order suffices** for coordinates — the unique finite expansion $v = \sum_{i \in F} a_i b_i$ already determines a well-defined coordinate vector (Definition 5.8.2) — and for the *entrywise* notion of triangularity, where one only compares a row index with a column index.
- **No order at all** is needed to define matrices or matrix multiplication: only **column-finiteness**. The matrix of a map and the product of two matrices are well-defined because each column has finite support, so each entry of a product is a support-finite sum over the contracted index — never an ordered computation (see the chapter on Matrices).
- **Well-ordering** is needed only for transfinite *constructions*: recovering an adapted basis from a complete flag (Section 5.9), echelon and rank normal forms by recursion (Section 8.2), and Gram–Schmidt orthogonalisation (inner product spaces). These are recursions that require well-foundedness in order to terminate.
- **No generality is lost** by defaulting to a total order: by Zorn’s lemma every basis admits a well-order, so requiring one excludes no space — it merely fixes more structure than the lower layers use. Defaulting to a total order also keeps the order *reversible*, which is what makes lower-triangularity statable (see the chapter on Endomorphisms).

Remark (Ordering Matters):

The same underlying set with different orderings gives *different* ordered bases. For example, $\{e_0, e_1\}$ in $\mathbb{R}^2$ yields two ordered bases: (e_0, e_1) and (e_1, e_0) . These give different coordinate representations for the same vector.

Definition 5.8.2 (Coordinates):

Let $\mathcal{B} = (b_i)_{i \in I}$ be an ordered basis for V (so I is totally ordered). For $v \in V$, the unique representation

$$v = \sum_{i \in F} a_i b_i \quad (F \subseteq I \text{ finite, } a_i \neq 0)$$

determines the **coordinate vector** $[v]^{\mathcal{B}} \in K^{(I)}$ defined by

$$[v]^{\mathcal{B}}(i) = \begin{cases} a_i & \text{if } i \in F \\ 0 & \text{otherwise} \end{cases}$$

When I is finite, we write $[v]^{\mathcal{B}} = \begin{pmatrix} a_0 \\ \vdots \\ a_{n-1} \end{pmatrix}_{\mathcal{B}} \in K^n$.

Example 5.8.3 (Coordinate Dependence on Ordering):

In $\mathbb{R}^2$, let $v = (3, 2)$. With the standard basis:

- $\mathcal{B}_1 = (e_0, e_1)$: $[v]^{\mathcal{B}_1} = \begin{pmatrix} 3 \\ 2 \end{pmatrix}_{\mathcal{B}_1}$
- $\mathcal{B}_2 = (e_1, e_0)$: $[v]^{\mathcal{B}_2} = \begin{pmatrix} 2 \\ 3 \end{pmatrix}_{\mathcal{B}_2}$

The coordinates depend on the **ordering**, not just the set of basis vectors.

Theorem 5.8.4 (Coordinate Bijection):

Let $\mathcal{B} = (b_i)_{i \in I}$ be an ordered basis for V . The coordinate map

$$\phi^{\mathcal{B}} : V \rightarrow K^{(I)}, \quad v \mapsto [v]^{\mathcal{B}}$$

is a bijection. (It is in fact a linear isomorphism; see Theorem 6.7.11.)

Proof:

Injectivity: If $[v]^{\mathcal{B}} = 0$, then all coefficients are zero, so $v = 0$.

Surjectivity: For any $f \in K^{(I)}$ with finite support F , the vector $v = \sum_{i \in F} f(i)b_i$ satisfies $[v]^{\mathcal{B}} = f$. QED

Remark (Three Layers of a Vector):

For a vector $v \in V$ with ordered basis $\mathcal{B} = (b_i)_{i \in I}$, we now have three distinct objects:

1. **The geometric object:** $v \in V$. Basis-free, lives in the abstract vector space.
2. **The coordinate representation:** $\phi^{\mathcal{B}}(v) \in K^{(I)}$. An element of the canonical vector space $K^{(I)}$, obtained via the K -linear bijection $\phi^{\mathcal{B}}$. Basis-dependent, but still an algebraic object — it is a vector, and K -linear maps can act on it.
3. **The tuple of scalars:** the function $i \mapsto a_i$ where $v = \sum_{i \in I} a_i b_i$. This is the same underlying set-theoretic object as (2), but viewed purely as data — a finitely-supported family of field elements, available for entrywise manipulation, printing, storage, and arithmetic.

Layers (1) and (2) are related by the coordinate map $\phi^{\mathcal{B}}$, a K -linear bijection. Layers (2) and (3) are the same underlying set-theoretic object, but viewed with different intent: $K^{(I)}$ as a vector space (where addition and scalar multiplication are the relevant operations) versus $K^{(I)}$ as a set of functions $I \rightarrow K$ (where evaluation, support, and entrywise manipulation are the relevant operations). We identify them and rely on context to determine which structure is active, just as we identify $\mathbb{R}^2$ (vector space) with $\mathbb{R} \times \mathbb{R}$ (set of pairs) in everyday use.

This pattern — geometric object, coordinate representation in a canonical space, extracted data — will recur at the level of linear maps in Section 7.4.

Proposition 5.8.5 (Canonical Basis of $K^{(I)}$):

For any set I , define $e_i \in K^{(I)}$ by $e_i(j) = \delta_{ij}$ (Kronecker delta). Then $(e_i)_{i \in I}$ is a basis for $K^{(I)}$, called the **canonical basis**.

Proof:

Generating: For any $f \in K^{(I)}$ with finite support F :

$$f = \sum_{i \in F} f(i) \cdot e_i$$

(Verify: both sides agree at every $j \in I$.)

Linear independence: Suppose $0 = \sum_{i \in F} a_i e_i$ for some finite F with $a_i \neq 0$. Evaluating at any $j \in F$: $0 = a_j$, contradiction. QED

Remark (Canonicity is Special to $K^{(I)}$):

The basis $(e_i)_{i \in I}$ is **canonical** because it is defined intrinsically by the function-space structure of $K^{(I)}$ itself — no arbitrary choice is involved. **An abstract K -vector space V has no canonical basis:** any basis of V requires a choice (equivalently, a choice of isomorphism $V \cong K^{(I)}$). Even $\mathbb{R}^n$ only has a “standard” basis because we keep track of the identification $\mathbb{R}^n = K^{(n)}$ where $n = \{0, 1, \dots, n-1\}$. The canonical basis exists *only* for spaces of the form $K^{(I)}$.

Corollary 5.8.6 (Dimension of $K^{(I)}$):

$$\dim_K (K^{(I)}) = |I|.$$

Example 5.8.7 (Canonical Bases):

Spaces of the form $K^{(I)}$ have a canonical basis. The following are instances:

- $K^n = K^{(n)}$ where $n = \{0, 1, \dots, n-1\}$: canonical basis $(e_0, e_1, \dots, e_{n-1})$
- $K^{(\kappa)}$ for a cardinal κ : canonical basis $(e_\alpha)_{\alpha < \kappa}$

Other vector spaces have *no* canonical basis. For instance, $K[X]$ has a natural ordered basis $(1, X, X^2, \dots)$, but this is only “natural” because we implicitly choose the isomorphism $K[X] \cong K^{(\mathbb{N})}$ sending $X^n \mapsto e_n$. A different isomorphism would yield a different basis. The basis is not canonical — only the isomorphism type is.

Remark (Cardinal Properties of $K^{(I)}$):

We study the cardinal properties of $K^{(I)}$:

1. **Lower bound:** Embeddings $k \mapsto k \cdot e_{i_0}$ and $i \mapsto e_i$ give $\max(|K|, |I|) \leq |K^{(I)}|$.
2. **Bijection:** $K^{(I)} \sim \bigcup_{n \in \mathbb{N}} (K \setminus \{0\})^n \times \binom{I}{n}$ (disjoint union; a vector is determined by its finite support and nonzero values).
3. **Finite case:** When both K and I are finite, $K^{(I)} = K^I$, so $|K^{(I)}| = |K|^{|I|}$.
4. **Infinite case:** When $\kappa = \max(|K|, |I|) \geq \aleph_0$:

$$|K^{(I)}| = \sum_{n \in \mathbb{N}} \kappa^n = \aleph_0 \cdot \kappa = \kappa$$

5.9 Flags and Chains of Subspaces

Remark (Motivation):

Having introduced ordered bases, we now study a related but distinct concept: ordered families of *subspaces*. Just as an ordered basis imposes a total order on vectors to produce coordinates, a *flag* imposes a total order on a chain of subspaces to capture the geometry of nested substructures. Flags arise naturally in the study of triangular and block-triangular matrix forms, Gaussian elimination, and Jordan normal forms.

5.9.1 Ordered Bases and Set-Theoretic Requirements

Remark (Total Order vs. Well-Order):

A *total order* on the basis already suffices for coordinates, and *no* order at all is needed to define matrices or matrix multiplication — only column-finiteness (see the chapter on Matrices). A *well-order* is needed only for the transfinite *constructions* of this section (recovering an adapted basis from a complete flag) and of later chapters (echelon and rank normal forms, Gram–Schmidt). For **finite-dimensional** spaces every total order is a well-order, so the distinction vanishes. See Remark 5.8 for the full accounting of what each layer of the theory requires.

5.9.2 Flags

Definition 5.9.1 (Flag):

A **flag** in a K -vector space V is a totally ordered set (chain) of subspaces ordered by strict inclusion. A **finite flag** of **length** $r + 1$ is a sequence

$$V_0 \subsetneq V_1 \subsetneq \cdots \subsetneq V_r$$

of $r + 1$ subspaces of V . For an **infinite flag**, one has a chain $(V_\alpha)_{\alpha \in A}$ indexed by a totally ordered set A , with $V_\alpha \subsetneq V_\beta$ whenever $\alpha < \beta$.

Equivalently, a flag is a chain in the poset $(\text{Sub}(V), \leq_K)$ of all subspaces of V ordered by inclusion.

Example 5.9.2 (Flags in $\mathbb{R}^2$):

In $V = \mathbb{R}^2$ (dimension 2), examples of flags:

- $\{0\} \subsetneq \mathbb{R}^2$ — length 2, a chain of two subspaces.
- $\{0\} \subsetneq \text{span}_K(e_0) \subsetneq \mathbb{R}^2$ — length 3, a chain of three subspaces where $\text{span}_K(e_0)$ is a line.
- $\text{span}_K(e_1) \subsetneq \mathbb{R}^2$ — length 2; a flag need not start at $\{0\}$ or end at V .

There are infinitely many flags of length 3 in $\mathbb{R}^2$: one for each line through the origin.

Example 5.9.3 (Flags in Polynomial Spaces):

Let $V = K[X]_{\leq 2}$, the space of polynomials of degree at most 2 (dimension 3). Flags of different lengths:

- Length 2: $\{0\} \subsetneq K[X]_{\leq 2}$ (jumps directly from $\{0\}$ to V).
- Length 3: $\{0\} \subsetneq \text{span}_K(1) \subsetneq K[X]_{\leq 2}$ (dimensions 0, 1, 3; a gap of 2 in the last step).
- Length 4: $\{0\} \subsetneq \text{span}_K(1) \subsetneq K[X]_{\leq 1} \subsetneq K[X]_{\leq 2}$ (dimensions 0, 1, 2, 3; each step increases by 1).

Definition 5.9.4 (Complete Flag):

A flag $\mathcal{F}$ in V is **complete** if it is a *maximal chain* in the poset of all subspaces of V : no subspace can be inserted into $\mathcal{F}$ at any position without violating strict inclusion.

In infinite dimensions a complete flag is, in addition, **continuous at limits**: being a maximal chain, it is closed under the unions and intersections of its subchains. (Indeed the union of any subchain is a subspace

comparable to every member, so maximality returns it to the chain; dually for intersections.) Consequently each *limit* member — one with no immediate predecessor — equals the union of its predecessors, and dually each member with no immediate successor equals the intersection of its successors; a *jump* member instead exceeds its predecessor by a single dimension. (In finite dimensions there are no limit members and this is automatic.) This is the order-theoretic content of a *complete nest* in operator theory.

Example 5.9.5 (Complete and Non-Complete Flags in $\mathbb{R}^3$):

In $V = \mathbb{R}^3$ (dimension 3), consider flags from $\{0\}$ to $\mathbb{R}^3$:

- **Non-complete, length 2:** $\{0\} \subsetneq \mathbb{R}^3$. Any line or plane can be inserted.
- **Non-complete, length 3 (dimension gap):** $\{0\} \subsetneq \text{span}_K(e_0) \subsetneq \mathbb{R}^3$ (dimensions 0, 1, 3). Any plane P with $\text{span}_K(e_0) \subsetneq P \subsetneq \mathbb{R}^3$ (e.g. $P = \text{span}_K(e_0, e_1)$) can be inserted between the last two terms. Not maximal.
- **Complete, length 4:** $\{0\} \subsetneq \text{span}_K(e_0) \subsetneq \text{span}_K(e_0, e_1) \subsetneq \mathbb{R}^3$ (dimensions 0, 1, 2, 3; each step +1; nothing can be inserted).
- **Another complete flag, length 4:** $\{0\} \subsetneq \text{span}_K(e_0 + e_1) \subsetneq \text{span}_K(e_0 + e_1, e_2) \subsetneq \mathbb{R}^3$.

Proposition 5.9.6 (Complete Flags in Finite Dimensions):

Let $\dim_K(V) = n \in \mathbb{N}$. A flag $V_0 \subsetneq V_1 \subsetneq \cdots \subsetneq V_r$ with $V_0 = \{0\}$ and $V_r = V$ is complete if and only if $r + 1 = n + 1$ (i.e. $r = n$) and $\dim_K(V_i) = i$ for all $i = 0, 1, \dots, n$. In particular, every step increases dimension by exactly 1.

Proof:

($\Rightarrow$) Suppose the flag is complete. Since each inclusion is strict, $\dim_K(V_{i+1}) > \dim_K(V_i)$. If $\dim_K(V_{i+1}) \geq \dim_K(V_i) + 2$ for some i , pick any subspace W with $V_i \subsetneq W \subsetneq V_{i+1}$ and $\dim_K(W) = \dim_K(V_i) + 1$ (such W exists by the basis extension theorem). Inserting W contradicts maximality. Hence $\dim_K(V_{i+1}) = \dim_K(V_i) + 1$ for all i , giving $r = n$ and $\dim_K(V_i) = i$.

($\Leftarrow$) Suppose $r = n$ and $\dim_K(V_i) = i$. If W satisfies $V_i \subsetneq W \leq_K V_{i+1}$, then $i < \dim_K(W) \leq i + 1$, so $\dim_K(W) = i + 1 = \dim_K(V_{i+1})$, hence $W = V_{i+1}$ by the equal-dimension corollary. No intermediate subspace fits. QED

Example 5.9.7 (Verifying Completeness in $\mathbb{R}^4$):

Two flags from $\{0\}$ to $\mathbb{R}^4$:

- $\mathcal{F}_1$: $\{0\} \subsetneq \text{span}_K(e_0) \subsetneq \text{span}_K(e_0, e_1) \subsetneq \text{span}_K(e_0, e_1, e_2) \subsetneq \mathbb{R}^4$. Dimensions: 0, 1, 2, 3, 4. Length = 5 = 4 + 1. Each step +1. **Complete.**
- $\mathcal{F}_2$: $\{0\} \subsetneq \text{span}_K(e_0) \subsetneq \text{span}_K(e_0, e_1, e_2) \subsetneq \mathbb{R}^4$. Dimensions: 0, 1, 3, 4. Length = 4. Step 1 $\rightarrow$ 3 jumps by 2: $\text{span}_K(e_0, e_1)$ can be inserted. **Not complete.**

5.9.3 Flags from Ordered Bases

Remark (Why the Naive Successor Chain Fails):

One is tempted to set $V_\alpha := \text{span}_K(\{b_i \mid i \leq \alpha\})$ for each $\alpha \in I$ and string these into a chain through successors α^+ . This works in finite dimensions but is **incomplete in infinite dimensions**: it presupposes that every index has a successor and that I has a minimum, and its maximality check only closes the *successor* gaps $V_\alpha \subsetneq V_{\alpha^+}$. At a *limit* index λ the open segment $\text{span}_K(\{b_i \mid i < \lambda\})$ lies strictly between all earlier V_α and V_λ and is insertable, so the chain is *not maximal*. This already fails for a basis of order type $\omega + 1$, and for every uncountable dimension. The cure is to index the flag by *initial segments* rather than by elements, which is uniform over any total order.

Proposition 5.9.8 (Every Ordered Basis Induces a Complete Flag; Any Total Order):

Let $\mathcal{B} = (b_i)_{i \in I}$ be an ordered basis of V , with I totally ordered. For each **initial segment** $L \subseteq I$ — a downward-closed subset, i.e. $i \in L$ and $j < i$ imply $j \in L$ — set

$$W_L := \text{span}_K(\{b_i \mid i \in L\}).$$

Then $\{W_L \mid L \text{ an initial segment of } I\}$ is a complete flag of V .

Proof:

Order-embedding. If $L \subsetneq L'$ are initial segments, pick $i \in L' \setminus L$; then $b_i \in W_{L'}$, while $b_i \notin W_L$ by linear independence (every element of W_L is a finite combination of basis vectors indexed in L , and $i \notin L$). Hence $W_L \subsetneq W_{L'}$. The initial segments of I , ordered by $\subseteq$, form a complete chain $\hat{I}$ — the cut completion of I — and $L \mapsto W_L$ embeds it into $\text{Sub}(V)$ as a chain. The least initial segment $L = \emptyset$ gives $W_\emptyset = \{0\}$, and the greatest $L = I$ gives $W_I = V$.

Continuity. Call L a *limit cut* if $L = \bigcup_{L'' \subsetneq L} L''$, the union of the strictly smaller initial segments. For such L ,

$$W_L = \text{span}_K(\{b_i \mid i \in L\}) = \bigcup_{L'' \subsetneq L} W_{L''},$$

because any vector of W_L uses only finitely many basis indices: each lies in some strictly smaller initial segment, and as those segments form a chain the finitely many of them all lie in their largest, a single $L'' \subsetneq L$. Thus the chain is closed under the unions of its subchains.

Maximality. Let $X \leq_K V$ be comparable (under $\leq_K$) to every W_L . Put

$$L^* := \bigcup \{L \mid W_L \leq_K X\},$$

a union of initial segments, hence itself an initial segment; by continuity $W_{L^*} = \bigcup \{W_L \mid W_L \leq_K X\} \leq_K X$. For every initial segment $L \supsetneq L^*$ we have $W_L \not\leq_K [K]X$ (otherwise L would be one of the segments unioned into L^*), so comparability forces $X \leq_K W_L$; hence $X \leq_K \bigcap_{L \supsetneq L^*} W_L$. Two cases:

- If $I \setminus L^*$ has a least element i , then $L^* \cup \{i\}$ is the smallest initial segment properly containing L^* , so

$$\bigcap_{L \supsetneq L^*} W_L = W_{L^* \cup \{i\}} = W_{L^*} \oplus \text{span}_K(b_i),$$

a one-dimensional extension of W_{L^*} . From $W_{L^*} \leq_K X \leq_K W_{L^*} \oplus \text{span}_K(b_i)$ we get $X \in \{W_{L^*}, W_{L^*} \oplus \text{span}_K(b_i)\}$ — in either case some W_L .

- Otherwise $I \setminus L^*$ has no least element. Suppose $v \in \bigcap_{L \supsetneq L^*} W_L$ but $v \notin W_{L^*}$; then the finite support of v meets $I \setminus L^*$ in a nonempty set, with least element m . As $I \setminus L^*$ has no least element, some $m' \in I \setminus L^*$ satisfies $m' < m$, so $L := \{x \in I \mid x < m\}$ is an initial segment with $L^* \subsetneq L$ and $m \notin L$. Then $v \in W_L = \text{span}_K(\{b_x \mid x < m\})$ forces the coefficient of b_m in v to vanish, contradicting m being in the support of v . Hence $\bigcap_{L \supsetneq L^*} W_L = W_{L^*}$ and $X = W_{L^*}$.

Either way X is already a member of the chain, so nothing can be inserted: the flag is maximal, hence complete. QED

Remark (The Jumps Recover the Basis):

The covering pairs of $\hat{I}$ are exactly $\{j \mid j < i\} < \{j \mid j \leq i\}$, one for each $i \in I$, and the corresponding jump $W_{\{j < i\}} \subsetneq W_{\{j \leq i\}}$ has one-dimensional quotient spanned by b_i . So the flag remembers $\mathcal{B}$: the basis vector b_i sits in the i -th jump.

Remark (Well-Ordered Specialization):

When I is well-ordered (order-isomorphic to an ordinal κ), the initial segments are exactly the sets $\{i \mid i < \beta\}$ for $\beta \leq \kappa$, and the cut-flag becomes

$$W_\beta = \text{span}_K(\{b_i \mid i < \beta\}), \quad W_0 = \{0\}, \quad W_\kappa = V,$$

$$W_{\beta+1} = W_\beta \oplus \text{span}_K(b_\beta), \quad W_\lambda = \bigcup_{\beta < \lambda} W_\beta \quad (\lambda \text{ a limit ordinal}).$$

This is the corrected form of the naive statement (Remark 5.9.3): use *strict* initial segments $\{i \mid i < \beta\}$ and include the limit clause $W_\lambda = \bigcup_{\beta < \lambda} W_\beta$ in the completeness proof.

Example 5.9.9 (Induced Flags):

The following ordered bases yield the following complete flags:

- Standard basis (e_0, e_1, e_2) of $\mathbb{R}^3$: $\{0\} \subsetneq \text{span}_K(e_0) \subsetneq \text{span}_K(e_0, e_1) \subsetneq \mathbb{R}^3$.
- Rotated basis $(e_0 + e_1, e_1, e_2)$ of $\mathbb{R}^3$: $\{0\} \subsetneq \text{span}_K(e_0 + e_1) \subsetneq \text{span}_K(e_0 + e_1, e_1) \subsetneq \mathbb{R}^3$.
- Canonical basis $(e_n)_{n \in \mathbb{N}}$ of $K^{(\mathbb{N})}$: $\{0\} \subsetneq \text{span}_K(e_0) \subsetneq \text{span}_K(e_0, e_1) \subsetneq \cdots$ (an infinite complete flag).

Example 5.9.10 (Cut-Flags in Infinite Dimensions):

The cut construction specialises cleanly:

- **Standard basis of K^n** : the initial segments are $\emptyset \subsetneq \{0\} \subsetneq \{0, 1\} \subsetneq \cdots \subsetneq \{0, \dots, n-1\}$, giving the standard complete flag of dimensions $0, 1, \dots, n$.
- **Canonical basis of $K^{(\mathbb{N})}$** : $\{0\} \subsetneq \text{span}_K(e_0) \subsetneq \text{span}_K(e_0, e_1) \subsetneq \cdots$, with the whole space reached as the union $\bigcup_{n \in \mathbb{N}} W_{\{0, \dots, n-1\}}$ at the top. There is *no top jump*: the top member is a limit, not a successor.
- **A $\mathbb{Q}$ -indexed basis $(b_q)_{q \in \mathbb{Q}}$** : the initial segments of $\mathbb{Q}$ are its Dedekind cuts, so the flag is indexed by $\hat{\mathbb{Q}} \approx \mathbb{R}$ (with endpoints), a continuum, even though the basis itself is countable. The jumps sit back at the rationals; the irrational cuts are continuous — gap-filling, with no jump and no basis vector.

5.9.4 From a Complete Flag Back to a Basis: Jumps**Definition 5.9.11 (Jump):**

In a complete flag, a **jump** is a covering pair $W^- < W$ (members with $W^- \subsetneq W$ and nothing strictly between). Maximality forces $\dim_K(W/W^-) = 1$. A **jump-vector** is a choice of one vector in $W \setminus W^-$ for each jump.

Proposition 5.9.12 (Jump-Vectors Are Linearly Independent):

The jump-vectors of any complete flag are linearly independent.

Proof:

Suppose $\sum_{k \in F} c_k v_k = 0$ with F finite, every $c_k \neq 0$, and the v_k distinct jump-vectors at jumps $W_k^- < W_k$. The members W_k are distinct (distinct jumps have distinct top members); among the finitely many of them, let W_m be the highest in the chain, so each other W_k is strictly below W_m . As the members form a chain, W_k is comparable to W_m^- , and being strictly below W_m with $W_m^- < W_m$ a covering pair forces $W_k \leq_K W_m^-$; hence $v_k \in W_m^-$ for $k \neq m$. Then

$$v_m = -c_m^{-1} \sum_{k \in F \setminus \{m\}} c_k v_k \in W_m^-,$$

contradicting $v_m \in W_m \setminus W_m^-$.

QED

Proposition 5.9.13 (When the Jumps Form a Basis):

The jump-vectors of a complete flag span V — and hence, with Proposition 5.9.12, form a basis *adapted* to the flag — if and only if the flag is **jump-spanned**: every member is the span of the jumps below it. In particular this holds in the following two cases:

1. **Well-ordered flags.** If the flag is well-ordered (no infinite strictly descending chain of members), then “peeling” terminates: given $v \neq 0$, let W be the least member containing it; W is a successor $W^- < W$ (a limit member is the union of its predecessors, so a vector in it already lies in a smaller member), so $v = cb + w$ with b the jump-vector at W and $w \in W^-$, and one repeats on w . Well-foundedness forbids an infinite descent, so v is expressed in finitely many jump-vectors.
2. **Cut-flags.** The flags of Proposition 5.9.8 are jump-spanned by construction (each W_L is spanned by the b_i with $i \in L$, i.e. by the jumps below it), and are exactly the complete flags arising from totally ordered bases.

Proof:

($\Leftarrow$) First note the flag’s greatest member is V : the union of all members is a subspace (a union of a chain of subspaces) comparable to every member, so by maximality it belongs to the chain as its top element, and that top must be V (otherwise V itself could be appended). Jump-spannedness applied to this top member gives that V is the span of all jump-vectors, so they generate V ; with Proposition 5.9.12 they form a basis, adapted to the flag since each jump-vector lies in its member.

($\Rightarrow$) Suppose the jump-vectors span V , and let W be a member. The jump-vectors whose jump is $\leq_K W$ clearly lie in W . Conversely, take $w \in W$ and write $w = \sum_{k \in F} c_k v_k$ over jump-vectors (F finite, every $c_k \neq 0$). Let v_M have the highest jump $W_M^- < W_M$ among them. If $W_M \not\leq [K]W$, then $W \subsetneq W_M$; as W is comparable to W_M^- (the members form a chain) and lies strictly below W_M with $W_M^- < W_M$ a covering pair, $W \leq_K W_M^-$. Every other v_k also lies in W_M^- (its jump is lower), so $v_M = c_M^{-1}(w - \sum_{k \in F \setminus \{M\}} c_k v_k) \in W_M^-$, contradicting $v_M \in W_M \setminus W_M^-$. Hence $W_M \leq_K W$, every jump in the expansion is $\leq_K W$, and w is a combination of jumps below W . Thus W is the span of the jumps below it. The two displayed cases are verified inline above. QED

Remark (The Jump-Spanned Hypothesis Is Genuine):

Not every complete flag is jump-spanned, so the equivalence above is not vacuous. A maximal chain in $\text{Sub}(V)$ need not be well-ordered: it can contain an order-*dense* interval — a subinterval order-isomorphic to a densely ordered family of subspaces, with no covering pairs at all. Across such an interval the dimension grows while there are *no* jumps, so the jump-vectors cannot reach those members and fail to span V ; correspondingly there is no least member containing a given vector, and “peeling” has nothing to start from. Thus jump-spannedness — automatic for cut-flags (Proposition 5.9.8) and for well-ordered flags — is a real restriction, and recovering a basis from an arbitrary complete flag genuinely requires it.

Remark (Round Trip; What Is Indexed by What):

The passage basis $\rightsquigarrow$ cut-flag $\rightsquigarrow$ jumps returns the original basis together with its order I . Note that the basis is indexed by the *jumps* of $\hat{I}$ — which biject with I — and *not* by all of $\hat{I}$: the gap-filling cuts are continuous (no jump, no vector). So the flag index $\hat{I}$ (Dedekind-complete) and the basis index I are genuinely different objects whenever I has gaps. Cardinality forces the gap: $\dim_K(V) = |I| \leq |\hat{I}|$, e.g. $|\mathbb{Q}| = \aleph_0$ versus $|\hat{\mathbb{Q}}| = \mathfrak{c}$ for the $\mathbb{Q}$ -indexed basis of Example 5.9.10.

Corollary 5.9.14 (Converse in Finite Dimensions):

In finite dimensions every complete flag $\{0\} = V_0 \subsetneq V_1 \subsetneq \cdots \subsetneq V_n = V$ is a finite chain, hence well-ordered and jump-spanned, so by Proposition 5.9.13 it admits an *adapted ordered basis* $(b_0, \dots, b_{n-1})$ with $V_{i+1} = \text{span}_K(\{b_0, \dots, b_i\})$, obtained by choosing $b_i \in V_{i+1} \setminus V_i$ at each step. Different choices give different adapted bases for the same flag.

5.10 Chapter Compendium

Chapter Compendium — Foundations of Vector Spaces:

Minkowski Operations:

- $A + B = \{a + b \mid a \in A, b \in B\}$, $L \cdot A = \{\lambda a \mid \lambda \in L, a \in A\}$

Monoid Structure: $(A + B) + C = A + (B + C)$, $A + B = B + A$, $\{0\} + A = A$, $\emptyset + A = \emptyset$ (absorbing)

Scalar Action: $\{1_K\} \cdot A = A$, $(LM)A = L(MA)$, $\emptyset \cdot A = L \cdot \emptyset = \emptyset$, $\{0\} \cdot A = L \cdot \{0\} = \{0\}$

Monotonicity: $A \subseteq B \Rightarrow A + C \subseteq B + C$, $LA \subseteq LB$; $L \subseteq M \Rightarrow LA \subseteq MA$

Distributivity over Unions: $A + \bigcup_i B_i = \bigcup_i (A + B_i)$, $L \cdot \bigcup_i A_i = \bigcup_i LA_i$

Sub-Distributivity: $L(A + B) \subseteq LA + LB$, $(L + M)A \subseteq LA + MA$

Full Distributivity (singletons): $\lambda(A + B) = \lambda A + \lambda B$, $(L + M)\{a\} = L\{a\} + M\{a\}$

Sub-Distributivity over Intersections: $A + \bigcap_i B_i \subseteq \bigcap_i (A + B_i)$, $\{a\} + \bigcap_i B_i = \bigcap_i (\{a\} + B_i)$ (singleton equality)

Subspace Criterion: $U \leq_K V \Leftrightarrow U \neq \emptyset$ and $\forall a, b \in K, u, v \in U: au + bv \in U$

Minkowski Characterization: U subspace $\Leftrightarrow U \neq \emptyset$ and $U + U = U$ and $KU = U \Leftrightarrow U \neq \emptyset$ and $KU + KU = U$

Operations: $\bigcap_{i \in I} U_i$ is subspace; $U \cup W$ subspace $\Leftrightarrow U \subseteq W$ or $W \subseteq U$

Span (equivalences): $\text{span}_K(S) = \bigcap \{U \leq_K V \mid S \subseteq U\} = \{\sum_{i \in n} a_i v_i \mid n \in \mathbb{N}, a_i \in K, v_i \in S\}$

Span Properties:

- $\text{span}_K(\emptyset) = \{0\}$, $S \subseteq \text{span}_K(S)$, $\text{span}_K(\text{span}_K(S)) = \text{span}_K(S)$
- Monotonicity: $S \subseteq T \Rightarrow \text{span}_K(S) \subseteq \text{span}_K(T)$
- Subspace iff $\text{span}_K(U) = U$
- $\text{span}_K(S \cup T) = \text{span}_K(\text{span}_K(S) \cup \text{span}_K(T))$
- $\sum_{i \in I} \text{span}_K(S_i) = \text{span}_K(\bigcup_{i \in I} S_i)$ (sum of spans = span of union)

Linear Independence (equivalent characterizations):

- (i) $\forall v \in L: \text{span}_K(\{v\}) \cap \text{span}_K(L \setminus \{v\}) = \{0\}$
- (i') $\forall v \in L: \text{span}_K(L \setminus \{v\}) \subsetneq \text{span}_K(L)$
- (ii) Every $v \in \text{span}_K(L)$ has unique representation
- (iii) Some $v \in \text{span}_K(L)$ has unique representation
- (iv) 0 has unique representation (only trivial: all $a_i = 0$)

Direct Sum (equivalent characterizations): $(U_i)_{i \in I}$ is direct sum family iff:

- (i) $\forall j \in I: U_j \cap \sum_{i \neq j} U_i = \{0\}$
- (ii) Every $v \in \sum_{i \in I} U_i$ has unique representation
- (iii) Some $v \neq 0$ in the sum has unique representation
- (iv) 0 has unique representation

Basis: B lin. indep. AND $\text{span}_K(B) = V \Leftrightarrow$ maximal lin. indep. $\Leftrightarrow$ minimal generating

Basis Extension (Zorn): L lin. indep. $\subseteq G$ generating $\Rightarrow \exists B$ basis: $L \subseteq B \subseteq G$

Steinitz Exchange: L lin. indep., S spanning $\Rightarrow |L| \leq |S|$

Basis Union Lemma: (U_i) direct sum, B_i basis of $U_i \Rightarrow \bigcup_{i \in I} B_i$ basis of $\bigoplus_{i \in I} U_i$

Dimension: $\dim_K(V) = |B|$ for any basis B ; $|L| \leq \dim_K(V) \leq |G|$

Grassmann Formula: $\dim_K(U + W) + \dim_K(U \cap W) = \dim_K(U) + \dim_K(W)$

Direct Sum Dimension: $\dim_K(\bigoplus_{i \in I} U_i) = \sum_{i \in I} \dim_K(U_i)$

Cardinal Properties: $|V| = \sum_{n \in \mathbb{N}} (|K| - 1)^n \cdot \binom{|B|}{n}$

- Finite: K and B both finite $\Rightarrow |V| = |K|^{|B|}$ (Newton binomial, $\binom{m}{n} = 0$ for $n > m$)
- Infinite: $\max(|K|, |B|) = \kappa \geq \aleph_0 \Rightarrow |V| = \kappa$

Coordinate Isomorphism: $\phi^B: V \xrightarrow{\sim} K^{(I)}$, $v \mapsto [v]^B$; $\dim_K(K^{(I)}) = |I|$

Key Dimensions: $\dim_K(K^n) = n$, $\dim_K(K[X]_{\leq n}) = n + 1$, $\dim_K(\text{Mat}_{m,n}(K)) = mn$, $\dim_K(K[X]) = \aleph_0$, $\dim_{\mathbb{Q}}(\mathbb{R}) = \mathfrak{c}$, $\dim_{\mathbb{R}}(C([0, 1])) = \mathfrak{c}$

CHAPTER 6

Linear Maps

6.1 Linear Transformations

Intuition (Chapter Overview):

Chapter 1 built the objects of linear algebra: vector spaces, subspaces, bases, and dimension. This chapter introduces the **morphisms** — functions between vector spaces that respect the linear structure. A linear map $T : U \rightarrow V$ is one that commutes with addition and scalar multiplication; equivalently, it sends linear combinations to linear combinations.

The theory of linear maps is organized around three themes:

- **Determination by basis values.** A linear map is completely specified by its values on a basis (the universal property). This is the fundamental reason that bases are useful: they reduce infinite data (the action of T on every vector) to finite data (the action on finitely many basis vectors).
- **Fundamental subspaces.** Every linear map T carries two canonical subspaces: its **kernel** $\ker(T) \leq_K U$ (what T collapses to zero) and its **range** $\text{Rng}(T) \leq_K V$ (what T actually hits). The interplay between these two subspaces governs the structure of T .
- **Injectivity, surjectivity, and their characterizations.** The classical properties of functions specialize to a rich web of equivalent conditions for linear maps, connecting set-theoretic, lattice-theoretic, dimensional, and categorical viewpoints.

The chapter culminates in the **rank-nullity theorem**, which relates the dimensions of the kernel and range and provides the quantitative backbone for much of what follows.

Intuition (The Category $K\text{-Vect}$):

Having defined vector spaces as the objects of study, we now define the **morphisms** between them. In category-theoretic terms, a K -vector space homomorphism is a function that preserves the vector space structure. The collection of all such morphisms from U to V is denoted

$$\text{Hom}_K(U, V) = \{T : U \rightarrow V \mid T \text{ is } K\text{-linear}\}$$

Together with composition of morphisms, this forms the **category $K\text{-Vect}$** of K -vector spaces and K -linear maps.

Definition 6.1.1 (Linear Transformation):

Let U and V be K -vector spaces. A function $T : U \rightarrow V$ is a **linear transformation** (or **linear map**, or **K -linear map**) if it preserves the vector space structure:

1. $\forall u, v \in U : T(u + v) = T(u) + T(v)$ (additivity)
2. $\forall a \in K : \forall u \in U : T(au) = aT(u)$ (homogeneity)

Equivalently, T is linear iff $\forall a, b \in K : \forall u, v \in U : T(au + bv) = aT(u) + bT(v)$.

Proposition 6.1.2 (Composition of Linear Maps):

If $T : U \rightarrow V$ and $S : V \rightarrow W$ are linear, then $S \circ T : U \rightarrow W$ is linear.

Proof:

For any $a, b \in K$ and $u_1, u_2 \in U$:

$$\begin{aligned} (S \circ T)(au_1 + bu_2) &= S(T(au_1 + bu_2)) \\ &= S(aT(u_1) + bT(u_2)) \\ &= aS(T(u_1)) + bS(T(u_2)) \\ &= a(S \circ T)(u_1) + b(S \circ T)(u_2) \end{aligned}$$

QED

Remark (Category Structure):

The composition of linear maps is associative (inherited from function composition), and Id_V serves as the identity morphism. Thus $K\text{-Vect}$ forms a category.

Proposition 6.1.3 (Basic Properties of Linear Maps):

Let $T : U \rightarrow V$ be a linear map. Then:

1. $T(0_U) = 0_V$
2. $T(-u) = -T(u)$ for all $u \in U$
3. $T(\sum_{i \in n} a_i u_i) = \sum_{i \in n} a_i T(u_i)$ for any finite linear combination

Proof:

(1): $T(0_U) = T(0 \cdot 0_U) = 0 \cdot T(0_U) = 0_V$.

(2): $T(-u) = T((-1) \cdot u) = (-1) \cdot T(u) = -T(u)$.

(3): By induction on n using additivity and homogeneity. QED

Example 6.1.4 (Basic Linear Maps):

Consider the following examples of linear maps:

1. **Scalar multiplication:** For any $a \in K$, the map $T : V \rightarrow V$ defined by $T(v) = av$ is linear (verify: $T(bu + cv) = a(bu + cv) = abu + acv = bT(u) + cT(v)$).
2. **Identity:** The map $\text{Id}_V : V \rightarrow V$ with $\text{Id}_V(v) = v$ is linear.
3. **Zero map:** The map $0 : U \rightarrow V$ with $0(u) = 0_V$ for all $u \in U$ is linear.
4. **Inclusion:** If $W \leq_K V$, the inclusion $\iota : W \hookrightarrow V$ is linear.
5. **Derivative:** The map $D : K[X] \rightarrow K[X]$ with $D(p) = p'$ (formal derivative) is linear.
6. **Evaluation:** For fixed $a \in K$, the map $\text{ev}_a : K[X] \rightarrow K$ with $\text{ev}_a(p) = p(a)$ is linear.
7. **Coordinate projection:** For any index set I and $i \in I$:
 - $\pi_i : K^I \rightarrow K$ with $\pi_i(f) = f(i)$ is linear
 - $\pi_i : K^{(I)} \rightarrow K$ with $\pi_i(f) = f(i)$ is linear (restriction to finite support)

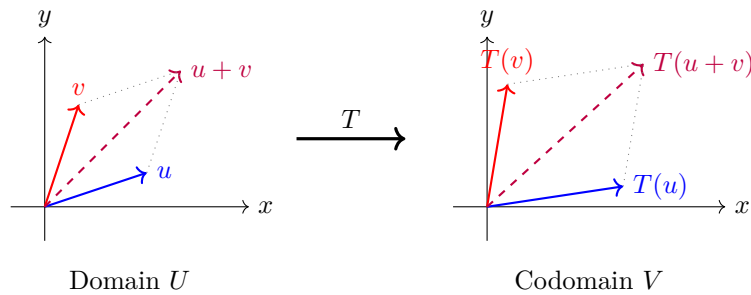


Figure 6.1: A linear map T preserves vector addition: the parallelogram law holds in both domain and codomain, ensuring $T(u + v) = T(u) + T(v)$.

Example 6.1.5 (Non-Linear Maps):

The following maps are **not** linear:

1. **Translation:** $T(v) = v + c$ for nonzero c (fails $T(0) = 0$).
2. **Squaring:** $T : \mathbb{R} \rightarrow \mathbb{R}$ with $T(x) = x^2$ (fails homogeneity: $T(2x) = 4x^2 \neq 2x^2$).
3. **Norm:** $T : \mathbb{R}^n \rightarrow \mathbb{R}$ with $T(v) = \|v\|$ (fails additivity in general).

6.2 Universal Property of Vector Spaces

Remark (Why the Universal Property Matters):

The universal property is the single most important structural fact about bases. It says that a linear map out of V is *completely and freely* determined by an arbitrary assignment of values on the basis vectors — no constraints, no compatibility conditions. This has two immediate practical consequences:

- **Construction:** To define a linear map $T : V \rightarrow W$, it suffices to specify $T(b)$ for each basis vector $b \in B$ and declare “extend linearly.” The universal property guarantees that a unique linear map with these values exists.
- **Equality:** To show two linear maps $S, T : V \rightarrow W$ are equal, it suffices to check $S(b) = T(b)$ for all $b \in B$. The universal property (uniqueness) does the rest.

In categorical language, a vector space with basis is a **free object**: the basis plays the role of “generators” with no “relations” imposed.

Theorem 6.2.1 (Universal Property of Vector Spaces (Free Object)):

Let V be a K -vector space with basis B , let W be any K -vector space, and let $f : B \rightarrow W$ be any function. Then there exists a **unique** linear map $T : V \rightarrow W$ such that $T|_B = f$.

$$\begin{array}{ccc} B & \xhookrightarrow{\iota} & V \\ & \searrow f & \downarrow \exists! T \\ & & W \end{array}$$

This characterizes vector spaces with bases as **free objects** in $K\text{-Vect}$: the vector space V is “freely generated” by the set B .

Proof:

Existence: Every $v \in V$ has a unique representation $v = \sum_{b \in F} a_b b$ for some finite $F \subseteq B$ with nonzero coefficients a_b . Define

$$T(v) = \sum_{b \in F} a_b f(b)$$

We verify T is linear. Let $v = \sum_b a_b b$ and $w = \sum_b c_b b$ (with zero coefficients for b not in the support). For $\alpha, \beta \in K$:

$$\begin{aligned} T(\alpha v + \beta w) &= T\left(\sum_b (\alpha a_b + \beta c_b) b\right) \\ &= \sum_b (\alpha a_b + \beta c_b) f(b) \\ &= \alpha \sum_b a_b f(b) + \beta \sum_b c_b f(b) \\ &= \alpha T(v) + \beta T(w) \end{aligned}$$

By construction, $T(b) = 1 \cdot f(b) = f(b)$ for all $b \in B$, so $T|_B = f$.

Uniqueness: Let $S : V \rightarrow W$ be linear with $S|_B = f$. For any $v = \sum_{b \in F} a_b b$:

$$S(v) = S\left(\sum_{b \in F} a_b b\right) = \sum_{b \in F} a_b S(b) = \sum_{b \in F} a_b f(b) = T(v)$$

Thus $S = T$.

QED

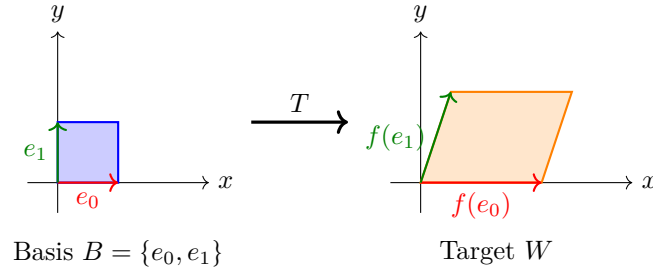


Figure 6.2: The universal property in action: specifying $f(e_0)$ and $f(e_1)$ uniquely determines the linear map T . The unit square spanned by $\{e_0, e_1\}$ maps to the parallelogram spanned by $\{f(e_0), f(e_1)\}$.

Theorem 6.2.2 (Equivalent Characterizations of Free Vector Spaces):

For a K -vector space V , the following are equivalent:

1. V has a basis
2. V satisfies the universal property: for some set $B \subseteq V$ and any $f : B \rightarrow W$, there exists a unique linear extension $T : V \rightarrow W$
3. V is an internal direct sum of one-dimensional subspaces: $V = \bigoplus_{i \in I} \text{span}_K(\{v_i\})$ for some family of nonzero vectors $(v_i)_{i \in I}$

Proof:

(1) $\Rightarrow$ (2): This is exactly the content of Theorem 6.2.1.

(2) $\Rightarrow$ (1): We show the set B satisfying the universal property must be a basis.

Linear independence: Suppose $\sum_{b \in F} a_b b = 0$ for some finite $F \subseteq B$ with not all $a_b = 0$. Pick $b_0 \in F$ with $a_{b_0} \neq 0$. Consider two maps from B to K :

- $f_1 : B \rightarrow K$ defined by $f_1(b) = 0$ for all $b \in B$
- $f_2 : B \rightarrow K$ defined by $f_2(b_0) = 1$ and $f_2(b) = 0$ for $b \neq b_0$

Let $T_1, T_2 : V \rightarrow K$ be the unique linear extensions. Then:

$$T_1 \left(\sum_{b \in F} a_b b \right) = \sum_{b \in F} a_b f_1(b) = 0$$

$$T_2 \left(\sum_{b \in F} a_b b \right) = \sum_{b \in F} a_b f_2(b) = a_{b_0} \neq 0$$

But $\sum a_b b = 0$, so $T_1(0) = 0 = T_2(0)$, contradiction. Thus B is linearly independent.

Generation: Consider the identity map $\text{Id}_V : V \rightarrow V$. The inclusion $\iota : B \hookrightarrow V$ extends to some linear $T : V \rightarrow V$ with $T|_B = \iota$. But Id_V also satisfies $\text{Id}_V|_B = \iota$. By uniqueness, $T = \text{Id}_V$. Since T is determined by its values on B , every $v \in V$ is a linear combination of elements of B . Thus B generates V .

(1) $\Leftrightarrow$ (3): If $B = \{b_i\}_{i \in I}$ is a basis, then $V = \bigoplus_{i \in I} \text{span}_K(\{b_i\})$ (internal direct sum decomposition). Conversely, if $V = \bigoplus_{i \in I} \text{span}_K(\{v_i\})$ with each $v_i \neq 0$, then $\{v_i\}_{i \in I}$ is a basis. QED

Corollary 6.2.3 (Every Vector Space is Free):

Every K -vector space is free, since every vector space has a basis (by Zorn's lemma).

Corollary 6.2.4 (Linear Maps Determined by Basis Values):

A linear map $T : V \rightarrow W$ is uniquely determined by its values on any basis of V . In particular, for finite-dimensional spaces, every linear map $T : K^n \rightarrow K^m$ is uniquely determined by the images of the standard basis vectors $T(e_0), \dots, T(e_{n-1}) \in K^m$.

Proposition 6.2.5 (Images and Preimages of Subspaces):

Let $T : U \rightarrow V$ be a linear map.

1. If $W \leq_K U$, then $T[W] \leq_K V$.
2. If $W \leq_K V$, then $T^{\leftarrow}[W] \leq_K U$.

Proof:

(1): We verify the subspace criterion.

- $T[W] \neq \emptyset$: Since $0_U \in W$, we have $T(0_U) = 0_V \in T[W]$.
- Closure: Let $v_1, v_2 \in T[W]$, so $v_i = T(w_i)$ for some $w_i \in W$. For any $a, b \in K$:

$$av_1 + bv_2 = aT(w_1) + bT(w_2) = T(aw_1 + bw_2) \in T[W]$$

since $aw_1 + bw_2 \in W$ (as W is a subspace).

(2): We verify the subspace criterion.

- $T^{\leftarrow}[W] \neq \emptyset$: Since $0_V \in W$ and $T(0_U) = 0_V$, we have $0_U \in T^{\leftarrow}[W]$.
- Closure: Let $u_1, u_2 \in T^{\leftarrow}[W]$, so $T(u_i) \in W$. For any $a, b \in K$:

$$T(au_1 + bu_2) = aT(u_1) + bT(u_2) \in W$$

since W is a subspace. Thus $au_1 + bu_2 \in T^{\leftarrow}[W]$.

QED

6.3 Fundamental Subspaces

Remark (Motivation):

Every linear map $T : U \rightarrow V$ naturally singles out two subspaces that encode the essential information about T :

- The **kernel** $\ker(T) = \{u \in U \mid T(u) = 0\}$: the set of vectors that T “kills.” The kernel measures the *non-injectivity* of T — the larger the kernel, the more information T destroys. In particular, T is injective if and only if $\ker(T) = \{0\}$.
- The **range** $\text{Rng}(T) = \{T(u) \mid u \in U\}$: the set of vectors that T actually produces. The range measures the *reach* of T — T is surjective if and only if $\text{Rng}(T) = V$.

The rank-nullity theorem (Theorem 6.8.1) will show that the dimensions of these two subspaces are complementary: $\dim(U) = \dim(\ker(T)) + \dim(\text{Rng}(T))$. Intuitively, the “budget” of dimensions in the domain is split between what T collapses and what T covers.

Definition 6.3.1 (Kernel):

The **kernel** (or **null space**) of a linear map $T : U \rightarrow V$ is

$$\ker(T) = T^{\leftarrow}[\{0_V\}] = \{u \in U \mid T(u) = 0\}$$

Definition 6.3.2 (Range):

The **range** (or **image**) of a linear map $T : U \rightarrow V$ is

$$\text{Rng}(T) = T[U] = \{T(u) \mid u \in U\}$$

Proposition 6.3.3 (Kernel and Range are Subspaces):

For any linear map $T : U \rightarrow V$:

1. $\ker(T) \leq_K U$
2. $\text{Rng}(T) \leq_K V$

Proof:

(1): This is a special case of Proposition 6.2.5(2) with $W = \{0_V\}$.

(2): This is a special case of Proposition 6.2.5(1) with $W = U$.

QED

Proposition 6.3.4 (Image-Preimage Formulas):

Let $T : U \rightarrow V$ be a linear map. For any subspace $W \leq_K U$ and any subspace $W' \leq_K V$:

1. $T^{\leftarrow}[T[W]] = W + \ker(T)$
2. $T[T^{\leftarrow}[W']] = W' \cap \text{Rng}(T)$

Proof:

(1): ($\supseteq$): If $u \in W$, then $T(u) \in T[W]$, so $u \in T^{\leftarrow}[T[W]]$. If $u \in \ker(T)$, then $T(u) = 0 \in T[W]$ (since $0 = T(0) \in T[W]$), so $u \in T^{\leftarrow}[T[W]]$. Thus $W + \ker(T) \subseteq T^{\leftarrow}[T[W]]$.

($\subseteq$): Let $u \in T^{\leftarrow}[T[W]]$, so $T(u) \in T[W]$, i.e. $T(u) = T(w)$ for some $w \in W$. Then $T(u - w) = 0$, so $u - w \in \ker(T)$, hence $u = w + (u - w) \in W + \ker(T)$.

(2): ($\subseteq$): Let $v \in T[T^{\leftarrow}[W']]$, so $v = T(u)$ for some $u \in T^{\leftarrow}[W']$, meaning $T(u) \in W'$. Thus $v = T(u) \in W'$ and $v \in \text{Rng}(T)$, so $v \in W' \cap \text{Rng}(T)$.

($\supseteq$): Let $v \in W' \cap \text{Rng}(T)$. Then $v \in W'$ and $v = T(u)$ for some $u \in U$. Since $T(u) = v \in W'$, we have $u \in T^{\leftarrow}[W']$, so $v = T(u) \in T[T^{\leftarrow}[W']]$.

QED

Remark:

Part (1) says: pulling back the image of W recovers W “up to the kernel.” Part (2) says: pushing forward the preimage of W' recovers only the part of W' that T actually hits.

Definition 6.3.5 (Rank and Nullity):

For a linear map $T : U \rightarrow V$:

- The **rank** of T is $\text{rank}(T) = \dim_K(\text{Rng}(T))$
- The **nullity** of T is $\text{nullity}(T) = \dim_K(\ker(T))$

Corollary 6.3.6 (Dimension Corollaries for Linear Maps):

Let $T : U \rightarrow V$ be a linear map. Then:

1. $\text{rank}(T) \leq \dim_K(V)$ (since $\text{Rng}(T) \leq_K V$)
2. $\text{nullity}(T) \leq \dim_K(U)$ (since $\ker(T) \leq_K U$)
3. $\ker(T) = \{0\}$ if and only if $\text{nullity}(T) = 0$ (by Corollary 5.7.16)
4. If V is finite-dimensional: $\text{Rng}(T) = V$ iff $\text{rank}(T) = \dim_K(V)$ (by Corollary 5.7.17)

Example 6.3.7 (Computing Kernel and Range):

Consider the following examples:

1. **Zero map** $0 : U \rightarrow V$: $\ker(0) = U$ and $\text{Rng}(0) = \{0\}$.
2. **Identity** Id_V : $\ker(\text{Id}_V) = \{0\}$ and $\text{Rng}(\text{Id}_V) = V$.
3. **Projection** $\pi_1 : K^2 \rightarrow K$ with $\pi_1(x, y) = x$: $\ker(\pi_1) = \{(0, y) \mid y \in K\}$ (the y -axis) and $\text{Rng}(\pi_1) = K$.
4. **Derivative** $D : K[X] \rightarrow K[X]$: $\ker(D) = K$ (constants) and $\text{Rng}(D) = K[X]$.

Proposition 6.3.8 (Span and Image Commute):

For a linear map $T : U \rightarrow V$ and any subset $S \subseteq U$:

$$\text{span}_K(T[S]) = T[\text{span}_K(S)]$$

In particular, if G generates U , then $T[G]$ generates $\text{Rng}(T)$.

Proof:

($\supseteq$): Let $v \in T[\text{span}_K(S)]$, so $v = T(u)$ for some $u \in \text{span}_K(S)$. Then $u = \sum_{i \in n} a_i s_i$ for some $a_i \in K$, $s_i \in S$. By linearity:

$$v = T(u) = T\left(\sum_{i \in n} a_i s_i\right) = \sum_{i \in n} a_i T(s_i) \in \text{span}_K(T[S])$$

($\subseteq$): $T[S] \subseteq T[\text{span}_K(S)]$, and the latter is a subspace (image of a subspace), so $\text{span}_K(T[S]) \subseteq T[\text{span}_K(S)]$. QED

Proposition 6.3.9 (Kernel and Range under Composition):

For linear maps $T : U \rightarrow V$ and $S : V \rightarrow W$:

1. $\ker(T) \subseteq \ker(S \circ T)$
2. $\text{Rng}(S \circ T) \subseteq \text{Rng}(S)$

Proof:

(1): If $u \in \ker(T)$, then $T(u) = 0$, so $(S \circ T)(u) = S(T(u)) = S(0) = 0$. Thus $u \in \ker(S \circ T)$.

(2): Every element of $\text{Rng}(S \circ T)$ has the form $S(T(u)) = S(v)$ where $v = T(u) \in V$. Thus $\text{Rng}(S \circ T) \subseteq \text{Rng}(S)$. QED

6.4 Hom Spaces as Vector Spaces

Remark (Motivation):

A striking feature of linear algebra is that the *maps* between vector spaces are themselves vectors. The set $\text{Hom}_K(U, V)$ of all linear maps $U \rightarrow V$ carries a natural vector space structure: we add maps pointwise and scale them pointwise, and the result is again linear. This “maps are vectors” principle is the source of much of the subject’s power:

- It allows us to *compute* with maps (add them, scale them, take linear combinations) just as we compute with vectors.
- It allows us to ask *dimensional* questions about maps: the space $\text{Hom}_K(U, V)$ has dimension $\dim(U) \cdot \dim(V)$ when both are finite-dimensional, giving a precise count of the “degrees of freedom” available in choosing a linear map.
- When source and target coincide ($U = V$), the set $\text{End}_K(V) = \text{Hom}_K(V, V)$ acquires a *multiplication* (composition), making it a K -algebra. This endomorphism algebra is the algebraic home of eigenvalues, minimal polynomials, and all of spectral theory.

Theorem 6.4.1 ($\text{Hom}_K(U, V)$ is a Vector Space):

For K -vector spaces U and V , the set $\text{Hom}_K(U, V)$ of all K -linear maps $T : U \rightarrow V$ is a K -vector space under pointwise operations:

$$\begin{aligned}(S + T)(u) &= S(u) + T(u) \\ (aT)(u) &= a \cdot T(u)\end{aligned}$$

Proof:

We verify all vector space axioms.

Closure under addition: For $S, T \in \text{Hom}_K(U, V)$, we show $S + T \in \text{Hom}_K(U, V)$:

$$\begin{aligned}(S + T)(au_1 + bu_2) &= S(au_1 + bu_2) + T(au_1 + bu_2) \\ &= aS(u_1) + bS(u_2) + aT(u_1) + bT(u_2) \\ &= a(S(u_1) + T(u_1)) + b(S(u_2) + T(u_2)) \\ &= a(S + T)(u_1) + b(S + T)(u_2)\end{aligned}$$

Closure under scalar multiplication: For $a \in K$ and $T \in \text{Hom}_K(U, V)$, we show $aT \in \text{Hom}_K(U, V)$:

$$\begin{aligned}(aT)(bu_1 + cu_2) &= a \cdot T(bu_1 + cu_2) \\ &= a \cdot (bT(u_1) + cT(u_2)) \\ &= abT(u_1) + acT(u_2) \\ &= b(aT)(u_1) + c(aT)(u_2)\end{aligned}$$

Additive identity: The zero map $0 : U \rightarrow V$ defined by $0(u) = 0_V$ is in $\text{Hom}_K(U, V)$, and $(T + 0)(u) = T(u) + 0_V = T(u)$, so $T + 0 = T$.

Additive inverse: For $T \in \text{Hom}_K(U, V)$, define $(-T)(u) = -T(u)$. Then $-T \in \text{Hom}_K(U, V)$ since $(-T)(au + bv) = -(aT(u) + bT(v)) = a(-T)(u) + b(-T)(v)$, and $(T + (-T))(u) = T(u) + (-T(u)) = 0_V$, so $T + (-T) = 0$.

Commutativity of addition: $(S + T)(u) = S(u) + T(u) = T(u) + S(u) = (T + S)(u)$, using commutativity in V .

Associativity of addition: $((R + S) + T)(u) = (R + S)(u) + T(u) = R(u) + S(u) + T(u) = R(u) + (S + T)(u) = (R + (S + T))(u)$.

Compatibility of scalar and field multiplication: $((ab)T)(u) = (ab)T(u) = a(bT(u)) = a((bT)(u)) = (a(bT))(u)$, so $(ab)T = a(bT)$.

Identity element of scalar multiplication: $(1_K T)(u) = 1_K \cdot T(u) = T(u)$, so $1_K T = T$.

Distributivity of scalar multiplication over vector addition: $(a(S+T))(u) = a(S+T)(u) = a(S(u) + T(u)) = aS(u) + aT(u) = (aS + aT)(u)$.

Distributivity of scalar multiplication over field addition: $((a+b)T)(u) = (a+b)T(u) = aT(u) + bT(u) = (aT)(u) + (bT)(u) = (aT + bT)(u)$. QED

6.5 Endomorphism Algebras

Definition 6.5.1 (Endomorphism Space):

$\text{End}_K(V) = \text{Hom}_K(V, V)$ is the space of all linear maps from V to itself. It carries an additional multiplication structure: composition of maps. That is, for $S, T \in \text{End}_K(V)$, we define $S \cdot T = S \circ T$.

Definition 6.5.2 (K -Algebra):

A K -algebra is a K -vector space A equipped with a bilinear multiplication $\cdot : A \times A \rightarrow A$ such that $(A, +, \cdot)$ is a ring and the following **compatibility** holds:

$$a(xy) = (ax)y = x(ay) \quad \forall a \in K, x, y \in A:$$

A K -algebra is **associative** if $(xy)z = x(yz)$, and **unital** if it has a multiplicative identity 1_A .

Theorem 6.5.3 ($\text{End}_K(V)$ is a Ring):

$\text{End}_K(V)$ is a ring with composition as multiplication:

- Addition: $(S + T)(v) = S(v) + T(v)$
- Multiplication: $(S \cdot T)(v) = S(T(v)) = (S \circ T)(v)$
- Zero: $0(v) = 0_V$
- Unity: $\text{Id}_V(v) = v$

Proof:

$(\text{End}_K(V), +)$ is abelian (Theorem 6.4.1).

Associativity: $(R \circ S) \cdot T = R \circ (S \circ T)$ (function composition).

Distributivity: $(R \circ (S + T))(v) = R(S(v) + T(v)) = R(S(v)) + R(T(v))$.

Unity: $\text{Id}_V \circ T = T \circ \text{Id}_V = T$.

QED

Theorem 6.5.4 ($\text{End}_K(V)$ is a K -Algebra):

$\text{End}_K(V)$ is a unital associative K -algebra.

Proof:

Vector space and ring structures are established. For compatibility:

$$((aS \circ T)(v)) = (aS)(T(v)) = a \cdot S(T(v)) = a \cdot (S \circ T)(v)$$

Similarly $(S \circ (aT))(v) = S(a \cdot T(v)) = a \cdot S(T(v))$.

QED

Theorem 6.5.5 (Non-Commutativity of $\text{End}_K(V)$):

If $\dim_K(V) \geq 2$, then $\text{End}_K(V)$ is non-commutative.

Proof:

Since $\dim_K(V) \geq 2$, there exist linearly independent vectors $v_0, v_1 \in V$. Extend $\{v_0, v_1\}$ to a basis B of V (by Zorn's lemma if V is infinite-dimensional).

Define $S, T \in \text{End}_K(V)$ via their action on B :

$$S(v_0) = v_1, \quad S(v_1) = 0, \quad S(b) = 0 \text{ for } b \in B \setminus \{v_0, v_1\}$$

$$T(v_0) = 0, \quad T(v_1) = v_0, \quad T(b) = 0 \text{ for } b \in B \setminus \{v_0, v_1\}$$

Extend linearly to all of V .

Compute $S \circ T$: $(S \circ T)(v_0) = S(T(v_0)) = S(0) = 0$. $(S \circ T)(v_1) = S(T(v_1)) = S(v_0) = v_1$.

Compute $T \circ S$: $(T \circ S)(v_0) = T(S(v_0)) = T(v_1) = v_0$. $(T \circ S)(v_1) = T(S(v_1)) = T(0) = 0$.

Thus $(S \circ T)(v_0) = 0 \neq v_0 = (T \circ S)(v_0)$, so $S \circ T \neq T \circ S$.

QED

6.6 Dimension of Hom Spaces

Theorem 6.6.1 (Dimension of $\text{Hom}_K(U, V)$ (Finite-Dimensional Case)):

Let U and V be finite-dimensional K -vector spaces with $\dim_K(U) = m$ and $\dim_K(V) = n$. Then

$$\dim_K(\text{Hom}_K(U, V)) = m \cdot n$$

Proof:

Let $(u_0, \dots, u_{m-1})$ be a basis of U and $(v_0, \dots, v_{n-1})$ be a basis of V . For each pair (i, j) with $0 \leq i \leq m-1$ and $0 \leq j \leq n-1$, define $T_{ij} \in \text{Hom}_K(U, V)$ by its values on the basis:

$$T_{ij}(u_k) = \delta_{ik} v_j = \begin{cases} v_j & \text{if } k = i \\ 0 & \text{if } k \neq i \end{cases}$$

and extend linearly. Each T_{ij} is well-defined and linear by the universal property of bases. We claim $\{T_{ij} \mid 0 \leq i \leq m-1, 0 \leq j \leq n-1\}$ is a basis for $\text{Hom}_K(U, V)$.

Spanning. Let $T \in \text{Hom}_K(U, V)$. Since $(v_0, \dots, v_{n-1})$ is a basis of V , for each $i \in \{0, \dots, m-1\}$ there exist unique scalars $a_{ij} \in K$ such that

$$T(u_i) = \sum_{j \in n} a_{ij} v_j$$

Define $S = \sum_{i \in m} \sum_{j \in n} a_{ij} T_{ij}$. Then for each basis vector u_k :

$$S(u_k) = \sum_{i \in m} \sum_{j \in n} a_{ij} T_{ij}(u_k) = \sum_{i \in m} \sum_{j \in n} a_{ij} \delta_{ik} v_j = \sum_{j \in n} a_{kj} v_j = T(u_k)$$

Since S and T agree on a basis of U , by the uniqueness of the extension, $S = T$. Hence $T = \sum_{i,j} a_{ij} T_{ij}$ is a linear combination of the T_{ij} 's.

Linear independence. Suppose $\sum_{i \in m} \sum_{j \in n} a_{ij} T_{ij} = 0$ (the zero map). Evaluating at u_k for each $k \in \{0, \dots, m-1\}$:

$$0 = \sum_{i \in m} \sum_{j \in n} a_{ij} T_{ij}(u_k) = \sum_{j \in n} a_{kj} v_j$$

Since $(v_0, \dots, v_{n-1})$ is linearly independent, $a_{kj} = 0$ for all $j \in \{0, \dots, n-1\}$. As this holds for every k , all coefficients $a_{ij} = 0$.

Thus $\{T_{ij}\}$ is a basis of $\text{Hom}_K(U, V)$ with $m \cdot n$ elements. QED

Corollary 6.6.2 (Dimension of End):

If $\dim_K(V) = n$, then $\dim_K(\text{End}_K(V)) = n^2$.

Proof:

$$\dim_K(\text{End}_K(V)) = \dim_K(\text{Hom}_K(V, V)) = n \cdot n = n^2. \quad \text{QED}$$

6.7 Injectivity, Surjectivity, and Bijectivity

Remark (Overview):

For a general function between sets, injectivity, surjectivity, and bijectivity are independent properties. For a *linear* map, however, these set-theoretic conditions have multiple equivalent reformulations involving kernels, images, dimensions, independence, spanning, lattice operations, and categorical properties.

The three theorems below collect these equivalences into exhaustive lists. Each list is proved by reducing every condition to its core case — a trivial kernel for injectivity, a full range for surjectivity — via a few short implication cycles through that core; bijectivity then follows by conjoining the two theorems. The key insight running through all three is:

- **Injectivity** is about *preserving independence*: an injective map does not collapse distinct directions. Its hallmark is a trivial kernel.
- **Surjectivity** is about *preserving generation*: a surjective map covers the entire target. Its hallmark is that every generating set maps to a generating set.
- **Bijectivity** combines both: it is about *preserving bases*, i.e., both independence and generation simultaneously.

In finite dimensions, injectivity and surjectivity become equivalent for maps between spaces of the same dimension (Theorem 6.7.7) — a consequence of the fact that on finite sets of the same cardinal injective functions are surjective and vice versa.

Theorem 6.7.1 (Characterizations of Injectivity):

For a linear map $T : U \rightarrow V$, the following are equivalent:

1. $\ker(T) = \{0\}$
2. T is injective
3. $T^{\leftarrow}[T[W]] = W$ for every subspace $W \leq_K U$
4. $W \mapsto T[W]$ is injective on the subspace lattice of U
5. $\dim_K(T[W]) = \dim_K(W)$ for every subspace $W \leq_K U$
6. $\dim_K(T[W]) = \dim_K(W)$ for every *finite-dimensional* subspace $W \leq_K U$
7. $\dim_K(T[W]) = \dim_K(W)$ for every *one-dimensional* subspace $W \leq_K U$ (equivalently, $T(v) \neq 0$ for every $v \neq 0$)
8. T sends every linearly independent set to a linearly independent set
9. T sends every basis of U to a linearly independent set
10. T sends some basis of U to a linearly independent set
11. T is a split monomorphism (has a left inverse): $\exists S : V \rightarrow U : S \circ T = \text{Id}_U$
12. T is monic (left-cancellable): $\forall S_1, S_2 : W \rightarrow U : T \circ S_1 = T \circ S_2 \implies S_1 = S_2$

Proof:

We show every condition is equivalent to the trivial-kernel condition (1) by proving three cycles through it:

$$(1) \Rightarrow (2) \Rightarrow (3) \Rightarrow (4) \Rightarrow (1), \quad (1) \Rightarrow (5) \Rightarrow (6) \Rightarrow (7) \Rightarrow (1), \quad (1) \Rightarrow (8) \Rightarrow (9) \Rightarrow (10) \Rightarrow (11) \Rightarrow (12) \Rightarrow (1).$$

- (1) $\Rightarrow$ (2): Let $T(u_1) = T(u_2)$. Then $T(u_1 - u_2) = 0$, so $u_1 - u_2 \in \ker(T) = \{0\}$, hence $u_1 = u_2$.
- (2) $\Rightarrow$ (3): The inclusion $W \subseteq T^{\leftarrow}[T[W]]$ is trivial. Conversely, if $u \in T^{\leftarrow}[T[W]]$, then $T(u) = T(w)$ for some $w \in W$. Injectivity gives $u = w \in W$.
- (3) $\Rightarrow$ (4): If $T[W_1] = T[W_2]$, then $W_1 = T^{\leftarrow}[T[W_1]] = T^{\leftarrow}[T[W_2]] = W_2$.
- (4) $\Rightarrow$ (1): Let $k \in \ker(T)$. Then $T[\text{span}_K(\{k\})] = \text{span}_K(\{T(k)\}) = \{0\} = T[\{0\}]$, so injectivity of $W \mapsto T[W]$ forces $\text{span}_K(\{k\}) = \{0\}$, i.e. $k = 0$. Hence $\ker(T) = \{0\}$.
- (1) $\Rightarrow$ (5): Let $W \leq_K U$ with basis B . By Proposition 6.3.8, $T[B]$ generates $T[W]$, and it is linearly independent: $\sum_{i \in n} a_i T(b_i) = 0$ gives $\sum_{i \in n} a_i b_i \in \ker(T) = \{0\}$, so all $a_i = 0$. Since $\ker(T) = \{0\}$ also makes T injective on B , we have $|T[B]| = |B|$; thus $T[B]$ is a basis of $T[W]$ and $\dim_K(T[W]) = \dim_K(W)$.
- (5) $\Rightarrow$ (6): Every finite-dimensional subspace is a subspace, so the equality in (5) holds in particular for every finite-dimensional $W \leq_K U$.
- (6) $\Rightarrow$ (7): Every one-dimensional subspace is finite-dimensional, so the equality in (6) holds in particular for every one-dimensional $W \leq_K U$.
- (7) $\Rightarrow$ (1): Item (7) says $T(v) \neq 0$ for every $v \neq 0$; equivalently, $T(v) = 0$ implies $v = 0$. Thus $\ker(T) = \{0\}$.
- (1) $\Rightarrow$ (8): Let $L \subseteq U$ be linearly independent and suppose $\sum_{i \in n} a_i T(\ell_i) = 0$ for distinct $\ell_i \in L$. Then $T(\sum_{i \in n} a_i \ell_i) = 0$, so $\sum_{i \in n} a_i \ell_i \in \ker(T) = \{0\}$; linear independence of L forces all $a_i = 0$. Hence $T[L]$ is linearly independent.
- (8) $\Rightarrow$ (9): Every basis is linearly independent.
- (9) $\Rightarrow$ (10): Every basis is some basis.
- (10) $\Rightarrow$ (11): Suppose some basis B is sent to a linearly independent set $T[B]$. Extend $T[B]$ to a basis $B' = T[B] \cup C$ for V (Theorem 5.7.6). Define $S : V \rightarrow U$ by $S(T(b)) = b$ for $b \in B$ and $S(c) = 0$ for $c \in C$, extended linearly (Theorem 6.2.1). Then $(S \circ T)(b) = b$ for all $b \in B$, so $S \circ T = \text{Id}_U$.
- (11) $\Rightarrow$ (12): Suppose $S \circ T = \text{Id}_U$ and $T \circ S_1 = T \circ S_2$. Composing with S on the left: $S_1 = (S \circ T) \circ S_1 = S \circ (T \circ S_1) = S \circ (T \circ S_2) = (S \circ T) \circ S_2 = S_2$.
- (12) $\Rightarrow$ (1): Let $\iota : \ker(T) \hookrightarrow U$ be the inclusion and $0 : \ker(T) \rightarrow U$ the zero map. For any $u \in \ker(T)$: $T(\iota(u)) = T(u) = 0 = T(0(u))$, so $T \circ \iota = T \circ 0$. By the monic property, $\iota = 0$, hence $\ker(T) = \{0\}$. QED

Remark (Split monomorphism and monomorphism, diagrammatically):

The two categorical characterizations have the following pictures. Condition (11), that T is a **split monomorphism**, asserts a left inverse (retraction) S making the triangle commute:

$$\begin{array}{ccc} U & \xrightarrow{T} & V \\ & \searrow & \downarrow S \\ & \text{Id}_U & U \end{array}$$

Condition (12), that T is a **monomorphism**, is left-cancellability of T against any parallel pair $S_1, S_2 : W \rightarrow U$:

$$W \xrightarrow[S_2]{S_1} U \xrightarrow{T} V \quad T \circ S_1 = T \circ S_2 \implies S_1 = S_2.$$

A retraction always makes T monic; over a field the converse holds as well, which is the substance of the equivalence (11) $\Leftrightarrow$ (12).

Theorem 6.7.2 (Characterizations of Surjectivity):

For a linear map $T : U \rightarrow V$, the following are equivalent:

1. $\text{Rng}(T) = V$
2. T is surjective
3. $T[T^\leftarrow[W]] = W$ for every subspace $W \leq_K V$
4. $W \mapsto T^\leftarrow[W]$ is injective on the subspace lattice of V
5. $\text{codim}_K(T^\leftarrow[W]; U) = \text{codim}_K(W; V)$ for every subspace $W \leq_K V$
6. $\text{codim}_K(T^\leftarrow[W]; U) = \text{codim}_K(W; V)$ for every *finite-codimensional* subspace $W \leq_K V$
7. $\text{codim}_K(T^\leftarrow[W]; U) = \text{codim}_K(W; V)$ for every subspace $W \leq_K V$ of *codimension one* (every hyperplane)
8. T sends every generating system of U to a generating system of V
9. T sends every basis of U to a generating system of V
10. T sends some basis of U to a generating system of V
11. T is a split epimorphism (has a right inverse): $\exists S : V \rightarrow U : T \circ S = \text{Id}_V$
12. T is epic (right-cancellable): $\forall S_1, S_2 : V \rightarrow U : S_1 \circ T = S_2 \circ T \implies S_1 = S_2$

Proof:

Dually, every condition is equivalent to the full-range condition (1) by three cycles through it:

$$(1) \Rightarrow (2) \Rightarrow (3) \Rightarrow (4) \Rightarrow (1), \quad (1) \Rightarrow (5) \Rightarrow (6) \Rightarrow (7) \Rightarrow (1), \quad (1) \Rightarrow (8) \Rightarrow (9) \Rightarrow (10) \Rightarrow (11) \Rightarrow (12) \Rightarrow (1).$$

(1) $\Rightarrow$ (2): This is the definition of surjectivity.

(2) $\Rightarrow$ (3): By Proposition 6.3.4(2), $T[T^\leftarrow[W]] = W \cap \text{Rng}(T) = W \cap V = W$.

(3) $\Rightarrow$ (4): If $T^\leftarrow[W_1] = T^\leftarrow[W_2]$, then $W_1 = T[T^\leftarrow[W_1]] = T[T^\leftarrow[W_2]] = W_2$.

(4) $\Rightarrow$ (1): Since $T^\leftarrow[\text{Rng}(T)] = U = T^\leftarrow[V]$, injectivity of $W \mapsto T^\leftarrow[W]$ forces $\text{Rng}(T) = V$.

(1) $\Rightarrow$ (5): Let $W \leq_K V$ and pick a complement C of $T^\leftarrow[W]$ in U .

Claim: $T[C]$ is a complement of W in V . If $T(c) \in W$ for $c \in C$, then $c \in T^\leftarrow[W] \cap C = \{0\}$. For any $v \in V$, $\text{Rng}(T) = V$ gives $v = T(w' + c) = T(w') + T(c)$ with $w' \in T^\leftarrow[W]$, $c \in C$, so $v \in W + T[C]$.

The map $T|_C : C \rightarrow T[C]$ is injective: if $T(c) = 0$, then $c \in \ker(T) \subseteq T^\leftarrow[W]$ (since $0 \in W$), so $c \in T^\leftarrow[W] \cap C = \{0\}$. By Theorem 6.7.1(5), $\dim_K(C) = \dim_K(T[C])$. Thus $\text{codim}_K(T^\leftarrow[W]; U) = \dim_K(C) = \dim_K(T[C]) = \text{codim}_K(W; V)$.

(5) $\Rightarrow$ (6): Every finite-codimensional subspace is a subspace, so the equality in (5) holds in particular for every finite-codimensional $W \leq_K V$.

(6) $\Rightarrow$ (7): Every subspace of codimension one is finite-codimensional, so the equality in (6) holds in particular for every codimension-one $W \leq_K V$.

(7) $\Rightarrow$ (1): Item (7) says every hyperplane $W \leq_K V$ has $\text{codim}_K(T^\leftarrow[W]; U) = 1$, so in particular $T^\leftarrow[W] \neq U$: some $u \in U$ has $T(u) \notin W$, whence $\text{Rng}(T) \not\subseteq W$. Thus $\text{Rng}(T)$ is contained in no hyperplane of V . But every proper subspace of V lies in a hyperplane (extend a basis of the subspace to a basis of V by Theorem 5.7.6 and omit one of the added vectors), so $\text{Rng}(T)$ is not proper; that is, $\text{Rng}(T) = V$.

(1) $\Rightarrow$ (8): Let G generate U . By Proposition 6.3.8, $\text{span}_K(T[G]) = T[\text{span}_K(G)] = \text{Rng}(T) = V$, so $T[G]$ generates V .

(8) $\Rightarrow$ (9): Every basis is a generating system.

(9) $\Rightarrow$ (10): Every basis is some basis.

(10) $\Rightarrow$ (11): Suppose some basis B of U maps to a generating system $T[B]$ for V . Let B_V be any basis for V . For each $b_V \in B_V$, write $b_V = \sum_{b \in F} a_b T(b)$ for some finite $F \subseteq B$. Define $S : V \rightarrow U$ by $S(b_V) = \sum_{b \in F} a_b b$ and extend linearly. Then $(T \circ S)(b_V) = b_V$ for all $b_V \in B_V$, so $T \circ S = \text{Id}_V$.

(11) $\Rightarrow$ (12): Suppose $T \circ S = \text{Id}_V$ and $S_1 \circ T = S_2 \circ T$. Composing with S on the right: $S_1 = S_1 \circ (T \circ S) = (S_1 \circ T) \circ S = (S_2 \circ T) \circ S = S_2 \circ (T \circ S) = S_2$.

(12) $\Rightarrow$ (1): Suppose T is epic. Pick a complement W of $\text{Rng}(T)$ in V , so $V = \text{Rng}(T) \oplus W$. Let $\pi : V \rightarrow W$ be the projection ($\pi(r + w) = w$). For any $u \in U$: $\pi(T(u)) = 0$ since $T(u) \in \text{Rng}(T)$. Thus $\pi \circ T = 0 \circ T$. By the epic property, $\pi = 0$, so $W = \{0\}$ and $\text{Rng}(T) = V$. QED

Remark (Split epimorphism and epimorphism, diagrammatically):

Dually to the injective case, condition (11), that T is a **split epimorphism**, asserts a right inverse (section) S making the triangle commute:

$$\begin{array}{ccc} V & \xrightarrow{S} & U \\ & \searrow \text{Id}_V & \downarrow T \\ & & V \end{array}$$

Condition (12), that T is an **epimorphism**, is right-cancellability of T against any parallel pair $S_1, S_2 : V \rightarrow W$:

$$U \xrightarrow{T} V \xrightarrow[S_2]{S_1} W \quad S_1 \circ T = S_2 \circ T \implies S_1 = S_2.$$

A section always makes T epic; over a field the converse holds as well, which is the substance of the equivalence (11) $\Leftrightarrow$ (12). The arrows are exactly those of the split-monomorphism diagram with their direction reversed — the formal expression of injectivity/surjectivity duality.

Theorem 6.7.3 (Characterizations of Bijectivity):

For a linear map $T : U \rightarrow V$, the following are equivalent. Apart from the defining condition (2), each is the conjunction of a characterization of injectivity (Theorem 6.7.1) with the corresponding characterization of surjectivity (Theorem 6.7.2):

1. $\ker(T) = \{0\}$ **and** $\text{Rng}(T) = V$
2. T is bijective
3. $T^\leftarrow[T[W]] = W$ for every $W \leq_K U$ **and** $T[T^\leftarrow[W']] = W'$ for every $W' \leq_K V$
4. $W \mapsto T[W]$ is injective on the subspace lattice of U **and** $W' \mapsto T^\leftarrow[W']$ is injective on the subspace lattice of V (equivalently, the two are mutually inverse lattice isomorphisms)
5. $\dim_K(T[W]) = \dim_K(W)$ for every $W \leq_K U$ **and** $\text{codim}_K(T^\leftarrow[W']; U) = \text{codim}_K(W'; V)$ for every $W' \leq_K V$
6. $\dim_K(T[W]) = \dim_K(W)$ for every *finite-dimensional* $W \leq_K U$ **and** $\text{codim}_K(T^\leftarrow[W']; U) = \text{codim}_K(W'; V)$ for every *finite-codimensional* $W' \leq_K V$
7. $\dim_K(T[W]) = \dim_K(W)$ for every *one-dimensional* $W \leq_K U$ **and** $\text{codim}_K(T^\leftarrow[W']; U) = \text{codim}_K(W'; V)$ for every *hyperplane* $W' \leq_K V$
8. T sends every basis of U to a basis of V
9. T sends some basis of U to a basis of V
10. T is an isomorphism: there is a *unique* linear map $T^{-1} : V \rightarrow U$ with $T^{-1} \circ T = \text{Id}_U$ and $T \circ T^{-1} = \text{Id}_V$
11. T is monic **and** epic

Proof:

Let I_k and S_k denote characterization (k) of Theorems 6.7.1 and 6.7.2; within each of those theorems all characterizations are equivalent. We prove the chain $(1) \Rightarrow (2) \Rightarrow (3) \Rightarrow (4) \Rightarrow (5) \Rightarrow (6) \Rightarrow (7) \Rightarrow (8) \Rightarrow (9) \Rightarrow (10) \Rightarrow (11) \Rightarrow (1)$, each step pairing an injectivity implication with the parallel surjectivity one.

(1) $\Rightarrow$ (2): $\ker(T) = \{0\}$ gives injectivity ($I_1 \Rightarrow I_2$) and $\text{Rng}(T) = V$ gives surjectivity ($S_1 \Rightarrow S_2$), so T is bijective.

(2) $\Rightarrow$ (3): Injectivity yields $T^\leftarrow[T[W]] = W$ for all $W \leq_K U$ ($I_2 \Rightarrow I_3$); surjectivity yields $T[T^\leftarrow[W']] = W'$ for all $W' \leq_K V$ ($S_2 \Rightarrow S_3$).

(3) $\Rightarrow$ (4): These identities make $W \mapsto T[W]$ and $W' \mapsto T^\leftarrow[W']$ injective on the respective subspace lattices ($I_3 \Rightarrow I_4$, $S_3 \Rightarrow S_4$); being mutually inverse by (3) and inclusion-preserving, they are then mutually inverse lattice isomorphisms.

(4) $\Rightarrow$ (5): Injectivity of $W \mapsto T[W]$ gives $\dim_K(T[W]) = \dim_K(W)$ for all W ($I_4 \Rightarrow I_5$); injectivity of $W' \mapsto T^\leftarrow[W']$ gives the corresponding codimension equality ($S_4 \Rightarrow S_5$).

(5) $\Rightarrow$ (6): Restrict the dimension equality to finite-dimensional W and the codimension equality to finite-codimensional W' ($I_5 \Rightarrow I_6$, $S_5 \Rightarrow S_6$).

(6) $\Rightarrow$ (7): Restrict further to one-dimensional W and to hyperplanes W' ($I_6 \Rightarrow I_7$, $S_6 \Rightarrow S_7$).

(7) $\Rightarrow$ (8): In Theorem 6.7.1, $I_7 \Rightarrow I_9$: T sends every basis to a linearly independent set. In Theorem 6.7.2, $S_7 \Rightarrow S_9$: T sends every basis to a generating system. Since a basis is an independent generating set, T sends every basis to a basis.

(8) $\Rightarrow$ (9): Every basis is some basis.

(9) $\Rightarrow$ (10): Let B be a basis of U with $T[B]$ a basis of V . Since $T[B]$ is linearly independent, $\ker(T) = \{0\}$ ($I_{10} \Rightarrow I_1$); since it generates V , $\text{Rng}(T) = V$ ($S_{10} \Rightarrow S_1$). Thus T is bijective, and its set-theoretic inverse

$T^{-1} : V \rightarrow U$ is linear: for $v_1, v_2 \in V$ and $a, b \in K$, putting $u_i = T^{-1}(v_i)$ gives $T(au_1 + bu_2) = av_1 + bv_2$, so $T^{-1}(av_1 + bv_2) = au_1 + bu_2 = aT^{-1}(v_1) + bT^{-1}(v_2)$. It is the unique two-sided inverse: if also $R \circ T = \text{Id}_U$ and $T \circ R = \text{Id}_V$, then $R = R \circ (T \circ T^{-1}) = (R \circ T) \circ T^{-1} = T^{-1}$.

(10) $\Rightarrow$ (11): T^{-1} is a left inverse, so T is monic ($I_{11} \Rightarrow I_{12}$), and a right inverse, so T is epic ($S_{11} \Rightarrow S_{12}$).

(11) $\Rightarrow$ (1): Monic gives $\ker(T) = \{0\}$ ($I_{12} \Rightarrow I_1$) and epic gives $\text{Rng}(T) = V$ ($S_{12} \Rightarrow S_1$). QED

Remark (Isomorphism, diagrammatically):

Bijectivity (condition (10)) fuses a retraction and a section into a single two-sided inverse: an **isomorphism** is a map T together with T^{-1} making both round trips the identity,

$$U \xrightleftharpoons[T^{-1}]{T} V \quad T^{-1} \circ T = \text{Id}_U, \quad T \circ T^{-1} = \text{Id}_V.$$

Equivalently, T is at once a split monomorphism and a split epimorphism, and its left and right inverses then necessarily coincide (the computation in step (9) $\Rightarrow$ (10) above).

Definition 6.7.4 (Inverse Linear Map):

Let $T : U \rightarrow V$ be a bijective linear map. Its **inverse linear map** $T^{-1} : V \rightarrow U$ is the unique linear map such that $T^{-1} \circ T = \text{Id}_U$ and $T \circ T^{-1} = \text{Id}_V$.

Definition 6.7.5 (Isomorphism):

A linear map $T : U \rightarrow V$ is an **isomorphism** if it has a two sided linear inverse. We write $U \cong V$ if there exists an isomorphism between U and V .

Corollary 6.7.6 (Dimension and Isomorphism):

Two vector spaces are isomorphic if and only if they have the same dimension. In particular, every n -dimensional K -vector space is isomorphic to K^n .

Proof:

($\Rightarrow$) If $T : U \rightarrow V$ is an isomorphism and B is a basis for U , then $T[B]$ is a basis for V (Theorem 6.7.3). Since T is bijective, $|B| = |T[B]|$. Thus $\dim_K(U) = \dim_K(V)$.

($\Leftarrow$) If $\dim_K(U) = \dim_K(V)$, let B_U and B_V be bases for U and V respectively. There exists a bijection $f : B_U \rightarrow B_V$. By the universal property (Theorem 6.2.1), f extends to a linear map $T : U \rightarrow V$. Since T sends the basis B_U to the basis B_V , T is an isomorphism. QED

Theorem 6.7.7 (Equal Dimension Equivalence):

Let U and V be **finite-dimensional** K -vector spaces with $\dim_K(U) = \dim_K(V)$, and let $T : U \rightarrow V$ be a linear map. Then the following are equivalent:

1. T is injective
2. T is surjective
3. T is bijective

Proof:

Let $n = \dim_K(U) = \dim_K(V)$, and let $B = \{b_0, \dots, b_{n-1}\}$ be a basis for U .

(1) $\Rightarrow$ (3): Suppose T is injective. By Theorem 6.7.1(8), $T[B]$ is linearly independent in V . Since $|T[B]| = n = \dim_K(V)$, it is a maximal linearly independent set, hence a basis for V (Proposition 5.7.5). Thus T is surjective, hence bijective.

(2) $\Rightarrow$ (3): Suppose T is surjective. By Theorem 6.7.2(9), $T[B]$ generates V . Since a generating set has at least $\dim_K(V) = n$ elements and $|T[B]| \leq n$, we have $|T[B]| = n$, so it is a minimal generating set, hence a basis. Thus $T[B]$ is linearly independent, so T is injective, hence bijective.

(3) $\Rightarrow$ (1), (2): Trivial. QED

Corollary 6.7.8 (Endomorphism Equivalence):

Let V be a **finite-dimensional** K -vector space and $T : V \rightarrow V$ a linear endomorphism. Then the following are equivalent:

1. T is injective
2. T is surjective
3. T is bijective

Proof:

Apply Theorem 6.7.7 with $U = V$.

QED

Example 6.7.9 (Counterexamples):

The theorem fails without the hypotheses:

1. **Infinite-dimensional, injective but not surjective:** The right-shift operator $R : K^{(\mathbb{N})} \rightarrow K^{(\mathbb{N})}$ defined by $R(a_0, a_1, a_2, \dots) = (0, a_0, a_1, a_2, \dots)$ is injective but not surjective ($(1, 0, 0, \dots) \notin \text{Rng}(R)$).
2. **Infinite-dimensional, surjective but not injective:** The left-shift operator $L : K^{(\mathbb{N})} \rightarrow K^{(\mathbb{N})}$ defined by $L(a_0, a_1, a_2, \dots) = (a_1, a_2, a_3, \dots)$ is surjective but not injective ($L(1, 0, 0, \dots) = (0, 0, \dots) = L(0, 0, \dots)$).
3. **Unequal dimensions** ($\dim_K(U) \neq \dim_K(V)$): What matters is not that domain and codomain differ as sets but that their dimensions do — here $\dim_K(K^n) = n \neq n+1 = \dim_K(K^{n+1})$. The inclusion $\iota : K^n \hookrightarrow K^{n+1}$, $\iota(x_0, \dots, x_{n-1}) = (x_0, \dots, x_{n-1}, 0)$ is injective but not surjective. The projection $\pi : K^{n+1} \rightarrow K^n$, $\pi(x_0, \dots, x_n) = (x_0, \dots, x_{n-1})$ is surjective but not injective.

Theorem 6.7.10 (Natural Isomorphism $\text{Hom}_K(K, V) \cong V$):

Let V be a K -vector space. The evaluation map

$$\varepsilon : \text{Hom}_K(K, V) \rightarrow V, \quad T \mapsto T(1_K)$$

is a K -linear isomorphism.

Proof:

Linearity: For $T, S \in \text{Hom}_K(K, V)$ and $\alpha, \beta \in K$:

$$\varepsilon(\alpha T + \beta S) = (\alpha T + \beta S)(1_K) = \alpha T(1_K) + \beta S(1_K) = \alpha \varepsilon(T) + \beta \varepsilon(S)$$

Injectivity: Suppose $\varepsilon(T) = T(1_K) = 0$. For any $\lambda \in K$:

$$T(\lambda) = T(\lambda \cdot 1_K) = \lambda T(1_K) = \lambda \cdot 0 = 0$$

Thus $T = 0$, so $\ker(\varepsilon) = \{0\}$.

Surjectivity: For any $v \in V$, define $T_v : K \rightarrow V$ by $T_v(\lambda) = \lambda v$. Then T_v is linear:

$$T_v(\alpha \lambda + \beta \mu) = (\alpha \lambda + \beta \mu)v = \alpha(\lambda v) + \beta(\mu v) = \alpha T_v(\lambda) + \beta T_v(\mu)$$

and $\varepsilon(T_v) = T_v(1_K) = 1_K \cdot v = v$.

QED

Remark:

This isomorphism is **natural**: it does not depend on any choice of basis. Every linear map $T : K \rightarrow V$ is completely determined by $T(1_K)$, since $T(\lambda) = \lambda T(1_K)$ by linearity. The inverse $\varepsilon^{-1} : V \rightarrow \text{Hom}_K(K, V)$ sends $v \mapsto T_v$ where $T_v(\lambda) = \lambda v$.

Remark (Function application is a special case of composition):

The isomorphism ε does more than count dimensions: it lets us regard each vector $v \in V$ as the linear map $T_v = \varepsilon^{-1}(v) : K \rightarrow V$, $\lambda \mapsto \lambda v$ — the *name* of v . Under this identification, **function application becomes composition**. For any linear map $T : V \rightarrow W$ and any $v \in V$, the composite $T \circ T_v : K \rightarrow W$ sends

$$1_K \mapsto T(T_v(1_K)) = T(v),$$

so $T \circ T_v = T_{T(v)}$, i.e. $\varepsilon(T \circ T_v) = T(v)$. Evaluating T at v is the same as post-composing the name T_v with T :

$$\underbrace{T(v)}_{\text{application}} \longleftrightarrow \underbrace{T \circ T_v}_{\text{composition}}.$$

Thus application is not a separate operation sitting beside composition; it is exactly the case of composition in which the source is the ground field K . This is the seed of a later fact: once linear maps are represented by matrices, the action of a matrix on a column vector $v \mapsto Tv$ is precisely this composition, and so “matrix times vector” is a special case of matrix multiplication rather than a new primitive.

Theorem 6.7.11 (Coordinate Isomorphism):

Let $\mathcal{B} = (b_i)_{i \in I}$ be an ordered basis for V . The coordinate map

$$\phi^{\mathcal{B}} : V \rightarrow K^{(I)}, \quad v \mapsto [v]^{\mathcal{B}}$$

is a K -linear isomorphism.

Proof:

Linearity: For $v = \sum_{i \in I} a_i b_i$ and $w = \sum_{i \in I} c_i b_i$ (with finite supports), and $\alpha, \beta \in K$:

$$\alpha v + \beta w = \sum_{i \in I} (\alpha a_i + \beta c_i) b_i$$

so $\phi^{\mathcal{B}}(\alpha v + \beta w) = (\alpha a_i + \beta c_i)_{i \in I} = \alpha(a_i)_{i \in I} + \beta(c_i)_{i \in I} = \alpha \phi^{\mathcal{B}}(v) + \beta \phi^{\mathcal{B}}(w)$.

Injectivity: If $\phi^{\mathcal{B}}(v) = 0$, then $v = \sum_{i \in I} 0 \cdot b_i = 0$.

Surjectivity: For any $(a_i)_{i \in I} \in K^{(I)}$ (finitely supported), $v = \sum_{i \in I} a_i b_i \in V$ satisfies $\phi^{\mathcal{B}}(v) = (a_i)_{i \in I}$. QED

Corollary 6.7.12 (Coordinate Bijection):

The coordinate map $\phi^{\mathcal{B}}$ is a bijection between V and $K^{(I)}$. In particular, $V \cong K^{(I)}$ as K -vector spaces.

Theorem 6.7.13 (Ordered Bases as Isomorphisms):

Let V be a K -vector space and I an index set. The assignment sending an ordered basis $\mathcal{B} = (b_i)_{i \in I}$ of V to the unique K -linear map

$$\psi_{\mathcal{B}} : K^{(I)} \rightarrow V, \quad \psi_{\mathcal{B}}(e_i) = b_i \quad (i \in I),$$

is a bijection

$$\{\text{ordered bases of } V \text{ indexed by } I\} \xrightarrow{\sim} \text{Iso}_K(K^{(I)}, V)$$

onto the set of K -linear isomorphisms $K^{(I)} \rightarrow V$, with inverse $\psi \mapsto (\psi(e_i))_{i \in I}$ (the image of the canonical basis). Taking $|I| = \dim_K(V)$, the ordered bases of V are thus in canonical bijection with the isomorphisms $K^{(\dim_K(V))} \rightarrow V$; moreover $\psi_{\mathcal{B}} = (\phi^{\mathcal{B}})^{-1}$ is the inverse of the coordinate isomorphism of Theorem 6.7.11.

Proof:

Well defined. Given an ordered basis $\mathcal{B} = (b_i)_{i \in I}$, the canonical basis $(e_i)_{i \in I}$ of $K^{(I)}$ determines a *unique* linear map $\psi_{\mathcal{B}}$ with $\psi_{\mathcal{B}}(e_i) = b_i$ (a linear map is freely determined by prescribing arbitrary values on a basis).

It sends the basis $(e_i)_{i \in I}$ to the basis $\mathcal{B}$, hence is an isomorphism by the basis characterization of bijectivity (Theorem 6.7.3); thus $\psi_{\mathcal{B}} \in \text{Iso}_K(K^{(I)}, V)$.

Inverse. Write $\Psi(\mathcal{B}) = \psi_{\mathcal{B}}$ and $\Theta(\psi) = (\psi(e_i))_{i \in I}$. For an isomorphism ψ , the family $(\psi(e_i))_{i \in I}$ is an ordered basis of V , since an isomorphism carries a basis to a basis (Theorem 6.7.3); so Θ is well defined. Now $\Theta(\Psi(\mathcal{B})) = (\psi_{\mathcal{B}}(e_i))_{i \in I} = (b_i)_{i \in I} = \mathcal{B}$, and for any isomorphism ψ the maps $\Psi(\Theta(\psi))$ and ψ agree on the basis $(e_i)_{i \in I}$, hence are equal. Therefore Ψ and Θ are mutually inverse, and Ψ is a bijection.

Finally, $\psi_{\mathcal{B}}(e_i) = b_i$ gives $\phi^{\mathcal{B}}(b_i) = e_i = \psi_{\mathcal{B}}^{-1}(b_i)$, so $\phi^{\mathcal{B}} = \psi_{\mathcal{B}}^{-1}$.

QED

Remark (Bases are a choice of coordinates):

Theorem 6.7.13 is the precise sense in which “choosing an ordered basis of V ” and “choosing an isomorphism $K^{(I)} \rightarrow V$ ” are the same act: a basis *is* an identification of V with a standard coordinate space. This identification is the foundation of the next chapter, where it turns linear maps into matrices.

6.8 The Rank-Nullity Theorem

Remark (Motivation):

The rank-nullity theorem is one of the most important results in linear algebra. It says that for any linear map $T : U \rightarrow V$, the dimension of the domain is the sum of two contributions:

- the **nullity** $\text{nullity}(T) = \dim(\ker(T))$: the number of independent directions that T collapses to zero,
- the **rank** $\text{rank}(T) = \dim(\text{Rng}(T))$: the number of independent directions that T produces in the target.

In other words, every dimension in the domain either “survives” (contributes to the range) or “dies” (is absorbed by the kernel), and nothing is lost in the accounting. This is the quantitative expression of the fact that a linear map cannot create or destroy dimensions independently — collapsing k dimensions in the domain automatically reduces the image by exactly k dimensions.

The proof constructs an explicit basis witnessing this decomposition: start with a basis for the kernel, extend it to a basis for U , and show the images of the extension vectors form a basis for the range.

Theorem 6.8.1 (Rank-Nullity Theorem):

For any linear map $T : U \rightarrow V$:

$$\dim_K(U) = \text{rank}(T) + \text{nullity}(T)$$

That is, $\dim_K(U) = \dim_K(\text{Rng}(T)) + \dim_K(\ker(T))$.

Proof:

We prove this via basis extension, not quotient spaces.

Let $B_K = \{k_1, k_2, \dots\}$ be a basis for $\ker(T)$ (possibly empty, possibly infinite). By the Basis Extension Theorem (Theorem 5.7.6), extend B_K to a basis $B = B_K \cup C$ for U , where $C = \{c_1, c_2, \dots\}$ and $B_K \cap C = \emptyset$.

Claim: $T[C] = \{T(c) \mid c \in C\}$ is a basis for $\text{Rng}(T)$.

Linear independence of $T[C]$: Suppose $\sum_{i \in n} a_i T(c_i) = 0$ for $c_i \in C$. Then $T(\sum_{i \in n} a_i c_i) = 0$, so $\sum_{i \in n} a_i c_i \in \ker(T) = \text{span}_K(B_K)$.

Write $\sum_{i \in n} a_i c_i = \sum_{j \in m} b_j k_j$ for some $k_j \in B_K$. Then $\sum_{i \in n} a_i c_i - \sum_{j \in m} b_j k_j = 0$. Since $B = B_K \cup C$ is a basis and $B_K \cap C = \emptyset$, all coefficients must be zero. In particular, all $a_i = 0$.

$T[C]$ generates $\text{Rng}(T)$: Let $v \in \text{Rng}(T)$, so $v = T(u)$ for some $u \in U$. Write $u = \sum_i a_i c_i + \sum_j b_j k_j$ (finite sums with $c_i \in C$, $k_j \in B_K$). Then:

$$v = T(u) = \sum_i a_i T(c_i) + \sum_j b_j T(k_j) = \sum_i a_i T(c_i) + 0 = \sum_i a_i T(c_i)$$

since $T(k_j) = 0$ for all $k_j \in B_K \subseteq \ker(T)$. Thus $v \in \text{span}_K(T[C])$.

Conclusion: We have shown $T[C]$ is a basis for $\text{Rng}(T)$.

Since T restricted to C is injective (if $T(c_1) = T(c_2)$ for $c_1, c_2 \in C$, then $c_1 - c_2 \in \ker(T)$, but $c_1 - c_2 \in \text{span}_K(C)$, and $\text{span}_K(C) \cap \ker(T) = \{0\}$ since $B = B_K \cup C$ is linearly independent with B_K a basis for $\ker(T)$).

Therefore:

$$\begin{aligned} \dim_K(U) &= |B| = |B_K| + |C| \\ &= \dim_K(\ker(T)) + |T[C]| \\ &= \text{nullity}(T) + \text{rank}(T) \end{aligned}$$

QED

Remark (Additive Form):

The rank-nullity theorem is stated in **additive form**: $\dim_K(U) = \text{rank}(T) + \text{nullity}(T)$.

This form is valid for **all** vector spaces, including infinite-dimensional ones, where cardinal addition is well-defined.

The subtraction forms require the **subtrahend** to be finite:

- $\text{rank}(T) = \dim_K(U) - \text{nullity}(T)$ is meaningful when $\text{nullity}(T)$ is finite
- $\text{nullity}(T) = \dim_K(U) - \text{rank}(T)$ is meaningful when $\text{rank}(T)$ is finite

Example 6.8.2 (Applying Rank-Nullity):

Consider the following applications:

1. **Derivative on polynomials:** Let $D : K[X]_{\leq n} \rightarrow K[X]_{\leq n-1}$ be the derivative. We have $\ker(D) = K$ (constants), so $\text{nullity}(D) = 1$. By rank-nullity: $\text{rank}(D) = (n+1) - 1 = n = \dim_K(K[X]_{\leq n-1})$. Thus D is surjective (but not injective).
2. **Projection:** Let $\pi : K^3 \rightarrow K^2$ be $\pi(x, y, z) = (x, y)$. Then $\ker(\pi) = \{(0, 0, z) \mid z \in K\}$, so $\text{nullity}(\pi) = 1$. By rank-nullity: $\text{rank}(\pi) = 3 - 1 = 2 = \dim_K(K^2)$. Thus π is surjective.
3. **Inclusion:** Let $\iota : K^2 \rightarrow K^3$ be $\iota(x, y) = (x, y, 0)$. Then $\ker(\iota) = \{0\}$, so $\text{nullity}(\iota) = 0$. By rank-nullity: $\text{rank}(\iota) = 2 - 0 = 2 < 3 = \dim_K(K^3)$. Thus ι is injective but not surjective.

6.9 Chapter Compendium

Chapter Compendium — Linear Transformations:

Definition: $T : U \rightarrow V$ linear iff $T(au + bv) = aT(u) + bT(v)$

Category $K\text{-Vect}$: $\text{Hom}_K(U, V) = \{T : U \rightarrow V \mid T \text{ linear}\}$

Composition: T, S linear $\Rightarrow S \circ T$ linear (associative, identity morphisms)

Basic Properties: $T(0) = 0$, $T(-v) = -T(v)$, $T(\sum a_i u_i) = \sum a_i T(u_i)$

Chapter Compendium — Universal Property (Free Objects):

For basis B of V , any $f : B \rightarrow W$ extends uniquely to linear $T : V \rightarrow W$

Equivalent (no isomorphism): V has basis $\Leftrightarrow$ universal property $\Leftrightarrow V = \bigoplus \text{span}(v_i)$

Corollary: Every vector space is free (has basis by Zorn's lemma)

Corollary: Linear maps determined by values on basis

Chapter Compendium — Subspaces Under Linear Maps:

Images and Preimages: If $W \leq_K U$, then $T[W] \leq_K V$. If $W \leq_K V$, then $T^{\leftarrow}[W] \leq_K U$.

Image-Preimage Formulas:

- $T^{\leftarrow}[T[W]] = W + \ker(T)$ for $W \leq_K U$
- $T[T^{\leftarrow}[W']] = W' \cap \text{Rng}(T)$ for $W' \leq_K V$

Span and Image Commute: $\text{span}_K(T[S]) = T[\text{span}_K(S)]$

Chapter Compendium — Fundamental Subspaces:

Kernel and Range:

- $\ker(T) = T^{\leftarrow}[\{0\}] = \{u \in U \mid T(u) = 0\} \leq_K U$
- $\text{Rng}(T) = T[U] = \{T(u) \mid u \in U\} \leq_K V$

Dimensions: $\text{rank}(T) = \dim_K(\text{Rng}(T))$, $\text{nullity}(T) = \dim_K(\ker(T))$

Chapter Compendium — Hom Spaces and Endomorphisms:

Vector Space: $\text{Hom}_K(U, V)$ is a K -vector space under pointwise operations

Natural Isomorphism: $\text{Hom}_K(K, V) \cong V$ via $T \mapsto T(1_K)$

Endomorphism Space: $\text{End}_K(V) = \text{Hom}_K(V, V)$ with multiplication $S \cdot T = S \circ T$

K -Algebra: $(\text{End}_K(V), +, \cdot)$ is a unital associative K -algebra; non-commutative when $\dim_K(V) \geq 2$

Chapter Compendium — Injectivity Characterizations:

For $T : U \rightarrow V$ linear, TFAE (one chain):

1. $\ker(T) = \{0\}$
2. T injective
3. $T^{\leftarrow}[T[W]] = W$ for every $W \leq_K U$
4. $W \mapsto T[W]$ injective on subspace lattice of U
5. $\dim_K(T[W]) = \dim_K(W)$ for every $W \leq_K U$
6. lin. indep. $\rightarrow$ lin. indep.
7. every basis $\rightarrow$ lin. indep.
8. some basis $\rightarrow$ lin. indep.
9. split mono ($\exists S$ $S \circ T = \text{Id}_U$)
10. monic (left-cancellable)

Chapter Compendium — Surjectivity Characterizations:

For $T : U \rightarrow V$ linear, TFAE (one chain):

1. $\text{Rng}(T) = V$

2. T surjective
3. $T[T^{\leftarrow}[W]] = W$ for every $W \leqslant_K V$
4. $W \mapsto T^{\leftarrow}[W]$ injective on subspace lattice of V
5. $\text{codim}_K(T^{\leftarrow}[W]; U) = \text{codim}_K(W; V)$ for every $W \leqslant_K V$
6. gen. set $\rightarrow$ gen. set
7. every basis $\rightarrow$ gen. set
8. some basis $\rightarrow$ gen. set
9. split epi ($\exists S T \circ S = \text{Id}_V$)
10. epic (right-cancellable)

Chapter Compendium — Bijectivity and Isomorphism:

Bijectivity TFAE (one chain): bijective $\Rightarrow$ preimage/image formulas $\Rightarrow$ lattice isomorphism $\Rightarrow$ $\dim + \text{codim}$ preserved $\Rightarrow$ every/some basis $\rightarrow$ basis $\Rightarrow$ isomorphism $\Rightarrow$ monic + epic $\Rightarrow$ bijective

Isomorphism: $U \cong V \Leftrightarrow \dim_K(U) = \dim_K(V)$

Coordinate Isomorphism: $\phi^{\mathcal{B}} : V \xrightarrow{\cong} K^{(I)}$ (depends on ordered basis $\mathcal{B}$)

Chapter Compendium — Rank-Nullity Theorem:

Theorem: $\dim_K(U) = \text{rank}(T) + \text{nullity}(T)$

Proof idea: Extend $\ker(T)$ basis to U basis; remaining vectors map to $\text{Rng}(T)$ basis

Equal Dimension Equivalence: For $T : U \rightarrow V$ with $\dim_K(U) = \dim_K(V)$ finite: injective $\Leftrightarrow$ surjective $\Leftrightarrow$ bijective

Endomorphism Corollary: For $T : V \rightarrow V$ with V finite-dimensional: injective $\Leftrightarrow$ surjective $\Leftrightarrow$ bijective

CHAPTER 7

Matrices

7.1 Matrices

7.1.1 Why Matrices?

Intuition:

The previous chapter built linear maps as *abstract* objects: a linear map $T : U \rightarrow V$ is a function respecting addition and scalar multiplication, and we proved theorems about such functions — about kernels, images, rank, injectivity — without ever writing a single number. That abstraction is exactly what makes the theory clean, but it is useless for *computation*. To actually evaluate T on a vector, to compose two maps, to invert one, or to solve $T(v) = w$, we need a concrete, mechanical surrogate for T — an array of scalars we can add and multiply by rote.

A **matrix** is that surrogate. The whole chapter is the disciplined construction of the dictionary between maps and their numerical surrogates: we show (i) that for maps between coordinate spaces the surrogate loses *no* information — the correspondence is a bijection — and (ii) that every operation on maps (composition, addition, scaling, later inversion) can be *transported* across the dictionary into a purely arithmetic operation on the surrogates. Once that is done, “computing with a linear map” becomes “computing with an array of scalars,” and the abstract guarantees of the previous chapter certify that the computation means what we want.

Goal:

Fix the field K . For index sets X (the *source* index) and Y (the *target* index) of *arbitrary* cardinality, we will:

1. define the set $\text{Mat}_{Y,X}(K)$ of $(Y \times X)$ -grids of scalars — a pure data structure, with no operations assumed — and its column-finite subset $\text{Mat}_{(Y),X}(K)$;
2. exhibit a canonical **bijection** between linear maps $K^{(X)} \rightarrow K^{(Y)}$ and the column-finite grids $\text{Mat}_{(Y),X}(K)$;
3. *transport* every operation — addition, scalar multiplication, and composition — from maps onto grids, and *derive* the familiar entry formulas as theorems rather than positing them.

The finite case — $X = n = \{0, \dots, n-1\}$ for the source and $Y = m = \{0, \dots, m-1\}$ for the target, giving maps $K^n \rightarrow K^m$ — is the one to keep in mind; there $K^{(X)} = K^n$ and every grid is automatically column-finite.

7.1.2 Linear Maps Between Coordinate Spaces and Their Grids

Recall that for an index set X the coordinate space $K^{(X)}$ is the space of *finitely supported* families $X \rightarrow K$, with canonical basis $(e_j)_{j \in X}$, where $e_j(i) = \delta_{ij}$. We begin with the data structure, deliberately stripped of all algebraic structure.

Definition 7.1.1 (Grids and column-finite grids):

Let X, Y be index sets. A $(Y \times X)$ -grid over K is a function

$$A : Y \times X \longrightarrow K.$$

We write $A_{ij} := A(i, j)$ for its **entries**, where $i \in Y$ is the **row index** and $j \in X$ the **column index**, and $A(-, j) : i \mapsto A_{ij}$ for the j -th **column**. The set of all grids is $\text{Mat}_{Y,X}(K) := K^{Y \times X}$. A grid A is **column-finite** if every column has finite support, $A(-, j) \in K^{(Y)}$ for all $j \in X$; the set of column-finite grids is

$$\text{Mat}_{(Y),X}(K) := \left\{ A \in \text{Mat}_{Y,X}(K) \mid A(-, j) \in K^{(Y)} \text{ for all } j \in X \right\}.$$

At this stage these are merely *sets of grids*: no addition, no multiplication, no interpretation as a map.

Remark (The convention $\text{Mat}_{Y,X}(K)$ for maps $K^{(X)} \rightarrow K^{(Y)}$, and the finite case):

The row index Y is written first and matches the *target* $K^{(Y)}$; the column index X matches the *source* $K^{(X)}$. This is the convention under which composition of maps reads as left-to-right multiplication of grids (Theorem 7.1.5). When X and Y are *finite*, say $X = n = \{0, \dots, n-1\}$ and $Y = m = \{0, \dots, m-1\}$, every column is automatically finitely supported, so $\text{Mat}_{(m),n}(K) = \text{Mat}_{m,n}(K)$ is the set of ordinary $m \times n$ matrices and $K^{(n)} = K^n$. Column-finiteness only bites when Y is infinite.

Remark (Column-finiteness is weaker than global finiteness):

Reading a grid one column at a time identifies $\text{Mat}_{(Y),X}(K)$ with the functions $X \rightarrow K^{(Y)}$, $j \mapsto A(-, j)$ — one finitely supported column per index j — so that $\text{Mat}_{(Y),X}(K) \cong (K^{(Y)})^X$. This is *not* the space $K^{(Y \times X)}$ of grids with **globally** finite support: column-finiteness constrains each column on its own and still permits infinitely many nonzero columns, whereas $K^{(Y \times X)}$ allows only finitely many nonzero entries in total. The two notions coincide when X is finite and diverge strictly once X is infinite. It is column-finiteness — not global finiteness — that matches the linear maps $K^{(X)} \rightarrow K^{(Y)}$, as the next theorem shows.

The link between a map and a grid rests on one elementary fact: a linear map out of $K^{(X)}$ is completely pinned down by what it does to the canonical basis, and each such value lies in $K^{(Y)}$, hence has finitely many nonzero coordinates.

Theorem 7.1.2 (The Map–Grid Bijection):

Let X, Y be index sets. The assignment

$$\Phi : \text{Hom}_K(K^{(X)}, K^{(Y)}) \longrightarrow \text{Mat}_{Y,X}(K), \quad \Phi(T)_{ij} := (T(e_j))_i,$$

whose j -th column is the image vector $T(e_j) \in K^{(Y)}$, takes values in $\text{Mat}_{(Y),X}(K)$ and is a *bijection*

$$\Phi : \text{Hom}_K(K^{(X)}, K^{(Y)}) \xrightarrow{\sim} \text{Mat}_{(Y),X}(K),$$

with inverse denoted Φ^{-1} .

Proof:

Φ *lands in* $\text{Mat}_{(Y),X}(K)$. For each T , the j -th column of $\Phi(T)$ is $T(e_j) \in K^{(Y)}$, which has finite support; hence $\Phi(T)$ is column-finite.

Injective. Suppose $\Phi(S) = \Phi(T)$. Then $S(e_j) = T(e_j)$ for every $j \in X$, since two vectors of $K^{(Y)}$ with equal coordinates are equal. A linear map is determined by its values on a basis: for arbitrary $x = \sum_{j \in X} x_j e_j \in K^{(X)}$ — a finite sum, as x has finite support —

$$S(x) = \sum_{j \in X} x_j S(e_j) = \sum_{j \in X} x_j T(e_j) = T(x),$$

so $S = T$.

Surjective. Let $A \in \text{Mat}_{(Y),X}(K)$. Each column $A(-, j)$ lies in $K^{(Y)}$, so there is a *unique* linear map $T_A : K^{(X)} \rightarrow K^{(Y)}$ with $T_A(e_j) := A(-, j)$ (a basis determines a linear map uniquely by prescribing its values on the basis). By construction $\Phi(T_A)_{ij} = (T_A(e_j))_i = A_{ij}$, i.e. $\Phi(T_A) = A$. Thus Φ is onto, and being injective and surjective it is a bijection. QED

Remark (The fundamental datum: rows index the target, columns the source):

The double index (i, j) is the central bookkeeping device of the chapter. The **column** index j selects a source basis vector e_j , and the corresponding column of $\Phi(T)$ is its image $T(e_j)$; the **row** index i then reads off the i -th coordinate of that image in the target. “Apply T to each input basis vector and stack the results as columns” is the entire recipe for $\Phi(T)$.

Remark (Convention: why $(T(e_j))_i$ and not $(T(e_i))_j$?:

The definition $\Phi(T)_{ij} = (T(e_j))_i$ is a *choice*. We could equally have declared the (i, j) -entry to be $(T(e_i))_j$, which would identify $\text{Hom}_K(K^{(X)}, K^{(Y)})$ with $\text{Mat}_{X,Y}(K)$ (an $X \times Y$ matrix) rather than with $\text{Mat}_{Y,X}(K)$ (a $Y \times X$ matrix). In some sense that alternative is more natural: the size of the matrix would then match the order (X, Y) in which source and target appear in $K^{(X)} \rightarrow K^{(Y)}$.

We adopt the swapped convention $\Phi(T)_{ij} = (T(e_j))_i$ instead, because we want a map $K^{(X)} \rightarrow K^{(Y)}$ to be recorded as a $Y \times X$ matrix — the *target* index i labelling rows, the *source* index j labelling columns. This is the standard convention. Under it the image $T(e_j)$ of the j -th source basis vector is exactly the j -th *column* of $\Phi(T)$, matching the recipe of Remark 7.1.2; the alternative would stack those images as rows instead. We follow the target-first ($Y \times X$) convention throughout.

Definition 7.1.3 (The identity grid):

The **identity grid** on an index set X is the image of the identity map under the bijection,

$$I_X := \Phi(\text{Id}_{K^{(X)}}) \in \text{Mat}_{(X),X}(K).$$

Its entries are the Kronecker delta, read straight off the definition of Φ :

$$(I_X)_{ij} = (\text{Id}_{K^{(X)}}(e_j))_i = (e_j)_i = \delta_{ij} = \begin{cases} 1 & i = j, \\ 0 & i \neq j, \end{cases}$$

so its j -th column is e_j . When $X = n$ is finite we write I_n for the $n \times n$ identity. Once composition is transported (Subsection 7.1.3), I_X is the two-sided multiplicative unit (Corollary 7.1.6) — as befits the grid of the identity map.

Remark (A one-column grid is a vector: the case $X = 1$):

The most degenerate source index makes the bijection reproduce a fact from the previous chapter. Take $X = 1 = \{0\}$, a one-element set, so that $K^{(X)} = K^{(\{0\})} = K$ with canonical basis the single vector $e_0 = 1$. A grid in $\text{Mat}_{(Y),1}(K)$ then has a *single* column, and Theorem 7.1.2 specialises to

$$\text{Hom}_K(K, K^{(Y)}) \xrightarrow{\sim} \text{Mat}_{(Y),1}(K), \quad T \mapsto T(1),$$

that one column being the image $T(1) \in K^{(Y)}$. Composing with the natural isomorphism $\text{Hom}_K(K, K^{(Y)}) \cong K^{(Y)}$ that evaluates a map at 1 (Theorem 6.7.10, applied to $V = K^{(Y)}$) collapses both sides to

$$\text{Mat}_{(Y),1}(K) \cong K^{(Y)} :$$

a one-column grid *is* a vector of $K^{(Y)}$, namely its single column, and conversely every vector is the one-column grid of its own coordinates. This is why a vector is recorded as a column. The same identification, carried out for an *abstract* space V through a chosen basis, is what makes the coordinate vector $[v]^B$ itself a column matrix.

Remark (Two structures on one set: grid versus map — insist on this):

Theorem 7.1.2 identifies two *a priori* different kinds of object on the same underlying data. It is worth dwelling on the distinction, because the rest of the chapter consists of moving back and forth across it.

- **The map view** ($T : K^{(X)} \rightarrow K^{(Y)}$). Here the object is a *morphism*: it can be evaluated on vectors, *composed* with other maps, and it has structural invariants (kernel, image, rank). These are the operations of linear algebra proper — they see the object as an arrow in the category of K -vector spaces.
- **The grid view** ($A \in \text{Mat}_{(Y),X}(K)$). Here the object is a *table of numbers*: a function on a rectangle of indices. It supports entrywise operations that ignore the linear-algebraic meaning entirely — reading off a single entry, extracting a row or column or diagonal, the entrywise (Hadamard) product $(A \circ B)_{ij} = A_{ij}B_{ij}$, the index swap $(i, j) \mapsto (j, i)$, sparsity patterns, reshaping.

The point of Φ is that these are two *coordinatisations* of one and the same information: the abstract map, its presentation as data, and the bridge between them. The same three-layer pattern governed vectors in the previous chapters — a geometric vector v , its coordinate tuple $[v]^{\mathcal{B}} = \phi^{\mathcal{B}}(v)$ given by the coordinate isomorphism $\phi^{\mathcal{B}}$ of an ordered basis, and the bridge $\phi^{\mathcal{B}}$ itself — and it will reappear in full force in a later section, when we coordinatise maps between *abstract* spaces. For maps $K^{(X)} \rightarrow K^{(Y)}$ the middle “coordinatisation” layer is invisible because these spaces already carry canonical bases; the table below records the general shape we are heading toward (here $\phi^{\mathcal{B}} : V \rightarrow K^{(X)}$ is the coordinate isomorphism of a basis $\mathcal{B}$ of V).

Level	Geometric object	Coordinatisation	Data structure
Scalar	$a \in K$	$a \in K$	a field element
Vector	$v \in V$	$\phi^{\mathcal{B}}(v) \in K^{(X)}$	tuple $(a_i)_{i \in X}$
Map	$T : U \rightarrow V$	$\phi^{\mathcal{B}_V} \circ T \circ \psi_{\mathcal{B}_U}$	grid $(A_{ij})_{(i,j) \in Y \times X}$

Until we make the identification official in a later section, we keep the two views *notationally distinct*: $\Phi(T)$ is the grid of a map, and $\Phi^{-1}(A)$ is the map of a grid. We do not yet speak of “the kernel of a matrix” or let a bare grid act on vectors; that licence is granted only after the identification.

7.1.3 Transport of Structure

The set $\text{Hom}_K(K^{(X)}, K^{(Y)})$ of linear maps is itself a K -vector space under the **pointwise** operations

$$(S + T)(x) := S(x) + T(x), \quad (\lambda T)(x) := \lambda T(x),$$

which are again linear (a linear combination of linear maps is linear, as established for linear maps), and linear maps can be *composed*. Using the bijection Φ of Theorem 7.1.2 — and its inverse Φ^{-1} — we **transport** each of these operations onto column-finite grids: we *define* the matrix operation to be whatever Φ turns the map operation into. Nothing is posited about grids directly; every entry formula below is then *derived*.

Definition 7.1.4 (Matrix operations, by transport along Φ):

Let Φ be the bijection of Theorem 7.1.2 and Φ^{-1} its inverse. For $A, B \in \text{Mat}_{(Y),X}(K)$ and $\lambda \in K$ define the **sum** and **scalar multiple** by transporting the pointwise operations on $\text{Hom}_K(K^{(X)}, K^{(Y)})$:

$$A + B := \Phi(\Phi^{-1}(A) + \Phi^{-1}(B)), \quad \lambda A := \Phi(\lambda \Phi^{-1}(A)).$$

For $A \in \text{Mat}_{(Y),Z}(K)$ and $B \in \text{Mat}_{(Z),X}(K)$ define the **product** by transporting composition,

$$AB := \Phi(\Phi^{-1}(A) \circ \Phi^{-1}(B)) \in \text{Mat}_{(Y),X}(K),$$

where $\Phi^{-1}(A) : K^{(Z)} \rightarrow K^{(Y)}$ and $\Phi^{-1}(B) : K^{(X)} \rightarrow K^{(Z)}$ compose to a map $K^{(X)} \rightarrow K^{(Y)}$, whose grid is again column-finite by Theorem 7.1.2.

Because they are transports of operations on a vector space along a bijection, these endow $\text{Mat}_{(Y),X}(K)$ with a K -vector space structure for which Φ is, *by construction*, a linear isomorphism; and the product inherits associativity, bilinearity, and units from composition. What remains is to compute what these transported operations *are* on entries. Note in particular that addition and scalar multiplication are obtained here *by transport*, not posited entrywise — the entrywise formulas are a conclusion.

Theorem 7.1.5 (Entry formulas):

The transported operations of Definition 7.1.4 are given on entries by

$$(A + B)_{ij} = A_{ij} + B_{ij}, \quad (\lambda A)_{ij} = \lambda A_{ij}, \quad (AB)_{ij} = \sum_{k \in Z} A_{ik} B_{kj},$$

where each product sum is finite because $B(-, j) \in K^{(Z)}$ has finite support, and the product AB is again column-finite. Moreover, for $A \in \text{Mat}_{(Y), X}(K)$ and $x \in K^{(X)}$, the map $\Phi^{-1}(A)$ acts by

$$(\Phi^{-1}(A)(x))_i = \sum_{j \in X} A_{ij} x_j,$$

a finite sum since x has finite support — the matrix–vector product.

Proof:

Action on a vector. Write $S = \Phi^{-1}(A)$, so $S(e_j) = A(-, j)$ has i -th coordinate A_{ij} . For $x = \sum_{j \in X} x_j e_j$ (finite support), linearity gives $S(x) = \sum_{j \in X} x_j S(e_j)$, whose i -th coordinate is $\sum_{j \in X} x_j A_{ij} = \sum_{j \in X} A_{ij} x_j$ (a finite sum).

Sum and scalar multiple. Let $S = \Phi^{-1}(A)$, $T = \Phi^{-1}(B)$. By definition $A + B = \Phi(S + T)$, so

$$(A + B)_{ij} = ((S + T)(e_j))_i = (S(e_j))_i + (T(e_j))_i = A_{ij} + B_{ij},$$

using additivity of coordinates in $K^{(Y)}$. Likewise $(\lambda A)_{ij} = ((\lambda S)(e_j))_i = \lambda (S(e_j))_i = \lambda A_{ij}$.

Product. Let $S = \Phi^{-1}(A) : K^{(Z)} \rightarrow K^{(Y)}$ and $T = \Phi^{-1}(B) : K^{(X)} \rightarrow K^{(Z)}$. By definition $AB = \Phi(S \circ T)$, whose j -th column is $(S \circ T)(e_j) = S(T(e_j)) = S(B(-, j))$. Now $B(-, j) = \sum_{k \in Z} B_{kj} e_k$ is a finite sum (column-finiteness of B), so by linearity of S and the action formula just proved,

$$(AB)_{ij} = \left(S(B(-, j)) \right)_i = \sum_{k \in Z} B_{kj} (S(e_k))_i = \sum_{k \in Z} B_{kj} A_{ik} = \sum_{k \in Z} A_{ik} B_{kj}.$$

Finally, $\{i \in Y \mid (AB)_{ij} \neq 0\} \subseteq \bigcup_{k \in \text{supp } B(-, j)} \{i \in Y \mid A_{ik} \neq 0\}$ is a finite union of finite sets, so $(AB)(-, j) \in K^{(Y)}$ and AB is column-finite, consistent with Definition 7.1.4. QED

Remark (The matrix–vector product is transport of structure too):

The action formula $(\Phi^{-1}(A)(x))_i = \sum_{j \in X} A_{ij} x_j$ reads like a new operation — a matrix acting on a vector — but it is transport of structure once more, with nothing extra to prove. By Remark 6.7, applying the map $\Phi^{-1}(A)$ to a vector $x \in K^{(X)}$ is composition with the name $T_x : K \rightarrow K^{(X)}$, $1_K \mapsto x$, of that vector: $\Phi^{-1}(A)(x) = \Phi^{-1}(A) \circ T_x$. Read through Φ , the vector x is the single-column grid $\Phi(T_x)$ — the one-column-grid-is-a-vector identification of Remark 7.1.2, the case $X = 1$ where $\text{Hom}_K(K, K^{(X)}) \cong K^{(X)}$ — and the matrix–vector product Ax is the matrix product of A with that column (Definition 7.1.4). Hence “matrix times vector” is not a separate primitive: it is matrix multiplication with a one-column factor — a special case of the matrix product — exactly as application is the one-argument case of composition.

Corollary 7.1.6 (Inherited algebraic laws):

Matrix addition and scalar multiplication make $\text{Mat}_{(Y), X}(K)$ a K -vector space, and Φ is a linear isomorphism $\text{Hom}_K(K^{(X)}, K^{(Y)}) \xrightarrow{\sim} \text{Mat}_{(Y), X}(K)$. Matrix multiplication is associative and bilinear wherever the index sets match, with the identity grid I_X (Definition 7.1.3) as two-sided unit:

$$(AB)C = A(BC), \quad A(B + C) = AB + AC, \quad (A + B)C = AC + BC, \quad I_Y A = A = A I_X.$$

In particular, for $X = Y$, Φ is an isomorphism of K -algebras $\text{End}_K(K^{(X)}) \xrightarrow{\sim} \text{Mat}_{(X), X}(K)$.

Proof:

Each law holds for the corresponding operation on maps — the pointwise operations make $\text{Hom}_K(K^{(X)}, K^{(Y)})$ a vector space, composition is associative and bilinear, and the identity map $\text{Id}_{K^{(X)}}$ is its unit. Transporting along the bijection Φ preserves every such identity: associativity follows from $(R \circ S) \circ T = R \circ (S \circ T)$ together with the definitions, and $\Phi(\text{Id}_{K^{(X)}}) = I_X$ since $(\text{Id}_{K^{(X)}}(e_j))_i = (e_j)_i = \delta_{ij}$. No entrywise computation is needed. QED

Remark (The general grid space and the action map):

Dropping column-finiteness, every grid $A \in \text{Mat}_{Y,X}(K)$ still defines a linear **action map**

$$\mu_A : K^{(X)} \rightarrow K^Y, \quad (\mu_A(x))_i := \sum_{j \in X} A_{ij} x_j,$$

the sum being finite because $x \in K^{(X)}$ has finite support. Column-finiteness of A is exactly the condition that μ_A land in $K^{(Y)}$ rather than the full product K^Y , i.e. that A represent a map $K^{(X)} \rightarrow K^{(Y)}$. Thus $\text{Mat}_{(Y),X}(K)$ is the correct home for matrices of maps between coordinate spaces at *any* cardinality; the larger $\text{Mat}_{Y,X}(K)$ becomes indispensable only once a topology lets one make sense of non-finite sums (the passage from algebra to functional analysis).

7.1.4 Practical Examples

We compute with the formulas just derived; everything below is elementary arithmetic in the finite case $K^n \rightarrow K^m$, where every grid is column-finite.

Example 7.1.7 (Reading off $\Phi(T)$):

Let $T : K^3 \rightarrow K^2$ be $T(x_0, x_1, x_2) = (x_0 + 2x_2, x_0 - x_1)$. Applying T to the canonical basis (e_0, e_1, e_2) of K^3 gives the columns:

$$T(e_0) = (1, 1), \quad T(e_1) = (0, -1), \quad T(e_2) = (2, 0), \quad \Phi(T) = \begin{pmatrix} 1 & 0 & 2 \\ 1 & -1 & 0 \end{pmatrix} \in \text{Mat}_{(2),3}(K).$$

The action formula recovers T : for $x = (x_0, x_1, x_2)$, $(\Phi^{-1}(\Phi(T))(x)) = (\sum_{j \in 3} \Phi(T)_{0j} x_j, \sum_{j \in 3} \Phi(T)_{1j} x_j) = (x_0 + 2x_2, x_0 - x_1) = T(x)$.

Example 7.1.8 (Composition is multiplication):

Let $T : K^3 \rightarrow K^2$ be as above and $S : K^2 \rightarrow K^2$, $S(y_0, y_1) = (y_1, y_0 + y_1)$, so $\Phi(S) = \begin{pmatrix} 0 & 1 \\ 1 & 1 \end{pmatrix}$. The composite $S \circ T : K^3 \rightarrow K^2$ has grid

$$\Phi(S \circ T) = \Phi(S) \Phi(T) = \begin{pmatrix} 0 & 1 \\ 1 & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 & 2 \\ 1 & -1 & 0 \end{pmatrix} = \begin{pmatrix} 1 & -1 & 0 \\ 2 & -1 & 2 \end{pmatrix}.$$

Checking directly: $S(T(x)) = S(x_0 + 2x_2, x_0 - x_1) = (x_0 - x_1, 2x_0 - x_1 + 2x_2)$, whose grid is exactly the product above. The shared index k in $(\Phi(S)\Phi(T))_{ij} = \sum_{k \in 2} \Phi(S)_{ik} \Phi(T)_{kj}$ is the intermediate space K^2 being cancelled — this is why the inner index sets must agree.

Example 7.1.9 (Associativity for free):

With $A = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$, $B = \begin{pmatrix} 2 & 0 \\ 1 & 3 \end{pmatrix}$, $C = \begin{pmatrix} 1 \\ 1 \end{pmatrix}$ one verifies $(AB)C = \begin{pmatrix} 3 & 3 \\ 1 & 3 \end{pmatrix} \begin{pmatrix} 1 \\ 1 \end{pmatrix} = \begin{pmatrix} 6 \\ 4 \end{pmatrix}$ and $A(BC) = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 2 \\ 4 \end{pmatrix} = \begin{pmatrix} 6 \\ 4 \end{pmatrix}$. The agreement is guaranteed in advance by Corollary 7.1.6: it is associativity of composition, read through Φ .

7.2 Matrices as Linear Maps

Intuition:

Up to this point we have been scrupulous: a matrix is a grid, a map is a map, and the bijection Φ of Theorem 7.1.2 ferries one to the other. We even refused to speak of “the kernel of a matrix” (Remark 7.1.2). That scruple has done its work — it forced us to prove that Φ is not merely a bijection of sets but an isomorphism carrying every operation faithfully (Corollary 7.1.6).

When two objects are linked by an isomorphism that (i) requires *no arbitrary choices* to write down and (ii) respects *all* the structure in sight, mathematics drops the distinction and writes a single symbol for both. The canonical bases of $K^{(X)}$ and $K^{(Y)}$ are built into those spaces, so Φ needs no input from us; this is exactly the licence to **identify** a map with its grid. From now on a matrix *is* a linear map, and a linear map between coordinate spaces *is* a matrix, with no translation step. This is the move that makes every map-theoretic notion — kernel, image, rank — available to matrices, and conversely lets a matrix be evaluated, composed, and inverted.

Definition 7.2.1 (The map–grid identification, $\text{Hom}_K(K^{(X)}, K^{(Y)}) = \text{Mat}_{(Y),X}(K)$):

For all index sets X, Y we henceforth **identify**, along the isomorphism Φ of Theorem 7.1.2 and Corollary 7.1.6,

$$\text{Hom}_K(K^{(X)}, K^{(Y)}) = \text{Mat}_{(Y),X}(K),$$

suppressing Φ and Φ^{-1} from the notation: a column-finite grid A and the linear map $\Phi^{-1}(A) = \mu_A : K^{(X)} \rightarrow K^{(Y)}$ are written by the same symbol A , and a map T and its grid $\Phi(T)$ by the same symbol T . Concretely:

- $A(x)$ (also written Ax) denotes the value of the map on $x \in K^{(X)}$, computed by the matrix–vector formula $(Ax)_i = \sum_{j \in X} A_{ij} x_j$ (Theorem 7.1.5);
- the product AB denotes *simultaneously* the composite of maps and the grid product, the two being equal by the definition of the latter (Definition 7.1.4);
- the j -th column of A is the image vector $A(e_j) \in K^{(Y)}$.

In the finite case this reads $\text{Hom}_K(K^n, K^m) = \text{Mat}_{m,n}(K)$, and for endomorphisms it is an identity of K -algebras $\text{End}_K(K^{(X)}) = \text{Mat}_{(X),X}(K)$.

Remark (Why “no questions asked” — and where it stops):

The identification is legitimate precisely because Φ is *canonical*: it is determined by the spaces $K^{(X)}, K^{(Y)}$ alone, through their built-in canonical bases, with no choice left to the reader. It is worth being clear in advance about where this licence ends. Later in this chapter we attach to a map between *abstract* spaces $U \rightarrow V$ a matrix $[T]_{\mathcal{B}_U}^{\mathcal{B}_V}$ (Section 7.4); that matrix is *not* canonically attached to T — it will depend on a choice of ordered bases $\mathcal{B}_U, \mathcal{B}_V$. So we do *not* identify an abstract map $T : U \rightarrow V$ with a grid. What we *will* be entitled to say is that, once bases are fixed, $[T]_{\mathcal{B}_U}^{\mathcal{B}_V}$ — by construction an element of $\text{Hom}_K(K^{(X)}, K^{(Y)})$ — already *is* a column-finite grid under the present identification, with no further bijection to invoke. The choice of bases is the only arbitrary input; the passage from the coordinatised map to its array is free.

The structural invariants of a linear map, defined in the previous chapter, now apply *verbatim* to a matrix.

Definition 7.2.2 (Kernel, image, rank, and nullity of a matrix):

For $A \in \text{Mat}_{(Y),X}(K)$, regarded as the map $A : K^{(X)} \rightarrow K^{(Y)}$, set

$$\ker(A) := \{x \in K^{(X)} \mid A(x) = 0\}, \quad \text{Rng}(A) := A[K^{(X)}] \subseteq K^{(Y)},$$

and $\text{rank}(A) := \dim_K(\text{Rng}(A))$, $\text{nullity}(A) := \dim_K(\ker(A))$. The subspace $\text{Rng}(A)$ is called the **column space** of A .

Proposition 7.2.3 (The image is the span of the columns):

For $A \in \text{Mat}_{(Y),X}(K)$,

$$\text{Rng}(A) = \text{span}_K(\{A(e_j) \mid j \in X\}),$$

the span of the columns of A ; consequently $\text{rank}(A)$ is the maximal number of linearly independent columns.

Proof:

Every $x \in K^{(X)}$ is a finite combination $x = \sum_{j \in X} x_j e_j$ (finite support), so by linearity $A(x) = \sum_{j \in X} x_j A(e_j)$ is a finite linear combination of columns; hence $\text{Rng}(A) \subseteq \text{span}_K(\{A(e_j) \mid j \in X\})$. Conversely each column $A(e_j)$ lies in $\text{Rng}(A)$, and $\text{Rng}(A)$ is a subspace, so it contains their span. The two inclusions give equality, and the dimension of a span is the maximal number of linearly independent vectors drawn from a spanning set. QED

Remark (Map-side and grid-side operations):

The identification stacks two layers of structure on a single object, and it pays to keep them apart — this is the distinction foretold in Remark 7.1.2, now made operative.

- **Map-side** (morphism structure): evaluation $A(x)$, the product/composite AB , kernel, image, rank, nullity, and — next — invertibility. These respect the linear-algebraic meaning: for instance $\text{Rng}(AB) \subseteq \text{Rng}(A)$ holds because it is a statement about composing maps, and the product is not a free choice — it *is* composition, read through Φ .
- **Grid-side** (data structure): the entrywise (Hadamard) product $(A \circ B)_{ij} = A_{ij} B_{ij}$, the index swap $((A)^\top)_{ji} = A_{ij}$ (transpose), extraction of a single row, column, or the diagonal, and sparsity bookkeeping. These manipulate the table of scalars and need carry *no* map-theoretic meaning: the Hadamard product of two maps is not in general a natural operation on maps at all, and the intrinsic meaning of the transpose — the induced map between dual spaces — must wait until those are built (a forward reference).

Now that both layers ride on one symbol, the only discipline required is to know which structure a given manipulation belongs to: composition and kernels concern the *arrow*; Hadamard products and transposes concern the *array*.

Example 7.2.4 (Kernel and column space of a concrete matrix):

Take $A = \begin{pmatrix} 1 & 0 & 2 \\ 1 & -1 & 0 \end{pmatrix} \in \text{Mat}_{2,3}(K)$ from Example 7.1.7, now read as the map $A : K^3 \rightarrow K^2$. Its columns

$$A(e_0) = (1, 1), \quad A(e_1) = (0, -1), \quad A(e_2) = (2, 0)$$

already span K^2 (the first two are independent), so $\text{Rng}(A) = K^2$ and $\text{rank}(A) = 2$. By rank-nullity (previous chapter) $\text{nullity}(A) = 3 - 2 = 1$. Concretely, $A(x) = 0$ means $x_0 + 2x_2 = 0$ and $x_0 - x_1 = 0$, i.e. $x_0 = x_1 = -2x_2$, so

$$\ker(A) = \text{span}_K(\{(-2, -2, 1)\}),$$

a line, confirming $\text{nullity}(A) = 1$. Every symbol here — $A(x)$, $\ker(A)$, $\text{Rng}(A)$, $\text{rank}(A)$ — is now applied directly to the array, with no Φ in sight.

7.3 Coordinates and Actions: A Pattern Beyond Linear Algebra

Intuition:

The bijection Φ of Section 7.1 did something very general: it pinned an object (a linear map) to a *standard model* (a grid) through an invertible dictionary (the assignment Φ , built from the canonical bases). Almost nothing in that move used linearity. The same shape recurs whenever a structured object is identified with a standard model through an invertible dictionary, and it always comes with the same subtlety: a transformation can act *through* the dictionary in *two* different ways — by moving the object, or by changing the dictionary — and the two are related, but not equal.

To see the pattern stripped of all linear-algebraic distraction, we first play it out for plain finite sets, where the “dictionary” is an *enumeration* and the transformations are *permutations*. Every phenomenon we will later meet for bases of vector spaces — the two actions, the four ways to translate between them, the conjugation relating them — appears already here, with permutations in place of invertible matrices.

7.3.1 The Set Picture: Enumerations and Permutations

Fix a finite set A with $|A| = n$ and write $n = \{0, 1, \dots, n-1\}$ for the standard n -element set (the von Neumann ordinal). The role played by an ordered basis is played here by an **enumeration**.

Definition 7.3.1 (Enumeration and lookup):

An **enumeration** of A is a bijection $\psi_B : n \rightarrow A$ (“the j -th element is $\psi_B(j)$ ”). Its inverse $\phi^B := \psi_B^{-1} : A \rightarrow n$ is the **lookup**, assigning to each element its *index* $\phi^B(x) \in n$. The two are inverse, $\phi^B \circ \psi_B = \text{Id}_n$ and $\psi_B \circ \phi^B = \text{Id}_A$ — precisely the relationship a parametrisation and a chart will have for a basis.

Two groups act on enumerations. Permutations of the *index* set n — the symmetric group $\text{Sym}(n)$ — relabel the slots; permutations of the *element* set A — the symmetric group $\text{Sym}(A)$ — move the contents. They act on opposite sides of ψ_B .

Proposition 7.3.2 (Two actions producing a new enumeration):

Let $\psi_B : n \rightarrow A$ be an enumeration.

1. For $P \in \text{Sym}(n)$, the composite $\psi_B \circ P : n \rightarrow A$ is again an enumeration. This is the **right action** of $\text{Sym}(n)$: it relabels the slots, leaving the elements of A untouched.
2. For $S \in \text{Sym}(A)$, the composite $S \circ \psi_B : n \rightarrow A$ is again an enumeration. This is the **left action** of $\text{Sym}(A)$: it permutes the elements of A .

Given a second enumeration ψ_C , there is a *unique* $P \in \text{Sym}(n)$ with $\psi_B \circ P = \psi_C$ and a *unique* $S \in \text{Sym}(A)$ with $S \circ \psi_B = \psi_C$; the two produce the same ψ_C , and they are related by **conjugation through the enumeration**:

$$S = \psi_B \circ P \circ \psi_B^{-1} = \psi_B \circ P \circ \phi^B.$$

Proof:

A composite of bijections is a bijection, giving (1) and (2). For the uniqueness, set $P := \psi_B^{-1} \circ \psi_C = \phi^B \circ \psi_C \in \text{Sym}(n)$ and $S := \psi_C \circ \psi_B^{-1} = \psi_C \circ \phi^B \in \text{Sym}(A)$; then $\psi_B \circ P = \psi_C = S \circ \psi_B$, and any P' with $\psi_B \circ P' = \psi_C$ equals P by cancelling the bijection ψ_B on the left (similarly for S). Eliminating ψ_C from $\psi_B \circ P = S \circ \psi_B$ gives $S = \psi_B \circ P \circ \psi_B^{-1}$. QED

Example 7.3.3 (A 3-cycle and a transposition on five elements):

Take $A = \{a, b, c, d, e\}$ and $n = 5 = \{0, 1, 2, 3, 4\}$, with the enumeration

$$\psi_B : 0 \mapsto a, 1 \mapsto b, 2 \mapsto c, 3 \mapsto d, 4 \mapsto e, \quad \phi^B : a \mapsto 0, b \mapsto 1, c \mapsto 2, d \mapsto 3, e \mapsto 4.$$

Let $P = (012)(34) \in \text{Sym}(5)$ — a 3-cycle and a transposition — so $P(0) = 1$, $P(1) = 2$, $P(2) = 0$, $P(3) = 4$, $P(4) = 3$; its inverse is $P^{-1} = (021)(34)$.

The right action (relabelling the slots). $\psi_C := \psi_B \circ P$ sends

$$0 \mapsto \psi_B(1) = b, \quad 1 \mapsto c, \quad 2 \mapsto a, \quad 3 \mapsto e, \quad 4 \mapsto d,$$

i.e. $\psi_C : 0 \mapsto b, 1 \mapsto c, 2 \mapsto a, 3 \mapsto e, 4 \mapsto d$. Nothing in A moved; we merely relabelled which slot holds which element.

The left action (permuting the elements). The unique $S \in \text{Sym}(A)$ with $S \circ \psi_B = \psi_C$ is $S = \psi_C \circ \phi^B$:

$$S(a) = \psi_C(0) = b, \quad S(b) = c, \quad S(c) = a, \quad S(d) = e, \quad S(e) = d,$$

that is $S = (abc)(de) \in \text{Sym}(A)$ — again a 3-cycle and a transposition. Here the elements genuinely move: a becomes b , and so on.

Same result, two routes. Both give the same new enumeration ψ_C : relabelling the slots by P and physically permuting the elements by S are indistinguishable at the level of the resulting enumeration. They are tied by conjugation, $S = \psi_C \circ P \circ \phi^B$: indeed $\psi_B \circ P \circ \phi^B(a) = \psi_B(P(0)) = \psi_B(1) = b = S(a)$, and likewise on the rest.

Remark (The four ways to translate, and what each does):

Around one re-enumeration $\mathcal{B} \rightsquigarrow \mathcal{C}$ there are exactly four invertible maps, in two conjugate pairs:

- $P = \phi^B \circ \psi_C \in \text{Sym}(n)$ — relabels slots forward, $\mathcal{B} \rightsquigarrow \mathcal{C}$; here $P = (012)(34)$.
- $P^{-1} = \phi^C \circ \psi_B = (021)(34)$ — relabels slots backward, $\mathcal{C} \rightsquigarrow \mathcal{B}$.
- $S = \psi_C \circ \phi^B \in \text{Sym}(A)$ — moves the elements forward; here $S = (abc)(de)$.
- $S^{-1} = \psi_B \circ \phi^C = (acb)(de)$ — moves the elements backward.

The slot maps P, P^{-1} live in $\text{Sym}(n)$ and act on the *right* of an enumeration; the element maps S, S^{-1} live in $\text{Sym}(A)$ and act on the *left*. Conjugation by the enumeration carries one side to the other, $S = \psi_B \circ P \circ \phi^B$ and $P = \phi^B \circ S \circ \psi_B$; in particular P and S always have the *same cycle type* (here both are a 3-cycle and a transposition).

Proposition 7.3.4 (The index of a fixed element transforms by the inverse):

Let $\mathcal{B}, \mathcal{C}, P, S$ be as in Proposition 7.3.2, so that $\psi_B \circ P = \psi_C = S \circ \psi_B$. Then the lookups satisfy

$$\phi^C = P^{-1} \circ \phi^B = \phi^B \circ S^{-1}, \quad \text{equivalently} \quad \phi^C(x) = \phi^B(S^{-1}(x)) \quad \text{for all } x \in A.$$

While the elements move forward by S , the index of a *fixed* element moves backward, by P^{-1} .

Proof:

Inverting $\psi_C = \psi_B \circ P$ gives $\phi^C = \psi_C^{-1} = P^{-1} \circ \psi_B^{-1} = P^{-1} \circ \phi^B$. Inverting $\psi_C = S \circ \psi_B$ gives $\phi^C = \psi_B^{-1} \circ S^{-1} = \phi^B \circ S^{-1}$. The two expressions are equal since both equal ψ_C^{-1} . QED

Remark (The five-element computation):

For Example 7.3.3 the lookups are tabulated below; the new index of x is the old index of the element S drags onto x .

element x	a	b	c	d	e
index $\phi^B(x)$	0	1	2	3	4
index $\phi^C(x)$	2	0	1	4	3

For instance $\phi^C(a) = 2 = \phi^B(c) = \phi^B(S^{-1}(a))$, since $S^{-1}(a) = c$.

The conjugation relation is exactly the commuting square

$$\begin{array}{ccc}
 A & \xrightarrow{S} & A \\
 \psi_{\mathcal{B}} \uparrow & & \uparrow \psi_{\mathcal{B}} \\
 n & \xrightarrow{P} & n
 \end{array}$$

7.3.2 The Same Pattern in Vector Spaces

Now replace the finite set A by a K -vector space V , the standard model n by a coordinate space $K^{(X)}$, an enumeration $\psi_{\mathcal{B}} : n \rightarrow A$ by a **frame** $\psi_{\mathcal{B}} : K^{(X)} \rightarrow V$, and the symmetric groups by the groups of invertible linear maps. Everything in Section 7.3.1 transcribes verbatim, with “bijection” replaced by “linear isomorphism.”

Definition 7.3.5 (Frame and chart of a basis):

Let V be a K -vector space with an ordered basis $\mathcal{B} = (b_j)_{j \in X}$. The **frame** (or parametrisation) of $\mathcal{B}$ is the linear isomorphism

$$\psi_{\mathcal{B}} : K^{(X)} \rightarrow V, \quad \psi_{\mathcal{B}}(e_j) = b_j,$$

and the **chart** (or coordinate map) is its inverse $\phi^{\mathcal{B}} := \psi_{\mathcal{B}}^{-1} : V \rightarrow K^{(X)}$, with $\phi^{\mathcal{B}}(v) = [v]^{\mathcal{B}}$ the coordinate tuple of v . As before $\phi^{\mathcal{B}} \circ \psi_{\mathcal{B}} = \text{Id}_{K^{(X)}}$, $\psi_{\mathcal{B}} \circ \phi^{\mathcal{B}} = \text{Id}_V$, and $v = \psi_{\mathcal{B}}([v]^{\mathcal{B}})$.

The two acting groups are now $\text{GL}(V)$, the invertible linear self-maps of V (the analogue of $\text{Sym}(A)$), and $\text{GL}(K^{(X)})$, the invertible linear self-maps of the coordinate space (the analogue of $\text{Sym}(n)$). In the finite-dimensional case $X = n$, Section 7.1 identified $\text{GL}(K^{(X)})$ with the invertible matrices $\text{GL}(K^n)$.

Proposition 7.3.6 (Two actions on frames, related by conjugation):

Let $\psi_{\mathcal{B}} : K^{(X)} \rightarrow V$ be a frame.

1. For $P \in \text{GL}(K^{(X)})$, the composite $\psi_{\mathcal{B}} \circ P$ is again a frame — the **right (passive)** action of $\text{GL}(K^{(X)})$, a re-coordinatisation.
2. For $S \in \text{GL}(V)$, the composite $S \circ \psi_{\mathcal{B}}$ is again a frame — the **left (active)** action of $\text{GL}(V)$, a genuine motion of vectors.

Given a second frame $\psi_{\mathcal{C}}$, there are unique $P = \phi^{\mathcal{B}} \circ \psi_{\mathcal{C}} \in \text{GL}(K^{(X)})$ and $S = \psi_{\mathcal{C}} \circ \phi^{\mathcal{B}} \in \text{GL}(V)$ with $\psi_{\mathcal{B}} \circ P = \psi_{\mathcal{C}} = S \circ \psi_{\mathcal{B}}$, related by conjugation through the frame,

$$S = \psi_{\mathcal{B}} \circ P \circ \phi^{\mathcal{B}}, \quad P = \phi^{\mathcal{B}} \circ S \circ \psi_{\mathcal{B}}.$$

Proof:

Word for word the proof of Proposition 7.3.2, with “bijection” replaced by “linear isomorphism”: composites of isomorphisms are isomorphisms, the displayed P and S satisfy the two equations, uniqueness is left/right cancellation of the isomorphism $\psi_{\mathcal{B}}$, and eliminating $\psi_{\mathcal{C}}$ gives the conjugation. QED

Remark (Physical transformations are active; changes of perspective are passive):

The two sides carry a standard interpretation in applications. An element $S \in \text{GL}(V)$ is a *physical* transformation: a rotation, reflection, shear, or deformation that actually moves the vectors of V . An element $P \in \text{GL}(K^{(X)})$ is a change of *perspective*: it does not touch the vectors at all, only the coordinates we use to name them. So

$$\text{GL}(V) \longleftrightarrow \text{active (physical) transformations}, \quad \text{GL}(K^{(X)}) \longleftrightarrow \text{passive (coordinate) transformations}.$$

The conjugation $S = \psi_B \circ P \circ \phi^B$ says these are two readings of one frame change: P describes it downstairs on coordinates, S performs it upstairs on vectors. (The labels “active” and “passive” are ours, tracking which side we deem physical; the mathematics privileges neither, just as for sets neither A nor n was preferred.)

Proposition 7.3.7 (Coordinates of a fixed vector transform by the inverse):

Let $\mathcal{B}, \mathcal{C}, P, S$ be as in Proposition 7.3.6, so that $\psi_B \circ P = \psi_C = S \circ \psi_B$. Then

$$\phi^C = P^{-1} \circ \phi^B = \phi^B \circ S^{-1}, \quad \text{so} \quad [v]^C = P^{-1} [v]^B = [S^{-1}v]^B \quad \text{for all } v \in V,$$

where $P^{-1} [v]^B$ denotes the matrix–vector action of Theorem 7.1.5. The active transformation S moves a vector by P in coordinates, whereas the change of perspective from $\mathcal{B}$ to $\mathcal{C}$ moves the coordinates of a *fixed* vector by P^{-1} .

Proof:

Inverting $\psi_C = \psi_B \circ P$ gives $\phi^C = P^{-1} \circ \phi^B$, and inverting $\psi_C = S \circ \psi_B$ gives $\phi^C = \phi^B \circ S^{-1}$; both equal ψ_C^{-1} . Applying these to a vector v yields $[v]^C = \phi^C(v) = P^{-1}\phi^B(v) = P^{-1}[v]^B$ and $\phi^C(v) = \phi^B(S^{-1}v) = [S^{-1}v]^B$. QED

Remark (The merry-go-round):

A fixed vector, re-coordinatised to the new frame, reads exactly as the old coordinates of the vector dragged backward by S^{-1} : turn the frame one way and a fixed point sweeps the other.

Example 7.3.8 (A rotation of the plane):

Let $V = \mathbb{R}^2$ with the standard frame $\mathcal{B}$, so $\psi_B = \text{Id}$ and $[v]^B = v$. Let $S \in \text{GL}(V)$ be the physical rotation by 90° counter-clockwise: $S(e_0) = e_1$, $S(e_1) = -e_0$. The frame it produces is $\mathcal{C} = (Se_0, Se_1) = (e_1, -e_0)$, and the corresponding passive coordinate map is $P = \phi^B \circ \psi_C = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix} \in \text{GL}(K^2)$ (its columns are $[e_1]^B$ and $[-e_0]^B$).

Take the fixed vector $v = (1, 1)$. Physically moving it gives $[Sv]^B = P [v]^B = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} 1 \\ 1 \end{pmatrix} = (-1, 1)$ — the arrow swung 90° counter-clockwise. Re-coordinatising the *same* fixed v to the rotated frame gives instead

$$[v]^C = P^{-1} [v]^B = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix} \begin{pmatrix} 1 \\ 1 \end{pmatrix} = (1, -1) = [S^{-1}v]^B,$$

the components reading as though v had rotated 90° clockwise. One matrix P , two opposite effects: P on a physically moved vector, P^{-1} on the coordinates of a fixed one.

The conjugation is again a commuting square, now of linear isomorphisms:

$$\begin{array}{ccc} V & \xrightarrow{S} & V \\ \psi_B \uparrow & & \uparrow \psi_B \\ K^{(X)} & \xrightarrow{P} & K^{(X)} \end{array}$$

Remark (Where this is heading):

The matrix $P = \phi^B \circ \psi_C$ attached to a frame change, and the way the operator S is read “in coordinates” through ψ_B , are the first instances of a single construction: the *matrix of a linear map in chosen bases*, developed in the next section. There we will see that every such P is the matrix of the *identity* read in two different bases, and that conjugation by ψ_B is exactly the passage between an operator and its matrix. For now we have only used invertible maps and their composites — no notion of “the matrix of an operator” has been invoked.

7.4 The Matrix of a Linear Map

Intuition:

Section 7.1 turned linear maps *between coordinate spaces*, $K^{(X)} \rightarrow K^{(Y)}$, into grids, because those spaces carry canonical bases. But the linear maps one meets in practice go between *abstract* spaces $U \rightarrow V$ — spaces of polynomials, of functions, of geometric arrows — which have no preferred basis at all. To compute with such a T we do the only thing available: *choose* an ordered basis of U and one of V , use them to coordinatise source and target, and read off the map $K^{(X)} \rightarrow K^{(Y)}$ that T becomes in those coordinates. That coordinatised map is the **matrix of T in the chosen bases**.

Two points organise everything below. First, the construction is just conjugation by the two charts, exactly the move that produced the transition maps of Section 7.3: those were the special case $T = \text{Id}_V$. Second, the object we build is presented as a *linear map* between coordinate spaces, an element of $\text{Hom}_K(K^{(X)}, K^{(Y)})$. Although such a map is by now *identified* with its grid of scalars (Section 7.2, via the bijection Φ of Theorem 7.1.2), every structural fact here (evaluation, composition, change of bases, invariance of rank) is cleanest as a statement about *composing maps*; so we deliberately reason at the level of maps and read the grid off only when computing a concrete example.

Throughout, U and V are K -vector spaces with ordered bases $\mathcal{B}_U = (b_j)_{j \in X}$ and $\mathcal{B}_V = (c_i)_{i \in Y}$, with frames $\psi_{\mathcal{B}_U} : K^{(X)} \rightarrow U$, $\psi_{\mathcal{B}_V} : K^{(Y)} \rightarrow V$ and charts $\phi^{\mathcal{B}_U} = \psi_{\mathcal{B}_U}^{-1}$, $\phi^{\mathcal{B}_V} = \psi_{\mathcal{B}_V}^{-1}$ (Definition 7.3.5).

Definition 7.4.1 (Matrix of a linear map):

Let $T : U \rightarrow V$ be K -linear. The **matrix of T with respect to $\mathcal{B}_U$ and $\mathcal{B}_V$** is the linear map

$$[T]_{\mathcal{B}_U}^{\mathcal{B}_V} := \phi^{\mathcal{B}_V} \circ T \circ \psi_{\mathcal{B}_U} \in \text{Hom}_K(K^{(X)}, K^{(Y)}),$$

the unique map making the coordinate square commute:

$$\begin{array}{ccc} U & \xrightarrow{T} & V \\ \phi^{\mathcal{B}_U} \downarrow & & \downarrow \phi^{\mathcal{B}_V} \\ K^{(X)} & \xrightarrow{[T]_{\mathcal{B}_U}^{\mathcal{B}_V}} & K^{(Y)} \end{array}$$

Commutativity reads $\phi^{\mathcal{B}_V} \circ T = [T]_{\mathcal{B}_U}^{\mathcal{B}_V} \circ \phi^{\mathcal{B}_U}$, i.e. $[T]_{\mathcal{B}_U}^{\mathcal{B}_V} = \phi^{\mathcal{B}_V} \circ T \circ \psi_{\mathcal{B}_U}$.

Remark (Reason with it as a map; read it as a grid):

The bracket notation $[T]_{\mathcal{B}_U}^{\mathcal{B}_V}$ names the single map $K^{(X)} \rightarrow K^{(Y)}$ through which T factors once source and target are coordinatised — and hence, under the identification of Section 7.2, *simultaneously* the column-finite grid $\Phi([T]_{\mathcal{B}_U}^{\mathcal{B}_V}) \in \text{Mat}_{(Y),X}(K)$ that one writes down. We nonetheless reason with it as a *map*, so that the theorems below are visibly statements about composing maps; the grid is read off only in concrete computations.

7.4.1 Columns and the Coordinate Formula

Proposition 7.4.2 (Action on the canonical basis):

For each $j \in X$,

$$[T]_{\mathcal{B}_U}^{\mathcal{B}_V}(e_j) = [T(b_j)]^{\mathcal{B}_V}.$$

That is, $[T]_{\mathcal{B}_U}^{\mathcal{B}_V}$ sends the canonical vector e_j to the coordinate vector of $T(b_j)$ in $\mathcal{B}_V$ — the data that, under the identification of maps with grids (Section 7.2), are the columns of its grid.

Proof:

Since $\psi_{\mathcal{B}_U}(e_j) = b_j$,

$$[T]_{\mathcal{B}_U}^{\mathcal{B}_V}(e_j) = \phi^{\mathcal{B}_V}(T(\psi_{\mathcal{B}_U}(e_j))) = \phi^{\mathcal{B}_V}(T(b_j)) = [T(b_j)]^{\mathcal{B}_V}. \quad \text{QED}$$

Proposition 7.4.3 (Coordinate formula):

For every $v \in U$,

$$[T(v)]^{\mathcal{B}_V} = [T]_{\mathcal{B}_U}^{\mathcal{B}_V}([v]^{\mathcal{B}_U}).$$

Equivalently, coordinatising commutes with applying T : read v in $\mathcal{B}_U$, apply the matrix, and obtain the coordinates of $T(v)$ in $\mathcal{B}_V$.

Proof:

Using $[v]^{\mathcal{B}_U} = \phi^{\mathcal{B}_U}(v)$ and $\psi_{\mathcal{B}_U} \circ \phi^{\mathcal{B}_U} = \text{Id}_U$,

$$[T]_{\mathcal{B}_U}^{\mathcal{B}_V}([v]^{\mathcal{B}_U}) = \phi^{\mathcal{B}_V}(T(\psi_{\mathcal{B}_U}(\phi^{\mathcal{B}_U}(v)))) = \phi^{\mathcal{B}_V}(T(v)) = [T(v)]^{\mathcal{B}_V}. \quad \text{QED}$$

Theorem 7.4.4 (Coordinatising Hom is a linear isomorphism):

The assignment $T \mapsto [T]_{\mathcal{B}_U}^{\mathcal{B}_V}$ is a K -linear isomorphism

$$\text{Hom}_K(U, V) \xrightarrow{\sim} \text{Hom}_K(K^{(X)}, K^{(Y)}),$$

with inverse $S \mapsto \psi_{\mathcal{B}_V} \circ S \circ \phi^{\mathcal{B}_U}$.

Proof:

Write $\Lambda(T) = \phi^{\mathcal{B}_V} \circ T \circ \psi_{\mathcal{B}_U}$ and $\Gamma(S) = \psi_{\mathcal{B}_V} \circ S \circ \phi^{\mathcal{B}_U}$. *Linearity:* for $T, T' \in \text{Hom}_K(U, V)$ and $\alpha, \beta \in K$, pre- and post-composition are linear, so $\Lambda(\alpha T + \beta T') = \alpha \Lambda(T) + \beta \Lambda(T')$. *Mutually inverse:* using $\psi_{\mathcal{B}_V} \circ \phi^{\mathcal{B}_V} = \text{Id}_V$ and $\phi^{\mathcal{B}_U} \circ \psi_{\mathcal{B}_U} = \text{Id}_{K^{(X)}}$,

$$\Gamma(\Lambda(T)) = \psi_{\mathcal{B}_V} \circ \phi^{\mathcal{B}_V} \circ T \circ \psi_{\mathcal{B}_U} \circ \phi^{\mathcal{B}_U} = T,$$

and symmetrically $\Lambda(\Gamma(S)) = S$. Hence Λ is a linear bijection. QED

7.4.2 Composition and Change of Bases

Theorem 7.4.5 (Composition is composition):

Let $T : U \rightarrow V$ and $S : V \rightarrow W$ be K -linear, with W carrying an ordered basis $\mathcal{B}_W$ indexed by Z . Then

$$[S \circ T]_{\mathcal{B}_U}^{\mathcal{B}_W} = [S]_{\mathcal{B}_V}^{\mathcal{B}_W} \circ [T]_{\mathcal{B}_U}^{\mathcal{B}_V}.$$

Proof:

Insert $\psi_{\mathcal{B}_V} \circ \phi^{\mathcal{B}_V} = \text{Id}_V$:

$$[S \circ T]_{\mathcal{B}_U}^{\mathcal{B}_W} = \phi^{\mathcal{B}_W} \circ S \circ T \circ \psi_{\mathcal{B}_U} = (\phi^{\mathcal{B}_W} \circ S \circ \psi_{\mathcal{B}_V}) \circ (\phi^{\mathcal{B}_V} \circ T \circ \psi_{\mathcal{B}_U}) = [S]_{\mathcal{B}_V}^{\mathcal{B}_W} \circ [T]_{\mathcal{B}_U}^{\mathcal{B}_V}. \quad \text{QED}$$

The cancellation $\psi_{\mathcal{B}_V} \circ \phi^{\mathcal{B}_V}$ is the middle space V being contracted, visible in the stacked coordinate squares:

$$\begin{array}{ccccc} U & \xrightarrow{T} & V & \xrightarrow{S} & W \\ \phi^{\mathcal{B}_U} \downarrow & & \downarrow \phi^{\mathcal{B}_V} & & \downarrow \phi^{\mathcal{B}_W} \\ K^{(X)} & \xrightarrow{[T]_{\mathcal{B}_U}^{\mathcal{B}_V}} & K^{(Y)} & \xrightarrow{[S]_{\mathcal{B}_V}^{\mathcal{B}_W}} & K^{(Z)} \end{array}$$

The transition maps of Section 7.3 are now recognised as a special case.

Remark (The matrix of the identity):

Let V have two ordered bases $\mathcal{B}$ and $\mathcal{C}$ (indexed by the same set X , since $|X| = \dim_K(V)$). Reading the identity $\text{Id}_V : V \rightarrow V$ through Definition 7.4.1, with $\mathcal{B}$ on the source and $\mathcal{C}$ on the target, introduces nothing new: it is the automorphism

$$[\text{Id}_V]_{\mathcal{B}}^{\mathcal{C}} = \phi^{\mathcal{C}} \circ \psi_{\mathcal{B}} \in \text{GL}\left(K^{(X)}\right).$$

By Proposition 7.4.2 its action on e_j is $[b_j]^{\mathcal{C}}$, and by Proposition 7.4.3 it converts coordinates, $[v]^{\mathcal{C}} = [\text{Id}_V]_{\mathcal{B}}^{\mathcal{C}}([v]^{\mathcal{B}})$. Such a matrix of the identity is what one calls a *change-of-basis* or *change-of-coordinates* matrix — a name for this special case, not a new object; the two names are told apart in Remark 7.4.2.

Remark (Transition matrices are matrices of the identity):

The map $P = \phi^{\mathcal{B}} \circ \psi_{\mathcal{C}}$ of Proposition 7.3.6 is precisely $[\text{Id}_V]_{\mathcal{C}}^{\mathcal{B}}$, the change-of-basis matrix from $\mathcal{C}$ to $\mathcal{B}$; its inverse $[\text{Id}_V]_{\mathcal{B}}^{\mathcal{C}}$ is the one in the other direction. Thus the entire “passive” side of Section 7.3 is the construction of this section applied to $T = \text{Id}_V$. Composing identities gives $[\text{Id}_V]_{\mathcal{C}}^{\mathcal{B}} \circ [\text{Id}_V]_{\mathcal{B}}^{\mathcal{C}} = [\text{Id}_V]_{\mathcal{B}}^{\mathcal{B}} = \text{Id}_{K^{(X)}}$, so change-of-basis matrices are invertible, with $([\text{Id}_V]_{\mathcal{B}}^{\mathcal{C}})^{-1} = [\text{Id}_V]_{\mathcal{C}}^{\mathcal{B}}$.

Remark (Change of basis versus change of coordinates):

The pair $(\mathcal{B}, \mathcal{C})$ determines *two* automorphisms of $K^{(X)}$, mutually inverse, and they are best told apart by what they *do* rather than by what they are called. Write $P = \phi^{\mathcal{B}} \circ \psi_{\mathcal{C}} = [\text{Id}_V]_{\mathcal{C}}^{\mathcal{B}}$ for the transition map of Section 7.3.

- P is the *change-of-basis matrix*: its columns are the new basis vectors read in the old basis, $P(e_j) = [c_j]^{\mathcal{B}}$, and it transports the frame, $\psi_{\mathcal{C}} = \psi_{\mathcal{B}} \circ P$. It acts on the *basis*, carrying $\mathcal{B}$ to $\mathcal{C}$.
- $P^{-1} = \phi^{\mathcal{C}} \circ \psi_{\mathcal{B}} = [\text{Id}_V]_{\mathcal{B}}^{\mathcal{C}}$ is the *change-of-coordinates matrix*: it converts the coordinates of a *fixed* vector, $[v]^{\mathcal{C}} = [\text{Id}_V]_{\mathcal{B}}^{\mathcal{C}}[v]^{\mathcal{B}}$ (Proposition 7.3.7). It acts on *coordinates*, carrying $\mathcal{B}$ -coordinates to $\mathcal{C}$ -coordinates.

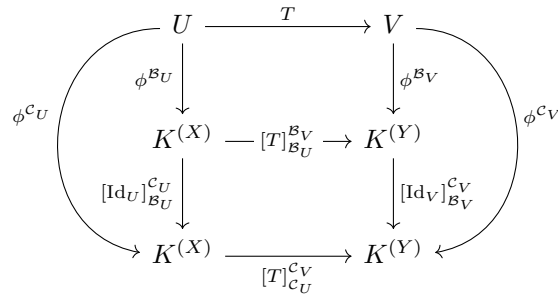
In this finer language the matrix $[\text{Id}_V]_{\mathcal{B}}^{\mathcal{C}}$ of Remark 7.4.2 is the change-of-*coordinates* matrix, notwithstanding the looser “change-of-basis” name in common use. But the names are the inessential part: they are not consistent across sources, and nothing rests on them. What is fixed is the *meaning* — a basis and the coordinates of a fixed vector transform *inversely* (Proposition 7.3.7), the basis covariantly and the components contravariantly — so the two matrices are mutual inverses. Confronted with any “change of basis from $\mathcal{B}$ to $\mathcal{C}$,” the question to ask is not what it is called but what it moves, the frame or the coordinates; those two run in opposite directions, and that opposition is the whole content.

Corollary 7.4.6 (Change-of-bases formula):

Let $T : U \rightarrow V$, let $\mathcal{B}_U, \mathcal{C}_U$ be ordered bases of U and $\mathcal{B}_V, \mathcal{C}_V$ ordered bases of V . Then

$$[T]_{\mathcal{B}_U}^{\mathcal{B}_V} = [\text{Id}_V]_{\mathcal{C}_V}^{\mathcal{B}_V} \circ [T]_{\mathcal{C}_U}^{\mathcal{C}_V} \circ [\text{Id}_U]_{\mathcal{B}_U}^{\mathcal{C}_U}.$$

The abstract map sits on top, and we descend through two coordinate layers:



At the top is the abstract map $T : U \rightarrow V$. The charts $\phi^{\mathcal{B}_U}, \phi^{\mathcal{B}_V}$ drop it onto the middle floor as its $\mathcal{B}$ -representation $[T]_{\mathcal{B}_U}^{\mathcal{B}_V}$; the transition matrices $[\text{Id}_U]_{\mathcal{B}_U}^{\mathcal{C}_U}, [\text{Id}_V]_{\mathcal{B}_V}^{\mathcal{C}_V}$ then carry it down one further step to the bottom floor as its $\mathcal{C}$ -representation $[T]_{\mathcal{C}_U}^{\mathcal{C}_V}$. The two long arrows $\phi^{\mathcal{C}_U}, \phi^{\mathcal{C}_V}$ are the direct charts onto the $\mathcal{C}$ -coordinates, and each side-triangle commutes because a chart followed by a transition is the other chart, e.g. $\phi^{\mathcal{C}_U} = [\text{Id}_U]_{\mathcal{B}_U}^{\mathcal{C}_U} \circ \phi^{\mathcal{B}_U}$. Reading off the lower square gives

$$[\text{Id}_V]_{\mathcal{B}_V}^{\mathcal{C}_V} \circ [T]_{\mathcal{B}_U}^{\mathcal{B}_V} = [T]_{\mathcal{C}_U}^{\mathcal{C}_V} \circ [\text{Id}_U]_{\mathcal{B}_U}^{\mathcal{C}_U},$$

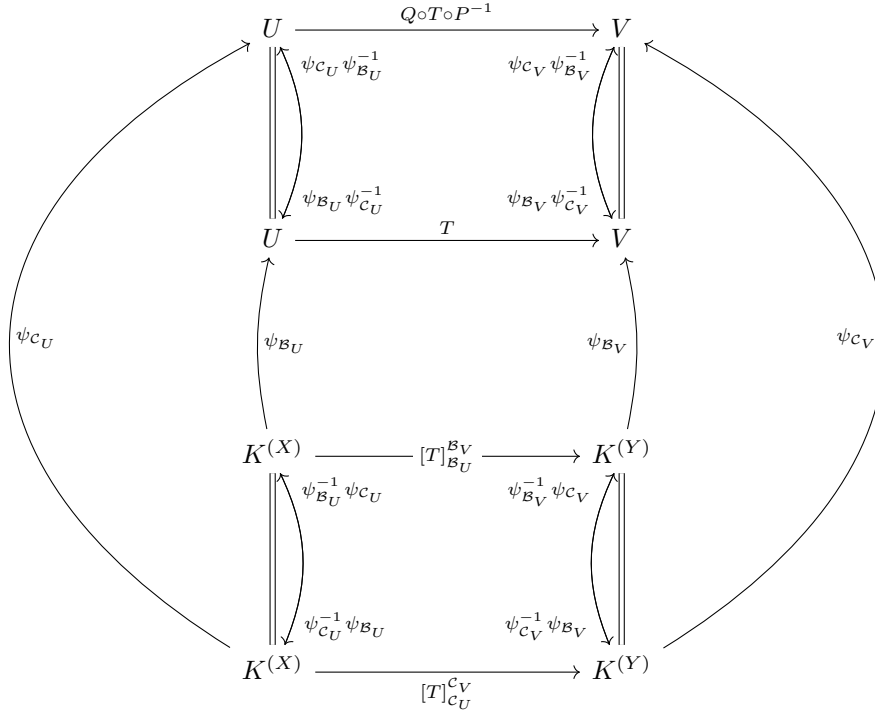
and since the transition matrices are invertible this rearranges to the displayed identity.

Proof:

Apply Theorem 7.4.5 to the factorisation $T = \text{Id}_V \circ T \circ \text{Id}_U$, routing through $\mathcal{C}_U$ on U and $\mathcal{C}_V$ on V : $[T]_{\mathcal{B}_U}^{\mathcal{B}_V} = [\text{Id}_V \circ T \circ \text{Id}_U]_{\mathcal{B}_U}^{\mathcal{B}_V} = [\text{Id}_V]_{\mathcal{B}_V}^{\mathcal{B}_V} \circ [T]_{\mathcal{C}_U}^{\mathcal{C}_V} \circ [\text{Id}_U]_{\mathcal{B}_U}^{\mathcal{C}_U}$. QED

Remark (Active and passive change of basis):

A change of basis wears two faces, displayed here together for source and target. The four corners on each side are the *same* space seen twice (the vertical $=$): once carrying the $\mathcal{B}$ -frame, once the $\mathcal{C}$ -frame. The small pair of arrows flanking each $=$ is the basis change together with its inverse.



Read the source column U ; the frames $\psi_{\mathcal{B}_U}, \psi_{\mathcal{C}_U} : K^{(X)} \rightarrow U$ install the two bases. Write $P := \psi_{\mathcal{C}_U} \psi_{\mathcal{B}_U}^{-1} \in \text{GL}(U)$ and $Q := \psi_{\mathcal{C}_V} \psi_{\mathcal{B}_V}^{-1} \in \text{GL}(V)$ for the upward arrows between the abstract copies.

- **Active** (*moves the frame*). The automorphism P carries each old basis vector to the new one, $b_j \mapsto c_j$ — it physically transforms the frame *inside* U (and Q does the same in V). This is the active reading of Section 7.3.
- **Passive** (*reexpresses coordinates*). The transition matrix $\psi_{\mathcal{B}_U}^{-1} \psi_{\mathcal{C}_U} = [\text{Id}_U]_{\mathcal{C}_U}^{\mathcal{B}_U} \in \text{GL}(K^{(X)})$ leaves U untouched and rewrites the coordinates of a fixed vector from $\mathcal{C}_U$ to $\mathcal{B}_U$ — the passive reading.

Active is passive, in coordinates. The two are one change seen from two sides: read the active automorphism in its *own* coordinates and the passive transition matrix reappears,

$$[P]_{\mathcal{C}_U}^{\mathcal{C}_U} = [\text{Id}_U]_{\mathcal{C}_U}^{\mathcal{B}_U}, \quad [Q]_{\mathcal{C}_V}^{\mathcal{C}_V} = [\text{Id}_V]_{\mathcal{C}_V}^{\mathcal{B}_V}.$$

So conjugating T by the active automorphisms (inside the spaces) is the very same operation as multiplying $[T]$ by the change-of-basis matrices (on coordinates).

The top arrow. The top edge is the actively transformed map $Q \circ T \circ P^{-1} : U \rightarrow V$. Read in the *new* $\mathcal{C}$ -bases it returns the *old* matrix,

$$[Q \circ T \circ P^{-1}]_{\mathcal{C}_U}^{\mathcal{C}_V} = [T]_{\mathcal{B}_U}^{\mathcal{B}_V},$$

because the active conjugation by P, Q exactly cancels the passive change of basis (Corollary 7.4.6). The asymmetry — P^{-1} on the source against Q on the target — is the contravariance of Remark 7.4.4: the source basis change enters inverted. A map of this form $Q \circ T \circ P^{-1}$ with $P \in \text{GL}(U)$, $Q \in \text{GL}(V)$ is exactly an *equivalence* of T (Section 8.1). The four horizontal arrows are thus $Q \circ T \circ P^{-1}$ and T above, and the two coordinate forms $[T]_{\mathcal{B}_U}^{\mathcal{B}_V}$, $[T]_{\mathcal{C}_U}^{\mathcal{C}_V}$ below — the diagram is Corollary 7.4.6 with the frames drawn in.

7.4.3 Invariance of Rank and Nullity

Theorem 7.4.7 (Rank and nullity are basis-independent):

For any $T : U \rightarrow V$ and any ordered bases $\mathcal{B}_U, \mathcal{B}_V$,

$$\text{rank}(T) = \text{rank}\left([T]_{\mathcal{B}_U}^{\mathcal{B}_V}\right), \quad \text{nullity}(T) = \text{nullity}\left([T]_{\mathcal{B}_U}^{\mathcal{B}_V}\right).$$

In particular all matrix representations of T have the same rank and nullity.

Proof:

The charts and frames are isomorphisms, and $[T]_{\mathcal{B}_U}^{\mathcal{B}_V} = \phi^{\mathcal{B}_V} \circ T \circ \psi_{\mathcal{B}_U}$. An isomorphism carries a subspace to a subspace of the same dimension, so $\text{Rng}\left([T]_{\mathcal{B}_U}^{\mathcal{B}_V}\right) = \phi^{\mathcal{B}_V}[\text{Rng}(T)]$ has the same dimension as $\text{Rng}(T)$, giving equality of ranks; and $\ker\left([T]_{\mathcal{B}_U}^{\mathcal{B}_V}\right) = \phi^{\mathcal{B}_V}[\ker(T)]$ has the same dimension as $\ker(T)$, giving equality of nullities. QED

Combined with the identification of a matrix with the map it is (Section 7.2), this pins the rank of an abstract map to the rank of *any* of its matrices.

Corollary 7.4.8 (Rank of an abstract map equals rank of its matrix):

For $T : U \rightarrow V$ with ordered bases $\mathcal{B}_U, \mathcal{B}_V$, the matrix $[T]_{\mathcal{B}_U}^{\mathcal{B}_V} \in \text{Mat}_{(Y),X}(K)$ satisfies

$$\text{rank}\left([T]_{\mathcal{B}_U}^{\mathcal{B}_V}\right) = \text{rank}(T), \quad \text{nullity}\left([T]_{\mathcal{B}_U}^{\mathcal{B}_V}\right) = \text{nullity}(T).$$

Proof:

Read as the linear map $K^{(X)} \rightarrow K^{(Y)}$ it is identified with (Definition 7.2.2), the matrix $[T]_{\mathcal{B}_U}^{\mathcal{B}_V}$ has rank and nullity equal to $\text{rank}(T)$ and $\text{nullity}(T)$ by Theorem 7.4.7. QED

7.4.4 Variance, and a Forward Reference

Remark (Covariance and contravariance (informal)):

Proposition 7.3.7 already records the basic asymmetry of a change of basis: under the transition matrix $P = [\text{Id}_V]_{\mathcal{C}}^{\mathcal{B}}$, the *basis vectors* transform one way while the *coordinate components* of a fixed vector transform by P^{-1} — the two move in opposite directions, and cancel on $v = \sum_{j \in X} [v]_j^{\mathcal{B}} b_j$. One calls quantities that

transform like the basis **covariant** and those that transform oppositely (like components) **contravariant**. We use these words only as informal labels for this opposite-direction behaviour; a precise definition — in terms of the dual space and covectors — must wait until that machinery is available, and is taken up in a later chapter. No dual bases or index-variance calculus are used here.

Remark (The frame as a row of vectors (forward reference)):

The frame $\psi_{\mathcal{B}}$ can be recorded as the *row of basis vectors* $(b_j)_{j \in X}$ — a “matrix with vector entries,” an object of a type we have not yet defined, whose product with a coordinate column $[v]^{\mathcal{B}}$ reconstructs $v = \sum_{j \in X} [v]_j^{\mathcal{B}} b_j$. Making this precise (and identifying the chart $\phi^{\mathcal{B}}$ with a dual object) requires tools developed later; we record the viewpoint now and return to it once those tools are in hand.

7.4.5 Examples

Example 7.4.9 (The matrix of differentiation):

Let $U = \{p \in K[x] : \deg p \leq 2\}$ with ordered basis $\mathcal{B}_U = (1, x, x^2)$ (so $b_0 = 1$, $b_1 = x$, $b_2 = x^2$), and $V = \{p \in K[x] : \deg p \leq 1\}$ with $\mathcal{B}_V = (1, x)$. Let $D : U \rightarrow V$ be the formal derivative, $D(p) = p'$. Applying D to the source basis,

$$D(1) = 0, \quad D(x) = 1, \quad D(x^2) = 2x,$$

so by Proposition 7.4.2,

$$[D]_{\mathcal{B}_U}^{\mathcal{B}_V}(e_0) = [0]^{\mathcal{B}_V} = (0, 0), \quad [D]_{\mathcal{B}_U}^{\mathcal{B}_V}(e_1) = [1]^{\mathcal{B}_V} = (1, 0), \quad [D]_{\mathcal{B}_U}^{\mathcal{B}_V}(e_2) = [2x]^{\mathcal{B}_V} = (0, 2).$$

Thus $[D]_{\mathcal{B}_U}^{\mathcal{B}_V} : K^3 \rightarrow K^2$ is the map $(a_0, a_1, a_2) \mapsto (a_1, 2a_2)$, i.e. the grid $\begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 2 \end{pmatrix}$ (Section 7.2). The coordinate formula checks out: for $p = a_0 + a_1x + a_2x^2$, $D(p) = a_1 + 2a_2x$ has coordinates $(a_1, 2a_2)$, matching $[D]_{\mathcal{B}_U}^{\mathcal{B}_V}(a_0, a_1, a_2)$.

Example 7.4.10 (Change of bases):

Let $U = V = \mathbb{R}^2$, $T(x, y) = (2x + y, x)$, with $\mathcal{B} = (e_0, e_1)$ standard and $\mathcal{C} = (e_0 + e_1, e_1)$. In the standard basis, $T(e_0) = (2, 1)$ and $T(e_1) = (1, 0)$, so (as a grid) $[T]_{\mathcal{B}}^{\mathcal{B}} = \begin{pmatrix} 2 & 1 \\ 1 & 0 \end{pmatrix}$. The change-of-basis matrices are $[\text{Id}]_{\mathcal{B}}^{\mathcal{C}} = \begin{pmatrix} 1 & 0 \\ -1 & 1 \end{pmatrix}$ (columns $[e_0]^{\mathcal{C}}, [e_1]^{\mathcal{C}}$) and its inverse $[\text{Id}]_{\mathcal{C}}^{\mathcal{B}} = \begin{pmatrix} 1 & 0 \\ 1 & 1 \end{pmatrix}$. By Corollary 7.4.6,

$$[T]_{\mathcal{C}}^{\mathcal{C}} = [\text{Id}]_{\mathcal{B}}^{\mathcal{C}} [T]_{\mathcal{B}}^{\mathcal{B}} [\text{Id}]_{\mathcal{C}}^{\mathcal{B}} = \begin{pmatrix} 1 & 0 \\ -1 & 1 \end{pmatrix} \begin{pmatrix} 2 & 1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 1 & 1 \end{pmatrix} = \begin{pmatrix} 3 & 1 \\ -2 & -1 \end{pmatrix}.$$

Both representations have rank 2 (the columns are independent), as Theorem 7.4.7 demands.

7.5 Invertibility

Intuition:

With maps and matrices now identified (Section 7.2), “inverting a matrix” means exactly “inverting the map it is,” and the arithmetic of Section 7.1 guarantees the inverse is again a matrix. One question then organises this section: how does inversion interact with the matrix of an *abstract* map? The answer is as clean as possible — inverting the map swaps the roles of the two bases and inverts the matrix — and it specialises to the fact that every change-of-basis matrix is invertible, its inverse being the change run backwards. (Which matrices represent one and the same map, the question this naturally raises next, is taken up in the following chapter, on equivalence and similarity.)

7.5.1 Invertible Matrices

Definition 7.5.1 (Invertible matrix):

A matrix $A \in \text{Mat}_{(Y),X}(K)$ is **invertible** if it is invertible as a linear map $A : K^{(X)} \rightarrow K^{(Y)}$ — that is, bijective. Its inverse map is again linear, hence a matrix $A^{-1} \in \text{Mat}_{(X),Y}(K)$ (the grid of A^{-1} under the identification of Definition 7.2.1), characterised by

$$A^{-1}A = I_X, \quad AA^{-1} = I_Y.$$

Remark (Two-sided inverse, and when one can exist):

Because the identification turns composition into the matrix product and the identity map into the identity grid (Corollary 7.1.6), A is invertible exactly when it has a two-sided multiplicative inverse B with $BA = I_X$ and $AB = I_Y$, and then $B = A^{-1}$ is unique (a two-sided inverse in a monoid is unique). An invertible A is an isomorphism $K^{(X)} \xrightarrow{\sim} K^{(Y)}$, so $|X| = |Y|$ (isomorphic coordinate spaces have canonical bases of equal cardinality); in the finite case $X = Y = n$ and A is a *square* $n \times n$ matrix. The invertible elements of the algebra $\text{End}_K(K^{(X)}) = \text{Mat}_{(X),X}(K)$ form, under the matrix product, the group $\text{GL}(K^{(X)})$ — written $\text{GL}(K^n)$ when $X = n$.

7.5.2 The Inverse of the Matrix of a Map

Theorem 7.5.2 (Inversion swaps the two bases):

Let $T : U \rightarrow V$ be an isomorphism, with ordered bases $\mathcal{B}_U$ of U and $\mathcal{B}_V$ of V . Then $[T]_{\mathcal{B}_U}^{\mathcal{B}_V}$ is invertible and

$$([T]_{\mathcal{B}_U}^{\mathcal{B}_V})^{-1} = [T^{-1}]_{\mathcal{B}_V}^{\mathcal{B}_U}.$$

Proof:

Since T is an isomorphism, $T^{-1} : V \rightarrow U$ is linear, so $[T^{-1}]_{\mathcal{B}_V}^{\mathcal{B}_U}$ is defined. By the composition theorem (Theorem 7.4.5) and the fact that the matrix of an identity in one basis is the identity grid (Remark 7.4.2, with $\Phi(\text{Id}) = I$ from Corollary 7.1.6),

$$[T^{-1}]_{\mathcal{B}_V}^{\mathcal{B}_U} \circ [T]_{\mathcal{B}_U}^{\mathcal{B}_V} = [T^{-1} \circ T]_{\mathcal{B}_U}^{\mathcal{B}_U} = [\text{Id}_U]_{\mathcal{B}_U}^{\mathcal{B}_U} = I_X,$$

and symmetrically $[T]_{\mathcal{B}_U}^{\mathcal{B}_V} \circ [T^{-1}]_{\mathcal{B}_V}^{\mathcal{B}_U} = [T \circ T^{-1}]_{\mathcal{B}_V}^{\mathcal{B}_V} = [\text{Id}_V]_{\mathcal{B}_V}^{\mathcal{B}_V} = I_Y$. By Remark 7.5.1 the two-sided inverse is unique, which is the claim. QED

Corollary 7.5.3 (Invertibility is detected by any matrix):

For $T : U \rightarrow V$ with ordered bases $\mathcal{B}_U, \mathcal{B}_V$, the map T is an isomorphism iff $[T]_{\mathcal{B}_U}^{\mathcal{B}_V}$ is invertible. In particular every change-of-basis matrix is invertible, with

$$([\text{Id}_V]_{\mathcal{B}}^{\mathcal{C}})^{-1} = [\text{Id}_V]_{\mathcal{C}}^{\mathcal{B}}.$$

Proof:

If T is an isomorphism, Theorem 7.5.2 exhibits the inverse. Conversely, $[T]_{\mathcal{B}_U}^{\mathcal{B}_V} = \phi^{\mathcal{B}_V} \circ T \circ \psi_{\mathcal{B}_U}$ with $\phi^{\mathcal{B}_V}, \psi_{\mathcal{B}_U}$ isomorphisms (Definition 7.4.1); if this composite is bijective then $T = \psi_{\mathcal{B}_V} \circ [T]_{\mathcal{B}_U}^{\mathcal{B}_V} \circ \phi^{\mathcal{B}_U}$ is a composite of bijections, hence bijective. The change-of-basis matrix is the case $T = \text{Id}_V$, an isomorphism, and its inverse is read off from Theorem 7.5.2. QED

7.5.3 Worked Example**Example 7.5.4 (The inverse of a matrix is the matrix of the inverse):**

Let $T : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ be the shear $T(x, y) = (x + y, y)$, an isomorphism with inverse $T^{-1}(x, y) = (x - y, y)$. In the standard basis $\mathcal{B}$,

$$[T]_{\mathcal{B}}^{\mathcal{B}} = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}, \quad [T^{-1}]_{\mathcal{B}}^{\mathcal{B}} = \begin{pmatrix} 1 & -1 \\ 0 & 1 \end{pmatrix},$$

and indeed $\begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & -1 \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$, confirming $([T]_{\mathcal{B}}^{\mathcal{B}})^{-1} = [T^{-1}]_{\mathcal{B}}^{\mathcal{B}}$ as Theorem 7.5.2 predicts.

CHAPTER 8

Subspace Representations

8.1 Equivalence and Similarity of Linear Maps

Intuition:

The matrix of a fixed linear map $T : U \rightarrow V$ is not unique: it depends on the ordered bases chosen for source and target. So which matrices represent one and the same map? The change-of-bases formula (Corollary 7.4.6) already answers in form — two matrices of T differ by invertible factors fore and aft — and isolating that relation, abstracted from any particular T , produces the notion of **equivalence** of linear maps (hence of matrices). Tightening it to a *single* change of basis on a self-map gives **similarity** of endomorphisms. The payoff is conceptual: the genuinely basis-free content of a map is precisely its equivalence class — for an endomorphism, its similarity class — so a quantity read off a matrix is an intrinsic invariant of the underlying map exactly when it depends only on that class. This chapter develops the two relations and their one-sided refinements; the *classification* that completes the story — normal forms, and the Gaussian elimination that computes them — follows in the chapters to come.

Equivalence and similarity are, at bottom, relations between *linear maps*: they say that two maps agree after a change of coordinates — on source and target independently for equivalence, by a single change for similarity. We define them at the level of maps, where they are most transparent. There is no separate notion for matrices: a matrix *is* a linear map between coordinate spaces (Section 7.2), and the invertible matrices are exactly the automorphisms $\mathrm{GL}(K^{(X)})$ (Remark 7.5.1), so each definition below applies to matrices with nothing added — equivalence of matrices simply *is* equivalence of the maps they are. We record the resulting matrix form inline as each definition is given.

Definition 8.1.1 (Equivalence of linear maps):

Linear maps $S, T : U \rightarrow V$ are **equivalent**, written $S \approx T$, if there are isomorphisms $P \in \mathrm{GL}(U)$ and $Q \in \mathrm{GL}(V)$ with

$$T = Q \circ S \circ P^{-1},$$

that is, the square

$$\begin{array}{ccc} U & \xrightarrow{S} & V \\ P \downarrow & & \downarrow Q \\ U & \xrightarrow{T} & V \end{array}$$

commutes ($Q \circ S = T \circ P$). One changes coordinates by P in the source and, independently, by Q in the target.

In coordinates. Taking $U = K^{(X)}$ and $V = K^{(Y)}$ — so that S, T are matrices $A, B \in \mathrm{Mat}_{(Y),X}(K)$ — the very same definition reads: $A \approx B$ iff there are invertible matrices $P \in \mathrm{Mat}_{(X),X}(K)$ and $Q \in \mathrm{Mat}_{(Y),Y}(K)$ with $A = QBP^{-1}$, where $\circ$ is the matrix product. This *is* the equivalence of matrices; it is not a fresh definition but the one above with coordinate spaces in place of U, V .

Before tying the two ends together by a single change (similarity, below), it pays to isolate each end on its own. Full equivalence reparametrises source and target at once; relaxing it to move only one at a time splits $\approx$ into two *one-sided* relations, of which it is the composite.

Definition 8.1.2 (Domain and codomain equivalence):

Let $S, T : U \rightarrow V$ be linear maps. They are:

- **domain-equivalent** (or **column-equivalent**), written $S \overset{\mathrm{col}}{\approx} T$, if $T = S \circ P^{-1}$ for some $P \in \mathrm{GL}(U)$ — only the source is reparametrised (the case $Q = \mathrm{Id}_V$ of Definition 8.1.1);
- **codomain-equivalent** (or **row-equivalent**), written $S \overset{\mathrm{row}}{\approx} T$, if $T = Q \circ S$ for some $Q \in \mathrm{GL}(V)$ — only the target is reparametrised (the case $P = \mathrm{Id}_U$).

Each is again an equivalence relation, by Proposition 8.1.5.

In coordinates. For matrices $A, B \in \text{Mat}_{(Y),X}(K)$ the same two definitions read: $A \overset{\text{col}}{\approx} B$ iff $A = BP^{-1}$ for an invertible $P \in \text{Mat}_{(X),X}(K)$ — right multiplication, remixing the *columns* (indexed by the source X) — and $A \overset{\text{row}}{\approx} B$ iff $A = QB$ for an invertible $Q \in \text{Mat}_{(Y),Y}(K)$ — left multiplication, remixing the *rows* (indexed by the target Y). Again these are the map relations applied to coordinate spaces, not new notions.

Proposition 8.1.3 (Equivalence is a domain step followed by a codomain step):

For linear maps $S, T : U \rightarrow V$ the following are equivalent.

1. $S \approx T$.
2. There is a map $R : U \rightarrow V$ with $S \overset{\text{col}}{\approx} R$ and $R \overset{\text{row}}{\approx} T$ — reparametrise the source, then the target.
3. There is a map $R' : U \rightarrow V$ with $S \overset{\text{row}}{\approx} R'$ and $R' \overset{\text{col}}{\approx} T$ — reparametrise the target, then the source.

Hence, as composites of relations,

$$\approx = \overset{\text{row}}{\approx} \circ \overset{\text{col}}{\approx} = \overset{\text{col}}{\approx} \circ \overset{\text{row}}{\approx} :$$

the two one-sided changes commute, and either order realises full equivalence. The same holds verbatim for matrices, with P, Q the invertible factors of $A = QBP^{-1}$.

Proof:

$S \approx T$ means $T = Q \circ S \circ P^{-1}$ with $P \in \text{GL}(U)$, $Q \in \text{GL}(V)$ (Definition 8.1.1). Associating the composite one way, set $R := S \circ P^{-1}$: then $S \overset{\text{col}}{\approx} R$ via P , and $T = Q \circ R$ gives $R \overset{\text{row}}{\approx} T$ via Q — this is (2). Associating the other way, set $R' := Q \circ S$: then $S \overset{\text{row}}{\approx} R'$ via Q , and $T = R' \circ P^{-1}$ gives $R' \overset{\text{col}}{\approx} T$ via P — this is (3). Conversely, substituting the intermediate map back into either chain reconstitutes $T = Q \circ S \circ P^{-1}$, so each of (2), (3) implies $S \approx T$. The two factorisations agree because composition is associative, $Q \circ (S \circ P^{-1}) = (Q \circ S) \circ P^{-1}$, which is exactly the commutativity of the two one-sided relations. QED

Remark (Reading the decomposition off the four-level diagram):

The two one-sided equivalences are the two independent *columns* of the four-level change-of-basis diagram (Remark 7.4.2): the source space U with its frames occupies the left column, the target V the right. Passing between the coordinate rows $[T]_{\mathcal{B}_U}^{\mathcal{B}_V}$ and $[T]_{\mathcal{C}_U}^{\mathcal{C}_V}$ is governed, on the right, by the source transition $[\text{Id}_U]_{\mathcal{C}_U}^{\mathcal{B}_U}$ and, on the left, by the target transition $[\text{Id}_V]_{\mathcal{B}_V}^{\mathcal{C}_V}$. Sliding down the *source* column alone — holding $\mathcal{B}_V = \mathcal{C}_V$ — turns $[T]_{\mathcal{B}_U}^{\mathcal{B}_V}$ into

$$[T]_{\mathcal{C}_U}^{\mathcal{B}_V} = [T]_{\mathcal{B}_U}^{\mathcal{B}_V} \circ [\text{Id}_U]_{\mathcal{C}_U}^{\mathcal{B}_U} ,$$

a domain (column) equivalence; sliding down the *target* column alone gives $[T]_{\mathcal{B}_U}^{\mathcal{C}_V} = [\text{Id}_V]_{\mathcal{B}_V}^{\mathcal{C}_V} \circ [T]_{\mathcal{B}_U}^{\mathcal{B}_V}$, a codomain (row) equivalence; performing both — in either order — is the full change-of-bases formula (Corollary 7.4.6), hence the full equivalence $[T]_{\mathcal{B}_U}^{\mathcal{B}_V} \approx [T]_{\mathcal{C}_U}^{\mathcal{C}_V}$ (of the matrices, as the maps they are). The commutativity of Proposition 8.1.3 is precisely the independence of the two columns: the source frame may be changed before or after the target frame, with the same result.

Definition 8.1.4 (Similarity of endomorphisms):

Endomorphisms $S, T \in \text{End}_K(V)$ are **similar** (or **conjugate**), written $S \sim T$, if there is an automorphism $Q \in \text{GL}(V)$ with

$$T = Q^{-1} \circ S \circ Q.$$

Similarity is the special case of equivalence in which source and target are changed by the *same* isomorphism ($U = V$, $P = Q$); it is precisely the conjugation action of $\text{GL}(V)$ on $\text{End}_K(V)$.

In coordinates. For square matrices $A, B \in \text{Mat}_{(X),X}(K)$ the same definition reads: $A \sim B$ iff $A = Q^{-1}BQ$ for an invertible $Q \in \text{Mat}_{(X),X}(K)$ — once more the map relation in coordinate spaces, the conjugation orbit of A under $\text{GL}(K^{(X)})$.

Proposition 8.1.5 (All four relations are equivalence relations):

Equivalence, column (domain) equivalence, and row (codomain) equivalence are equivalence relations on $\text{Hom}_K(U, V)$, and similarity is an equivalence relation on $\text{End}_K(V)$.

Proof:

We prove it for $\approx$ and obtain the other three as special cases. Reflexivity: $T = \text{Id}_V \circ T \circ \text{Id}_U^{-1}$, witnessed by the identity pair $(P, Q) = (\text{Id}_U, \text{Id}_V)$. Symmetry: from $T = Q \circ S \circ P^{-1}$ we get $S = Q^{-1} \circ T \circ P = Q^{-1} \circ T \circ (P^{-1})^{-1}$, witnessed by the inverse pair (P^{-1}, Q^{-1}) . Transitivity: if $T = Q \circ S \circ P^{-1}$ and $S = Q' \circ R \circ (P')^{-1}$, then $T = (Q \circ Q') \circ R \circ (P \circ P')^{-1}$, witnessed by the composite pair $(P \circ P', Q \circ Q')$, using $(P \circ P')^{-1} = (P')^{-1} \circ P^{-1}$ and that composites of isomorphisms are isomorphisms.

The only facts used are that the automorphism pairs (P, Q) contain the identity pair and are closed under componentwise inverse and composition. Each of the other three relations is $\approx$ with (P, Q) confined to a subset of $\text{GL}(U) \times \text{GL}(V)$ that is again closed under these three operations:

- $\overset{\text{row}}{\approx}$ (row): $P = \text{Id}_U$, i.e. the subset $\{\text{Id}_U\} \times \text{GL}(V)$;
- $\overset{\text{col}}{\approx}$ (column): $Q = \text{Id}_V$, i.e. the subset $\text{GL}(U) \times \{\text{Id}_V\}$;
- $\sim$ (when $U = V$): $P = Q$, i.e. the diagonal $\{(R, R) \mid R \in \text{GL}(V)\}$, giving the conjugation $T = Q \circ S \circ Q^{-1}$.

Each subset contains $(\text{Id}_U, \text{Id}_V)$ and is closed under inverses and composition, so the three displayed witnesses stay inside it. The same computation therefore proves each of the four relations reflexive, symmetric, and transitive. QED

Since the matrix relations are nothing but these map relations restricted to coordinate spaces, the content still to be settled is how they connect a *fixed abstract* map to its coordinate matrices in *arbitrary* bases. The next two theorems are that bridge: a map relation holds iff the corresponding relation holds between coordinate matrices, in *any* bases; and conversely two matrices are related iff they present a single map.

Theorem 8.1.6 (Equivalence in coordinates):

Let $S, T : U \rightarrow V$ be linear maps.

1. For *any* ordered bases $\mathcal{B}_U$ of U and $\mathcal{B}_V$ of V ,

$$S \approx T \iff [S]_{\mathcal{B}_U}^{\mathcal{B}_V} \approx [T]_{\mathcal{B}_U}^{\mathcal{B}_V}.$$

2. Two matrices $A, B \in \text{Mat}_{(Y),X}(K)$ are equivalent iff they are the matrices of one common map: $A = [R]_{\mathcal{B}_U}^{\mathcal{B}_V}$ and $B = [R]_{\mathcal{C}_U}^{\mathcal{C}_V}$ for some $R : U \rightarrow V$ and ordered bases.

Proof:

(1). If $T = Q \circ S \circ P^{-1}$ with $P \in \text{GL}(U)$, $Q \in \text{GL}(V)$, then the composition theorem (Theorem 7.4.5) and the inversion rule (Theorem 7.5.2) give

$$[T]_{\mathcal{B}_U}^{\mathcal{B}_V} = [Q]_{\mathcal{B}_U}^{\mathcal{B}_V} \circ [S]_{\mathcal{B}_U}^{\mathcal{B}_V} \circ ([P]_{\mathcal{B}_U}^{\mathcal{B}_U})^{-1},$$

an equivalence of the two matrices via the invertible factors $[Q]_{\mathcal{B}_U}^{\mathcal{B}_V}$ and $[P]_{\mathcal{B}_U}^{\mathcal{B}_U}$ (Corollary 7.5.3). Conversely, if $[S]_{\mathcal{B}_U}^{\mathcal{B}_V}$ and $[T]_{\mathcal{B}_U}^{\mathcal{B}_V}$ are equivalent matrices, their invertible factors are — again by Corollary 7.5.3 — the

matrices $[Q]_{\mathcal{B}_V}^{\mathcal{B}_V}$ and $[P]_{\mathcal{B}_U}^{\mathcal{B}_U}$ of automorphisms $Q \in \text{GL}(V)$ and $P \in \text{GL}(U)$; reading the display backwards gives $T = Q \circ S \circ P^{-1}$, i.e. $S \approx T$.

(2). ($\Leftarrow$) The change-of-bases formula (Corollary 7.4.6) gives $A = [\text{Id}_V]_{\mathcal{C}_V}^{\mathcal{B}_V} B [\text{Id}_U]_{\mathcal{B}_U}^{\mathcal{C}_U}$, a product QBP^{-1} of B with invertible transition matrices, so $A \approx B$. ($\Rightarrow$) If $A = QBP^{-1}$, take $R = B : K^{(X)} \rightarrow K^{(Y)}$ with $\mathcal{C}_U, \mathcal{C}_V$ the canonical bases (so $[R]_{\mathcal{C}_U}^{\mathcal{C}_V} = B$); the columns of P^{-1} and Q^{-1} are ordered bases $\mathcal{B}_U, \mathcal{B}_V$ with $[\text{Id}]_{\mathcal{B}_U}^{\mathcal{C}_U} = P^{-1}$ and $[\text{Id}]_{\mathcal{C}_V}^{\mathcal{B}_V} = Q$, whence $[R]_{\mathcal{B}_U}^{\mathcal{B}_V} = QBP^{-1} = A$. QED

Theorem 8.1.7 (Similarity in coordinates):

Let $S, T \in \text{End}_K(V)$ be endomorphisms.

1. For *any* ordered basis $\mathcal{B}$ of V ,

$$S \sim T \iff [S]_{\mathcal{B}}^{\mathcal{B}} \sim [T]_{\mathcal{B}}^{\mathcal{B}}.$$

2. Square matrices $A, B \in \text{Mat}_{(X),X}(K)$ are similar iff they are the matrices of one common endomorphism: $A = [R]_{\mathcal{B}}^{\mathcal{B}}$ and $B = [R]_{\mathcal{C}}^{\mathcal{C}}$ for some $R \in \text{End}_K(V)$ and ordered bases $\mathcal{B}, \mathcal{C}$ of V .

Proof:

Both parts repeat the previous proof with one basis on each side and $P = Q$. For (1), if $T = Q^{-1} \circ S \circ Q$ then $[T]_{\mathcal{B}}^{\mathcal{B}} = ([Q]_{\mathcal{B}}^{\mathcal{B}})^{-1} [S]_{\mathcal{B}}^{\mathcal{B}} [Q]_{\mathcal{B}}^{\mathcal{B}}$ with $[Q]_{\mathcal{B}}^{\mathcal{B}}$ invertible, and conversely. For (2), the ($\Leftarrow$) direction is Corollary 7.4.6 with a single basis per side, yielding $A = [\text{Id}_V]_{\mathcal{C}}^{\mathcal{B}} B [\text{Id}_V]_{\mathcal{B}}^{\mathcal{C}} = Q^{-1} B Q$; the ($\Rightarrow$) direction takes $R = B$ on $K^{(X)}$ with $\mathcal{C}$ canonical and $\mathcal{B}$ the columns of Q . QED

The sharp statement is at the level of the *subspaces* themselves, and it splits cleanly along the two one-sided relations: a row (codomain) change leaves the kernel untouched, a column (domain) change leaves the range untouched. The numerical invariants — rank and nullity — are then mere shadows.

Proposition 8.1.8 (Kernel is codomain-invariant; range is domain-invariant):

Let $A, B \in \text{Mat}_{(Y),X}(K)$.

1. If $A \overset{\text{row}}{\approx} B$ then $\ker(A) = \ker(B)$: the kernel depends only on the codomain (row) class.
2. If $A \overset{\text{col}}{\approx} B$ then $\text{Rng}(A) = \text{Rng}(B)$: the range depends only on the domain (column) class.

Proof:

(1). Here $A = QB$ with Q invertible, in particular injective; so for every $x \in K^{(X)}$, $A(x) = Q(B(x)) = 0 \iff B(x) = 0$, that is $\ker(A) = \ker(B)$.

(2). Here $A = BP^{-1}$ with P^{-1} invertible, in particular surjective; so

$$\text{Rng}(A) = A[K^{(X)}] = B[P^{-1}[K^{(X)}]] = B[K^{(X)}] = \text{Rng}(B). \quad \text{QED}$$

Corollary 8.1.9 (Rank and nullity are equivalence invariants):

If $A \approx B$ then $\text{rank}(A) = \text{rank}(B)$ and $\text{nullity}(A) = \text{nullity}(B)$.

Proof:

By Proposition 8.1.3 factor the equivalence as a column step followed by a row step: there is C with $B \overset{\text{col}}{\approx} C$ and $C \overset{\text{row}}{\approx} A$, namely $C = BP^{-1}$ and $A = QC$.

- *Rank.* The column step fixes the range exactly, $\text{Rng}(C) = \text{Rng}(B)$ (Proposition 8.1.8(2)); the row step $A = QC$ moves it by the isomorphism Q , $\text{Rng}(A) = Q[\text{Rng}(C)]$, preserving its dimension. Hence $\text{rank}(A) = \text{rank}(C) = \text{rank}(B)$.

- *Nullity.* The row step fixes the kernel exactly, $\ker(A) = \ker(C)$ (Proposition 8.1.8(1)); the column step $C = BP^{-1}$ moves it by the isomorphism P , $\ker(C) = P[\ker(B)]$, preserving its dimension. Hence $\text{nullity}(A) = \text{nullity}(C) = \text{nullity}(B)$.

QED

Remark (The basis-free content: the equivalence and similarity classes):

Theorem 8.1.6 says the set of all matrices of a fixed map $T : U \rightarrow V$ is exactly one equivalence class; for an endomorphism, insisting on one basis for both source and target (Theorem 8.1.7) refines this to a single similarity class. So a quantity computed from a matrix is an intrinsic invariant of the underlying map precisely when it depends only on the class, not on the chosen representative. We have already isolated one invariant of the equivalence class — the rank (Corollary 8.1.9), which by the forthcoming normal-form theorem turns out to be a *complete* invariant of equivalence. The similarity classes are finer, and their classification needs invariants special to the endomorphism case — the **determinant**, the **trace**, and the **characteristic polynomial**. These are constructed in later chapters and there shown to depend only on the similarity class; we name them here only to signpost where the theory is heading.

8.1.1 Worked Example

Example 8.1.10 (Two matrices of one endomorphism are similar):

Return to Example 7.4.10: $T(x, y) = (2x + y, x)$ on $\mathbb{R}^2$, with $\mathcal{B}$ standard and $\mathcal{C} = (e_0 + e_1, e_1)$. There we found $[T]_{\mathcal{B}}^{\mathcal{B}} = \begin{pmatrix} 2 & 1 \\ 1 & 0 \end{pmatrix}$ and $[T]_{\mathcal{C}}^{\mathcal{C}} = \begin{pmatrix} 3 & 1 \\ -2 & -1 \end{pmatrix}$. With $Q = [\text{Id}]_{\mathcal{B}}^{\mathcal{C}} = \begin{pmatrix} 1 & 0 \\ -1 & 1 \end{pmatrix}$ and $Q^{-1} = \begin{pmatrix} 1 & 0 \\ 1 & 1 \end{pmatrix}$,

$$Q^{-1} [T]_{\mathcal{C}}^{\mathcal{C}} Q = \begin{pmatrix} 1 & 0 \\ 1 & 1 \end{pmatrix} \begin{pmatrix} 3 & 1 \\ -2 & -1 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ -1 & 1 \end{pmatrix} = \begin{pmatrix} 2 & 1 \\ 1 & 0 \end{pmatrix} = [T]_{\mathcal{B}}^{\mathcal{B}},$$

so $[T]_{\mathcal{B}}^{\mathcal{B}} \sim [T]_{\mathcal{C}}^{\mathcal{C}}$ — exactly part (2) of Theorem 8.1.7, the two matrices of the one endomorphism T . Both have rank 2 (Corollary 8.1.9); looking ahead, they will also share the finer similarity invariants — determinant, trace, characteristic polynomial — once those are defined.

8.2 Reduction: Gaussian Elimination and Normal Forms

Remark (Roadmap — section under construction):

This section completes the basis-free analysis begun in Section 8.1. Two maps are equivalent exactly when they share a rank, and a suitable pair of bases reduces any $T : U \rightarrow V$ to a *partial identity* of size $\text{rank}(T)$ — the **rank normal form**, the complete invariant of equivalence (foretold in Remark 8.1). In finite dimensions the form is *computed* by **Gaussian elimination**, under the dictionary

row operations = codomain (left) change of basis, column operations = domain (right) change of basis,

and matrix inversion is the special case of solving $Av_j = e_j$ for the unknown columns v_j of A^{-1} , one block-column at a time. The injectivity, surjectivity, and bijectivity normal forms $\left(\begin{pmatrix} I \\ 0 \end{pmatrix}, \begin{pmatrix} I & 0 \end{pmatrix}, I\right)$ are the extreme cases. The full development, together with triangular/flag form, is deferred to the next implementation pass; the porting checklist is recorded in the source of this file.

CHAPTER 9

Quotient Spaces and Isomorphism Theorems

9.1 Quotient Space Construction

Intuition (Chapter Overview):

Subspaces are the substructures of a vector space; quotient spaces are the **factor structures**. Given a subspace $W \leq_K V$, the quotient V/W is the vector space obtained by “collapsing W to zero” — declaring two vectors equivalent if they differ by an element of W . The resulting equivalence classes (cosets $v + W$) inherit well-defined vector space operations, and the canonical projection $\pi : V \rightarrow V/W$ is a surjective linear map whose kernel is exactly W .

Quotient spaces serve two fundamental roles:

- **Factoring linear maps.** Every linear map $T : V \rightarrow X$ with $W \subseteq \ker(T)$ factors uniquely through V/W . This universal property makes quotients the natural tool for analyzing maps by “dividing out” their kernel.
- **Structural theorems.** The four isomorphism theorems — direct analogues of the classical theorems for groups and rings — describe how subspaces, sums, intersections, and nested quotients interact. These theorems reduce complex structural questions to manageable calculations.

Geometrically, if $V = \mathbb{R}^3$ and W is a plane through the origin, the quotient V/W is a one-dimensional space whose “points” are the planes parallel to W . Each coset $v + W$ is a translate of W , and the quotient records only how far a vector is displaced from W in the transverse direction.

Remark (Intuition):

Given a subspace $W \leq_K V$, we want to build a new vector space in which every vector in W becomes zero — we “collapse” W to a single point. The price is that we can no longer distinguish vectors that differ only by an element of W : if $u - v \in W$, then u and v become the same object in the quotient. The equivalence classes $v + W = \{v + w \mid w \in W\}$ are the **cosets** of W , and they are the “points” of the quotient space V/W .

Geometrically, each coset is a translate of W — a parallel copy of the subspace W displaced by the vector v . The quotient space is the set of all such parallel copies, equipped with natural addition and scalar multiplication defined through representatives. For instance:

- If $V = \mathbb{R}^2$ and W is a line through the origin, the cosets are lines parallel to W . The quotient $\mathbb{R}^2/W$ is one-dimensional: it parameterizes how far a point is from W in the transverse direction.
- If $V = \mathbb{R}^3$ and W is a plane through the origin, the cosets are planes parallel to W , and $\mathbb{R}^3/W \cong \mathbb{R}$.
- If $V = K[X]$ and $W = \text{span}_K(\{X^n \mid n \geq 2\})$, the quotient identifies two polynomials that agree up to degree 1 — it retains only the constant and linear terms.

The key technical point is **well-definedness**: the operations on cosets must not depend on which representative we choose. This is exactly where the hypothesis that W is a subspace (not just a subset) is used.

Definition 9.1.1 (Quotient Equivalence Relation):

Let $W \leq_K V$. Define an equivalence relation on V by:

$$u \sim_W v \iff u - v \in W$$

The equivalence class of v is the **coset** $v + W = \{v + w \mid w \in W\}$.

Proposition 9.1.2 (Equivalence Relation and Cosets):

The relation $\sim_W$ is an equivalence relation on V , and the equivalence class of v is the coset $v + W = \{v + w \mid w \in W\}$ for any $v \in V$.

Proof:

We verify the three properties:

- **Reflexivity:** $v - v = 0 \in W$, so $v \sim_W v$.
- **Symmetry:** If $u - v \in W$, then $v - u = -(u - v) \in W$ since W is a subspace.
- **Transitivity:** If $u - v \in W$ and $v - w \in W$, then $u - w = (u - v) + (v - w) \in W$ since W is closed under addition.

Equivalence class equals coset: Let $[v]$ denote the equivalence class of v and $v + W = \{v + w \mid w \in W\}$.

$$\begin{aligned}
 u \in [v] &\Leftrightarrow u \sim_W v \\
 &\Leftrightarrow u - v \in W \\
 &\Leftrightarrow \exists w \in W : u - v = w \\
 &\Leftrightarrow \exists w \in W : u = v + w \\
 &\Leftrightarrow u \in v + W
 \end{aligned}$$

Thus $[v] = v + W$.

QED

Definition 9.1.3 (Quotient Space):

Let $W \leq_K V$. The **quotient space** V/W is the set of all cosets of W in V :

$$V/W = \{v + W \mid v \in V\} = \{[v] \mid v \in V\}$$

We write $v + W$ (coset notation) and $[v]$ (equivalence class notation) interchangeably for the same element of V/W .

The vector space operations are defined through representatives:

$$\begin{aligned}
 u + W + v + W &= (u + v) + W & \text{i.e., } [u] + [v] &= [u + v] \\
 a \cdot v + W &= (av) + W & \text{i.e., } a \cdot [v] &= [av]
 \end{aligned}$$

for all $u, v \in V$ and $a \in K$. More generally, $a[u] + b[v] = [au + bv]$.

Theorem 9.1.4 (Vector Space Structure):

The operations on V/W are well-defined, and V/W is a K -vector space.

Proof:

Well-definedness of addition: Suppose $u_1 + W = u_2 + W$ and $v_1 + W = v_2 + W$, i.e., $u_1 - u_2 \in W$ and $v_1 - v_2 \in W$. Then $(u_1 + v_1) - (u_2 + v_2) = (u_1 - u_2) + (v_1 - v_2) \in W$, so $(u_1 + v_1) + W = (u_2 + v_2) + W$.

Well-definedness of scalar multiplication: Suppose $v_1 + W = v_2 + W$, i.e., $v_1 - v_2 \in W$. Then $av_1 - av_2 = a(v_1 - v_2) \in W$ since W is a subspace. Thus $(av_1) + W = (av_2) + W$.

Vector space axioms: The axioms follow from those of V via representatives. For example:

- Zero element: $[0] = 0 + W = W$ satisfies $[v] + [0] = [v + 0] = [v]$.
- Additive inverse: $-[v] = [-v]$ since $[v] + [-v] = [0]$.
- Associativity, commutativity, distributivity all follow by computing with representatives and using well-definedness.

QED

Definition 9.1.5 (Canonical Projection):

The **canonical projection** (or **quotient map**) is the linear map:

$$\pi : V \rightarrow V/W, \quad \pi(v) = v + W = [v]$$

Proposition 9.1.6 (Properties of Canonical Projection):

The canonical projection $\pi : V \rightarrow V/W$ satisfies:

1. π is linear
2. π is surjective
3. $\ker(\pi) = W$

Proof:

(1) **Linearity:** For $u, v \in V$ and $a, b \in K$:

$$\pi(au + bv) = [au + bv] = a[u] + b[v] = a\pi(u) + b\pi(v)$$

(2) **Surjectivity:** Every element of V/W has the form $[v] = \pi(v)$ for some $v \in V$.

(3) **Kernel:**

$$\begin{aligned} v \in \ker(\pi) &\Leftrightarrow \pi(v) = [0] \\ &\Leftrightarrow [v] = [0] \\ &\Leftrightarrow v - 0 \in W \\ &\Leftrightarrow v \in W \end{aligned}$$

QED

9.2 Examples of Quotient Spaces

Example 9.2.1 ($\mathbb{R}^2$ Quotient by a Line):

Let $V = \mathbb{R}^2$ and $L = \text{span}_{\mathbb{R}}(\{(1, 1)\})$ be the line $y = x$ through the origin.

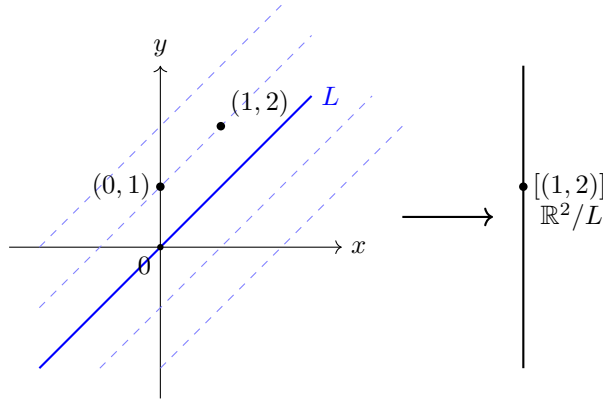
The cosets are parallel lines: for any $(a, b) \in \mathbb{R}^2$,

$$(a, b) + L = \{(a + t, b + t) \mid t \in \mathbb{R}\}$$

This is the line $y = x + (b - a)$.

Two points (a_1, b_1) and (a_2, b_2) are in the same coset iff $b_1 - a_1 = b_2 - a_2$, i.e., iff they have the same “signed distance” from the line L .

The quotient $\mathbb{R}^2/L$ is 1-dimensional, isomorphic to $\mathbb{R}$ via the map $(a, b) + L \mapsto b - a$.



Example 9.2.2 ($\mathbb{R}^3$ Quotient by a Plane):

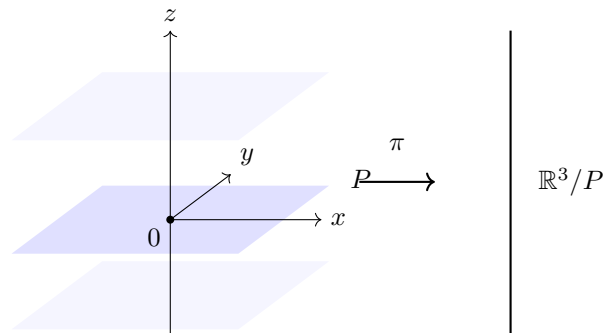
Let $V = \mathbb{R}^3$ and $P = \{(x, y, 0) \mid x, y \in \mathbb{R}\}$ be the xy -plane.

The cosets are planes parallel to P : for any $(a, b, c) \in \mathbb{R}^3$,

$$(a, b, c) + P = \{(a + s, b + t, c) \mid s, t \in \mathbb{R}\}$$

This is the plane at height $z = c$.

Two points are in the same coset iff they have the same z -coordinate. The quotient $\mathbb{R}^3/P$ is 1-dimensional, isomorphic to $\mathbb{R}$ (the z -axis) via $(a, b, c) + P \mapsto c$.



Example 9.2.3 (Polynomials mod X^2):

Let $V = K[X]$ and $W = \text{span}_K(\{X^n \mid n \geq 2\})$ be the subspace of polynomials with zero constant and linear terms.

Two polynomials $p, q \in K[X]$ satisfy $p \sim_W q$ iff $p - q \in W$, i.e., iff p and q have the same constant and linear coefficients.

Each coset is determined by its “remainder modulo X^2 ”:

$$(a_0 + a_1X + a_2X^2 + \cdots) + W = (a_0 + a_1X) + W$$

The quotient space has basis $\{[1], [X]\}$, so $\dim_K (K[X]/W) = 2$.

9.3 Dimension of Quotient Spaces

Remark (Motivation):

Since V/W is obtained by collapsing a $\dim(W)$ -dimensional piece of V to zero, we expect the quotient to have dimension $\dim(V) - \dim(W)$. The quotient dimension formula makes this precise: $\dim(W) + \dim(V/W) = \dim(V)$. This is stated in additive form so that it is valid even when the dimensions are infinite cardinals, where subtraction may not be meaningful.

The proof is constructive: start with a basis for W , extend it to a basis for V , and show that the cosets of the extension vectors form a basis for the quotient. This is the same “extend and project” strategy used in the direct proof of rank-nullity (Theorem 6.8.1), and the resemblance is not coincidental — the first isomorphism theorem will show that rank-nullity is a corollary of the quotient dimension formula.

Theorem 9.3.1 (Quotient Dimension Formula):

Let $W \leq_K V$. Then:

$$\dim_K(W) + \dim_K(V/W) = \dim_K(V)$$

In particular, for finite-dimensional V :

$$\dim_K(V/W) = \dim_K(V) - \dim_K(W)$$

Proof:

Let $B_W = \{w_i\}_{i \in I}$ be a basis for W . By the Basis Extension Theorem (Theorem 5.7.6), extend B_W to a basis $B = B_W \cup \{v_j\}_{j \in J}$ for V , where $B_W \cap \{v_j\}_{j \in J} = \emptyset$.

We claim that $\{[v_j]\}_{j \in J}$ is a basis for V/W .

Generating: Let $[v] \in V/W$ with $v = \sum_{i \in F_1} a_i w_i + \sum_{j \in F_2} b_j v_j$ for finite $F_1 \subseteq I$, $F_2 \subseteq J$. Then:

$$[v] = \left[\sum_{i \in F_1} a_i w_i + \sum_{j \in F_2} b_j v_j \right] = \sum_{i \in F_1} a_i [w_i] + \sum_{j \in F_2} b_j [v_j] = \sum_{j \in F_2} b_j [v_j]$$

since $[w_i] = [0]$ for all $w_i \in W$.

Linear independence: Suppose $\sum_{j \in F} b_j [v_j] = [0]$ for some finite $F \subseteq J$. Then $\left[\sum_{j \in F} b_j v_j \right] = [0]$, which means $\sum_{j \in F} b_j v_j \in W$.

Write $\sum_{j \in F} b_j v_j = \sum_{i \in G} a_i w_i$ for some finite $G \subseteq I$. Rearranging: $\sum_{j \in F} b_j v_j - \sum_{i \in G} a_i w_i = 0$.

Since B is linearly independent and $\{w_i\}_{i \in G} \cup \{v_j\}_{j \in F} \subseteq B$, all coefficients must be zero. In particular, all $b_j = 0$.

Thus $\{[v_j]\}_{j \in J}$ is a basis for V/W , hence:

$$|B_W| + \dim_K(V/W) = |B_W| + |J| = |B|$$

That is, $\dim_K(W) + \dim_K(V/W) = \dim_K(V)$.

In finite dimensions, we can subtract to obtain $\dim_K(V/W) = \dim_K(V) - \dim_K(W)$. QED

9.4 Universal Property of Quotient Spaces

Remark (Motivation):

The universal property of quotient spaces is the dual of the universal property of free vector spaces (Theorem 6.2.1). Where the free property says “maps *out of* V are determined by values on a basis,” the quotient property says “maps *out of* V/W correspond exactly to maps out of V that kill W .”

In categorical language, the canonical projection $\pi : V \rightarrow V/W$ is the **cokernel** of the inclusion $W \hookrightarrow V$: it is the universal map that makes the inclusion zero. This universal characterization means that V/W is determined up to unique isomorphism by the pair (V, W) — any other space with the same factoring property is canonically isomorphic to V/W .

In practice, the universal property is the main tool for constructing maps out of quotient spaces: rather than working with cosets directly, define a map on V , verify that it kills W , and invoke the theorem.

The universal property characterizes the quotient space by its relationship to linear maps whose kernel contains the subspace W .

$$\begin{array}{ccc} V & \xrightarrow{\pi} & V/W \\ & \searrow T & \downarrow \exists! \tilde{T} \\ & & X \end{array}$$

Theorem 9.4.1 (Universal Property of Quotient Spaces):

Let $W \leq_K V$ and $T : V \rightarrow X$ be a linear map. Then:

1. There exists a linear map $\tilde{T} : V/W \rightarrow X$ with $T = \tilde{T} \circ \pi$ if and only if $W \subseteq \ker(T)$.
2. If such $\tilde{T}$ exists, it is unique.
3. $\tilde{T}$ is surjective if and only if T is surjective.
4. $\tilde{T}$ is injective if and only if $\ker(T) = W$.
5. $\tilde{T}$ is an isomorphism if and only if T is surjective and $\ker(T) = W$.

Proof:

(1) Existence:

($\Rightarrow$) Suppose $\tilde{T}$ exists with $T = \tilde{T} \circ \pi$. Then:

$$\ker(T) = \ker(\tilde{T} \circ \pi) \supseteq \ker(\pi) = W$$

Thus $W \subseteq \ker(T)$.

($\Leftarrow$) Suppose $W \subseteq \ker(T)$. Define $\tilde{T} : V/W \rightarrow X$ by:

$$\tilde{T}(v + W) = T(v)$$

Well-defined: If $v_1 + W = v_2 + W$, then $v_1 - v_2 \in W \subseteq \ker(T)$. Thus $T(v_1) - T(v_2) = T(v_1 - v_2) = 0$, so $T(v_1) = T(v_2)$.

Linear: For $u + W, v + W \in V/W$ and $a, b \in K$:

$$\begin{aligned} \tilde{T}(au + W + bv + W) &= \tilde{T}(au + bv + W) \\ &= T(au + bv) \\ &= aT(u) + bT(v) \\ &= a\tilde{T}(u + W) + b\tilde{T}(v + W) \end{aligned}$$

Factorization: For any $v \in V$: $\tilde{T}(\pi(v)) = \tilde{T}(v + W) = T(v)$. Thus $\tilde{T} \circ \pi = T$.

(2) Uniqueness: Suppose $\tilde{T}_1$ and $\tilde{T}_2$ both satisfy $\tilde{T}_1 \circ \pi = T = \tilde{T}_2 \circ \pi$.

Since π is surjective (an epimorphism), and $\tilde{T}_1 \circ \pi = \tilde{T}_2 \circ \pi$, it follows that $\tilde{T}_1 = \tilde{T}_2$ by the cancellation property of epimorphisms.

(3) Surjectivity: Since $T = \tilde{T} \circ \pi$ and π is surjective:

$$\text{Rng}(T) = \text{Rng}(\tilde{T} \circ \pi) = \tilde{T}[\text{Rng}(\pi)] = \tilde{T}[V/W] = \text{Rng}(\tilde{T})$$

Thus T is surjective if and only if $\tilde{T}$ is surjective.

(4) Injectivity:

($\Rightarrow$) Suppose $\tilde{T}$ is injective (a monomorphism). We have $W \subseteq \ker(T)$ by (1). Conversely, let $v \in \ker(T)$. Then $\tilde{T}(\pi(v)) = T(v) = 0 = \tilde{T}(0)$. Since $\tilde{T}$ is a monomorphism, $\pi(v) = 0$, i.e., $v \in \ker(\pi) = W$. Thus $\ker(T) \subseteq W$, so $\ker(T) = W$.

($\Leftarrow$) Suppose $\ker(T) = W$. Let $\tilde{T}(v_1 + W) = \tilde{T}(v_2 + W)$. Then $T(v_1) = T(v_2)$, so $v_1 - v_2 \in \ker(T) = W$. Thus $v_1 + W = v_2 + W$, so $\tilde{T}$ is injective.

(5) Isomorphism: Combine (3) and (4): $\tilde{T}$ is bijective (hence an isomorphism, by Theorem 6.7.3) if and only if T is surjective and $\ker(T) = W$. QED

Remark (Application of Universal Property):

The universal property provides a systematic way to:

1. **Factor maps:** Given $T : V \rightarrow X$ with $W \subseteq \ker(T)$, we can factor $T = \tilde{T} \circ \pi$ through the quotient.
2. **Construct maps out of quotients:** To define a linear map from V/W , it suffices to define a map on V that vanishes on W .
3. **Identify when quotients are isomorphic:** $V/W \cong X$ iff there exists surjective $T : V \rightarrow X$ with $\ker(T) = W$.

9.5 Isomorphism Theorems

Remark (Overview):

The four isomorphism theorems are the structural backbone of the subject. They describe how quotient spaces interact with subspaces, sums, intersections, and nested inclusions. Each theorem identifies a natural isomorphism between two quotient spaces that might appear unrelated at first glance.

- The **First Isomorphism Theorem** ($U/\ker(T) \cong \text{Rng}(T)$) says that every linear map factors as a surjection followed by an injection, with the quotient by the kernel as the intermediate space. This is the most frequently used of the four theorems.
- The **Second Isomorphism Theorem** ($(U+W)/W \cong U/(U \cap W)$) compares two ways of “modding out” by W : starting from $U+W$ or starting from U . As a corollary, it yields the Grassmann dimension formula.
- The **Third Isomorphism Theorem** ($(V/U)/(W/U) \cong V/W$) says that taking quotients in two stages is the same as taking the quotient in one step — “quotienting by a quotient.”
- The **Fourth Isomorphism Theorem** (Lattice Correspondence) establishes a bijection between the subspaces of V/W and the subspaces of V that contain W . This is a lattice isomorphism: it preserves sums, intersections, and inclusions.

All proofs follow the same strategy: define a linear map, identify its kernel and image, and invoke the universal property of quotient spaces (Theorem 9.4.1).

9.5.1 First Isomorphism Theorem

$$\begin{array}{ccc} U & \xrightarrow{T} & V \\ \pi \downarrow & \nearrow \tilde{T} & \\ U/\ker(T) & \xrightarrow{\cong} & \end{array}$$

Theorem 9.5.1 (First Isomorphism Theorem):

For any linear map $T : U \rightarrow V$:

$$U/\ker(T) \cong \text{Rng}(T)$$

Proof:

Apply the Universal Property (Theorem 9.4.1) with $W = \ker(T)$ and $X = V$.

Since $\ker(T) \subseteq \ker(T)$, there exists a unique linear map $\tilde{T} : U/\ker(T) \rightarrow V$ such that $T = \tilde{T} \circ \pi$.

By part (4) of the universal property: $\tilde{T}$ is injective since $\ker(T) = \ker(T)$.

The image of $\tilde{T}$ equals the image of T :

$$\text{Rng}(\tilde{T}) = \{\tilde{T}([u]) \mid u \in U\} = \{T(u) \mid u \in U\} = \text{Rng}(T)$$

Thus $\tilde{T} : U/\ker(T) \rightarrow \text{Rng}(T)$ is a bijective linear map, hence an isomorphism.

QED

Corollary 9.5.2 (Rank-Nullity via First Isomorphism Theorem):

For any linear map $T : U \rightarrow V$:

$$\dim_K(\ker(T)) + \dim_K(\text{Rng}(T)) = \dim_K(U)$$

That is: $\text{nullity}(T) + \text{rank}(T) = \dim_K(U)$.

Proof:

By the First Isomorphism Theorem: $U/\ker(T) \cong \text{Rng}(T)$.

Thus $\dim_K(U/\ker(T)) = \dim_K(\text{Rng}(T))$.

By the Quotient Dimension Formula (Theorem 9.3.1):

$$\dim_K(\ker(T)) + \dim_K(U/\ker(T)) = \dim_K(U)$$

Substituting: $\dim_K(\ker(T)) + \dim_K(\text{Rng}(T)) = \dim_K(U)$.

QED

Remark (Two Proofs of Rank-Nullity):

We now have two proofs of the Rank-Nullity Theorem:

1. **Direct proof** (Theorem 6.8.1): Extend a basis of $\ker(T)$ and show the images of the extension vectors form a basis of $\text{Rng}(T)$.
2. **Via isomorphism theorem** (Corollary 9.5.2): Use $U/\ker(T) \cong \text{Rng}(T)$ and the quotient dimension formula.

Both approaches are valuable and highlight different aspects of the structure.

Example 9.5.3 (First Isomorphism Theorem in $\mathbb{R}^3$):

Define $T: \mathbb{R}^3 \rightarrow \mathbb{R}^2$ by $T(x, y, z) = (x + y, y + z)$.

Kernel: $T(x, y, z) = (0, 0)$ iff $x = -y$ and $z = -y$, so $\ker(T) = \text{span}_{\mathbb{R}}(\{(1, -1, 1)\})$, a line through the origin.

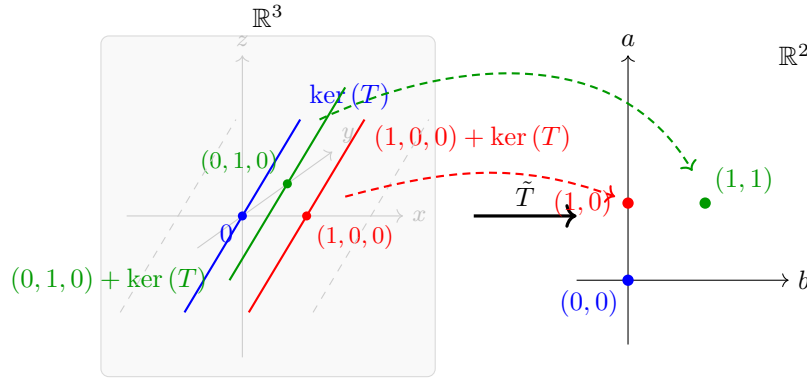
Range: T is surjective since $T(1, 0, 0) = (1, 0)$ and $T(0, 0, 1) = (0, 1)$, so $\text{Rng}(T) = \mathbb{R}^2$.

The isomorphism explicitly: Each coset in $\mathbb{R}^3/\ker(T)$ is a line parallel to $(1, -1, 1)$. The induced map $\tilde{T}: \mathbb{R}^3/\ker(T) \rightarrow \mathbb{R}^2$ sends

$$(x, y, z) + \ker(T) \mapsto (x + y, y + z)$$

For instance, $(1, 0, 0) + \ker(T) = \{(1 + t, -t, t) \mid t \in \mathbb{R}\}$ maps to $(1, 0)$, and every point on that line maps to $(1, 0)$.

Dimension check: $\dim_{\mathbb{R}}(\mathbb{R}^3/\ker(T)) = 3 - 1 = 2 = \dim_{\mathbb{R}}(\mathbb{R}^2)$.

**9.5.2 Second Isomorphism Theorem**

$$\begin{array}{ccc}
 & & U + W \\
 & \nearrow & \downarrow \pi_W \\
 U & \xrightarrow{\iota} & (U + W)/W \\
 \uparrow & & \uparrow \cong \\
 U \cap W & \xrightarrow{\pi_{U \cap W}} & U/(U \cap W)
 \end{array}$$

Theorem 9.5.4 (Second Isomorphism Theorem):

Let $U, W \leq_K V$. Then:

$$(U + W)/W \cong U/(U \cap W)$$

Proof:

Consider the composition:

$$T : U \xrightarrow{\iota} U + W \xrightarrow{\pi_W} (U + W)/W$$

where ι is the inclusion and π_W is the canonical projection.

T is surjective: For any $[u + w] \in (U + W)/W$ with $u \in U$ and $w \in W$:

$$[u + w] = [u] + [w] = [u] + [0] = [u] = T(u)$$

since $w \in W$ implies $[w] = [0]$.

$\ker(T) = U \cap W$:

$$\begin{aligned} u \in \ker(T) &\Leftrightarrow \pi_W(\iota(u)) = [0] \\ &\Leftrightarrow [u] = [0] \text{ in } (U + W)/W \\ &\Leftrightarrow u \in W \\ &\Leftrightarrow u \in U \cap W \end{aligned}$$

By the First Isomorphism Theorem:

$$U/\ker(T) = U/(U \cap W) \cong \text{Rng}(T) = (U + W)/W$$

QED

Example 9.5.5 (Second Isomorphism Theorem in $\mathbb{R}^3$):

Let $V = \mathbb{R}^3$, $U = \{(x, y, 0) \mid x, y \in \mathbb{R}\}$ (the xy -plane), and $W = \{(x, 0, z) \mid x, z \in \mathbb{R}\}$ (the xz -plane).

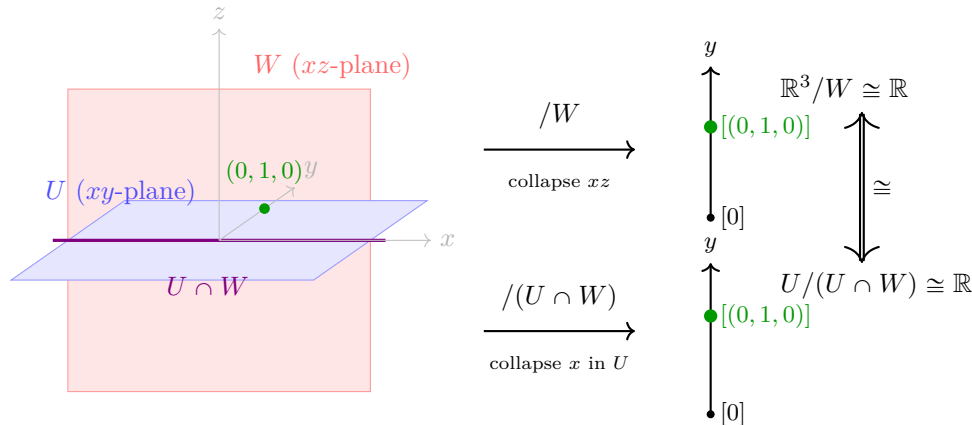
Then $U + W = \mathbb{R}^3$ and $U \cap W = \{(x, 0, 0) \mid x \in \mathbb{R}\}$ (the x -axis).

The theorem says $(U + W)/W \cong U/(U \cap W)$, i.e., $\mathbb{R}^3/W \cong U/(U \cap W)$.

Left side: $\mathbb{R}^3/W$ collapses the xz -plane to zero. Each coset $(a, b, c) + W = \{(a + s, b, c + t) \mid s, t \in \mathbb{R}\}$ is determined by b alone, so $\mathbb{R}^3/W \cong \mathbb{R}$ (the y -direction).

Right side: $U/(U \cap W)$ collapses the x -axis within the xy -plane. Each coset $(a, b, 0) + U \cap W = \{(a + s, b, 0) \mid s \in \mathbb{R}\}$ is determined by b alone, so $U/(U \cap W) \cong \mathbb{R}$ (again the y -direction).

Both quotients collapse to the same copy of $\mathbb{R}$ —two different ways of “projecting onto the y -axis.”



$$\begin{array}{ccc}
 & U + W = \mathbb{R}^3 & \\
 \swarrow & & \searrow \\
 U \text{ (} xy\text{-plane)} & & W \text{ (} xz\text{-plane)} \\
 \searrow & & \swarrow \\
 & U \cap W \text{ (} x\text{-axis)} &
 \end{array}
 \quad
 \begin{array}{ccc}
 & \xrightarrow{/W} & \mathbb{R}^3/W \cong \mathbb{R} \\
 & & \updownarrow \cong \\
 & \xrightarrow{/(U \cap W)} & U/(U \cap W) \cong \mathbb{R}
 \end{array}$$

Corollary 9.5.6 (Grassmann Formula via Second Isomorphism Theorem):

Let $U, W \leq_K V$. Then:

$$\dim_K (U + W) + \dim_K (U \cap W) = \dim_K (U) + \dim_K (W)$$

In finite dimensions, this rearranges to $\dim_K (U + W) = \dim_K (U) + \dim_K (W) - \dim_K (U \cap W)$.

Proof:

By the Second Isomorphism Theorem, $(U + W)/W \cong U/(U \cap W)$, so the two quotients have a common dimension

$$\delta := \dim_K ((U + W)/W) = \dim_K (U/(U \cap W))$$

By the Quotient Dimension Formula (applied twice):

$$\begin{aligned}
 \dim_K (U + W) &= \dim_K (W) + \delta \\
 \dim_K (U) &= \dim_K (U \cap W) + \delta
 \end{aligned}$$

Substituting both into the desired left-hand side:

$$\dim_K (U + W) + \dim_K (U \cap W) = \dim_K (W) + \delta + \dim_K (U \cap W) = \dim_K (W) + \dim_K (U) \quad \text{QED}$$

9.5.3 Third Isomorphism Theorem**Remark (Motivation):**

Suppose we have nested subspaces $U \subseteq W \subseteq V$ and want to form the quotient V/W . We can do this in one step, or in two: first collapse U (getting V/U), then collapse the image of W (getting $(V/U)/(W/U)$). The third isomorphism theorem says these two procedures give the same result. Intuitively, it does not matter whether we collapse a large subspace all at once or in stages — the final quotient depends only on what was collapsed in total.

$$\begin{array}{ccc}
 V & \xrightarrow{\pi_U} & V/U \\
 \pi_W \downarrow & & \downarrow \bar{\pi} \\
 V/W & \xleftarrow{\cong} & (V/U)/(W/U)
 \end{array}$$

Theorem 9.5.7 (Third Isomorphism Theorem):

Let $U \subseteq W \subseteq V$ be subspaces. Then:

$$(V/U)/(W/U) \cong V/W$$

where $W/U \leq_K V/U$ is identified with its image under the natural inclusion.

Proof:

Note that $W/U = \{w + U \mid w \in W\}$ is naturally a subspace of V/U .

Consider the canonical projection $\pi_W : V \rightarrow V/W$.

We apply the Universal Property to factor π_W through V/U .

Since $U \subseteq W = \ker(\pi_W)$, there exists a unique linear map $\bar{\pi} : V/U \rightarrow V/W$ with $\pi_W = \bar{\pi} \circ \pi_U$.

Explicitly: $\bar{\pi}(v + U) = v + W$ for all $v \in V$.

$\bar{\pi}$ is surjective: For any $v + W \in V/W$: $\bar{\pi}(v + U) = v + W$.

$\ker(\bar{\pi}) = W/U$:

$$\begin{aligned} v + U \in \ker(\bar{\pi}) &\Leftrightarrow \bar{\pi}(v + U) = 0 + W \\ &\Leftrightarrow v + W = 0 + W \\ &\Leftrightarrow v \in W \\ &\Leftrightarrow v + U \in W/U \end{aligned}$$

By the First Isomorphism Theorem applied to $\bar{\pi}$:

$$(V/U) / \ker(\bar{\pi}) = (V/U) / (W/U) \cong \text{Rng}(\bar{\pi}) = V/W$$

QED

Corollary 9.5.8 (Dimension Cancellation):

Let $U \subseteq W \subseteq V$ be subspaces. Then:

$$\dim_K(V/U) = \dim_K(V/W) + \dim_K(W/U)$$

Proof:

By the Third Isomorphism Theorem: $(V/U) / (W/U) \cong V/W$.

By the Quotient Dimension Formula:

$$\dim_K(W/U) + \dim_K((V/U) / (W/U)) = \dim_K(V/U)$$

Since $\dim_K((V/U) / (W/U)) = \dim_K(V/W)$:

$$\dim_K(W/U) + \dim_K(V/W) = \dim_K(V/U)$$

QED

Example 9.5.9 (Third Isomorphism Theorem in $\mathbb{R}^3$):

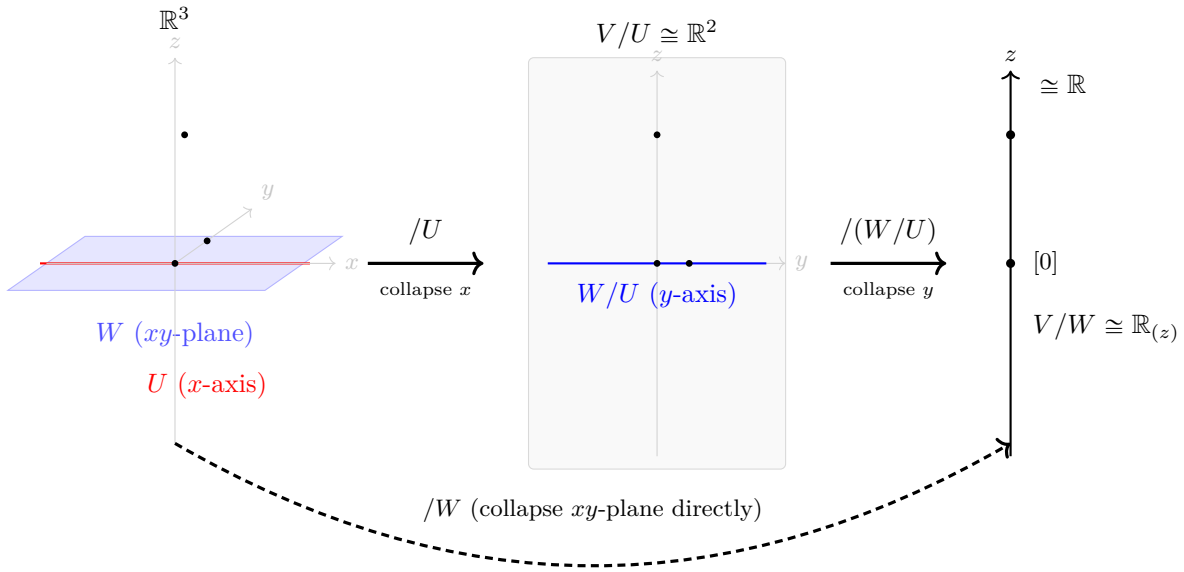
Let $V = \mathbb{R}^3$, $U = \text{span}_{\mathbb{R}}(\{e_0\})$ (the x -axis), and $W = \text{span}_{\mathbb{R}}(\{e_0, e_1\})$ (the xy -plane), so $U \subseteq W \subseteq V$.

Stage 1 — collapse the x -axis: $V/U = \mathbb{R}^3/x\text{-axis}$ is 2-dimensional, parameterized by (y, z) . The image of W is $W/U = \{(0, y, 0) + U \mid y \in \mathbb{R}\}$, a 1-dimensional subspace (the “ y -line” in the quotient).

Stage 2 — collapse that line: $(V/U) / (W/U)$ further collapses the y -direction, leaving a 1-dimensional space parameterized by z alone.

All at once: $V/W = \mathbb{R}^3/xy\text{-plane}$ is also 1-dimensional, parameterized by z .

Collapsing in two stages (x -axis first, then y -direction) gives the same result as collapsing the entire xy -plane at once.

**Remark (Intuition for Third Isomorphism Theorem):**

The third isomorphism theorem says that “quotients of quotients” simplify. Intuitively: if we first collapse U to a point, then collapse the image of W , we get the same result as collapsing W directly.

In finite dimensions with $\dim_K(V) = n$, $\dim_K(W) = m$, $\dim_K(U) = k$ (where $k \leq m \leq n$):

$$\dim_K(V/U) = n - k$$

$$\dim_K(W/U) = m - k$$

$$\dim_K((V/U)/(W/U)) = (n - k) - (m - k) = n - m = \dim_K(V/W)$$

9.5.4 Fourth Isomorphism Theorem (Lattice Correspondence)**Remark (Motivation):**

The previous theorems constructed specific isomorphisms between quotient spaces. The fourth theorem is different in character: it is a **structural correspondence** between the subspaces of a quotient V/W and the subspaces of V that contain W . This is the linear-algebraic analogue of the correspondence theorem for groups and rings.

The practical import is that we can study the internal structure of a quotient space without ever leaving V : every subspace $X \leq_K V/W$ “lifts” to a unique subspace $U \leq_K V$ with $W \subseteq U$, and the operations of sum and intersection are preserved. This makes the fourth isomorphism theorem an indispensable tool for inductive arguments and for analyzing chains of subspaces (flags) in quotient spaces.

Theorem 9.5.10 (Fourth Isomorphism Theorem):

Let $W \leq_K V$. There is a bijective correspondence between:

- Subspaces of V/W
- Subspaces of V containing W

Equivalently, the subspace lattice of V/W is isomorphic to the sublattice of $\text{Sub}(V)$ bounded below by W :

$$\text{Sub}(V/W) \cong \{U \in \text{Sub}(V) \mid W \subseteq U\}$$

The correspondence is given by:

$$\begin{aligned}\Phi : \{U \in \text{Sub}(V) \mid W \subseteq U\} &\rightarrow \text{Sub}(V/W) \\ U &\mapsto U/W = \{u + W \mid u \in U\}\end{aligned}$$

with inverse:

$$\begin{aligned}\Psi : \text{Sub}(V/W) &\rightarrow \{U \in \text{Sub}(V) \mid W \subseteq U\} \\ X &\mapsto \pi^{\leftarrow}[X] = \{v \in V \mid v + W \in X\}\end{aligned}$$

Moreover, Φ is a lattice isomorphism — it preserves:

1. Inclusion: $U_1 \subseteq U_2 \Leftrightarrow \Phi(U_1) \subseteq \Phi(U_2)$
2. Join: $\Phi(U_1 + U_2) = \Phi(U_1) + \Phi(U_2)$
3. Meet: $\Phi(U_1 \cap U_2) = \Phi(U_1) \cap \Phi(U_2)$
4. Dimension: $\dim_K(\Phi(U)) = \dim_K(U/W)$

Proof:

We prove (a) well-definedness of both maps, (b) that they are mutual inverses, and (c) that both are lattice homomorphisms.

(a) Well-definedness.

Φ is well-defined: If $U \leq_K V$ with $W \subseteq U$, then $U/W = \{u + W \mid u \in U\}$ is a subspace of V/W . Indeed: $0 + W \in U/W$, and $au_1 + W + bu_2 + W = au_1 + bu_2 + W \in U/W$ for $u_1, u_2 \in U$ and $a, b \in K$.

Ψ is well-defined: For any subspace $X \leq_K V/W$, the preimage $\pi^{\leftarrow}[X]$ is a subspace of V (Proposition 6.2.5). Since $0 + W \in X$ (because X is a subspace), we have $W = \pi^{\leftarrow}[\{0 + W\}] \subseteq \pi^{\leftarrow}[X]$. Thus $W \subseteq \pi^{\leftarrow}[X]$.

(b) Φ and Ψ are mutual inverses.

$\Psi \circ \Phi = \text{Id}_{\{U \in \text{Sub}(V) \mid W \subseteq U\}}$: For $U \leq_K V$ with $W \subseteq U$:

$$\begin{aligned}\Psi(\Phi(U)) &= \Psi(U/W) \\ &= \{v \in V \mid v + W \in U/W\} \\ &= \{v \in V \mid \exists u \in U : v + W = u + W\} \\ &= \{v \in V \mid \exists u \in U : v - u \in W\} \\ &= \{v \in V \mid v \in U + W\} \\ &= U + W = U\end{aligned}$$

The last equality holds because $W \subseteq U$ implies $U + W = U$.

$\Phi \circ \Psi = \text{Id}_{\text{Sub}(V/W)}$: For $X \leq_K V/W$:

$$\begin{aligned}\Phi(\Psi(X)) &= \Phi(\pi^{\leftarrow}[X]) \\ &= \{v + W \mid v \in \pi^{\leftarrow}[X]\} \\ &= \{v + W \mid v + W \in X\} \\ &= X\end{aligned}$$

(c) Both Φ and Ψ are lattice homomorphisms.

We prove that each map preserves inclusion, join (sum), and meet (intersection).

Φ preserves inclusion: If $U_1 \subseteq U_2$, then every coset $u + W$ with $u \in U_1$ also has $u \in U_2$, so $\Phi(U_1) \subseteq \Phi(U_2)$. Conversely, if $\Phi(U_1) \subseteq \Phi(U_2)$, apply Ψ (which is order-preserving as a preimage) to get $U_1 = \Psi(\Phi(U_1)) \subseteq \Psi(\Phi(U_2)) = U_2$.

Φ preserves joins:

$$\begin{aligned} v + W \in \Phi(U_1 + U_2) &\Leftrightarrow v \in U_1 + U_2 \\ &\Leftrightarrow v = u_1 + u_2 \text{ for some } u_1 \in U_1, u_2 \in U_2 \\ &\Leftrightarrow v + W = u_1 + W + u_2 + W \in \Phi(U_1) + \Phi(U_2) \end{aligned}$$

Φ preserves meets: ($\subseteq$): If $v \in U_1 \cap U_2$, then $v + W \in \Phi(U_1)$ and $v + W \in \Phi(U_2)$, so $v + W \in \Phi(U_1) \cap \Phi(U_2)$.

($\supseteq$): If $v + W \in \Phi(U_1) \cap \Phi(U_2)$, then $v \in \Psi(\Phi(U_1)) \cap \Psi(\Phi(U_2)) = U_1 \cap U_2$, so $v + W \in \Phi(U_1 \cap U_2)$.

Ψ preserves inclusion: If $X_1 \subseteq X_2$, then $\Psi(X_1) = \pi^{\leftarrow}[X_1] \subseteq \pi^{\leftarrow}[X_2] = \Psi(X_2)$, since preimages preserve inclusion. Conversely, if $\Psi(X_1) \subseteq \Psi(X_2)$, apply Φ to get $X_1 = \Phi(\Psi(X_1)) \subseteq \Phi(\Psi(X_2)) = X_2$.

Ψ preserves joins: We show $\Psi(X_1 + X_2) = \Psi(X_1) + \Psi(X_2)$.

($\supseteq$): If $v_1 \in \Psi(X_1)$ and $v_2 \in \Psi(X_2)$, then $v_1 + W \in X_1$ and $v_2 + W \in X_2$. So $\pi(v_1 + v_2) = v_1 + W + v_2 + W \in X_1 + X_2$, hence $v_1 + v_2 \in \Psi(X_1 + X_2)$.

($\subseteq$): If $v \in \Psi(X_1 + X_2)$, then $v + W \in X_1 + X_2$, so $v + W = \bar{x}_1 + \bar{x}_2$ for some $\bar{x}_1 \in X_1$, $\bar{x}_2 \in X_2$. Since π is surjective, pick $v_1, v_2 \in V$ with $\pi(v_1) = \bar{x}_1$ and $\pi(v_2) = \bar{x}_2$. Then $v_1 \in \Psi(X_1)$, $v_2 \in \Psi(X_2)$, and $\pi(v - v_1 - v_2) = 0 + W$, so $w := v - v_1 - v_2 \in W \subseteq \Psi(X_1)$. Therefore $v = (v_1 + w) + v_2 \in \Psi(X_1) + \Psi(X_2)$.

Ψ preserves meets: We show $\Psi(X_1 \cap X_2) = \Psi(X_1) \cap \Psi(X_2)$.

($\subseteq$): If $v \in \Psi(X_1 \cap X_2)$, then $v + W \in X_1 \cap X_2$, so $v + W \in X_1$ and $v + W \in X_2$. Thus $v \in \Psi(X_1) \cap \Psi(X_2)$.

($\supseteq$): If $v \in \Psi(X_1) \cap \Psi(X_2)$, then $v + W \in X_1$ and $v + W \in X_2$, so $v + W \in X_1 \cap X_2$, hence $v \in \Psi(X_1 \cap X_2)$.

Dimension: Follows from $\Phi(U) = U/W$ by definition. QED

Example 9.5.11 (Fourth Isomorphism Theorem in $\mathbb{R}^3$):

Let $V = \mathbb{R}^3$ and $W = \text{span}_{\mathbb{R}}(\{e_0\})$ (the x -axis). Then $V/W \cong \mathbb{R}^2$ (parameterized by (y, z)).

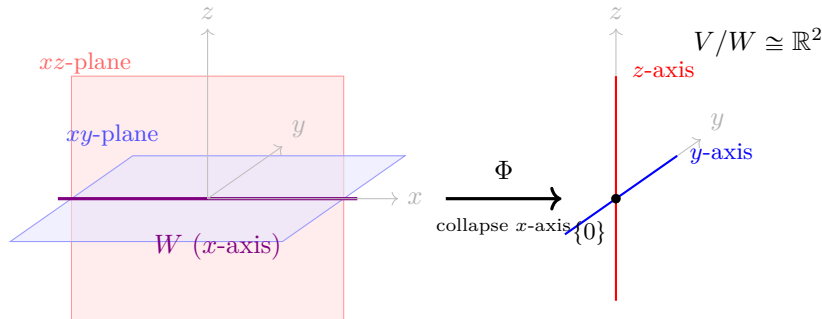
The subspaces of V containing W are:

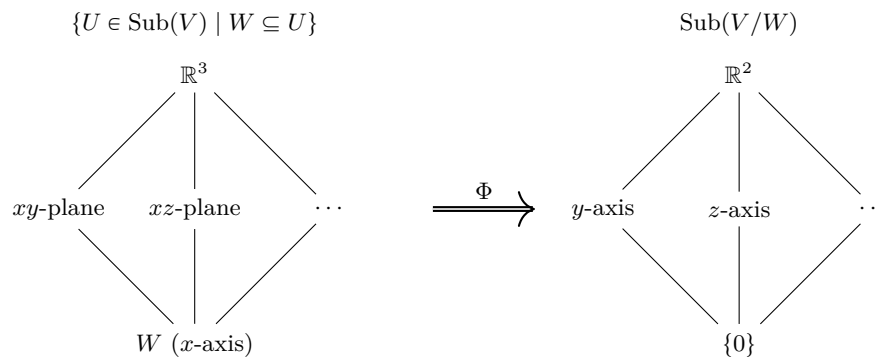
- W itself (the x -axis)
- Every plane through the origin containing the x -axis (e.g., the xy -plane, the xz -plane, $\text{span}_{\mathbb{R}}(\{e_0, e_1 + e_2\})$, ...)
- $\mathbb{R}^3$

The subspaces of $V/W \cong \mathbb{R}^2$ are:

- $\{0\}$
- Every line through the origin in $\mathbb{R}^2$
- $\mathbb{R}^2$

The correspondence Φ maps: $W \mapsto \{0\}$, each plane containing W to the line it becomes after collapsing the x -axis, and $\mathbb{R}^3 \mapsto \mathbb{R}^2$. For instance, the xy -plane maps to the y -axis, the xz -plane maps to the z -axis.



**Remark (Lattice Interpretation):**

The fourth isomorphism theorem says that the subspace lattice $\text{Sub}(V/W)$ is isomorphic to the **principal filter** $\uparrow W = \{U \in \text{Sub}(V) \mid W \subseteq U\}$ in $\text{Sub}(V)$ —that is, the portion of the lattice of V that lies above W .

Quotienting by W “cuts off” everything below W in the lattice and relabels what remains. The lattice operations are preserved:

- Join (supremum): $U_1 \vee U_2 = U_1 + U_2$
- Meet (infimum): $U_1 \wedge U_2 = U_1 \cap U_2$

This is particularly useful when analyzing quotient spaces by examining the subspace structure above W : every question about subspaces of V/W can be translated into a question about subspaces of V containing W , and vice versa.

9.6 Chapter Compendium

Chapter Compendium — Quotient Space Construction:

Equivalence Relation: $u \sim_W v \Leftrightarrow u - v \in W$

Coset: $v + W = \{v + w \mid w \in W\} = [v]$

Operations: $(u + W) + (v + W) = (u + v) + W, \quad a \cdot (v + W) = av + W$

Canonical Projection: $\pi : V \rightarrow V/W, \pi(v) = v + W$ (linear epimorphism, $\ker(\pi) = W$)

Chapter Compendium — Dimension Formula:

Additive Form (all dimensions):

$$\dim_K(W) + \dim_K(V/W) = \dim_K(V)$$

Subtraction Form (finite-dimensional only): $\dim_K(V/W) = \dim_K(V) - \dim_K(W)$

Proof idea: Extend basis of W to basis of V ; images of extension vectors form basis of quotient

Chapter Compendium — Universal Property of Quotient Spaces:

$$\begin{array}{ccc} V & \xrightarrow{\pi} & V/W \\ & \searrow T & \downarrow \tilde{T} \\ & & X \end{array}$$

Existence: $\tilde{T}$ exists with $T = \tilde{T} \circ \pi \Leftrightarrow W \subseteq \ker(T)$

Uniqueness: If exists, $\tilde{T}$ is unique (since π is an epimorphism)

Properties: $\tilde{T}$ surjective $\Leftrightarrow T$ surjective; $\tilde{T}$ injective $\Leftrightarrow \ker(T) = W$; $\tilde{T}$ isomorphism $\Leftrightarrow T$ surjective and $\ker(T) = W$

Chapter Compendium — Isomorphism Theorems:

First Isomorphism Theorem: $U/\ker(T) \cong \text{Rng}(T)$

Corollary (Rank-Nullity via iso): $\text{nullity}(T) + \text{rank}(T) = \dim_K(U)$

Second Isomorphism Theorem: For $U, W \leq_K V$: $(U + W)/W \cong U/(U \cap W)$

Corollary (Grassmann): $\dim_K(U + W) + \dim_K(U \cap W) = \dim_K(U) + \dim_K(W)$

Third Isomorphism Theorem: For $U \subseteq W \subseteq V$: $(V/U)/(W/U) \cong V/W$

Fourth Isomorphism Theorem (Lattice Correspondence): $\{U \leq_K V \mid W \subseteq U\} \xleftrightarrow{1:1} \{X \leq_K V/W\}$ via $U \mapsto U/W$ and $X \mapsto \pi^{-1}[X]$.

Equivalently: $\text{Sub}(V/W) \cong \{U \in \text{Sub}(V) \mid W \subseteq U\}$ (the sublattice of $\text{Sub}(V)$ bounded below by W , i.e., the principal filter $\uparrow W$).

Preserves: inclusion, join (+), meet ($\cap$), dimension

Chapter Compendium — Critical Counterexample: Cardinal vs Dimension:

Let $W = \text{span}_K(\{X^n \mid n \geq 2\}) \leq_K K[X]$

Cardinal arithmetic: $\dim_K(W) = \aleph_0$ and $\aleph_0 + 1 = \aleph_0 = \dim_K(K[X])$

But $\dim_K(K[X]/W) = 2 \neq 1$, so W is **NOT** a hyperplane

Lesson: In infinite dimensions, use $\dim_K(V/W) = 1$, not cardinal subtraction

Chapter Compendium — Key Examples:

Finite-Dim: $\mathbb{R}^2/L \cong \mathbb{R}$ (parallel lines), $\mathbb{R}^3/P \cong \mathbb{R}$ (parallel planes)

Infinite-Dim: $K[X]/\text{span}_K(\{X^n \mid n \geq 2\}) \cong K^2$

CHAPTER 10

Products, Hom Spaces, and Dual Spaces

10.1 Direct Products and Coproducts

Remark (Categorical Perspective):

In category theory, products and coproducts are defined by **universal properties**, not by explicit constructions. The construction then follows as the unique (up to isomorphism) object satisfying the universal property. We follow this approach here.

10.1.1 The Categorical Product

Definition 10.1.1 (Categorical Product):

Let $(V_i)_{i \in I}$ be a family of K -vector spaces. A **product** of this family is a pair $(P, (\pi_j)_{j \in I})$ where P is a K -vector space and $\pi_j : P \rightarrow V_j$ are linear maps (called **projections**), satisfying:

Universal Property: For any K -vector space W and family of linear maps $(f_j : W \rightarrow V_j)_{j \in I}$, there exists a **unique** linear map $f : W \rightarrow P$ such that $\pi_j \circ f = f_j$ for all $j \in I$.

$$\begin{array}{ccccc} & & W & & \\ & f_j \swarrow & \downarrow \exists! f & \searrow f_k & \\ V_j & \xleftarrow{\pi_j} & P & \xrightarrow{\pi_k} & V_k \end{array}$$

Remark (Uniqueness up to Isomorphism):

A standard categorical argument shows that if $(P, (\pi_j))$ and $(P', (\pi'_j))$ both satisfy the product universal property, then there exists a unique isomorphism $\phi : P \rightarrow P'$ with $\pi'_j \circ \phi = \pi_j$ for all j . We omit the proof as it is a general categorical fact.

Theorem 10.1.2 (Construction of Product):

The product is realized by the set of choice functions:

$$\prod_{i \in I} V_i = \left\{ f : I \rightarrow \bigcup_{i \in I} V_i \mid \forall i \in I : f(i) \in V_i \right\}$$

with pointwise operations $(f + g)(i) = f(i) + g(i)$, $(af)(i) = a \cdot f(i)$, and projections $\pi_j(f) = f(j)$.

We often write f as a tuple $(v_i)_{i \in I}$ where $v_i = f(i)$.

Proof:

Vector space structure: We verify that $\prod V_i$ is a K -vector space with pointwise operations.

Contains zero: Define $0 : I \rightarrow \bigcup V_i$ by $0(i) = 0_{V_i}$. This satisfies $0(i) \in V_i$ for all i , so $0 \in \prod V_i$.

Closure under addition: For $f, g \in \prod V_i$, define $(f + g)(i) = f(i) + g(i)$. Since $f(i), g(i) \in V_i$ and V_i is a vector space, $f(i) + g(i) \in V_i$. Thus $f + g \in \prod V_i$.

Closure under scalar multiplication: For $a \in K$ and $f \in \prod V_i$, define $(af)(i) = a \cdot f(i)$. Since $f(i) \in V_i$ and V_i is a vector space, $a \cdot f(i) \in V_i$. Thus $af \in \prod V_i$.

Additive inverse: For $f \in \prod V_i$, define $(-f)(i) = -f(i)$. Then $(f + (-f))(i) = f(i) + (-f(i)) = 0_{V_i} = 0(i)$, so $f + (-f) = 0$.

Remaining axioms: Commutativity, associativity, distributivity, and scalar identity all follow pointwise from the corresponding properties in each V_i .

Projections are linear: For $f, g \in \prod V_i$ and $a, b \in K$:

$$\begin{aligned} \pi_j(af + bg) &= (af + bg)(j) \\ &= a \cdot f(j) + b \cdot g(j) \\ &= a \cdot \pi_j(f) + b \cdot \pi_j(g) \end{aligned}$$

Thus π_j is linear.

Projections are surjective: For any $v \in V_j$, we construct $f \in \prod V_i$ with $\pi_j(f) = v$. Define $f : I \rightarrow \bigcup V_i$ by:

$$f(i) = \begin{cases} v & \text{if } i = j \\ 0_{V_i} & \text{otherwise} \end{cases}$$

This satisfies $f(i) \in V_i$ for all i , so $f \in \prod V_i$. Moreover, $\pi_j(f) = f(j) = v$. Thus π_j is surjective.

Universal property: Given a family of linear maps $(f_j : W \rightarrow V_j)_{j \in I}$, we construct the unique linear map $\Phi : W \rightarrow \prod V_i$ satisfying $\pi_j \circ \Phi = f_j$ for all j .

Definition: For $w \in W$, define $\Phi(w) : I \rightarrow \bigcup V_i$ by $\Phi(w)(j) = f_j(w)$. Since $f_j(w) \in V_j$ for all j , we have $\Phi(w) \in \prod V_i$.

Linearity: For $w_1, w_2 \in W$ and $a, b \in K$:

$$\begin{aligned} \Phi(aw_1 + bw_2)(j) &= f_j(aw_1 + bw_2) \\ &= a \cdot f_j(w_1) + b \cdot f_j(w_2) \quad (\text{since } f_j \text{ is linear}) \\ &= a \cdot \Phi(w_1)(j) + b \cdot \Phi(w_2)(j) \\ &= (a \cdot \Phi(w_1) + b \cdot \Phi(w_2))(j) \end{aligned}$$

Since this holds for all j , we have $\Phi(aw_1 + bw_2) = a\Phi(w_1) + b\Phi(w_2)$. Thus Φ is linear.

Factorization: For any $w \in W$ and $j \in I$:

$$(\pi_j \circ \Phi)(w) = \pi_j(\Phi(w)) = \Phi(w)(j) = f_j(w)$$

Thus $\pi_j \circ \Phi = f_j$ for all j .

Uniqueness: Suppose $\Psi : W \rightarrow \prod V_i$ is another linear map satisfying $\pi_j \circ \Psi = f_j$ for all j . For any $w \in W$ and $j \in I$:

$$\Psi(w)(j) = \pi_j(\Psi(w)) = (\pi_j \circ \Psi)(w) = f_j(w) = \Phi(w)(j)$$

Since $\Psi(w)$ and $\Phi(w)$ agree on all components j , they are equal as functions. Thus $\Psi(w) = \Phi(w)$ for all w , so $\Psi = \Phi$. QED

10.1.2 The Categorical Coproduct

Definition 10.1.3 (Categorical Coproduct):

A **coproduct** of $(V_i)_{i \in I}$ is a pair $(C, (\iota_j)_{j \in I})$ where C is a K -vector space and $\iota_j : V_j \rightarrow C$ are linear maps (called **injections**), satisfying:

Universal Property: For any K -vector space W and family of linear maps $(g_j : V_j \rightarrow W)_{j \in I}$, there exists a **unique** linear map $g : C \rightarrow W$ such that $g \circ \iota_j = g_j$ for all $j \in I$.

$$\begin{array}{ccccc} V_j & \xrightarrow{\iota_j} & C & \xleftarrow{\iota_k} & V_k \\ & \searrow g_j & \downarrow \exists! g & \swarrow g_k & \\ & & W & & \end{array}$$

Remark (Uniqueness up to Isomorphism):

Similarly to products, if $(C, (\iota_j))$ and $(C', (\iota'_j))$ both satisfy the coproduct universal property, then there exists a unique isomorphism $\psi : C \rightarrow C'$ with $\psi \circ \iota_j = \iota'_j$ for all j . This is a general categorical fact.

Theorem 10.1.4 (Construction of Coproduct):

The coproduct (external direct sum) is realized by functions with finite support:

$$\coprod_{i \in I} V_i = \left\{ f : I \rightarrow \bigcup_{i \in I} V_i \mid f(i) \in V_i \text{ and } f^{\leftarrow} \left[\bigcup_{i \in I} (V_i \setminus \{0\}) \right] \text{ is finite} \right\}$$

Equivalently, the set of all $(v_i)_{i \in I}$ such that $\{i \in I \mid v_i \neq 0\}$ is finite.

The injection $\iota_j(v)$ is the function with $\iota_j(v)(j) = v$ and $\iota_j(v)(i) = 0$ for $i \neq j$.

Proof:

Subspace of product: We verify that $\coprod V_i$ is a subspace of $\prod V_i$.

Contains zero: The zero function $0(i) = 0_{V_i}$ has empty support.

Closure under addition: Let $f, g \in \coprod V_i$ with finite supports F, G . Then $(f + g)(i) = f(i) + g(i)$, and $(f + g)(i) \neq 0$ implies $f(i) \neq 0$ or $g(i) \neq 0$, so the support of $f + g$ is contained in $F \cup G$, which is finite.

Closure under scalar multiplication: For $a \in K$ and f with support F , $(af)(i) = a \cdot f(i)$. If $a \neq 0$, then $(af)(i) \neq 0$ iff $f(i) \neq 0$, so the support of af equals F . If $a = 0$, then $af = 0$ has empty support.

Injections are linear: For $v, w \in V_j$ and $a, b \in K$, we verify:

$$\begin{aligned} \iota_j(av + bw)(i) &= \begin{cases} av + bw & \text{if } i = j \\ 0 & \text{otherwise} \end{cases} \\ (a \cdot \iota_j(v) + b \cdot \iota_j(w))(i) &= a \cdot \iota_j(v)(i) + b \cdot \iota_j(w)(i) \\ &= \begin{cases} av + bw & \text{if } i = j \\ 0 & \text{otherwise} \end{cases} \end{aligned}$$

Thus $\iota_j(av + bw) = a \cdot \iota_j(v) + b \cdot \iota_j(w)$, proving linearity.

Injections are injective: Suppose $\iota_j(v) = 0$, i.e., the zero function. Then:

$$v = \iota_j(v)(j) = 0(j) = 0_{V_j}$$

Thus $\ker(\iota_j) = \{0\}$, proving injectivity.

Universal property: Given $(g_j : V_j \rightarrow W)_{j \in I}$, for $f \in \coprod V_i$ with finite support $F = \{j \in I \mid f(j) \neq 0\}$, define:

$$g(f) = \sum_{j \in F} g_j(f(j))$$

Well-defined: The support F is finite, so the sum has finitely many terms.

Linearity: Let $f, h \in \coprod V_i$ with supports F, G , and let $a, b \in K$. Then $af + bh$ has support $\subseteq F \cup G$. For any $j \in F \cup G$:

$$\begin{aligned} g(af + bh) &= \sum_{j \in F \cup G} g_j((af + bh)(j)) \\ &= \sum_{j \in F \cup G} g_j(af(j) + bh(j)) \\ &= \sum_{j \in F \cup G} (a \cdot g_j(f(j)) + b \cdot g_j(h(j))) \\ &= a \sum_{j \in F \cup G} g_j(f(j)) + b \sum_{j \in F \cup G} g_j(h(j)) \\ &= a \cdot g(f) + b \cdot g(h) \end{aligned}$$

where we used that $g_j(f(j)) = g_j(0) = 0$ for $j \notin F$ (since g_j is linear).

Factorization: For $v \in V_j$, the function $\iota_j(v)$ has support $\{j\}$ (assuming $v \neq 0$):

$$g(\iota_j(v)) = \sum_{k \in \{j\}} g_k(\iota_j(v)(k)) = g_j(v)$$

Thus $g \circ \iota_j = g_j$ for all j .

Uniqueness: Suppose $h : \coprod V_i \rightarrow W$ satisfies $h \circ \iota_j = g_j$ for all j . For $f \in \coprod V_i$ with support F , we can write $f = \sum_{j \in F} \iota_j(f(j))$ (this is a finite sum since F is finite). By linearity of h :

$$h(f) = h\left(\sum_{j \in F} \iota_j(f(j))\right) = \sum_{j \in F} h(\iota_j(f(j))) = \sum_{j \in F} g_j(f(j)) = g(f)$$

Thus $h = g$, proving uniqueness. QED

Example 10.1.5 (Constant Family: $V_i = V$ for all i):

When all factors are the same space V , the product and coproduct have special notation:

$$\prod_{i \in I} V = V^I, \quad \coprod_{i \in I} V = V^{(I)}$$

Product V^I : The set of all functions $f : I \rightarrow V$, with pointwise operations. This is the space of I -indexed sequences in V .

Coproduct $V^{(I)}$: The set of all functions $f : I \rightarrow V$ with finite support. This is the space of I -indexed sequences in V with only finitely many non-zero terms.

Proof of Example 10.1.5:

Product V^I : By Theorem 10.1.2:

$$\prod_{i \in I} V = \left\{ f : I \rightarrow \bigcup_{i \in I} V \mid \forall i \in I : f(i) \in V \right\}$$

Since each $V_i = V$, the disjoint union $\bigcup_{i \in I} V = V$ (all copies are the same). Thus the condition $f(i) \in V$ is automatic for any function $f : I \rightarrow V$. Hence $\prod_{i \in I} V = \{f : I \rightarrow V\} = V^I$.

Coproduct $V^{(I)}$: By Theorem 10.1.4:

$$\coprod_{i \in I} V = \{f \in V^I \mid \{i \in I \mid f(i) \neq 0\} \text{ is finite}\}$$

This is precisely the definition of $V^{(I)}$: functions from I to V with finite support. QED

10.1.3 Finite Case: Biproduct Structure

Theorem 10.1.6 (Finite Case: Product Equals Coproduct):

If $|I| \in \mathbb{N}$, then $\prod_{i \in I} V_i \cong \coprod_{i \in I} V_i$. In fact, the two constructions coincide as sets: every tuple has finite support.

$$\begin{array}{ccccc}
 V_j & \xrightarrow{\iota_j} & \prod V_i & \xrightleftharpoons[\psi]{\phi} & \prod V_i & \xrightarrow{\pi_k} & V_k \\
 \parallel & & & & & & \parallel \\
 V_j & \xrightarrow{\quad \delta_{jk} \quad} & & & & & V_k
 \end{array}$$

The diagram commutes: $\pi_k \circ \phi \circ \iota_j = \delta_{jk}$, and ϕ, ψ are inverse isomorphisms.

Proof (Categorical):

We give a purely categorical proof that works in any additive category.

Step 1: $K\text{-Vect}$ is an additive category. An *additive category* is a category where:

1. There exists a *zero object* 0 (both initial and terminal)
2. Every hom-set $\text{Hom}(X, Y)$ is an abelian group under addition
3. Composition distributes over addition: $(f + g) \circ h = f \circ h + g \circ h$ and $h \circ (f + g) = h \circ f + h \circ g$

$K\text{-Vect}$ satisfies these: the trivial space $\{0\}$ is the zero object, morphisms can be added pointwise, and composition is bilinear.

Step 2: Existence of zero morphisms. For any V, W , the *zero morphism* $0_{VW} : V \rightarrow W$ is the unique composition $V \rightarrow 0 \rightarrow W$. In $K\text{-Vect}$, this is the zero linear map $0_{VW}(v) = 0_W$.

Step 3: Define the canonical morphism ϕ . We construct $\phi : \coprod_{j \in n} V_j \rightarrow \prod_{i \in n} V_i$ using both universal properties.

Step 3a: By the coproduct universal property, to define ϕ it suffices to specify the compositions $\phi \circ \iota_j : V_j \rightarrow \prod V_i$ for each $j \in \{0, \dots, n-1\}$.

Step 3b: By the product universal property, each $\phi \circ \iota_j$ is uniquely determined by specifying $\pi_i \circ (\phi \circ \iota_j) : V_j \rightarrow V_i$ for each i .

Step 3c: We define these component maps as the Kronecker delta:

$$\pi_i \circ \phi \circ \iota_j = \delta_{ij} = \begin{cases} \text{Id}_{V_i} & \text{if } i = j \\ 0_{V_j V_i} & \text{if } i \neq j \end{cases}$$

This uniquely determines $\phi \circ \iota_j$ for each j , which in turn uniquely determines ϕ by the coproduct universal property.

Step 4: Construct the inverse ψ . In an additive category, we can sum morphisms with common domain and codomain. Define:

$$\psi = \sum_{k \in n} \iota_k \circ \pi_k : \prod V_i \rightarrow \coprod V_j$$

Why well-defined: Each $\iota_k \circ \pi_k : \prod V_i \rightarrow \coprod V_j$ is a morphism in $\text{Hom}(\prod V_i, \coprod V_j)$. The sum of n morphisms is well-defined in an additive category. This requires n to be finite.

Step 5: Verify $\psi \circ \phi = \text{Id}_{\coprod V_i}$.

Strategy: By the uniqueness part of the coproduct universal property, it suffices to show that $(\psi \circ \phi) \circ \iota_j = \iota_j$ for all j .

Computation: For each $j \in \{0, \dots, n-1\}$:

$$\begin{aligned}
 (\psi \circ \phi) \circ \iota_j &= \psi \circ (\phi \circ \iota_j) && \text{(associativity)} \\
 &= \left(\sum_{k \in n} \iota_k \circ \pi_k \right) \circ (\phi \circ \iota_j) && \text{(definition of } \psi) \\
 &= \sum_{k \in n} (\iota_k \circ \pi_k \circ \phi \circ \iota_j) && \text{(distributivity in additive category)} \\
 &= \sum_{k \in n} (\iota_k \circ (\pi_k \circ \phi \circ \iota_j)) && \text{(associativity)} \\
 &= \sum_{k \in n} \iota_k \circ \delta_{kj} && \text{(Step 3c)} \\
 &= \iota_j \circ \text{Id}_{V_j} + \sum_{k \neq j} \iota_k \circ 0_{V_j V_k} && \text{(split sum)} \\
 &= \iota_j + 0 = \iota_j && (\iota_k \circ 0 = 0 \text{ in additive category})
 \end{aligned}$$

Thus $(\psi \circ \phi) \circ \iota_j = \iota_j$ for all j , so $\psi \circ \phi = \text{Id}_{\prod V_i}$.

Step 6: Verify $\phi \circ \psi = \text{Id}_{\prod V_i}$.

Strategy: By the uniqueness part of the product universal property, it suffices to show that $\pi_i \circ (\phi \circ \psi) = \pi_i$ for all i .

Computation: For each $i \in \{0, \dots, n-1\}$:

$$\begin{aligned}
 \pi_i \circ (\phi \circ \psi) &= \pi_i \circ \phi \circ \left(\sum_{k \in n} \iota_k \circ \pi_k \right) && \text{(definition of } \psi) \\
 &= \sum_{k \in n} (\pi_i \circ \phi \circ \iota_k \circ \pi_k) && \text{(distributivity)} \\
 &= \sum_{k \in n} ((\pi_i \circ \phi \circ \iota_k) \circ \pi_k) && \text{(associativity)} \\
 &= \sum_{k \in n} \delta_{ik} \circ \pi_k && \text{(Step 3c)} \\
 &= \text{Id}_{V_i} \circ \pi_i + \sum_{k \neq i} 0_{V_k V_i} \circ \pi_k && \text{(split sum)} \\
 &= \pi_i + 0 = \pi_i && (0 \circ f = 0)
 \end{aligned}$$

Thus $\pi_i \circ (\phi \circ \psi) = \pi_i$ for all i , so $\phi \circ \psi = \text{Id}_{\prod V_i}$.

Conclusion: Since ϕ has a two-sided inverse ψ , it is an isomorphism. This result **requires finiteness** because:

- $\psi = \sum_{k \in n} \iota_k \circ \pi_k$ is a finite sum of morphisms
- For infinite I , we cannot form such a sum in general categories
- In $K\text{-Vect}$ with infinite I : the image of ψ would have finite support, but $\prod V_i$ contains elements with infinite support

QED

10.1.4 Dimensions of Products and Coproducts

Theorem 10.1.7 (Dimension of Product and Coproduct):

Let $(V_i)_{i \in I}$ be a family of K -vector spaces with $\dim_K(V_i) = d_i$.

1. $\dim_K(\coprod_{i \in I} V_i) = \sum_{i \in I} d_i$ (sum of cardinals, finite for countable families with finite-dim factors)
2. For finite I with $|I| = n \in \mathbb{N}$: $\dim_K(\prod_{i \in n} V_i) = \sum_{i \in n} d_i$
3. For infinite I : $\dim_K(\prod_{i \in I} V_i)$ can be strictly larger than $\sum_{i \in I} d_i$

Proof:

(1) For each i , let B_i be a basis of V_i . Then $\{\iota_i(b) \mid i \in I, b \in B_i\}$ is a basis of $\coprod V_i$.

Linear independence: If $\sum_{j \in F} \sum_{b \in B_j} a_{j,b} \iota_j(b) = 0$, then projecting to each component shows $\sum_{b \in B_j} a_{j,b} b = 0$ in V_j . By linear independence of B_j , all $a_{j,b} = 0$.

Spanning: Every $f \in \coprod V_i$ with support F can be written as $\sum_{j \in F} \iota_j(f(j))$, and each $f(j) \in \text{span}_K(B_j)$.

(2) Follows from (1) since product = coproduct when I is finite.

(3) Consider $I = \mathbb{N}$ and $V_i = K$ for all i . Then $\prod_{i \in \mathbb{N}} K = K^{\mathbb{N}}$ has dimension $|K|^{\aleph_0}$, while $\coprod_{i \in \mathbb{N}} K = K^{(\mathbb{N})}$ has dimension $\aleph_0$. QED

Remark (Dimension Consequences for Constant Families):

For the constant family $V_i = V$ with $\dim_K(V) = d$:

1. $\dim_K(V^{(I)}) = d \cdot |I|$ (sum of dimensions)
2. For finite I with $|I| = n$: $\dim_K(V^n) = nd$ and $V^n = V^{(n)}$
3. For infinite I : $\dim_K(V^I) > d \cdot |I|$ (strict inequality)

Special case $V = K$: $K^{(I)}$ has dimension $|I|$ with basis $\{\delta_i\}_{i \in I}$ (indicator functions). K^I has dimension $|K|^{|I|}$ when I is infinite.

Infinite case: For infinite I , product and coproduct are **not** isomorphic because $\coprod V_i$ has finite support while $\prod V_i$ contains elements with infinite support.

Theorem 10.1.8 (Dimension of Infinite Products):

Let I be an infinite index set and $(V_i)_{i \in I}$ be a family of non-trivial K -vector spaces. Then:

$$\dim_K\left(\prod_{i \in I} V_i\right) = \left|\prod_{i \in I} V_i\right| = \prod_{i \in I} |V_i|$$

Proof:

Let $P = \prod_{i \in I} V_i$ and $d = \dim_K(P)$.

Step 1: Cardinality bounds. Since each V_i is non-trivial, $|V_i| \geq |K| \geq 2$ (any field has at least 2 elements). Thus:

$$|P| = \prod_{i \in I} |V_i| \geq 2^{|I|}$$

Since I is infinite, $|I| \geq \aleph_0$, so $|P| \geq 2^{\aleph_0}$ is infinite.

For any infinite vector space P over K :

$$|P| = \max(|K|, d)$$

This gives the upper bound: $d \leq |P|$.

Step 2: Case A ($|P| > |K|$). If $|P| > |K|$, then from $|P| = \max(|K|, d)$ we get $d = |P|$.

Step 3: Case B ($|P| = |K|$). We must show $d \geq |K|$ by constructing a linearly independent set of size $|K|$.

Construction: Since I is infinite, there exists an injection $\nu : \mathbb{N} \hookrightarrow I$. For each $\alpha \in K$, define $u_\alpha \in P$ by:

$$u_\alpha(x) = \begin{cases} \alpha^n & \text{if } x = \nu(n) \text{ for some } n \in \mathbb{N} \\ 0 & \text{otherwise} \end{cases}$$

This is the “geometric sequence in K ” embedded along the image of ν . Let $S = \{u_\alpha \mid \alpha \in K\}$. Then $|S| = |K|$.

Linear independence: Suppose $\sum_{j \in m} c_j u_{\alpha_j} = 0$ for distinct $\alpha_0, \dots, \alpha_{m-1}$. Evaluating at coordinates $\nu(0), \nu(1), \dots, \nu(m-1)$:

$$\sum_{j \in m} c_j \alpha_j^n = 0 \quad \text{for } n = 0, 1, \dots, m-1$$

In matrix form: $\mathbf{V} \cdot \mathbf{c} = 0$ where $\mathbf{V}$ is the Vandermonde matrix with entries $\mathbf{V}_{nj} = \alpha_j^n$.

The Vandermonde determinant is:

$$\det(\mathbf{V}) = \prod_{0 \leq r < s \leq m-1} (\alpha_s - \alpha_r) \neq 0$$

since the α_j are distinct. Thus $\mathbf{V}$ is invertible, so $\mathbf{c} = 0$.

Therefore S is linearly independent, giving $d \geq |K|$.

Step 4: Conclusion. Combining $|K| \leq d \leq |P| = |K|$, we get $d = |P|$.

In both cases, $\dim_K(P) = |P| = \prod_{i \in I} |V_i|$.

QED

Corollary 10.1.9 (Dimension of Power Spaces):

For a non-trivial K -vector space V and infinite index set I :

$$\dim_K(V^I) = |V^I| = |V|^{|I|}$$

In particular, for infinite I : $\dim_K(K^I) = |K|^{|I|}$.

Proof:

Apply Theorem 10.1.8 with $V_i = V$ for all $i \in I$. Then $\prod_{i \in I} |V_i| = |V|^{|I|}$.

QED

Proposition 10.1.10 (Trivial Factors Do Not Contribute):

Let $(V_i)_{i \in I}$ be a family of K -vector spaces and let $J = \{i \in I \mid V_i \neq \{0\}\}$ be the set of indices with non-trivial factors. Then:

1. $\prod_{i \in I} V_i \cong \prod_{j \in J} V_j$ (natural isomorphism)
2. $\coprod_{i \in I} V_i \cong \coprod_{j \in J} V_j$ (natural isomorphism)

In particular, trivial factors can be “removed” without changing the product or coproduct.

Proof:

(1) Product isomorphism: Define $\phi : \prod_{i \in I} V_i \rightarrow \prod_{j \in J} V_j$ by restriction:

$$\phi(f) = f|_J \quad (\text{where } f|_J(j) = f(j) \text{ for } j \in J)$$

ϕ is well-defined and linear: For $f \in \prod_{i \in I} V_i$, we have $f(j) \in V_j$ for each $j \in J$, so $f|_J \in \prod_{j \in J} V_j$. Linearity: $(af + bg)|_J = a(f|_J) + b(g|_J)$.

ϕ is injective: Suppose $\phi(f) = 0$, i.e., $f(j) = 0$ for all $j \in J$. For $i \notin J$, we have $V_i = \{0\}$, so $f(i) = 0$ automatically. Thus $f = 0$.

ϕ is surjective: Let $g \in \prod_{j \in J} V_j$. Define $f \in \prod_{i \in I} V_i$ by:

$$f(i) = \begin{cases} g(i) & \text{if } i \in J \\ 0 & \text{if } i \notin J \end{cases} \quad (\text{this is the only element of } V_i = \{0\})$$

Then $\phi(f) = f|_J = g$.

(2) Coproduct isomorphism: Define $\psi : \coprod_{i \in I} V_i \rightarrow \coprod_{j \in J} V_j$ by restriction:

$$\psi(f) = f|_J$$

ψ is well-defined: If f has finite support $F \subseteq I$, then $f|_J$ has support $F \cap J$, which is also finite. For $i \in F \setminus J$: $f(i) \in V_i = \{0\}$, so $f(i) = 0$, contradicting $i \in F$. Thus $F \subseteq J$, so $f|_J$ has the same non-zero values.

ψ is linear: Same argument as for ϕ .

ψ is an isomorphism: Injectivity and surjectivity follow by the same arguments as for ϕ , since any element of the coproduct already has support in J . QED

Corollary 10.1.11 (Dimension with Potentially Trivial Factors):

Let I be an index set and $(V_i)_{i \in I}$ be a family of K -vector spaces. Let $J = \{i \in I \mid V_i \neq \{0\}\}$. Then:

1. If $J = \emptyset$: $\prod_{i \in I} V_i = \coprod_{i \in I} V_i = \{0\}$.
2. If J is finite: $\dim_K (\prod V_i) = \dim_K (\coprod V_i) = \sum_{j \in J} \dim_K (V_j)$.
3. If J is infinite: $\dim_K (\prod V_i) = \prod_{j \in J} |V_j|$ and $\dim_K (\coprod V_i) = \sum_{j \in J} \dim_K (V_j)$.

Proof:

Apply Proposition 10.1.10 to reduce to the non-trivial subfamily $(V_j)_{j \in J}$, then apply Theorem 10.1.7 (finite case) or Theorem 10.1.8 (infinite case). QED

10.2 Internal vs External Direct Sums

Theorem 10.2.1 (Internal-External Equivalence):

Let $(U_i)_{i \in I}$ be a family of subspaces of V . The following are equivalent:

1. The sum $\sum_{i \in I} U_i$ is a direct sum (internal)
2. The natural map $\phi : \coprod_{i \in I} U_i \rightarrow V$ defined by $\phi((u_i)) = \sum_{i \in F} u_i$ is an isomorphism onto $\sum_{i \in I} U_i$

Proof:

Construction of ϕ : Define the inclusions $\iota_j^V : U_j \hookrightarrow V$ for each $j \in I$. By the coproduct universal property (Theorem 10.1.4), there exists a unique linear map $\phi : \coprod_{i \in I} U_i \rightarrow V$ such that $\phi \circ \iota_j = \iota_j^V$ for all j . Explicitly, for $f \in \coprod U_i$ with finite support F :

$$\phi(f) = \sum_{j \in F} \iota_j^V(f(j)) = \sum_{j \in F} f(j) \in V$$

ϕ is well-defined: Each $f(j) \in U_j \subseteq V$, so the sum $\sum_{j \in F} f(j)$ is a finite sum in V . The result is independent of how we write f , since if $f = \sum_{j \in F} \iota_j(f(j))$, then $\phi(f) = \sum_{j \in F} f(j)$ by the coproduct universal property.

ϕ is linear: This follows from the coproduct construction: ϕ is the unique map induced by the linear inclusions ι_j^V , hence ϕ is linear.

Alternatively, for $f, g \in \coprod U_i$ with supports F, G and $a, b \in K$:

$$\begin{aligned} \phi(af + bg) &= \sum_{j \in F \cup G} (af + bg)(j) \\ &= \sum_{j \in F \cup G} (af(j) + bg(j)) \\ &= a \sum_{j \in F \cup G} f(j) + b \sum_{j \in F \cup G} g(j) \\ &= a \cdot \phi(f) + b \cdot \phi(g) \end{aligned}$$

Image of ϕ : We claim $\text{Rng}(\phi) = \sum_{i \in I} U_i$.

$\text{Rng}(\phi) \subseteq \sum U_i$: For any $f \in \coprod U_i$, $\phi(f) = \sum_{j \in F} f(j)$ where each $f(j) \in U_j$. Thus $\phi(f) \in \sum U_i$.

$\sum U_i \subseteq \text{Rng}(\phi)$: Any $v \in \sum U_i$ can be written as $v = \sum_{j \in F} u_j$ for some finite F and $u_j \in U_j$. Define $f \in \coprod U_i$ by $f(j) = u_j$ for $j \in F$ and $f(i) = 0$ otherwise. Then $\phi(f) = v$.

(1) $\Rightarrow$ (2): Direct sum implies ϕ is an isomorphism

Assume $\sum U_i$ is a direct sum, i.e., every $v \in \sum U_i$ has a **unique** representation as $v = \sum_{j \in F} u_j$ with $u_j \in U_j$.

We already showed ϕ is linear and $\text{Rng}(\phi) = \sum U_i$ (surjective onto image). It remains to show ϕ is injective.

Injectivity: Suppose $\phi(f) = 0$ for some $f \in \coprod U_i$ with support F . Then $\sum_{j \in F} f(j) = 0$ with $f(j) \in U_j$. By the unique representation property (direct sum), the only way to write 0 as a sum from the U_j is with all summands zero. Thus $f(j) = 0$ for all $j \in F$, which means $f = 0$. Hence $\ker(\phi) = \{0\}$, proving injectivity.

(2) $\Rightarrow$ (1): ϕ isomorphism implies direct sum

Assume $\phi : \coprod U_i \rightarrow \sum U_i$ is an isomorphism (in particular, injective).

We must show that if $\sum_{j \in F} u_j = 0$ with $u_j \in U_j$ (all but finitely many zero), then $u_j = 0$ for all j .

Define $f \in \coprod U_i$ by $f(j) = u_j$ for $j \in F$ and $f(i) = 0$ otherwise. Then $\phi(f) = \sum_{j \in F} u_j = 0$.

Since ϕ is injective, $\phi(f) = 0$ implies $f = 0$. Thus $f(j) = u_j = 0$ for all $j \in F$.

This proves the unique representation property, so $\sum U_i$ is a direct sum.

QED

Corollary 10.2.2 (Basis as Direct Sum):

$(v_i)_{i \in I}$ is a basis for V iff $V = \bigoplus_{i \in I} \text{span}_K(\{v_i\})$.

10.3 Hom-Product Isomorphisms

Theorem 10.3.1 (Contravariance in First Argument):

For any family $(U_i)_{i \in I}$ and vector space V :

$$\text{Hom}_K \left(\coprod_{i \in I} U_i, V \right) \cong \prod_{i \in I} \text{Hom}_K (U_i, V)$$

Proof:

Definition of Φ : Define $\Phi : \text{Hom}_K (\coprod U_i, V) \rightarrow \prod \text{Hom}_K (U_i, V)$ by

$$\Phi(T) = (T \circ \iota_i)_{i \in I}$$

For each i , $T \circ \iota_i : U_i \rightarrow V$ is a composition of linear maps, hence linear. Thus $\Phi(T) \in \prod \text{Hom}_K (U_i, V)$.

Φ is well-defined: For any $T \in \text{Hom}_K (\coprod U_i, V)$, the tuple $(T \circ \iota_i)_{i \in I}$ is a function from I to $\bigcup \text{Hom}_K (U_i, V)$ with $(T \circ \iota_i) \in \text{Hom}_K (U_i, V)$ for each i . This is exactly the definition of an element in the product.

Φ is linear: For $S, T \in \text{Hom}_K (\coprod U_i, V)$ and $a, b \in K$:

$$\begin{aligned} \Phi(aS + bT) &= ((aS + bT) \circ \iota_i)_{i \in I} \\ &= (aS \circ \iota_i + bT \circ \iota_i)_{i \in I} && \text{(linearity of composition)} \\ &= a(S \circ \iota_i)_{i \in I} + b(T \circ \iota_i)_{i \in I} && \text{(pointwise operations)} \\ &= a \cdot \Phi(S) + b \cdot \Phi(T) \end{aligned}$$

Φ is injective: Suppose $\Phi(T) = 0$, i.e., $T \circ \iota_i = 0$ for all $i \in I$. We must show $T = 0$.

Let $f \in \coprod U_i$ be arbitrary with finite support $F = \{j \in I \mid f(j) \neq 0\}$. By the coproduct construction, $f = \sum_{j \in F} \iota_j(f(j))$. Then:

$$\begin{aligned} T(f) &= T \left(\sum_{j \in F} \iota_j(f(j)) \right) \\ &= \sum_{j \in F} T(\iota_j(f(j))) && (T \text{ is linear}) \\ &= \sum_{j \in F} (T \circ \iota_j)(f(j)) && \text{(definition of composition)} \\ &= \sum_{j \in F} 0(f(j)) && (T \circ \iota_j = 0 \text{ by assumption}) \\ &= \sum_{j \in F} 0_V = 0_V \end{aligned}$$

Since $T(f) = 0$ for all $f \in \coprod U_i$, we have $T = 0$. Thus $\ker(\Phi) = \{0\}$, proving injectivity.

Φ is surjective: Let $(f_i)_{i \in I} \in \prod \text{Hom}_K (U_i, V)$ be arbitrary. By the coproduct universal property (Definition 10.1.3), there exists a unique linear map $T : \coprod U_i \rightarrow V$ such that $T \circ \iota_i = f_i$ for all i .

Explicitly, for $g \in \coprod U_i$ with support F :

$$T(g) = \sum_{j \in F} f_j(g(j))$$

We verify $\Phi(T) = (f_i)_{i \in I}$:

$$\Phi(T) = (T \circ \iota_i)_{i \in I} = (f_i)_{i \in I}$$

since $T \circ \iota_i = f_i$ by construction. Thus Φ is surjective.

Conclusion: Φ is a linear bijection, hence an isomorphism.

QED

Theorem 10.3.2 (Covariance in Second Argument):

For any vector space U and family $(V_i)_{i \in I}$:

$$\text{Hom}_K \left(U, \prod_{i \in I} V_i \right) \cong \prod_{i \in I} \text{Hom}_K (U, V_i)$$

Proof:

Definition of Ψ : Define $\Psi : \text{Hom}_K (U, \prod V_i) \rightarrow \prod \text{Hom}_K (U, V_i)$ by

$$\Psi(T) = (\pi_i \circ T)_{i \in I}$$

For each i , $\pi_i \circ T : U \rightarrow V_i$ is a composition of linear maps, hence linear. Thus $\Psi(T) \in \prod \text{Hom}_K (U, V_i)$.

Ψ is linear: For $S, T \in \text{Hom}_K (U, \prod V_i)$ and $a, b \in K$:

$$\begin{aligned} \Psi(aS + bT) &= (\pi_i \circ (aS + bT))_{i \in I} \\ &= (a \cdot (\pi_i \circ S) + b \cdot (\pi_i \circ T))_{i \in I} && (\pi_i \text{ is linear}) \\ &= a(\pi_i \circ S)_{i \in I} + b(\pi_i \circ T)_{i \in I} && (\text{pointwise operations}) \\ &= a \cdot \Psi(S) + b \cdot \Psi(T) \end{aligned}$$

Ψ is injective: Suppose $\Psi(T) = 0$, i.e., $\pi_i \circ T = 0$ for all $i \in I$. We must show $T = 0$.

Let $u \in U$ be arbitrary. We have $T(u) \in \prod V_i$, so $T(u)$ is a function from I to $\bigcup V_i$. For each $i \in I$:

$$T(u)(i) = \pi_i(T(u)) = (\pi_i \circ T)(u) = 0(u) = 0_{V_i}$$

Since $T(u)(i) = 0$ for all i , the function $T(u)$ is the zero function in $\prod V_i$. Thus $T(u) = 0$ for all $u \in U$, proving $T = 0$. Hence $\ker(\Psi) = \{0\}$, proving injectivity.

Ψ is surjective: Let $(g_i)_{i \in I} \in \prod \text{Hom}_K (U, V_i)$ be arbitrary. By the product universal property (Definition 10.1.1), there exists a unique linear map $T : U \rightarrow \prod V_i$ such that $\pi_i \circ T = g_i$ for all i .

Explicitly, for $u \in U$, define $T(u) : I \rightarrow \bigcup V_i$ by:

$$T(u)(i) = g_i(u) \in V_i$$

T is well-defined: For each u , $T(u)$ is a function with $T(u)(i) \in V_i$, so $T(u) \in \prod V_i$.

T is linear: For $u, v \in U$ and $a, b \in K$, and for each $i \in I$:

$$\begin{aligned} T(au + bv)(i) &= g_i(au + bv) = a \cdot g_i(u) + b \cdot g_i(v) \\ &= a \cdot T(u)(i) + b \cdot T(v)(i) = (aT(u) + bT(v))(i) \end{aligned}$$

Thus $T(au + bv) = aT(u) + bT(v)$.

We verify $\Psi(T) = (g_i)_{i \in I}$:

$$(\pi_i \circ T)(u) = \pi_i(T(u)) = T(u)(i) = g_i(u)$$

Thus $\pi_i \circ T = g_i$ for all i , so $\Psi(T) = (g_i)_{i \in I}$.

Conclusion: Ψ is a linear bijection, hence an isomorphism. QED

Theorem 10.3.3 (Finite-Dimensional Covariance):

If $\dim_K (U) \in \mathbb{N}$, then:

$$\text{Hom}_K \left(U, \prod_{i \in I} V_i \right) \cong \prod_{i \in I} \text{Hom}_K (U, V_i)$$

Proof:

Let $\dim_K(U) = n$ with basis $\mathcal{B} = (u_0, \dots, u_{n-1})$.

Definition of Θ : Define $\Theta : \text{Hom}_K(U, \coprod V_i) \rightarrow \coprod \text{Hom}_K(U, V_i)$ by

$$\Theta(T) = (\pi_i \circ T)_{i \in I}$$

For each i , $\pi_i \circ T : U \rightarrow V_i$ is a composition of linear maps, hence linear.

$\Theta(T)$ has finite support (crucial for well-definedness): For each basis vector u_k , we have $T(u_k) \in \coprod V_i$, which has finite support

$$F_k = \{i \in I \mid \pi_i(T(u_k)) \neq 0\}$$

The total support is $F = \bigcup_{k \in n} F_k$, which is a finite union of finite sets, hence finite.

For any $i \notin F$ and any $u = \sum_{k \in n} a_k u_k \in U$:

$$\begin{aligned} (\pi_i \circ T)(u) &= \pi_i \left(T \left(\sum_{k \in n} a_k u_k \right) \right) \\ &= \pi_i \left(\sum_{k \in n} a_k T(u_k) \right) && (T \text{ is linear}) \\ &= \sum_{k \in n} a_k \cdot \pi_i(T(u_k)) && (\pi_i \text{ is linear}) \\ &= \sum_{k \in n} a_k \cdot 0 = 0 && (i \notin F_k \text{ for all } k) \end{aligned}$$

Thus $\pi_i \circ T = 0$ for all $i \notin F$, so $\Theta(T) \in \coprod \text{Hom}_K(U, V_i)$.

Θ is linear: For $S, T \in \text{Hom}_K(U, \coprod V_i)$ and $a, b \in K$:

$$\begin{aligned} \Theta(aS + bT) &= (\pi_i \circ (aS + bT))_{i \in I} \\ &= (a \cdot (\pi_i \circ S) + b \cdot (\pi_i \circ T))_{i \in I} && (\pi_i \text{ is linear}) \\ &= a(\pi_i \circ S)_{i \in I} + b(\pi_i \circ T)_{i \in I} && (\text{pointwise operations}) \\ &= a \cdot \Theta(S) + b \cdot \Theta(T) \end{aligned}$$

Θ is injective: Suppose $\Theta(T) = 0$, i.e., $\pi_i \circ T = 0$ for all $i \in I$. We must show $T = 0$.

Let $u \in U$ be arbitrary. We have $T(u) \in \coprod V_i$. For each $i \in I$:

$$T(u)(i) = \pi_i(T(u)) = (\pi_i \circ T)(u) = 0(u) = 0_{V_i}$$

Since $T(u)(i) = 0$ for all i , the function $T(u)$ is the zero function. Thus $T(u) = 0$ for all $u \in U$, proving $T = 0$. Hence $\ker(\Theta) = \{0\}$, proving injectivity.

Θ is surjective: Let $(f_i)_{i \in I} \in \coprod \text{Hom}_K(U, V_i)$ with finite support F .

Define $T : U \rightarrow \coprod V_i$ by: for each $u \in U$,

$$T(u)(i) = f_i(u) \in V_i$$

$T(u) \in \coprod$: The support of $T(u)$ is $\{i \in I \mid f_i(u) \neq 0\} \subseteq F$, which is finite.

T is linear: For $u, v \in U$ and $a, b \in K$, and for each $i \in I$:

$$\begin{aligned} T(au + bv)(i) &= f_i(au + bv) = a \cdot f_i(u) + b \cdot f_i(v) \\ &= a \cdot T(u)(i) + b \cdot T(v)(i) = (aT(u) + bT(v))(i) \end{aligned}$$

Thus $T(au + bv) = aT(u) + bT(v)$.

We verify $\Theta(T) = (f_i)_{i \in I}$:

$$(\pi_i \circ T)(u) = \pi_i(T(u)) = T(u)(i) = f_i(u)$$

Thus $\pi_i \circ T = f_i$ for all i , so $\Theta(T) = (f_i)_{i \in I}$.

Conclusion: Θ is a linear bijection, hence an isomorphism.

Note: This proof requires $\dim_K(U)$ to be finite because finite-dimensionality ensures that $F = \bigcup_{k \in \mathbb{N}} F_k$ is a finite union. For infinite-dimensional U , this union could be infinite, and $\Theta(T)$ might not have finite support. QED

Remark (Categorical Interpretation):

These isomorphisms express that $\text{Hom}_K(-, V)$ is contravariant (coproducts $\rightarrow$ products), $\text{Hom}_K(U, -)$ is covariant (products $\rightarrow$ products), and for finite-dimensional U , $\text{Hom}_K(U, -)$ also sends coproducts to coproducts.

10.4 Free Vector Space Characterizations

Theorem 10.4.1 (Equivalent Characterizations):

For a K -vector space V with set $B \subseteq V$, the following are equivalent:

1. B is a basis for V
2. The universal property: any $f : B \rightarrow W$ extends uniquely to $T : V \rightarrow W$
3. $V \cong \coprod_{b \in B} \text{span}_K(\{b\})$
4. $V \cong K^{(B)}$

Proof:

(1) $\Leftrightarrow$ (2): Theorem 6.2.1.

(1) $\Rightarrow$ (3): By Theorem 10.2.1, since $V = \bigoplus_{b \in B} \text{span}_K(\{b\})$ (internal).

(3) $\Rightarrow$ (4): Each $\text{span}_K(\{b\}) \cong K$, so $\coprod_{b \in B} \text{span}_K(\{b\}) \cong \coprod_{b \in B} K = K^{(B)}$.

(4) $\Rightarrow$ (1): $K^{(B)}$ has basis $\{\delta_b\}_{b \in B}$. Any isomorphism $\psi : K^{(B)} \rightarrow V$ sends this to a basis of V . QED

Corollary 10.4.2 (Coordinate Representation):

For any basis B of V , the coordinate map $\phi^B : V \rightarrow K^{(B)}$ is an isomorphism. For $\dim_K(V) = n \in \mathbb{N}$: $V \cong K^n$.

10.5 Dimension of Hom Spaces

Proposition 10.5.1 ($\text{Hom}_K(K, V) \cong V$):

For any K -vector space V , $\text{Hom}_K(K, V) \cong V$ via:

$$\Phi : \text{Hom}_K(K, V) \rightarrow V, \quad \Phi(T) = T(1_K)$$

with inverse $\Phi^{-1}(v) = (a \mapsto av)$.

Proof:

Φ is well-defined: For $T \in \text{Hom}_K(K, V)$, $T(1_K) \in V$.

Φ is linear: $\Phi(aS + bT) = (aS + bT)(1_K) = aS(1_K) + bT(1_K) = a\Phi(S) + b\Phi(T)$.

Φ^{-1} is well-defined: For $v \in V$, define $T_v : K \rightarrow V$ by $T_v(a) = av$. Then $T_v(ab + cd) = (ab + cd)v = a(bv) + c(dv) = aT_v(b) + cT_v(d)$, so $T_v \in \text{Hom}_K(K, V)$.

Inverse: $\Phi(\Phi^{-1}(v)) = \Phi(T_v) = T_v(1_K) = 1_K \cdot v = v$. $\Phi^{-1}(\Phi(T)) = \Phi^{-1}(T(1_K)) = T_{T(1_K)}$. For any $a \in K$: $T_{T(1_K)}(a) = a \cdot T(1_K) = T(a \cdot 1_K) = T(a)$. Thus $T_{T(1_K)} = T$. QED

Remark (Asymmetry: $\text{Hom}_K(V, K)$ is different):

Unlike $\text{Hom}_K(K, V) \cong V$ which holds for all V , the space $\text{Hom}_K(V, K)$ is **not** isomorphic to V in general:

- For finite-dimensional V : $\dim_K(\text{Hom}_K(V, K)) = \dim_K(V)$, but the isomorphism depends on a choice of basis (non-canonical).
- For infinite-dimensional V : $\dim_K(\text{Hom}_K(V, K)) > \dim_K(V)$, so no isomorphism exists.

Corollary 10.5.2 ($\text{Hom}_K(K, K) \cong K$):

$\text{Hom}_K(K, K) \cong K$.

Proof:

Apply Proposition 10.5.1 with $V = K$. QED

Theorem 10.5.3 (Isomorphism for $\text{Hom}_K(U, V)$):

Let U, V be K -vector spaces with bases B_U and B_V respectively.

If $\dim_K(V)$ is finite, then:

$$\begin{aligned}
 \text{Hom}_K(U, V) &\cong \text{Hom}_K\left(K^{(B_U)}, V\right) && (U \cong K^{(B_U)}) \\
 &\cong \prod_{b \in B_U} \text{Hom}_K(K, V) && (\text{contravariance: coproduct} \rightarrow \text{product}) \\
 &\cong \prod_{b \in B_U} \text{Hom}_K\left(K, K^{(B_V)}\right) && (V \cong K^{(B_V)}) \\
 &\cong \prod_{b \in B_U} \prod_{b' \in B_V} \text{Hom}_K(K, K) && (\text{covariance, } \dim_K(V) \text{ finite}) \\
 &\cong \prod_{b \in B_U} \prod_{b' \in B_V} K && (\text{Corollary 10.5.2}) \\
 &\cong \prod_{b \in B_U} K^{(\dim_K(V))} && \left(\prod_{b' \in B_V} K \cong K^{(|B_V|)}\right)
 \end{aligned}$$

Thus $\text{Hom}_K(U, V) \cong \prod_{b \in B_U} K^{(\dim_K(V))}$, which has dimension $|K^{(\dim_K(V))}|^{\dim_K(U)}$.

Corollary 10.5.4 (Finite-Dimensional Case):

If $\dim_K(U) = m \in \mathbb{N}$ and $\dim_K(V) = n \in \mathbb{N}$, then:

$$\dim_K(\text{Hom}_K(U, V)) = m \cdot n$$

Proof:

By Theorem 10.5.3, $\text{Hom}_K(U, V) \cong \prod_{i \in m} K^{(n)}$. Since both m and n are finite, finite products and coproducts coincide (biproduct), so $\prod_{i \in m} K^{(n)} \cong K^{m \times n}$ and thus $\dim_K(\text{Hom}_K(U, V)) = m \cdot n$. QED

Example 10.5.5 (Dimension of $\text{Hom}_K(K^2, K^3)$):

$$\dim_K(\text{Hom}_K(K^2, K^3)) = 2 \cdot 3 = 6.$$

This can be seen directly: a linear map $K^2 \rightarrow K^3$ is determined by where it sends the basis vectors $(1, 0)$ and $(0, 1)$. Each can go to any vector in K^3 , giving $3 + 3 = 6$ free parameters.

Corollary 10.5.6 (Dimension of End):

If $\dim_K(V) = n \in \mathbb{N}$, then $\dim_K(\text{End}_K(V)) = n^2$.

Proof:

$$\dim_K(\text{End}_K(V)) = \dim_K(\text{Hom}_K(V, V)) = n \cdot n = n^2.$$

QED

10.6 Dual Spaces

Definition 10.6.1 (Dual Space):

The **dual space** of a K -vector space V is:

$$V^{\odot} = \text{Hom}_K(V, K)$$

Elements of $V^{\odot}$ are called **linear functionals** or **covectors**.

Theorem 10.6.2 (Dimension of Dual Space):

Let V be a K -vector space with basis B .

1. If $\dim_K(V) = n \in \mathbb{N}$, then $\dim_K(V^{\odot}) = n$
2. If $\dim_K(V)$ is infinite, then $\dim_K(V^{\odot}) > \dim_K(V)$

Proof:

By definition, $V^{\odot} = \text{Hom}_K(V, K)$.

(1) **Finite case:** By Theorem 10.5.3 with codomain K (where $\dim_K(K) = 1$):

$$\text{Hom}_K(V, K) \cong \prod_{b \in B} K^{(1)} = \prod_{b \in B} K = K^B$$

Applying Corollary 10.5.4 with $U = V$ and $\dim_K(K) = 1$:

$$\dim_K(V^{\odot}) = \dim_K(\text{Hom}_K(V, K)) = n \cdot 1 = n$$

(2) **Infinite case:** Let $|B| = \kappa$ be infinite. By Theorem 10.5.3:

$$V^{\odot} = \text{Hom}_K(V, K) \cong \prod_{b \in B} K^{(1)} = K^B$$

This is a product over an infinite index set with non-trivial factors.

By Theorem 10.1.8:

$$\dim_K(V^{\odot}) = |K^B| = |K|^{\kappa}$$

By Cantor's theorem, $|K|^{\kappa} > \kappa = \dim_K(V)$ (since $|K| \geq 2$).

QED

10.7 Dual Basis

Definition 10.7.1 (Dual Basis):

For ordered basis $\mathcal{B} = (b_0, \dots, b_{n-1})$ of finite-dimensional V , the **dual basis** $\mathcal{B}^\vee = (b^0, \dots, b^{n-1})$ consists of linear functionals defined by:

$$b^i(b_j) = \delta_{ij} = \begin{cases} 1 & \text{if } i = j \\ 0 & \text{if } i \neq j \end{cases}$$

Theorem 10.7.2 (Dual Basis is Linearly Independent):

The dual basis $\mathcal{B}^\vee = (b^0, \dots, b^{n-1})$ is linearly independent in $V^\vee$.

Proof:

Suppose $\sum_{i \in n} a_i b^i = 0$ (the zero functional).

For each $j \in \{0, \dots, n-1\}$, evaluate at b_j :

$$0 = \left(\sum_{i \in n} a_i b^i \right) (b_j) = \sum_{i \in n} a_i \cdot b^i(b_j) = \sum_{i \in n} a_i \cdot \delta_{ij} = a_j$$

Thus $a_j = 0$ for all j , proving linear independence. QED

Theorem 10.7.3 (Dual Basis is a Basis iff Finite-Dimensional):

$\mathcal{B}^\vee$ is a basis of $V^\vee$ if and only if $\dim_K(V) \in \mathbb{N}$.

Proof:

($\Rightarrow$) If $\mathcal{B}^\vee$ is a basis of $V^\vee$, then $\dim_K(V^\vee) = |\mathcal{B}^\vee| = |\mathcal{B}| = \dim_K(V)$. By Theorem 10.6.2, this implies $\dim_K(V)$ is finite.

($\Leftarrow$) Suppose $\dim_K(V) = n \in \mathbb{N}$. By Theorem 10.7.2, $\mathcal{B}^\vee$ is a linearly independent set of n elements in $V^\vee$. Since $\dim_K(V^\vee) = n$ by Theorem 10.6.2, any n linearly independent elements form a basis. QED

Example 10.7.4 (Functional Not in Span of Dual Basis):

Let $V = K[X]$ with basis $\mathcal{B} = (1, X, X^2, X^3, \dots)$ (monomials).

The **dual basis** consists of the coefficient extraction functionals $\delta_n : K[X] \rightarrow K$ defined by:

$$\delta_n \left(\sum_{i=0}^m a_i X^i \right) = a_n$$

These satisfy $\delta_n(X^k) = \delta_{nk}$ (Kronecker delta).

The span of the dual basis: Any finite linear combination $\sum_{i=0}^N c_i \delta_i$ only “sees” the first $N+1$ coefficients of a polynomial. Specifically:

$$\left(\sum_{i=0}^N c_i \delta_i \right) (p) = \sum_{i=0}^N c_i a_i \quad \text{for } p = \sum_{j \in \mathbb{N}} a_j X^j$$

Evaluation at 1: Define $\text{ev}_1 : K[X] \rightarrow K$ by:

$$\text{ev}_1(p) = p(1) = \sum_{i=0}^{\deg p} a_i$$

This functional sums **all** coefficients of the polynomial.

Claim: $\text{ev}_1 \notin \text{span}_K(\{\delta_0, \delta_1, \delta_2, \dots\})$.

Proof: Suppose $\text{ev}_1 = \sum_{i=0}^N c_i \delta_i$ for some finite sum. Evaluate at X^{N+1} :

$$\text{ev}_1(X^{N+1}) = 1^{N+1} = 1$$

But:

$$\left(\sum_{i=0}^N c_i \delta_i \right) (X^{N+1}) = \sum_{i=0}^N c_i \cdot \delta_i(X^{N+1}) = \sum_{i=0}^N c_i \cdot 0 = 0$$

Contradiction. Therefore ev_1 cannot be written as a finite linear combination of the dual basis elements.

Proposition 10.7.5 (Expansion in Dual Basis):

For finite-dimensional V with basis $\mathcal{B}$, every $\varphi \in V^*$ satisfies:

$$\varphi = \sum_{i \in n} \varphi(b_i) \cdot b^i$$

Proof:

Let $\psi = \sum_{i \in n} \varphi(b_i) \cdot b^i$. We show $\varphi = \psi$.

For each basis element b_j :

$$\psi(b_j) = \sum_{i \in n} \varphi(b_i) \cdot b^i(b_j) = \sum_{i \in n} \varphi(b_i) \cdot \delta_{ij} = \varphi(b_j)$$

Since φ and ψ agree on all basis elements, and both are linear maps, they are equal (a linear map is determined by its values on a basis). QED

Theorem 10.7.6 (Non-Canonical Isomorphism):

For $\dim_K(V) = n \in \mathbb{N}$, choosing a basis $\mathcal{B} = (b_0, \dots, b_{n-1})$ determines an isomorphism:

$$\psi_{\mathcal{B}} : V \rightarrow V^*, \quad b_i \mapsto b^i$$

This isomorphism **depends on the choice of basis** (hence "non-canonical").

Proof:

Well-defined: By Theorem 6.2.1, $\psi_{\mathcal{B}}$ extends uniquely to a linear map on all of V .

Injectivity: Suppose $\psi_{\mathcal{B}}(v) = 0$ for some $v = \sum_{i \in n} a_i b_i$. Then $\psi_{\mathcal{B}}(v) = \sum_{i \in n} a_i b^i = 0$. By linear independence of the dual basis (Theorem 10.7.2), $a_i = 0$ for all i . Thus $v = 0$.

Surjectivity: $\dim_K(V) = n = \dim_K(V^*)$ by Theorem 10.6.2. An injective linear map between spaces of equal finite dimension is surjective.

Basis-dependence: A different basis $\mathcal{B}'$ yields a different dual basis $\mathcal{B}'^*$ and hence a different isomorphism $\psi_{\mathcal{B}'} \neq \psi_{\mathcal{B}}$. QED

10.8 Annihilators

Definition 10.8.1 (Annihilator):

For $S \subseteq V$, the **annihilator** of S is:

$$\text{Ann}(S) = \left\{ \varphi \in V^{\otimes} \mid \forall v \in S: \varphi(v) = 0 \right\}$$

Proposition 10.8.2 (Annihilator is a Subspace):

For any $S \subseteq V$, $\text{Ann}(S) \leq_K V^{\otimes}$.

Proof:

Non-empty: The zero functional $0 \in V^{\otimes}$ satisfies $0(v) = 0$ for all $v \in S$, so $0 \in \text{Ann}(S)$.

Closed under addition: Let $\varphi, \psi \in \text{Ann}(S)$. For any $v \in S$:

$$(\varphi + \psi)(v) = \varphi(v) + \psi(v) = 0 + 0 = 0$$

Thus $\varphi + \psi \in \text{Ann}(S)$.

Closed under scalar multiplication: Let $\varphi \in \text{Ann}(S)$ and $a \in K$. For any $v \in S$:

$$(a\varphi)(v) = a \cdot \varphi(v) = a \cdot 0 = 0$$

Thus $a\varphi \in \text{Ann}(S)$. QED

Proposition 10.8.3 (Annihilator Antitonicity):

If $S \subseteq T \subseteq V$, then $\text{Ann}(T) \subseteq \text{Ann}(S)$.

Proof:

Let $\varphi \in \text{Ann}(T)$. By definition of annihilator, $\varphi(v) = 0$ for all $v \in T$.

Since $S \subseteq T$, every element of S is also in T . Therefore $\varphi(v) = 0$ for all $v \in S$.

By definition of $\text{Ann}(S)$, this means $\varphi \in \text{Ann}(S)$. QED

Proposition 10.8.4 (Annihilator and Span):

For any $S \subseteq V$, $\text{Ann}(S) = \text{Ann}(\text{span}_K(S))$.

Proof:

($\supseteq$) Since $S \subseteq \text{span}_K(S)$, by Proposition 10.8.3 we have $\text{Ann}(\text{span}_K(S)) \subseteq \text{Ann}(S)$.

($\subseteq$) Let $\varphi \in \text{Ann}(S)$. We must show $\varphi(w) = 0$ for all $w \in \text{span}_K(S)$.

Any $w \in \text{span}_K(S)$ can be written as $w = \sum_{i \in k} a_i s_i$ for some $a_i \in K$, $s_i \in S$, $k \in \mathbb{N}$.

By linearity of φ :

$$\varphi(w) = \varphi\left(\sum_{i \in k} a_i s_i\right) = \sum_{i \in k} a_i \varphi(s_i)$$

Since $s_i \in S$ and $\varphi \in \text{Ann}(S)$, we have $\varphi(s_i) = 0$ for all i . Thus:

$$\varphi(w) = \sum_{i \in k} a_i \cdot 0 = 0$$

Therefore $\varphi \in \text{Ann}(\text{span}_K(S))$. QED

Proposition 10.8.5 (Annihilator of Zero):

$\text{Ann}(\{0\}) = V^{\otimes}$.

Proof:

We show both inclusions.

($\supseteq$) Let $\varphi \in V^*$. For the unique element $0 \in \{0\}$:

$$\varphi(0) = \varphi(0 \cdot 0) = 0 \cdot \varphi(0) = 0$$

Thus φ vanishes on every element of $\{0\}$, so $\varphi \in \text{Ann}(\{0\})$.

($\subseteq$) $\text{Ann}(\{0\}) \subseteq V^*$ by definition. QED

Proposition 10.8.6 (Annihilator of Entire Space):

$\text{Ann}(V) = \{0\}$.

Proof:

We show both inclusions.

($\supseteq$) The zero functional $0 \in V^*$ satisfies $0(v) = 0$ for all $v \in V$, so $0 \in \text{Ann}(V)$.

($\subseteq$) Let $\varphi \in \text{Ann}(V)$. Then $\varphi(v) = 0$ for all $v \in V$. This means φ is the zero map $V \rightarrow K$, i.e., $\varphi = 0$. Thus $\varphi \in \{0\}$. QED

Theorem 10.8.7 (Annihilator Dimension):

For $U \leq_K V$ with $\dim_K(V) = n \in \mathbb{N}$:

$$\dim_K(\text{Ann}(U)) + \dim_K(U) = \dim_K(V)$$

Proof:

Let $\dim_K(U) = m$ and choose a basis $(u_0, \dots, u_{m-1})$ for U . Extend to a basis $(u_0, \dots, u_{m-1}, v_m, \dots, v_{n-1})$ for V . Let $(u^0, \dots, u^{m-1}, v^m, \dots, v^{n-1})$ be the dual basis.

Claim: $\text{Ann}(U) = \text{span}_K(\{v^m, \dots, v^{n-1}\})$.

($\supseteq$) For $j \geq m$ and $i < m$: $v^j(u_i) = \delta_{ji} = 0$. Thus $v^j \in \text{Ann}(U)$.

($\subseteq$) Let $\varphi \in \text{Ann}(U)$. By Proposition 10.7.5:

$$\varphi = \sum_{i \in m} \varphi(u_i) \cdot u^i + \sum_{j=m}^{n-1} \varphi(v_j) \cdot v^j$$

Since $\varphi \in \text{Ann}(U)$, we have $\varphi(u_i) = 0$ for $i < m$. Thus $\varphi = \sum_{j=m}^{n-1} \varphi(v_j) \cdot v^j \in \text{span}_K(\{v^m, \dots, v^{n-1}\})$.

Therefore $\dim_K(\text{Ann}(U)) = n - m = \dim_K(V) - \dim_K(U)$. QED

Corollary 10.8.8 (Annihilator and Quotient):

$\text{Ann}(U) \cong V/U^*$ for $\dim_K(V) \in \mathbb{N}$.

Proof:

Define $\Phi : \text{Ann}(U) \rightarrow V/U^*$ by $\Phi(\varphi) = \tilde{\varphi}$, where $\tilde{\varphi}(v + U) = \varphi(v)$.

Well-defined: If $v_1 + U = v_2 + U$, then $v_1 - v_2 \in U$. Since $\varphi \in \text{Ann}(U)$, $\varphi(v_1 - v_2) = 0$, so $\varphi(v_1) = \varphi(v_2)$.

Linearity: $\tilde{\varphi}(av + U + bw + U) = \varphi(av + bw) = a\varphi(v) + b\varphi(w)$.

Injectivity: If $\Phi(\varphi) = 0$, then $\varphi(v) = 0$ for all $v \in V$, so $\varphi = 0$.

Surjectivity: By Theorem 10.8.7, $\dim_K(\text{Ann}(U)) = \dim_K(V) - \dim_K(U) = \dim_K(V/U) = \dim_K(V/U^*)$. QED

Injective + equal finite dimensions $\Rightarrow$ surjective.

10.9 Hyperplanes

Remark:

This section is moved here from Quotient Spaces as hyperplanes are naturally characterized via linear functionals and annihilators.

Definition 10.9.1 (Hyperplane):

A subspace $H \leq_K V$ is a **hyperplane** if $\dim_K(V/H) = 1$.

Theorem 10.9.2 (Hyperplane Characterizations):

For $H \leq_K V$, the following are equivalent:

1. H is a hyperplane (i.e., $\dim_K(V/H) = 1$)
2. $\exists 1\text{-dim } L \leq V : V = H \oplus L$
3. $H = \ker(f)$ for some nonzero $f \in V^*$
4. $\dim_K(\text{Ann}(H)) = 1$

For $\dim_K(V) = n \in \mathbb{N}$: these are equivalent to $\dim_K(H) = n - 1$.

Proof:

(1) $\Leftrightarrow$ (2): $\dim_K(V/H) = 1$ iff $\exists v_0 \notin H : V = H + \text{span}_K(\{v_0\})$. Since $H \cap \text{span}_K(\{v_0\}) = \{0\}$ (else $v_0 \in H$), this sum is direct.

(1) $\Rightarrow$ (3): The projection $\pi : V \rightarrow V/H$ composed with an isomorphism $V/H \cong K$ gives $f : V \rightarrow K$ with $\ker(f) = H$.

(3) $\Rightarrow$ (1): For nonzero f , pick v_0 with $f(v_0) \neq 0$. Scale so $f(v_0) = 1$. Then $V = \ker(f) \oplus \text{span}_K(\{v_0\})$, and the induced map $V/\ker(f) \rightarrow K$ is an isomorphism.

(3) $\Leftrightarrow$ (4):

($\Rightarrow$) If $H = \ker(f)$ for nonzero f , then $\text{span}_K(\{f\}) \subseteq \text{Ann}(H)$. Conversely, if $g \in \text{Ann}(H)$, then $\ker(f) \subseteq \ker(g)$. For any $v \in V$, write $v = h + tv_0$ with $h \in H$ and $f(v_0) = 1$. Then $g(v) = g(h) + tg(v_0) = tg(v_0) = f(v) \cdot g(v_0)$, so $g = g(v_0)f \in \text{span}_K(\{f\})$. Thus $\dim_K(\text{Ann}(H)) = 1$.

($\Leftarrow$) If $\dim_K(\text{Ann}(H)) = 1$, pick nonzero $f \in \text{Ann}(H)$. Then $H \subseteq \ker(f)$. If $H \subsetneq \ker(f)$, pick $v \in \ker(f) \setminus H$. Define $g \in V^*$ with $g(v) = 1$ and $g|_H = 0$ (extend a basis of $H \cup \{v\}$). Then $g \in \text{Ann}(H)$ but $g \notin \text{span}_K(\{f\})$ (since $f(v) = 0$ but $g(v) = 1$), contradicting $\dim_K(\text{Ann}(H)) = 1$. So $H = \ker(f)$.

Finite-dim: $\dim_K(H) + \dim_K(V/H) = n$, so $\dim_K(V/H) = 1$ iff $\dim_K(H) = n - 1$. QED

Example 10.9.3 (Hyperplanes in Finite Dimensions):

In K^n , consider the linear functional $f : K^n \rightarrow K$ defined by $f(x_0, \dots, x_{n-1}) = \sum_{i \in n} x_i$. Then:

$$H = \ker(f) = \{(x_0, \dots, x_{n-1}) \in K^n \mid x_0 + x_1 + \dots + x_{n-1} = 0\}$$

is a hyperplane with $\dim_K(H) = n - 1$.

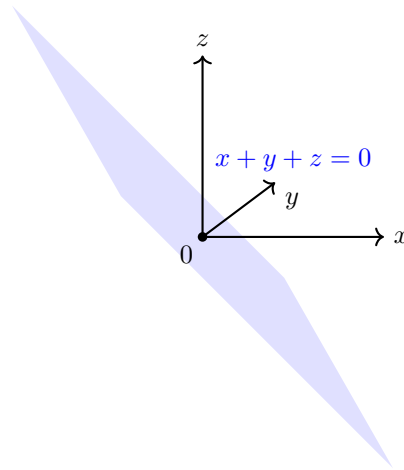


Figure 10.1: A hyperplane in $\mathbb{R}^3$: the plane $x + y + z = 0$ through the origin.

Example 10.9.4 (Hyperplanes in Infinite Dimensions):

The following are examples of hyperplanes in infinite-dimensional spaces:

1. In $K[X]$ (polynomials), let $\text{ev}_0 : K[X] \rightarrow K$ be evaluation at 0:

$$H = \ker(\text{ev}_0) = \{p \in K[X] \mid p(0) = 0\}$$

This is the set of polynomials with zero constant term.

2. In $K^{(\mathbb{N})}$ (finite support sequences), define $f : K^{(\mathbb{N})} \rightarrow K$ by $f((a_n)) = \sum_{n \in \mathbb{N}} a_n$ (a finite sum). Then:

$$H = \ker(f) = \left\{ (a_n) \in K^{(\mathbb{N})} \mid \sum_{n \in \mathbb{N}} a_n = 0 \right\}$$

Example 10.9.5 (Cardinal vs Dimension: A Critical Counterexample):

Let $V = K[X]$ with basis $\{1, X, X^2, \dots\}$, so $\dim_K(V) = \aleph_0$.

Let $W = \text{span}_K(\{X^n \mid n \geq 2\})$ (polynomials with no constant or linear term). Then $\dim_K(W) = \aleph_0$ as well.

Cardinal arithmetic suggests a hyperplane: $\aleph_0 + 1 = \aleph_0 = \dim_K(V)$, so naively $\dim_K(W) + 1 = \dim_K(V)$.

But W is NOT a hyperplane:

$$V/W \cong \text{span}_K(\{1, X\}) \quad \text{has} \quad \dim_K(V/W) = 2 \neq 1$$

Lesson: In infinite dimensions, the condition “ $\dim_K(W) + 1 = \dim_K(V)$ ” (in the cardinal sense) does NOT imply W is a hyperplane. Always use the definition: $\dim_K(V/W) = 1$.

10.10 Dual of a Linear Map

Definition 10.10.1 (Dual Map):

For $T : U \rightarrow V$, the **dual map** is:

$$T^{\smiley} : V^{\smiley} \rightarrow U^{\smiley}, \quad T^{\smiley}(\varphi) = \varphi \circ T$$

$$\begin{array}{ccc} U & \xrightarrow{T} & V \\ \varphi \circ T \downarrow & & \downarrow \varphi \\ K & \xlongequal{\quad} & K \end{array}$$

Theorem 10.10.2 (Functorial Properties):

$$1. \text{Id}_V^{\smiley} = \text{Id}_{V^{\smiley}}$$

$$2. (S \circ T)^{\smiley} = T^{\smiley} \circ S^{\smiley} \text{ (contravariant)}$$

$$3. (aS + bT)^{\smiley} = aS^{\smiley} + bT^{\smiley}$$

Theorem 10.10.3 (Injectivity/Surjectivity Duality):

For a linear map $T : U \rightarrow V$:

$$1. T^{\smiley} \text{ is injective} \Leftrightarrow T \text{ is surjective}$$

$$2. T^{\smiley} \text{ is surjective} \Leftrightarrow T \text{ is injective}$$

Proof:

(1) $T^{\smiley} \text{ inj} \Leftrightarrow T \text{ surj}$:

($\Leftarrow$) If T surjective and $T^{\smiley}(\varphi) = \varphi \circ T = 0$, then for any $v = T(u)$: $\varphi(v) = \varphi(T(u)) = 0$. Thus $\varphi = 0$.

($\Rightarrow$) If T not surjective, pick $v_0 \notin \text{Rng}(T)$, $v_0 \neq 0$. Extend $\{v_0\}$ to basis, define φ with $\varphi(v_0) = 1$, $\varphi|_{\text{Rng}(T)} = 0$. Then $\varphi \neq 0$ but $T^{\smiley}(\varphi) = 0$.

(2) $T^{\smiley} \text{ surj} \Leftrightarrow T \text{ inj}$:

($\Leftarrow$) Let T be injective. For any $\psi \in U^{\smiley}$, we need $\varphi \in V^{\smiley}$ with $\varphi \circ T = \psi$.

Since T is injective, we can define $\varphi_0 : \text{Rng}(T) \rightarrow K$ by $\varphi_0(T(u)) = \psi(u)$. This is well-defined (injectivity) and linear.

Choose a complement: $V = \text{Rng}(T) \oplus W$ (by extending a basis of $\text{Rng}(T)$). Define $\varphi(v) = \varphi_0(v_R)$ where $v = v_R + v_W$. Then $T^{\smiley}(\varphi) = \psi$.

($\Rightarrow$) If T not injective, pick $u_0 \in \ker(T)$, $u_0 \neq 0$. Define $\psi \in U^{\smiley}$ with $\psi(u_0) = 1$ (extend $\{u_0\}$ to basis).

If $T^{\smiley}(\varphi) = \psi$ for some φ , then $\psi(u_0) = \varphi(T(u_0)) = \varphi(0) = 0 \neq 1$. Contradiction.

QED

10.11 Bidual and Canonical Isomorphism

Definition 10.11.1 (Evaluation Map):

The **evaluation map** is:

$$\eta_V : V \rightarrow V^{\smile\smile}, \quad \eta_V(v)(\varphi) = \varphi(v)$$

Theorem 10.11.2 (η_V is Always Injective):

$\ker(\eta_V) = \{0\}$ for any V .

Proof:

If $\eta_V(v) = 0$ and $v \neq 0$, extend $\{v\}$ to basis, define $\varphi_0(v) = 1$. Then $\varphi_0(v) = 1 \neq 0 = \eta_V(v)(\varphi_0)$. Contradiction. QED

Theorem 10.11.3 (Canonical Isomorphism):

$$\eta_V : V \xrightarrow{\cong} V^{\smile\smile} \text{ iff } \dim_K(V) \in \mathbb{N}.$$

Proof:

($\Leftarrow$) If $\dim_K(V) = n \in \mathbb{N}$, then $\dim_K(V^{\smile\smile}) = \dim_K(V^{\smile}) = n$. Injective + equal dimensions $\Rightarrow$ isomorphism.

($\Rightarrow$) If infinite-dim, $\dim_K(V^{\smile\smile}) > \dim_K(V^{\smile}) > \dim_K(V)$. Cannot be surjective. QED

Remark (Canonicity):

η_V is **canonical**: no basis choice needed.

Naturality: For $T : U \rightarrow V$, the following commutes:

$$\begin{array}{ccc} U & \xrightarrow{\eta_U} & U^{\smile\smile} \\ \downarrow T & & \downarrow T^{\smile\smile} \\ V & \xrightarrow{\eta_V} & V^{\smile\smile} \end{array}$$

Proof: $(T^{\smile\smile}(\eta_U(u)))(\psi) = \eta_U(u)(T^{\smile\smile}(\psi)) = (\psi \circ T)(u) = \psi(T(u)) = \eta_V(T(u))(\psi)$.

10.12 Chapter Compendium

Chapter Compendium — Products and Coproducts:

Product: $\prod_{i \in I} V_i$ with projections π_j

Coproduct: $\coprod_{i \in I} V_i$ = finite-support, with injections ι_j

Finite I : product = coproduct (biproduct)

Dimensions: $\dim(\prod_{i \in I} V_i) = \sum_{i \in I} \dim(V_i)$

Infinite I : $\dim(\prod_{i \in I} V_i) = \prod_{i \in I} |V_i|$ (for non-trivial factors)

Chapter Compendium — Hom-Product Isomorphisms:

$\text{Hom}_K(\prod U_i, V) \cong \prod \text{Hom}_K(U_i, V)$ (contravariant)

$\text{Hom}_K(U, \prod V_i) \cong \prod \text{Hom}_K(U, V_i)$ (covariant)

$\text{Hom}_K(U, \prod V_i) \cong \prod \text{Hom}_K(U, V_i)$ (finite-dim U)

Chapter Compendium — Hom Spaces and K -Algebras:

$\text{Hom}_K(U, V)$ is a K -vector space (all 10 axioms verified pointwise)

K -algebra: vector space + ring with compatibility $a(xy) = (ax)y = x(ay)$

$\text{End}_K(V)$: unital associative K -algebra, non-commutative when $\dim(V) \geq 2$

Non-commutativity: explicit S, T with $S(v_1) = v_2, T(v_2) = v_1 \Rightarrow ST \neq TS$

$\text{Hom}_K(K, V) \cong V; \quad \text{Hom}_K(K, K) \cong K$

Chapter Compendium — Hom Dimension (Finite $\dim(V)$):

Full Chain:

$$\text{Hom}(U, V) \cong \prod_{B_U} \text{Hom}\left(K, K^{(B_V)}\right) \cong \prod_{B_U} \prod_{B_V} K \cong \prod_{B_U} K^{(\dim(V))}$$

Dimension: $|K^{(\dim(V))}|^{\dim(U)}$; finite case: $\dim(\text{Hom}(U, V)) = m \cdot n$

Chapter Compendium — Dual Space:

$V^{\odot} = \text{Hom}_K(V, K) \cong \prod_B K^{(1)} = K^B$

$\dim(V^{\odot}) = \dim(V)$ (finite); $\dim(V^{\odot}) = |K|^{\kappa} > \kappa$ (infinite, Cantor)

Dual basis: $b^i(b_j) = \delta_{ij}$; spans $V^{\odot}$ iff finite-dim

Not in span example: $\text{ev}_1 \notin \text{span}(\{\delta_n\})$ on $K[X]$

$V \cong V^{\odot}$ non-canonical (basis-dependent)

Chapter Compendium — Annihilators:

$\text{Ann}(S) = \left\{ \varphi \in V^{\odot} \mid \varphi|_S = 0 \right\}$

$\text{Ann}(S) = \text{Ann}(\text{span}(S)); \quad \text{Ann}(\{0\}) = V^{\odot}; \quad \text{Ann}(V) = \{0\}$

$\dim(\text{Ann}(U)) + \dim(U) = \dim(V)$ (finite-dim)

$\text{Ann}(U) \cong V/U^{\odot}$ (finite-dim)

Chapter Compendium — Hyperplanes:

Equivalent characterizations:

(i) $\dim(V/H) = 1$

(ii) $\exists 1\text{-dim } L \text{ s.t. } L \oplus H = V$

(iii) $H = \ker(f)$ for nonzero $f \in V^{\odot}$

(iv) $\dim(\text{Ann}(H)) = 1$

Finite-dim only: $\dim(H) = \dim(V) - 1$

Chapter Compendium — Dual Map and Bidual:

$$T^{\odot} : V^{\odot} \rightarrow U^{\odot}, \quad T^{\odot}(\varphi) = \varphi \circ T \text{ (contravariant)}$$

$$T^{\odot} \text{ inj} \Leftrightarrow T \text{ surj}; \quad T^{\odot} \text{ surj} \Leftrightarrow T \text{ inj (all dimensions)}$$

$$\eta_V : V \hookrightarrow V^{\odot\odot} \text{ canonical; iso iff } \dim(V) \in \mathbb{N}$$

$$\text{Naturality: } T^{\odot\odot} \circ \eta_U = \eta_V \circ T$$

CHAPTER 11

Endomorphisms and Spectral Theory

CHAPTER 12

Multilinear Algebra and Determinants

CHAPTER 13

Normed and Inner Product Spaces

Part III

Algebra III

CHAPTER 14

3. Prime and Irreducible elements

Definition 14.0.1 (Primes and Irreducibles in Integral Domains):

Let A be an integral domain and let $U(A) = \{a \in A \mid \exists b \in A : ab = 1\}$ be the set of invertible elements (units).

1. An element $p \in A$ is called **prime** if $p \notin U(A) \cup 0$ and

$$\forall a, b \in A : p \mid ab \implies p \mid a \text{ or } p \mid b$$

2. An element $r \in A$ is called **irreducible** if $r \notin U(A) \cup 0$ and

$$\forall a, b \in A : r = ab \implies a \in U(A) \text{ or } b \in U(A)$$

In other words, a prime element divides a product only if it divides at least one of the factors, while an irreducible element cannot be written as a product of two non-units.

Remark: 1. If $a \mid r$ where r is irreducible, then a is associated with r or $a \in U(A)$ (i.e., a is associated in divisibility with r or with 1).

2. Every prime element is irreducible. That is, if p is prime, then p is irreducible.

Proof of Remark 1:

Suppose $a \mid r$ where r is irreducible. Then $\exists b \in A : r = ab$.

Since r is irreducible, by definition we have $a \in U(A)$ or $b \in U(A)$.

Case 1: If $a \in U(A)$, then a is associated with 1, and we are done.

Case 2: If $b \in U(A)$, then b is invertible, so $\exists b^{-1} \in A : bb^{-1} = 1$. Multiplying both sides of $r = ab$ by b^{-1} , we get:

$$r \cdot b^{-1} = ab \cdot b^{-1} = a(bb^{-1}) = a \cdot 1 = a$$

Therefore $a = rb^{-1}$, which shows that a and r are associated (they differ by the unit b^{-1}).

In either case, a is associated with r or $a \in U(A)$. QED

Proof of Remark 2:

Let p be a prime element. We need to show that p is irreducible.

By definition, $p \notin U(A) \cup \{0\}$. Now suppose $p = ab$ for some $a, b \in A$. We need to show that $a \in U(A)$ or $b \in U(A)$.

Since $p = ab$, we have $p \mid ab$. Since p is prime, by definition $p \mid a$ or $p \mid b$.

Case 1: Suppose $p \mid a$. Then $\exists c \in A : a = pc$. Substituting into $p = ab$:

$$p = ab = (pc)b = p(cb)$$

Since A is an integral domain and $p \neq 0$, we can cancel p to obtain $1 = cb$. Therefore $b \in U(A)$.

Case 2: Suppose $p \mid b$. By symmetric reasoning (with roles of a and b reversed), we obtain $a \in U(A)$.

In either case, $a \in U(A)$ or $b \in U(A)$, which shows that p is irreducible. QED

Example 14.0.2:

Consider the ring $\mathbb{Z}[\sqrt{-5}] = \{a + b\sqrt{-5} \mid a, b \in \mathbb{Z}\}$, which is an integral domain. We will show that each element in $\{2, 3, 1 + \sqrt{-5}, 1 - \sqrt{-5}\}$ is irreducible but not prime.

Define the norm function $N : \mathbb{Z}[\sqrt{-5}] \rightarrow \mathbb{Z}_{>0}$ by $N(a + b\sqrt{-5}) = a^2 + 5b^2$. Note that N is multiplicative: $N(\alpha\beta) = N(\alpha)N(\beta)$ for all $\alpha, \beta \in \mathbb{Z}[\sqrt{-5}]$.

Also observe that $U(\mathbb{Z}[\sqrt{-5}]) = \{-1, 1\}$, since if $\alpha \in U(\mathbb{Z}[\sqrt{-5}])$, then $\exists \beta \in \mathbb{Z}[\sqrt{-5}] : \alpha\beta = 1$, which implies $N(\alpha)N(\beta) = N(1) = 1$. Since $N(\alpha), N(\beta) \in \mathbb{Z}_{>0}$ and their product is 1, we must have $N(\alpha) = 1$. If $\alpha = a + b\sqrt{-5}$, then $a^2 + 5b^2 = 1$ forces $b = 0$ and $a = \pm 1$.

Irreducibility:

For each $q \in \{2, 3, 1 + \sqrt{-5}, 1 - \sqrt{-5}\}$, we have:

$$\begin{aligned} N(2) &= 4, \\ N(3) &= 9, \\ N(1 + \sqrt{-5}) &= 1 + 5 = 6, \\ N(1 - \sqrt{-5}) &= 1 + 5 = 6. \end{aligned}$$

Suppose $q = \alpha\beta$ for some $\alpha, \beta \in \mathbb{Z}[\sqrt{-5}]$. Then $N(q) = N(\alpha)N(\beta)$.

- For $q = 2$: We have $4 = N(\alpha)N(\beta)$. The possible factorizations are $4 = 1 \cdot 4 = 2 \cdot 2$. If $N(\alpha) = 1$, then $\alpha \in U(\mathbb{Z}[\sqrt{-5}])$. If $N(\alpha) = 2$, then $a^2 + 5b^2 = 2$ for $\alpha = a + b\sqrt{-5}$, which has no integer solutions. If $N(\alpha) = 4$, then $N(\beta) = 1$, so $\beta \in U(\mathbb{Z}[\sqrt{-5}])$. Thus 2 is irreducible.
- For $q = 3$: We have $9 = N(\alpha)N(\beta)$. The possible factorizations are $9 = 1 \cdot 9 = 3 \cdot 3$. If $N(\alpha) = 1$, then $\alpha \in U(\mathbb{Z}[\sqrt{-5}])$. If $N(\alpha) = 3$, then $a^2 + 5b^2 = 3$, which has no integer solutions. If $N(\alpha) = 9$, then $N(\beta) = 1$, so $\beta \in U(\mathbb{Z}[\sqrt{-5}])$. Thus 3 is irreducible.
- For $q = 1 \pm \sqrt{-5}$: We have $6 = N(\alpha)N(\beta)$. The factorizations are $6 = 1 \cdot 6 = 2 \cdot 3$. We've shown that $a^2 + 5b^2 = 2$ and $a^2 + 5b^2 = 3$ have no integer solutions. If $N(\alpha) = 1$ or $N(\beta) = 1$, then one factor is a unit. Thus $1 \pm \sqrt{-5}$ are irreducible.

Not Prime:

Observe that in $\mathbb{Z}[\sqrt{-5}]$, we have the factorization:

$$6 = 2 \cdot 3 = (1 + \sqrt{-5})(1 - \sqrt{-5})$$

We can verify: $(1 + \sqrt{-5})(1 - \sqrt{-5}) = 1 - (i\sqrt{5})^2 = 1 - (-5) = 6$.

- **2 is not prime:** Since $2 \mid 6 = (1 + \sqrt{-5})(1 - \sqrt{-5})$, but $2 \nmid (1 + \sqrt{-5})$ and $2 \nmid (1 - \sqrt{-5})$.
To see $2 \nmid (1 + \sqrt{-5})$: if $1 + \sqrt{-5} = 2\alpha$ for some $\alpha = a + b\sqrt{-5}$, then $1 + \sqrt{-5} = 2a + 2b\sqrt{-5}$, giving $1 = 2a$ and $1 = 2b$, which is impossible in $\mathbb{Z}$.
- **3 is not prime:** Since $3 \mid 6 = (1 + \sqrt{-5})(1 - \sqrt{-5})$, but $3 \nmid (1 + \sqrt{-5})$ and $3 \nmid (1 - \sqrt{-5})$.
To see $3 \nmid (1 + \sqrt{-5})$: if $1 + \sqrt{-5} = 3\alpha$ for some $\alpha = a + b\sqrt{-5}$, then $1 + \sqrt{-5} = 3a + 3b\sqrt{-5}$, giving $1 = 3a$ and $1 = 3b$, which is impossible in $\mathbb{Z}$.
- **$1 + \sqrt{-5}$ is not prime:** Since $(1 + \sqrt{-5}) \mid 6 = 2 \cdot 3$, but $(1 + \sqrt{-5}) \nmid 2$ and $(1 + \sqrt{-5}) \nmid 3$.
To see $(1 + \sqrt{-5}) \nmid 2$: if $2 = (1 + \sqrt{-5})\alpha$ for $\alpha = a + b\sqrt{-5}$, then $N(2) = N(1 + \sqrt{-5})N(\alpha)$ gives $4 = 6N(\alpha)$, impossible since $N(\alpha) \in \mathbb{Z}_{>0}$.
- **$1 - \sqrt{-5}$ is not prime:** By symmetric reasoning with $(1 - \sqrt{-5}) \mid 6 = 2 \cdot 3$.

This example demonstrates that in general integral domains, the converse of Remark 2 is false: irreducible elements need not be prime.

Lemma 14.0.3:

Let A be an integral domain and let a, b, c be elements in A such that $\text{lcm}(a, b)$ exists and $\text{lcm}(a, b) = ab$. If $a \mid bc$ then $a \mid c$.

Proof:

Suppose $a \mid bc$. Then $\exists d \in A : bc = ad$.

Since $\text{lcm}(a, b) = ab$, the element ab is a common multiple of both a and b , and it is the least such multiple (up to associates).

Since $a \mid bc$, we have that bc is also a common multiple of a and b (as $b \mid bc$ trivially). By the definition of lcm , we have $ab \mid bc$, so $\exists e \in A : bc = abe$.

From $bc = ad$ and $bc = abe$, we obtain $ad = abe$.

Since A is an integral domain and $a \neq 0$ (as a divides something non-zero), we can cancel a to get $d = be$.

Substituting back into $bc = ad$:

$$bc = a(be) = (ab)e$$

Canceling b (which is non-zero since $bc \neq 0$), we obtain $c = ae$.

Therefore $a \mid c$, as required.

QED

Proposition 14.0.4 (Primes and Irreducibles coincide in GCD Domains):

Let A be a GCD domain. Then an element $r \in A$ is irreducible if and only if r is prime.

Proof:

We already know from Remark 2 that every prime element is irreducible in any integral domain. Thus it suffices to show that in a GCD domain, every irreducible element is prime.

Let r be an irreducible element in A . We need to show that r is prime, i.e., $\forall a, b \in A : r \mid ab \implies r \mid a \text{ or } r \mid b$.

Suppose $r \mid ab$ for some $a, b \in A$. Since A is a GCD domain, $\text{gcd}(r, a)$ exists. Let $d = \text{gcd}(r, a)$.

Then $d \mid r$. Since r is irreducible, by Remark 1, either d is associated with r or $d \in U(A)$.

Case 1: Suppose d is associated with r . Then $d \sim r$, so $r \mid d$. Since $d \mid a$, we have $r \mid a$, and we are done.

Case 2: Suppose $d \in U(A)$. Then $\text{gcd}(r, a) \sim 1$, which means r and a are coprime.

Since r and a are coprime, we have $\text{lcm}(r, a) \sim ra$ (in GCD domains, $\text{gcd}(x, y) \cdot \text{lcm}(x, y) \sim xy$).

Now, since $r \mid ab$ and $a \mid ab$, we have that ab is a common multiple of r and a . Therefore $\text{lcm}(r, a) \mid ab$.

Since $\text{lcm}(r, a) \sim ra$, we can assume without loss of generality that $\text{lcm}(r, a) = ra$ (by replacing r with an associate if necessary).

By the lemma, since $\text{lcm}(r, a) = ra$ and $r \mid ab$, we conclude that $r \mid b$.

In either case, $r \mid a$ or $r \mid b$, which shows that r is prime.

QED

Part IV

Measure Theory

CHAPTER 15

Set-Theoretic Foundations

15.1 π -Systems

Definition 15.1.1 (π -System):

A collection $\mathcal{P} \subseteq \mathcal{P}(X)$ is a π -system if it is closed under finite intersections:

$$A, B \in \mathcal{P} \implies A \cap B \in \mathcal{P}$$

Intuition:

A π -system is the minimal structure capturing “intersection-closed” collections. The name comes from “product” (intersection corresponds to logical AND). π -systems are important because:

- They arise naturally: intervals, rectangles, open sets form π -systems.
- They uniquely determine measures: if two σ -finite measures agree on a π -system, they agree on the generated σ -algebra.

Example 15.1.2:

The collection of intervals $(a, b]$ on $\mathbb{R}$ forms a π -system, since $(a, b] \cap (c, d] = (\max(a, c), \min(b, d)]$.

Lemma 15.1.3 (Intersection of π -Systems):

Let $\{\mathcal{P}_i\}_{i \in I}$ be an arbitrary family of π -systems on X . Then $\bigcap_{i \in I} \mathcal{P}_i$ is a π -system.

Proof:

Let $\mathcal{P} = \bigcap_{i \in I} \mathcal{P}_i$. Let $A, B \in \mathcal{P}$. Then $A, B \in \mathcal{P}_i$ for all i . Since each $\mathcal{P}_i$ is a π -system, $A \cap B \in \mathcal{P}_i$ for all i . Thus $A \cap B \in \mathcal{P}$. QED

Definition 15.1.4 (Generated π -System):

For $\mathcal{F} \subseteq \mathcal{P}(X)$, the π -system generated by $\mathcal{F}$, denoted $\pi(\mathcal{F})$, is:

$$\pi(\mathcal{F}) = \bigcap \{ \mathcal{P} \mid \mathcal{P} \text{ is a } \pi\text{-system on } X \text{ and } \mathcal{F} \subseteq \mathcal{P} \}$$

Proposition 15.1.5 (Properties of Generated π -Systems):

Let $\mathcal{F} \subseteq \mathcal{P}(X)$. Then:

1. $\pi(\mathcal{F})$ is a π -system (by the Intersection Lemma)
2. $\mathcal{F} \subseteq \pi(\mathcal{F})$ (extension)
3. If $\mathcal{P}$ is a π -system and $\mathcal{F} \subseteq \mathcal{P}$, then $\pi(\mathcal{F}) \subseteq \mathcal{P}$ (minimality)

Proof:

Property 1: By the Intersection Lemma. The intersection is nonempty since $\mathcal{P}(X)$ is a π -system containing $\mathcal{F}$.

Property 2: Every $\mathcal{P}$ in the intersection satisfies $\mathcal{F} \subseteq \mathcal{P}$. Thus $\mathcal{F} \subseteq \bigcap \mathcal{P} = \pi(\mathcal{F})$.

Property 3: If $\mathcal{P} \supseteq \mathcal{F}$ is a π -system, then $\mathcal{P}$ appears in the intersection. Hence $\pi(\mathcal{F}) \subseteq \mathcal{P}$. QED

Corollary 15.1.6 (Monotonicity and Idempotence):

Let $\mathcal{F}, \mathcal{G} \subseteq \mathcal{P}(X)$. Then:

1. If $\mathcal{F} \subseteq \mathcal{G}$, then $\pi(\mathcal{F}) \subseteq \pi(\mathcal{G})$ (monotonicity)
2. $\pi(\pi(\mathcal{F})) = \pi(\mathcal{F})$ (idempotence)

Proof:

Monotonicity: Let $\mathcal{F} \subseteq \mathcal{G}$. Then $\pi(\mathcal{G})$ is a π -system containing $\mathcal{G} \supseteq \mathcal{F}$. By minimality (Property 3), $\pi(\mathcal{F}) \subseteq \pi(\mathcal{G})$.

Idempotence: $\pi(\mathcal{F})$ is a π -system, so by minimality, $\pi(\pi(\mathcal{F})) \subseteq \pi(\mathcal{F})$. By extension (Property 2), $\pi(\mathcal{F}) \subseteq \pi(\pi(\mathcal{F}))$. Thus equality. QED

Theorem 15.1.7 (π -System Induction Principle):

Let $\mathcal{F} \subseteq \mathcal{P}(X)$ and let P be a property of subsets of X . Suppose:

1. $\forall F \in \mathcal{F} : P(F)$
2. $P(A)$ and $P(B) \implies P(A \cap B)$

Then $\forall A \in \pi(\mathcal{F}) : P(A)$.

Proof:

Let $\mathcal{M} = \{A \subseteq X \mid P(A)\}$. By hypothesis (2), $\mathcal{M}$ is a π -system. By hypothesis (1), $\mathcal{F} \subseteq \mathcal{M}$. By minimality, $\pi(\mathcal{F}) \subseteq \mathcal{M}$. QED

Proposition 15.1.8 (Constructive Description of $\pi(\mathcal{F})$):

Define $\mathcal{F}_0 = \mathcal{F}$ and for $n \geq 0$:

$$\mathcal{F}_{n+1} = \mathcal{F}_n \cup \{A \cap B \mid A, B \in \mathcal{F}_n\}$$

Then $\pi(\mathcal{F}) = \bigcup_{n \in \mathbb{N}} \mathcal{F}_n$.

Proof:

Let $\mathcal{G} = \bigcup_{n \in \mathbb{N}} \mathcal{F}_n$.

$\mathcal{G}$ is a π -system: Let $A, B \in \mathcal{G}$. Then $A \in \mathcal{F}_m$ and $B \in \mathcal{F}_n$ for some m, n . Let $k = \max(m, n)$. Then $A, B \in \mathcal{F}_k$, so $A \cap B \in \mathcal{F}_{k+1} \subseteq \mathcal{G}$.

$\mathcal{F} \subseteq \mathcal{G}$: Since $\mathcal{F} = \mathcal{F}_0 \subseteq \mathcal{G}$.

$\mathcal{G}$ is minimal: Let $\mathcal{P}$ be any π -system with $\mathcal{F} \subseteq \mathcal{P}$. We show $\mathcal{F}_n \subseteq \mathcal{P}$ for all n by induction.

Base: $\mathcal{F}_0 = \mathcal{F} \subseteq \mathcal{P}$.

Step: Assume $\mathcal{F}_n \subseteq \mathcal{P}$. If $A, B \in \mathcal{F}_n \subseteq \mathcal{P}$, then $A \cap B \in \mathcal{P}$ since $\mathcal{P}$ is a π -system. Thus $\mathcal{F}_{n+1} \subseteq \mathcal{P}$.

Hence $\mathcal{G} = \bigcup_{n \in \mathbb{N}} \mathcal{F}_n \subseteq \mathcal{P}$, proving $\mathcal{G}$ is minimal among π -systems containing $\mathcal{F}$.

By the three properties, $\mathcal{G} = \pi(\mathcal{F})$. QED

Theorem 15.1.9 (Cardinality of Generated π -Systems):

For any $\mathcal{F} \subseteq \mathcal{P}(X)$:

$$|\pi(\mathcal{F})| \leq \max(|\mathcal{F}|, \aleph_0)$$

Proof:

Let $\kappa = \max(|\mathcal{F}|, \aleph_0)$. We show $|\mathcal{F}_n| \leq \kappa$ for all n by induction.

Base: $|\mathcal{F}_0| = |\mathcal{F}| \leq \kappa$.

Step: Assume $|\mathcal{F}_n| \leq \kappa$. Each element of $\mathcal{F}_{n+1} \setminus \mathcal{F}_n$ is of the form $A \cap B$ for $A, B \in \mathcal{F}_n$. Thus $|\mathcal{F}_{n+1}| \leq |\mathcal{F}_n| + |\mathcal{F}_n|^2 \leq \kappa + \kappa^2 = \kappa$.

Conclusion: $|\pi(\mathcal{F})| = |\bigcup_{n \in \mathbb{N}} \mathcal{F}_n| \leq \aleph_0 \cdot \kappa = \kappa$.

QED

15.2 Algebras of Sets

Definition 15.2.1 (Algebra of Sets):

Let X be a set. A collection $\mathcal{A} \subseteq \mathcal{P}(X)$ is an **algebra** (or **field** or **Boolean algebra**) of sets if:

1. $\emptyset \in \mathcal{A}$
2. If $A \in \mathcal{A}$, then $A^c = X \setminus A \in \mathcal{A}$
3. If $A, B \in \mathcal{A}$, then $A \cup B \in \mathcal{A}$

Intuition:

An algebra captures “finite set operations”: complements, finite unions, and finite intersections. Every algebra is a π -system (closed under finite intersections via De Morgan). Algebras are sufficient for defining *contents* (finitely additive measures).

Proposition 15.2.2 (Properties of Algebras):

Let $\mathcal{A}$ be an algebra on X . Then:

1. $X \in \mathcal{A}$
2. $\mathcal{A}$ is a π -system (closed under finite intersections)
3. If $A, B \in \mathcal{A}$, then $A \setminus B \in \mathcal{A}$

Proof:

Property 1: $X = \emptyset^c \in \mathcal{A}$.

Property 2: $A \cap B = (A^c \cup B^c)^c \in \mathcal{A}$ by De Morgan.

Property 3: $A \setminus B = A \cap B^c \in \mathcal{A}$ by Property 2.

QED

Lemma 15.2.3 (Intersection of Algebras):

Let $\{\mathcal{A}_i\}_{i \in I}$ be an arbitrary family of algebras on X . Then $\bigcap_{i \in I} \mathcal{A}_i$ is an algebra.

Proof:

Let $\mathcal{A} = \bigcap_{i \in I} \mathcal{A}_i$.

Axiom 1: $\emptyset \in \mathcal{A}_i$ for all i , so $\emptyset \in \mathcal{A}$.

Axiom 2: Let $A \in \mathcal{A}$. Then $A \in \mathcal{A}_i$ for all i . Since each $\mathcal{A}_i$ is an algebra, $A^c \in \mathcal{A}_i$ for all i . Thus $A^c \in \mathcal{A}$.

Axiom 3: Let $A, B \in \mathcal{A}$. Then $A, B \in \mathcal{A}_i$ for all i . Since each $\mathcal{A}_i$ is an algebra, $A \cup B \in \mathcal{A}_i$ for all i . Thus $A \cup B \in \mathcal{A}$.

QED

Definition 15.2.4 (Generated Algebra):

For $\mathcal{F} \subseteq \mathcal{P}(X)$, the **algebra generated by $\mathcal{F}$** , denoted $\mathcal{A}(\mathcal{F})$, is:

$$\mathcal{A}(\mathcal{F}) = \bigcap \{ \mathcal{A} \mid \mathcal{A} \text{ is an algebra on } X \text{ and } \mathcal{F} \subseteq \mathcal{A} \}$$

Proposition 15.2.5 (Properties of Generated Algebras):

Let $\mathcal{F} \subseteq \mathcal{P}(X)$. Then:

1. $\mathcal{A}(\mathcal{F})$ is an algebra (by the Intersection Lemma)
2. $\mathcal{F} \subseteq \mathcal{A}(\mathcal{F})$ (extension)
3. If $\mathcal{A}$ is an algebra and $\mathcal{F} \subseteq \mathcal{A}$, then $\mathcal{A}(\mathcal{F}) \subseteq \mathcal{A}$ (minimality)

Proof:

Property 1: By the Intersection Lemma. The intersection is nonempty since $\mathcal{P}(X)$ is an algebra containing $\mathcal{F}$.

Property 2: Every $\mathcal{A}$ in the intersection satisfies $\mathcal{F} \subseteq \mathcal{A}$. Thus $\mathcal{F} \subseteq \bigcap \mathcal{A} = \mathcal{A}(\mathcal{F})$.

Property 3: If $\mathcal{A} \supseteq \mathcal{F}$ is an algebra, then $\mathcal{A}$ appears in the intersection. Hence $\mathcal{A}(\mathcal{F}) \subseteq \mathcal{A}$. QED

Corollary 15.2.6 (Monotonicity and Idempotence):

Let $\mathcal{F}, \mathcal{G} \subseteq \mathcal{P}(X)$. Then:

1. If $\mathcal{F} \subseteq \mathcal{G}$, then $\mathcal{A}(\mathcal{F}) \subseteq \mathcal{A}(\mathcal{G})$ (monotonicity)
2. $\mathcal{A}(\mathcal{A}(\mathcal{F})) = \mathcal{A}(\mathcal{F})$ (idempotence)

Proof:

Monotonicity: Let $\mathcal{F} \subseteq \mathcal{G}$. Then $\mathcal{A}(\mathcal{G})$ is an algebra containing $\mathcal{G} \supseteq \mathcal{F}$. By minimality, $\mathcal{A}(\mathcal{F}) \subseteq \mathcal{A}(\mathcal{G})$.

Idempotence: $\mathcal{A}(\mathcal{F})$ is an algebra, so by minimality, $\mathcal{A}(\mathcal{A}(\mathcal{F})) \subseteq \mathcal{A}(\mathcal{F})$. By extension, $\mathcal{A}(\mathcal{F}) \subseteq \mathcal{A}(\mathcal{A}(\mathcal{F}))$. Thus equality. QED

Theorem 15.2.7 (Algebra Induction Principle):

Let $\mathcal{F} \subseteq \mathcal{P}(X)$ and let P be a property of subsets of X . Suppose:

1. $P(\emptyset)$
2. $\forall F \in \mathcal{F} : P(F)$
3. $P(A) \implies P(A^c)$
4. $P(A) \text{ and } P(B) \implies P(A \cup B)$

Then $\forall A \in \mathcal{A}(\mathcal{F}) : P(A)$.

Proof:

Let $\mathcal{M} = \{A \subseteq X \mid P(A)\}$. By hypotheses (1), (3), (4), $\mathcal{M}$ is an algebra. By hypothesis (2), $\mathcal{F} \subseteq \mathcal{M}$. By minimality, $\mathcal{A}(\mathcal{F}) \subseteq \mathcal{M}$. QED

Proposition 15.2.8 (Constructive Description of $\mathcal{A}(\mathcal{F})$):

Define $\mathcal{F}_0 = \mathcal{F} \cup \{\emptyset, X\}$ and for $n \geq 0$:

$$\mathcal{F}_{n+1} = \mathcal{F}_n \cup \{A^c \mid A \in \mathcal{F}_n\} \cup \{A \cup B \mid A, B \in \mathcal{F}_n\}$$

Then $\mathcal{A}(\mathcal{F}) = \bigcup_{n \in \mathbb{N}} \mathcal{F}_n$.

Proof:

Let $\mathcal{G} = \bigcup_{n \in \mathbb{N}} \mathcal{F}_n$.

$\mathcal{G}$ is an algebra:

Axiom 1: $\emptyset \in \mathcal{F}_0 \subseteq \mathcal{G}$.

Axiom 2: Let $A \in \mathcal{G}$. Then $A \in \mathcal{F}_n$ for some n . By construction, $A^c \in \mathcal{F}_{n+1} \subseteq \mathcal{G}$.

Axiom 3: Let $A, B \in \mathcal{G}$. Then $A \in \mathcal{F}_m$ and $B \in \mathcal{F}_n$ for some m, n . Let $k = \max(m, n)$. Then $A, B \in \mathcal{F}_k$, so $A \cup B \in \mathcal{F}_{k+1} \subseteq \mathcal{G}$.

$\mathcal{F} \subseteq \mathcal{G}$: Since $\mathcal{F} \subseteq \mathcal{F}_0 \subseteq \mathcal{G}$.

$\mathcal{G}$ is minimal: Let $\mathcal{A}$ be any algebra with $\mathcal{F} \subseteq \mathcal{A}$. We show $\mathcal{F}_n \subseteq \mathcal{A}$ for all n by induction.

Base: $\mathcal{F}_0 = \mathcal{F} \cup \{\emptyset, X\} \subseteq \mathcal{A}$ since $\mathcal{F} \subseteq \mathcal{A}$ and $\emptyset, X \in \mathcal{A}$.

Step: Assume $\mathcal{F}_n \subseteq \mathcal{A}$. If $A \in \mathcal{F}_n \subseteq \mathcal{A}$, then $A^c \in \mathcal{A}$. If $A, B \in \mathcal{F}_n \subseteq \mathcal{A}$, then $A \cup B \in \mathcal{A}$. Thus $\mathcal{F}_{n+1} \subseteq \mathcal{A}$.

Hence $\mathcal{G} = \bigcup_{n \in \mathbb{N}} \mathcal{F}_n \subseteq \mathcal{A}$, proving minimality.

By the three properties, $\mathcal{G} = \mathcal{A}(\mathcal{F})$. QED

Theorem 15.2.9 (Cardinality of Generated Algebras):

For any $\mathcal{F} \subseteq \mathcal{P}(X)$:

$$|\mathcal{A}(\mathcal{F})| \leq \max(|\mathcal{F}|, \aleph_0)$$

Proof:

Let $\kappa = \max(|\mathcal{F}|, \aleph_0)$. We show $|\mathcal{F}_n| \leq \kappa$ for all n by induction.

Base: $|\mathcal{F}_0| = |\mathcal{F} \cup \{\emptyset, X\}| \leq \kappa + 2 = \kappa$.

Step: Assume $|\mathcal{F}_n| \leq \kappa$. Each element of $\mathcal{F}_{n+1}$ is either in $\mathcal{F}_n$, a complement of an element of $\mathcal{F}_n$, or a union of two elements of $\mathcal{F}_n$. Thus $|\mathcal{F}_{n+1}| \leq |\mathcal{F}_n| + |\mathcal{F}_n| + |\mathcal{F}_n|^2 \leq \kappa + \kappa + \kappa^2 = \kappa$.

Conclusion: $|\mathcal{A}(\mathcal{F})| = |\bigcup_{n \in \mathbb{N}} \mathcal{F}_n| \leq \aleph_0 \cdot \kappa = \kappa$. QED

15.3 Monotone Classes

Definition 15.3.1 (Monotone Class):

A collection $\mathcal{N} \subseteq \mathcal{P}(X)$ is a **monotone class** if:

1. If $(A_n)_{n \in \mathbb{N}} \subseteq \mathcal{N}$ with $A_n \subseteq A_{n+1}$, then $\bigcup_{n \in \mathbb{N}} A_n \in \mathcal{N}$
2. If $(A_n)_{n \in \mathbb{N}} \subseteq \mathcal{N}$ with $A_n \supseteq A_{n+1}$, then $\bigcap_{n \in \mathbb{N}} A_n \in \mathcal{N}$

Intuition:

A monotone class captures “closure under monotone limits.” Monotone sequences (increasing or decreasing) have their limits in the class. This is significant because:

- Continuity of measures is inherently about monotone sequences: $\mu(A_n) \rightarrow \mu(A)$ when $A_n \nearrow A$ or $A_n \searrow A$.
- The Monotone Class Lemma shows that an algebra generates the same σ -algebra as it does monotone class.

Lemma 15.3.2 (Intersection of Monotone Classes):

Let $\{\mathcal{N}_i\}_{i \in I}$ be an arbitrary family of monotone classes on X . Then $\bigcap_{i \in I} \mathcal{N}_i$ is a monotone class.

Proof:

Let $\mathcal{N} = \bigcap_{i \in I} \mathcal{N}_i$.

Axiom 1: Let $(A_n)_{n \in \mathbb{N}} \subseteq \mathcal{N}$ with $A_n \nearrow$. Then each $A_n \in \mathcal{N}_i$ for all i . Since each $\mathcal{N}_i$ is a monotone class, $\bigcup_{n \in \mathbb{N}} A_n \in \mathcal{N}_i$ for all i . Thus $\bigcup_{n \in \mathbb{N}} A_n \in \mathcal{N}$.

Axiom 2: Let $(A_n)_{n \in \mathbb{N}} \subseteq \mathcal{N}$ with $A_n \searrow$. Then each $A_n \in \mathcal{N}_i$ for all i . Since each $\mathcal{N}_i$ is a monotone class, $\bigcap_{n \in \mathbb{N}} A_n \in \mathcal{N}_i$ for all i . Thus $\bigcap_{n \in \mathbb{N}} A_n \in \mathcal{N}$. QED

Definition 15.3.3 (Generated Monotone Class):

For $\mathcal{F} \subseteq \mathcal{P}(X)$, the **monotone class generated by $\mathcal{F}$** , denoted $\mathcal{N}(\mathcal{F})$, is:

$$\mathcal{N}(\mathcal{F}) = \bigcap \{ \mathcal{N} \mid \mathcal{N} \text{ is a monotone class on } X \text{ and } \mathcal{F} \subseteq \mathcal{N} \}$$

Proposition 15.3.4 (Properties of Generated Monotone Classes):

Let $\mathcal{F} \subseteq \mathcal{P}(X)$. Then:

1. $\mathcal{N}(\mathcal{F})$ is a monotone class (by the Intersection Lemma)
2. $\mathcal{F} \subseteq \mathcal{N}(\mathcal{F})$ (extension)
3. If $\mathcal{N}$ is a monotone class and $\mathcal{F} \subseteq \mathcal{N}$, then $\mathcal{N}(\mathcal{F}) \subseteq \mathcal{N}$ (minimality)

Proof:

Property 1: By the Intersection Lemma. The intersection is nonempty since $\mathcal{P}(X)$ is a monotone class containing $\mathcal{F}$.

Property 2: Every $\mathcal{N}$ in the intersection satisfies $\mathcal{F} \subseteq \mathcal{N}$. Thus $\mathcal{F} \subseteq \bigcap \mathcal{N} = \mathcal{N}(\mathcal{F})$.

Property 3: If $\mathcal{N} \supseteq \mathcal{F}$ is a monotone class, then $\mathcal{N}$ appears in the intersection. Hence $\mathcal{N}(\mathcal{F}) \subseteq \mathcal{N}$. QED

Corollary 15.3.5 (Monotonicity and Idempotence):

Let $\mathcal{F}, \mathcal{G} \subseteq \mathcal{P}(X)$. Then:

1. If $\mathcal{F} \subseteq \mathcal{G}$, then $\mathcal{N}(\mathcal{F}) \subseteq \mathcal{N}(\mathcal{G})$ (monotonicity)
2. $\mathcal{N}(\mathcal{N}(\mathcal{F})) = \mathcal{N}(\mathcal{F})$ (idempotence)

Proof:

Monotonicity: Let $\mathcal{F} \subseteq \mathcal{G}$. Then $\mathcal{N}(\mathcal{G})$ is a monotone class containing $\mathcal{G} \supseteq \mathcal{F}$. By minimality, $\mathcal{N}(\mathcal{F}) \subseteq \mathcal{N}(\mathcal{G})$.

Idempotence: $\mathcal{N}(\mathcal{F})$ is a monotone class, so by minimality, $\mathcal{N}(\mathcal{N}(\mathcal{F})) \subseteq \mathcal{N}(\mathcal{F})$. By extension, $\mathcal{N}(\mathcal{F}) \subseteq \mathcal{N}(\mathcal{N}(\mathcal{F}))$. Thus equality. QED

Theorem 15.3.6 (Monotone Class Induction Principle):

Let $\mathcal{F} \subseteq \mathcal{P}(X)$ and let P be a property of subsets of X . Suppose:

1. $\forall F \in \mathcal{F}: P(F)$
2. $\forall n \in \mathbb{N}: P(A_n) \text{ and } A_n \nearrow A \implies P(A)$
3. $\forall n \in \mathbb{N}: P(A_n) \text{ and } A_n \searrow A \implies P(A)$

Then $\forall A \in \mathcal{N}(\mathcal{F}): P(A)$.

Proof:

Let $\mathcal{M} = \{A \subseteq X \mid P(A)\}$. By hypotheses (2) and (3), $\mathcal{M}$ is a monotone class. By hypothesis (1), $\mathcal{F} \subseteq \mathcal{M}$. By minimality, $\mathcal{N}(\mathcal{F}) \subseteq \mathcal{M}$. QED

Proposition 15.3.7 (Constructive Description of $\mathcal{N}(\mathcal{F})$):

Define $\mathcal{F}_0 = \mathcal{F}$ and for $n \geq 0$:

$$\mathcal{F}_{n+1} = \mathcal{F}_n \cup \left\{ \bigcup_{k \in \mathbb{N}} A_k \mid (A_k)_{k \in \mathbb{N}} \subseteq \mathcal{F}_n, A_k \nearrow \right\} \cup \left\{ \bigcap_{k \in \mathbb{N}} A_k \mid (A_k)_{k \in \mathbb{N}} \subseteq \mathcal{F}_n, A_k \searrow \right\}$$

Then $\mathcal{N}(\mathcal{F}) = \bigcup_{n \in \mathbb{N}} \mathcal{F}_n$.

Proof:

Let $\mathcal{G} = \bigcup_{n \in \mathbb{N}} \mathcal{F}_n$.

$\mathcal{G}$ is a monotone class:

Axiom 1: Let $(A_k)_{k \in \mathbb{N}} \subseteq \mathcal{G}$ with $A_k \nearrow$. Each $A_k \in \mathcal{F}_{n_k}$ for some n_k . Let $m = \sup_k n_k$. If m is finite, all $A_k \in \mathcal{F}_m$, so $\bigcup_k A_k \in \mathcal{F}_{m+1} \subseteq \mathcal{G}$. If $m = \infty$, we proceed by a diagonal argument: the limit of monotone sequences from $\mathcal{G}$ remains in $\mathcal{G}$ since each finite truncation is handled.

Axiom 2: Similar reasoning for decreasing sequences.

$\mathcal{F} \subseteq \mathcal{G}$: Since $\mathcal{F} = \mathcal{F}_0 \subseteq \mathcal{G}$.

$\mathcal{G}$ is minimal: Let $\mathcal{N}$ be any monotone class with $\mathcal{F} \subseteq \mathcal{N}$. We show $\mathcal{F}_n \subseteq \mathcal{N}$ for all n by induction.

Base: $\mathcal{F}_0 = \mathcal{F} \subseteq \mathcal{N}$.

Step: Assume $\mathcal{F}_n \subseteq \mathcal{N}$. If $(A_k) \subseteq \mathcal{F}_n \subseteq \mathcal{N}$ with $A_k \nearrow$, then $\bigcup_k A_k \in \mathcal{N}$. Similarly for $\bigcap_k$. Thus $\mathcal{F}_{n+1} \subseteq \mathcal{N}$.

Hence $\mathcal{G} = \bigcup_{n \in \mathbb{N}} \mathcal{F}_n \subseteq \mathcal{N}$, proving minimality.

By the three properties, $\mathcal{G} = \mathcal{N}(\mathcal{F})$. QED

Theorem 15.3.8 (Cardinality of Generated Monotone Classes):

For any $\mathcal{F} \subseteq \mathcal{P}(X)$:

$$|\mathcal{N}(\mathcal{F})| \leq \max(|\mathcal{F}|, \aleph_0)^{\aleph_0}$$

Proof:

Let $\kappa = \max(|\mathcal{F}|, \aleph_0)$. We show $|\mathcal{F}_n| \leq \kappa^{\aleph_0}$ for all n by induction.

Base: $|\mathcal{F}_0| = |\mathcal{F}| \leq \kappa \leq \kappa^{\aleph_0}$.

Step: Assume $|\mathcal{F}_n| \leq \kappa^{\aleph_0}$. Each element of $\mathcal{F}_{n+1} \setminus \mathcal{F}_n$ is a monotone limit of a sequence from $\mathcal{F}_n$. The number of such sequences is at most $|\mathcal{F}_n|^{\aleph_0} \leq (\kappa^{\aleph_0})^{\aleph_0} = \kappa^{\aleph_0}$. Thus $|\mathcal{F}_{n+1}| \leq \kappa^{\aleph_0}$.

Conclusion: $|\mathcal{N}(\mathcal{F})| = |\bigcup_{n \in \mathbb{N}} \mathcal{F}_n| \leq \aleph_0 \cdot \kappa^{\aleph_0} = \kappa^{\aleph_0}$. QED

15.4 σ -Algebras

Definition 15.4.1 (σ -Algebra):

Let X be a set. A collection $\mathcal{M} \subseteq \mathcal{P}(X)$ is a σ -algebra if:

1. $\emptyset \in \mathcal{M}$
2. If $A \in \mathcal{M}$, then $A^c \in \mathcal{M}$
3. If $(A_n)_{n \in \mathbb{N}} \subseteq \mathcal{M}$, then $\bigcup_{n \in \mathbb{N}} A_n \in \mathcal{M}$

Intuition:

A σ -algebra extends an algebra by allowing *countable* unions, not just finite ones. This seemingly small upgrade has profound consequences:

- **Limits:** In analysis, limits are inherently countable operations. Closure under countable unions means limits of measurable sets remain measurable.
- **Measure theory:** σ -additivity of measures (the defining property) requires countable additivity, which only makes sense on a σ -algebra.
- **The gap:** There exist algebras that are not σ -algebras. For example, the collection of finite or co-finite subsets of $\mathbb{N}$ is an algebra but not a σ -algebra (consider $\bigcup_{n \in \mathbb{N}} \{2n\}$).

Remark:

Every σ -algebra is an algebra, but not conversely.

Proposition 15.4.2 (Properties of σ -Algebras):

Let $\mathcal{M}$ be a σ -algebra on X . Then:

1. $X \in \mathcal{M}$
2. If $(A_n)_{n \in \mathbb{N}} \subseteq \mathcal{M}$, then $\bigcap_{n \in \mathbb{N}} A_n \in \mathcal{M}$
3. If $A, B \in \mathcal{M}$, then $A \setminus B \in \mathcal{M}$

Proof:

Property 1: By axiom 1, $\emptyset \in \mathcal{M}$. By axiom 2, $X = \emptyset^c \in \mathcal{M}$.

Property 2: Let $(A_n)_{n \in \mathbb{N}} \subseteq \mathcal{M}$. By axiom 2, each $A_n^c \in \mathcal{M}$. By axiom 3, $\bigcup_{n \in \mathbb{N}} A_n^c \in \mathcal{M}$. By axiom 2 again, $(\bigcup_{n \in \mathbb{N}} A_n^c)^c \in \mathcal{M}$. By De Morgan's law, this equals $\bigcap_{n \in \mathbb{N}} A_n$.

Property 3: Let $A, B \in \mathcal{M}$. We have $A \setminus B = A \cap B^c$. Since $B^c \in \mathcal{M}$ by axiom 2, and $A \cap B^c$ can be written as a countable intersection (specifically, $A \cap B^c = A \cap B^c \cap X \cap X \cap \dots$), we have $A \setminus B \in \mathcal{M}$ by property 2. QED

Lemma 15.4.3 (Intersection of σ -Algebras):

Let $\{\mathcal{M}_i\}_{i \in I}$ be an arbitrary family of σ -algebras on X . Then $\bigcap_{i \in I} \mathcal{M}_i$ is a σ -algebra.

Proof:

Let $\mathcal{M} = \bigcap_{i \in I} \mathcal{M}_i$.

Axiom 1: $\emptyset \in \mathcal{M}_i$ for all i , so $\emptyset \in \mathcal{M}$.

Axiom 2: Let $A \in \mathcal{M}$. Then $A \in \mathcal{M}_i$ for all i . Since each $\mathcal{M}_i$ is a σ -algebra, $A^c \in \mathcal{M}_i$ for all i . Thus $A^c \in \mathcal{M}$.

Axiom 3: Let $(A_n)_{n \in \mathbb{N}} \subseteq \mathcal{M}$. Then each $A_n \in \mathcal{M}_i$ for all i . Since each $\mathcal{M}_i$ is closed under countable unions, $\bigcup_{n \in \mathbb{N}} A_n \in \mathcal{M}_i$ for all i . Thus $\bigcup_{n \in \mathbb{N}} A_n \in \mathcal{M}$. QED

Definition 15.4.4 (Generated σ -Algebra):

Let $\mathcal{F} \subseteq \mathcal{P}(X)$. The σ -algebra generated by $\mathcal{F}$, denoted $\sigma(\mathcal{F})$, is:

$$\sigma(\mathcal{F}) = \bigcap \{ \mathcal{M} \mid \mathcal{M} \text{ is a } \sigma\text{-algebra on } X \text{ and } \mathcal{F} \subseteq \mathcal{M} \}$$

Proposition 15.4.5 (Properties of Generated σ -Algebras):

Let $\mathcal{F} \subseteq \mathcal{P}(X)$. Then:

1. $\sigma(\mathcal{F})$ is a σ -algebra (by the Intersection Lemma)
2. $\mathcal{F} \subseteq \sigma(\mathcal{F})$ (extension)
3. If $\mathcal{M}$ is a σ -algebra and $\mathcal{F} \subseteq \mathcal{M}$, then $\sigma(\mathcal{F}) \subseteq \mathcal{M}$ (minimality)

Monotonicity ($\mathcal{F} \subseteq \mathcal{G} \implies \sigma(\mathcal{F}) \subseteq \sigma(\mathcal{G})$) and idempotence ($\sigma(\sigma(\mathcal{F})) = \sigma(\mathcal{F})$) follow trivially from these three properties.

Proof:

Property 1: By the Intersection Lemma, since $\mathcal{P}(X)$ is a σ -algebra containing $\mathcal{F}$, the intersection is nonempty and is a σ -algebra.

Property 2: Every $\mathcal{M}$ in the intersection satisfies $\mathcal{F} \subseteq \mathcal{M}$. Thus $\mathcal{F} \subseteq \bigcap \mathcal{M} = \sigma(\mathcal{F})$.

Property 3: If $\mathcal{M} \supseteq \mathcal{F}$ is a σ -algebra, then $\mathcal{M}$ appears in the intersection. Hence $\sigma(\mathcal{F}) \subseteq \mathcal{M}$. QED

Theorem 15.4.6 (σ -Algebra Induction Principle):

Let $\mathcal{F} \subseteq \mathcal{P}(X)$ and let P be a property of subsets of X . Suppose:

1. $P(\emptyset)$
2. $\forall F \in \mathcal{F}: P(F)$
3. $P(A) \implies P(A^c)$
4. $\forall n \in \mathbb{N}: P(A_n) \implies P(\bigcup_{n \in \mathbb{N}} A_n)$

Then $\forall A \in \sigma(\mathcal{F}): P(A)$.

Proof:

Let $\mathcal{M} = \{A \subseteq X \mid P(A)\}$. By hypotheses (1), (3), (4), $\mathcal{M}$ is a σ -algebra. By hypothesis (2), $\mathcal{F} \subseteq \mathcal{M}$. By minimality, $\sigma(\mathcal{F}) \subseteq \mathcal{M}$. QED

15.4.1 The Borel Hierarchy

The Borel hierarchy provides a constructive description of $\sigma(\mathcal{F})$ using transfinite induction.

Definition 15.4.7 (Borel Hierarchy):

For $\mathcal{F} \subseteq \mathcal{P}(X)$, define by transfinite recursion on ordinals $\alpha < \omega_1$:

- $\mathcal{F}_0 = \mathcal{F} \cup \{\emptyset, X\}$
- For successor ordinals: $\mathcal{F}_{\alpha+1} = \{A \subseteq X \mid A \text{ or } A^c \text{ is a countable union } \bigcup_{n \in \mathbb{N}} A_n \text{ with each } A_n \in \mathcal{F}_\alpha\}$
- For limit ordinals: $\mathcal{F}_\alpha = \bigcup_{\beta < \alpha} \mathcal{F}_\beta$

Theorem 15.4.8 (Constructive Description of $\sigma(\mathcal{F})$):

$$\sigma(\mathcal{F}) = \bigcup_{\alpha < \omega_1} \mathcal{F}_\alpha.$$

Proof:

Step 1: $\bigcup_{\alpha < \omega_1} \mathcal{F}_\alpha$ is a σ -algebra.

Contains $\emptyset$: $\emptyset \in \mathcal{F}_0 \subseteq \bigcup_{\alpha < \omega_1} \mathcal{F}_\alpha$ by definition.

Closure under complements: Let $A \in \bigcup_{\alpha < \omega_1} \mathcal{F}_\alpha$, so $A \in \mathcal{F}_\alpha$ for some $\alpha < \omega_1$. By definition of $\mathcal{F}_{\alpha+1}$, $A^c \in \mathcal{F}_{\alpha+1} \subseteq \bigcup_{\alpha < \omega_1} \mathcal{F}_\alpha$.

Closure under countable unions: Let $(A_n)_{n \in \mathbb{N}} \subseteq \bigcup_{\alpha < \omega_1} \mathcal{F}_\alpha$. For each n , choose $\alpha_n < \omega_1$ with $A_n \in \mathcal{F}_{\alpha_n}$. Let $\gamma = \sup_{n \in \mathbb{N}} \alpha_n$. Since ω_1 is the first uncountable ordinal and $\{\alpha_n\}_{n \in \mathbb{N}}$ is countable, we have $\gamma < \omega_1$. Each $A_n \in \mathcal{F}_{\alpha_n} \subseteq \mathcal{F}_\gamma$. Thus $\bigcup_{n \in \mathbb{N}} A_n \in \mathcal{F}_{\gamma+1} \subseteq \bigcup_{\alpha < \omega_1} \mathcal{F}_\alpha$.

Step 2: $\mathcal{F} \subseteq \bigcup_{\alpha < \omega_1} \mathcal{F}_\alpha$.

This is immediate since $\mathcal{F} \subseteq \mathcal{F}_0 \subseteq \bigcup_{\alpha < \omega_1} \mathcal{F}_\alpha$.

Step 3: Minimality by transfinite induction.

Let $\mathcal{M}$ be any σ -algebra with $\mathcal{F} \subseteq \mathcal{M}$. We prove $\mathcal{F}_\alpha \subseteq \mathcal{M}$ for all $\alpha < \omega_1$ by transfinite induction on α .

Base ($\alpha = 0$): $\mathcal{F}_0 = \mathcal{F} \cup \{\emptyset, X\} \subseteq \mathcal{M}$ since $\mathcal{F} \subseteq \mathcal{M}$ by hypothesis and $\emptyset, X \in \mathcal{M}$ for any σ -algebra.

Successor ($\alpha + 1$): Assume $\mathcal{F}_\alpha \subseteq \mathcal{M}$. Let $A \in \mathcal{F}_{\alpha+1}$. By definition, either A or A^c equals $\bigcup_{n \in \mathbb{N}} A_n$ for some $(A_n)_{n \in \mathbb{N}} \subseteq \mathcal{F}_\alpha$. Since $\mathcal{F}_\alpha \subseteq \mathcal{M}$ and $\mathcal{M}$ is a σ -algebra, $\bigcup_{n \in \mathbb{N}} A_n \in \mathcal{M}$. If $A = \bigcup_{n \in \mathbb{N}} A_n$, then $A \in \mathcal{M}$. If $A^c = \bigcup_{n \in \mathbb{N}} A_n$, then $A^c \in \mathcal{M}$, hence $A \in \mathcal{M}$.

Limit (α limit): $\mathcal{F}_\alpha = \bigcup_{\beta < \alpha} \mathcal{F}_\beta$. By the induction hypothesis, $\mathcal{F}_\beta \subseteq \mathcal{M}$ for all $\beta < \alpha$. Thus $\mathcal{F}_\alpha = \bigcup_{\beta < \alpha} \mathcal{F}_\beta \subseteq \mathcal{M}$.

Conclusion: $\bigcup_{\alpha < \omega_1} \mathcal{F}_\alpha \subseteq \mathcal{M}$, proving minimality.

By Properties 1, 2, 3: $\bigcup_{\alpha < \omega_1} \mathcal{F}_\alpha = \sigma(\mathcal{F})$. QED

Theorem 15.4.9 (Cardinality of Generated σ -Algebras):

For any $\mathcal{F} \subseteq \mathcal{P}(X)$:

$$|\sigma(\mathcal{F})| \leq \max(|\mathcal{F}|, \aleph_0)^{\aleph_0}$$

In particular:

- If $\mathcal{F}$ is countable, then $|\sigma(\mathcal{F})| \leq 2^{\aleph_0}$.
- If $|\mathcal{F}| = \kappa$ for infinite κ , then $|\sigma(\mathcal{F})| \leq \kappa^{\aleph_0}$.

Proof:

Let $\kappa = \max(|\mathcal{F}|, \aleph_0)$. We show $|\mathcal{F}_\alpha| \leq \kappa^{\aleph_0}$ for all $\alpha < \omega_1$ by transfinite induction.

Base ($\alpha = 0$): $|\mathcal{F}_0| = |\mathcal{F} \cup \{\emptyset, X\}| \leq \kappa + 2 = \kappa \leq \kappa^{\aleph_0}$.

Successor ($\alpha + 1$): Assume $|\mathcal{F}_\alpha| \leq \kappa^{\aleph_0}$. Each element of $\mathcal{F}_{\alpha+1}$ is determined by:

- A countable sequence from $\mathcal{F}_\alpha$, i.e., an element of $(\kappa^{\aleph_0})^{\aleph_0} = \kappa^{\aleph_0}$
- A choice of whether to take the union or its complement (2 choices)

Thus $|\mathcal{F}_{\alpha+1}| \leq 2 \cdot \kappa^{\aleph_0} = \kappa^{\aleph_0}$.

Limit (α limit): $|\mathcal{F}_\alpha| = |\bigcup_{\beta < \alpha} \mathcal{F}_\beta| \leq |\alpha| \cdot \kappa^{\aleph_0}$. Since $\alpha < \omega_1$, we have $|\alpha| \leq \aleph_0$. Thus $|\mathcal{F}_\alpha| \leq \aleph_0 \cdot \kappa^{\aleph_0} = \kappa^{\aleph_0}$.

Conclusion: $|\sigma(\mathcal{F})| = |\bigcup_{\alpha < \omega_1} \mathcal{F}_\alpha| \leq |\omega_1| \cdot \kappa^{\aleph_0} = \aleph_1 \cdot \kappa^{\aleph_0} = \kappa^{\aleph_0}$. QED

Definition 15.4.10 (Borel σ -Algebra):

Let $(X, \mathcal{T})$ be a topological space. The **Borel σ -algebra** is $\mathcal{B}(X) = \sigma(\mathcal{T})$, i.e., the σ -algebra generated by the open sets. Elements of $\mathcal{B}(X)$ are called **Borel sets**.

Proposition 15.4.11:

On $\mathbb{R}$, the Borel σ -algebra can be generated by:

1. Open intervals: $\sigma(\{(a, b) \mid a, b \in \mathbb{R}\}) = \mathcal{B}(\mathbb{R})$
2. Half-open intervals: $\sigma(\{(a, b] \mid a, b \in \mathbb{R}\}) = \mathcal{B}(\mathbb{R})$
3. Rays: $\sigma(\{(a, \infty) \mid a \in \mathbb{R}\}) = \mathcal{B}(\mathbb{R})$

Proof:

(1) Every open set in $\mathbb{R}$ is a countable union of open intervals. Thus $\sigma(\text{open intervals}) \supseteq \sigma(\text{open sets}) = \mathcal{B}(\mathbb{R})$. The reverse is clear since open intervals are open.

(2) $(a, b] = \bigcap_{n \in \mathbb{N}} (a, b + 1/n)$, so half-open intervals are Borel. Conversely, $(a, b) = \bigcup_{n \in \mathbb{N}} (a, b - 1/n]$.

(3) (a, ∞) is open, so rays are Borel. Conversely, $(a, b) = (a, \infty) \setminus [b, \infty) = (a, \infty) \cap (-\infty, b)$ where $(-\infty, b) = \bigcup_{n \in \mathbb{N}} (-n, b)$ and $(-n, b) = (-n, \infty) \setminus [b, \infty)$. QED

15.5 Set Decomposition Lemmas

The following lemmas allow us to transform arbitrary well-ordered families of sets into families with special structure (disjoint or monotone) while preserving the union or intersection.

Lemma 15.5.1 (Disjointification of Well-Ordered Families):

Let λ be an ordinal and $(A_\alpha)_{\alpha < \lambda}$ be a family of sets indexed by ordinals less than λ . Define:

$$B_\alpha = A_\alpha \setminus \bigcup_{\beta < \alpha} A_\beta$$

Then:

1. $(B_\alpha)_{\alpha < \lambda}$ are pairwise disjoint
2. Each $B_\alpha \subseteq A_\alpha$
3. $\bigcup_{\alpha < \lambda} A_\alpha = \bigcup_{\alpha < \lambda} B_\alpha$

Proof:

Property 1: Let $\beta < \alpha < \lambda$. We have $B_\alpha = A_\alpha \setminus \bigcup_{\gamma < \alpha} A_\gamma \subseteq A_\alpha \setminus A_\beta$. Since $B_\beta \subseteq A_\beta$, we have $B_\alpha \cap B_\beta \subseteq (A_\alpha \setminus A_\beta) \cap A_\beta = \emptyset$.

Property 2: By definition, $B_\alpha = A_\alpha \setminus \bigcup_{\beta < \alpha} A_\beta \subseteq A_\alpha$.

Property 3: $(\supseteq)$ Each $B_\alpha \subseteq A_\alpha$, so $\bigcup_{\alpha < \lambda} B_\alpha \subseteq \bigcup_{\alpha < \lambda} A_\alpha$.

$(\subseteq)$ Let $x \in \bigcup_{\alpha < \lambda} A_\alpha$. Define $\gamma = \min\{\alpha < \lambda \mid x \in A_\alpha\}$. This minimum exists since the ordinals are well-ordered. Then $x \in A_\gamma$ and $x \notin A_\beta$ for all $\beta < \gamma$. Thus $x \in A_\gamma \setminus \bigcup_{\beta < \gamma} A_\beta = B_\gamma \subseteq \bigcup_{\alpha < \lambda} B_\alpha$. QED

Lemma 15.5.2 (Monotonization to Increasing Family):

Let λ be an ordinal and $(A_\alpha)_{\alpha < \lambda}$ be a family of sets. Define:

$$C_\alpha = \bigcup_{\beta \leq \alpha} A_\beta$$

Then:

1. $(C_\alpha)_{\alpha < \lambda}$ is increasing: $\alpha < \beta \implies C_\alpha \subseteq C_\beta$
2. Each $A_\alpha \subseteq C_\alpha$
3. $\bigcup_{\alpha < \lambda} A_\alpha = \bigcup_{\alpha < \lambda} C_\alpha$

Proof:

Property 1: If $\alpha < \beta$, then $\{\gamma \mid \gamma \leq \alpha\} \subseteq \{\gamma \mid \gamma \leq \beta\}$, so $C_\alpha = \bigcup_{\gamma \leq \alpha} A_\gamma \subseteq \bigcup_{\gamma \leq \beta} A_\gamma = C_\beta$.

Property 2: $A_\alpha \subseteq \bigcup_{\beta \leq \alpha} A_\beta = C_\alpha$.

Property 3: $(\subseteq)$ Each $A_\alpha \subseteq C_\alpha$, so $\bigcup_{\alpha < \lambda} A_\alpha \subseteq \bigcup_{\alpha < \lambda} C_\alpha$.

$(\supseteq)$ Each $C_\alpha = \bigcup_{\beta \leq \alpha} A_\beta \subseteq \bigcup_{\alpha < \lambda} A_\alpha$. QED

Lemma 15.5.3 (Monotonization to Decreasing Family):

Let λ be an ordinal and $(A_\alpha)_{\alpha < \lambda}$ be a family of sets. Define:

$$D_\alpha = \bigcap_{\beta \leq \alpha} A_\beta$$

Then:

1. $(D_\alpha)_{\alpha < \lambda}$ is decreasing: $\alpha < \beta \implies D_\alpha \supseteq D_\beta$
2. Each $D_\alpha \subseteq A_\alpha$
3. $\bigcap_{\alpha < \lambda} A_\alpha = \bigcap_{\alpha < \lambda} D_\alpha$

Proof:

Property 1: If $\alpha < \beta$, then $\{\gamma \mid \gamma \leq \alpha\} \subseteq \{\gamma \mid \gamma \leq \beta\}$, so $D_\alpha = \bigcap_{\gamma \leq \alpha} A_\gamma \supseteq \bigcap_{\gamma \leq \beta} A_\gamma = D_\beta$.

Property 2: $D_\alpha = \bigcap_{\beta \leq \alpha} A_\beta \subseteq A_\alpha$.

Property 3: $(\supseteq)$ Each $D_\alpha \supseteq \bigcap_{\beta < \lambda} A_\beta$, so $\bigcap_{\alpha < \lambda} D_\alpha \supseteq \bigcap_{\alpha < \lambda} A_\alpha$.

$(\subseteq)$ For each $\gamma < \lambda$, we have $D_\alpha \subseteq A_\gamma$ whenever $\alpha \geq \gamma$. Thus $\bigcap_{\alpha < \lambda} D_\alpha \subseteq A_\gamma$. Since γ is arbitrary, $\bigcap_{\alpha < \lambda} D_\alpha \subseteq \bigcap_{\gamma < \lambda} A_\gamma$. QED

Remark:

These decompositions are fundamental in measure theory. For countable families:

- **Disjointification** reduces countable unions to disjoint countable unions, essential for σ -additivity: $\mu(\bigcup_{n \in \mathbb{N}} A_n) = \mu(\bigcup_{n \in \mathbb{N}} B_n) = \sum_{n \in \mathbb{N}} \mu(B_n)$.
- **Monotonization** allows application of measure continuity: $\mu(\bigcup_{n \in \mathbb{N}} A_n) = \mu(\bigcup_{n \in \mathbb{N}} C_n) = \lim_{n \rightarrow \infty} \mu(C_n)$.

The transfinite versions are used in set-theoretic constructions such as the Borel hierarchy.

15.6 λ -Systems (Dynkin Systems)

Definition 15.6.1 (λ -System (First Formulation)):

A collection $\mathcal{L} \subseteq \mathcal{P}(X)$ is a λ -system (or **Dynkin system**) if:

- (I.1) $X \in \mathcal{L}$
- (I.2) If $A, B \in \mathcal{L}$ and $A \subseteq B$, then $B \setminus A \in \mathcal{L}$
- (I.3) If $(A_n)_{n \in \mathbb{N}} \subseteq \mathcal{L}$ with $A_n \subseteq A_{n+1}$, then $\bigcup_{n \in \mathbb{N}} A_n \in \mathcal{L}$

Definition 15.6.2 (λ -System (Second Formulation)):

Equivalently, $\mathcal{L} \subseteq \mathcal{P}(X)$ is a λ -system if:

- (II.1) $\emptyset \in \mathcal{L}$
- (II.2) If $A \in \mathcal{L}$, then $A^c \in \mathcal{L}$
- (II.3) If $(A_n)_{n \in \mathbb{N}} \subseteq \mathcal{L}$ are pairwise disjoint, then $\bigcup_{n \in \mathbb{N}} A_n \in \mathcal{L}$

Proposition 15.6.3 (Equivalence of Formulations):

The two formulations of λ -system are equivalent.

Proof:

(I) $\implies$ (II):

(II.1): $\emptyset = X \setminus X \in \mathcal{L}$ by (I.2) since $X \in \mathcal{L}$ and $X \subseteq X$.

(II.2): $A^c = X \setminus A \in \mathcal{L}$ by (I.2) since $A \subseteq X$.

(II.3): Let $(A_n)_{n \in \mathbb{N}}$ be pairwise disjoint. Define $B_n = \bigcup_{k=1}^n A_k$. Then $B_n \subseteq B_{n+1}$ (increasing). We show each $B_n \in \mathcal{L}$ by induction on n :

- $B_1 = A_1 \in \mathcal{L}$.
- $B_{n+1} = B_n \cup A_{n+1}$. From (II.2), $B_n^c \in \mathcal{L}$. Since $A_{n+1} \subseteq B_n^c$ (disjointness), we have $B_n^c \setminus A_{n+1} = (B_n \cup A_{n+1})^c \in \mathcal{L}$ by (I.2). Thus $B_{n+1} = B_n \cup A_{n+1} \in \mathcal{L}$ by (II.2).

By (I.3), $\bigcup_{n \in \mathbb{N}} A_n = \bigcup_{n \in \mathbb{N}} B_n \in \mathcal{L}$.

(II) $\Rightarrow$ (I):

(I.1): $X = \emptyset^c \in \mathcal{L}$ by (II.2).

(I.2): Let $A \subseteq B$ with $A, B \in \mathcal{L}$. We have $B \setminus A = B \cap A^c$. Consider the disjoint decomposition:

$$B = A \cup (B \setminus A), \quad B^c, \quad A \setminus B = \emptyset \quad (\text{since } A \subseteq B)$$

Actually, $B \setminus A$, A , and B^c partition X . By (II.2), $B^c \in \mathcal{L}$. Now $X = A \cup (B \setminus A) \cup B^c$, i.e., these three are pairwise disjoint. By (II.3), $A \cup (B \setminus A) \cup B^c = X \in \mathcal{L}$. Actually, we need a different approach:

Define $C = B \setminus A$. We must show $C \in \mathcal{L}$. Note $C^c = A \cup B^c$. We have A and B^c are disjoint (since $A \subseteq B$). By (II.3), $A \cup B^c \in \mathcal{L}$. Thus $C^c \in \mathcal{L}$, so $C \in \mathcal{L}$ by (II.2).

(I.3): Let $A_n \nearrow A$. By the Disjointification Lemma, define $B_1 = A_1$ and $B_n = A_n \setminus A_{n-1}$ for $n \geq 2$. Then $(B_n)_{n \in \mathbb{N}}$ are pairwise disjoint and $\bigcup_{n \in \mathbb{N}} B_n = A$. Each $B_n \in \mathcal{L}$ by (I.2) (which we just proved). By (II.3), $A \in \mathcal{L}$. QED

Intuition:

A λ -system is weaker than a σ -algebra:

- Formulation I: closure under *proper* differences (when $A \subseteq B$) and *increasing* countable unions.
- Formulation II: closure under complements and *disjoint* countable unions.

Neither requires closure under arbitrary unions or intersections. The key insight: a λ -system that is *also* a π -system must be a σ -algebra. This is the content of Dynkin's π - λ theorem.

Proposition 15.6.4:

Every σ -algebra is a λ -system.

Proof:

Let $\mathcal{M}$ be a σ -algebra.

Axiom 1: $X = \emptyset^c \in \mathcal{M}$.

Axiom 2: $B \setminus A = B \cap A^c \in \mathcal{M}$.

Axiom 3: Countable unions are in $\mathcal{M}$ by definition. QED

Lemma 15.6.5 (Intersection of λ -Systems):

Let $\{\mathcal{L}_i\}_{i \in I}$ be an arbitrary family of λ -systems on X . Then $\bigcap_{i \in I} \mathcal{L}_i$ is a λ -system.

Proof:

Let $\mathcal{L} = \bigcap_{i \in I} \mathcal{L}_i$.

Axiom 1: $X \in \mathcal{L}_i$ for all i , so $X \in \mathcal{L}$.

Axiom 2: Let $A, B \in \mathcal{L}$ with $A \subseteq B$. Then $A, B \in \mathcal{L}_i$ for all i . Since each $\mathcal{L}_i$ is a λ -system, $B \setminus A \in \mathcal{L}_i$ for all i . Thus $B \setminus A \in \mathcal{L}$.

Axiom 3: Let $(A_n)_{n \in \mathbb{N}} \subseteq \mathcal{L}$ with $A_n \nearrow$. Then each $A_n \in \mathcal{L}_i$ for all i . Since each $\mathcal{L}_i$ is a λ -system, $\bigcup_{n \in \mathbb{N}} A_n \in \mathcal{L}_i$ for all i . Thus $\bigcup_{n \in \mathbb{N}} A_n \in \mathcal{L}$. QED

Definition 15.6.6 (Generated λ -System):

For $\mathcal{F} \subseteq \mathcal{P}(X)$, the λ -system generated by $\mathcal{F}$, denoted $\lambda(\mathcal{F})$, is:

$$\lambda(\mathcal{F}) = \bigcap \{ \mathcal{L} \mid \mathcal{L} \text{ is a } \lambda\text{-system on } X \text{ and } \mathcal{F} \subseteq \mathcal{L} \}$$

Proposition 15.6.7 (Properties of Generated λ -Systems):

Let $\mathcal{F} \subseteq \mathcal{P}(X)$. Then:

1. $\lambda(\mathcal{F})$ is a λ -system (by the Intersection Lemma)
2. $\mathcal{F} \subseteq \lambda(\mathcal{F})$ (extension)
3. If $\mathcal{L}$ is a λ -system and $\mathcal{F} \subseteq \mathcal{L}$, then $\lambda(\mathcal{F}) \subseteq \mathcal{L}$ (minimality)

Proof:

Property 1: By the Intersection Lemma. The intersection is nonempty since $\mathcal{P}(X)$ is a λ -system containing $\mathcal{F}$.

Property 2: Every $\mathcal{L}$ in the intersection satisfies $\mathcal{F} \subseteq \mathcal{L}$. Thus $\mathcal{F} \subseteq \bigcap \mathcal{L} = \lambda(\mathcal{F})$.

Property 3: If $\mathcal{L} \supseteq \mathcal{F}$ is a λ -system, then $\mathcal{L}$ appears in the intersection. Hence $\lambda(\mathcal{F}) \subseteq \mathcal{L}$. QED

Corollary 15.6.8 (Monotonicity and Idempotence):

Let $\mathcal{F}, \mathcal{G} \subseteq \mathcal{P}(X)$. Then:

1. If $\mathcal{F} \subseteq \mathcal{G}$, then $\lambda(\mathcal{F}) \subseteq \lambda(\mathcal{G})$ (monotonicity)
2. $\lambda(\lambda(\mathcal{F})) = \lambda(\mathcal{F})$ (idempotence)

Proof:

Monotonicity: Let $\mathcal{F} \subseteq \mathcal{G}$. Then $\lambda(\mathcal{G})$ is a λ -system containing $\mathcal{G} \supseteq \mathcal{F}$. By minimality, $\lambda(\mathcal{F}) \subseteq \lambda(\mathcal{G})$.

Idempotence: $\lambda(\mathcal{F})$ is a λ -system, so by minimality, $\lambda(\lambda(\mathcal{F})) \subseteq \lambda(\mathcal{F})$. By extension, $\lambda(\mathcal{F}) \subseteq \lambda(\lambda(\mathcal{F}))$. Thus equality. QED

Theorem 15.6.9 (λ -System Induction Principle):

Let $\mathcal{F} \subseteq \mathcal{P}(X)$ and let P be a property of subsets of X . Suppose:

1. $P(X)$
2. $\forall F \in \mathcal{F} : P(F)$
3. $P(A)$ and $P(B)$ and $A \subseteq B \implies P(B \setminus A)$
4. $\forall n \in \mathbb{N} : P(A_n)$ and $A_n \nearrow A \implies P(A)$

Then $\forall A \in \lambda(\mathcal{F}) : P(A)$.

Proof:

Let $\mathcal{M} = \{A \subseteq X \mid P(A)\}$. By hypotheses (1), (3), (4), $\mathcal{M}$ is a λ -system. By hypothesis (2), $\mathcal{F} \subseteq \mathcal{M}$. By minimality, $\lambda(\mathcal{F}) \subseteq \mathcal{M}$. QED

Proposition 15.6.10 (Constructive Description of $\lambda(\mathcal{F})$):

Define $\mathcal{F}_0 = \mathcal{F} \cup \{X\}$ and for $n \geq 0$:

$$\mathcal{F}_{n+1} = \mathcal{F}_n \cup \{B \setminus A \mid A, B \in \mathcal{F}_n, A \subseteq B\} \cup \left\{ \bigcup_{k \in \mathbb{N}} A_k \mid (A_k)_{k \in \mathbb{N}} \subseteq \mathcal{F}_n, A_k \nearrow \right\}$$

Then $\lambda(\mathcal{F}) = \bigcup_{n \in \mathbb{N}} \mathcal{F}_n$.

Proof:

Let $\mathcal{G} = \bigcup_{n \in \mathbb{N}} \mathcal{F}_n$.

$\mathcal{G}$ is a λ -system:

Axiom 1: $X \in \mathcal{F}_0 \subseteq \mathcal{G}$.

Axiom 2: Let $A, B \in \mathcal{G}$ with $A \subseteq B$. Then $A \in \mathcal{F}_m$ and $B \in \mathcal{F}_n$ for some m, n . Let $k = \max(m, n)$. Then $A, B \in \mathcal{F}_k$, so $B \setminus A \in \mathcal{F}_{k+1} \subseteq \mathcal{G}$.

Axiom 3: Let $(A_k)_{k \in \mathbb{N}} \subseteq \mathcal{G}$ with $A_k \nearrow$. Each $A_k \in \mathcal{F}_{n_k}$ for some n_k . If all n_k are bounded by some m , then all $A_k \in \mathcal{F}_m$, so $\bigcup_k A_k \in \mathcal{F}_{m+1} \subseteq \mathcal{G}$.

$\mathcal{F} \subseteq \mathcal{G}$: Since $\mathcal{F} \subseteq \mathcal{F}_0 \subseteq \mathcal{G}$.

$\mathcal{G}$ is **minimal**: Let $\mathcal{L}$ be any λ -system with $\mathcal{F} \subseteq \mathcal{L}$. We show $\mathcal{F}_n \subseteq \mathcal{L}$ for all n by induction.

Base: $\mathcal{F}_0 = \mathcal{F} \cup \{X\} \subseteq \mathcal{L}$ since $\mathcal{F} \subseteq \mathcal{L}$ and $X \in \mathcal{L}$.

Step: Assume $\mathcal{F}_n \subseteq \mathcal{L}$. If $A, B \in \mathcal{F}_n \subseteq \mathcal{L}$ with $A \subseteq B$, then $B \setminus A \in \mathcal{L}$. If $(A_k) \subseteq \mathcal{F}_n \subseteq \mathcal{L}$ with $A_k \nearrow$, then $\bigcup_k A_k \in \mathcal{L}$. Thus $\mathcal{F}_{n+1} \subseteq \mathcal{L}$.

Hence $\mathcal{G} = \bigcup_{n \in \mathbb{N}} \mathcal{F}_n \subseteq \mathcal{L}$, proving minimality.

By the three properties, $\mathcal{G} = \lambda(\mathcal{F})$. QED

Theorem 15.6.11 (Cardinality of Generated λ -Systems):

For any $\mathcal{F} \subseteq \mathcal{P}(X)$:

$$|\lambda(\mathcal{F})| \leq \max(|\mathcal{F}|, \aleph_0)^{\aleph_0}$$

Proof:

Let $\kappa = \max(|\mathcal{F}|, \aleph_0)$. We show $|\mathcal{F}_n| \leq \kappa^{\aleph_0}$ for all n by induction.

Base: $|\mathcal{F}_0| = |\mathcal{F} \cup \{X\}| \leq \kappa + 1 = \kappa \leq \kappa^{\aleph_0}$.

Step: Assume $|\mathcal{F}_n| \leq \kappa^{\aleph_0}$. Elements of $\mathcal{F}_{n+1} \setminus \mathcal{F}_n$ are either:

- Proper differences $B \setminus A$: at most $|\mathcal{F}_n|^2 \leq (\kappa^{\aleph_0})^2 = \kappa^{\aleph_0}$ such elements.
- Increasing unions of sequences from $\mathcal{F}_n$: at most $|\mathcal{F}_n|^{\aleph_0} \leq (\kappa^{\aleph_0})^{\aleph_0} = \kappa^{\aleph_0}$ such elements.

Thus $|\mathcal{F}_{n+1}| \leq \kappa^{\aleph_0}$.

Conclusion: $|\lambda(\mathcal{F})| = |\bigcup_{n \in \mathbb{N}} \mathcal{F}_n| \leq \aleph_0 \cdot \kappa^{\aleph_0} = \kappa^{\aleph_0}$. QED

15.7 Monotone Class Lemma

Theorem 15.7.1 (Monotone Class Lemma):

If $\mathcal{A}$ is an algebra on X , then $\sigma(\mathcal{A}) = \mathcal{N}(\mathcal{A})$.

Intuition:

The Monotone Class Lemma says:

Starting from an algebra, monotone limits give you everything.

Why is this useful? When proving two σ -algebras are equal, it sometimes suffices to check that a certain property holds for an algebra and is preserved under monotone limits. This is particularly natural when dealing with measure continuity, where monotone sequences arise naturally.

The proof strategy: show the generated monotone class has enough structure (complementation + unions) to be a σ -algebra, then minimality forces equality.

Proof:

We proceed in three steps. Our goal is to show $\mathcal{N}(\mathcal{A}) = \sigma(\mathcal{A})$.

Step 1: One inclusion is immediate.

Every σ -algebra is a monotone class (as shown earlier). Thus $\sigma(\mathcal{A})$ is a monotone class containing $\mathcal{A}$. Since $\mathcal{N}(\mathcal{A})$ is the *smallest* monotone class containing $\mathcal{A}$:

$$\mathcal{N}(\mathcal{A}) \subseteq \sigma(\mathcal{A})$$

Step 2: Showing $\mathcal{N}(\mathcal{A})$ is an algebra.

For the reverse inclusion, we show $\mathcal{N}(\mathcal{A})$ is a σ -algebra. We first prove it is an algebra.

Step 2a: Closure under complements.

Define:

$$\mathcal{G} = \{A \in \mathcal{N}(\mathcal{A}) \mid A^c \in \mathcal{N}(\mathcal{A})\}$$

We show $\mathcal{G}$ is a monotone class.

Increasing sequences: Let $A_n \nearrow A$ with each $A_n \in \mathcal{G}$. Then:

- Each $A_n^c \in \mathcal{N}(\mathcal{A})$ (since $A_n \in \mathcal{G}$)
- $A_n^c \searrow A^c$ (taking complements reverses the direction)
- Since $\mathcal{N}(\mathcal{A})$ is closed under decreasing limits, $A^c \in \mathcal{N}(\mathcal{A})$

Thus $A \in \mathcal{G}$.

Decreasing sequences: Let $A_n \searrow A$ with each $A_n \in \mathcal{G}$. Then:

- Each $A_n^c \in \mathcal{N}(\mathcal{A})$
- $A_n^c \nearrow A^c$
- Since $\mathcal{N}(\mathcal{A})$ is closed under increasing limits, $A^c \in \mathcal{N}(\mathcal{A})$

Thus $A \in \mathcal{G}$.

Since $\mathcal{A}$ is an algebra, it is closed under complements, so $\mathcal{A} \subseteq \mathcal{G}$. Since $\mathcal{G}$ is a monotone class containing $\mathcal{A}$:

$$\mathcal{N}(\mathcal{A}) \subseteq \mathcal{G} \subseteq \mathcal{N}(\mathcal{A})$$

Therefore $\mathcal{N}(\mathcal{A}) = \mathcal{G}$, proving closure under complements.

Step 2b: Closure under finite unions.

We use a double bootstrap similar to the π - λ theorem.

First bootstrap: For fixed $B \in \mathcal{A}$, define:

$$\mathcal{H}_B = \{A \in \mathcal{N}(\mathcal{A}) \mid A \cup B \in \mathcal{N}(\mathcal{A})\}$$

$\mathcal{H}_B$ is a monotone class because:

- If $A_n \nearrow A$ with $A_n \in \mathcal{H}_B$, then $A_n \cup B \nearrow A \cup B$. Since each $A_n \cup B \in \mathcal{N}(\mathcal{A})$ and $\mathcal{N}(\mathcal{A})$ is closed under increasing limits, $A \cup B \in \mathcal{N}(\mathcal{A})$.
- Similarly for decreasing sequences.

Since $\mathcal{A}$ is an algebra, $A \cup B \in \mathcal{A} \subseteq \mathcal{N}(\mathcal{A})$ for any $A \in \mathcal{A}$. Thus $\mathcal{A} \subseteq \mathcal{H}_B$, so $\mathcal{N}(\mathcal{A}) = \mathcal{H}_B$.

Second bootstrap: For fixed $A \in \mathcal{N}(\mathcal{A})$, define:

$$\mathcal{H}_A = \{B \in \mathcal{N}(\mathcal{A}) \mid A \cup B \in \mathcal{N}(\mathcal{A})\}$$

$\mathcal{H}_A$ is a monotone class by the same reasoning. From the first bootstrap, for any $B \in \mathcal{A}$, we have $A \cup B \in \mathcal{N}(\mathcal{A})$ (since $A \in \mathcal{N}(\mathcal{A}) = \mathcal{H}_B$). Thus $\mathcal{A} \subseteq \mathcal{H}_A$, so $\mathcal{N}(\mathcal{A}) = \mathcal{H}_A$.

Conclusion of Step 2: For any $A, B \in \mathcal{N}(\mathcal{A})$, we have $A \cup B \in \mathcal{N}(\mathcal{A})$. Together with closure under complements and the fact that $\emptyset \in \mathcal{A} \subseteq \mathcal{N}(\mathcal{A})$, this shows $\mathcal{N}(\mathcal{A})$ is an algebra.

Step 3: An algebra that is also a monotone class is a σ -algebra.

Let $(A_n)_{n \in \mathbb{N}} \subseteq \mathcal{N}(\mathcal{A})$. Define:

$$B_n = \bigcup_{i=1}^n A_i$$

Since $\mathcal{N}(\mathcal{A})$ is an algebra, each $B_n \in \mathcal{N}(\mathcal{A})$ (finite unions). The sequence (B_n) is increasing: $B_n \nearrow \bigcup_{n \in \mathbb{N}} A_n$. Since $\mathcal{N}(\mathcal{A})$ is a monotone class (closed under increasing limits):

$$\bigcup_{n \in \mathbb{N}} A_n \in \mathcal{N}(\mathcal{A})$$

Thus $\mathcal{N}(\mathcal{A})$ is closed under countable unions. Combined with closure under complements and $\emptyset \in \mathcal{N}(\mathcal{A})$, it is a σ -algebra.

Conclusion: Since $\mathcal{N}(\mathcal{A})$ is a σ -algebra containing $\mathcal{A}$:

$$\sigma(\mathcal{A}) \subseteq \mathcal{N}(\mathcal{A})$$

Combined with Step 1: $\mathcal{N}(\mathcal{A}) = \sigma(\mathcal{A})$.

QED

15.8 Dynkin's π - λ Theorem

Theorem 15.8.1 (Dynkin's π - λ Theorem):

If $\mathcal{P}$ is a π -system and $\mathcal{L}$ is a λ -system with $\mathcal{P} \subseteq \mathcal{L}$, then $\sigma(\mathcal{P}) \subseteq \mathcal{L}$.

Equivalently: If $\mathcal{P}$ is a π -system, then $\lambda(\mathcal{P}) = \sigma(\mathcal{P})$.

Intuition:

Why is this theorem powerful? It allows us to prove that two σ -finite measures agree by checking only a π -system, which is often much simpler. For example:

- To show Lebesgue measure is unique, we only need to check it on intervals.
- To prove independence, we only need to check products of generating sets.

The bootstrap strategy: The proof uses a clever “bootstrapping” argument. We want to show $\lambda(\mathcal{P})$ is closed under intersections. We cannot verify this directly, so we:

1. First show: every element of $\lambda(\mathcal{P})$ intersects nicely with elements of $\mathcal{P}$.
2. Then upgrade: every element of $\lambda(\mathcal{P})$ intersects nicely with *all* elements of $\lambda(\mathcal{P})$.

Proof:

We proceed in four steps. Our goal is to show $\lambda(\mathcal{P}) = \sigma(\mathcal{P})$.

Step 1: One inclusion is immediate.

Every σ -algebra is a λ -system, so $\sigma(\mathcal{P})$ is the smallest σ -algebra containing $\mathcal{P}$, and it is also a λ -system. Since $\lambda(\mathcal{P})$ is the *smallest* λ -system containing $\mathcal{P}$, and $\sigma(\mathcal{P})$ is one such λ -system:

$$\lambda(\mathcal{P}) \subseteq \sigma(\mathcal{P})$$

Step 2: A λ -system that is also a π -system is a σ -algebra.

We prove this as a lemma within the proof. Let $\mathcal{L}$ be both a λ -system and a π -system. We verify the σ -algebra axioms:

- (1) $\emptyset \in \mathcal{L}$: Since $\mathcal{L}$ is a λ -system, $X \in \mathcal{L}$. Thus $\emptyset = X \setminus X \in \mathcal{L}$ (proper difference).
- (2) *Closure under complements*: For any $A \in \mathcal{L}$, since $A \subseteq X$, we have $A^c = X \setminus A \in \mathcal{L}$ (proper difference).
- (3) *Closure under countable unions*: First, we show $\mathcal{L}$ is closed under finite unions. For $A, B \in \mathcal{L}$:

$$A \cup B = (A^c \cap B^c)^c$$

Since $A^c, B^c \in \mathcal{L}$ (by closure under complements) and $A^c \cap B^c \in \mathcal{L}$ (by π -system property), we have $A \cup B \in \mathcal{L}$. By induction, $\mathcal{L}$ is closed under all finite unions.

Now let $(A_n)_{n \in \mathbb{N}} \subseteq \mathcal{L}$. By the Disjointification Lemma, define $B_1 = A_1$ and $B_n = A_n \setminus \bigcup_{k=1}^{n-1} A_k$ for $n \geq 2$. Then $(B_n)_{n \in \mathbb{N}}$ are pairwise disjoint and $\bigcup_{n \in \mathbb{N}} A_n = \bigcup_{n \in \mathbb{N}} B_n$.

Each $B_n \in \mathcal{L}$: For $n = 1$, $B_1 = A_1 \in \mathcal{L}$. For $n \geq 2$, let $C_n = \bigcup_{k=1}^{n-1} A_k \in \mathcal{L}$ (finite union). Then $A_n \cap C_n \in \mathcal{L}$ (by π -system property), and $B_n = A_n \setminus (A_n \cap C_n) \in \mathcal{L}$ (proper difference, since $A_n \cap C_n \subseteq A_n$).

By the Monotonization Lemma, $S_n = \bigcup_{k=1}^n B_k$ forms an increasing sequence with $S_n \nearrow \bigcup_{n \in \mathbb{N}} B_n$. Each $S_n \in \mathcal{L}$ (finite union). By the increasing union property of λ -systems, $\bigcup_{n \in \mathbb{N}} A_n = \bigcup_{n \in \mathbb{N}} B_n \in \mathcal{L}$.

Thus $\mathcal{L}$ is a σ -algebra. $\square$

Strategy: By this lemma, to show $\lambda(\mathcal{P})$ is a σ -algebra, it suffices to show $\lambda(\mathcal{P})$ is a π -system, i.e., closed under finite intersections.

Step 3: The first bootstrap—intersecting with the generating π -system.

Define the collection:

$$\mathcal{G}_1 = \{A \in \lambda(\mathcal{P}) \mid \forall P \in \mathcal{P} : A \cap P \in \lambda(\mathcal{P})\}$$

We claim $\mathcal{G}_1$ is a λ -system.

Verification of λ -system axioms for $\mathcal{G}_1$:

Axiom 1 ($X \in \mathcal{G}_1$): For any $P \in \mathcal{P}$, we have $X \cap P = P \in \mathcal{P} \subseteq \lambda(\mathcal{P})$. Thus $X \in \mathcal{G}_1$.

Axiom 2 (Proper differences): Let $A, B \in \mathcal{G}_1$ with $A \subseteq B$. We must show $(B \setminus A) \cap P \in \lambda(\mathcal{P})$ for all $P \in \mathcal{P}$.

Note that $(B \setminus A) \cap P = (B \cap P) \setminus (A \cap P)$. Since $A, B \in \mathcal{G}_1$, both $A \cap P$ and $B \cap P$ are in $\lambda(\mathcal{P})$. Moreover, $A \subseteq B$ implies $A \cap P \subseteq B \cap P$. Since $\lambda(\mathcal{P})$ is a λ -system, $(B \cap P) \setminus (A \cap P) \in \lambda(\mathcal{P})$.

Axiom 3 (Increasing unions): Let $A_n \nearrow A$ with each $A_n \in \mathcal{G}_1$. For any $P \in \mathcal{P}$:

$$A_n \cap P \nearrow A \cap P$$

Since each $A_n \cap P \in \lambda(\mathcal{P})$ and $\lambda(\mathcal{P})$ is closed under increasing unions, we have $A \cap P \in \lambda(\mathcal{P})$. Thus $A \in \mathcal{G}_1$.

Conclusion of Step 3: Since $\mathcal{P}$ is a π -system, for any $P', P \in \mathcal{P}$, we have $P' \cap P \in \mathcal{P} \subseteq \lambda(\mathcal{P})$. Thus $\mathcal{P} \subseteq \mathcal{G}_1$. Since $\mathcal{G}_1 \subseteq \lambda(\mathcal{P})$ by definition and $\mathcal{G}_1$ is a λ -system containing $\mathcal{P}$:

$$\lambda(\mathcal{P}) \subseteq \mathcal{G}_1 \subseteq \lambda(\mathcal{P})$$

Therefore $\lambda(\mathcal{P}) = \mathcal{G}_1$.

Step 4: The second bootstrap—full closure under intersections.

Define:

$$\mathcal{G}_2 = \{A \in \lambda(\mathcal{P}) \mid \forall B \in \lambda(\mathcal{P}) : A \cap B \in \lambda(\mathcal{P})\}$$

By the same argument as Step 3, $\mathcal{G}_2$ is a λ -system (replace $P \in \mathcal{P}$ with $B \in \lambda(\mathcal{P})$ throughout).

From Step 3, we know $\lambda(\mathcal{P}) = \mathcal{G}_1$. This means: for any $P \in \mathcal{P}$ and any $B \in \lambda(\mathcal{P})$, we have $P \cap B \in \lambda(\mathcal{P})$ (since $B \in \mathcal{G}_1$).

This shows $\mathcal{P} \subseteq \mathcal{G}_2$. Since $\mathcal{G}_2$ is a λ -system:

$$\lambda(\mathcal{P}) \subseteq \mathcal{G}_2 \subseteq \lambda(\mathcal{P})$$

Therefore $\lambda(\mathcal{P}) = \mathcal{G}_2$.

Conclusion: For all $A, B \in \lambda(\mathcal{P})$, we have $A \cap B \in \lambda(\mathcal{P})$. Thus $\lambda(\mathcal{P})$ is a π -system. Being both a π -system and a λ -system, it is a σ -algebra.

Since $\lambda(\mathcal{P})$ is a σ -algebra containing $\mathcal{P}$:

$$\sigma(\mathcal{P}) \subseteq \lambda(\mathcal{P})$$

Combined with Step 1: $\lambda(\mathcal{P}) = \sigma(\mathcal{P})$.

QED

Chapter Compendium — Set-Theoretic Foundations:**Set Systems Hierarchy:**

- **π -System:** Closed under finite intersections.
- **Algebra:** Contains $\emptyset$, closed under complements and finite unions.
- **Monotone Class:** Closed under increasing and decreasing countable limits.
- **σ -Algebra:** Contains $\emptyset$, closed under complements and countable unions.
- **λ -System:** Contains X , closed under proper differences and increasing unions.

Generated Systems: For each type, $\text{Gen}(\mathcal{F})$ is the intersection of all systems of that type containing $\mathcal{F}$.

Cardinality Bounds:

- $|\pi(\mathcal{F})|, |\mathcal{A}(\mathcal{F})| \leq \max(|\mathcal{F}|, \aleph_0)$
- $|\mathcal{N}(\mathcal{F})|, |\lambda(\mathcal{F})|, |\sigma(\mathcal{F})| \leq \max(|\mathcal{F}|, \aleph_0)^{\aleph_0}$

Key Theorems:

- **Monotone Class Lemma:** $\sigma(\mathcal{A}) = \mathcal{N}(\mathcal{A})$ for any algebra $\mathcal{A}$.
- **Dynkin π - λ :** If $\mathcal{P}$ is a π -system and $\mathcal{L}$ is a λ -system with $\mathcal{P} \subseteq \mathcal{L}$, then $\sigma(\mathcal{P}) \subseteq \mathcal{L}$. Equivalently: $\lambda(\mathcal{P}) = \sigma(\mathcal{P})$.

Set Decomposition: Any family $(A_\alpha)_{\alpha < \lambda}$ can be transformed into:

- **Disjoint family** (B_α) with same union, where $B_\alpha = A_\alpha \setminus \bigcup_{\beta < \alpha} A_\beta$.
- **Increasing family** (C_α) with same union, where $C_\alpha = \bigcup_{\beta \leq \alpha} A_\beta$.
- **Decreasing family** (D_α) with same intersection, where $D_\alpha = \bigcap_{\beta \leq \alpha} A_\beta$.

CHAPTER 16

Contents

16.1 Contentable Spaces

Definition 16.1.1 (Contentable Space):

A **contentable space** is a pair $(X, \mathcal{A})$ where X is a set and $\mathcal{A}$ is an algebra on X .

Intuition:

A contentable space is the natural domain for defining finitely additive measures (contents). The algebra structure provides closure under finite set operations, which is sufficient for defining finite additivity. This is analogous to how a measurable space (set + σ -algebra) is the natural domain for σ -additive measures.

Definition 16.1.2 (Contentable Subspace):

Let $(X, \mathcal{A})$ be a contentable space and $Y \subseteq X$. The **contentable subspace** induced on Y is $(Y, \mathcal{A}_Y)$ where:

$$\mathcal{A}_Y = \{A \cap Y \mid A \in \mathcal{A}\}$$

Proposition 16.1.3:

If $(X, \mathcal{A})$ is a contentable space and $Y \subseteq X$, then $\mathcal{A}_Y$ is an algebra on Y .

Proof:

Contains $\emptyset$: $\emptyset = \emptyset \cap Y \in \mathcal{A}_Y$.

Contains Y : $Y = X \cap Y \in \mathcal{A}_Y$.

Closure under complements: Let $A \cap Y \in \mathcal{A}_Y$ with $A \in \mathcal{A}$. Then $Y \setminus (A \cap Y) = (X \setminus A) \cap Y$. Since $\mathcal{A}$ is an algebra, $X \setminus A \in \mathcal{A}$, so $(X \setminus A) \cap Y \in \mathcal{A}_Y$.

Closure under unions: Let $A_1 \cap Y, A_2 \cap Y \in \mathcal{A}_Y$. Then $(A_1 \cap Y) \cup (A_2 \cap Y) = (A_1 \cup A_2) \cap Y \in \mathcal{A}_Y$. QED

16.2 Definition of Content

Definition 16.2.1 (Content / Premeasure):

Let $(X, \mathcal{A})$ be a contentable space. A function $\mu : \mathcal{A} \rightarrow [0, \infty]$ is a **content** (also called a **finitely additive measure** or **premeasure**) if:

1. $\mu(\emptyset) = 0$
2. **Finite additivity:** If $A, B \in \mathcal{A}$ are disjoint, then $\mu(A \cup B) = \mu(A) + \mu(B)$

Intuition:

A content assigns “sizes” to sets in a way that respects finite disjoint unions. This is the precursor to a measure, but lacks the crucial σ -additivity needed for countable operations. Contents arise naturally:

- Jordan content on elementary sets measures “nice” bounded subsets of $\mathbb{R}^n$.
- Every measure restricted to an algebra is a content.

The properties of contents (monotonicity, subadditivity) follow from finite additivity alone.

Proposition 16.2.2 (Properties of Contents):

Let μ be a content on an algebra $\mathcal{A}$. Then:

1. **Monotonicity:** $A \subseteq B \implies \mu(A) \leq \mu(B)$
2. **Subtractivity:** If $A \subseteq B$ and $\mu(A) < \infty$, then $\mu(B \setminus A) = \mu(B) - \mu(A)$
3. **Finite subadditivity:** $\mu(A \cup B) \leq \mu(A) + \mu(B)$

Proof:

Property 1 (Monotonicity): Let $A \subseteq B$. We can write B as a disjoint union:

$$B = A \cup (B \setminus A)$$

By finite additivity: $\mu(B) = \mu(A) + \mu(B \setminus A)$. Since $\mu(B \setminus A) \geq 0$, we have $\mu(B) \geq \mu(A)$.

Property 2 (Subtractivity): Let $A \subseteq B$ with $\mu(A) < \infty$. From the disjoint decomposition $B = A \cup (B \setminus A)$:

$$\mu(B) = \mu(A) + \mu(B \setminus A)$$

Since $\mu(A) < \infty$, we can subtract: $\mu(B \setminus A) = \mu(B) - \mu(A)$.

Property 3 (Finite subadditivity): Write $A \cup B$ as the disjoint union:

$$A \cup B = A \cup (B \setminus A)$$

By finite additivity: $\mu(A \cup B) = \mu(A) + \mu(B \setminus A)$. By monotonicity (since $B \setminus A \subseteq B$): $\mu(B \setminus A) \leq \mu(B)$. Thus:

$$\mu(A \cup B) \leq \mu(A) + \mu(B)$$

QED

16.3 Jordan Content on $\mathbb{R}^n$

Goal:

Our goal is to define a notion of “size” or “content” for bounded subsets of $\mathbb{R}^n$ that:

- Assigns rectangles their expected volume
- Is finitely additive on disjoint sets
- Can measure sets more general than rectangles

The Jordan content accomplishes this for “nice” sets—those whose boundary is negligible. This motivates the later development of Lebesgue measure, which can handle more pathological sets.

Definition 16.3.1 (Rectangle):

A **rectangle** (or **box**) in $\mathbb{R}^n$ is a set of the form:

$$R = [a_1, b_1] \times [a_2, b_2] \times \cdots \times [a_n, b_n]$$

The **volume** of R is $\text{vol}(R) = \prod_{i=1}^n (b_i - a_i)$.

Definition 16.3.2 (Elementary Set):

An **elementary set** is a finite union of rectangles. The collection of elementary sets is denoted $\mathcal{E}$.

Proposition 16.3.3 (Elementary Sets Form an Algebra):

$\mathcal{E}$ is an algebra on $\mathbb{R}^n$ (within any fixed bounding rectangle).

Proof:

Contains $\emptyset$: The empty union is $\emptyset \in \mathcal{E}$.

Closure under complements: Let $E = \bigcup_{i=1}^k R_i$ be an elementary set. Then:

$$E^c = \left(\bigcup_{i=1}^k R_i \right)^c = \bigcap_{i=1}^k R_i^c$$

Each R_i^c (within a bounding box) is a finite union of rectangles (obtained by removing a rectangle from a box). The intersection of finitely many such sets is again a finite union of rectangles.

Closure under finite unions: By definition, finite unions of elementary sets are finite unions of rectangles, hence elementary. QED

Proposition 16.3.4:

Every elementary set can be written as a finite disjoint union of rectangles.

Proof:

Proceed by induction on the number of rectangles in the union. Base case is trivial.

For the inductive step, let $E = R \cup E'$ where E' is a disjoint union of rectangles $R_1, \dots, R_k$.

Decompose R into disjoint pieces: $R = (R \setminus \bigcup R_i) \cup \bigcup (R \cap R_i)$.

Each $R \cap R_i$ is a rectangle (intersection of rectangles). The set $R \setminus \bigcup R_i$ can be written as a finite disjoint union of rectangles by slicing along the faces of each R_i . QED

Definition 16.3.5 (Jordan Content):

The **Jordan content** c on $\mathcal{E}$ is defined by:

$$c(E) = \sum_{i=1}^k \text{vol}(R_i)$$

where $E = \bigcup_{i=1}^k R_i$ is a disjoint decomposition into rectangles.

Proposition 16.3.6:

Jordan content is well-defined and is a content on $\mathcal{E}$.

Proof:

Well-defined: If $E = \bigcup R_i = \bigcup S_j$ are two disjoint decompositions, then $R_i = \bigcup_j (R_i \cap S_j)$. Since volume is additive on disjoint rectangles:

$$\sum \text{vol}(R_i) = \sum_i \sum_j \text{vol}(R_i \cap S_j) = \sum_j \sum_i \text{vol}(R_i \cap S_j) = \sum \text{vol}(S_j)$$

Content: $c(\emptyset) = 0$ (empty sum). For disjoint $E, F \in \mathcal{E}$, write $E = \bigcup R_i$, $F = \bigcup S_j$. Then $E \cup F = (\bigcup R_i) \cup (\bigcup S_j)$, so $c(E \cup F) = c(E) + c(F)$. QED

16.4 Inner and Outer Content

Definition 16.4.1 (Outer Content (Axiomatic)):

An **outer content** on $\mathcal{P}(X)$ is a function $\mu^* : \mathcal{P}(X) \rightarrow [0, \infty]$ satisfying:

- (O1) $\mu^*(\emptyset) = 0$
- (O2) **Monotonicity:** $A \subseteq B \implies \mu^*(A) \leq \mu^*(B)$
- (O3) **Finite subadditivity:** $\mu^*(A \cup B) \leq \mu^*(A) + \mu^*(B)$

Proposition 16.4.2 (Subadditivity on Disjoint Sets):

For an outer content μ^* , the following are equivalent:

1. $\mu^*(A \cup B) \leq \mu^*(A) + \mu^*(B)$ for all A, B
2. $\mu^*(A \cup B) \leq \mu^*(A) + \mu^*(B)$ for all disjoint A, B

Proof:

(1) $\implies$ (2): Immediate.

(2) $\implies$ (1): Let A, B be arbitrary. Then $A \cup B = A \cup (B \setminus A)$ where A and $B \setminus A$ are disjoint. By hypothesis:

$$\mu^*(A \cup B) = \mu^*(A \cup (B \setminus A)) \leq \mu^*(A) + \mu^*(B \setminus A)$$

By monotonicity, $\mu^*(B \setminus A) \leq \mu^*(B)$. Thus $\mu^*(A \cup B) \leq \mu^*(A) + \mu^*(B)$. QED

Definition 16.4.3 (Inner Content (Axiomatic)):

An **inner content** on $\mathcal{P}(X)$ is a function $\mu_* : \mathcal{P}(X) \rightarrow [0, \infty]$ satisfying:

- (I1) $\mu_*(\emptyset) = 0$
- (I2) **Monotonicity:** $A \subseteq B \implies \mu_*(A) \leq \mu_*(B)$
- (I3) **Finite superadditivity on disjoint sets:** If $A \cap B = \emptyset$, then $\mu_*(A \cup B) \geq \mu_*(A) + \mu_*(B)$

Remark:

The asymmetry between outer and inner content reflects their definitions via infima and suprema:

- Outer content uses inf of covers, leading to subadditivity: covering $A \cup B$ separately may use more total volume.
- Inner content uses sup of inscribed sets, leading to superadditivity: disjoint inscribed sets can be combined.

16.4.1 Jordan Inner and Outer Content

Definition 16.4.4 (Jordan Outer Content):

For any bounded $A \subseteq \mathbb{R}^n$, the **Jordan outer content** is:

$$m^*(A) = \inf\{m(E) \mid E \in \mathcal{E}, A \subseteq E\}$$

Definition 16.4.5 (Jordan Inner Content):

For any bounded $A \subseteq \mathbb{R}^n$, the **Jordan inner content** is:

$$m_*(A) = \sup\{m(E) \mid E \in \mathcal{E}, E \subseteq A\}$$

Proposition 16.4.6 (Jordan Contents Satisfy the Axioms):

The Jordan outer content $m^*(\cdot)$ satisfies (O1)–(O3), and the Jordan inner content $m_*(\cdot)$ satisfies (I1)–(I3).

Proof:**Outer content axioms:**

(O1) $m^*(\emptyset) = \inf\{m(E) \mid \emptyset \subseteq E\} = m(\emptyset) = 0$.

(O2) If $A \subseteq B$, then any cover of B is a cover of A . Thus $m^*(A) \leq m^*(B)$.

(O3) If $A \subseteq E_1$ and $B \subseteq E_2$, then $A \cup B \subseteq E_1 \cup E_2$ and $m(E_1 \cup E_2) \leq m(E_1) + m(E_2)$ (finite subadditivity of Jordan content). Taking infima gives $m^*(A \cup B) \leq m^*(A) + m^*(B)$.

Inner content axioms:

(I1) $m_*(\emptyset) = \sup\{m(E) \mid E \subseteq \emptyset\} = m(\emptyset) = 0$.

(I2) If $A \subseteq B$, then any set contained in A is also contained in B . Thus $m_*(A) \leq m_*(B)$.

(I3) Let $A \cap B = \emptyset$. If $E_1 \subseteq A$ and $E_2 \subseteq B$, then $E_1 \cap E_2 = \emptyset$ and $E_1 \cup E_2 \subseteq A \cup B$. Thus $m(E_1) + m(E_2) = m(E_1 \cup E_2) \leq m_*(A \cup B)$. Taking suprema over E_1 and E_2 gives $m_*(A) + m_*(B) \leq m_*(A \cup B)$. QED

Proposition 16.4.7 (Properties of Jordan Contents):

For bounded sets $A, B \subseteq \mathbb{R}^n$:

1. $m_*(A) \leq m^*(A)$

Proof:

For any $E_1 \subseteq A \subseteq E_2$ with $E_1, E_2 \in \mathcal{E}$, we have $m(E_1) \leq m(E_2)$ (monotonicity of Jordan content). Taking supremum over E_1 and infimum over E_2 gives $m_*(A) \leq m^*(A)$. QED

Definition 16.4.8 (Jordan Measurable):

A bounded set A is **Jordan measurable** if $m_*(A) = m^*(A)$. The common value is the **Jordan content** $m(A)$.

Theorem 16.4.9 (Characterization of Jordan Measurability):

A bounded set A is Jordan measurable if and only if for every $\epsilon > 0$, there exist elementary sets $E_1 \subseteq A \subseteq E_2$ with $m(E_2 \setminus E_1) < \epsilon$.

Proof:

($\Rightarrow$) Given $\epsilon > 0$, by definition of sup and inf, there exist $E_1 \subseteq A$ with $m(E_1) > m_*(A) - \epsilon/2$ and $E_2 \supseteq A$ with $m(E_2) < m^*(A) + \epsilon/2$. Since $m_*(A) = m^*(A)$:

$$m(E_2 \setminus E_1) = m(E_2) - m(E_1) < \epsilon$$

($\Leftarrow$) We have $m_*(A) \geq m(E_1)$ and $m^*(A) \leq m(E_2)$. Thus:

$$m^*(A) - m_*(A) \leq m(E_2) - m(E_1) = m(E_2 \setminus E_1) < \epsilon$$

Since ϵ is arbitrary, $m_*(A) = m^*(A)$.

QED

Theorem 16.4.10 (Boundary Characterization):

A bounded set A is Jordan measurable if and only if $m^*(\partial A) = 0$.

Proof:

Note that $\partial A = \bar{A} \setminus A^\circ$, where $\bar{A}$ is the closure and A° is the interior.

($\Rightarrow$) Assume A is Jordan measurable. For any $\epsilon > 0$, there exist elementary sets $E_1 \subseteq A \subseteq E_2$ with $m(E_2 \setminus E_1) < \epsilon$.

Since $E_1 \subseteq A$, we have $E_1^\circ \subseteq A^\circ$. Since $A \subseteq E_2$, we have $\bar{A} \subseteq \bar{E}_2$.

Thus $\partial A \subseteq \bar{E}_2 \setminus E_1^\circ \subseteq E_2 \setminus E_1$ (for appropriate elementary sets).

So $m^*(\partial A) \leq m(E_2 \setminus E_1) < \epsilon$. Since ϵ is arbitrary, $m^*(\partial A) = 0$.

($\Leftarrow$) Assume $m^*(\partial A) = 0$. For any $\epsilon > 0$, cover ∂A by an elementary set E with $m(E) < \epsilon$.

Take $E_1 = A^\circ \setminus E$ (elementary approximation of interior) and $E_2 = \bar{A} \cup E$.

Then $E_1 \subseteq A \subseteq E_2$ and $m(E_2 \setminus E_1) \leq m(E) < \epsilon$.

QED

Proposition 16.4.11 (Jordan Measurable Sets Form an Algebra):

The collection $\mathcal{J}$ of Jordan measurable bounded sets forms an algebra on $\mathbb{R}^n$ (within any fixed bounding rectangle).

Proof:

Contains $\emptyset$: $m_*(\emptyset) = 0 = m^*(\emptyset)$, so $\emptyset \in \mathcal{J}$.

Closure under complements: Let $A \in \mathcal{J}$ be bounded within a rectangle R . Then $A^c \cap R$ is bounded. We claim $A^c \cap R \in \mathcal{J}$.

For any $\epsilon > 0$, there exist elementary sets $E_1 \subseteq A \subseteq E_2$ with $m(E_2 \setminus E_1) < \epsilon$. Then $E_2^c \subseteq A^c \subseteq E_1^c$, so:

$$m^*(A^c \cap R) - m_*(A^c \cap R) \leq m((E_1^c \cap R) \setminus (E_2^c \cap R)) = m(E_2 \setminus E_1) < \epsilon$$

Since ϵ is arbitrary, $A^c \cap R \in \mathcal{J}$.

Closure under finite unions: Let $A, B \in \mathcal{J}$. We use the boundary characterization. Since $\partial A \cup B \subseteq \partial A \cup \partial B$:

$$m^*(\partial A \cup B) \leq m^*(\partial A) + m^*(\partial B) = 0 + 0 = 0$$

Thus $A \cup B \in \mathcal{J}$.

QED

Proposition 16.4.12 (Null Sets are Jordan Measurable):

If $m^*(A) = 0$, then A is Jordan measurable with $m(A) = 0$.

Proof:

We have $0 \leq m_*(A) \leq m^*(A) = 0$. Thus $m_*(A) = m^*(A) = 0$, so A is Jordan measurable with $m(A) = 0$.

QED

Corollary 16.4.13 (Subsets of Null Sets):

If $m^*(B) = 0$ and $A \subseteq B$, then A is Jordan measurable with $m(A) = 0$.

Proof:

By monotonicity of outer content, $m^*(A) \leq m^*(B) = 0$. Thus $m^*(A) = 0$, and by the previous proposition, A is Jordan measurable with $m(A) = 0$.

QED

Chapter Compendium:

Contentable Spaces: A pair $(X, \mathcal{A})$ where $\mathcal{A}$ is an algebra on X .

Content: A function $\mu : \mathcal{A} \rightarrow [0, \infty]$ with $\mu(\emptyset) = 0$ and finite additivity.

Content Properties:

- Monotonicity: $A \subseteq B \implies \mu(A) \leq \mu(B)$
- Subtractivity: $A \subseteq B, \mu(A) < \infty \implies \mu(B \setminus A) = \mu(B) - \mu(A)$
- Finite subadditivity: $\mu(A \cup B) \leq \mu(A) + \mu(B)$

Outer Content Axioms (O1–O3):

- (O1) $\mu^*(\emptyset) = 0$
- (O2) Monotonicity
- (O3) Finite subadditivity

Inner Content Axioms (I1–I3):

- (I1) $\mu_*(\emptyset) = 0$
- (I2) Monotonicity
- (I3) Superadditivity on disjoint sets

Jordan Content:

- Outer: $m^*(A) = \inf\{m(E) \mid A \subseteq E, E \in \mathcal{E}\}$
- Inner: $m_*(A) = \sup\{m(E) \mid E \subseteq A, E \in \mathcal{E}\}$
- Measurable: $m_*(A) = m^*(A)$

Key Theorems:

- A is Jordan measurable $\iff m^*(\partial A) = 0$
- Jordan measurable sets form an algebra
- Null sets and their subsets are Jordan measurable

CHAPTER 17

The Riemann Integral

17.1 Riemann/Jordan Simple Functions

Definition 17.1.1 (Riemann Simple Function):

A function $s : R \rightarrow \mathbb{R}$ defined on a rectangle $R \subseteq \mathbb{R}^n$ is **Riemann simple** (or **step function**) if:

1. s has finite image
2. For any subset $B \subseteq \mathbb{R}$, the preimage $s^{-1}(B)$ is a finite disjoint union of rectangles

Equivalently, $s = \sum_{i=1}^k \alpha_i \mathbb{1}_{R_i}$ where R_i are disjoint rectangles.

Definition 17.1.2 (Integral of Riemann Simple Function):

For a Riemann simple function $s = \sum_{i=1}^k \alpha_i \mathbb{1}_{R_i}$:

$$\int_R s \, dc = \sum_{i=1}^k \alpha_i \cdot c(R_i)$$

17.2 Upper and Lower Riemann Sums

Definition 17.2.1 (Partition):

A **partition** P of a rectangle $R = [a_1, b_1] \times \cdots \times [a_n, b_n]$ is a collection of subrectangles $\{R_j\}$ such that:

1. $R = \bigcup_j R_j$
2. The interiors of distinct R_j are disjoint

Definition 17.2.2 (Upper and Lower Sums):

For a bounded function $f : R \rightarrow \mathbb{R}$ and partition $P = \{R_j\}$:

$$\underline{S}(f, P) = \sum_j \left(\inf_{x \in R_j} f(x) \right) \cdot c(R_j)$$

$$\overline{S}(f, P) = \sum_j \left(\sup_{x \in R_j} f(x) \right) \cdot c(R_j)$$

Proposition 17.2.3:

For any partitions P, Q of R :

$$\underline{S}(f, P) \leq \overline{S}(f, Q)$$

Proof:

Let $P \vee Q$ be the common refinement. Then:

$$\underline{S}(f, P) \leq \underline{S}(f, P \vee Q) \leq \overline{S}(f, P \vee Q) \leq \overline{S}(f, Q)$$

QED

17.3 Riemann Integral

Definition 17.3.1 (Upper and Lower Riemann Integrals):

$$\begin{aligned}\int_{\underline{R}} f \, dc &= \sup_P \underline{S}(f, P) \\ \int_{\overline{R}} f \, dc &= \inf_P \overline{S}(f, P)\end{aligned}$$

Definition 17.3.2 (Riemann Integrable):

f is **Riemann integrable** on R if $\int_{\underline{R}} f = \int_{\overline{R}} f$. The common value is $\int_R f \, dc$.

Theorem 17.3.3 (Characterization of Riemann Integrability):

f is Riemann integrable if and only if for every $\epsilon > 0$, there exists a partition P with:

$$\overline{S}(f, P) - \underline{S}(f, P) < \epsilon$$

Proof:

($\Rightarrow$) Given $\epsilon > 0$, choose partitions P_1, P_2 with $\underline{S}(f, P_1) > \int f - \epsilon/2$ and $\overline{S}(f, P_2) < \int f + \epsilon/2$. For $P = P_1 \vee P_2$:

$$\overline{S}(f, P) - \underline{S}(f, P) \leq \overline{S}(f, P_2) - \underline{S}(f, P_1) < \int f - \int f + \epsilon = \epsilon$$

($\Leftarrow$) For any $\epsilon > 0$:

$$\int f - \int f \leq \overline{S}(f, P) - \underline{S}(f, P) < \epsilon$$

QED

17.4 Riemann Double Integral

Definition 17.4.1 (Iterated Integral):

For $f : R_1 \times R_2 \rightarrow \mathbb{R}$ where $R_i \subseteq \mathbb{R}^{n_i}$, the **iterated integral** is:

$$\int_{R_1} \left(\int_{R_2} f(x, y) \, dc(y) \right) dc(x)$$

when both inner integrals exist.

Theorem 17.4.2 (Fubini for Riemann Integral):

If $f : R_1 \times R_2 \rightarrow \mathbb{R}$ is Riemann integrable, and for each $x \in R_1$ the function $y \mapsto f(x, y)$ is Riemann integrable on R_2 , then:

$$\int_{R_1 \times R_2} f \, dc = \int_{R_1} \left(\int_{R_2} f(x, y) \, dc(y) \right) dc(x)$$

Proof:

Let $P = P_1 \times P_2$ be a partition of $R_1 \times R_2$ with P_i partitions of R_i .

For each subrectangle $S_1 \times S_2$ in P :

$$\inf_{S_1 \times S_2} f \leq \inf_{x \in S_1} \left(\inf_{y \in S_2} f(x, y) \right) \leq \inf_{x \in S_1} \int_{S_2} f(x, y) dc(y) / c(S_2)$$

Summing over subrectangles:

$$\underline{S}(f, P) \leq \underline{S}(g, P_1)$$

where $g(x) = \int_{R_2} f(x, y) dc(y)$.

Similarly $\overline{S}(g, P_1) \leq \overline{S}(f, P)$.

Since f is Riemann integrable, $\overline{S}(f, P) - \underline{S}(f, P) \rightarrow 0$, forcing g to be Riemann integrable with the stated equality. QED

17.5 Limitations of Riemann Integration

Example 17.5.1 (Dirichlet Function):

The function $\mathbb{1}_{\mathbb{Q} \cap [0,1]}$ is not Riemann integrable. For any partition:

$$\underline{S} = 0, \quad \overline{S} = 1$$

since every interval contains both rationals and irrationals.

Remark:

Riemann integration has fundamental limitations:

1. Cannot integrate many discontinuous functions
2. Poor behavior under limits (no good convergence theorems)
3. Requires bounded functions on bounded domains

These motivate the development of Lebesgue integration.

Chapter Compendium:

Riemann Simple Function: $s : R \rightarrow \mathbb{R}$ with finite image and preimages that are finite disjoint unions of rectangles.

Form: $s = \sum_{i=1}^k \alpha_i \mathbb{1}_{R_i}$.

Partition: Collection of subrectangles $\{R_j\}$ covering R with disjoint interiors.

Upper and Lower Sums:

$$\underline{S}(f, P) = \sum_j \left(\inf_{R_j} f \right) c(R_j), \quad \overline{S}(f, P) = \sum_j \left(\sup_{R_j} f \right) c(R_j)$$

Riemann Integrals:

$$\int_{\underline{R}} f = \sup_P \underline{S}(f, P), \quad \int_{\overline{R}} f = \inf_P \overline{S}(f, P)$$

Riemann Integrable: f is Riemann integrable iff $\int_{\underline{R}} f = \int_{\overline{R}} f$.

Integrability Criterion: f is Riemann integrable iff $\forall \epsilon > 0 : \exists P : \overline{S}(f, P) - \underline{S}(f, P) < \epsilon$.

Fubini for Riemann: If $f : R_1 \times R_2 \rightarrow \mathbb{R}$ is Riemann integrable and $y \mapsto f(x, y)$ is Riemann integrable for each x :

$$\int_{R_1 \times R_2} f dc = \int_{R_1} \left(\int_{R_2} f(x, y) dc(y) \right) dc(x)$$

Limitations: Cannot handle Dirichlet function, no good convergence theorems, requires bounded functions on bounded domains.

CHAPTER 18

Measures

18.1 Measurable Spaces

Definition 18.1.1 (Measurable Space):

A **measurable space** is a pair $(X, \mathcal{M})$ where X is a set and $\mathcal{M}$ is a σ -algebra on X .

Intuition:

A measurable space is the natural domain for defining σ -additive measures. The σ -algebra structure provides closure under countable set operations, which is essential for defining σ -additivity and proving convergence theorems.

Definition 18.1.2 (Measurable Subspace):

Let $(X, \mathcal{M})$ be a measurable space and $Y \subseteq X$. The **measurable subspace** induced on Y is $(Y, \mathcal{M}_Y)$ where:

$$\mathcal{M}_Y = \{A \cap Y \mid A \in \mathcal{M}\}$$

Proposition 18.1.3:

If $(X, \mathcal{M})$ is a measurable space and $Y \subseteq X$, then $\mathcal{M}_Y$ is a σ -algebra on Y .

Proof:

Contains $\emptyset$: $\emptyset = \emptyset \cap Y \in \mathcal{M}_Y$.

Contains Y : $Y = X \cap Y \in \mathcal{M}_Y$.

Closure under complements: Let $A \cap Y \in \mathcal{M}_Y$ with $A \in \mathcal{M}$. Then $Y \setminus (A \cap Y) = (X \setminus A) \cap Y$. Since $\mathcal{M}$ is a σ -algebra, $X \setminus A \in \mathcal{M}$, so $(X \setminus A) \cap Y \in \mathcal{M}_Y$.

Closure under countable unions: Let $(A_n \cap Y)_{n \in \mathbb{N}} \subseteq \mathcal{M}_Y$. Then:

$$\bigcup_{n \in \mathbb{N}} (A_n \cap Y) = \left(\bigcup_{n \in \mathbb{N}} A_n \right) \cap Y \in \mathcal{M}_Y$$

since $\bigcup_{n \in \mathbb{N}} A_n \in \mathcal{M}$.

QED

18.2 Definition of Measure

Definition 18.2.1 (Measure):

Let $(X, \mathcal{M})$ be a measurable space. A function $\mu : \mathcal{M} \rightarrow [0, \infty]$ is a **measure** if:

1. $\mu(\emptyset) = 0$

2. **σ -additivity:** For any sequence of pairwise disjoint sets $(A_n)_{n \in \mathbb{N}} \subseteq \mathcal{M}$:

$$\mu \left(\bigcup_{n \in \mathbb{N}} A_n \right) = \sum_{n \in \mathbb{N}} \mu(A_n)$$

The triple $(X, \mathcal{M}, \mu)$ is called a **measure space**.

Proposition 18.2.2 (Properties of Measures):

Let μ be a measure on $(X, \mathcal{M})$. Then:

1. **Finite additivity:** μ is a content
2. **Monotonicity:** $A \subseteq B \implies \mu(A) \leq \mu(B)$
3. **σ -subadditivity:** $\mu(\bigcup_{n \in \mathbb{N}} A_n) \leq \sum_{n \in \mathbb{N}} \mu(A_n)$
4. **Continuity from below:** If $A_n \nearrow A$, then $\mu(A) = \lim_{n \rightarrow \infty} \mu(A_n)$
5. **Continuity from above:** If $A_n \searrow A$ and $\mu(A_k) < \infty$ for some k , then $\mu(A) = \lim_{n \rightarrow \infty} \mu(A_n)$

Proof:

(1) **Finite additivity:** Take $A_3 = A_4 = \dots = \emptyset$ in the σ -additivity.

(2) **Monotonicity:** $\mu(B) = \mu(A) + \mu(B \setminus A) \geq \mu(A)$.

(3) **σ -subadditivity:** By the Disjointification Lemma, define $B_1 = A_1$, $B_n = A_n \setminus \bigcup_{i=1}^{n-1} A_i$ for $n \geq 2$. Then $(B_n)_{n \in \mathbb{N}}$ are pairwise disjoint, $\bigcup_{n \in \mathbb{N}} A_n = \bigcup_{n \in \mathbb{N}} B_n$, and $B_n \subseteq A_n$ for all n . Thus:

$$\mu \left(\bigcup_{n \in \mathbb{N}} A_n \right) = \mu \left(\bigcup_{n \in \mathbb{N}} B_n \right) = \sum_{n \in \mathbb{N}} \mu(B_n) \leq \sum_{n \in \mathbb{N}} \mu(A_n)$$

where the last inequality uses monotonicity: $\mu(B_n) \leq \mu(A_n)$.

(4) **Continuity from below:** By the Disjointification Lemma, define $B_1 = A_1$, $B_n = A_n \setminus A_{n-1}$ for $n \geq 2$. Then $(B_n)_{n \in \mathbb{N}}$ are pairwise disjoint and $A_N = \bigcup_{n=1}^N B_n$ for each N . Since $A = \bigcup_{n \in \mathbb{N}} A_n = \bigcup_{n \in \mathbb{N}} B_n$, by σ -additivity:

$$\mu(A) = \sum_{n \in \mathbb{N}} \mu(B_n) = \lim_{N \rightarrow \infty} \sum_{n=1}^N \mu(B_n)$$

Since $A_N = \bigcup_{n=1}^N B_n$ is a disjoint union, finite additivity gives $\sum_{n=1}^N \mu(B_n) = \mu(A_N)$. Therefore:

$$\mu(A) = \lim_{N \rightarrow \infty} \mu(A_N)$$

(5) **Continuity from above:** WLOG assume $\mu(A_k) < \infty$ for some k (by hypothesis). Define $C_n = A_k \setminus A_n$ for $n \geq k$. Then $C_n \nearrow A_k \setminus A$, so:

$$\mu(A_k \setminus A) = \lim_{n \rightarrow \infty} \mu(C_n) = \lim_{n \rightarrow \infty} (\mu(A_k) - \mu(A_n))$$

Thus $\mu(A) = \mu(A_k) - \mu(A_k \setminus A) = \lim_{n \rightarrow \infty} \mu(A_n)$. QED

Example 18.2.3:

The finiteness condition in continuity from above is essential. Let $A_n = [n, \infty) \subseteq \mathbb{R}$ with Lebesgue measure. Then $A_n \searrow \emptyset$ but $\lambda(A_n) = \infty$ for all n .

18.3 Outer Measures

Definition 18.3.1 (Outer Measure):

A function $\mu^* : \mathcal{P}(X) \rightarrow [0, \infty]$ is an **outer measure** if:

1. $\mu^*(\emptyset) = 0$
2. **Monotonicity:** $A \subseteq B \implies \mu^*(A) \leq \mu^*(B)$
3. **σ -subadditivity:** $\mu^*(\bigcup_{n \in \mathbb{N}} A_n) \leq \sum_{n \in \mathbb{N}} \mu^*(A_n)$

Definition 18.3.2 (Lebesgue Outer Measure):

For $A \subseteq \mathbb{R}^n$:

$$\lambda^*(A) = \inf \left\{ \sum_{k=1}^{\infty} \text{vol}(R_k) \mid A \subseteq \bigcup_{k=1}^{\infty} R_k, R_k \text{ open rectangles} \right\}$$

Proposition 18.3.3:

λ^* is an outer measure on $\mathbb{R}^n$.

Proof:

(1) $\lambda^*(\emptyset) = 0$: Cover $\emptyset$ by rectangles of arbitrarily small total volume.

(2) **Monotonicity:** If $A \subseteq B$, any cover of B is a cover of A , so $\lambda^*(A) \leq \lambda^*(B)$.

(3) **σ -subadditivity:** Let (A_n) be a sequence. For each n , given $\epsilon > 0$, choose open rectangles $\{R_{n,k}\}_k$ covering A_n with $\sum_k \text{vol}(R_{n,k}) < \lambda^*(A_n) + \epsilon/2^n$.

Then $\{R_{n,k}\}_{n,k}$ covers $\bigcup A_n$ and:

$$\sum_{n,k} \text{vol}(R_{n,k}) < \sum_n (\lambda^*(A_n) + \epsilon/2^n) = \sum_n \lambda^*(A_n) + \epsilon$$

Thus $\lambda^*(\bigcup A_n) \leq \sum \lambda^*(A_n) + \epsilon$. Since ϵ is arbitrary, done.

QED

18.4 Complete Measures and Completion

Definition 18.4.1 (Complete Measure Space):

A measure space $(X, \mathcal{M}, \mu)$ is **complete** if every subset of a null set is measurable:

$$\mu(N) = 0 \text{ and } E \subseteq N \implies E \in \mathcal{M}$$

Theorem 18.4.2 (Characterization of Complete Measures):

Let $(X, \mathcal{M}, \mu)$ be a measure space. The following are equivalent:

1. $(X, \mathcal{M}, \mu)$ is complete
2. Every subset of a null set is measurable
3. If $A \subseteq E \subseteq B$ with $A, B \in \mathcal{M}$ and $\mu(B \setminus A) = 0$, then $E \in \mathcal{M}$

Proof:

(1) $\Leftrightarrow$ (2): This is the definition of completeness.

(1) $\Rightarrow$ (3): Suppose $(X, \mathcal{M}, \mu)$ is complete. Let $A \subseteq E \subseteq B$ with $\mu(B \setminus A) = 0$.

We can write $E = A \cup (E \setminus A)$. Since $E \setminus A \subseteq B \setminus A$ and $\mu(B \setminus A) = 0$, by completeness $E \setminus A \in \mathcal{M}$.

Since $A \in \mathcal{M}$ and $\mathcal{M}$ is closed under unions, $E = A \cup (E \setminus A) \in \mathcal{M}$.

(3) $\Rightarrow$ (1): Let $N \in \mathcal{M}$ with $\mu(N) = 0$, and let $E \subseteq N$.

Take $A = \emptyset$ and $B = N$. Then $\emptyset \subseteq E \subseteq N$ and $\mu(N \setminus \emptyset) = \mu(N) = 0$. By (3), $E \in \mathcal{M}$. QED

Remark:

Characterization (3) is called the “squeezed set” criterion: any set sandwiched between two measurable sets differing by a null set is itself measurable.

Definition 18.4.3 (Completion of a Measure Space):

Let $(X, \mathcal{M}, \mu)$ be a measure space. Define the **completed σ -algebra**:

$$\overline{\mathcal{M}}^\mu = \{A \cup N \mid A \in \mathcal{M}, \exists M \in \mathcal{M} : N \subseteq M \text{ and } \mu(M) = 0\}$$

Define the **completed measure** $\overline{\mu} : \overline{\mathcal{M}}^\mu \rightarrow [0, \infty]$ by:

$$\overline{\mu}(A \cup N) = \mu(A)$$

Proposition 18.4.4 (Equivalent Characterization of Completion):

$$\overline{\mathcal{M}}^\mu = \{E \subseteq X \mid \exists A, B \in \mathcal{M} : A \subseteq E \subseteq B \text{ and } \mu(B \setminus A) = 0\}$$

Proof:

($\subseteq$): Let $E = A \cup N \in \overline{\mathcal{M}}^\mu$ where $A \in \mathcal{M}$ and $N \subseteq M$ for some null set $M \in \mathcal{M}$.

Take the inner approximation $A \in \mathcal{M}$ and outer approximation $B = A \cup M \in \mathcal{M}$. Then $A \subseteq E = A \cup N \subseteq A \cup M = B$, and $\mu(B \setminus A) = \mu(M \setminus A) \leq \mu(M) = 0$.

($\supseteq$): Let $A \subseteq E \subseteq B$ with $A, B \in \mathcal{M}$ and $\mu(B \setminus A) = 0$.

Write $E = A \cup (E \setminus A)$. Note that $N = E \setminus A \subseteq B \setminus A$ where $\mu(B \setminus A) = 0$. Thus $E = A \cup N \in \overline{\mathcal{M}}^\mu$. QED

Proposition 18.4.5 ($\overline{\mu}$ is Well-Defined):

If $E = A_1 \cup N_1 = A_2 \cup N_2$ with $A_1, A_2 \in \mathcal{M}$ and N_1, N_2 subsets of null sets in $\mathcal{M}$, then $\mu(A_1) = \mu(A_2)$.

Proof:

We have $A_1 \subseteq A_1 \cup N_1 = A_2 \cup N_2 \subseteq A_2 \cup M_2$ where $N_2 \subseteq M_2$ and $\mu(M_2) = 0$. Thus:

$$A_1 \setminus A_2 \subseteq (A_2 \cup M_2) \setminus A_2 = M_2 \setminus A_2 \subseteq M_2$$

Since $A_1 \setminus A_2 \in \mathcal{M}$ and $\mu(M_2) = 0$, monotonicity gives $\mu(A_1 \setminus A_2) = 0$.

By symmetry, $\mu(A_2 \setminus A_1) = 0$. Therefore:

$$\mu(A_1) = \mu(A_1 \cap A_2) + \mu(A_1 \setminus A_2) = \mu(A_1 \cap A_2) = \mu(A_1 \cap A_2) + \mu(A_2 \setminus A_1) = \mu(A_2)$$

QED

Proposition 18.4.6:

$\overline{\mathcal{M}}^\mu$ is a σ -algebra and $\overline{\mu}$ is a measure.

Proof: $\overline{\mathcal{M}}^\mu$ is a σ -algebra:Contains $\emptyset$: $\emptyset = \emptyset \cup \emptyset \in \overline{\mathcal{M}}^\mu$.Closure under complements: If $E = A \cup N$ with $N \subseteq M$ (null), then $E^c = (A \cup N)^c = A^c \cap N^c$. One can verify $E^c = A^c \cap M^c \cup N'$ for appropriate $N' \subseteq M$.Closure under countable unions: If $E_n = A_n \cup N_n$ with $N_n \subseteq M_n$, then $\bigcup_{n \in \mathbb{N}} E_n = \bigcup_{n \in \mathbb{N}} A_n \cup \bigcup_{n \in \mathbb{N}} N_n$ where $\bigcup_{n \in \mathbb{N}} N_n \subseteq \bigcup_{n \in \mathbb{N}} M_n$ is a subset of a null set. $\overline{\mu}$ is a measure: $\overline{\mu}(\emptyset) = 0$: $\overline{\mu}(\emptyset) = \mu(\emptyset) = 0$. σ -additivity: Let $(E_n)_{n \in \mathbb{N}}$ be pairwise disjoint sets in $\overline{\mathcal{M}}^\mu$ with $E_n = A_n \cup N_n$ where $N_n \subseteq M_n$ and $\mu(M_n) = 0$.Define $A'_1 = A_1$ and $A'_n = A_n \setminus \bigcup_{i=1}^{n-1} A_i$ for $n \geq 2$. Then $(A'_n)_{n \in \mathbb{N}}$ are pairwise disjoint and $A'_n \subseteq A_n$.Since $A_n \setminus A'_n \subseteq \bigcup_{i=1}^{n-1} A_i \cap A_n \subseteq \bigcup_{i=1}^{n-1} E_i \cap E_n \subseteq \bigcup_{i=1}^{n-1} N_i$ (using disjointness of E_n), and $\bigcup_{i=1}^{n-1} N_i \subseteq \bigcup_{i=1}^{n-1} M_i$ is null, we have $\mu(A_n \setminus A'_n) = 0$.Thus $\mu(A_n) = \mu(A'_n)$ and:

$$\overline{\mu}\left(\bigcup_{n \in \mathbb{N}} E_n\right) = \mu\left(\bigcup_{n \in \mathbb{N}} A_n\right) = \mu\left(\bigcup_{n \in \mathbb{N}} A'_n\right) = \sum_{n \in \mathbb{N}} \mu(A'_n) = \sum_{n \in \mathbb{N}} \mu(A_n) = \sum_{n \in \mathbb{N}} \overline{\mu}(E_n)$$

QED

Proposition 18.4.7 (Extension Property): $\mathcal{M} \subseteq \overline{\mathcal{M}}^\mu$ and $\overline{\mu}|_{\mathcal{M}} = \mu$.**Proof:**For any $A \in \mathcal{M}$, we have $A = A \cup \emptyset$ where $\emptyset \subseteq \emptyset$ and $\mu(\emptyset) = 0$. Thus $A \in \overline{\mathcal{M}}^\mu$.Moreover, $\overline{\mu}(A) = \overline{\mu}(A \cup \emptyset) = \mu(A)$.

QED

Proposition 18.4.8: $(X, \overline{\mathcal{M}}^\mu, \overline{\mu})$ is a complete measure space.**Proof:**Let $E \subseteq N'$ where $\overline{\mu}(N') = 0$. Write $N' = A \cup N$ with $\mu(A) = 0$ and $N \subseteq M$ for some null $M \in \mathcal{M}$.Then $E \subseteq N' \subseteq A \cup M$ where $\mu(A \cup M) = 0$. So $E = \emptyset \cup E$ with $E \subseteq A \cup M$, thus $E \in \overline{\mathcal{M}}^\mu$.

QED

Theorem 18.4.9 (Minimality of Completion):If $(X, \mathcal{N}, \nu)$ is a complete measure space with $\mathcal{M} \subseteq \mathcal{N}$ and $\nu|_{\mathcal{M}} = \mu$, then $(X, \overline{\mathcal{M}}^\mu, \overline{\mu})$ is a measurable subspace of $(X, \mathcal{N}, \nu)$.That is, $(X, \overline{\mathcal{M}}^\mu, \overline{\mu})$ is the smallest complete extension of $(X, \mathcal{M}, \mu)$.**Proof:**We show $\overline{\mathcal{M}}^\mu \subseteq \mathcal{N}$ and $\nu|_{\overline{\mathcal{M}}^\mu} = \overline{\mu}$.Let $E = A \cup N \in \overline{\mathcal{M}}^\mu$ where $A \in \mathcal{M}$ and $N \subseteq M$ for some $M \in \mathcal{M}$ with $\mu(M) = 0$.Since $\mathcal{M} \subseteq \mathcal{N}$, we have $A, M \in \mathcal{N}$. Since $\nu|_{\mathcal{M}} = \mu$, we have $\nu(M) = \mu(M) = 0$.Since $(X, \mathcal{N}, \nu)$ is complete and $N \subseteq M$ with $\nu(M) = 0$, we have $N \in \mathcal{N}$. Therefore $E = A \cup N \in \mathcal{N}$.Since $A \cap N \subseteq N \subseteq M$ and $\nu(M) = 0$, completeness gives $A \cap N \in \mathcal{N}$ with $\nu(A \cap N) = 0$. Thus:

$$\nu(E) = \nu(A \cup N) = \nu(A) + \nu(N \setminus A) = \nu(A) = \mu(A) = \overline{\mu}(E)$$

QED

Corollary 18.4.10:

If $(X, \mathcal{M}, \mu)$ is already complete, then it equals its completion:

$$\overline{\mathcal{M}}^\mu = \mathcal{M} \quad \text{and} \quad \overline{\mu} = \mu$$

Proof:

We already have $\mathcal{M} \subseteq \overline{\mathcal{M}}^\mu$ by the extension property.

For the reverse inclusion, let $E = A \cup N \in \overline{\mathcal{M}}^\mu$ with $N \subseteq M$ and $\mu(M) = 0$. Since $(X, \mathcal{M}, \mu)$ is complete and $N \subseteq M$ with $\mu(M) = 0$, we have $N \in \mathcal{M}$. Thus $E = A \cup N \in \mathcal{M}$.

For $\overline{\mu} = \mu$: Let $A \in \mathcal{M}$. Then $A = A \cup \emptyset$ in the completion representation, so $\overline{\mu}(A) = \mu(A)$. QED

18.5 Carathéodory's Theorem

Definition 18.5.1 (Carathéodory Measurability):

A set $E \subseteq X$ is μ^* -measurable if for every $A \subseteq X$:

$$\mu^*(A) = \mu^*(A \cap E) + \mu^*(A \cap E^c)$$

Intuition:

The Carathéodory condition says: E is measurable if it “splits” every test set A additively. This is a strong requirement—by subadditivity, we always have $\mu^*(A) \leq \mu^*(A \cap E) + \mu^*(A \cap E^c)$. The condition demands equality for *all* test sets A .

Why this definition? It ensures:

- Measurable sets form a σ -algebra (not obvious from simpler definitions)
- The restriction of μ^* to measurable sets is automatically σ -additive
- The construction is complete (subsets of null sets are measurable)

Theorem 18.5.2 (Carathéodory):

Let μ^* be an outer measure on X . Let $\mathcal{M}(\mu^*)$ be the collection of μ^* -measurable sets. Then:

1. $\mathcal{M}(\mu^*)$ is a σ -algebra
2. $\mu^*|_{\mathcal{M}(\mu^*)}$ is a complete measure

Proof:

We proceed in four steps to verify both claims.

Step 1: Closure under complements.

The Carathéodory condition is symmetric in E and E^c : the equation $\mu^*(A) = \mu^*(A \cap E) + \mu^*(A \cap E^c)$ is unchanged when E is replaced by E^c . Thus $E \in \mathcal{M}(\mu^*)$ iff $E^c \in \mathcal{M}(\mu^*)$.

Step 2: Closure under finite unions.

Let $E, F \in \mathcal{M}(\mu^*)$. We show $E \cup F \in \mathcal{M}(\mu^*)$.

For any test set $A \subseteq X$, we must verify:

$$\mu^*(A) = \mu^*(A \cap (E \cup F)) + \mu^*(A \cap (E \cup F)^c)$$

Since E is measurable, we can split A :

$$\mu^*(A) = \mu^*(A \cap E) + \mu^*(A \cap E^c)$$

Since F is measurable, we can split $A \cap E^c$:

$$\mu^*(A \cap E^c) = \mu^*(A \cap E^c \cap F) + \mu^*(A \cap E^c \cap F^c)$$

Substituting and using $(E \cup F)^c = E^c \cap F^c$:

$$\mu^*(A) = \mu^*(A \cap E) + \mu^*(A \cap E^c \cap F) + \mu^*(A \cap (E \cup F)^c)$$

By subadditivity: $\mu^*(A \cap E) + \mu^*(A \cap E^c \cap F) \geq \mu^*(A \cap (E \cup F))$.

Thus $\mu^*(A) \geq \mu^*(A \cap (E \cup F)) + \mu^*(A \cap (E \cup F)^c)$.

The reverse inequality holds by subadditivity applied to $A = (A \cap (E \cup F)) \cup (A \cap (E \cup F)^c)$. Hence $E \cup F \in \mathcal{M}(\mu^*)$.

Step 3: Closure under countable unions and σ -additivity.

Let $(E_n)_{n \in \mathbb{N}}$ be pairwise disjoint sets in $\mathcal{M}(\mu^*)$. Define $F_n = \bigcup_{i=1}^n E_i$.

Claim: For any set A , we have $\mu^*(A \cap F_n) = \sum_{i=1}^n \mu^*(A \cap E_i)$.

Proof by induction: The base case $n = 1$ is trivial. Assume the claim holds for n . Since E_{n+1} is measurable and $E_{n+1} \cap F_n = \emptyset$ (disjointness):

$$\mu^*(A \cap F_{n+1}) = \mu^*(A \cap F_{n+1} \cap E_{n+1}) + \mu^*(A \cap F_{n+1} \cap E_{n+1}^c)$$

Since $E_{n+1} \subseteq F_{n+1}$ and $F_n = F_{n+1} \cap E_{n+1}^c$:

$$\mu^*(A \cap F_{n+1}) = \mu^*(A \cap E_{n+1}) + \mu^*(A \cap F_n) = \mu^*(A \cap E_{n+1}) + \sum_{i=1}^n \mu^*(A \cap E_i)$$

Now let $E = \bigcup_{n \in \mathbb{N}} E_n$. Since each $F_n \in \mathcal{M}(\mu^*)$ (by Step 2) and $F_n \subseteq E$:

$$\mu^*(A) = \mu^*(A \cap F_n) + \mu^*(A \cap F_n^c) \geq \sum_{i=1}^n \mu^*(A \cap E_i) + \mu^*(A \cap E^c)$$

Taking $n \rightarrow \infty$:

$$\mu^*(A) \geq \sum_{i=1}^{\infty} \mu^*(A \cap E_i) + \mu^*(A \cap E^c) \geq \mu^*(A \cap E) + \mu^*(A \cap E^c)$$

The reverse inequality holds by subadditivity. Thus $E \in \mathcal{M}(\mu^*)$.

Setting $A = E$ (so $A \cap E^c = \emptyset$), we get $\mu^*(E) = \sum_{i=1}^{\infty} \mu^*(E_i)$, proving σ -additivity.

Step 4: Completeness.

Let $\mu^*(N) = 0$ and $E \subseteq N$. We show $E \in \mathcal{M}(\mu^*)$.

For any test set A :

$$\mu^*(A \cap E) \leq \mu^*(E) \leq \mu^*(N) = 0$$

Thus $\mu^*(A \cap E) = 0$. By subadditivity:

$$\mu^*(A) \leq \mu^*(A \cap E) + \mu^*(A \cap E^c) = 0 + \mu^*(A \cap E^c) = \mu^*(A \cap E^c)$$

But also $\mu^*(A \cap E^c) \leq \mu^*(A)$ by monotonicity. Thus $\mu^*(A) = \mu^*(A \cap E^c) = \mu^*(A \cap E) + \mu^*(A \cap E^c)$, so $E \in \mathcal{M}(\mu^*)$. QED

18.6 Lebesgue Measure

Definition 18.6.1:

The **Lebesgue measure** λ on $\mathbb{R}^n$ is the restriction of λ^* to $\mathcal{M}(\lambda^*)$, the Lebesgue measurable sets.

Theorem 18.6.2:

Every Borel set is Lebesgue measurable: $\mathcal{B}(\mathbb{R}^n) \subseteq \mathcal{M}(\lambda^*)$.

For any rectangle R : $\lambda(R) = \text{vol}(R)$.

Proof:

Step 1: Open sets are measurable. Let G be open. For any test set A :

Since λ^* is defined using open covers, for any $\epsilon > 0$ there exist open rectangles $\{R_k\}$ with $A \subseteq \bigcup R_k$ and $\sum \text{vol}(R_k) < \lambda^*(A) + \epsilon$.

Using the fact that G is open to split each R_k appropriately, one shows:

$$\lambda^*(A \cap G) + \lambda^*(A \cap G^c) \leq \lambda^*(A) + \epsilon$$

Since ϵ is arbitrary, $G \in \mathcal{M}(\lambda^*)$.

Step 2: Since $\mathcal{M}(\lambda^*)$ is a σ -algebra containing all open sets, and $\mathcal{B}(\mathbb{R}^n) = \sigma(\text{open sets})$, we have $\mathcal{B}(\mathbb{R}^n) \subseteq \mathcal{M}(\lambda^*)$.

Step 3: For rectangles, $\lambda^*(R) \leq \text{vol}(R)$ by covering with R itself. The reverse follows from σ -subadditivity and convergence arguments. QED

18.7 Regularity

Theorem 18.7.1 (Outer Regularity):

For any $A \subseteq \mathbb{R}^n$:

$$\lambda^*(A) = \inf\{\lambda(G) \mid G \text{ open}, G \supseteq A\}$$

Proof:

Let $\alpha = \inf\{\lambda(G) \mid G \text{ open}, G \supseteq A\}$.

Since $\lambda^*(A) \leq \lambda(G)$ for any open $G \supseteq A$, we have $\lambda^*(A) \leq \alpha$.

For the reverse: given $\epsilon > 0$, by definition of λ^* , there exist open rectangles $\{R_k\}$ with $A \subseteq \bigcup R_k$ and $\sum \text{vol}(R_k) < \lambda^*(A) + \epsilon$.

Let $G = \bigcup R_k$, which is open. Then $\lambda(G) \leq \sum \text{vol}(R_k) < \lambda^*(A) + \epsilon$.

Thus $\alpha \leq \lambda^*(A) + \epsilon$, so $\alpha = \lambda^*(A)$. QED

Theorem 18.7.2 (Inner Regularity):

For any Lebesgue measurable A :

$$\lambda(A) = \sup\{\lambda(K) \mid K \text{ compact}, K \subseteq A\}$$

Proof:

Case 1: A bounded. Let $A \subseteq [-M, M]^n$.

By outer regularity of $A^c \cap [-M, M]^n$, for any $\epsilon > 0$ there exists open $G \supseteq A^c \cap [-M, M]^n$ with $\lambda(G) < \lambda(A^c \cap [-M, M]^n) + \epsilon$.

Let $K = [-M, M]^n \setminus G$. Then K is closed and bounded (hence compact), $K \subseteq A$, and:

$$\lambda(K) = \lambda([-M, M]^n) - \lambda(G) > \lambda(A) - \epsilon$$

Case 2: A unbounded. Write $A = \bigcup_{n \in \mathbb{N}} (A \cap [-n, n]^n)$ and use continuity from below. QED

18.8 Completion

Theorem 18.8.1:

The Lebesgue measure is the completion of the Borel measure on $\mathbb{R}^n$.

Equivalently: $A \in \mathcal{M}(\lambda^*)$ iff $A = B \cup N$ where $B \in \mathcal{B}(\mathbb{R}^n)$ and $\lambda^*(N) = 0$.

Proof:

($\Leftarrow$) If $B \in \mathcal{B}(\mathbb{R}^n)$ and $\lambda^*(N) = 0$, then B is measurable (Borel $\subseteq$ Lebesgue) and N is measurable by completeness of λ . Thus $B \cup N$ is measurable.

($\Rightarrow$) Let $A \in \mathcal{M}(\lambda^*)$. By outer regularity, for each n there exists open $G_n \supseteq A$ with $\lambda(G_n) < \lambda(A) + 1/n$.

Let $B = \bigcap_{n \in \mathbb{N}} G_n$. Then $B \in \mathcal{B}(\mathbb{R}^n)$, $A \subseteq B$, and $\lambda(B) = \lambda(A)$.

Let $N = B \setminus A$. Then $\lambda(N) = \lambda(B) - \lambda(A) = 0$ and $A = B \setminus N$.

Since $A \subseteq B$, we can write $A = (B \setminus N) = B \cap N^c$. Rearranging: $B = A \cup N$ where $\lambda^*(N) = 0$. QED

18.9 Uniqueness of Measures

Intuition:

The π - λ (Dynkin) theorem provides a powerful tool for proving that two measures agree on an entire σ -algebra: one only needs to verify agreement on a much smaller π -system that generates the σ -algebra.

Lemma 18.9.1 (Uniqueness for Finite Measures):

Let μ, ν be finite measures on $(X, \mathcal{M})$ with $\mu(X) = \nu(X) < \infty$. If $\mathcal{P} \subseteq \mathcal{M}$ is a π -system with $\sigma(\mathcal{P}) = \mathcal{M}$ and $\mu(P) = \nu(P)$ for all $P \in \mathcal{P}$, then $\mu = \nu$.

Proof:

Define:

$$\mathcal{L} = \{A \in \mathcal{M} \mid \mu(A) = \nu(A)\}$$

We show $\mathcal{L}$ is a λ -system.

(1) $X \in \mathcal{L}$: By hypothesis, $\mu(X) = \nu(X)$.

(2) **Closure under proper differences:** Let $A, B \in \mathcal{L}$ with $A \subseteq B$.

Since $\mu(B) = \nu(B) < \infty$ and $\mu(A) = \nu(A)$:

$$\mu(B \setminus A) = \mu(B) - \mu(A) = \nu(B) - \nu(A) = \nu(B \setminus A)$$

Thus $B \setminus A \in \mathcal{L}$.

(3) **Closure under increasing unions:** Let $(A_n)_{n \in \mathbb{N}} \subseteq \mathcal{L}$ with $A_n \nearrow A$.

By continuity from below:

$$\mu(A) = \lim_{n \rightarrow \infty} \mu(A_n) = \lim_{n \rightarrow \infty} \nu(A_n) = \nu(A)$$

Thus $A \in \mathcal{L}$.

Since $\mathcal{P} \subseteq \mathcal{L}$ (by hypothesis) and $\mathcal{L}$ is a λ -system, the Dynkin π - λ theorem gives:

$$\mathcal{M} = \sigma(\mathcal{P}) = \mathcal{L}$$

Therefore $\mu = \nu$ on all of $\mathcal{M}$. QED

Theorem 18.9.2 (Uniqueness for σ -Finite Measures):

Let μ, ν be σ -finite measures on $(X, \mathcal{M})$. If $\mathcal{P} \subseteq \mathcal{M}$ is a π -system with $\sigma(\mathcal{P}) = \mathcal{M}$ and:

1. $\mu(P) = \nu(P)$ for all $P \in \mathcal{P}$
2. There exists an increasing sequence $(E_n)_{n \in \mathbb{N}} \subseteq \mathcal{P}$ with $E_n \nearrow X$ and $\mu(E_n) = \nu(E_n) < \infty$

Then $\mu = \nu$.

Proof:

For each $n \in \mathbb{N}$, define measures on $(E_n, \mathcal{M}_{E_n})$ by restriction:

$$\mu_n(A) = \mu(A \cap E_n), \quad \nu_n(A) = \nu(A \cap E_n)$$

These are finite measures with $\mu_n(E_n) = \nu_n(E_n) < \infty$.

The collection $\mathcal{P}_n = \{P \cap E_n \mid P \in \mathcal{P}\}$ is a π -system on E_n (since $E_n \in \mathcal{P}$), and $\sigma(\mathcal{P}_n) = \mathcal{M}_{E_n}$.

For any $P \in \mathcal{P}$:

$$\mu_n(P \cap E_n) = \mu(P \cap E_n) = \nu(P \cap E_n) = \nu_n(P \cap E_n)$$

since $P \cap E_n \in \mathcal{P}$ (as $\mathcal{P}$ is a π -system containing E_n).

By the finite measure lemma, $\mu_n = \nu_n$ on $\mathcal{M}_{E_n}$.

For any $A \in \mathcal{M}$, we have $A \cap E_n \nearrow A$. By continuity from below:

$$\mu(A) = \lim_{n \rightarrow \infty} \mu(A \cap E_n) = \lim_{n \rightarrow \infty} \nu(A \cap E_n) = \nu(A)$$

QED

Corollary 18.9.3 (Uniqueness of Lebesgue Measure):

The Lebesgue measure on $\mathbb{R}^n$ is the unique σ -finite Borel measure satisfying $\lambda(R) = \text{vol}(R)$ for all rectangles R .

Proof:

The rectangles form a π -system generating $\mathcal{B}(\mathbb{R}^n)$. Take $E_n = [-n, n]^n$, which are rectangles with finite volume.

Any two measures agreeing on rectangles agree on all Borel sets by the theorem.

QED

Chapter Compendium:

Measures: A function $\mu : \mathcal{M} \rightarrow [0, \infty]$ is a measure if $\mu(\emptyset) = 0$ and $\mu(\bigcup_n A_n) = \sum_n \mu(A_n)$ for disjoint sequences.

Key Properties:

- **Monotonicity:** $A \subseteq B \implies \mu(A) \leq \mu(B)$
- **σ -subadditivity:** $\mu(\bigcup_n A_n) \leq \sum_n \mu(A_n)$
- **Continuity from below:** $A_n \nearrow A \implies \mu(A_n) \rightarrow \mu(A)$
- **Continuity from above:** $A_n \searrow A, \mu(A_k) < \infty \implies \mu(A_n) \rightarrow \mu(A)$

Outer Measures: A function $\mu^* : \mathcal{P}(X) \rightarrow [0, \infty]$ with $\mu^*(\emptyset) = 0$, monotonicity, and σ -subadditivity. The Lebesgue outer measure is defined by:

$$\lambda^*(A) = \inf \left\{ \sum_k \text{vol}(R_k) \mid A \subseteq \bigcup_k R_k \right\}$$

Carathéodory Construction: A set E is μ^* -measurable if for all A : $\mu^*(A) = \mu^*(A \cap E) + \mu^*(A \cap E^c)$. The measurable sets form a complete σ -algebra.

Completion: $\overline{\mathcal{M}}^\mu = \{A \cup N \mid A \in \mathcal{M}, N \subseteq M, \mu(M) = 0\}$. This is the smallest complete extension of $(X, \mathcal{M}, \mu)$.

Regularity: For Lebesgue measure:

- **Outer:** $\lambda^*(A) = \inf\{\lambda(G) \mid G \supseteq A, G \text{ open}\}$
- **Inner:** $\lambda(A) = \sup\{\lambda(K) \mid K \subseteq A, K \text{ compact}\}$

Uniqueness (π - λ Theorem): If σ -finite measures agree on a π -system $\mathcal{P}$ with $\sigma(\mathcal{P}) = \mathcal{M}$, they agree on all of $\mathcal{M}$.

CHAPTER 19

Measurable Functions

19.1 Definition and Basic Properties

Intuition:

Measurable functions are to measure theory what continuous functions are to topology: they are the “structure-preserving” maps. A function $f : X \rightarrow Y$ is measurable if it “pulls back” measurable sets to measurable sets, i.e., $f^{-1}(A) \in \mathcal{M}$ whenever $A \in \mathcal{N}$.

Why this definition?

- It ensures that $\{x \mid f(x) \in A\}$ is measurable, so we can meaningfully ask “what is the measure of points where f takes values in A ?”
- It makes the integral well-defined: we need to integrate over level sets like $\{f > a\}$, which must be measurable.
- It is preserved under composition, limits, and algebraic operations.

Definition 19.1.1 (Measurable Function):

Let $(X, \mathcal{M})$ and $(Y, \mathcal{N})$ be measurable spaces. A function $f : X \rightarrow Y$ is **measurable** if:

$$\forall A \in \mathcal{N} : f^{-1}(A) \in \mathcal{M}$$

Proposition 19.1.2 (Monotonicity of Measurability):

Let $f : X \rightarrow Y$ be a function.

1. If f is $(\mathcal{M}, \mathcal{N})$ -measurable, $\mathcal{M} \subseteq \mathcal{M}'$, and $\mathcal{N}' \subseteq \mathcal{N}$, then f is $(\mathcal{M}', \mathcal{N}')$ -measurable.
2. Any function $f : X \rightarrow Y$ is $(\mathcal{P}(X), \mathcal{N})$ -measurable for any $\mathcal{N}$.
3. Any function $f : X \rightarrow Y$ is $(\mathcal{M}, \{\emptyset, Y\})$ -measurable for any $\mathcal{M}$.

Proof:

- (1) Let $A \in \mathcal{N}'$. Since $\mathcal{N}' \subseteq \mathcal{N}$, we have $A \in \mathcal{N}$. By $(\mathcal{M}, \mathcal{N})$ -measurability, $f^{-1}(A) \in \mathcal{M} \subseteq \mathcal{M}'$.
- (2) For any $A \in \mathcal{N}$, we have $f^{-1}(A) \subseteq X$, so $f^{-1}(A) \in \mathcal{P}(X)$.
- (3) We only need to check $f^{-1}(\emptyset) = \emptyset \in \mathcal{M}$ and $f^{-1}(Y) = X \in \mathcal{M}$.

QED

19.1.1 Pullback and Pushforward σ -Algebras

Definition 19.1.3 (Pullback σ -Algebra):

Let $f : X \rightarrow Y$ be a function and $\mathcal{N}$ a σ -algebra on Y . The **pullback σ -algebra** (or **inverse image σ -algebra**) is:

$$f^*(\mathcal{N}) = \{f^{-1}(A) \mid A \in \mathcal{N}\}$$

This is the coarsest σ -algebra on X making f measurable.

Proposition 19.1.4:

$f^*(\mathcal{N})$ is a σ -algebra on X .

Proof:

Contains $\emptyset$: $\emptyset = f^{-1}(\emptyset) \in f^*(\mathcal{N})$ since $\emptyset \in \mathcal{N}$.

Closure under complements: Let $f^{-1}(A) \in f^*(\mathcal{N})$ where $A \in \mathcal{N}$. Then $(f^{-1}(A))^c = f^{-1}(A^c)$. Since $\mathcal{N}$ is a σ -algebra, $A^c \in \mathcal{N}$, so $f^{-1}(A^c) \in f^*(\mathcal{N})$.

Closure under countable unions: Let $(f^{-1}(A_n))_{n \in \mathbb{N}} \subseteq f^*(\mathcal{N})$ where $A_n \in \mathcal{N}$. Then:

$$\bigcup_{n \in \mathbb{N}} f^{-1}(A_n) = f^{-1}\left(\bigcup_{n \in \mathbb{N}} A_n\right)$$

Since $\mathcal{N}$ is a σ -algebra, $\bigcup_{n \in \mathbb{N}} A_n \in \mathcal{N}$, so $f^{-1}(\bigcup_{n \in \mathbb{N}} A_n) \in f^*(\mathcal{N})$. QED

Definition 19.1.5 (Pushforward σ -Algebra):

Let $f : X \rightarrow Y$ be a function and $\mathcal{M}$ a σ -algebra on X . The **pushforward σ -algebra** is:

$$f_*(\mathcal{M}) = \{A \subseteq Y \mid f^{-1}(A) \in \mathcal{M}\}$$

This is the finest σ -algebra on Y making f measurable.

Proposition 19.1.6:

$f_*(\mathcal{M})$ is a σ -algebra on Y .

Proof:

Contains $\emptyset$: $f^{-1}(\emptyset) = \emptyset \in \mathcal{M}$.

Closure under complements: If $A \in f_*(\mathcal{M})$, then $f^{-1}(A) \in \mathcal{M}$, so $f^{-1}(A^c) = (f^{-1}(A))^c \in \mathcal{M}$, thus $A^c \in f_*(\mathcal{M})$.

Closure under countable unions: If $(A_n)_{n \in \mathbb{N}} \subseteq f_*(\mathcal{M})$, then $f^{-1}(\bigcup_{n \in \mathbb{N}} A_n) = \bigcup_{n \in \mathbb{N}} f^{-1}(A_n) \in \mathcal{M}$. QED

19.1.2 Pushforward Measure

Definition 19.1.7 (Pushforward Measure):

Let $(X, \mathcal{M}, \mu)$ be a measure space, $(Y, \mathcal{N})$ a measurable space, and $f : X \rightarrow Y$ a measurable function. The **pushforward measure** $f_*\mu : \mathcal{N} \rightarrow [0, \infty]$ is defined by:

$$(f_*\mu)(A) = \mu(f^{-1}(A)) \quad \text{for } A \in \mathcal{N}$$

Proposition 19.1.8:

$f_*\mu$ is a measure on $(Y, \mathcal{N})$.

Proof:

$$(f_*\mu)(\emptyset) = 0: (f_*\mu)(\emptyset) = \mu(f^{-1}(\emptyset)) = \mu(\emptyset) = 0.$$

σ -additivity: Let $(A_n)_{n \in \mathbb{N}}$ be pairwise disjoint in $\mathcal{N}$. Then $(f^{-1}(A_n))_{n \in \mathbb{N}}$ are pairwise disjoint in $\mathcal{M}$, so:

$$(f_*\mu)\left(\bigcup_{n \in \mathbb{N}} A_n\right) = \mu\left(f^{-1}\left(\bigcup_{n \in \mathbb{N}} A_n\right)\right) = \mu\left(\bigcup_{n \in \mathbb{N}} f^{-1}(A_n)\right) = \sum_{n \in \mathbb{N}} \mu(f^{-1}(A_n)) = \sum_{n \in \mathbb{N}} (f_*\mu)(A_n)$$

QED

Remark (Pullback Measure):

There is no natural “pullback measure” for a general measurable function $f : X \rightarrow Y$. The issue is that the direct image $f(A)$ of a measurable set $A \in \mathcal{M}$ need not be measurable in $\mathcal{N}$.

However, if $f : X \rightarrow Y$ is a bijection with both f and f^{-1} measurable (a **measurable isomorphism**), then the pullback of f can be defined as the pushforward of f^{-1} :

$$f^*\nu = (f^{-1})_*\nu$$

Proposition 19.1.9 (Characterization for Real Functions):

$f : X \rightarrow \mathbb{R}$ is measurable iff any of the following equivalent conditions hold:

1. $f^{-1}((a, \infty)) \in \mathcal{M}$ for all $a \in \mathbb{R}$
2. $f^{-1}([a, \infty)) \in \mathcal{M}$ for all $a \in \mathbb{R}$
3. $f^{-1}((-\infty, a)) \in \mathcal{M}$ for all $a \in \mathbb{R}$
4. $f^{-1}((-\infty, a]) \in \mathcal{M}$ for all $a \in \mathbb{R}$

Proof:

These four families generate $\mathcal{B}(\mathbb{R})$. Thus any condition implies $f^{-1}(B) \in \mathcal{M}$ for all $B \in \mathcal{B}(\mathbb{R})$.

The equivalences follow from: $(a, \infty)^c = (-\infty, a]$, $[a, \infty) = \bigcap_{n \in \mathbb{N}} (a - 1/n, \infty)$, etc.

QED

Proposition 19.1.10 (Closure Properties):

If f, g are measurable and $c \in \mathbb{R}$:

1. $cf, f + g, fg, |f|, \max(f, g), \min(f, g)$ are measurable
2. If (f_n) are measurable, then $\sup_n f_n, \inf_n f_n, \limsup f_n, \liminf f_n$ are measurable

Proof:

(1) For cf : $(cf)^{-1}((a, \infty)) = f^{-1}((a/c, \infty))$ (case $c > 0$).

For $f + g$: $(f + g)^{-1}((a, \infty)) = \bigcup_{r \in \mathbb{Q}} (f^{-1}((r, \infty)) \cap g^{-1}((a - r, \infty)))$.

For fg : use $fg = \frac{1}{4}((f + g)^2 - (f - g)^2)$ and h^2 is measurable since $(h^2)^{-1}((a, \infty)) = h^{-1}((\sqrt{a}, \infty)) \cup h^{-1}((-\infty, -\sqrt{a}))$ for $a \geq 0$.

For $\max$: $(\max(f, g))^{-1}((a, \infty)) = f^{-1}((a, \infty)) \cup g^{-1}((a, \infty))$.

(2) $(\sup_n f_n)^{-1}((a, \infty)) = \bigcup_{n \in \mathbb{N}} f_n^{-1}((a, \infty)) \in \mathcal{M}$. Similarly for $\inf$. Then $\limsup f_n = \inf_k \sup_{n \geq k} f_n$.

QED

19.2 Simple Functions

Definition 19.2.1 (Simple Function):

A function $s : X \rightarrow [0, \infty)$ is **simple** if its image is finite.

Definition 19.2.2 (Canonical Form of Simple Function):

Let $s : X \rightarrow [0, \infty)$ be a simple function with image $\{\alpha_1, \dots, \alpha_n\}$ where α_i are distinct. The **canonical form** is:

$$s = \sum_{i=1}^n \alpha_i \mathbb{1}_{A_i}$$

where $A_i = s^{-1}(\{\alpha_i\})$ are the level sets.

Proposition 19.2.3 (Uniqueness of Canonical Form):

The canonical form of a simple function is unique.

Proof:

Let $s : X \rightarrow [0, \infty)$ be a simple function. Suppose we have two canonical representations:

$$s = \sum_{i=1}^n \alpha_i \mathbb{1}_{A_i} = \sum_{j=1}^m \beta_j \mathbb{1}_{B_j}$$

where α_i are distinct, β_j are distinct, and A_i, B_j are the corresponding level sets.

Step 1: The number of values coincides. The image of s is a set-theoretic property of s itself:

$$\text{Im}(s) = \{s(x) \mid x \in X\}$$

In the first representation, $\text{Im}(s) = \{\alpha_1, \dots, \alpha_n\}$. In the second representation, $\text{Im}(s) = \{\beta_1, \dots, \beta_m\}$. Thus $\{\alpha_1, \dots, \alpha_n\} = \{\beta_1, \dots, \beta_m\}$, so $n = m$ and (after reordering) $\alpha_i = \beta_i$ for all i .

Step 2: The level sets coincide. For each i , the level set is uniquely determined by s and α_i :

$$A_i = s^{-1}(\{\alpha_i\}) = \{x \in X \mid s(x) = \alpha_i\}$$

Since $\alpha_i = \beta_i$, we have $A_i = s^{-1}(\{\alpha_i\}) = s^{-1}(\{\beta_i\}) = B_i$.

QED

Proposition 19.2.4 (Measurability Criterion for Simple Functions):

A simple function $s : X \rightarrow [0, \infty)$ is measurable if and only if each level set $A_i = s^{-1}(\{\alpha_i\})$ is measurable.

Proof:

($\Rightarrow$) Suppose s is measurable. Each singleton $\{\alpha_i\} \subseteq \mathbb{R}$ is a Borel set. Thus $A_i = s^{-1}(\{\alpha_i\})$ is measurable.

($\Leftarrow$) Suppose each A_i is measurable. For any Borel set $B \subseteq [0, \infty)$:

$$s^{-1}(B) = \bigcup_{\alpha_i \in B} A_i$$

Since this is a finite union of measurable sets (at most n terms), it is measurable.

QED

19.3 σ -Simple Functions

Definition 19.3.1 (σ -Simple Function):

A function $s : X \rightarrow [0, \infty]$ is **σ -simple** if its image is at most countable.

Definition 19.3.2 (Canonical Form of σ -Simple Function):

Let $s : X \rightarrow [0, \infty]$ be σ -simple with image $\{\alpha_i\}_{i \in I}$ where $I \subseteq \mathbb{N}$ and α_i are distinct. The **canonical form** is:

$$s = \sum_{i \in I} \alpha_i \mathbb{1}_{A_i}$$

where $A_i = s^{-1}(\{\alpha_i\})$ are the level sets.

Proposition 19.3.3 (Uniqueness of Canonical Form):

The canonical form of a σ -simple function is unique.

Proof:

Let $s : X \rightarrow [0, \infty]$ be σ -simple. Suppose we have two canonical representations:

$$s = \sum_{i \in I} \alpha_i \mathbb{1}_{A_i} = \sum_{j \in J} \beta_j \mathbb{1}_{B_j}$$

where α_i are distinct, β_j are distinct, and A_i, B_j are the corresponding level sets.

Step 1: The values coincide. The image of s is uniquely determined:

$$\text{Im}(s) = \{s(x) \mid x \in X\} = \{\alpha_i\}_{i \in I} = \{\beta_j\}_{j \in J}$$

Thus (after bijective reindexing) the values are the same.

Step 2: The level sets coincide. For each value $\alpha \in \text{Im}(s)$, the level set is uniquely determined:

$$s^{-1}(\{\alpha\}) = \{x \in X \mid s(x) = \alpha\}$$

This depends only on s and α , not on the representation. QED

Proposition 19.3.4 (Measurability Criterion for σ -Simple Functions):

A σ -simple function $s : X \rightarrow [0, \infty]$ is measurable if and only if each level set $A_i = s^{-1}(\{\alpha_i\})$ is measurable.

Proof:

($\Rightarrow$) Suppose s is measurable. Each singleton $\{\alpha_i\}$ is a Borel set in $[0, \infty]$. Thus $A_i = s^{-1}(\{\alpha_i\})$ is measurable.

($\Leftarrow$) Suppose each A_i is measurable. For any Borel set $B \subseteq [0, \infty]$:

$$s^{-1}(B) = \bigcup_{\alpha_i \in B} A_i$$

Since this is a countable union of measurable sets, it is measurable. QED

19.4 σ -Simple Function Approximation

Theorem 19.4.1 (Increasing σ -Simple Approximation):

Let $f : X \rightarrow [0, \infty]$ be measurable. Then there exists an increasing sequence of σ -simple functions $(s_n)_{n \in \mathbb{N}}$ such that $s_n \nearrow f$ uniformly.

Proof:

Construction: Define for each $n \in \mathbb{N}$:

$$s_n(x) = \begin{cases} \lfloor 2^n f(x) \rfloor / 2^n & \text{if } f(x) < \infty \\ \infty & \text{if } f(x) = \infty \end{cases}$$

Step 1: s_n is σ -simple. The image of s_n is $\{k/2^n \mid k \in \mathbb{N}_0\} \cup \{\infty\}$, which is countable. Each level set $s_n^{-1}(\{k/2^n\}) = f^{-1}([k/2^n, (k+1)/2^n))$ is measurable.

Step 2: $s_n \leq f$. For $f(x) < \infty$: $s_n(x) = \lfloor 2^n f(x) \rfloor / 2^n \leq f(x)$ by definition of floor. For $f(x) = \infty$: $s_n(x) = \infty = f(x)$.

Step 3: (s_n) is increasing. Let $k = \lfloor 2^n f(x) \rfloor$, so $k \leq 2^n f(x) < k+1$. Then $2k \leq 2^{n+1} f(x) < 2k+2$, so $\lfloor 2^{n+1} f(x) \rfloor \geq 2k$. Thus:

$$s_{n+1}(x) = \frac{\lfloor 2^{n+1} f(x) \rfloor}{2^{n+1}} \geq \frac{2k}{2^{n+1}} = \frac{k}{2^n} = s_n(x)$$

Step 4: $s_n \rightarrow f$ uniformly. For $f(x) < \infty$: $|f(x) - s_n(x)| = f(x) - \lfloor 2^n f(x) \rfloor / 2^n < 1/2^n$. For $f(x) = \infty$: $s_n(x) = f(x)$, so there is nothing to prove.

Thus $\|f - s_n\|_\infty \leq 1/2^n \rightarrow 0$.

QED

Theorem 19.4.2 (Decreasing σ -Simple Approximation):

Let $f : X \rightarrow [0, \infty]$ be measurable. Then there exists a decreasing sequence of σ -simple functions $(t_n)_{n \in \mathbb{N}}$ such that $t_n \searrow f$ uniformly.

Proof:

Construction: Define for each $n \in \mathbb{N}$:

$$t_n(x) = \begin{cases} \lceil 2^n f(x) \rceil / 2^n & \text{if } f(x) < \infty \\ \infty & \text{if } f(x) = \infty \end{cases}$$

Step 1: t_n is σ -simple. The image is $\{k/2^n \mid k \in \mathbb{N}_0\} \cup \{\infty\}$, which is countable.

Step 2: $t_n \geq f$. For $f(x) < \infty$: $t_n(x) = \lceil 2^n f(x) \rceil / 2^n \geq f(x)$ by definition of ceiling. For $f(x) = \infty$: $t_n(x) = \infty = f(x)$.

Step 3: (t_n) is decreasing. Let $k = \lceil 2^n f(x) \rceil$, so $k-1 < 2^n f(x) \leq k$. Then $2k-2 < 2^{n+1} f(x) \leq 2k$, so $\lceil 2^{n+1} f(x) \rceil \leq 2k$. Thus:

$$t_{n+1}(x) = \frac{\lceil 2^{n+1} f(x) \rceil}{2^{n+1}} \leq \frac{2k}{2^{n+1}} = \frac{k}{2^n} = t_n(x)$$

Step 4: $t_n \rightarrow f$ uniformly. For $f(x) < \infty$: $|t_n(x) - f(x)| = \lceil 2^n f(x) \rceil / 2^n - f(x) < 1/2^n$. For $f(x) = \infty$: $t_n(x) = f(x)$, so there is nothing to prove.

Thus $\|t_n - f\|_\infty \leq 1/2^n \rightarrow 0$.

QED

Remark:

Unlike simple function approximation, σ -simple approximation is **always uniform** because we allow the value ∞ in the approximating functions. This handles unbounded functions without needing to truncate.

19.5 Simple Function Approximation

Theorem 19.5.1 (Increasing Simple Function Approximation):

Let $f : X \rightarrow [0, \infty]$ be measurable. Then there exists an increasing sequence $(s_n)_{n \in \mathbb{N}}$ of simple functions such that $s_n \nearrow f$ pointwise.

Proof:

Construction: Define for each $n \in \mathbb{N}$:

$$s_n(x) = \frac{\lfloor 2^n f(x) \rfloor}{2^n} \cdot \mathbb{1}_{\{f < 2^n\}}(x) + 2^n \cdot \mathbb{1}_{\{f \geq 2^n\}}(x)$$

Step 1: s_n is simple. The image of s_n is $\{k/2^n \mid 0 \leq k \leq 2^{2^n}\} \cup \{2^n\}$, which is finite.

Step 2: $s_n \leq f$. If $f(x) < 2^n$: $s_n(x) = \lfloor 2^n f(x) \rfloor / 2^n \leq f(x)$. If $f(x) \geq 2^n$: $s_n(x) = 2^n \leq f(x)$.

Step 3: (s_n) is increasing. Let $k = \lfloor 2^n f(x) \rfloor$, so $k \leq 2^n f(x) < k + 1$. Then $2k \leq 2^{n+1} f(x) < 2k + 2$, so $\lfloor 2^{n+1} f(x) \rfloor \geq 2k$. Thus $s_{n+1}(x) \geq 2k/2^{n+1} = k/2^n = s_n(x)$. The cap also increases: $2^{n+1} > 2^n$.

Step 4: $s_n \rightarrow f$ pointwise. If $f(x) < \infty$: For n large enough that $2^n > f(x)$, we have $|f(x) - s_n(x)| < 1/2^n \rightarrow 0$. If $f(x) = \infty$: $s_n(x) = 2^n \rightarrow \infty = f(x)$. QED

Proposition 19.5.2 (Uniform Convergence Characterization):

The convergence $s_n \rightarrow f$ is uniform if and only if f is bounded.

Proof:

($\Leftarrow$) Bounded implies uniform: Suppose $f \leq M$ for some $M < \infty$. For n large enough that $2^n > M$, we have $f(x) < 2^n$ for all x , so:

$$|f(x) - s_n(x)| = f(x) - \lfloor 2^n f(x) \rfloor / 2^n < 1/2^n$$

Thus $\|f - s_n\|_\infty < 1/2^n \rightarrow 0$.

($\Rightarrow$) Uniform implies bounded: Suppose $s_n \rightarrow f$ uniformly. Then for some N , $\|f - s_N\|_\infty < 1$. Since s_N is simple, it is bounded: $s_N \leq C$ for some $C < \infty$. Thus $f \leq s_N + 1 \leq C + 1 < \infty$. QED

Proposition 19.5.3 (Decreasing Simple Function Approximation):

Let $f : X \rightarrow [0, \infty]$ be measurable. Then f can be approximated by a decreasing sequence of simple functions $t_n \searrow f$ if and only if f is bounded.

Proof:

($\Rightarrow$) Decreasing simple approximation implies bounded: If $t_n \searrow f$ with t_n simple, then each t_n is bounded. Since $t_1 \geq f$, we have $f \leq \|t_1\|_\infty < \infty$.

($\Leftarrow$) Bounded implies decreasing approximation: Suppose $f \leq M$. Define:

$$t_n(x) = \frac{\lfloor 2^n f(x) \rfloor}{2^n}$$

Then t_n is simple (values in $\{k/2^n \mid 0 \leq k \leq \lceil 2^n M \rceil\}$), $t_n \geq f$, t_n decreases (by similar argument to increasing case), and $\|t_n - f\|_\infty \leq 1/2^n \rightarrow 0$. QED

Remark (Why Simple Approximation Cannot Always Be Uniform):

For unbounded f , simple functions—which take only finitely many values—cannot approximate f uniformly because:

- Any simple function s satisfies $s \leq \max(\text{Im}(s)) < \infty$

- If $f(x_n) \rightarrow \infty$, then $|f(x_n) - s(x_n)| \rightarrow \infty$ for any fixed simple s

The σ -simple setting avoids this by allowing ∞ as a value.

Chapter Compendium:

Measurable Functions: $f : X \rightarrow Y$ is measurable if $f^{-1}(A) \in \mathcal{M}$ for all $A \in \mathcal{N}$. For $f : X \rightarrow \mathbb{R}$, equivalent to $\{f > a\} \in \mathcal{M}$ for all a .

σ -Algebras on Functions:

- **Pullback:** $f^*(\mathcal{N}) = \{f^{-1}(A) \mid A \in \mathcal{N}\}$ (coarsest making f measurable)
- **Pushforward:** $f_*(\mathcal{M}) = \{A \subseteq Y \mid f^{-1}(A) \in \mathcal{M}\}$ (finest making f measurable)

Pushforward Measure: $(f_*\mu)(A) = \mu(f^{-1}(A))$ is a measure on $(Y, \mathcal{N})$.

Closure Properties: If f, g measurable: $cf, f + g, fg, |f|, \max, \min$ are measurable. Also $\sup_n f_n, \inf_n f_n, \limsup f_n, \liminf f_n$ are measurable.

Function Classes:

- **Simple:** Finite image in $[0, +\infty)$. Form: $s = \sum_{i=1}^n \alpha_i \mathbb{1}_{A_i}$
- **σ -Simple:** Countable image in $[0, +\infty]$. Form: $s = \sum_{i \in I} \alpha_i \mathbb{1}_{A_i}$

Approximation Theorems: For measurable $f : X \rightarrow [0, \infty]$:

- **Increasing simple** $s_n \nearrow f$ pointwise (uniform if f bounded)
- **Increasing σ -simple** $s_n \nearrow f$ uniformly always
- **Decreasing σ -simple** $t_n \searrow f$ uniformly always
- **Decreasing simple** $t_n \searrow f$ exists only if f is bounded

CHAPTER 20

The Lebesgue Integral

Intuition:

The Lebesgue integral extends the Riemann integral to a much larger class of functions by “slicing horizontally” rather than “vertically”. Instead of partitioning the domain and summing $f(\text{sample point}) \times (\text{interval length})$, we partition the **range** and sum $\text{value} \times \mu(\text{level set})$.

Key advantages over Riemann integration:

- Works for highly discontinuous functions (e.g., Dirichlet function)
- Powerful convergence theorems (MCT, DCT, Fatou) that fail for Riemann
- Defines L^p spaces as complete normed vector spaces

Our construction proceeds: simple functions $\rightarrow$ σ -simple functions $\rightarrow$ non-negative measurable functions $\rightarrow$ integrable functions.

20.1 Integration of Positive Functions

Definition 20.1.1 (Integral of Simple Function):

For simple $s = \sum_{i=1}^n \alpha_i \mathbb{1}_{A_i}$ on $(X, \mathcal{M}, \mu)$:

$$\int_X s \, d\mu = \sum_{i=1}^n \alpha_i \mu(A_i)$$

with convention $0 \cdot \infty = 0$.

Definition 20.1.2 (Integral of σ -Simple Function):

For σ -simple $s = \sum_{i=1}^{\infty} \alpha_i \mathbb{1}_{A_i}$ on $(X, \mathcal{M}, \mu)$:

$$\int_X s \, d\mu = \sum_{i=1}^{\infty} \alpha_i \mu(A_i)$$

with convention $0 \cdot \infty = 0$. The sum is always well-defined in $[0, \infty]$.

Proposition 20.1.3 (Properties of the Integral for σ -Simple Functions):

Let $s, t : X \rightarrow [0, \infty]$ be measurable σ -simple functions and $c \geq 0$. Then:

1. **Linearity:** $\int_X (s + t) \, d\mu = \int_X s \, d\mu + \int_X t \, d\mu$ and $\int_X (cs) \, d\mu = c \int_X s \, d\mu$

2. **Monotonicity in function:** $s \leq t \implies \int_X s \, d\mu \leq \int_X t \, d\mu$
3. **Zero characterization:** $\int_X s \, d\mu = 0 \iff s = 0 \text{ a.e.}$
4. **Empty set:** $\int_{\emptyset} s \, d\mu = 0$
5. **Monotonicity in set:** $A \subseteq B \implies \int_A s \, d\mu \leq \int_B s \, d\mu$
6. **Countable additivity:** If $(A_k)_{k=1}^{\infty}$ are pairwise disjoint measurable sets:

$$\int_{\bigcup_{k=1}^{\infty} A_k} s \, d\mu = \sum_{k=1}^{\infty} \int_{A_k} s \, d\mu$$

Proof:

Let $s = \sum_{n=1}^{\infty} \alpha_n \mathbb{1}_{B_n}$ and $t = \sum_{m=1}^{\infty} \beta_m \mathbb{1}_{C_m}$ be the canonical forms.

(1) Linearity:

Scalar multiplication: The function cs has canonical form $\sum_{n=1}^{\infty} (c\alpha_n) \mathbb{1}_{B_n}$, so:

$$\int_X cs \, d\mu = \sum_{n=1}^{\infty} c\alpha_n \mu(B_n) = c \sum_{n=1}^{\infty} \alpha_n \mu(B_n) = c \int_X s \, d\mu$$

Addition: Consider the common refinement $\{B_n \cap C_m\}_{n,m \in \mathbb{N}}$. These sets partition X and on $B_n \cap C_m$, we have $(s+t)(x) = \alpha_n + \beta_m$.

Thus $(s+t)$ can be written as $\sum_{n,m} (\alpha_n + \beta_m) \mathbb{1}_{B_n \cap C_m}$, and:

$$\begin{aligned} \int_X (s+t) \, d\mu &= \sum_{n,m} (\alpha_n + \beta_m) \mu(B_n \cap C_m) \\ &= \sum_{n,m} \alpha_n \mu(B_n \cap C_m) + \sum_{n,m} \beta_m \mu(B_n \cap C_m) \end{aligned}$$

For the first sum: $\sum_{n,m} \alpha_n \mu(B_n \cap C_m) = \sum_n \alpha_n \sum_m \mu(B_n \cap C_m) = \sum_n \alpha_n \mu(B_n) = \int_X s \, d\mu$.

Similarly for the second sum. Thus $\int_X (s+t) = \int_X s + \int_X t$.

(2) Monotonicity:

Suppose $s \leq t$ pointwise. Consider the common refinement $\{B_n \cap C_m\}_{n,m}$.

For (n, m) such that $B_n \cap C_m \neq \emptyset$: on this set, $s(x) = \alpha_n$ and $t(x) = \beta_m$, so $\alpha_n \leq \beta_m$.

Thus:

$$\begin{aligned} \int_X s \, d\mu &= \sum_{n,m} \alpha_n \mu(B_n \cap C_m) \\ &\leq \sum_{n,m} \beta_m \mu(B_n \cap C_m) = \int_X t \, d\mu \end{aligned}$$

(3) Zero characterization:

$(\implies)$ Suppose $\int_X s \, d\mu = \sum_{n=1}^{\infty} \alpha_n \mu(B_n) = 0$.

Since all terms are non-negative, each term is zero. For $\alpha_n > 0$, we must have $\mu(B_n) = 0$. Thus $\{s > 0\} = \bigcup_{\alpha_n > 0} B_n$ is a null set.

$(\impliedby)$ If $s = 0$ a.e., then for each n with $\alpha_n > 0$, we have $\mu(B_n) = 0$. Thus $\int_X s \, d\mu = \sum_{n=1}^{\infty} \alpha_n \mu(B_n) = 0$.

(4) **Empty set:** $\int_{\emptyset} s \, d\mu = \sum_{n=1}^{\infty} \alpha_n \mu(B_n \cap \emptyset) = 0$.

(5) **Set monotonicity:** If $A \subseteq B$, then $B_n \cap A \subseteq B_n \cap B$, so:

$$\int_A s \, d\mu = \sum_{n=1}^{\infty} \alpha_n \mu(B_n \cap A) \leq \sum_{n=1}^{\infty} \alpha_n \mu(B_n \cap B) = \int_B s \, d\mu$$

(6) **Countable additivity:**

$$\int_{\bigcup_{k=1}^{\infty} A_k} s \, d\mu = \sum_{n=1}^{\infty} \alpha_n \mu\left(B_n \cap \bigcup_{k=1}^{\infty} A_k\right) = \sum_{n=1}^{\infty} \alpha_n \sum_{k=1}^{\infty} \mu(B_n \cap A_k)$$

Since all terms are non-negative, we can interchange sums (Tonelli):

$$= \sum_{k=1}^{\infty} \sum_{n=1}^{\infty} \alpha_n \mu(B_n \cap A_k) = \sum_{k=1}^{\infty} \int_{A_k} s \, d\mu$$

QED

Definition 20.1.4 (Lower and Upper Integrals):

For any function $f : X \rightarrow [0, \infty]$ (not necessarily measurable), we define:

Lower Simple Integral:

$$\int_X f \, d\mu = \sup \left\{ \int_X s \, d\mu \mid s \text{ simple measurable, } 0 \leq s \leq f \right\}$$

Lower σ -Simple Integral:

$$\int_X f \, d\mu = \sup \left\{ \int_X s \, d\mu \mid s \text{ } \sigma\text{-simple measurable, } 0 \leq s \leq f \right\}$$

Upper σ -Simple Integral:

$$\overline{\int}_X f \, d\mu = \inf \left\{ \int_X t \, d\mu \mid t \text{ } \sigma\text{-simple measurable, } t \geq f \right\}$$

Upper Simple Integral:

$$\overline{\int}_X f \, d\mu = \inf \left\{ \int_X t \, d\mu \mid t \text{ simple measurable, } t \geq f \right\}$$

Proposition 20.1.5 (Ordering):

For any function $f : X \rightarrow [0, \infty]$:

$$\int f \leq \int f \leq \int f \leq \overline{\int} f$$

Proof:

The outer inequalities follow since simple $\subseteq \sigma$ -simple (larger class for sup, smaller for inf). The middle inequality follows since $s \leq f \leq t$ implies $\int s \leq \int t$ by monotonicity. QED

Proposition 20.1.6 (Lower Integrals Equality):

For any function $f : X \rightarrow [0, \infty]$:

$$\int_{\underline{}} f = {}_{\sigma} \int f$$

Proof:

($\leq$) Immediate since simple $\subseteq \sigma$ -simple, so the supremum over simple functions is at most the supremum over σ -simple functions.

($\geq$) Let s be σ -simple measurable with $0 \leq s \leq f$. We show $\int_{\underline{}} f \geq \int s$.

Let $s = \sum_{n=1}^{\infty} \alpha_n \mathbb{1}_{A_n}$ be the canonical form. Define the truncated simple functions:

$$s_N = \sum_{n=1}^N \alpha_n \mathbb{1}_{A_n}$$

Then s_N is simple, $s_N \leq s \leq f$, and by monotone convergence of series:

$$\int s_N = \sum_{n=1}^N \alpha_n \mu(A_n) \nearrow \sum_{n=1}^{\infty} \alpha_n \mu(A_n) = \int s$$

For any $\epsilon > 0$, there exists N such that $\int s_N > \int s - \epsilon$. Since $s_N \leq f$ and s_N is simple: $\int_{\underline{}} f \geq \int s_N > \int s - \epsilon$.

Since ϵ is arbitrary, $\int_{\underline{}} f \geq \int s$. Taking supremum over σ -simple $s \leq f$: $\int_{\underline{}} f \geq {}_{\sigma} \int f$. QED

Proposition 20.1.7 (Upper Integrals Equality):

For any function $f : X \rightarrow [0, \infty]$:

$${}_{\sigma} \int f = \int_{\overline{}} f \iff f \text{ is bounded a.e. and } \mu(\{f > 0\}) < \infty$$

Proof:

($\Leftarrow$) Suppose $f \leq M$ a.e. for some $M < \infty$ and $\mu(\{f > 0\}) < \infty$.

We have ${}_{\sigma} \int f \leq \int_{\overline{}} f$ (from ordering, since simple $\subseteq \sigma$ -simple).

For the reverse: let t be σ -simple measurable with $t \geq f$, and let $\epsilon > 0$. Define truncated simple functions:

$$t_N = \sum_{n=1}^N \alpha_n \mathbb{1}_{A_n}$$

where $t = \sum_{n=1}^{\infty} \alpha_n \mathbb{1}_{A_n}$ is the canonical form.

Since f is bounded a.e. by M and has finite support, we can choose t such that t is also bounded and has finite support (take $t = M \cdot \mathbb{1}_{\{f > 0\}}$ as an upper bound).

Then for N large enough, $t_N \geq f$ (on the finite support) and $\int t_N < \int t + \epsilon$. Thus $\int_{\overline{}} f \leq \int t_N < \int t + \epsilon$.

Taking infimum: $\int_{\overline{}} f \leq {}_{\sigma} \int f$.

($\Rightarrow$) We prove the contrapositive.

Case 1: If f is unbounded on a set of positive measure, i.e., $\mu(\{f > n\}) > 0$ for all n , then any simple $t \geq f$ must satisfy $t \geq n$ on $\{f > n\}$, so $\int t \geq n \cdot \mu(\{f > n\}) \rightarrow \infty$. Thus $\int_{\overline{}} f = \infty$.

However, we can take σ -simple $t = f$ itself (if measurable) or approximate, giving finite ${}_{\sigma} \int f < \infty$ in many cases. So ${}_{\sigma} \int f \neq \int_{\overline{}} f$.

Case 2: If $\mu(\{f > 0\}) = \infty$, then any simple $t \geq f$ must be positive on an infinite measure set. For $t \geq c > 0$ on this set, $\int t = \infty$. Thus $\int_{\overline{}} f = \infty$, while ${}_{\sigma} \int f$ may be finite. QED

Theorem 20.1.8 (Equality of Integrals for Measurable Functions):

For any measurable $f : X \rightarrow [0, \infty]$:

$$\int f = \sigma \int f = \sigma \int f$$

We denote this common value by $\int f d\mu$, the **Lebesgue integral**.

Proof:

The equality $\int f = \sigma \int f$ holds for all f (by the previous proposition).

It remains to show $\sigma \int f = \sigma \int f$ for measurable f .

Let $\epsilon > 0$. By uniform σ -simple approximation with 2^n , there exist σ -simple $s_n \nearrow f$ and $t_n \searrow f$ uniformly. For n large enough: $\|t_n - s_n\|_\infty < \epsilon$, so:

$$\int t_n - \int s_n = \int (t_n - s_n) \leq \epsilon \cdot \mu(\text{supp}(f))$$

Thus $\sigma \int f \leq \int t_n \leq \int s_n + \epsilon \cdot \mu(\text{supp}(f)) \leq \sigma \int f + \epsilon \cdot \mu(\text{supp}(f))$.

Taking $\epsilon \rightarrow 0$: $\sigma \int f \leq \sigma \int f$.

Combined with $\sigma \int f \leq \sigma \int f$ (from ordering), we have equality.

QED

Definition 20.1.9 (Lebesgue Integral):

For measurable $f : X \rightarrow [0, \infty]$:

$$\int_X f d\mu = \int f = \sigma \int f = \sigma \int f$$

20.2 Monotone Convergence Theorem

Intuition:

MCT is the cornerstone of Lebesgue integration theory. It says: for **non-negative** functions increasing pointwise, the integral of the limit equals the limit of the integrals.

Why non-negativity? Without it, $\infty - \infty$ issues arise. Why increasing? Decreasing sequences may fail (e.g., $f_n = \mathbb{1}_{[n, \infty)}$ on $\mathbb{R}$ with Lebesgue measure: $\int f_n = \infty$ but $\int \lim f_n = 0$).

MCT unlocks powerful techniques:

- Interchange of $\sum$ and $\int$ for non-negative terms
- Definition of the integral via simple function approximation
- Proofs of Fatou and DCT

Theorem 20.2.1 (MCT / Beppo-Levi):

Let (f_n) be measurable with $0 \leq f_n \nearrow f$ pointwise. Then:

$$\lim_{n \rightarrow \infty} \int_X f_n d\mu = \int_X f d\mu$$

Proof:

Since $f_n \leq f$, monotonicity gives $\int_X f_n d\mu \leq \int_X f d\mu$, so $\lim \int_X f_n d\mu \leq \int_X f d\mu$.
 For the reverse, let s be simple with $0 \leq s \leq f$ and $0 < c < 1$. Define:

$$E_n = \{x \mid f_n(x) \geq c \cdot s(x)\}$$

Then $E_n \nearrow X$ since $f_n \nearrow f \geq s > cs$. Thus:

$$\int_X f_n d\mu \geq \int_{E_n} f_n d\mu \geq c \int_{E_n} s d\mu$$

By continuity from below, $\int_{E_n} s d\mu \rightarrow \int_X s d\mu$. So $\lim \int_X f_n d\mu \geq c \int_X s d\mu$.

Taking $c \rightarrow 1$ and supremum over s : $\lim \int_X f_n d\mu \geq \int_X f d\mu$. QED

Corollary 20.2.2:

For measurable $f_n \geq 0$: $\int_X \sum_{n=1}^{\infty} f_n = \sum_{n=1}^{\infty} \int_X f_n$.

Proof:

Let $S_N = \sum_{n=1}^N f_n$. Then $S_N \nearrow \sum_{n=1}^{\infty} f_n$. By MCT: $\int_X \sum f_n = \lim_N \int_X S_N = \lim_N \sum_{n=1}^N \int_X f_n = \sum_{n=1}^{\infty} \int_X f_n$. QED

20.3 Fatou's Lemma

Intuition:

Fatou's Lemma is a “one-sided MCT” for sequences that aren't necessarily monotone. It states: $\int(\liminf f_n) \leq \liminf(\int f_n)$.

The inequality can be strict: mass can “escape to infinity” in the limit. For example, $f_n = \mathbb{1}_{[n, n+1]}$ on $\mathbb{R}$: each $\int f_n = 1$, but $\liminf f_n = 0$.

Fatou is remarkably useful because it requires **no dominating function**—only non-negativity. This makes it applicable when DCT fails.

Theorem 20.3.1 (Fatou's Lemma):

If (f_n) are measurable with $f_n \geq 0$:

$$\int_X \liminf_{n \rightarrow \infty} f_n d\mu \leq \liminf_{n \rightarrow \infty} \int_X f_n d\mu$$

Proof:

Let $g_k = \inf_{n \geq k} f_n$. Then $g_k \nearrow \liminf_{n \rightarrow \infty} f_n$ and $g_k \leq f_k$. By MCT:

$$\int_X \liminf_{n \rightarrow \infty} f_n d\mu = \lim_{k \rightarrow \infty} \int_X g_k d\mu \leq \liminf_{k \rightarrow \infty} \int_X f_k d\mu$$

QED

20.4 Integrable Functions

Definition 20.4.1:

For measurable $f : X \rightarrow \mathbb{R}$, define $f^+ = \max(f, 0)$ and $f^- = \max(-f, 0)$.

f is **integrable** ($f \in L^1(\mu)$) if $\int_X f^+ d\mu < \infty$ and $\int_X f^- d\mu < \infty$.

The integral is: $\int_X f d\mu = \int_X f^+ d\mu - \int_X f^- d\mu$.

Proposition 20.4.2:

$f \in L^1(\mu)$ iff $\int_X |f| d\mu < \infty$. Moreover: $|\int_X f| \leq \int_X |f|$.

Proof:

$|f| = f^+ + f^-$, so $\int_X |f| = \int_X f^+ + \int_X f^- < \infty$ iff both are finite, iff $f \in L^1$.

Triangle inequality: $|\int_X f| = |\int_X f^+ - \int_X f^-| \leq \int_X f^+ + \int_X f^- = \int_X |f|$.

QED

20.5 Dominated Convergence Theorem

Intuition:

DCT is the most practical convergence theorem: if $f_n \rightarrow f$ pointwise and $|f_n| \leq g$ for some integrable g , then $\int f_n \rightarrow \int f$.

Unlike MCT, DCT works for **signed** functions and **non-monotone** convergence. The price is the dominating function $g \in L^1$, which prevents mass from escaping to infinity.

DCT is the go-to tool for:

- Differentiating under the integral sign (Leibniz rule)
- Interchanging limits and integrals in applications
- Proving continuity of parameter-dependent integrals

Theorem 20.5.1 (DCT / Lebesgue):

Let (f_n) be measurable with $f_n \rightarrow f$ a.e. If there exists $g \in L^1(\mu)$ with $|f_n| \leq g$ a.e. for all n , then:

$$f \in L^1(\mu) \quad \text{and} \quad \lim_{n \rightarrow \infty} \int_X |f_n - f| d\mu = 0$$

In particular: $\lim_{n \rightarrow \infty} \int_X f_n d\mu = \int_X f d\mu$.

Proof:

Since $|f| = \lim |f_n| \leq g$ a.e., we have $f \in L^1$.

Apply Fatou to $g - f_n \geq 0$ and $g + f_n \geq 0$:

$$\begin{aligned} \int_X (g - f) d\mu &\leq \liminf_{n \rightarrow \infty} \int_X (g - f_n) d\mu = \int_X g d\mu - \limsup_{n \rightarrow \infty} \int_X f_n d\mu \\ \int_X (g + f) d\mu &\leq \liminf_{n \rightarrow \infty} \int_X (g + f_n) d\mu = \int_X g d\mu + \liminf_{n \rightarrow \infty} \int_X f_n d\mu \end{aligned}$$

Thus $\limsup_{n \rightarrow \infty} \int_X f_n d\mu \leq \int_X f d\mu \leq \liminf_{n \rightarrow \infty} \int_X f_n d\mu$, so $\lim_{n \rightarrow \infty} \int_X f_n d\mu = \int_X f d\mu$.

For $\int_X |f_n - f| d\mu \rightarrow 0$: apply Fatou to $2g - |f_n - f| \geq 0$.

QED

Corollary 20.5.2 (DCT Implies L^1 Convergence):

If the hypotheses of DCT hold, then:

$$\int_X |f_n - f| d\mu \rightarrow 0 \implies \int_X f_n d\mu \rightarrow \int_X f d\mu$$

In fact, the converse implication $\int_X |f_n - f| d\mu \rightarrow 0$ follows from DCT by applying it to the sequence $|f_n - f|$ (which converges to 0 a.e. and is dominated by $2g$).

20.6 Change of Variables

Intuition:

The change of variables formula relates integrals via the pushforward measure. If we “transport” a measure μ on X to a measure $f_*\mu$ on Y via a measurable function $f : X \rightarrow Y$, then integrating g over Y with respect to $f_*\mu$ is the same as integrating $g \circ f$ over X with respect to μ .

This generalizes classical substitution rules and is fundamental for probability theory (where it underlies the law of the unconscious statistician).

Theorem 20.6.1 (Change of Variables / Pushforward Formula):

Let $(X, \mathcal{M}, \mu)$ be a measure space, $(Y, \mathcal{N})$ a measurable space, and $f : X \rightarrow Y$ a measurable function. Let $f_*\mu$ be the pushforward measure on Y .

For any measurable function $g : Y \rightarrow [0, \infty]$:

$$\int_Y g d(f_*\mu) = \int_X (g \circ f) d\mu$$

More generally, if $g : Y \rightarrow \overline{\mathbb{R}}$ is measurable and either $g \geq 0$ or $g \in L^1(f_*\mu)$, then the equality holds (with $g \in L^1(f_*\mu) \iff g \circ f \in L^1(\mu)$).

Proof:

We prove this in three steps: characteristic functions, simple functions, and general non-negative measurable functions.

Step 1: Characteristic functions.

Let $g = \mathbb{1}_A$ for some $A \in \mathcal{N}$. The left-hand side is:

$$\int_Y \mathbb{1}_A d(f_*\mu) = (f_*\mu)(A) = \mu(f^{-1}(A))$$

The right-hand side is:

$$\int_X (\mathbb{1}_A \circ f) d\mu = \int_X \mathbb{1}_{f^{-1}(A)} d\mu = \mu(f^{-1}(A))$$

Thus both sides are equal.

Step 2: Simple functions.

Let $g = \sum_{i=1}^n \alpha_i \mathbb{1}_{A_i}$ be a simple function on Y . By linearity:

$$\begin{aligned} \int_Y g d(f_*\mu) &= \sum_{i=1}^n \alpha_i (f_*\mu)(A_i) \\ &= \sum_{i=1}^n \alpha_i \mu(f^{-1}(A_i)) \quad (\text{by Step 1}) \\ &= \int_X \left(\sum_{i=1}^n \alpha_i \mathbb{1}_{f^{-1}(A_i)} \right) d\mu \\ &= \int_X (g \circ f) d\mu \end{aligned}$$

Step 3: General non-negative measurable functions.

Let $g : Y \rightarrow [0, \infty]$ be measurable. By the simple function approximation theorem, there exist simple functions $s_n \nearrow g$ pointwise on Y .

Then $s_n \circ f \nearrow g \circ f$ pointwise on X .

By MCT on both sides:

$$\begin{aligned} \int_Y g d(f_*\mu) &= \lim_{n \rightarrow \infty} \int_Y s_n d(f_*\mu) \quad (\text{MCT on } Y) \\ &= \lim_{n \rightarrow \infty} \int_X (s_n \circ f) d\mu \quad (\text{Step 2}) \\ &= \int_X (g \circ f) d\mu \quad (\text{MCT on } X) \end{aligned}$$

Step 4: General integrable functions.

For g with indefinite sign, apply the non-negative case to g^+ and g^- :

$$\int_Y g d(f_*\mu) = \int_Y g^+ d(f_*\mu) - \int_Y g^- d(f_*\mu) = \int_X (g^+ \circ f) d\mu - \int_X (g^- \circ f) d\mu$$

Since $(g^+ \circ f) = (g \circ f)^+$ and $(g^- \circ f) = (g \circ f)^-$:

$$= \int_X (g \circ f)^+ d\mu - \int_X (g \circ f)^- d\mu = \int_X (g \circ f) d\mu$$

The integrability equivalence follows: $g \in L^1(f_*\mu) \iff \int_Y |g| d(f_*\mu) < \infty \iff \int_X |g \circ f| d\mu < \infty \iff g \circ f \in L^1(\mu)$. QED

20.7 Riemann vs Lebesgue

Theorem 20.7.1 (Lebesgue Criterion):

A bounded function $f : [a, b] \rightarrow \mathbb{R}$ is Riemann integrable iff f is continuous a.e. (the set of discontinuities has Lebesgue measure 0).

If f is Riemann integrable, then f is Lebesgue integrable and:

$$\int_a^b f(x) dx = \int_{[a,b]} f d\lambda$$

Proof:

Let $D_f = \{x \in [a, b] \mid f \text{ is discontinuous at } x\}$.

Oscillation: For $x \in [a, b]$, define the oscillation:

$$\omega_f(x) = \lim_{\delta \rightarrow 0^+} \left(\sup_{|y-x| < \delta} f(y) - \inf_{|y-x| < \delta} f(y) \right)$$

Then $x \in D_f$ iff $\omega_f(x) > 0$. Note $D_f = \bigcup_{n \in \mathbb{N}} \{x \mid \omega_f(x) \geq 1/n\}$.

($\Rightarrow$): Assume f is Riemann integrable. For any $\epsilon > 0$, there exists partition P with $\bar{S}(f, P) - \underline{S}(f, P) < \epsilon$.

Let $A_\epsilon = \{x \mid \omega_f(x) \geq \epsilon\}$. The contribution to $\bar{S} - \underline{S}$ from intervals meeting A_ϵ is $\geq \epsilon \cdot \lambda^*(A_\epsilon \setminus \text{endpoints})$.

Thus $\lambda^*(A_\epsilon) \leq (\bar{S} - \underline{S})/\epsilon \rightarrow 0$ as partition refines. So $\lambda(A_\epsilon) = 0$ for all $\epsilon > 0$, hence $\lambda(D_f) = 0$.

($\Leftarrow$): Assume $\lambda(D_f) = 0$. Given $\epsilon > 0$, cover D_f by intervals of total length $< \epsilon$. On the compact set outside this cover, f is continuous hence uniformly continuous.

Choose partition fine enough that oscillation on each subinterval is $< \epsilon$ (except those meeting the cover). Then $\bar{S} - \underline{S} < \epsilon(b-a) + 2M\epsilon$ where $M = \sup |f|$.

Equality of integrals: For Riemann integrable f , the upper/lower Riemann sums are Lebesgue integrals of simple functions that converge a.e. to f . By DCT (with dominating function M), the Lebesgue integral equals the Riemann integral. QED

Chapter Compendium:

Integral Construction:

1. **Simple:** $\int s \, d\mu = \sum_i \alpha_i \mu(A_i)$ for $s = \sum_i \alpha_i \mathbb{1}_{A_i}$
2. **σ -Simple:** Same formula, countable sum
3. **Non-negative measurable:** $\int f = \sup\{\int s \mid s \text{ simple, } s \leq f\}$
4. **Integrable:** $f \in L^1$ if $\int |f| < \infty$; then $\int f = \int f^+ - \int f^-$

Integral Properties (for σ -simple and measurable functions):

- **Linearity:** $\int(\alpha f + \beta g) = \alpha \int f + \beta \int g$
- **Monotonicity:** $f \leq g \implies \int f \leq \int g$
- **Triangle inequality:** $|\int f| \leq \int |f|$
- **Monotonicity in set:** $A \subseteq B \implies \int_A f \leq \int_B f$ (for $f \geq 0$)
- **Countable additivity in set:** $\int_{\bigsqcup_n A_n} f = \sum_n \int_{A_n} f$ (for $f \geq 0$)
- **Zero characterization:** $\int f = 0$ and $f \geq 0 \implies f = 0$ a.e.

Convergence Theorems:

- **MCT:** $0 \leq f_n \nearrow f \implies \int f_n \rightarrow \int f$
- **Fatou:** $f_n \geq 0 \implies \int \liminf f_n \leq \liminf \int f_n$
- **DCT:** $f_n \rightarrow f$ a.e., $|f_n| \leq g \in L^1 \implies \int |f_n - f| \rightarrow 0$ and $\int f_n \rightarrow \int f$

MCT Corollary: $\int \sum_n f_n = \sum_n \int f_n$ for $f_n \geq 0$.

Change of Variables: For measurable $f : X \rightarrow Y$ and $g \geq 0$:

$$\int_Y g \, d(f_*\mu) = \int_X (g \circ f) \, d\mu$$

Riemann vs Lebesgue: A bounded $f : [a, b] \rightarrow \mathbb{R}$ is Riemann integrable iff continuous a.e. If Riemann integrable, then Lebesgue integrable with equal integrals.

CHAPTER 21

Product Measures

21.1 Product σ -Algebras

Intuition:

Given measurable spaces $(X, \mathcal{M})$ and $(Y, \mathcal{N})$, we want to define a natural σ -algebra on $X \times Y$ that:

1. Contains “rectangles” $A \times B$ for measurable $A \subseteq X$ and $B \subseteq Y$
2. Makes the canonical projections $\pi_X : X \times Y \rightarrow X$ and $\pi_Y : X \times Y \rightarrow Y$ measurable
3. Is the **smallest** σ -algebra satisfying these properties

It turns out these conditions are equivalent: the σ -algebra generated by rectangles is precisely the smallest σ -algebra making projections measurable.

Definition 21.1.1 (Canonical Projections):

For measurable spaces $(X, \mathcal{M})$ and $(Y, \mathcal{N})$, the **canonical projections** are:

$$\begin{aligned}\pi_X : X \times Y &\rightarrow X, & \pi_X(x, y) &= x \\ \pi_Y : X \times Y &\rightarrow Y, & \pi_Y(x, y) &= y\end{aligned}$$

Definition 21.1.2 (Product σ -Algebra):

Let $(X, \mathcal{M})$ and $(Y, \mathcal{N})$ be measurable spaces. The **product σ -algebra** is:

$$\mathcal{M} \otimes \mathcal{N} = \sigma(\{A \times B \mid A \in \mathcal{M}, B \in \mathcal{N}\})$$

Theorem 21.1.3 (Characterization via Projections):

The product σ -algebra is the smallest σ -algebra on $X \times Y$ making both canonical projections measurable:

$$\mathcal{M} \otimes \mathcal{N} = \sigma(\pi_X^{-1}(\mathcal{M}) \cup \pi_Y^{-1}(\mathcal{N}))$$

where $\pi_X^{-1}(\mathcal{M}) = \{\pi_X^{-1}(A) \mid A \in \mathcal{M}\}$ and similarly for π_Y .

Proof:

Let $\mathcal{S} = \sigma(\pi_X^{-1}(\mathcal{M}) \cup \pi_Y^{-1}(\mathcal{N}))$ be the smallest σ -algebra making projections measurable.

Step 1: $\mathcal{M} \otimes \mathcal{N} \subseteq \mathcal{S}$.

Every rectangle $A \times B$ with $A \in \mathcal{M}$, $B \in \mathcal{N}$ satisfies:

$$A \times B = \pi_X^{-1}(A) \cap \pi_Y^{-1}(B) \in \mathcal{S}$$

since both $\pi_X^{-1}(A), \pi_Y^{-1}(B) \in \mathcal{S}$ and $\mathcal{S}$ is closed under intersections.

Since $\mathcal{M} \otimes \mathcal{N}$ is the σ -algebra *generated* by rectangles, and $\mathcal{S}$ contains all rectangles and is a σ -algebra, we have $\mathcal{M} \otimes \mathcal{N} \subseteq \mathcal{S}$.

Step 2: $\mathcal{S} \subseteq \mathcal{M} \otimes \mathcal{N}$.

For any $A \in \mathcal{M}$:

$$\pi_X^{-1}(A) = A \times Y \in \mathcal{M} \otimes \mathcal{N}$$

since $A \times Y$ is a rectangle (with $B = Y \in \mathcal{N}$).

Similarly, $\pi_Y^{-1}(B) = X \times B \in \mathcal{M} \otimes \mathcal{N}$ for all $B \in \mathcal{N}$.

Thus $\pi_X^{-1}(\mathcal{M}) \cup \pi_Y^{-1}(\mathcal{N}) \subseteq \mathcal{M} \otimes \mathcal{N}$.

Since $\mathcal{S}$ is the *smallest* σ -algebra containing these sets, and $\mathcal{M} \otimes \mathcal{N}$ is a σ -algebra containing them, we have $\mathcal{S} \subseteq \mathcal{M} \otimes \mathcal{N}$.

Conclusion: $\mathcal{M} \otimes \mathcal{N} = \mathcal{S}$.

QED

Corollary 21.1.4:

The canonical projections π_X and π_Y are $(\mathcal{M} \otimes \mathcal{N})$ -measurable.

Proof:

By the characterization, $\mathcal{M} \otimes \mathcal{N}$ contains $\pi_X^{-1}(A)$ for all $A \in \mathcal{M}$ and $\pi_Y^{-1}(B)$ for all $B \in \mathcal{N}$.

QED

Proposition 21.1.5 (Product Borel σ -Algebra — General Case):

For arbitrary topological spaces X and Y :

$$\mathcal{B}(X) \otimes \mathcal{B}(Y) \subseteq \mathcal{B}(X \times Y)$$

where $X \times Y$ is given the product topology.

Proof:

The canonical projections $\pi_X : X \times Y \rightarrow X$ and $\pi_Y : X \times Y \rightarrow Y$ are continuous with respect to the product topology (by definition of product topology).

Step 1: Projections are Borel measurable.

Since π_X is continuous and $\mathcal{B}(X)$ is generated by open sets: for any open $U \subseteq X$, $\pi_X^{-1}(U)$ is open in $X \times Y$, hence in $\mathcal{B}(X \times Y)$.

Thus $\pi_X^{-1}(A) \in \mathcal{B}(X \times Y)$ for all $A \in \mathcal{B}(X)$. Similarly for π_Y .

Step 2: Rectangles are Borel.

For any $A \in \mathcal{B}(X)$ and $B \in \mathcal{B}(Y)$:

$$A \times B = \pi_X^{-1}(A) \cap \pi_Y^{-1}(B) \in \mathcal{B}(X \times Y)$$

since $\mathcal{B}(X \times Y)$ is closed under intersections.

Step 3: Conclude.

Since $\mathcal{B}(X) \otimes \mathcal{B}(Y)$ is the σ -algebra *generated* by rectangles $\{A \times B\}$, and all rectangles are in $\mathcal{B}(X \times Y)$, we have:

$$\mathcal{B}(X) \otimes \mathcal{B}(Y) \subseteq \mathcal{B}(X \times Y)$$

QED

Proposition 21.1.6 (Product Borel σ -Algebra — Second Countable Case):

Let X and Y be second countable topological spaces. Then:

$$\mathcal{B}(X \times Y) = \mathcal{B}(X) \otimes \mathcal{B}(Y)$$

Proof:

We already have $\mathcal{B}(X) \otimes \mathcal{B}(Y) \subseteq \mathcal{B}(X \times Y)$ from the previous proposition. It remains to show the reverse inclusion.

Step 1: Identify countable bases.

Since X is second countable, there exists a countable base $\mathcal{B}_X = \{U_n\}_{n \in \mathbb{N}}$ for the topology on X . Similarly, Y has a countable base $\mathcal{B}_Y = \{V_m\}_{m \in \mathbb{N}}$.

Step 2: The product topology has a countable base.

The collection $\mathcal{B} = \{U_n \times V_m \mid n, m \in \mathbb{N}\}$ is a countable base for the product topology on $X \times Y$. Indeed:

- $\mathcal{B}$ is countable since $\mathbb{N} \times \mathbb{N}$ is countable.
- For any open set $W \subseteq X \times Y$ and any point $(x, y) \in W$, there exist open sets $U \subseteq X$ and $V \subseteq Y$ with $(x, y) \in U \times V \subseteq W$ (by definition of product topology). Since $\mathcal{B}_X$ and $\mathcal{B}_Y$ are bases, there exist $U_n \ni x$ with $U_n \subseteq U$ and $V_m \ni y$ with $V_m \subseteq V$. Thus $(x, y) \in U_n \times V_m \subseteq W$.

Step 3: Open sets are countable unions of base elements.

Every open set $W \subseteq X \times Y$ can be written as:

$$W = \bigcup_{(n,m) \in I} (U_n \times V_m)$$

for some countable index set $I \subseteq \mathbb{N} \times \mathbb{N}$.

Step 4: Base elements are in $\mathcal{B}(X) \otimes \mathcal{B}(Y)$.

Each U_n is open in X , hence $U_n \in \mathcal{B}(X)$. Similarly, $V_m \in \mathcal{B}(Y)$. Therefore:

$$U_n \times V_m \in \mathcal{B}(X) \otimes \mathcal{B}(Y)$$

since this is precisely a measurable rectangle.

Step 5: Open sets are in $\mathcal{B}(X) \otimes \mathcal{B}(Y)$.

Since every open set W is a countable union of sets in $\mathcal{B}(X) \otimes \mathcal{B}(Y)$, and σ -algebras are closed under countable unions, we have:

$$W \in \mathcal{B}(X) \otimes \mathcal{B}(Y)$$

for every open $W \subseteq X \times Y$.

Step 6: Conclude.

The Borel σ -algebra $\mathcal{B}(X \times Y)$ is generated by open sets in $X \times Y$. Since all open sets are in $\mathcal{B}(X) \otimes \mathcal{B}(Y)$ (which is a σ -algebra), we have:

$$\mathcal{B}(X \times Y) = \sigma(\text{open sets in } X \times Y) \subseteq \mathcal{B}(X) \otimes \mathcal{B}(Y)$$

Combined with the previous proposition: $\mathcal{B}(X \times Y) = \mathcal{B}(X) \otimes \mathcal{B}(Y)$.

QED

Remark:

The reverse inclusion $\mathcal{B}(X \times Y) \subseteq \mathcal{B}(X) \otimes \mathcal{B}(Y)$ may fail for general topological spaces that are not second countable. The key to the proof above is that open sets in the product are *countable* unions of rectangles—this uses second countability essentially.

Proposition 21.1.7:

$$\mathcal{B}(\mathbb{R}^{n+m}) = \mathcal{B}(\mathbb{R}^n) \otimes \mathcal{B}(\mathbb{R}^m).$$

Proof:

Let $\mathcal{R} = \{A \times B \mid A \in \mathcal{B}(\mathbb{R}^n), B \in \mathcal{B}(\mathbb{R}^m)\}$. We show $\sigma(\mathcal{R}) = \mathcal{B}(\mathbb{R}^{n+m})$.

($\subseteq$) The projections $\pi_1 : \mathbb{R}^{n+m} \rightarrow \mathbb{R}^n$ and $\pi_2 : \mathbb{R}^{n+m} \rightarrow \mathbb{R}^m$ are continuous, hence $(\mathcal{B}(\mathbb{R}^{n+m}))$ -measurable. For $A \in \mathcal{B}(\mathbb{R}^n)$, $B \in \mathcal{B}(\mathbb{R}^m)$: $A \times B = \pi_1^{-1}(A) \cap \pi_2^{-1}(B) \in \mathcal{B}(\mathbb{R}^{n+m})$.

($\supseteq$) Open sets in $\mathbb{R}^{n+m}$ are countable unions of open rectangles. Each open rectangle is in $\sigma(\mathcal{R})$ since open sets are Borel. QED

21.2 Sections of Sets and Functions

Intuition:

To construct the product measure, we need to “slice” sets in $X \times Y$ along one coordinate. For a set $E \subseteq X \times Y$, we can fix an $x \in X$ and ask: “Which y values are paired with x in E ?” This gives the x -section $E_x \subseteq Y$. Similarly, fixing y gives the y -section $E^y \subseteq X$.

The key insight is that if E is measurable in the product σ -algebra, then its sections are measurable in the respective factor σ -algebras. This allows us to integrate over sections to construct the product measure.

Definition 21.2.1 (Sections of Sets):

Let $E \subseteq X \times Y$. For $x \in X$ and $y \in Y$, define:

- The x -section (or **horizontal section**) of E :

$$E_x = \{y \in Y \mid (x, y) \in E\} \subseteq Y$$

- The y -section (or **vertical section**) of E :

$$E^y = \{x \in X \mid (x, y) \in E\} \subseteq X$$

Example 21.2.2:

For a rectangle $A \times B$ with $A \subseteq X$ and $B \subseteq Y$:

$$(A \times B)_x = \begin{cases} B & \text{if } x \in A \\ \emptyset & \text{if } x \notin A \end{cases} \quad (A \times B)^y = \begin{cases} A & \text{if } y \in B \\ \emptyset & \text{if } y \notin B \end{cases}$$

Definition 21.2.3 (Sections of Functions):

Let $f : X \times Y \rightarrow \mathbb{R}$. For $x \in X$ and $y \in Y$, define:

- The x -section of f : $f_x : Y \rightarrow \overline{\mathbb{R}}$, $f_x(y) = f(x, y)$
- The y -section of f : $f^y : X \rightarrow \overline{\mathbb{R}}$, $f^y(x) = f(x, y)$

Remark:

For a set $E \subseteq X \times Y$ and its characteristic function $\mathbb{1}_E$:

$$(\mathbb{1}_E)_x = \mathbb{1}_{E_x} \quad \text{and} \quad (\mathbb{1}_E)^y = \mathbb{1}_{E^y}$$

Proposition 21.2.4 (Properties of Sections):

Let $E, F \subseteq X \times Y$ and $(E_n)_{n \in \mathbb{N}}$ be a sequence of subsets. Then:

1. $(E^c)_x = (E_x)^c$ and $(E^c)^y = (E^y)^c$
2. $(E \cup F)_x = E_x \cup F_x$ and $(E \cup F)^y = E^y \cup F^y$
3. $(E \cap F)_x = E_x \cap F_x$ and $(E \cap F)^y = E^y \cap F^y$
4. $(\bigcup_{n \in \mathbb{N}} E_n)_x = \bigcup_{n \in \mathbb{N}} (E_n)_x$ and similarly for y -sections
5. $(\bigcap_{n \in \mathbb{N}} E_n)_x = \bigcap_{n \in \mathbb{N}} (E_n)_x$ and similarly for y -sections

Proof:

These follow directly from the definitions. We prove (1) and (4), the others being similar.

Property 1:

$$\begin{aligned} (E^c)_x &= \{y \in Y \mid (x, y) \in E^c\} \\ &= \{y \in Y \mid (x, y) \notin E\} \\ &= \{y \in Y \mid y \notin E_x\} \\ &= Y \setminus E_x = (E_x)^c \end{aligned}$$

Property 4:

$$\begin{aligned} \left(\bigcup_{n \in \mathbb{N}} E_n \right)_x &= \left\{ y \in Y \mid (x, y) \in \bigcup_{n \in \mathbb{N}} E_n \right\} \\ &= \{y \in Y \mid \exists n \in \mathbb{N} : (x, y) \in E_n\} \\ &= \{y \in Y \mid \exists n \in \mathbb{N} : y \in (E_n)_x\} \\ &= \bigcup_{n \in \mathbb{N}} (E_n)_x \end{aligned}$$

QED

Theorem 21.2.5 (Measurability of Sections):

Let $(X, \mathcal{M})$ and $(Y, \mathcal{N})$ be measurable spaces. If $E \in \mathcal{M} \otimes \mathcal{N}$, then:

1. $\forall x \in X : E_x \in \mathcal{N}$ (all x -sections are $\mathcal{N}$ -measurable)
2. $\forall y \in Y : E^y \in \mathcal{M}$ (all y -sections are $\mathcal{M}$ -measurable)

Proof:

We prove (1); the proof of (2) is symmetric.

Strategy: We use the σ -algebra induction principle. Define:

$$\mathcal{G} = \{E \in \mathcal{M} \otimes \mathcal{N} \mid \forall x \in X : E_x \in \mathcal{N}\}$$

We show $\mathcal{G}$ is a σ -algebra containing all rectangles, hence $\mathcal{G} = \mathcal{M} \otimes \mathcal{N}$.

Step 1: Rectangles are in $\mathcal{G}$.

Let $E = A \times B$ with $A \in \mathcal{M}$, $B \in \mathcal{N}$. Then:

$$E_x = (A \times B)_x = \begin{cases} B & \text{if } x \in A \\ \emptyset & \text{if } x \notin A \end{cases}$$

Since $B \in \mathcal{N}$ and $\emptyset \in \mathcal{N}$, we have $E_x \in \mathcal{N}$ for all x . Thus $A \times B \in \mathcal{G}$.

Step 2: $\mathcal{G}$ is a σ -algebra.

Contains $\emptyset$: $\emptyset_x = \emptyset \in \mathcal{N}$ for all x , so $\emptyset \in \mathcal{G}$.

Closed under complements: Let $E \in \mathcal{G}$. Then $(E^c)_x = (E_x)^c \in \mathcal{N}$ (since $\mathcal{N}$ is closed under complements). Thus $E^c \in \mathcal{G}$.

Closed under countable unions: Let $(E_n)_{n \in \mathbb{N}} \subseteq \mathcal{G}$. Then:

$$\left(\bigcup_{n \in \mathbb{N}} E_n \right)_x = \bigcup_{n \in \mathbb{N}} (E_n)_x \in \mathcal{N}$$

since each $(E_n)_x \in \mathcal{N}$ and $\mathcal{N}$ is closed under countable unions.

Step 3: Conclude.

Since $\mathcal{G}$ is a σ -algebra containing all rectangles $\{A \times B\}$, and $\mathcal{M} \otimes \mathcal{N}$ is the smallest such σ -algebra:

$$\mathcal{M} \otimes \mathcal{N} \subseteq \mathcal{G}$$

But $\mathcal{G} \subseteq \mathcal{M} \otimes \mathcal{N}$ by definition. Hence $\mathcal{G} = \mathcal{M} \otimes \mathcal{N}$.

QED

21.3 Product Measures

Intuition:

Given measure spaces $(X, \mathcal{M}, \mu)$ and $(Y, \mathcal{N}, \nu)$, we want to define a measure on the product σ -algebra that assigns measure $\mu(A) \cdot \nu(B)$ to each rectangle $A \times B$.

The construction uses sections: for a measurable set $E \subseteq X \times Y$, we integrate the “slice measures” $\nu(E_x)$ over x . The key technical challenges are:

1. Showing $x \mapsto \nu(E_x)$ is measurable
2. Showing this construction yields a measure
3. Proving uniqueness

We use σ -finiteness to ensure uniqueness via the π - λ theorem.

Lemma 21.3.1 (Measurability of Section Measures):

Let $(X, \mathcal{M}, \mu)$ and $(Y, \mathcal{N}, \nu)$ be σ -finite measure spaces. For $E \in \mathcal{M} \otimes \mathcal{N}$:

1. The function $\varphi : X \rightarrow [0, \infty]$, $\varphi(x) = \nu(E_x)$ is $\mathcal{M}$ -measurable
2. The function $\psi : Y \rightarrow [0, \infty]$, $\psi(y) = \mu(E^y)$ is $\mathcal{N}$ -measurable

Proof:

We prove (1); the proof of (2) is symmetric. We proceed in two stages: first for finite measures, then extend to σ -finite.

Case 1: ν is a finite measure.

Define:

$$\mathcal{D} = \{E \in \mathcal{M} \otimes \mathcal{N} \mid x \mapsto \nu(E_x) \text{ is } \mathcal{M}\text{-measurable}\}$$

We show $\mathcal{D}$ is a λ -system containing all rectangles.

$\mathcal{D}$ contains rectangles:

For $E = A \times B$ with $A \in \mathcal{M}$, $B \in \mathcal{N}$:

$$\nu(E_x) = \nu((A \times B)_x) = \begin{cases} \nu(B) & \text{if } x \in A \\ 0 & \text{if } x \notin A \end{cases} = \nu(B) \cdot \mathbb{1}_A(x)$$

This is $\mathcal{M}$ -measurable since $\mathbb{1}_A$ is measurable.

$\mathcal{D}$ is a λ -system:

Axiom 1: $(X \times Y)_x = Y$, so $\nu((X \times Y)_x) = \nu(Y) < \infty$ (constant function). Thus $X \times Y \in \mathcal{D}$.

Axiom 2: Let $E, F \in \mathcal{D}$ with $E \subseteq F$. Then:

$$\nu((F \setminus E)_x) = \nu(F_x \setminus E_x) = \nu(F_x) - \nu(E_x)$$

(using that $E_x \subseteq F_x$ and ν is finite). The right-hand side is the difference of two measurable functions, hence measurable. Thus $F \setminus E \in \mathcal{D}$.

Axiom 3: Let $(E_n)_{n \in \mathbb{N}} \subseteq \mathcal{D}$ with $E_n \nearrow E$. Then $(E_n)_x \nearrow E_x$ for each x . By continuity from below:

$$\nu(E_x) = \lim_{n \rightarrow \infty} \nu((E_n)_x)$$

The right-hand side is the pointwise limit of measurable functions, hence measurable. Thus $E \in \mathcal{D}$.

Apply Dynkin π - λ theorem:

The rectangles $\mathcal{R} = \{A \times B \mid A \in \mathcal{M}, B \in \mathcal{N}\}$ form a π -system (since $(A_1 \times B_1) \cap (A_2 \times B_2) = (A_1 \cap A_2) \times (B_1 \cap B_2)$).

We have shown $\mathcal{R} \subseteq \mathcal{D}$ and $\mathcal{D}$ is a λ -system. By the π - λ theorem:

$$\mathcal{M} \otimes \mathcal{N} = \sigma(\mathcal{R}) = \lambda(\mathcal{R}) \subseteq \mathcal{D}$$

Hence $\mathcal{D} = \mathcal{M} \otimes \mathcal{N}$.

Case 2: ν is σ -finite.

Write $Y = \bigcup_{n \in \mathbb{N}} Y_n$ with $Y_n \nearrow$ and $\nu(Y_n) < \infty$.

For each n , the restricted measure $\nu_n = \nu|_{Y_n}$ is finite. Define $E_n = E \cap (X \times Y_n)$. By Case 1, $x \mapsto \nu((E_n)_x)$ is measurable.

Note that $(E_n)_x = E_x \cap Y_n \nearrow E_x$ as $n \rightarrow \infty$. By continuity from below:

$$\nu(E_x) = \lim_{n \rightarrow \infty} \nu(E_x \cap Y_n) = \lim_{n \rightarrow \infty} \nu((E_n)_x)$$

This is the limit of measurable functions, hence measurable.

QED

Theorem 21.3.2 (Construction of Product Measure via x -Sections):

Let $(X, \mathcal{M}, \mu)$ and $(Y, \mathcal{N}, \nu)$ be σ -finite measure spaces. Define $\rho_x : \mathcal{M} \otimes \mathcal{N} \rightarrow [0, \infty]$ by:

$$\rho_x(E) = \int_X \nu(E_x) d\mu(x)$$

Then ρ_x is a measure on $\mathcal{M} \otimes \mathcal{N}$ satisfying:

$$\rho_x(A \times B) = \mu(A) \cdot \nu(B)$$

for all $A \in \mathcal{M}$, $B \in \mathcal{N}$.

Proof:

We verify the measure axioms for ρ_x .

Step 1: $\rho_x(\emptyset) = 0$.

We have $\emptyset_x = \emptyset$ for all x , so $\nu(\emptyset_x) = 0$. Thus:

$$\rho_x(\emptyset) = \int_X 0 d\mu(x) = 0$$

Step 2: σ -additivity.

Let $(E_n)_{n \in \mathbb{N}}$ be pairwise disjoint sets in $\mathcal{M} \otimes \mathcal{N}$. Let $E = \bigsqcup_{n \in \mathbb{N}} E_n$.

For each $x \in X$, the sections $(E_n)_x$ are pairwise disjoint in Y (since if $y \in (E_i)_x \cap (E_j)_x$ with $i \neq j$, then $(x, y) \in E_i \cap E_j = \emptyset$, contradiction).

Moreover, $E_x = \bigsqcup_{n \in \mathbb{N}} (E_n)_x$. By σ -additivity of ν :

$$\nu(E_x) = \sum_{n \in \mathbb{N}} \nu((E_n)_x)$$

Now by the Monotone Convergence Theorem (applied to the partial sums):

$$\begin{aligned} \rho_x(E) &= \int_X \nu(E_x) d\mu(x) \\ &= \int_X \sum_{n \in \mathbb{N}} \nu((E_n)_x) d\mu(x) \\ &= \sum_{n \in \mathbb{N}} \int_X \nu((E_n)_x) d\mu(x) \quad (\text{by MCT}) \\ &= \sum_{n \in \mathbb{N}} \rho_x(E_n) \end{aligned}$$

Step 3: Value on rectangles.

For $E = A \times B$ with $A \in \mathcal{M}$, $B \in \mathcal{N}$:

$$\nu(E_x) = \nu((A \times B)_x) = \nu(B) \cdot \mathbb{1}_A(x)$$

Thus:

$$\rho_x(A \times B) = \int_X \nu(B) \cdot \mathbb{1}_A(x) d\mu(x) = \nu(B) \cdot \mu(A)$$

QED

Theorem 21.3.3 (Construction of Product Measure via y -Sections):

Let $(X, \mathcal{M}, \mu)$ and $(Y, \mathcal{N}, \nu)$ be σ -finite measure spaces. Define $\rho_y : \mathcal{M} \otimes \mathcal{N} \rightarrow [0, \infty]$ by:

$$\rho_y(E) = \int_Y \mu(E^y) d\nu(y)$$

Then ρ_y is a measure on $\mathcal{M} \otimes \mathcal{N}$ satisfying:

$$\rho_y(A \times B) = \mu(A) \cdot \nu(B)$$

Proof:

Symmetric to the proof for ρ_x .

QED

Theorem 21.3.4 (Uniqueness of Product Measure):

Let $(X, \mathcal{M}, \mu)$ and $(Y, \mathcal{N}, \nu)$ be σ -finite measure spaces. Then $\rho_x = \rho_y$. Moreover, there exists a **unique** measure $\mu \otimes \nu$ on $\mathcal{M} \otimes \mathcal{N}$ such that:

$$(\mu \otimes \nu)(A \times B) = \mu(A) \cdot \nu(B)$$

for all $A \in \mathcal{M}$, $B \in \mathcal{N}$.

Proof:

The rectangles $\mathcal{R} = \{A \times B \mid A \in \mathcal{M}, B \in \mathcal{N}\}$ form a π -system generating $\mathcal{M} \otimes \mathcal{N}$:

$$(A_1 \times B_1) \cap (A_2 \times B_2) = (A_1 \cap A_2) \times (B_1 \cap B_2) \in \mathcal{R}$$

Both ρ_x and ρ_y are σ -finite measures on $\mathcal{M} \otimes \mathcal{N}$ (since (X, μ) and (Y, ν) are σ -finite) that agree on $\mathcal{R}$:

$$\rho_x(A \times B) = \mu(A) \cdot \nu(B) = \rho_y(A \times B)$$

By the Uniqueness Theorem for σ -finite measures agreeing on a π -system: $\rho_x = \rho_y$ on all of $\mathcal{M} \otimes \mathcal{N}$.

The same argument shows any measure satisfying the rectangle property must equal $\rho_x = \rho_y$.

QED

Definition 21.3.5 (Product Measure):

The **product measure** $\mu \otimes \nu$ on $(X \times Y, \mathcal{M} \otimes \mathcal{N})$ is the unique measure satisfying:

$$(\mu \otimes \nu)(A \times B) = \mu(A) \cdot \nu(B)$$

Equivalently:

$$(\mu \otimes \nu)(E) = \int_X \nu(E_x) d\mu(x) = \int_Y \mu(E^y) d\nu(y)$$

21.4 Tonelli's Theorem

Intuition:

Tonelli's Theorem allows us to evaluate integrals over product spaces by iterating one-dimensional integrals. The key insight is that for **non-negative** measurable functions, no integrability assumptions are needed—the iterated integrals always equal the product integral (though all three may be infinite).

The theorem proceeds by:

1. Verifying for characteristic functions of measurable sets

2. Extending to simple functions by linearity
3. Extending to general non-negative functions via monotone approximation

The non-negativity is essential: it allows us to apply MCT freely without worrying about $\infty - \infty$ issues.

Theorem 21.4.1 (Tonelli):

Let $(X, \mathcal{M}, \mu)$ and $(Y, \mathcal{N}, \nu)$ be σ -finite measure spaces. Let $f : X \times Y \rightarrow [0, \infty]$ be $(\mathcal{M} \otimes \mathcal{N})$ -measurable. Then:

1. For each $x \in X$, the x -section $f_x : Y \rightarrow [0, \infty]$ is $\mathcal{N}$ -measurable
2. For each $y \in Y$, the y -section $f^y : X \rightarrow [0, \infty]$ is $\mathcal{M}$ -measurable
3. The function $x \mapsto \int_Y f(x, y) d\nu(y)$ is $\mathcal{M}$ -measurable
4. The function $y \mapsto \int_X f(x, y) d\mu(x)$ is $\mathcal{N}$ -measurable
5. The iterated integrals equal the product integral:

$$\int_{X \times Y} f d(\mu \otimes \nu) = \int_X \left(\int_Y f(x, y) d\nu(y) \right) d\mu(x) = \int_Y \left(\int_X f(x, y) d\mu(x) \right) d\nu(y)$$

Proof:

We prove the theorem in four steps, building from simple cases to the general result.

Step 1: Measurability of sections.

Claim: For any $(\mathcal{M} \otimes \mathcal{N})$ -measurable function $f : X \times Y \rightarrow [0, \infty]$, each section f_x is $\mathcal{N}$ -measurable and each section f^y is $\mathcal{M}$ -measurable.

Proof of claim: For any $a \in [0, \infty]$:

$$f_x^{-1}([0, a]) = \{y \in Y \mid f(x, y) \leq a\} = \{y \in Y \mid (x, y) \in f^{-1}([0, a])\}$$

Since f is $(\mathcal{M} \otimes \mathcal{N})$ -measurable, $f^{-1}([0, a]) \in \mathcal{M} \otimes \mathcal{N}$. By the Measurability of Sections theorem, $(f^{-1}([0, a]))_x \in \mathcal{N}$. But $(f^{-1}([0, a]))_x = f_x^{-1}([0, a])$, so f_x is $\mathcal{N}$ -measurable.

The argument for f^y is symmetric.

Step 2: Characteristic functions of measurable sets.

Let $E \in \mathcal{M} \otimes \mathcal{N}$ and $f = \mathbb{1}_E$.

Computing the inner integral: For each $x \in X$:

$$\int_Y \mathbb{1}_E(x, y) d\nu(y) = \int_Y \mathbb{1}_{E_x}(y) d\nu(y) = \nu(E_x)$$

where $E_x = \{y \in Y \mid (x, y) \in E\}$ is the x -section of E .

Measurability of the inner integral: By the Measurability of Section Measures lemma, $x \mapsto \nu(E_x)$ is $\mathcal{M}$ -measurable.

Computing the iterated integral:

$$\int_X \left(\int_Y \mathbb{1}_E(x, y) d\nu(y) \right) d\mu(x) = \int_X \nu(E_x) d\mu(x) = (\mu \otimes \nu)(E)$$

by definition of the product measure.

Product integral:

$$\int_{X \times Y} \mathbb{1}_E d(\mu \otimes \nu) = (\mu \otimes \nu)(E)$$

Thus the iterated integral equals the product integral for characteristic functions.

By symmetry, the same holds integrating first over X :

$$\int_Y \left(\int_X \mathbb{1}_E(x, y) d\mu(x) \right) d\nu(y) = \int_Y \mu(E^y) d\nu(y) = (\mu \otimes \nu)(E)$$

Step 3: Simple functions.

Let $s = \sum_{i=1}^n c_i \mathbb{1}_{E_i}$ be a simple function in canonical form, where $c_i \geq 0$ and $E_i \in \mathcal{M} \otimes \mathcal{N}$ are pairwise disjoint.

Measurability: By Step 1, each section s_x is $\mathcal{N}$ -measurable.

Inner integral: By linearity of the integral:

$$\int_Y s(x, y) d\nu(y) = \sum_{i=1}^n c_i \int_Y \mathbb{1}_{E_i}(x, y) d\nu(y) = \sum_{i=1}^n c_i \nu((E_i)_x)$$

This is a finite sum of $\mathcal{M}$ -measurable functions (by Step 2), hence $\mathcal{M}$ -measurable.

Iterated integral: By linearity of the integral (justified since $c_i \geq 0$):

$$\begin{aligned} \int_X \left(\int_Y s d\nu \right) d\mu &= \int_X \sum_{i=1}^n c_i \nu((E_i)_x) d\mu(x) \\ &= \sum_{i=1}^n c_i \int_X \nu((E_i)_x) d\mu(x) \\ &= \sum_{i=1}^n c_i (\mu \otimes \nu)(E_i) \quad (\text{by Step 2}) \\ &= \int_{X \times Y} s d(\mu \otimes \nu) \end{aligned}$$

By symmetry, the same holds integrating first over X .

Step 4: General non-negative measurable functions.

Let $f : X \times Y \rightarrow [0, \infty]$ be $(\mathcal{M} \otimes \mathcal{N})$ -measurable.

By the Simple Function Approximation Theorem, there exists a sequence $(s_n)_{n \in \mathbb{N}}$ of simple functions with $0 \leq s_n \nearrow f$ pointwise.

Measurability of the inner integral: For each x , we have $0 \leq (s_n)_x \nearrow f_x$ pointwise on Y . By MCT:

$$\int_Y f_x d\nu = \lim_{n \rightarrow \infty} \int_Y (s_n)_x d\nu$$

Define $\varphi_n(x) = \int_Y (s_n)_x d\nu$ and $\varphi(x) = \int_Y f_x d\nu$.

Each φ_n is $\mathcal{M}$ -measurable (by Step 3). Since $s_n \nearrow f$, we have $\varphi_n(x) \nearrow \varphi(x)$ for each x (by MCT). Thus $\varphi = \lim_n \varphi_n$ is $\mathcal{M}$ -measurable (as the pointwise limit of measurable functions).

Equality of integrals: By Step 3, for each n :

$$\int_{X \times Y} s_n d(\mu \otimes \nu) = \int_X \varphi_n d\mu$$

Taking limits as $n \rightarrow \infty$:

- **Left side:** By MCT on $(X \times Y, \mu \otimes \nu)$:

$$\lim_{n \rightarrow \infty} \int_{X \times Y} s_n d(\mu \otimes \nu) = \int_{X \times Y} f d(\mu \otimes \nu)$$

- **Right side:** Since $\varphi_n \nearrow \varphi$, by MCT on (X, μ) :

$$\lim_{n \rightarrow \infty} \int_X \varphi_n d\mu = \int_X \varphi d\mu = \int_X \left(\int_Y f d\nu \right) d\mu$$

Therefore:

$$\int_{X \times Y} f d(\mu \otimes \nu) = \int_X \left(\int_Y f(x, y) d\nu(y) \right) d\mu(x)$$

By symmetry (exchanging roles of X and Y):

$$\int_{X \times Y} f d(\mu \otimes \nu) = \int_Y \left(\int_X f(x, y) d\mu(x) \right) d\nu(y)$$

This completes the proof.

QED

Corollary 21.4.2:

If one of the iterated integrals of a non-negative measurable function is finite, then all three integrals (two iterated, one product) are equal and finite.

Remark:

Tonelli's Theorem does **not** require integrability. The equality holds even when all integrals are infinite. However, when applying the theorem to functions that may change sign, we must first apply it to $|f|$ to verify integrability before using Fubini.

21.5 Fubini's Theorem

Intuition:

Fubini's Theorem extends Tonelli to **integrable** functions that may take both positive and negative values. The key difference is that we need the integrability condition $f \in L^1(\mu \otimes \nu)$ to ensure the iterated integrals are well-defined (avoiding $\infty - \infty$).

The strategy is:

1. Apply Tonelli to $|f|$ to establish that sections are integrable a.e.
2. Decompose $f = f^+ - f^-$ and apply Tonelli to each part
3. Subtract, using that both parts have finite integrals

Theorem 21.5.1 (Fubini):

Let $(X, \mathcal{M}, \mu)$ and $(Y, \mathcal{N}, \nu)$ be σ -finite measure spaces. Let $f : X \times Y \rightarrow \overline{\mathbb{R}}$ be $(\mathcal{M} \otimes \mathcal{N})$ -measurable with $f \in L^1(\mu \otimes \nu)$. Then:

1. There exists a μ -null set $N \subseteq X$ such that for all $x \in X \setminus N$: the section $f_x \in L^1(\nu)$

2. There exists a ν -null set $M \subseteq Y$ such that for all $y \in Y \setminus M$: the section $f^y \in L^1(\mu)$

3. The function

$$h(x) = \begin{cases} \int_Y f(x, y) d\nu(y) & \text{if } x \in X \setminus N \\ 0 & \text{if } x \in N \end{cases}$$

is in $L^1(\mu)$ (and the choice of value on N does not affect the integral)

4. Similarly for $y \mapsto \int_X f(x, y) d\mu(x)$

5. The iterated integrals equal the product integral:

$$\int_{X \times Y} f d(\mu \otimes \nu) = \int_{X \setminus N} \left(\int_Y f(x, y) d\nu(y) \right) d\mu(x) = \int_{Y \setminus M} \left(\int_X f(x, y) d\mu(x) \right) d\nu(y)$$

Remark (Completeness Issue):

The product of complete measure spaces need not be complete. Even if $(X, \mathcal{M}, \mu)$ and $(Y, \mathcal{N}, \nu)$ are complete, the product measure space $(X \times Y, \mathcal{M} \otimes \mathcal{N}, \mu \otimes \nu)$ may fail to be complete.

This matters because the section f_x may fail to be $\mathcal{N}$ -measurable for x in a μ -null set N —even though f is $(\mathcal{M} \otimes \mathcal{N})$ -measurable. The issue is that if $N \times Y$ is a null set in the product, its “slices” $\{x\} \times Y$ for $x \in N$ may contain non-measurable subsets of Y (since the product σ -algebra does not include all subsets of null sets unless completed).

This is why we must state the theorem with explicit null sets N and M , and integrate over $X \setminus N$ and $Y \setminus M$ respectively.

Proof:

We proceed in four steps, carefully handling the exceptional null sets.

Step 1: Apply Tonelli to $|f|$.

Since $|f| : X \times Y \rightarrow [0, \infty]$ is non-negative and $(\mathcal{M} \otimes \mathcal{N})$ -measurable, Tonelli’s Theorem applies:

$$\int_{X \times Y} |f| d(\mu \otimes \nu) = \int_X \left(\int_Y |f(x, y)| d\nu(y) \right) d\mu(x)$$

By hypothesis, $f \in L^1(\mu \otimes \nu)$, so:

$$\int_{X \times Y} |f| d(\mu \otimes \nu) < \infty$$

Step 2: Identify the exceptional null set and establish a.e. integrability.

Define $g : X \rightarrow [0, \infty]$ by:

$$g(x) = \int_Y |f(x, y)| d\nu(y) = \int_Y |f_x| d\nu$$

By Tonelli, g is $\mathcal{M}$ -measurable and:

$$\int_X g d\mu = \int_{X \times Y} |f| d(\mu \otimes \nu) < \infty$$

Define the exceptional null set:

$$N = \{x \in X \mid g(x) = \infty\} = \left\{x \in X \mid \int_Y |f_x| d\nu = \infty\right\}$$

Claim: $\mu(N) = 0$.

Proof: If $\mu(N) > 0$, then:

$$\int_X g \, d\mu \geq \int_N g \, d\mu = \int_N \infty \, d\mu = \infty \cdot \mu(N) = \infty$$

contradicting $\int_X g \, d\mu < \infty$.

Therefore, for all $x \in X \setminus N$:

$$\int_Y |f(x, y)| \, d\nu(y) < \infty$$

which means $f_x \in L^1(\nu)$ for all $x \in X \setminus N$.

Important: We claim integrability for **all** $x \in X \setminus N$, not merely “ μ -a.e. x ”. The distinction matters because the inner integral is well-defined precisely on $X \setminus N$.

By symmetry, there exists a ν -null set $M \subseteq Y$ such that $f^y \in L^1(\mu)$ for all $y \in Y \setminus M$.

Step 3: Define the inner integral function on $X \setminus N$.

For $x \in X \setminus N$, define:

$$h(x) = \int_Y f(x, y) \, d\nu(y)$$

This is well-defined since $f_x \in L^1(\nu)$.

Extend h to all of X by setting $h(x) = 0$ for $x \in N$. (The specific value on N is irrelevant since $\mu(N) = 0$.)

h is $\mathcal{M}$ -measurable:

Write $f = f^+ - f^-$ where $f^+ = \max(f, 0) \geq 0$ and $f^- = \max(-f, 0) \geq 0$.

Define:

$$h^+(x) = \int_Y f^+(x, y) \, d\nu(y), \quad h^-(x) = \int_Y f^-(x, y) \, d\nu(y)$$

By Tonelli, h^+ and h^- are $\mathcal{M}$ -measurable (as functions $X \rightarrow [0, \infty]$).

For $x \in X \setminus N$: $h(x) = h^+(x) - h^-(x)$, where both terms are finite.

For $x \in N$: $h(x) = 0$.

Since $N \in \mathcal{M}$ (as $N = g^{-1}(\{\infty\})$ and g is $\mathcal{M}$ -measurable), the function h is $\mathcal{M}$ -measurable.

$h \in L^1(\mu)$:

For $x \in X \setminus N$:

$$|h(x)| = \left| \int_Y f_x \, d\nu \right| \leq \int_Y |f_x| \, d\nu = g(x)$$

For $x \in N$: $|h(x)| = 0 \leq g(x)$ (trivially, though $g(x) = \infty$ on N).

Thus $|h| \leq g$ on all of X , and:

$$\int_X |h| \, d\mu \leq \int_X g \, d\mu < \infty$$

so $h \in L^1(\mu)$.

Step 4: Compute the iterated integral using positive and negative parts.

Since $f = f^+ - f^-$ with $f^+, f^- \geq 0$, by Tonelli:

$$\int_{X \times Y} f^+ \, d(\mu \otimes \nu) = \int_X h^+(x) \, d\mu(x)$$

$$\int_{X \times Y} f^- \, d(\mu \otimes \nu) = \int_X h^-(x) \, d\mu(x)$$

Both sides are finite:

$$\int_{X \times Y} f^+ d(\mu \otimes \nu) \leq \int_{X \times Y} |f| d(\mu \otimes \nu) < \infty$$

and similarly for f^- .

Subtracting: Since both integrals are finite, we can subtract:

$$\begin{aligned} \int_{X \times Y} f d(\mu \otimes \nu) &= \int_{X \times Y} f^+ d(\mu \otimes \nu) - \int_{X \times Y} f^- d(\mu \otimes \nu) \\ &= \int_X h^+(x) d\mu - \int_X h^-(x) d\mu \\ &= \int_X (h^+(x) - h^-(x)) d\mu \quad (\text{since both are in } L^1(\mu)) \end{aligned}$$

Now, on $X \setminus N$: $h^+(x) - h^-(x) = h(x) = \int_Y f_x d\nu$.

On N : $\mu(N) = 0$, so the integrand's value there is irrelevant.

Therefore:

$$\int_{X \times Y} f d(\mu \otimes \nu) = \int_X h(x) d\mu = \int_{X \setminus N} \left(\int_Y f(x, y) d\nu(y) \right) d\mu(x)$$

(The last equality uses that $h = 0$ on N and $\mu(N) = 0$, so integrating over X or $X \setminus N$ gives the same result.)

By symmetry:

$$\int_{X \times Y} f d(\mu \otimes \nu) = \int_{Y \setminus M} \left(\int_X f(x, y) d\mu(x) \right) d\nu(y)$$

This completes the proof.

QED

Corollary 21.5.2:

If f is $(\mathcal{M} \otimes \mathcal{N})$ -measurable and one of the iterated integrals $\int_X (\int_Y |f| d\nu) d\mu$ or $\int_Y (\int_X |f| d\mu) d\nu$ is finite, then $f \in L^1(\mu \otimes \nu)$ and Fubini applies.

Proof:

By Tonelli, the iterated integral of $|f|$ equals $\int_{X \times Y} |f| d(\mu \otimes \nu)$. If this is finite, then $f \in L^1(\mu \otimes \nu)$. QED

Remark (When Fubini Fails Without Integrability):

The integrability condition in Fubini is essential. Consider $X = Y = [0, 1]$ with Lebesgue measure and $f(x, y) = \frac{x^2 - y^2}{(x^2 + y^2)^2}$ for $(x, y) \neq (0, 0)$. Then the two iterated integrals exist but are **not equal**:

$$\int_{[0,1]} \left(\int_{[0,1]} f(x, y) d\lambda(y) \right) d\lambda(x) = \frac{\pi}{4}, \quad \int_{[0,1]} \left(\int_{[0,1]} f(x, y) d\lambda(x) \right) d\lambda(y) = -\frac{\pi}{4}$$

This happens because $f \notin L^1(\lambda \otimes \lambda)$.

21.6 Consequences of Fubini-Tonelli

Intuition:

Fubini-Tonelli is a powerful tool that enables us to derive several classical results by viewing one-dimensional problems as sections of two-dimensional product spaces. Two beautiful applications are:

- **Young's Inequality:** Derived by computing areas in the plane
- **Layer Cake Formula:** Expresses integrals via the distribution function

21.6.1 Young's Inequality

Theorem 21.6.1 (Young's Inequality — Integral Form):

Let $\varphi : [0, \infty) \rightarrow [0, \infty)$ be continuous, strictly increasing, with $\varphi(0) = 0$ and $\lim_{t \rightarrow \infty} \varphi(t) = \infty$. Let $\psi = \varphi^{-1}$ be its inverse. Then:

$$ab \leq \int_{[0,a]} \varphi(t) d\lambda(t) + \int_{[0,b]} \psi(s) d\lambda(s)$$

with equality iff $b = \varphi(a)$.

Proof:

Step 1: Set up the regions.

Consider the rectangle $R = [0, a] \times [0, b] \subseteq \mathbb{R}^2$. Define:

$$E_1 = \{(t, s) \in \mathbb{R}^2 \mid 0 \leq t \leq a, 0 \leq s \leq \varphi(t)\}$$

$$E_2 = \{(t, s) \in \mathbb{R}^2 \mid 0 \leq s \leq b, 0 \leq t \leq \psi(s)\}$$

Step 2: Show $R \subseteq E_1 \cup E_2$.

Let $(t, s) \in R$. If $s \leq \varphi(t)$, then $(t, s) \in E_1$. If $s > \varphi(t)$, then $t < \psi(s)$ (since φ is strictly increasing), so $(t, s) \in E_2$.

Step 3: Compute the measures using Tonelli.

By Tonelli's theorem:

$$\lambda^2(E_1) = \int_{[0,a]} \varphi(t) d\lambda(t), \quad \lambda^2(E_2) = \int_{[0,b]} \psi(s) d\lambda(s)$$

Step 4: Conclude.

Since $R \subseteq E_1 \cup E_2$:

$$ab = \lambda^2(R) \leq \lambda^2(E_1 \cup E_2) \leq \lambda^2(E_1) + \lambda^2(E_2) = \int_{[0,a]} \varphi(t) d\lambda(t) + \int_{[0,b]} \psi(s) d\lambda(s)$$

Step 5: Equality case.

Equality holds iff $R = E_1 \cup E_2$ and $E_1 \cap E_2$ has measure zero. This occurs iff $b = \varphi(a)$, since then the point (a, b) lies on the graph of φ , and E_1, E_2 partition R (up to the graph, which has measure zero). QED

Corollary 21.6.2 (Young's Inequality for p -powers):

Let $1 < p, q < \infty$ with $\frac{1}{p} + \frac{1}{q} = 1$. Then for all $a, b \geq 0$:

$$ab \leq \frac{a^p}{p} + \frac{b^q}{q}$$

with equality iff $a^p = b^q$.

Proof:

Apply the integral form with $\varphi(t) = t^{p-1}$, so $\psi(s) = s^{q-1}$ (since $(p-1)(q-1) = 1$ follows from $\frac{1}{p} + \frac{1}{q} = 1$).

Compute $\int_{[0,a]} t^{p-1} d\lambda(t) = \frac{a^p}{p}$ and $\int_{[0,b]} s^{q-1} d\lambda(s) = \frac{b^q}{q}$.

Equality holds iff $b = \varphi(a) = a^{p-1}$, i.e., $b^q = a^{(p-1)q} = a^p$. QED

21.6.2 Layer Cake Formula

Theorem 21.6.3 (Layer Cake Representation):

Let $(X, \mathcal{M}, \mu)$ be a measure space and $f : X \rightarrow [0, \infty]$ be measurable. Then:

$$\int_X f \, d\mu = \int_{[0, \infty)} \mu(\{f > t\}) \, d\lambda(t)$$

where λ is Lebesgue measure on $[0, \infty)$.

Intuition:

The layer cake formula says we can compute the integral of a non-negative function by “stacking horizontal slices”. For each height $t \geq 0$, we measure the set $\{f > t\}$ (where the function exceeds t), and integrate these measures.

Geometrically, if we think of f as defining a “cake” with height $f(x)$ at each point x , then instead of integrating vertically (summing $f(x) \cdot d\mu(x)$), we integrate horizontally (summing $\mu(\{f > t\}) \cdot dt$).

This is Cavalieri’s principle for measure spaces!

Proof:

We apply Tonelli’s theorem on the product space $X \times [0, \infty)$ with measures μ and λ .

Step 1: Define the subgraph.

Let $E = \{(x, t) \in X \times [0, \infty) \mid 0 \leq t < f(x)\}$ be the “subgraph” of f .

Step 2: Show E is measurable.

Define $g : X \times [0, \infty) \rightarrow \mathbb{R}$ by $g(x, t) = f(x) - t$. Since f is measurable and projection onto the second coordinate is measurable:

$$E = g^{-1}((0, \infty)) = \{(x, t) \mid f(x) > t\} \in \mathcal{M} \otimes \mathcal{B}([0, \infty))$$

Step 3: Compute $(\mu \otimes \lambda)(E)$ via x -sections (Tonelli).

For fixed $x \in X$, the x -section is:

$$E_x = \{t \in [0, \infty) \mid t < f(x)\} = [0, f(x))$$

Thus $\lambda(E_x) = f(x)$ (with convention $\lambda([0, \infty)) = \infty$).

By Tonelli:

$$(\mu \otimes \lambda)(E) = \int_X \lambda(E_x) \, d\mu(x) = \int_X f(x) \, d\mu(x)$$

Step 4: Compute $(\mu \otimes \lambda)(E)$ via t -sections (Tonelli).

For fixed $t \in [0, \infty)$, the t -section is:

$$E^t = \{x \in X \mid f(x) > t\} = \{f > t\}$$

Thus $\mu(E^t) = \mu(\{f > t\})$.

By Tonelli:

$$(\mu \otimes \lambda)(E) = \int_{[0, \infty)} \mu(E^t) \, d\lambda(t) = \int_{[0, \infty)} \mu(\{f > t\}) \, d\lambda(t)$$

Step 5: Conclude.

Both expressions equal $(\mu \otimes \lambda)(E)$, so:

$$\int_X f \, d\mu = \int_{[0, \infty)} \mu(\{f > t\}) \, d\lambda(t)$$

QED

Remark:

The formula also holds with $\{f \geq t\}$ instead of $\{f > t\}$:

$$\int_X f \, d\mu = \int_{[0, \infty)} \mu(\{f \geq t\}) \, d\lambda(t)$$

This is because $\{f > t\}$ and $\{f \geq t\}$ differ only on $\{f = t\}$, which forms a family of disjoint sets as t varies, hence has λ -measure zero in the integral.

Corollary 21.6.4 (Power Formula):

For $f \geq 0$ measurable and $p > 0$:

$$\int_X f^p \, d\mu = p \int_{[0, \infty)} t^{p-1} \mu(\{f > t\}) \, d\lambda(t)$$

Proof:

Apply the layer cake formula to f^p :

$$\int_X f^p \, d\mu = \int_{[0, \infty)} \mu(\{f^p > s\}) \, d\lambda(s)$$

Substitute $s = t^p$, so $ds = pt^{p-1} dt$:

$$= \int_{[0, \infty)} \mu(\{f > t\}) \cdot pt^{p-1} \, d\lambda(t) = p \int_{[0, \infty)} t^{p-1} \mu(\{f > t\}) \, d\lambda(t)$$

QED

Chapter Compendium:

Product σ -Algebra: For measurable spaces $(X, \mathcal{M})$ and $(Y, \mathcal{N})$:

$$\mathcal{M} \otimes \mathcal{N} = \sigma(\{A \times B \mid A \in \mathcal{M}, B \in \mathcal{N}\})$$

Borel σ -Algebras:

- General: $\mathcal{B}(X) \otimes \mathcal{B}(Y) \subseteq \mathcal{B}(X \times Y)$
- Second countable: $\mathcal{B}(X) \otimes \mathcal{B}(Y) = \mathcal{B}(X \times Y)$

Sections: For $E \subseteq X \times Y$:

- x -section: $E_x = \{y \in Y \mid (x, y) \in E\}$
- y -section: $E^y = \{x \in X \mid (x, y) \in E\}$

If $E \in \mathcal{M} \otimes \mathcal{N}$, then $E_x \in \mathcal{N}$ and $E^y \in \mathcal{M}$.

Product Measure: For σ -finite $(X, \mathcal{M}, \mu)$ and $(Y, \mathcal{N}, \nu)$:

$$(\mu \otimes \nu)(E) = \int_X \nu(E_x) \, d\mu(x) = \int_Y \mu(E^y) \, d\nu(y)$$

Uniquely characterized by $(\mu \otimes \nu)(A \times B) = \mu(A) \cdot \nu(B)$.

Tonelli (Non-negative): For $f \geq 0$ measurable:

$$\int_{X \times Y} f \, d(\mu \otimes \nu) = \int_X \left(\int_Y f(x, y) \, d\nu \right) d\mu = \int_Y \left(\int_X f(x, y) \, d\mu \right) d\nu$$

No integrability assumption needed.

Fubini (Integrable): For $f \in L^1(\mu \otimes \nu)$: same equality holds, with iterated integrals over $X \setminus N$ and $Y \setminus M$ where N, M are null sets.

Completeness Issue: Product of complete spaces need not be complete. Sections may fail to be measurable on null sets, necessitating integration over $X \setminus N$.

Young's Inequality (Fubini Consequence):

$$ab \leq \int_{[0,a]} \varphi(t) d\lambda(t) + \int_{[0,b]} \psi(s) d\lambda(s)$$

where $\psi = \varphi^{-1}$. For p -powers: $ab \leq \frac{a^p}{p} + \frac{b^q}{q}$ with $\frac{1}{p} + \frac{1}{q} = 1$.

Layer Cake Formula: For $f \geq 0$ measurable:

$$\int_X f d\mu = \int_{[0,\infty)} \mu(\{f > t\}) d\lambda(t)$$

Power Formula: For $f \geq 0$ and $p > 0$:

$$\int_X f^p d\mu = p \int_{[0,\infty)} t^{p-1} \mu(\{f > t\}) d\lambda(t)$$

CHAPTER 22

The Radon-Nikodym Theorem

Intuition:

The Radon-Nikodym theorem is one of the central results in measure theory. It characterizes when one measure can be “written in terms of” another via a density function. If ν is absolutely continuous with respect to μ (meaning $\mu(A) = 0 \implies \nu(A) = 0$), then there exists a measurable function f such that:

$$\nu(A) = \int_A f d\mu$$

This function f is called the **Radon-Nikodym derivative** of ν with respect to μ , denoted $\frac{d\nu}{d\mu}$. The notation suggests an analogy with the Fundamental Theorem of Calculus: just as $F(x) = \int_{[a,x]} f(t)d\lambda(t)$ implies $F'(x) = f(x)$, we have $\nu(A) = \int_A f d\mu$ with $f = \frac{d\nu}{d\mu}$.

Key applications include:

- **Probability:** Conditional expectation, likelihood ratios, change of measure
- **Analysis:** Characterizing absolutely continuous functions
- **Ergodic theory:** Invariant measures and entropy

22.1 Signed Measures

Definition 22.1.1 (Signed Measure):

Let $(X, \mathcal{M})$ be a measurable space. A **signed measure** is a function $\nu : \mathcal{M} \rightarrow [-\infty, +\infty]$ such that:

1. $\nu(\emptyset) = 0$
2. ν takes at most one of the values $+\infty$ or $-\infty$
3. **σ -additivity:** If $(A_n)_{n \in \mathbb{N}}$ are pairwise disjoint, then:

$$\nu \left(\bigsqcup_{n=1}^{\infty} A_n \right) = \sum_{n=1}^{\infty} \nu(A_n)$$

where the series converges absolutely if $|\nu(\bigsqcup A_n)| < \infty$.

Definition 22.1.2 (Positive and Negative Sets):

Let ν be a signed measure on $(X, \mathcal{M})$.

- A set $P \in \mathcal{M}$ is **positive** for ν if $\forall A \in \mathcal{M}, A \subseteq P : \nu(A) \geq 0$
- A set $N \in \mathcal{M}$ is **negative** for ν if $\forall A \in \mathcal{M}, A \subseteq N : \nu(A) \leq 0$

- A set $A \in \mathcal{M}$ is **null** for ν if $\forall B \in \mathcal{M}, B \subseteq A: \nu(B) = 0$

Theorem 22.1.3 (Hahn Decomposition):

Let ν be a signed measure on $(X, \mathcal{M})$. There exist sets $P, N \in \mathcal{M}$ such that:

1. $X = P \sqcup N$ (disjoint union)
2. P is positive for ν
3. N is negative for ν

The decomposition is unique up to null sets: if $X = P' \sqcup N'$ is another Hahn decomposition, then $P \Delta P'$ and $N \Delta N'$ are null for ν .

Proof:

Without loss of generality, assume ν does not take the value $+\infty$ (otherwise consider $-\nu$).

Step 1: Construct the negative set N .

Let $\alpha = \inf\{\nu(A) \mid A \in \mathcal{M}\}$. Note $-\infty \leq \alpha \leq 0$ (since $\nu(\emptyset) = 0$).

Choose a sequence (A_n) with $\nu(A_n) \rightarrow \alpha$. We will refine these to get disjoint sets that capture the “most negative” parts.

Step 2: Extract a negative set from each A_n .

Claim: For any $A \in \mathcal{M}$, there exists a negative set $N_A \subseteq A$ with $\nu(N_A) \leq \nu(A)$.

Proof of claim: If A is negative, take $N_A = A$. Otherwise, there exists $B_1 \subseteq A$ with $\nu(B_1) > 0$. Let $n_1 = \min\{n \in \mathbb{N} \mid \exists B \subseteq A: \nu(B) \geq 1/n\}$ and choose B_1 with $\nu(B_1) \geq 1/n_1$.

Apply inductively to $A \setminus B_1, A \setminus (B_1 \cup B_2), \dots$. The process terminates (else $\sum \nu(B_k) = \infty$ contradicting that $\nu(A)$ is finite) or $A \setminus \bigcup B_k$ is negative. Either way, we get $N_A = A \setminus \bigcup B_k$ with $\nu(N_A) = \nu(A) - \sum \nu(B_k) \leq \nu(A)$.

Step 3: Construct N .

Apply the claim to each A_n to get negative sets $N_n \subseteq A_n$ with $\nu(N_n) \leq \nu(A_n)$. Let:

$$N = \bigcup_{n=1}^{\infty} N_n$$

We show N is negative. Let $B \subseteq N$. Define $B_n = B \cap N_n \setminus \bigcup_{k < n} N_k$. Then (B_n) are disjoint, $B = \bigsqcup B_n$, and each $B_n \subseteq N_n$ (negative), so $\nu(B_n) \leq 0$. Thus:

$$\nu(B) = \sum_{n=1}^{\infty} \nu(B_n) \leq 0$$

Step 4: Show $P = X \setminus N$ is positive.

Suppose not: there exists $A \subseteq P$ with $\nu(A) < 0$. By Step 2, there exists a negative set $N_A \subseteq A$ with $\nu(N_A) \leq \nu(A) < 0$.

Then $N \cup N_A$ is negative (union of negative sets that is disjoint from N_A), and:

$$\nu(N \cup N_A) = \nu(N) + \nu(N_A) < \nu(N) \leq \alpha$$

contradicting the definition of α .

Step 5: Uniqueness.

Let $X = P' \sqcup N'$ be another Hahn decomposition. Then $P \cap N' \subseteq P$ (positive) and $P \cap N' \subseteq N'$ (negative), so $P \cap N'$ is null. Similarly $P' \cap N$ is null. Thus $P \Delta P' = (P \cap N') \cup (P' \cap N)$ is null. QED

Definition 22.1.4 (Jordan Decomposition):

Let ν be a signed measure with Hahn decomposition $X = P \sqcup N$. The **positive part** and **negative part** are:

$$\nu^+(A) = \nu(A \cap P), \quad \nu^-(A) = -\nu(A \cap N)$$

The **total variation** is $|\nu| = \nu^+ + \nu^-$. The **Jordan decomposition** is:

$$\nu = \nu^+ - \nu^-$$

Proposition 22.1.5:

The measures ν^+ , ν^- , and $|\nu|$ are (unsigned) measures, and:

1. $\nu^+ \perp \nu^-$ (mutually singular: P supports ν^+ , N supports ν^-)
2. The Jordan decomposition is the unique decomposition $\nu = \mu_1 - \mu_2$ with $\mu_1 \perp \mu_2$

22.2 Absolute Continuity and Singularity

Definition 22.2.1 (Absolute Continuity):

Let μ and ν be measures (or signed measures) on $(X, \mathcal{M})$. We say ν is **absolutely continuous** with respect to μ , written $\nu \ll \mu$, if:

$$\forall A \in \mathcal{M} : \mu(A) = 0 \implies \nu(A) = 0$$

Definition 22.2.2 (Mutual Singularity):

Measures μ and ν are **mutually singular**, written $\mu \perp \nu$, if there exist disjoint sets $A, B \in \mathcal{M}$ with $X = A \sqcup B$ such that μ is concentrated on A and ν is concentrated on B :

$$\forall E \in \mathcal{M} : \mu(E) = \mu(E \cap A) \text{ and } \nu(E) = \nu(E \cap B)$$

Proposition 22.2.3 (Characterization of Absolute Continuity for Finite Measures):

Let μ be a measure and ν a finite measure on $(X, \mathcal{M})$. Then $\nu \ll \mu$ if and only if:

$$\forall \epsilon > 0 : \exists \delta > 0 : \mu(A) < \delta \implies \nu(A) < \epsilon$$

Proof:

($\Leftarrow$) If $\mu(A) = 0 < \delta$, then $\nu(A) < \epsilon$ for all ϵ , so $\nu(A) = 0$.

($\Rightarrow$) Suppose the ϵ - δ condition fails. Then there exists $\epsilon > 0$ and sets A_n with $\mu(A_n) < 2^{-n}$ but $\nu(A_n) \geq \epsilon$.

Let $B_n = \bigcup_{k \geq n} A_k$ and $B = \bigcap_n B_n$. Then $\mu(B_n) < 2^{1-n}$, so $\mu(B) = 0$. But $\nu(B_n) \geq \epsilon$ and $B_n \searrow B$, so by continuity from above (using $\nu(B_1) < \infty$), $\nu(B) \geq \epsilon > 0$.

This contradicts $\nu \ll \mu$. QED

22.3 The Radon-Nikodym Theorem

Theorem 22.3.1 (Radon-Nikodym):

Let $(X, \mathcal{M}, \mu)$ be a σ -finite measure space, and let ν be a σ -finite signed measure on $(X, \mathcal{M})$ with $\nu \ll \mu$. Then there exists a measurable function $f : X \rightarrow \mathbb{R}$ such that:

$$\nu(A) = \int_A f d\mu \quad \forall A \in \mathcal{M}:$$

The function f is unique μ -a.e., and is called the **Radon-Nikodym derivative** of ν with respect to μ , denoted:

$$f = \frac{d\nu}{d\mu}$$

If ν is a (non-negative) measure, then $f \geq 0$ μ -a.e.

Intuition:

The proof uses a Hilbert space argument (von Neumann's approach). The key observation is that the functional $\varphi : L^2(\mu + \nu) \rightarrow \mathbb{R}$ defined by $\varphi(g) = \int g d\nu$ is bounded, so by Riesz representation, it must come from an inner product with some function $h \in L^2(\mu + \nu)$. Unwinding the construction yields the Radon-Nikodym derivative.

Proof:

We prove the case where ν is a finite (non-negative) measure; the σ -finite and signed cases follow by decomposition.

Step 1: Set up the Hilbert space.

Let $\lambda = \mu + \nu$. Consider the Hilbert space $H = L^2(\lambda)$ with inner product:

$$\langle f, g \rangle = \int_X fg d\lambda$$

Step 2: Define a bounded linear functional.

Define $\varphi : H \rightarrow \mathbb{R}$ by:

$$\varphi(g) = \int_X g d\nu$$

This is well-defined since $\nu \leq \lambda$, and linear. To show boundedness:

$$|\varphi(g)| \leq \int_X |g| d\nu \leq \int_X |g| d\lambda \leq \sqrt{\lambda(X)} \cdot \|g\|_{L^2(\lambda)}$$

by Cauchy-Schwarz. Thus φ is bounded with $\|\varphi\| \leq \sqrt{\lambda(X)}$.

Step 3: Apply Riesz representation.

By the Riesz representation theorem for Hilbert spaces, there exists $h \in L^2(\lambda)$ such that:

$$\varphi(g) = \langle g, h \rangle = \int_X gh d\lambda \quad \forall g \in H:$$

Step 4: Show $0 \leq h \leq 1$ λ -a.e.

$h \geq 0$: Let $A = \{h < 0\}$ and $g = \mathbb{1}_A$. Then:

$$0 \leq \nu(A) = \varphi(g) = \int_A h d\lambda \leq 0$$

So $\int_A h d\lambda = 0$, which implies $h = 0$ λ -a.e. on A , contradicting $h < 0$ on A unless $\lambda(A) = 0$.

$h \leq 1$: Let $B = \{h > 1\}$ and $g = \mathbb{1}_B$. Then:

$$\nu(B) = \int_B h d\lambda > \lambda(B) = \mu(B) + \nu(B)$$

So $\mu(B) < 0$, impossible unless $\mu(B) = 0$, hence $\lambda(B) = \nu(B)$, forcing $\int_B h d\lambda = \lambda(B)$, so $h = 1$ a.e. on B , contradiction.

Step 5: Construct the Radon-Nikodym derivative.

We have $\int_X g d\nu = \int_X gh d\lambda = \int_X gh d\mu + \int_X gh d\nu$.

Rearranging: $\int_X g(1-h) d\nu = \int_X gh d\mu$.

Let $C = \{h = 1\}$. On C , we have $(1-h) = 0$, so for $g = \mathbb{1}_C$: $0 = \int_C h d\mu = \mu(C)$.

Thus $\mu(C) = 0$, so we can define $f = \frac{h}{1-h}$ on $X \setminus C$ and $f = 0$ on C .
For any $A \in \mathcal{M}$, take $g = \mathbb{1}_A \cdot \frac{1}{1-h}$ (well-defined a.e.):

$$\int_A \frac{1-h}{1-h} d\nu = \int_A \frac{h}{1-h} d\mu$$

$$\nu(A) = \int_A f d\mu$$

Step 6: Uniqueness.

If f_1 and f_2 both satisfy $\nu(A) = \int_A f_i d\mu$, then $\int_A (f_1 - f_2) d\mu = 0$ for all A . This implies $f_1 = f_2$ μ -a.e. QED

Theorem 22.3.2 (Lebesgue Decomposition):

Let μ and ν be σ -finite measures on $(X, \mathcal{M})$. There exist unique measures ν_a and ν_s such that:

1. $\nu = \nu_a + \nu_s$
2. $\nu_a \ll \mu$
3. $\nu_s \perp \mu$

Proof:

Apply Radon-Nikodym to ν with respect to $\mu + \nu$. There exists f with $0 \leq f \leq 1$ such that:

$$\nu(A) = \int_A f d(\mu + \nu)$$

Let $S = \{f = 1\}$. Define:

$$\nu_a(A) = \nu(A \cap S^c), \quad \nu_s(A) = \nu(A \cap S)$$

Then $\nu_s \perp \mu$ (supported on S where $\mu(S) = 0$ by similar argument to Step 5 above).

For $\nu_a \ll \mu$: if $\mu(A) = 0$, then on $A \cap S^c$ (where $f < 1$):

$$\nu(A \cap S^c) = \int_{A \cap S^c} f d(\mu + \nu) = \int_{A \cap S^c} f d\nu$$

This gives $(1-f)d\nu = 0$ on $A \cap S^c$, so $\nu_a(A) = 0$.

QED

22.4 Properties of the Radon-Nikodym Derivative

Proposition 22.4.1 (Chain Rule):

Let μ, ν, λ be σ -finite measures with $\nu \ll \mu \ll \lambda$. Then:

$$\frac{d\nu}{d\lambda} = \frac{d\nu}{d\mu} \cdot \frac{d\mu}{d\lambda} \quad \lambda\text{-a.e.}$$

Proof:

For any $A \in \mathcal{M}$:

$$\nu(A) = \int_A \frac{d\nu}{d\mu} d\mu = \int_A \frac{d\nu}{d\mu} \cdot \frac{d\mu}{d\lambda} d\lambda$$

By uniqueness, $\frac{d\nu}{d\lambda} = \frac{d\nu}{d\mu} \cdot \frac{d\mu}{d\lambda}$.

QED

Proposition 22.4.2 (Integration Formula):

Let $\nu \ll \mu$ with $f = \frac{d\nu}{d\mu}$. For any measurable $g : X \rightarrow [0, \infty]$:

$$\int_X g d\nu = \int_X g \cdot f d\mu$$

The same holds for integrable g (with the usual sign conventions).

Proof:

For $g = \mathbb{1}_A$, this is the definition of Radon-Nikodym derivative. Extend to simple functions by linearity, then to non-negative measurable functions by MCT. QED

Chapter Compendium:

Signed Measures: $\nu : \mathcal{M} \rightarrow [-\infty, +\infty]$ with $\nu(\emptyset) = 0$, at most one infinite value, and σ -additivity.

Hahn Decomposition: Every signed measure ν admits $X = P \sqcup N$ where P is positive and N is negative for ν . Unique up to null sets.

Jordan Decomposition: $\nu = \nu^+ - \nu^-$ where $\nu^+(A) = \nu(A \cap P)$, $\nu^-(A) = -\nu(A \cap N)$. Total variation: $|\nu| = \nu^+ + \nu^-$.

Absolute Continuity: $\nu \ll \mu$ means $\mu(A) = 0 \implies \nu(A) = 0$.

Mutual Singularity: $\mu \perp \nu$ means $X = A \sqcup B$ with μ on A , ν on B .

Radon-Nikodym Theorem: If $\nu \ll \mu$ (σ -finite), then $\exists \frac{d\nu}{d\mu} : \nu(A) = \int_A \frac{d\nu}{d\mu} d\mu$.

Lebesgue Decomposition: $\nu = \nu_a + \nu_s$ with $\nu_a \ll \mu$ and $\nu_s \perp \mu$.

Chain Rule: $\frac{d\nu}{d\lambda} = \frac{d\nu}{d\mu} \cdot \frac{d\mu}{d\lambda}$.

Integration: $\int g d\nu = \int g \cdot \frac{d\nu}{d\mu} d\mu$.

CHAPTER 23

L^p Spaces

23.1 Definitions

Definition 23.1.1 (Pre- L^p Space):

For $1 \leq p < \infty$, the **pre- L^p space** (or **curly $\mathcal{L}^p$**) is:

$$\mathcal{L}^p(\mu) = \left\{ f : X \rightarrow \mathbb{R} \text{ measurable} \mid \int_X |f|^p d\mu < \infty \right\}$$

with $\|f\|_p = \left(\int_X |f|^p d\mu \right)^{1/p}$.

Proposition 23.1.2 ($\mathcal{L}^p$ is a Vector Space):

$\mathcal{L}^p(\mu)$ is a vector subspace of the space of measurable functions $X \rightarrow \mathbb{R}$.

Proof:

Since $\mathcal{L}^p \subseteq \{f : X \rightarrow \mathbb{R} \text{ measurable}\}$ (which is a vector space), we only need to verify the subspace properties.

Non-empty: The zero function $0 \in \mathcal{L}^p$ since $\int_X 0^p d\mu = 0 < \infty$.

Closure under scalar multiplication:

Let $f \in \mathcal{L}^p$ and $c \in \mathbb{R}$. Then:

$$\int_X |cf|^p d\mu = |c|^p \int_X |f|^p d\mu < \infty$$

So $cf \in \mathcal{L}^p$.

Closure under addition:

Let $f, g \in \mathcal{L}^p$. We first show that for all $a, b \geq 0$:

$$|a + b|^p \leq 2^p \max(a, b)^p \leq 2^p(a^p + b^p)$$

Indeed: $|a + b| \leq 2 \max(a, b)$, so $|a + b|^p \leq 2^p \max(a, b)^p \leq 2^p(a^p + b^p)$.

Applying this pointwise and integrating:

$$\int_X |f + g|^p d\mu \leq 2^p \left(\int_X |f|^p d\mu + \int_X |g|^p d\mu \right) < \infty$$

So $f + g \in \mathcal{L}^p$.

QED

Definition 23.1.3 (L^p Space):

The L^p space is the quotient:

$$L^p(\mu) = \mathcal{L}^p(\mu)/\sim$$

where $f \sim g$ iff $f = g$ a.e. . The equivalence class of f is denoted $[f]$.

The L^p norm is well-defined on equivalence classes: $\|[f]\|_p = \|f\|_p$.

Remark:

We prove inequalities for $\mathcal{L}^p$ first. The corresponding results for L^p follow immediately since the seminorm $\|\cdot\|_p$ is constant on equivalence classes.

23.2 Young's Inequality

Remark:

Young's Inequality is proved in the Product Measures chapter as a consequence of Fubini-Tonelli theorem using a beautiful geometric argument. We state the result here for use in proving Hölder's inequality.

Theorem 23.2.1 (Young's Inequality for p -powers):

Let $1 < p, q < \infty$ with $\frac{1}{p} + \frac{1}{q} = 1$. Then for all $a, b \geq 0$:

$$ab \leq \frac{a^p}{p} + \frac{b^q}{q}$$

with equality iff $a^p = b^q$.

23.3 Hölder's Inequality

Theorem 23.3.1 (Hölder's Inequality for $\mathcal{L}^p$):

Let $1 < p, q < \infty$ with $\frac{1}{p} + \frac{1}{q} = 1$. For $f \in \mathcal{L}^p(\mu)$ and $g \in \mathcal{L}^q(\mu)$:

$$\int_X |fg| d\mu \leq \|f\|_p \cdot \|g\|_q$$

In particular, $fg \in \mathcal{L}^1(\mu)$.

Equality holds iff $|f|^p$ and $|g|^q$ are proportional a.e. , i.e., there exist constants $\alpha, \beta \geq 0$ (not both zero) such that $\alpha|f|^p = \beta|g|^q$ a.e.

Proof:

Case 1: $\|f\|_p = 0$ or $\|g\|_q = 0$.

Then $f = 0$ a.e. or $g = 0$ a.e., so $fg = 0$ a.e. and both sides equal 0.

Case 2: $\|f\|_p, \|g\|_q > 0$.

Step 1: Normalize. Define:

$$\tilde{f} = \frac{f}{\|f\|_p}, \quad \tilde{g} = \frac{g}{\|g\|_q}$$

Then $\|\tilde{f}\|_p = \|\tilde{g}\|_q = 1$.

Step 2: Apply Young's inequality pointwise.

For each $x \in X$, apply Young's inequality with $a = |\tilde{f}(x)|$ and $b = |\tilde{g}(x)|$:

$$|\tilde{f}(x)| \cdot |\tilde{g}(x)| \leq \frac{|\tilde{f}(x)|^p}{p} + \frac{|\tilde{g}(x)|^q}{q}$$

Step 3: Integrate.

$$\int_X |\tilde{f}\tilde{g}| d\mu \leq \frac{1}{p} \int_X |\tilde{f}|^p d\mu + \frac{1}{q} \int_X |\tilde{g}|^q d\mu = \frac{1}{p} \cdot 1 + \frac{1}{q} \cdot 1 = 1$$

Step 4: Rescale.

Since $\tilde{f}\tilde{g} = \frac{fg}{\|f\|_p \|g\|_q}$:

$$\frac{1}{\|f\|_p \|g\|_q} \int_X |fg| d\mu \leq 1$$

Thus $\int_X |fg| d\mu \leq \|f\|_p \|g\|_q$.

Step 5: Equality case.

Equality holds iff Young's inequality is an equality a.e., i.e., $|\tilde{f}(x)|^p = |\tilde{g}(x)|^q$ a.e.

This means $\frac{|f(x)|^p}{\|f\|_p^p} = \frac{|g(x)|^q}{\|g\|_q^q}$ a.e., i.e., $|f|^p$ and $|g|^q$ are proportional a.e.

QED

Corollary 23.3.2 (Hölder's Inequality for L^p):

Let $1 < p, q < \infty$ with $\frac{1}{p} + \frac{1}{q} = 1$. For $[f] \in L^p(\mu)$ and $[g] \in L^q(\mu)$:

$$\|[f][g]\|_1 \leq \|f\|_p \cdot \|g\|_q$$

Proof:

Immediate from the $\mathcal{L}^p$ version, since norms are constant on equivalence classes.

QED

23.4 Minkowski's Inequality

Theorem 23.4.1 (Minkowski's Inequality for $\mathcal{L}^p$):

For $1 \leq p < \infty$ and $f, g \in \mathcal{L}^p(\mu)$:

$$\|f + g\|_p \leq \|f\|_p + \|g\|_p$$

Equality holds for $p > 1$ iff f and g are proportional with the same sign a.e., i.e., there exist $\alpha, \beta \geq 0$ (not both zero) such that $\alpha f = \beta g$ a.e.

Proof:

Case $p = 1$: Direct from triangle inequality.

$$\int_X |f + g| d\mu \leq \int_X (|f| + |g|) d\mu = \int_X |f| d\mu + \int_X |g| d\mu$$

Case $p > 1$:

Step 1: Decompose $|f + g|^p$.

$$|f + g|^p = |f + g| \cdot |f + g|^{p-1} \leq (|f| + |g|) \cdot |f + g|^{p-1}$$

Thus:

$$|f + g|^p \leq |f| \cdot |f + g|^{p-1} + |g| \cdot |f + g|^{p-1}$$

Step 2: Integrate.

$$\int_X |f + g|^p d\mu \leq \int_X |f| \cdot |f + g|^{p-1} d\mu + \int_X |g| \cdot |f + g|^{p-1} d\mu$$

Step 3: Apply Hölder to each term.

Let $q = \frac{p}{p-1}$ (so $\frac{1}{p} + \frac{1}{q} = 1$). Note that $|f + g|^{p-1} \in \mathcal{L}^q$ with $\| |f + g|^{p-1} \|_q = \|f + g\|_p^{p-1}$.

By Hölder:

$$\int_X |f| \cdot |f + g|^{p-1} d\mu \leq \|f\|_p \cdot \|f + g\|_p^{p-1}$$

Similarly for g .

Step 4: Combine.

$$\|f + g\|_p^p \leq (\|f\|_p + \|g\|_p) \cdot \|f + g\|_p^{p-1}$$

Step 5: Divide.

If $\|f + g\|_p = 0$, the inequality holds trivially. Otherwise, divide both sides by $\|f + g\|_p^{p-1}$:

$$\|f + g\|_p \leq \|f\|_p + \|g\|_p$$

Step 6: Equality case.

Equality requires:

1. Triangle inequality: $|f + g| = |f| + |g|$ a.e. (same sign)
2. Hölder equality: $|f|^p \propto |f + g|^{(p-1)q} = |f + g|^p$ a.e.

Together, these imply f and g are proportional with the same sign a.e.

QED

Corollary 23.4.2 (Minkowski's Inequality for L^p):

For $1 \leq p < \infty$ and $[f], [g] \in L^p(\mu)$:

$$\|[f] + [g]\|_p \leq \|[f]\|_p + \|[g]\|_p$$

Proof:

Immediate from the $\mathcal{L}^p$ version.

QED

Remark:

Minkowski's inequality shows that $\|\cdot\|_p$ is a seminorm on $\mathcal{L}^p$ and a norm on L^p .

23.5 Completeness

Theorem 23.5.1 (Riesz-Fischer):

$(L^p(\mu), \|\cdot\|_p)$ is a Banach space (complete normed vector space).

Proof:

Let (f_n) be a Cauchy sequence in L^p .

Step 1: Extract a rapidly converging subsequence.

Choose a subsequence (f_{n_k}) such that:

$$\|f_{n_{k+1}} - f_{n_k}\|_p < 2^{-k}$$

Step 2: Define auxiliary function.

Let $g = \sum_{k=1}^{\infty} |f_{n_{k+1}} - f_{n_k}|$. By Minkowski:

$$\|g\|_p \leq \sum_{k=1}^{\infty} \|f_{n_{k+1}} - f_{n_k}\|_p < \sum_{k=1}^{\infty} 2^{-k} = 1$$

So $g \in \mathcal{L}^p$, hence $g < \infty$ a.e.

Step 3: Pointwise convergence.

Since $g < \infty$ a.e., the series $\sum (f_{n_{k+1}} - f_{n_k})$ converges absolutely a.e.

Define $f = \lim_{k \rightarrow \infty} f_{n_k}$ (exists a.e.). We have:

$$f = f_{n_1} + \sum_{k=1}^{\infty} (f_{n_{k+1}} - f_{n_k})$$

Step 4: $f \in \mathcal{L}^p$.

By Fatou's lemma:

$$\int_X |f|^p d\mu \leq \liminf_{k \rightarrow \infty} \int_X |f_{n_k}|^p d\mu < \infty$$

Step 5: Convergence in norm.

By DCT (dominance by g): $\|f_{n_k} - f\|_p \rightarrow 0$.

Since (f_n) is Cauchy, for any $\epsilon > 0$ there exists N such that $\|f_n - f_m\|_p < \epsilon/2$ for $n, m \geq N$.

Choose k large enough that $n_k \geq N$ and $\|f_{n_k} - f\|_p < \epsilon/2$. Then for $n \geq N$:

$$\|f_n - f\|_p \leq \|f_n - f_{n_k}\|_p + \|f_{n_k} - f\|_p < \epsilon$$

QED

Chapter Compendium:

L^p Spaces: For $1 \leq p < \infty$:

$$\mathcal{L}^p(\mu) = \{f \text{ measurable} \mid \int |f|^p < \infty\}, \quad L^p(\mu) = \mathcal{L}^p(\mu)/\sim$$

where $f \sim g$ iff $f = g$ a.e. The norm is $\|f\|_p = (\int |f|^p)^{1/p}$.

Young's Inequality: For $\frac{1}{p} + \frac{1}{q} = 1$:

$$ab \leq \frac{a^p}{p} + \frac{b^q}{q}$$

Equality iff $a^p = b^q$.

Hölder's Inequality: For $f \in \mathcal{L}^p$, $g \in \mathcal{L}^q$ with $\frac{1}{p} + \frac{1}{q} = 1$:

$$\int |fg| \leq \|f\|_p \cdot \|g\|_q$$

Equality iff $|f|^p \propto |g|^q$ a.e.

Minkowski's Inequality: For $f, g \in \mathcal{L}^p$:

$$\|f + g\|_p \leq \|f\|_p + \|g\|_p$$

This shows $\|\cdot\|_p$ is a norm on L^p .

Riesz-Fischer Theorem: $(L^p, \|\cdot\|_p)$ is a Banach space (complete).

Proof Strategy for Completeness:

1. Extract rapidly converging subsequence: $\|f_{n_{k+1}} - f_{n_k}\|_p < 2^{-k}$
2. Define $g = \sum |f_{n_{k+1}} - f_{n_k}|$, show $g \in \mathcal{L}^p$
3. Pointwise limit f exists a.e. (series converges absolutely)
4. Apply Fatou to show $f \in \mathcal{L}^p$, DCT for norm convergence

CHAPTER 24

Convergence in Measure

24.1 Definition

Definition 24.1.1:

$f_n \rightarrow f$ in measure if for every $\alpha > 0$:

$$\lim_{n \rightarrow \infty} \mu(\{x \mid |f_n(x) - f(x)| \geq \alpha\}) = 0$$

24.2 Relations Between Convergence Modes

Proposition 24.2.1: 1. L^p convergence $\implies$ convergence in measure

2. A.e. convergence + finite measure $\implies$ convergence in measure

Proof:

(1) Chebyshev: $\mu(|f_n - f| \geq \alpha) \leq \frac{1}{\alpha^p} \|f_n - f\|_p^p \rightarrow 0$.

(2) By Egorov's theorem (uniform convergence on large sets).

QED

24.3 Riesz's Theorem

Theorem 24.3.1:

If $f_n \rightarrow f$ in measure, there exists a subsequence $f_{n_k} \rightarrow f$ a.e.

Proof:

Choose n_k such that $\mu(|f_{n_k} - f| \geq 2^{-k}) < 2^{-k}$.

Let $E_k = \{|f_{n_k} - f| \geq 2^{-k}\}$. Then $\sum \mu(E_k) < \infty$.

By Borel-Cantelli: a.e. x is in only finitely many E_k , so $f_{n_k}(x) \rightarrow f(x)$.

QED

24.4 Vitali Convergence

Definition 24.4.1:

(f_n) is **uniformly integrable** if for every $\epsilon > 0$, there exists $\delta > 0$ such that:

$$\mu(A) < \delta \implies \sup_n \int_A |f_n| < \epsilon$$

Theorem 24.4.2 (Vitali):

Let $\mu(X) < \infty$ and $f_n \rightarrow f$ in measure. Then $f_n \rightarrow f$ in L^1 iff (f_n) is uniformly integrable.

Proof:

($\Rightarrow$) Suppose $f_n \rightarrow f$ in L^1 . Given $\epsilon > 0$, for large n , $\|f_n - f\|_1 < \epsilon/3$.

Since $f \in L^1$, there exists $\delta > 0$ such that $\mu(A) < \delta \implies \int_A |f| < \epsilon/3$.

For the finitely many remaining $f_1, \dots, f_N$, reduce δ so $\int_A |f_k| < \epsilon$ for $k \leq N$ when $\mu(A) < \delta$.

For $n > N$: $\int_A |f_n| \leq \int_A |f_n - f| + \int_A |f| < \epsilon/3 + \epsilon/3 < \epsilon$.

($\Leftarrow$) Suppose (f_n) is uniformly integrable and $f_n \rightarrow f$ in measure.

By Riesz, some subsequence $f_{n_k} \rightarrow f$ a.e. By Fatou, $f \in L^1$.

Given $\epsilon > 0$, choose δ from uniform integrability. By Egorov, there exists E with $\mu(E^c) < \delta$ and $f_n \rightarrow f$ uniformly on E .

Then:

$$\int |f_n - f| = \int_E |f_n - f| + \int_{E^c} |f_n - f| \leq \mu(E) \cdot \sup_E |f_n - f| + \int_{E^c} |f_n| + \int_{E^c} |f|$$

The first term $\rightarrow 0$. The last two are $< 2\epsilon$ by uniform integrability.

QED

CHAPTER 25

Applications

25.1 Differentiation Under the Integral

Theorem 25.1.1 (Leibniz Rule):

Let $f : I \times X \rightarrow \mathbb{R}$ where $I \subseteq \mathbb{R}$ is an interval. If:

1. $t \mapsto f(t, x)$ is differentiable for each x
2. $|\frac{\partial f}{\partial t}(t, x)| \leq g(x)$ for some $g \in L^1(\mu)$

Then $\phi(t) = \int_X f(t, x) d\mu(x)$ is differentiable and:

$$\phi'(t) = \int_X \frac{\partial f}{\partial t}(t, x) d\mu(x)$$

Proof:

Apply DCT to the difference quotients $\frac{f(t+h, x) - f(t, x)}{h}$.

QED

25.2 Gaussian Integral

Theorem 25.2.1:

$$\int_{\mathbb{R}} e^{-x^2} d\lambda(x) = \sqrt{\pi}.$$

Proof:

Let $I = \int_{\mathbb{R}} e^{-x^2} d\lambda(x)$. Then:

$$I^2 = \int_{\mathbb{R}^2} e^{-(x^2+y^2)} d(\lambda \otimes \lambda) = \int_{[0, 2\pi]} \int_{[0, \infty)} e^{-r^2} r d\lambda(r) d\lambda(\theta) = 2\pi \cdot \frac{1}{2} = \pi$$

Thus $I = \sqrt{\pi}$.

QED

CHAPTER 26

Exercises

26.1 σ -Algebras and Borel Sets

1. Prove that $\mathcal{B}(\mathbb{R}) = \sigma(\{(a, b) \mid a < b\})$.
2. Show $\mathbb{Q} \in \mathcal{B}(\mathbb{R})$ and $\mathbb{R} \setminus \mathbb{Q} \in \mathcal{B}(\mathbb{R})$.
3. Let $\mathcal{F} = \{(a, \infty) \mid a \in \mathbb{R}\}$. Prove $\sigma(\mathcal{F}) = \mathcal{B}(\mathbb{R})$.
4. Show that the collection of half-open intervals $\{(a, b] \mid a \leq b\}$ forms a π -system.
5. Prove the Cantor set is Borel measurable with $\lambda(C) = 0$.
6. Show that $\{f^{-1}(B) \mid B \in \mathcal{B}(\mathbb{R})\}$ is a σ -algebra for any $f : X \rightarrow \mathbb{R}$.
7. Prove that every π -system closed under complements is an algebra.
8. Show that $\sigma(\mathcal{F}_1 \cup \mathcal{F}_2) = \sigma(\sigma(\mathcal{F}_1) \cup \mathcal{F}_2)$.
9. Prove that $|\mathcal{B}(\mathbb{R})| = |\mathbb{R}|$ (cardinality of continuum).
10. Show that $\mathcal{B}(\mathbb{R}^n) = \sigma(\text{open balls})$.

26.2 Measures

1. Prove σ -subadditivity from σ -additivity using disjointification.
2. Let $A_n = [n, \infty)$. Show $\lim \lambda(A_n) \neq \lambda(\cap A_n)$. Why doesn't continuity from above apply?
3. Prove continuity from below directly from σ -additivity.
4. Let μ be counting measure on $\mathbb{N}$. Find $\mu(\{n \mid n \text{ is even}\})$.
5. Show that if $\mu(A_n) = 0$ for all n , then $\mu(\cup A_n) = 0$.
6. Prove that $\mu(A \triangle B) = 0$ defines an equivalence relation.
7. Show that $\mu(\liminf A_n) \leq \liminf \mu(A_n)$ (Fatou for sets).
8. Prove Borel-Cantelli: if $\sum \mu(A_n) < \infty$, then $\mu(\limsup A_n) = 0$.
9. Let μ be finite. Prove $\mu(\limsup A_n) \geq \limsup \mu(A_n)$ is false in general.
10. Construct a measure on $\mathbb{N}$ that is not σ -finite.
11. Prove that Dirac measure δ_x is a measure.
12. Show that $\mu(A) = \sum_{n \in A} 2^{-n}$ is a finite measure on $\mathbb{N}$.

26.3 Carathéodory and Lebesgue Measure

1. Verify that every interval $[a, b]$ is λ^* -measurable.
2. Prove $\lambda^*(A) = \inf\{\lambda(G) \mid G \text{ open}, G \supseteq A\}$.
3. Construct a Lebesgue measurable set that is not Borel.
4. Show $\mathcal{B}(\mathbb{R}) \subsetneq \text{Leb}(\mathbb{R}) \subsetneq \mathcal{P}(\mathbb{R})$.
5. Prove that the completion of Borel measure is Lebesgue measure.
6. Show $\lambda([0, 1] \cap \mathbb{Q}) = 0$ and $\lambda([0, 1] \setminus \mathbb{Q}) = 1$.
7. Prove that λ is translation-invariant: $\lambda(A + x) = \lambda(A)$.
8. Show $\lambda(cA) = |c|^n \lambda(A)$ for $A \subseteq \mathbb{R}^n$.
9. Construct a non-measurable set using the axiom of choice.
10. Prove that every set of positive measure contains a non-measurable subset.
11. Show that $\lambda^*(A) = 0$ iff A is contained in a Borel null set.
12. Prove Lebesgue's density theorem: $\lim_{r \rightarrow 0} \frac{\lambda(A \cap B_r(x))}{\lambda(B_r(x))} = 1$ for a.e. $x \in A$.

26.4 Riemann vs Lebesgue

1. Prove the Dirichlet function $\mathbb{1}_{\mathbb{Q} \cap [0, 1]}$ is Lebesgue integrable with $\int = 0$.
2. Show it is not Riemann integrable.
3. Find a function continuous a.e. that is Riemann integrable.
4. Construct f bounded with uncountable discontinuity set but still Riemann integrable.
5. Prove: Riemann integrable $\implies$ Lebesgue integrable with same value.
6. Show that $f(x) = \sin(1/x)$ on $(0, 1]$, $f(0) = 0$ is Riemann integrable.
7. Prove the Thomae function is Riemann integrable.
8. Show that $f_n(x) = x^n$ on $[0, 1]$ converges pointwise but $\int_0^1 f_n \not\rightarrow \int_0^1 \lim f_n$ in Riemann sense.
9. Compute $\int_0^1 \mathbb{1}_C dx$ where C is the Cantor set.
10. Show that if f is monotone on $[a, b]$, then f is Riemann integrable.

26.5 Measurable Functions

1. Show that continuous functions are Borel measurable.
2. Prove that monotone functions on $\mathbb{R}$ are Borel measurable.
3. Show $f \cdot g$ measurable when f, g measurable (direct proof).
4. Prove that $\{f = g\}$ is measurable when f, g are measurable.
5. Show that $\limsup f_n$ is measurable when each f_n is measurable.
6. Prove that the composition of measurable functions is measurable.
7. Show that $f^{-1}(\{y\})$ is measurable for each y iff f is measurable (false in general).
8. Prove Lusin's theorem: measurable f is continuous on a large set.
9. Show that every measurable $f \geq 0$ is a limit of simple functions.
10. Prove Egorov's theorem: a.e. convergence implies almost uniform convergence on finite measure spaces.

26.6 Convergence Theorems

1. Compute $\lim_{n \rightarrow \infty} \int_0^1 nx^{n-1} dx$. Does MCT apply? DCT?
2. Let $f_n = n \mathbb{1}_{[0, 1/n]}$. Show $f_n \rightarrow 0$ a.e. but $\int f_n = 1$.
3. Use MCT to prove $\int \sum f_n = \sum \int f_n$ for $f_n \geq 0$.
4. Apply Fatou to find $\liminf \int f_n$ when $f_n = \mathbb{1}_{[n, n+1]}$.
5. Use DCT to compute $\lim \int_0^\infty \frac{\sin(x/n)}{x} e^{-x} dx$.
6. Compute $\lim_{n \rightarrow \infty} \int_0^n (1 - x/n)^n e^{x/2} dx$.
7. Show $\lim_{n \rightarrow \infty} \int_0^\infty \frac{n \sin(x/n)}{x(1+x^2)} dx = \pi/2$.
8. Prove: if $f_n \rightarrow f$ in L^1 and $f_n \rightarrow g$ a.e., then $f = g$ a.e.
9. Show that MCT fails without monotonicity.
10. Compute $\lim_{n \rightarrow \infty} \int_0^1 \frac{nx}{1+n^2x^2} dx$.
11. Prove the reverse Fatou lemma with additional hypothesis.
12. Show $\int_0^\infty e^{-x^2} dx = \sqrt{\pi}/2$ using Fubini.

26.7 L^p Spaces

1. Let $f_n = n^{1/p} \mathbb{1}_{[1/n, 2/n]}$. Show $\|f_n\|_p = 1$ but $f_n \rightarrow 0$ pointwise.
2. Prove Young's inequality: $ab \leq \frac{a^p}{p} + \frac{b^q}{q}$.
3. Use Hölder to show $\|f\|_1 \leq \|f\|_2 \mu(X)^{1/2}$ for finite measure.
4. Prove L^2 convergence implies L^1 convergence on finite measure spaces.
5. Find $f \in L^1([0, 1])$ but $f \notin L^2([0, 1])$.
6. Show $L^p \subseteq L^q$ for $p > q$ on finite measure spaces.
7. Prove that L^2 is a Hilbert space with inner product $\langle f, g \rangle = \int f \bar{g}$.
8. Show that simple functions are dense in L^p .
9. Prove that $C_c(\mathbb{R})$ is dense in $L^p(\mathbb{R})$ for $p < \infty$.
10. Find the dual exponent q for $p = 3/2$.
11. Show $\|f\|_\infty = \lim_{p \rightarrow \infty} \|f\|_p$ on finite measure spaces.
12. Prove that L^∞ is not separable on $[0, 1]$.

26.8 Product Measures

1. Use Fubini to compute $\int_0^1 \int_0^1 \frac{x-y}{(x+y)^3} dy dx$.
2. Compute $\int_{\mathbb{R}^2} e^{-(x^2+y^2)} dx dy$ using polar coordinates.
3. Show iterated integrals may differ if $f \notin L^1$.
4. Prove $\mathcal{B}(\mathbb{R}^2) = \mathcal{B}(\mathbb{R}) \otimes \mathcal{B}(\mathbb{R})$.
5. Compute $\int_0^\infty \frac{\sin x}{x} dx$ using parameter integrals.
6. Use Fubini to prove $\int_0^\infty \frac{e^{-ax} - e^{-bx}}{x} dx = \ln(b/a)$.
7. Compute $\int_0^1 \int_0^1 \frac{1}{1-xy} dx dy$ (hint: expand as series).
8. Show that $(\mu \otimes \nu) \otimes \rho = \mu \otimes (\nu \otimes \rho)$.
9. Prove Cavalieri's principle for Lebesgue measure.
10. Compute the volume of the n -ball using Fubini.

26.9 Convergence in Measure

1. Show L^p convergence implies convergence in measure.
2. Find (f_n) converging in measure but not a.e.
3. Prove Riesz: convergence in measure has a.e. convergent subsequence.
4. Define uniform integrability. Give an example.
5. State and apply Vitali's theorem.
6. Show that a.e. convergence does not imply convergence in measure (infinite measure).
7. Prove: $f_n \rightarrow f$ in measure and $g_n \rightarrow g$ in measure implies $f_n + g_n \rightarrow f + g$ in measure.
8. Show that $f_n \rightarrow f$ in measure iff every subsequence has a further subsequence converging a.e.
9. Prove that uniform integrability + convergence in measure implies L^1 convergence.
10. Give an example where $f_n \rightarrow f$ in L^1 but not uniformly.

26.10 Parameter Integrals

1. Prove $\phi(t) = \int_0^\infty e^{-tx} \sin x \, dx$ is differentiable for $t > 0$.
2. Compute $\int_0^\infty \frac{e^{-x} - e^{-2x}}{x} dx$ using parameter integrals.
3. Prove $\Gamma(n+1) = n!$ where $\Gamma(s) = \int_0^\infty x^{s-1} e^{-x} dx$.
4. Justify differentiation under the integral for $\int_0^1 x^t dx$.
5. Use Leibniz rule to compute $\frac{d}{dt} \int_0^t f(t, x) dx$.
6. Compute $\int_0^\infty \frac{\ln x}{x^2 - 1} dx$.
7. Prove $\Gamma(1/2) = \sqrt{\pi}$.
8. Use parameter integrals to compute $\int_0^\infty \frac{\sin x}{x} dx = \pi/2$.
9. Prove the beta function identity: $B(a, b) = \frac{\Gamma(a)\Gamma(b)}{\Gamma(a+b)}$.
10. Compute $\int_0^\infty x^n e^{-x^2} dx$ in terms of Γ .

26.11 Modes of Convergence

1. Construct (f_n) on $[0, 1]$ with $f_n \rightarrow 0$ in measure but $f_n(x) \not\rightarrow 0$ for any x .
2. Show: a.e. convergence + domination implies L^p convergence.
3. Prove that on finite measure spaces, L^p convergence implies L^q convergence for $q < p$.
4. Show that uniform convergence implies L^p convergence on finite measure spaces.

5. Prove that $f_n \rightarrow f$ uniformly iff $\|f_n - f\|_\infty \rightarrow 0$.
6. Show the "typewriter sequence" converges in measure but not a.e.
7. Prove: L^1 convergence + a.e. convergence to g implies $f = g$ a.e.
8. Give an example of L^p convergence without a.e. convergence.

26.12 Simple Functions and Integration

1. Show that $\int(f + g) = \int f + \int g$ for integrable f, g .
2. Prove that $\int cf = c \int f$ for $c \in \mathbb{R}$.
3. Show that $f \leq g$ a.e. implies $\int f \leq \int g$.
4. Prove that $|\int f| \leq \int |f|$ (triangle inequality).
5. Show that $\int f = 0$ for all measurable E implies $f = 0$ a.e.
6. Prove Chebyshev's inequality: $\mu(\{|f| \geq \alpha\}) \leq \frac{1}{\alpha} \int |f|$.
7. Show that if $\int |f|^p < \infty$ and $\int |f|^q < \infty$, then $\int |f|^r < \infty$ for $p < r < q$.
8. Prove Markov's inequality: $\mu(\{f \geq \alpha\}) \leq \frac{\int f}{\alpha}$ for $f \geq 0$.

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